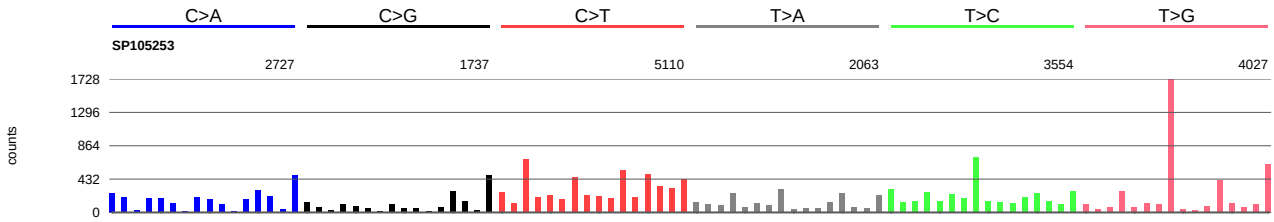
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SP105253
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5110
2063
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4027
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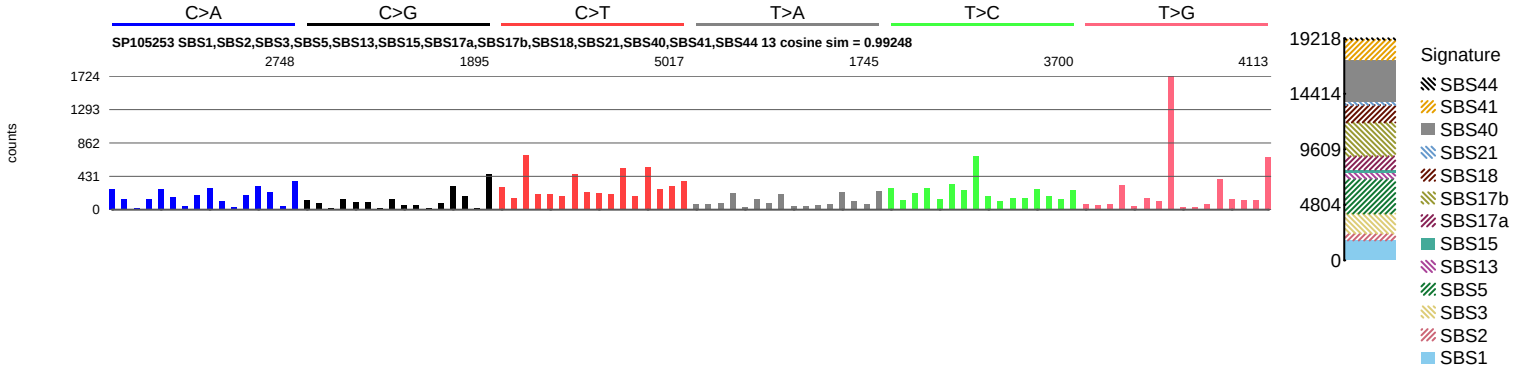
864
432
0
counts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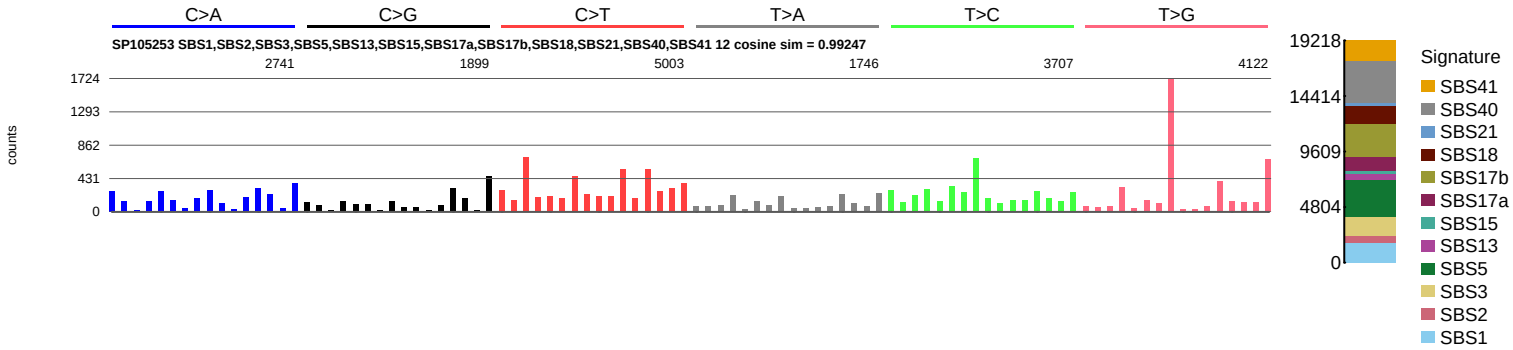
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 13 cosine sim = 0.99248
2748
1895
5017
1745
3700
4113
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 12 cosine sim = 0.99247
2741
1899
5003
1746
3707
4122
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
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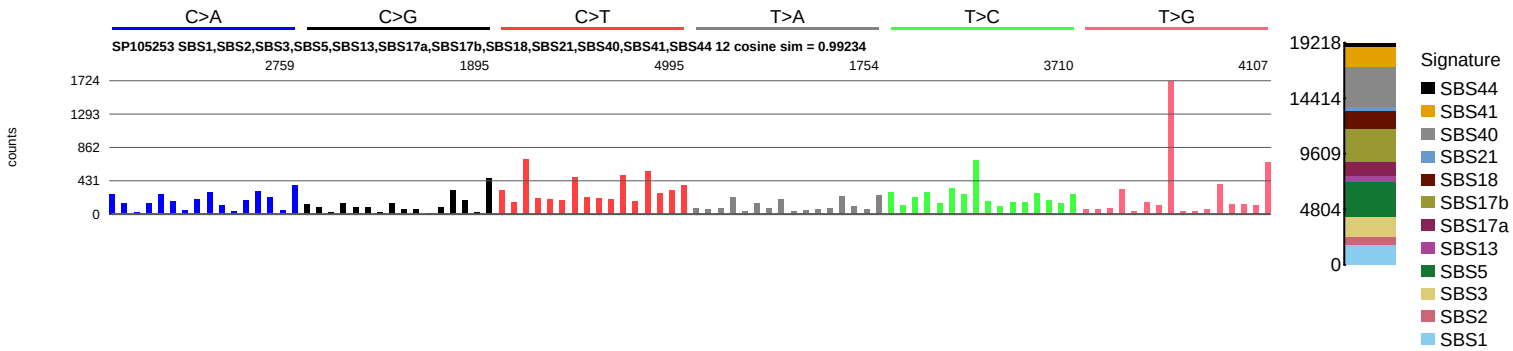
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99234
2759
1895
4995
1754
3710
4107
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
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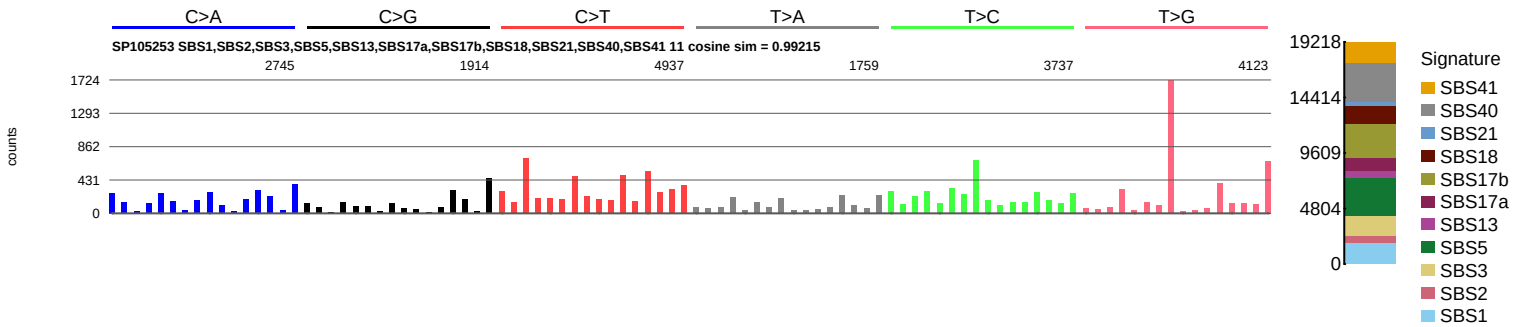
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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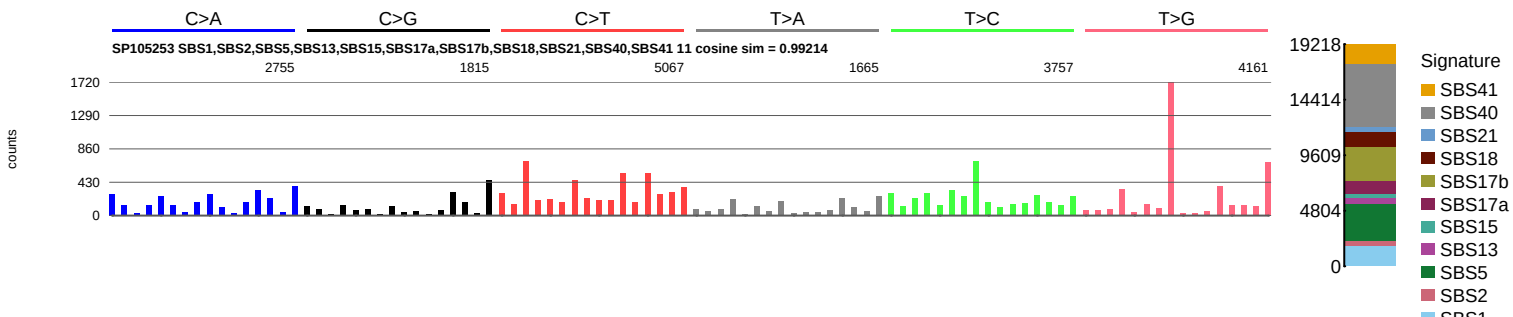
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99215
2745
1914
4937
1759
3737
4123
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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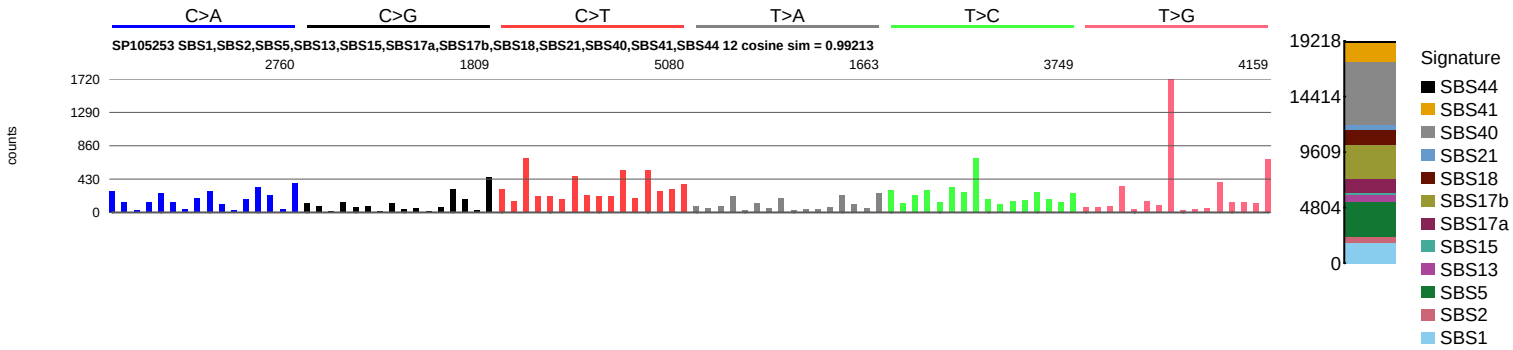
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99214
2755
1815
5067
1665
3757
4161
1720
1290
860
430
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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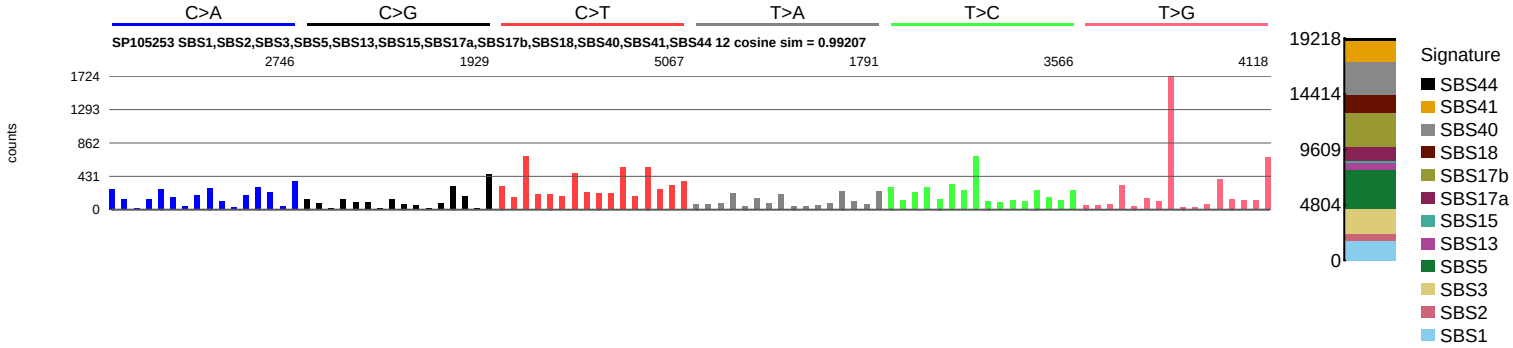
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99213
2760
1809
5080
1663
3749
4159
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
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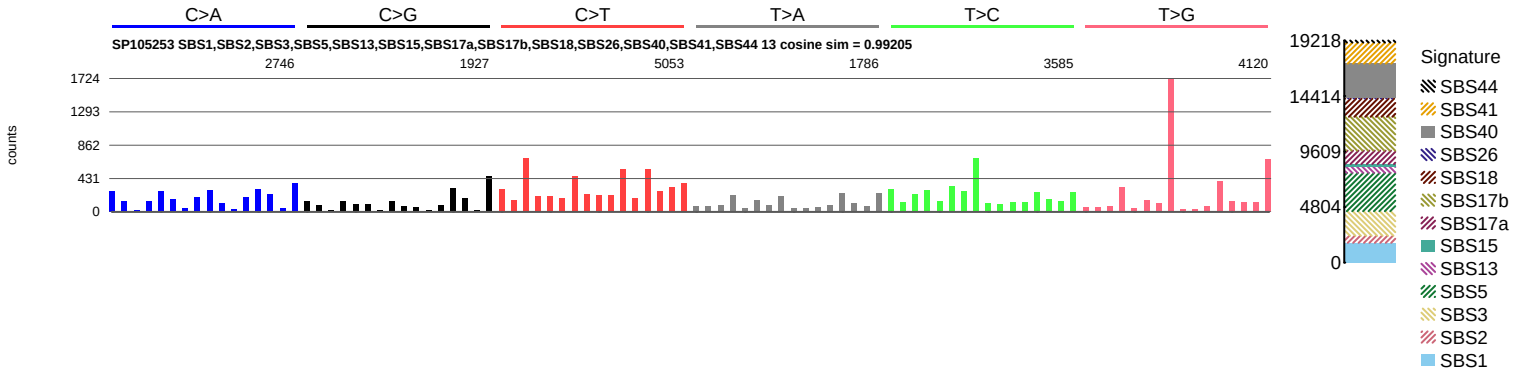
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 12 cosine sim = 0.99207
2746
1929
5067
1791
3566
4118
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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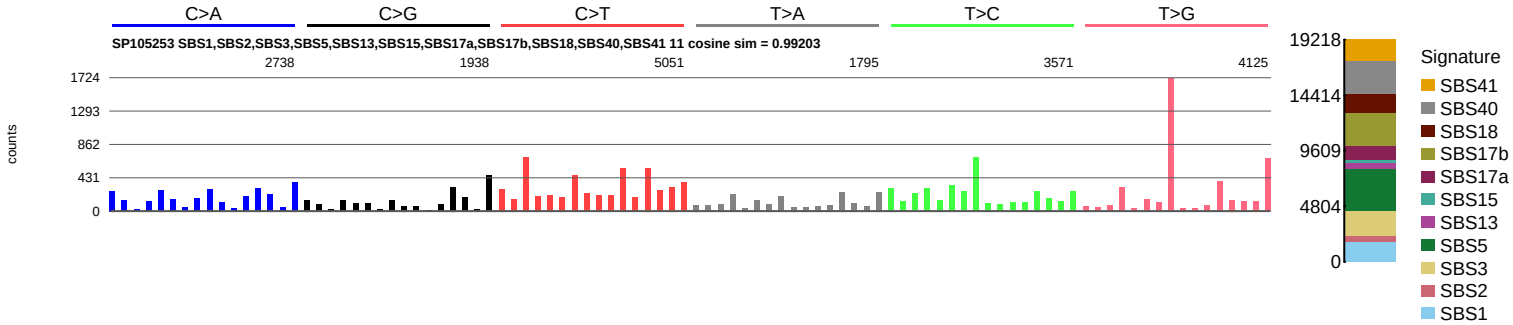
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.99205
2746
1927
5053
1786
3585
4120
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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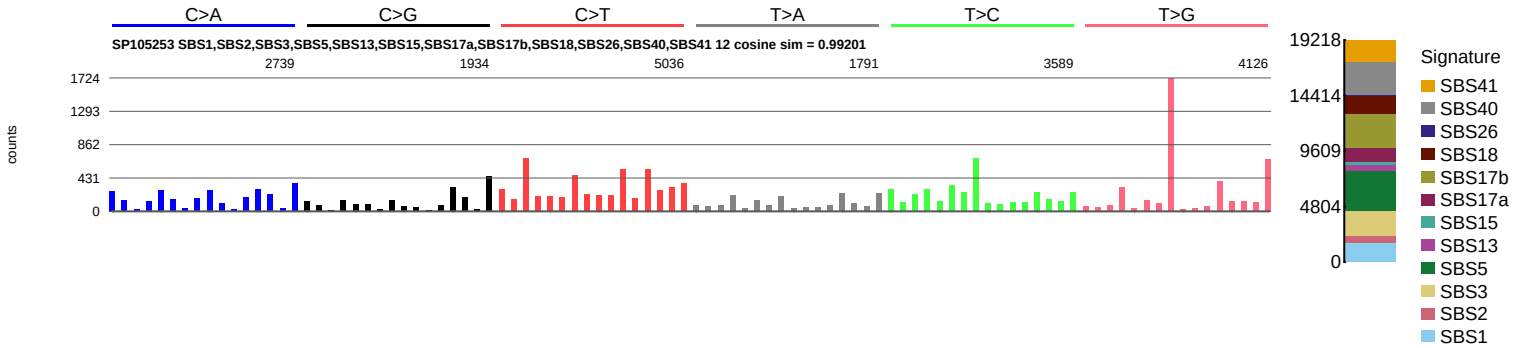
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 11 cosine sim = 0.99203
2738
1938
5051
1795
3571
4125
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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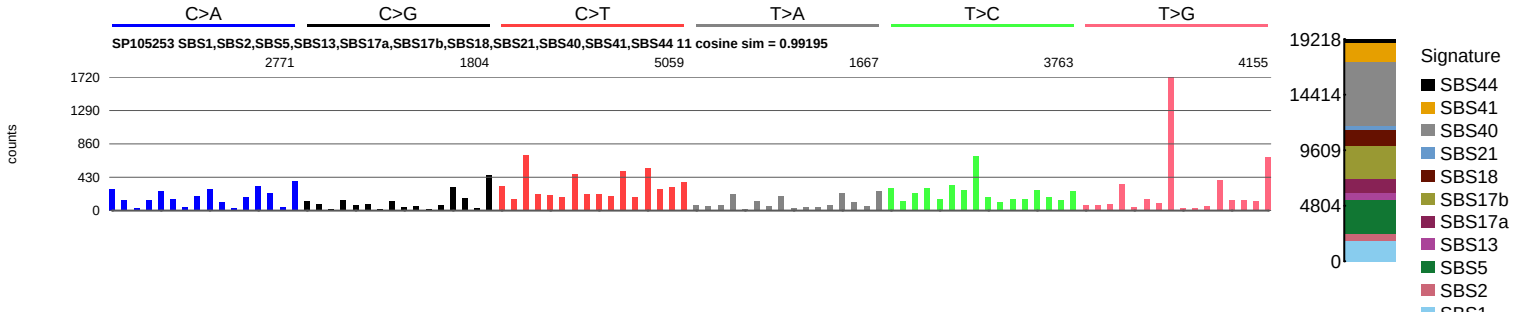
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 12 cosine sim = 0.99201
2739
1934
5036
1791
3589
4126
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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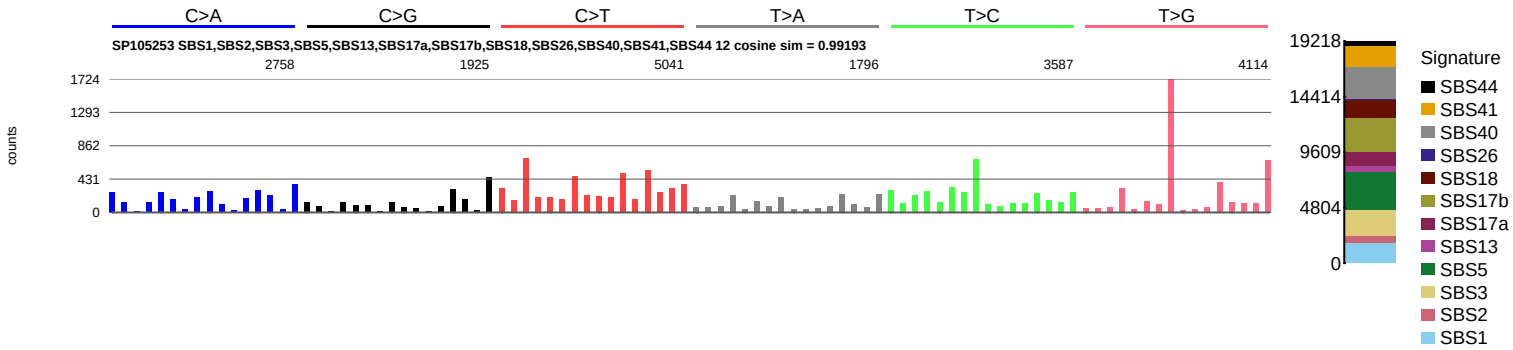
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99195
2771
1804
5059
1667
3763
4155
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2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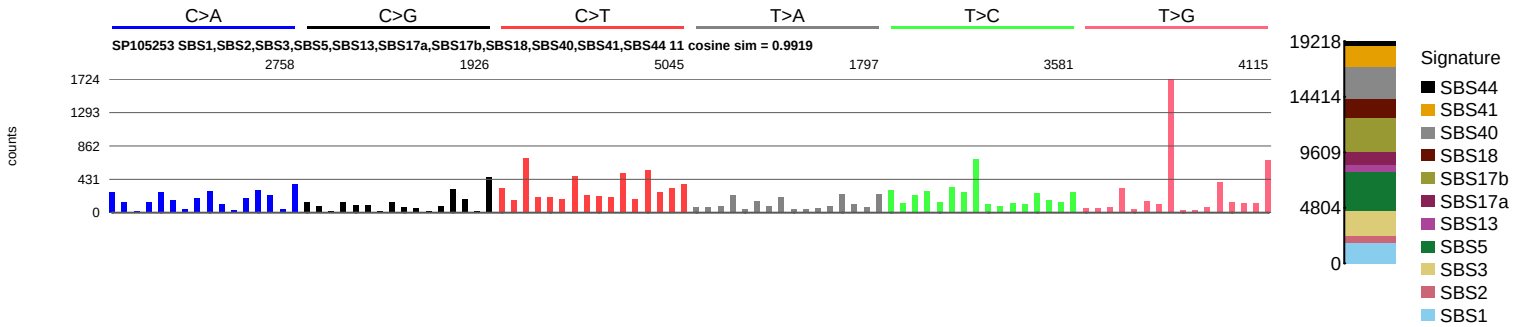
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99193
2758
1925
5041
1796
3587
4114
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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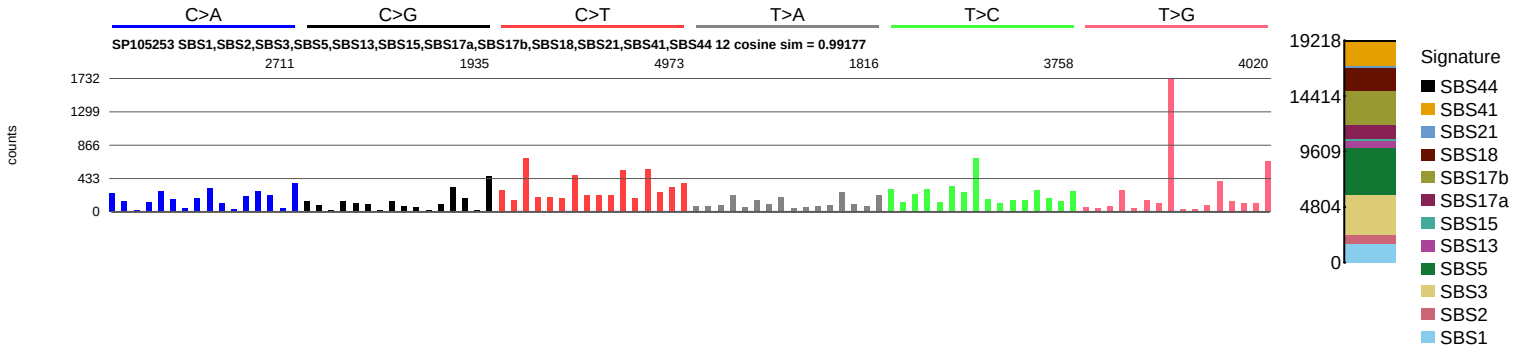
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.9919
2758
1926
5045
1797
3581
4115
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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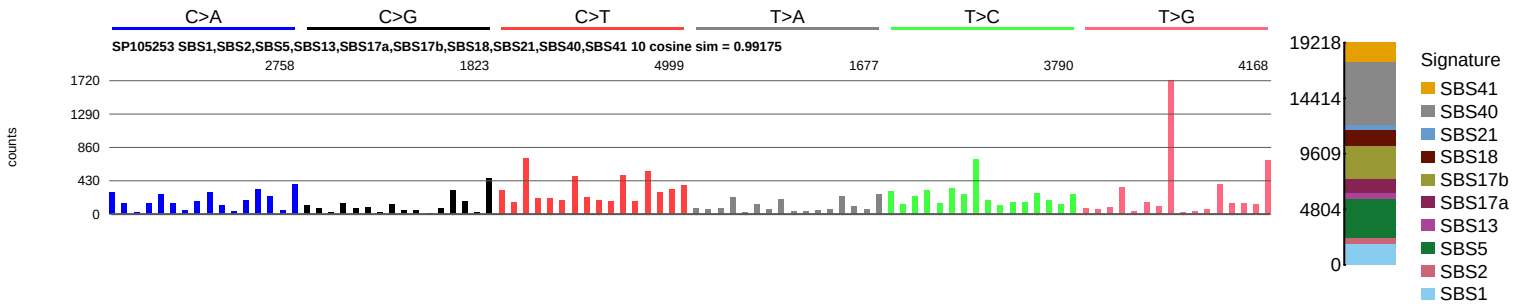
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 12 cosine sim = 0.99177
2711
1935
4973
1816
3758
4020
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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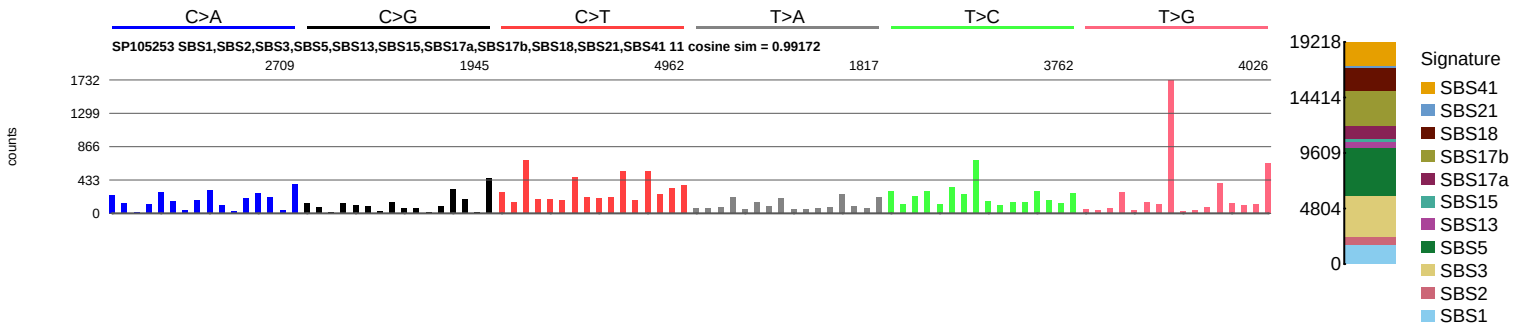
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99175
2758
1823
4999
1677
3790
4168
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
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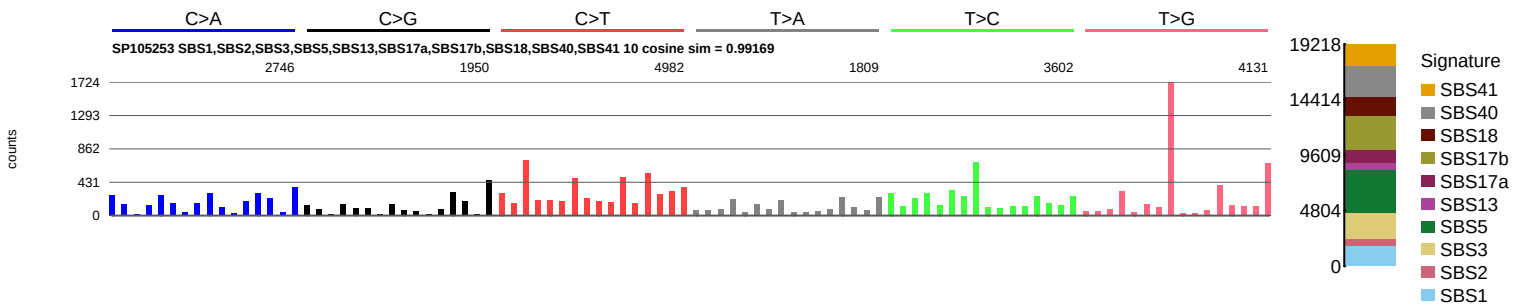
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99172
2709
1945
4962
1817
3762
4026
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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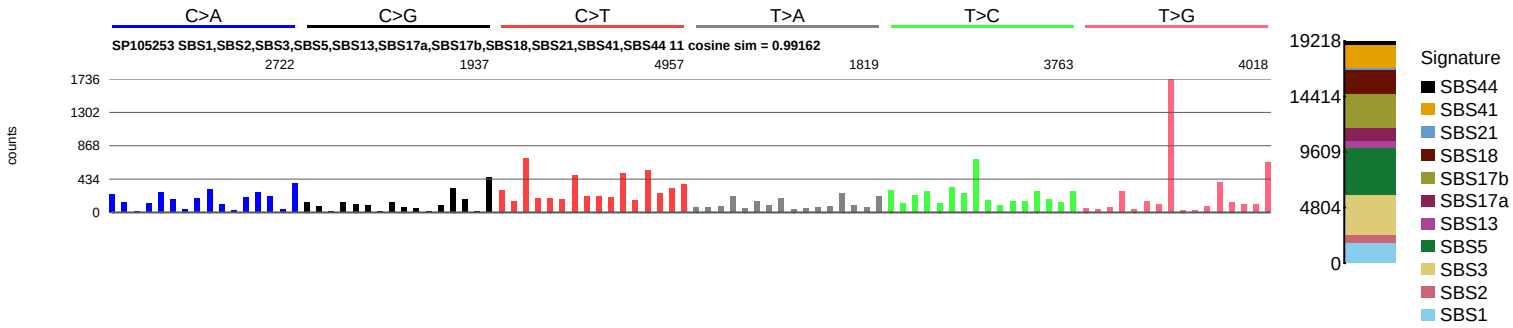
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.99169
2746
1950
4982
1809
3602
4131
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
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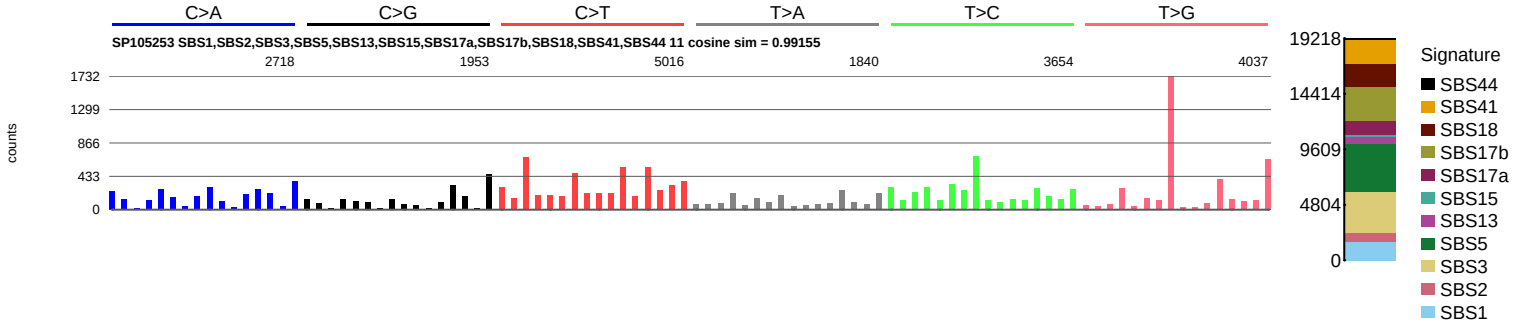
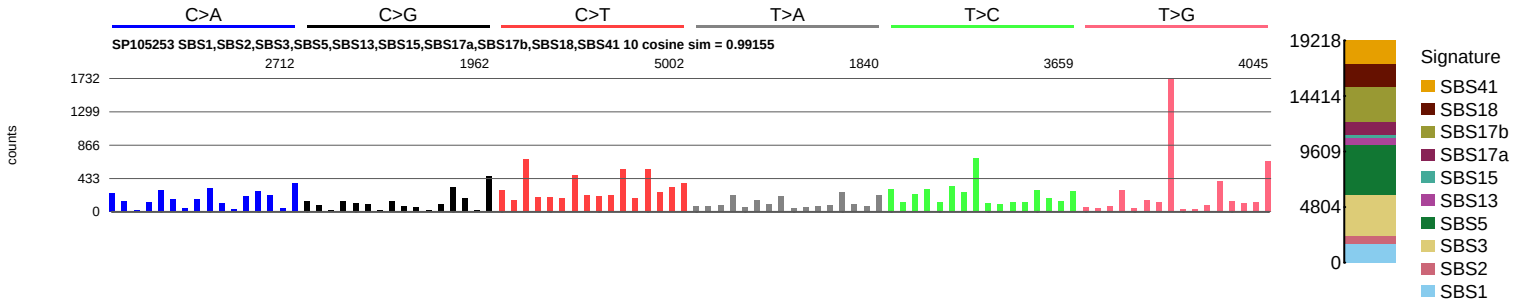
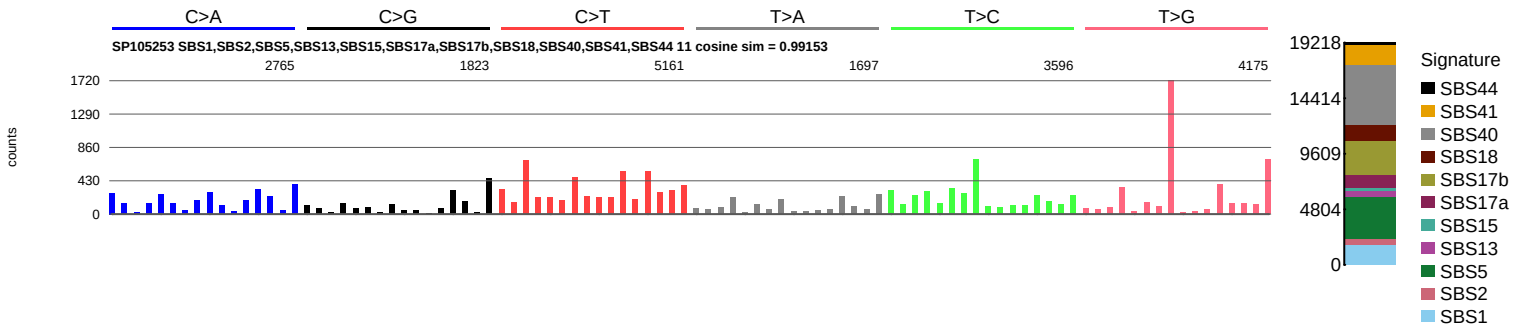
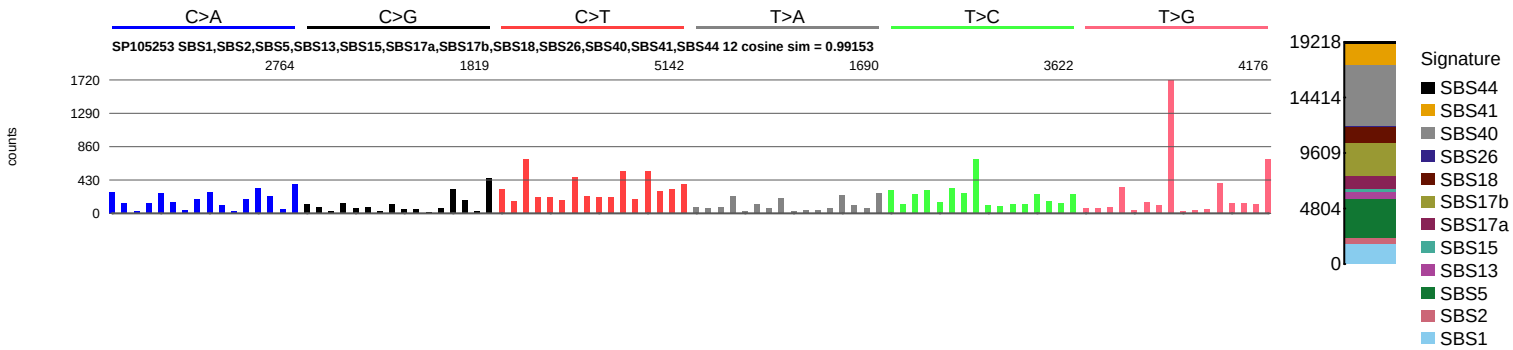
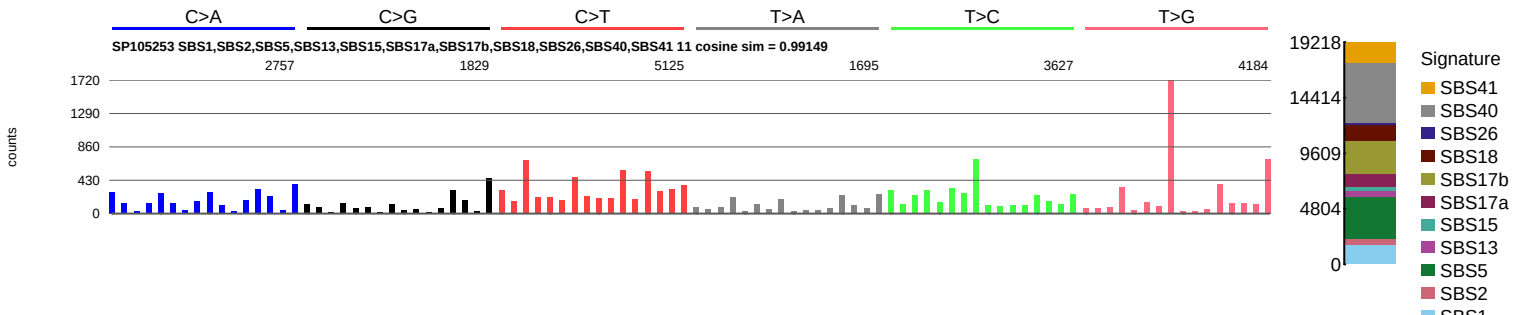
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.99162
2722
1937
4957
1819
3763
4018
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 11 cosine sim = 0.99155
2718
1953
5016
1840
3654
4037
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99155
2712
1962
5002
1840
3659
4045
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.99153
2765
1823
5161
1697
3596
4175
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99153
2764
1819
5142
1690
3622
4176
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.99149
2757
1829
5125
1695
3627
4184
counts
1720
1290
860
430
0
19218
14414
9609
4804
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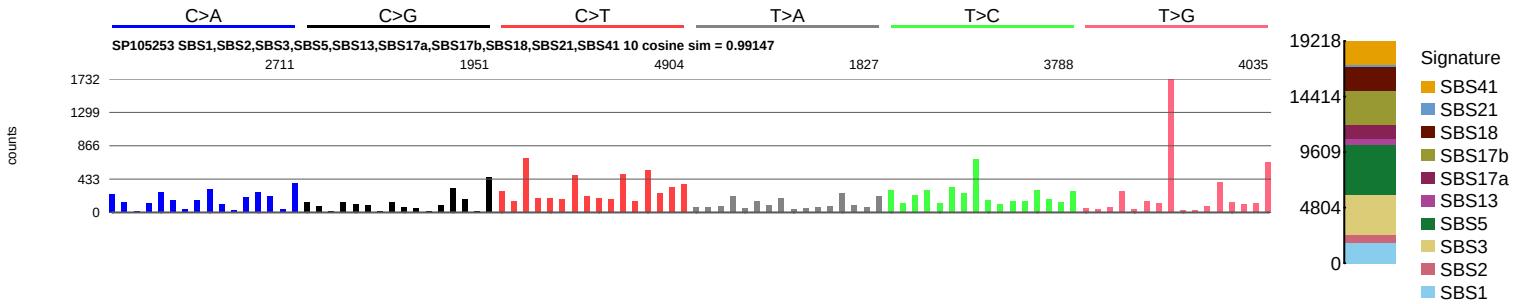
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99147
2711
1951
4904
1827
3788
4035
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
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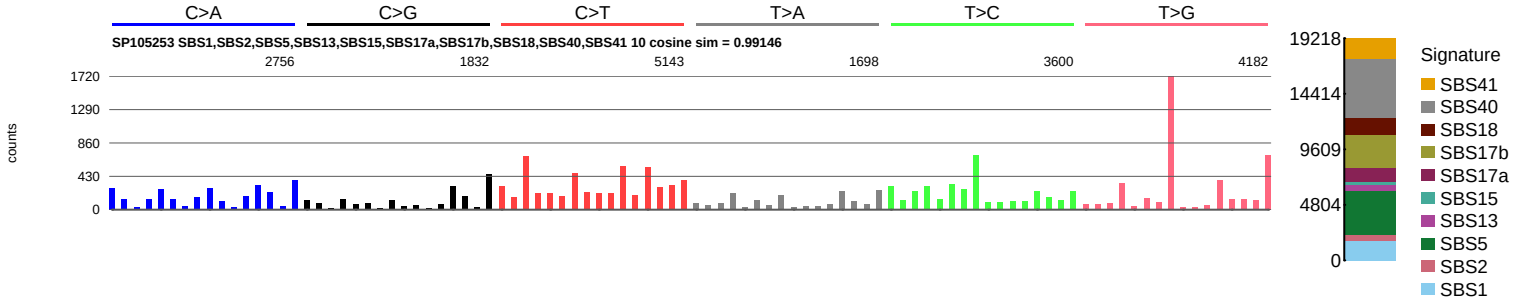
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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2756
1832
5143
1698
3600
4182
counts
1720
1290
860
430
0
19218
14414
9609
4804
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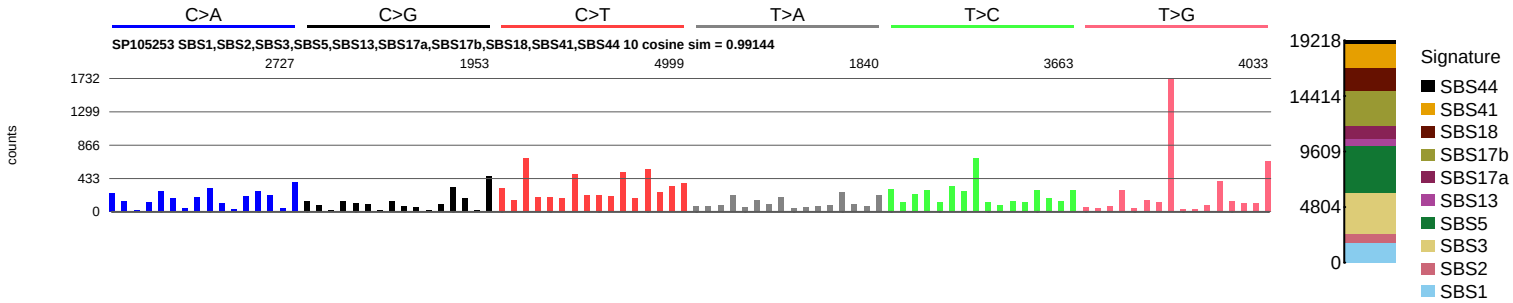
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.99144
2727
1953
4999
1840
3663
4033
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
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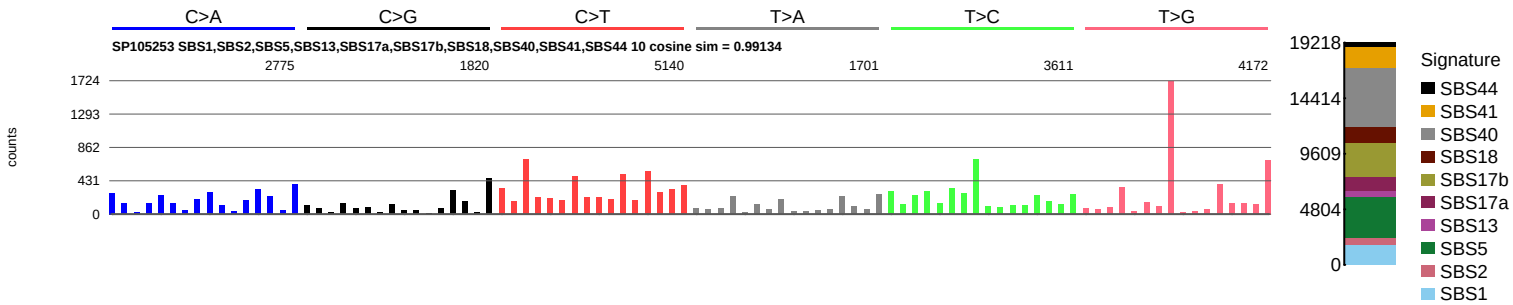
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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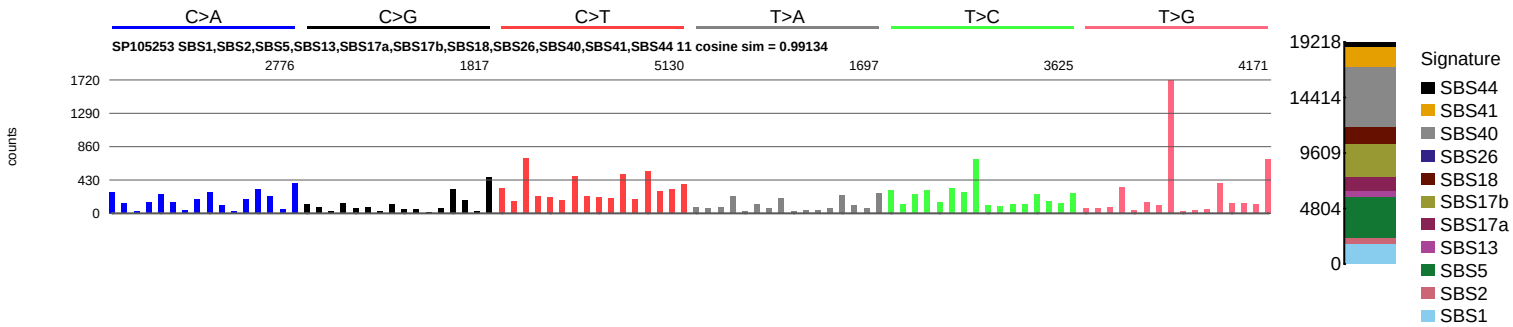
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.99134
2775
1820
5140
1701
3611
4172
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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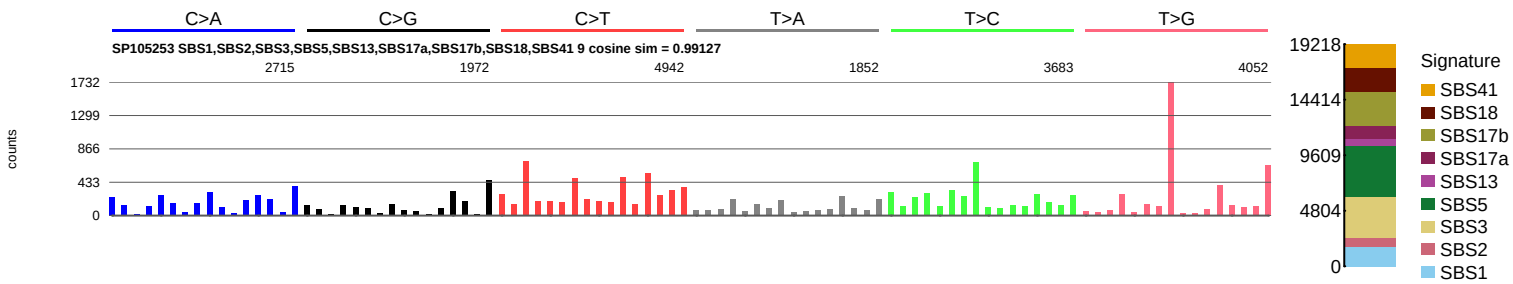
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99134
2776
1817
5130
1697
3625
4171
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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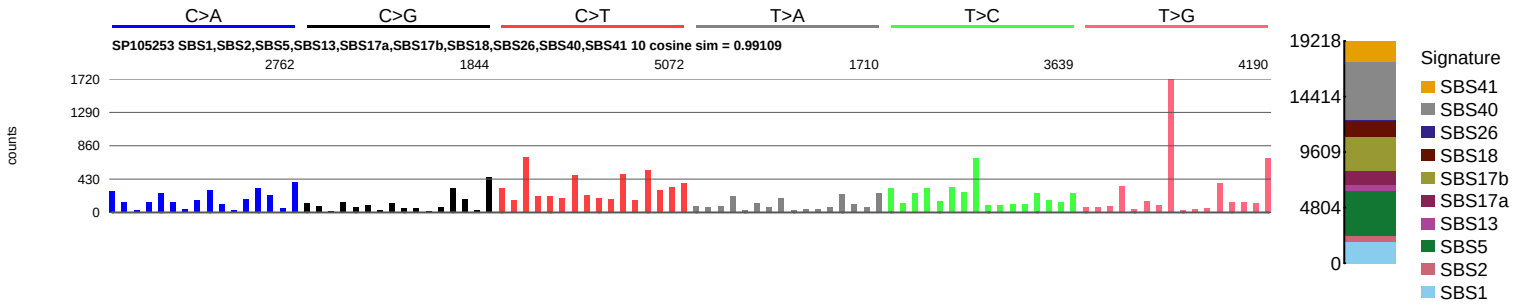
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99127
2715
1972
4942
1852
3683
4052
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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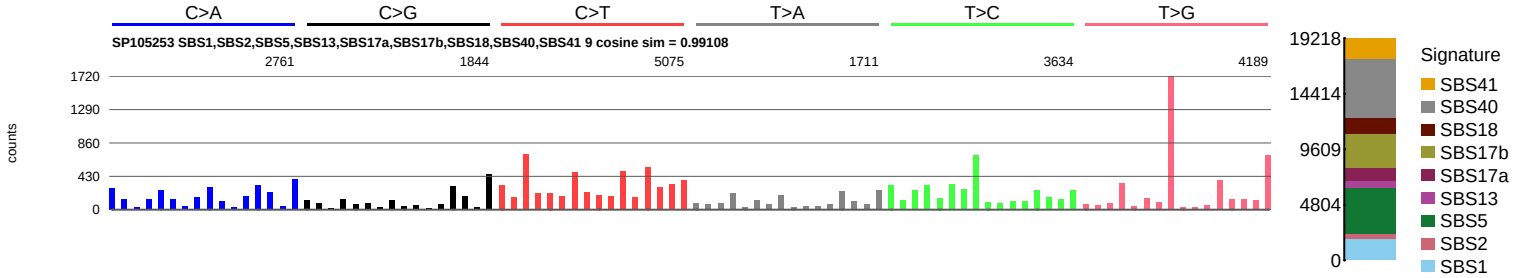
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99109
2762
1844
5072
1710
3639
4190
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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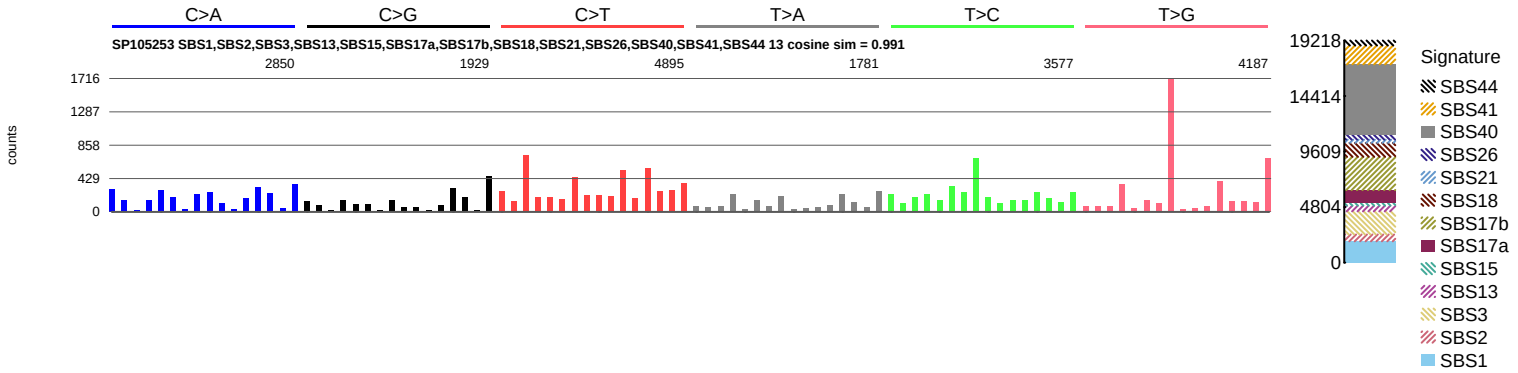
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99108
2761
1844
5075
1711
3634
4189
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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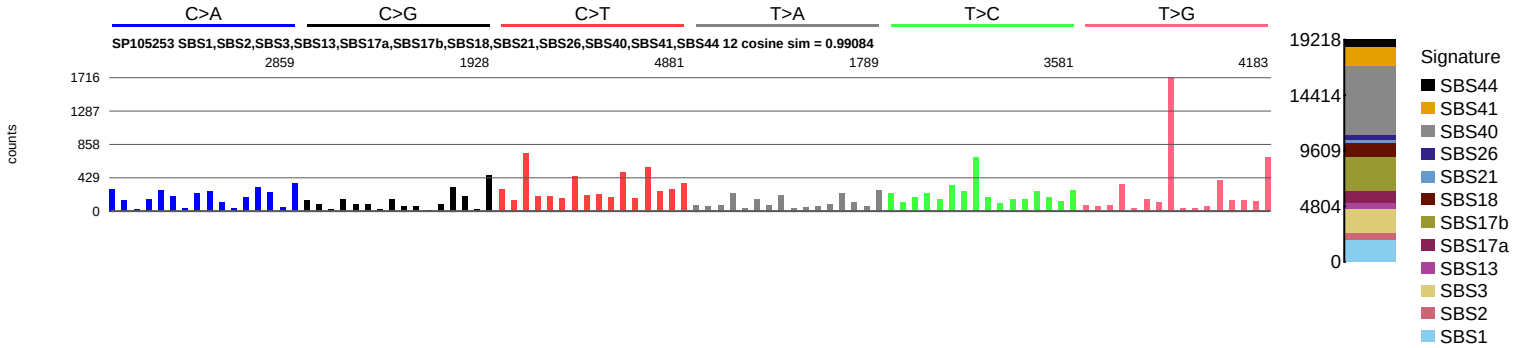
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 13 cosine sim = 0.991
2850
1929
4895
1781
3577
4187
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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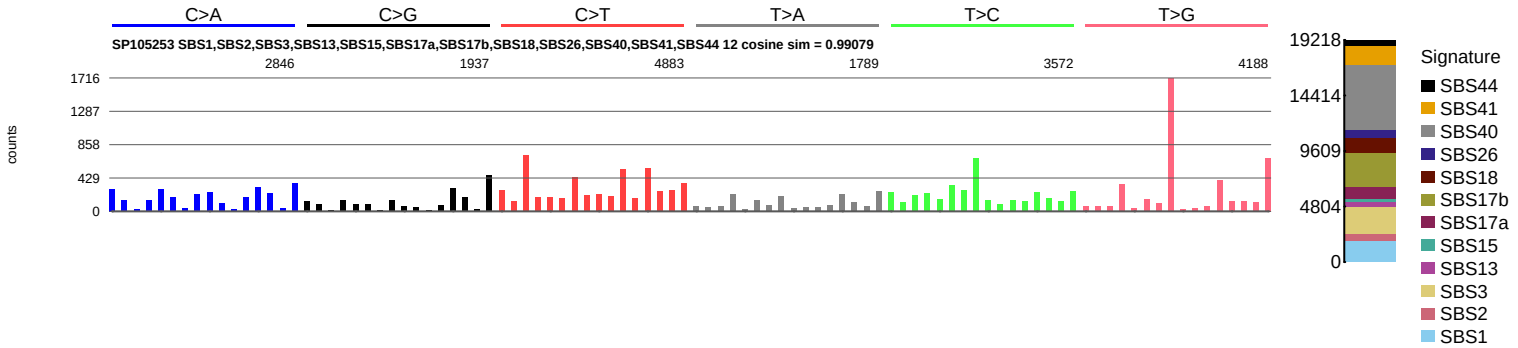
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99084
2859
1928
4881
1789
3581
4183
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
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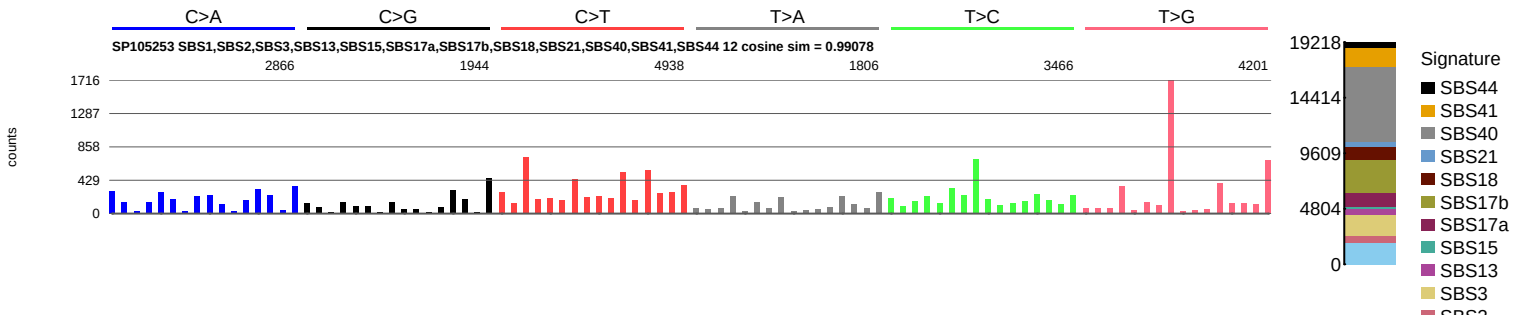
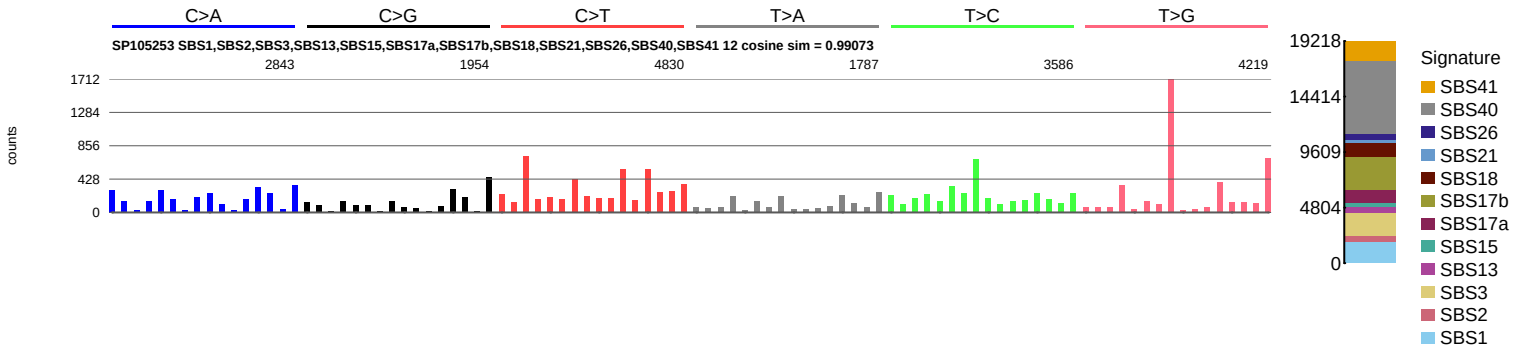
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99079
2846
1937
4883
1789
3572
4188
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.99078
2866
1944
4938
1806
3466
4201
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3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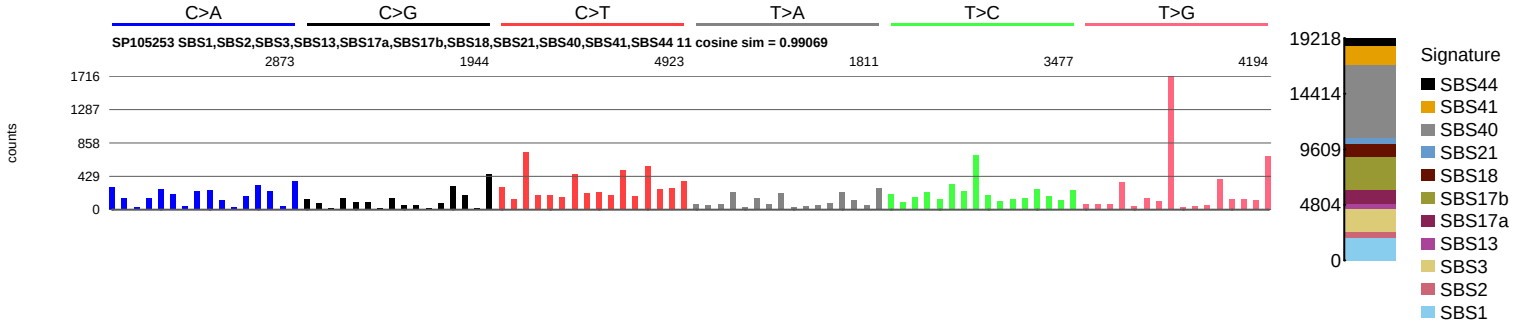
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 12 cosine sim = 0.99073
2843
1954
4830
1787
3586
4219
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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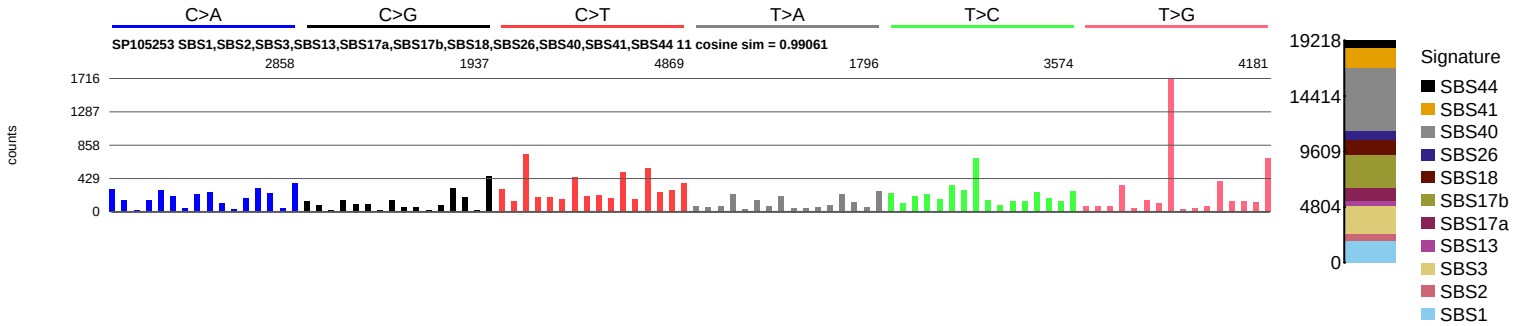
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99069
2873
1944
4923
1811
3477
4194
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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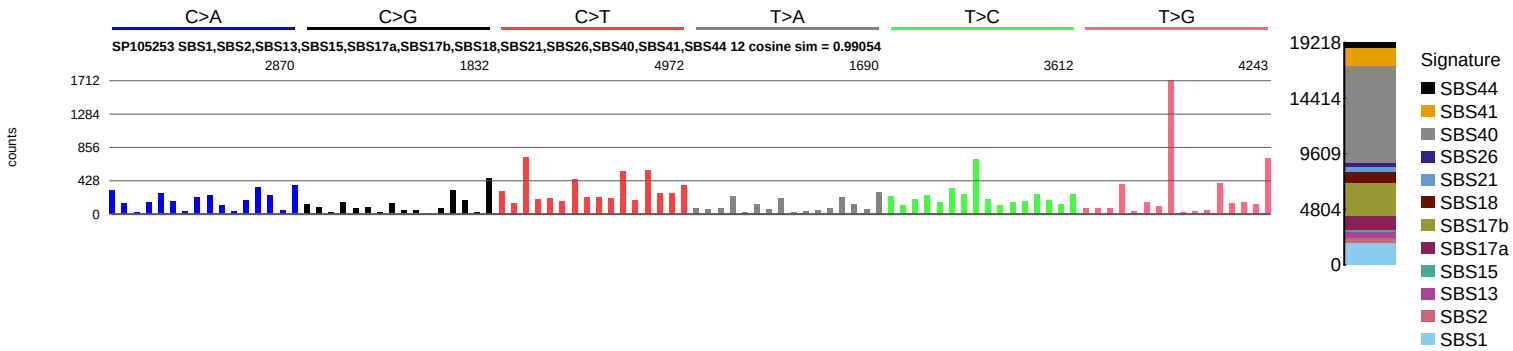
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99061
2858
1937
4869
1796
3574
4181
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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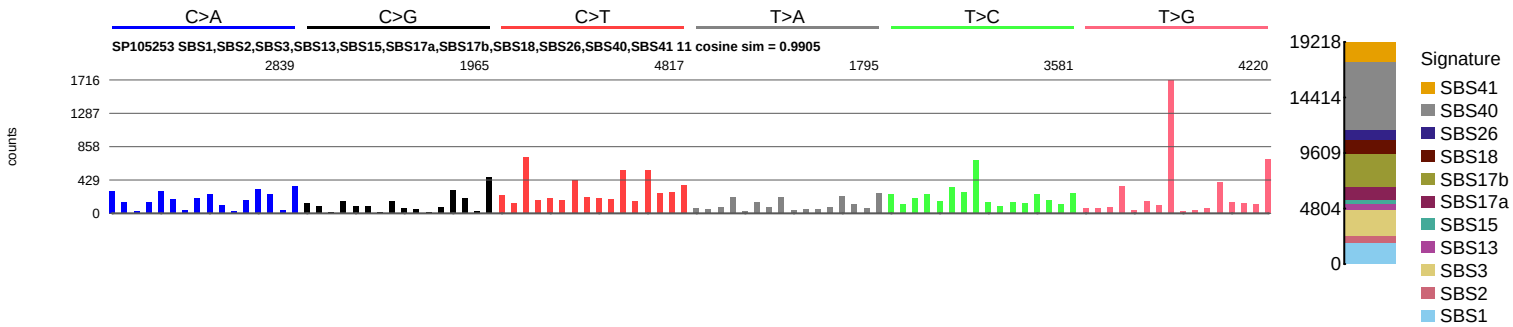
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.99054
2870
1832
4972
1690
3612
4243
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.9905
2839
1965
4817
1795
3581
4220
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
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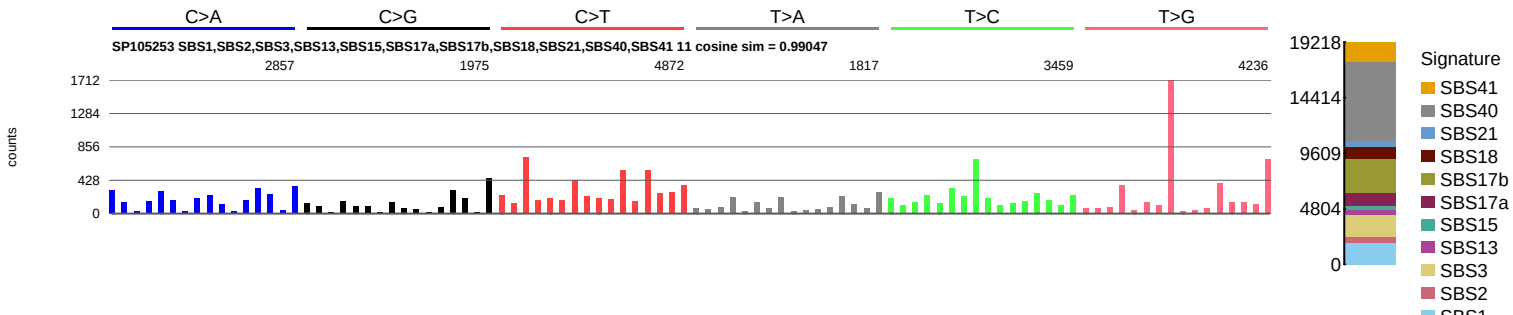
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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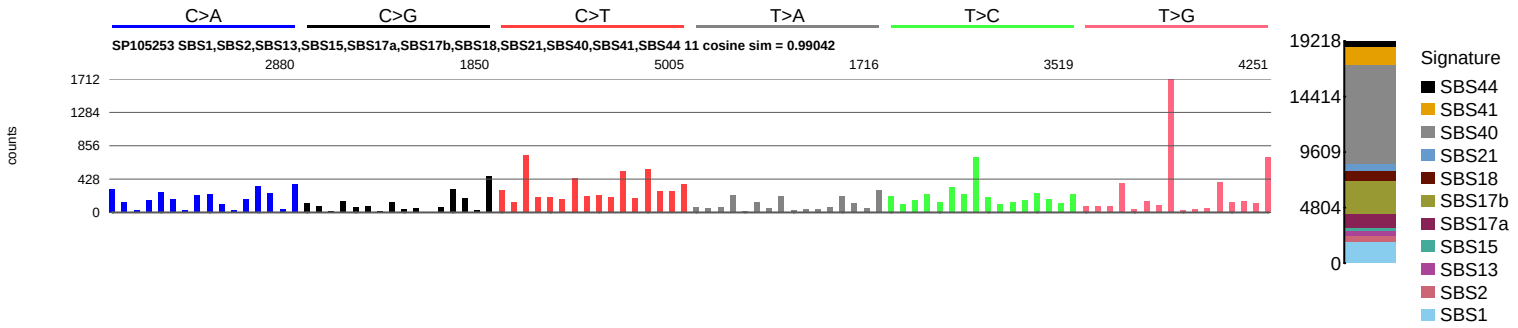
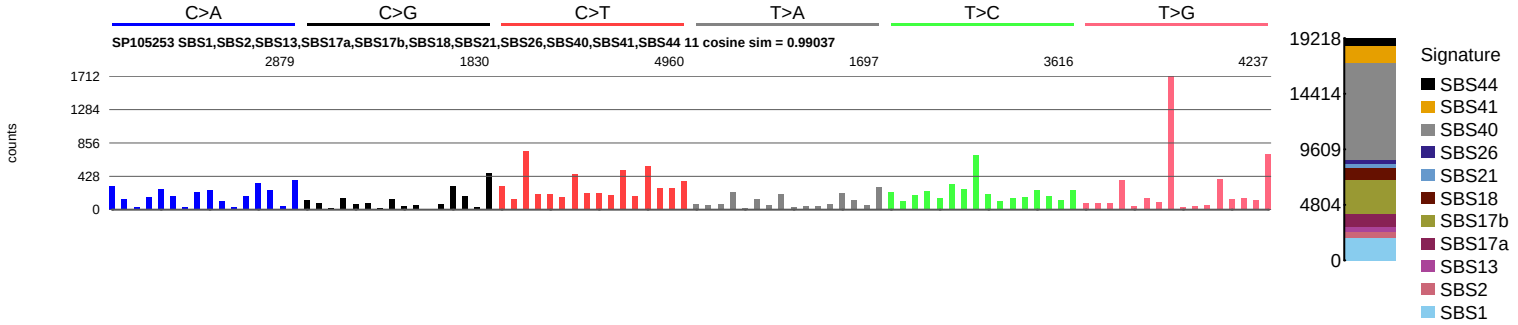
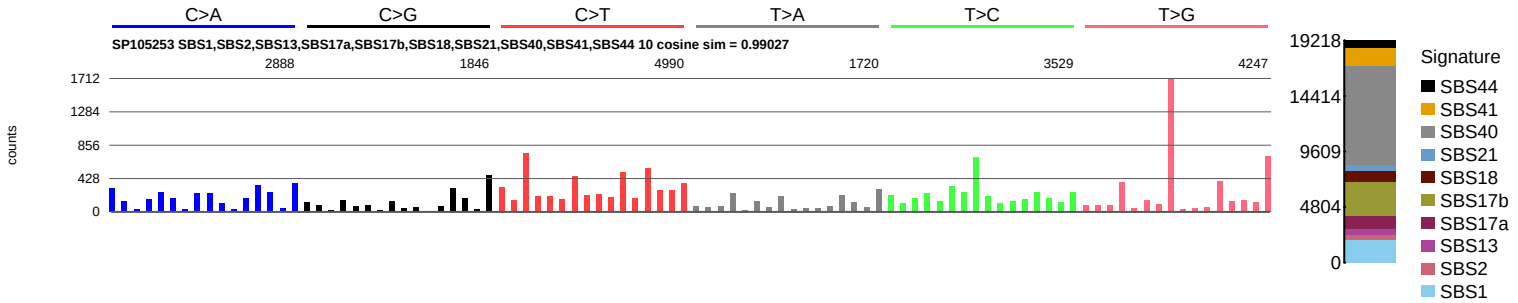
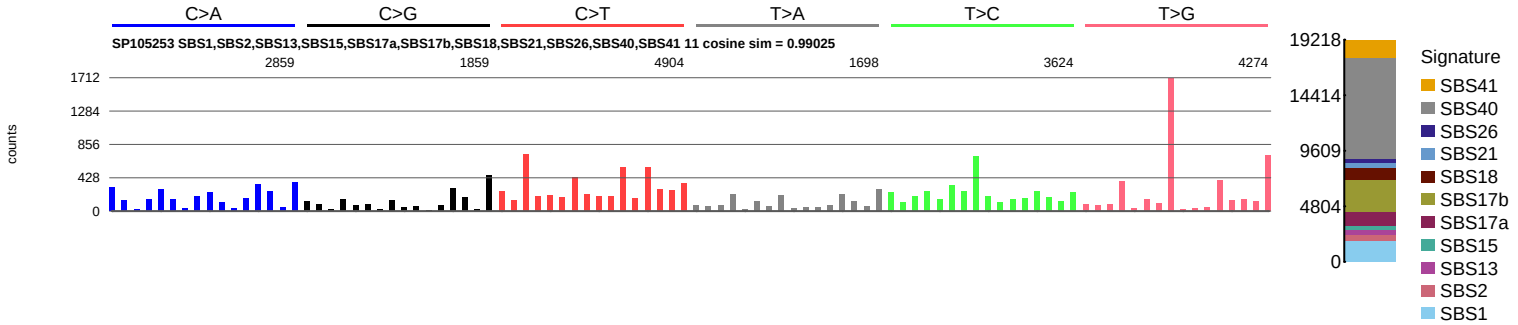
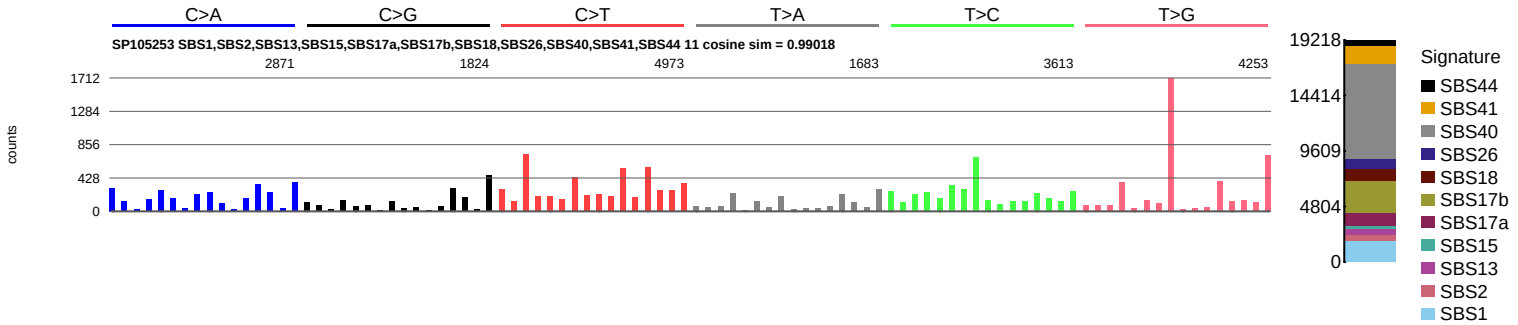
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.99047
2857
1975
4872
1817
3459
4236
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.99042
2880
1850
5005
1716
3519
4251
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99037
2879
1830
4960
1697
3616
4237
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.99027
2888
1846
4990
1720
3529
4247
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.99025
2859
1859
4904
1698
3624
4274
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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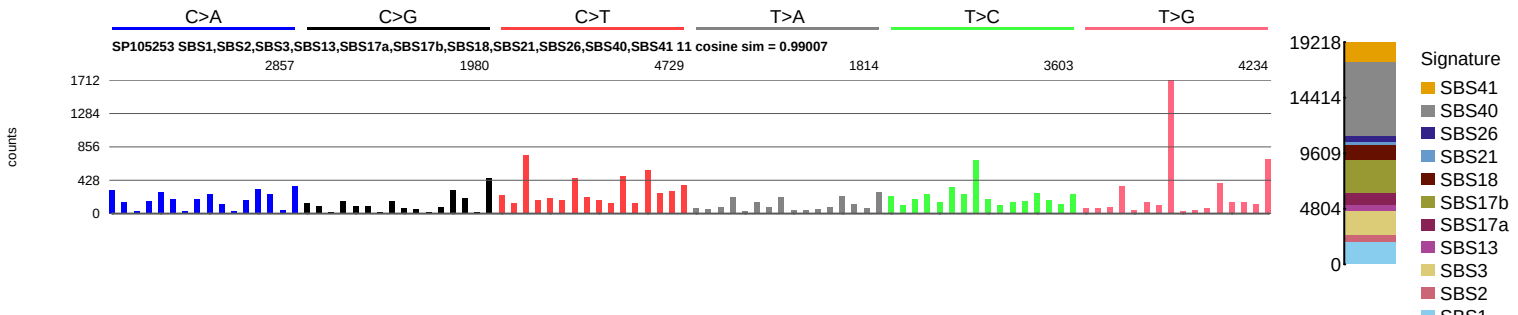
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.99018
2871
1824
4973
1683
3613
4253
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.99007
2857
1980
4729
1814
3603
4234
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
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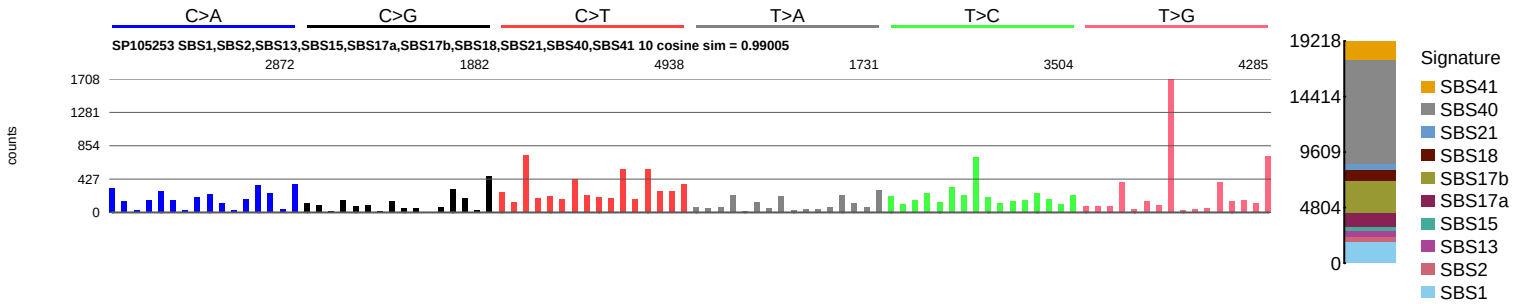
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2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SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99005
2872
1882
4938
1731
3504
4285
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
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SBS21
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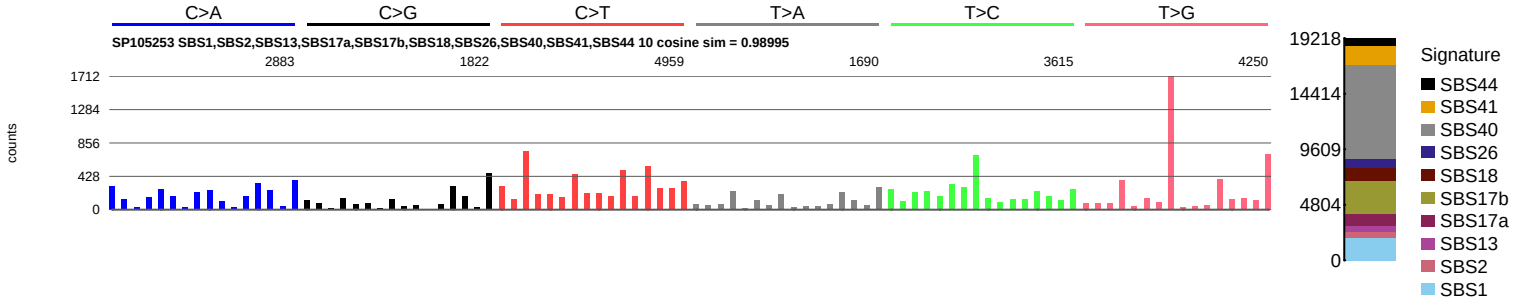
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
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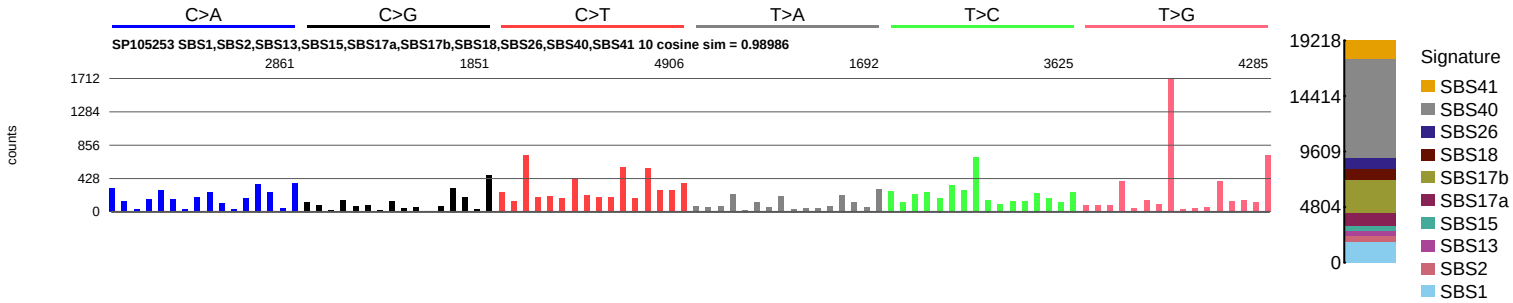
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98995
2883
1822
4959
1690
3615
4250
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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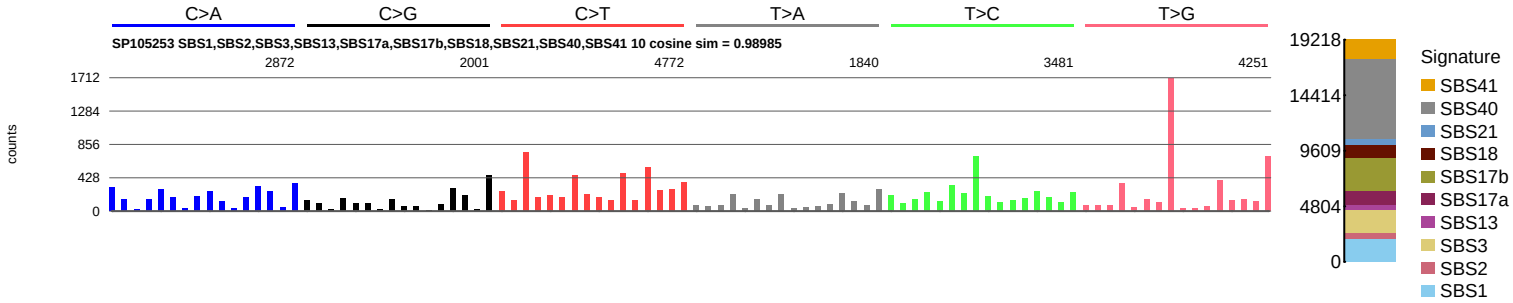
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98986
2861
1851
4906
1692
3625
4285
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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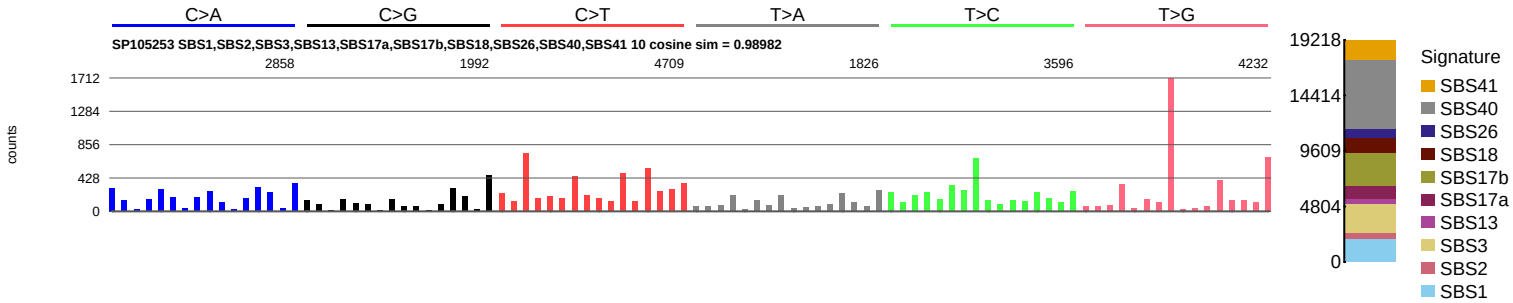
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98985
2872
2001
4772
1840
3481
4251
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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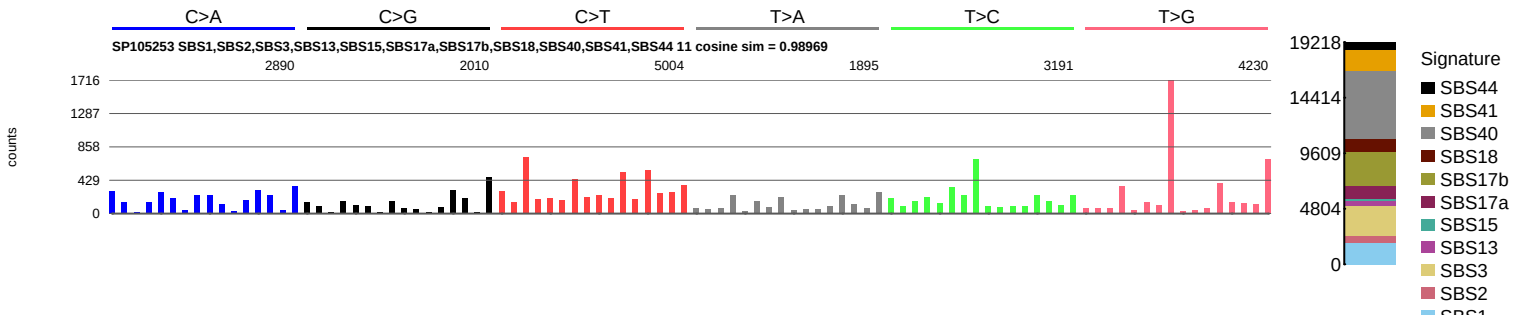
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98982
2858
1992
4709
1826
3596
4232
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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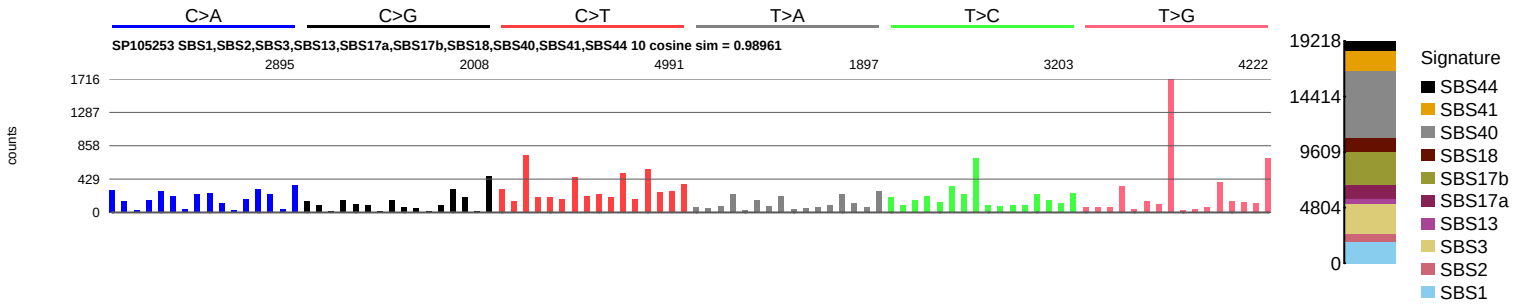
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98969
2890
2010
5004
1895
3191
4230
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2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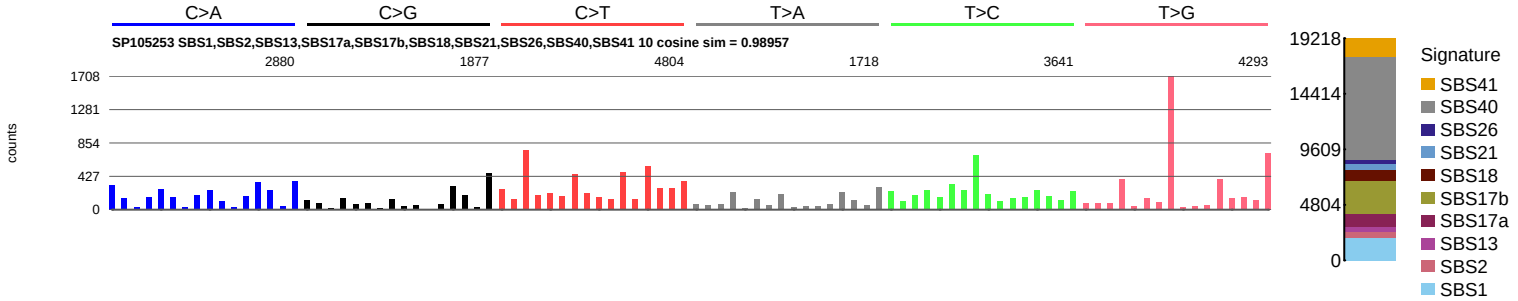
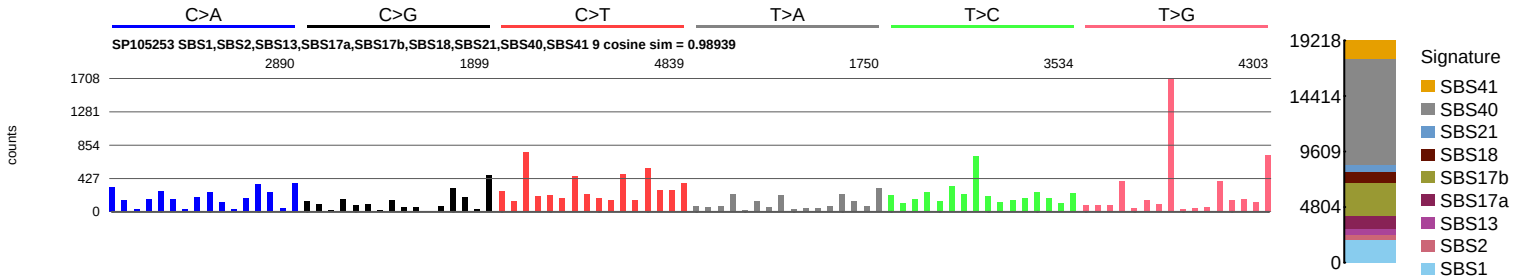
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98961
2895
2008
4991
1897
3203
4222
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98957
2880
1877
4804
1718
3641
4293
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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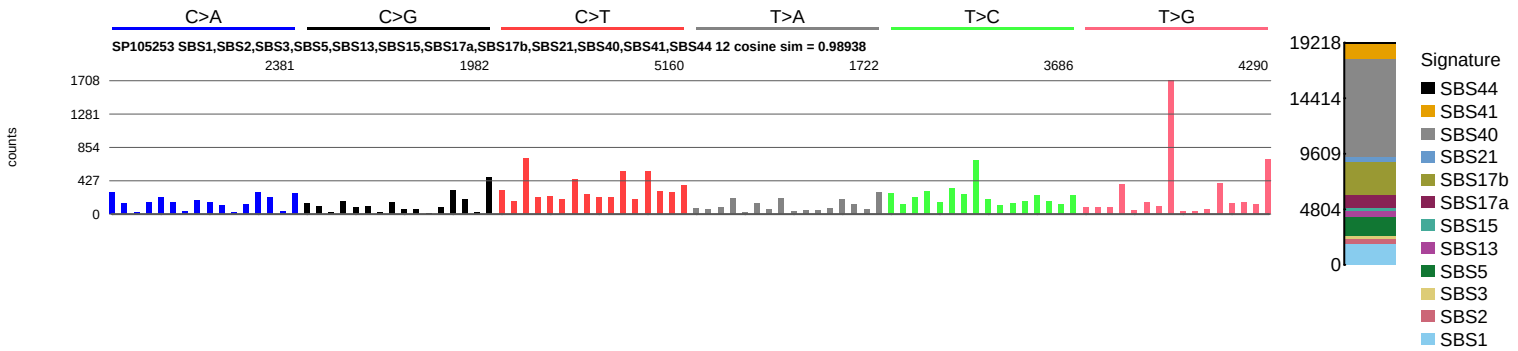
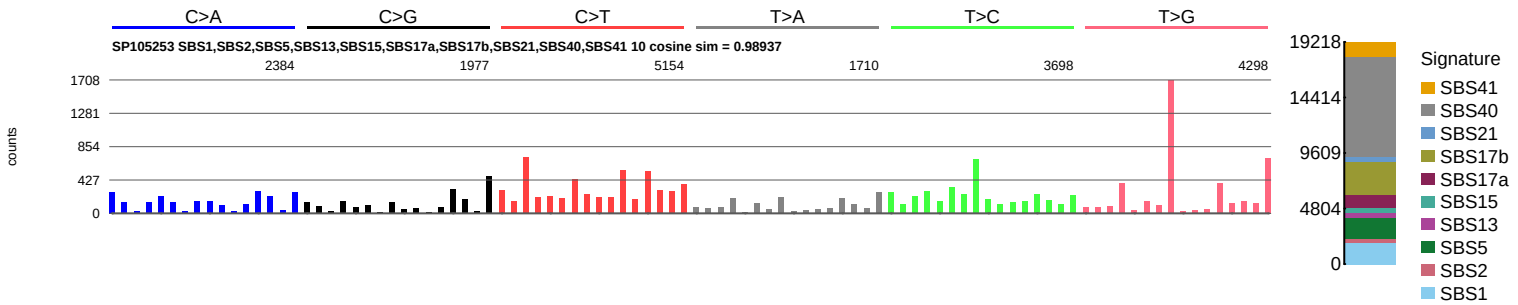
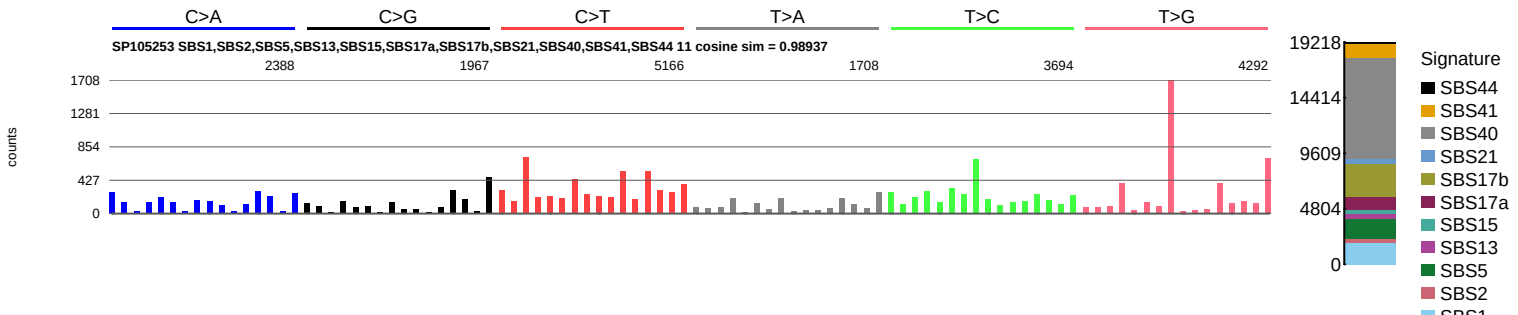
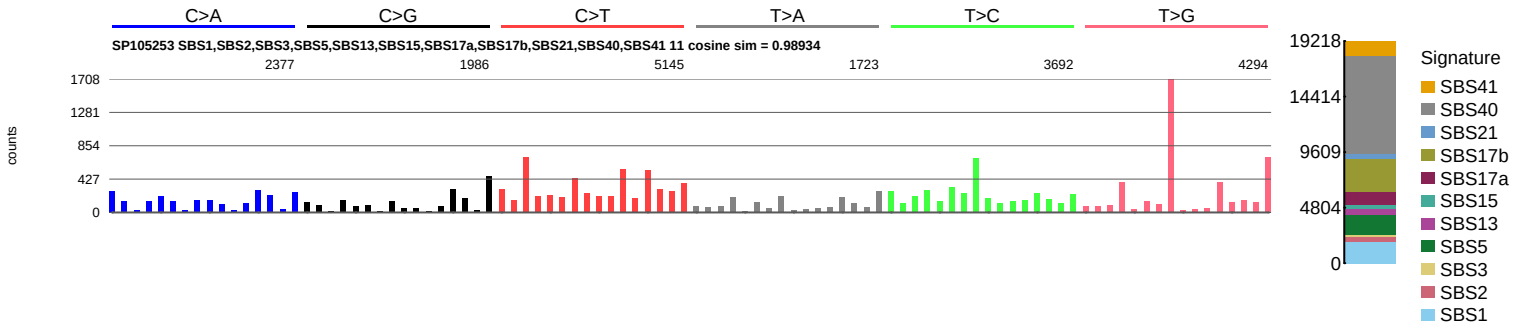
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98939
2890
1899
4839
1750
3534
4303
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 12 cosine sim = 0.98938
2381
1982
5160
1722
3686
4290
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98937
2384
1977
5154
1710
3698
4298
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98937
2388
1967
5166
1708
3694
4292
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2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C>G
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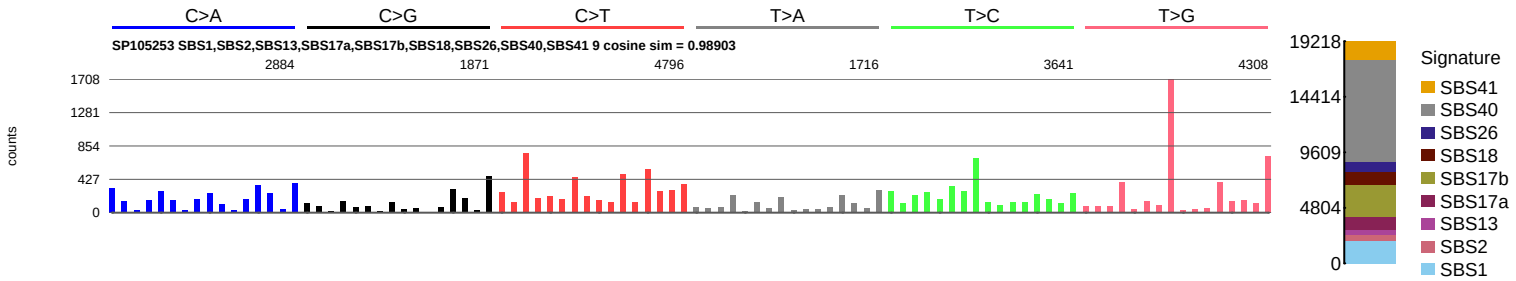
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 11 cosine sim = 0.98934
2377
1986
5145
1723
3692
4294
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98903
2884
1871
4796
1716
3641
4308
counts
1708
1281
854
427
0
19218
14414
9609
4804
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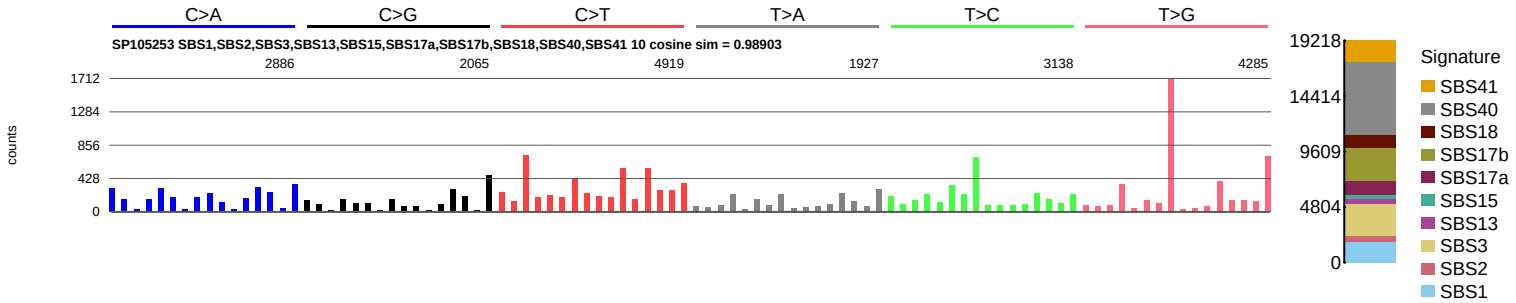
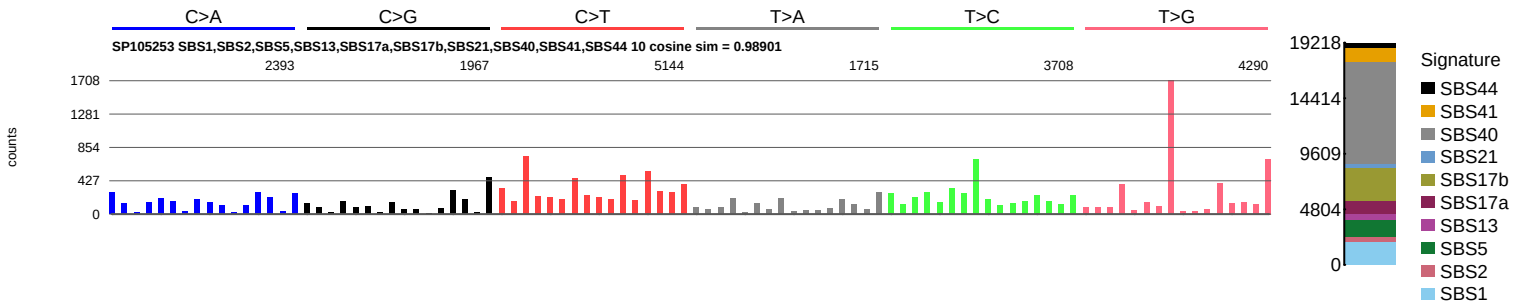
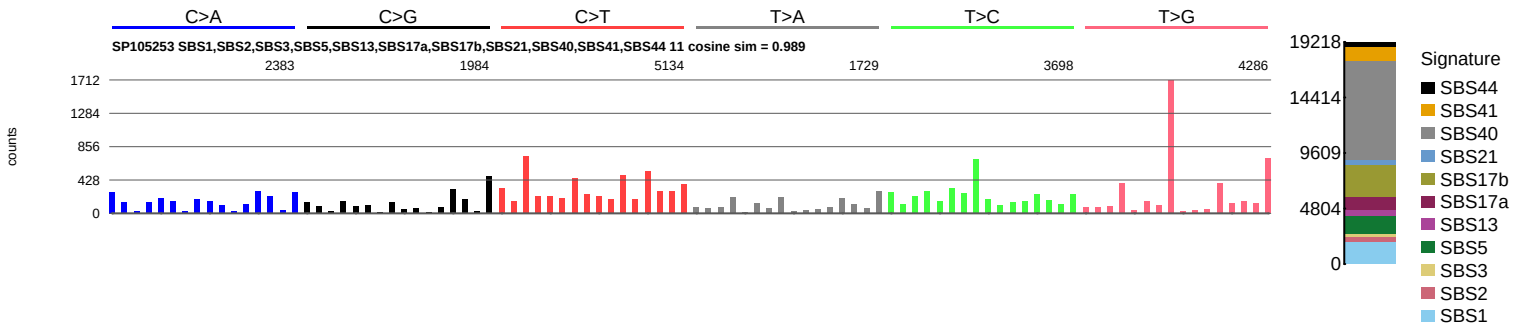
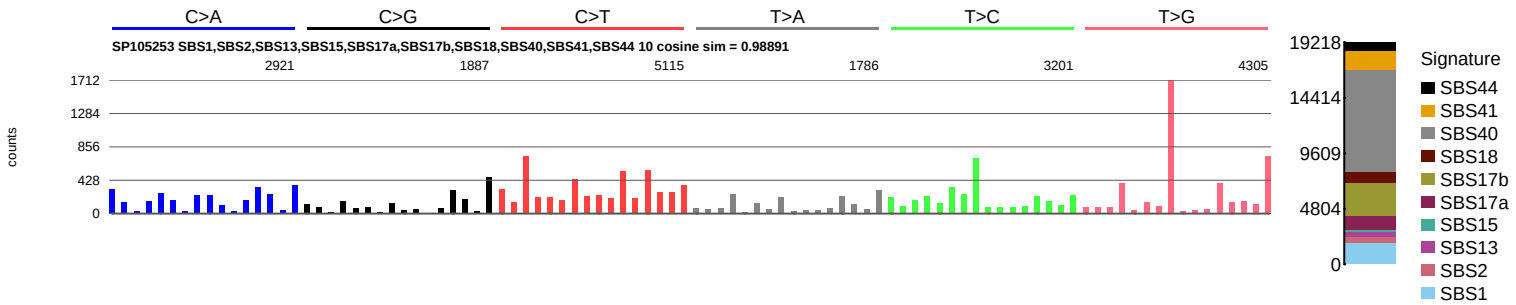
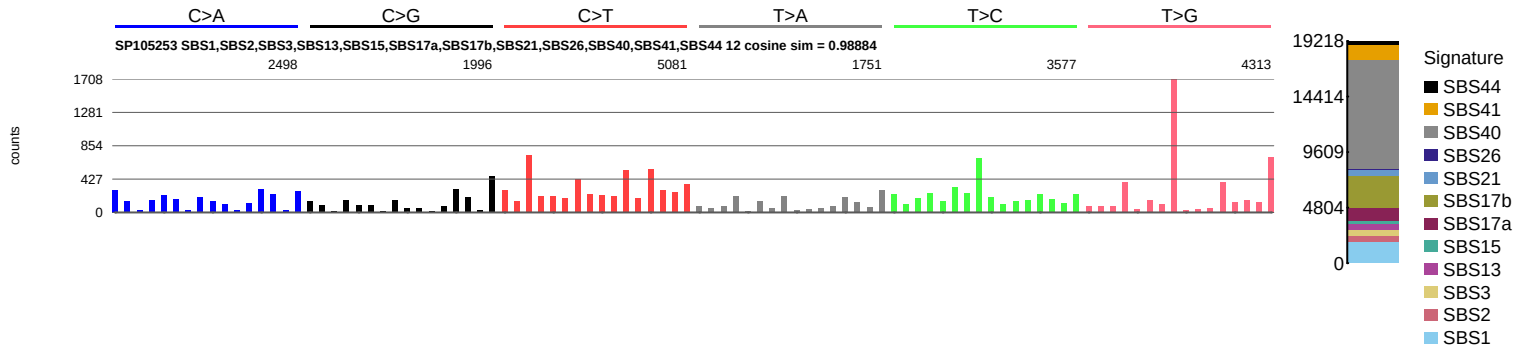
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

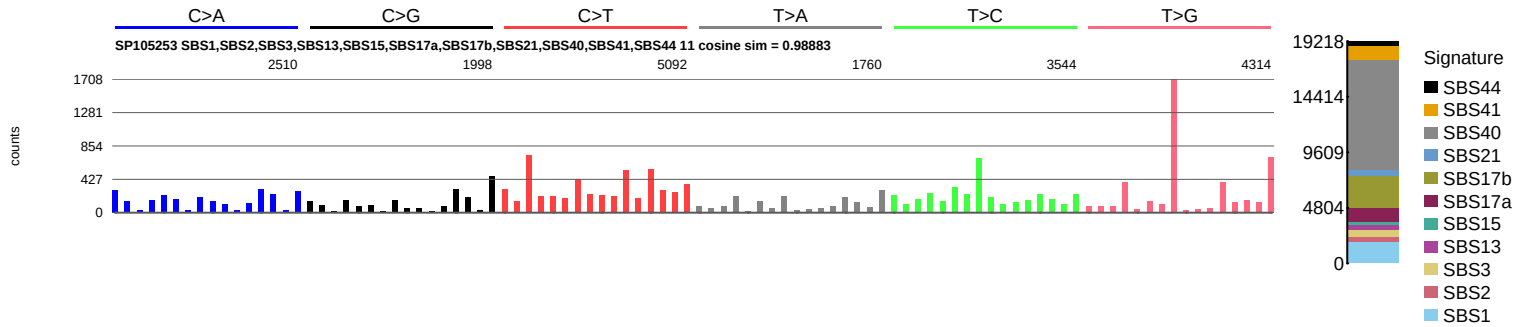

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.98903
2886
2065
4919
1927
3138
4285
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

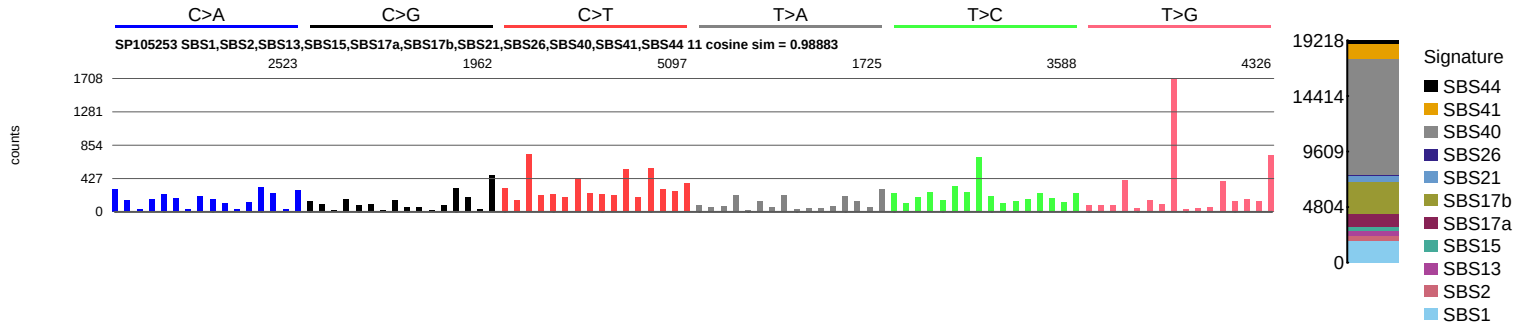

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98901
2393
1967
5144
1715
3708
4290
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

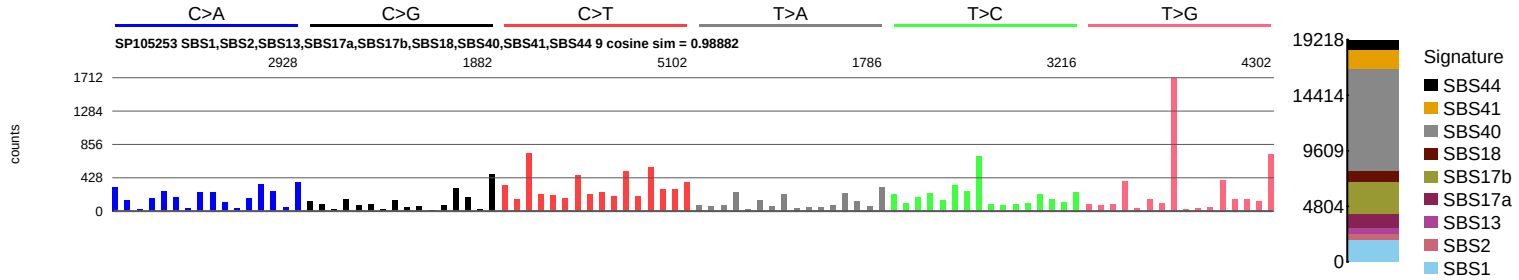

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.989
2383
1984
5134
1729
3698
4286
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98891
2921
1887
5115
1786
3201
4305
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98884
2498
1996
5081
1751
3577
4313
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98883
2510
1998
5092
1760
3544
4314
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98883
2523
1962
5097
1725
3588
4326
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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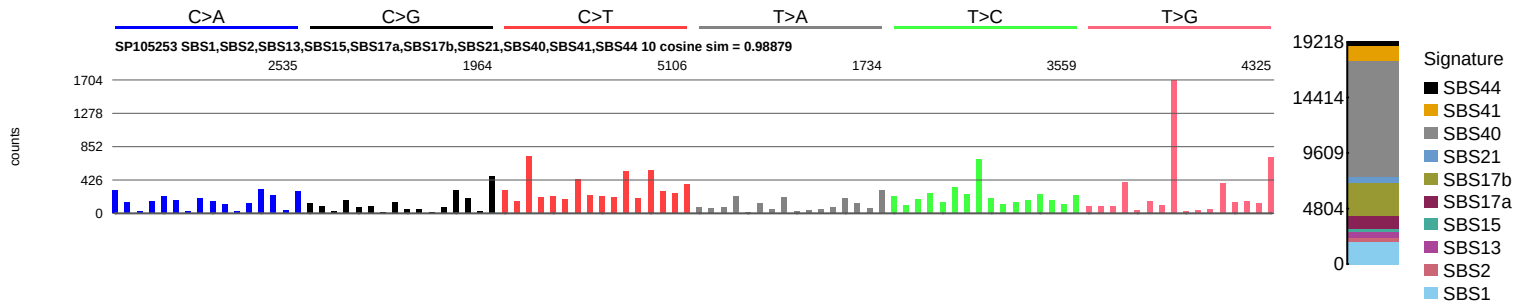
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98882
2928
1882
5102
1786
3216
4302
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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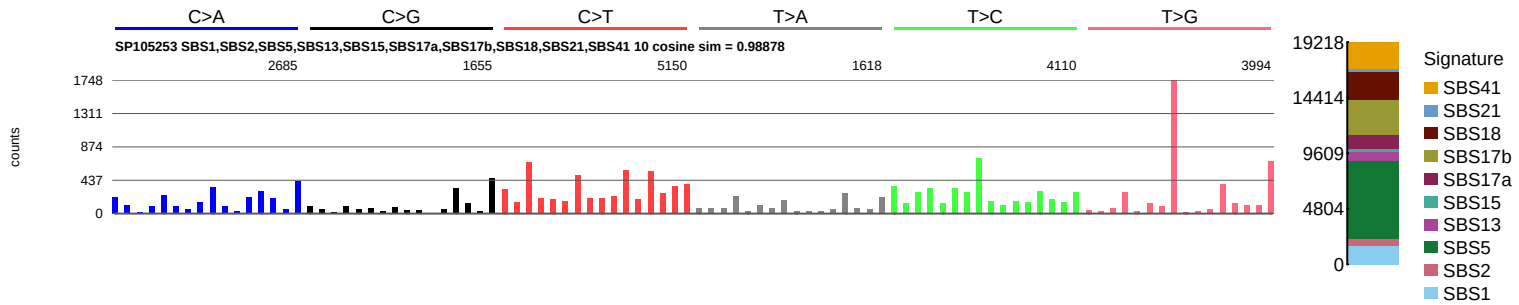
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98879
2535
1964
5106
1734
3559
4325
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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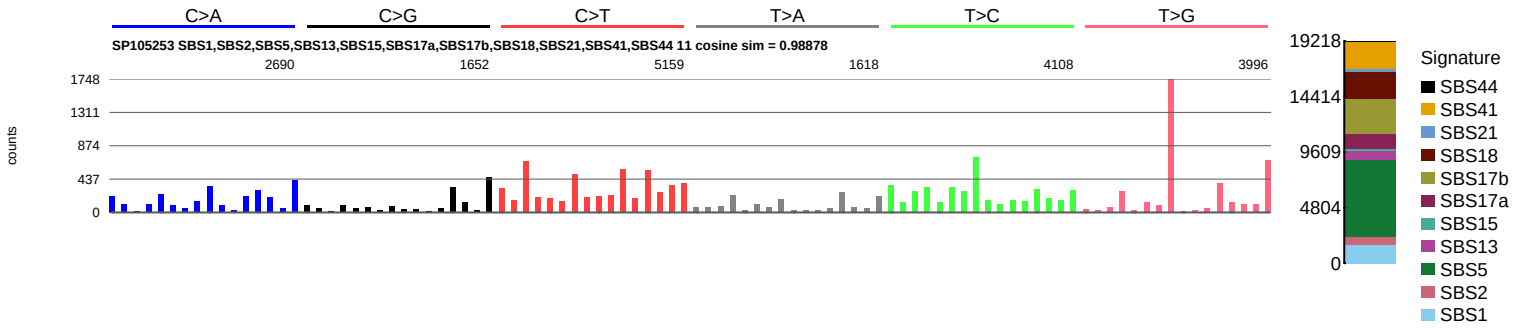
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98878
2685
1655
5150
1618
4110
3994
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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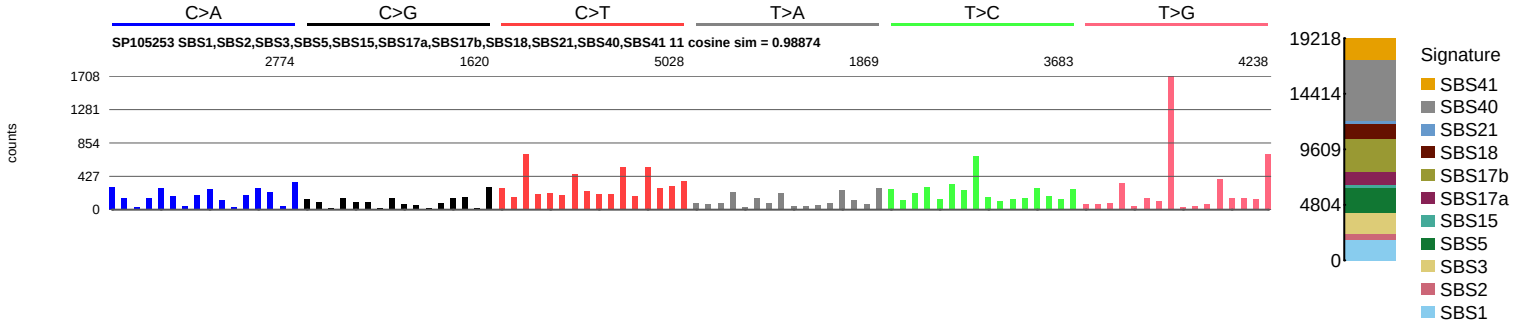
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98878
2690
1652
5159
1618
4108
3996
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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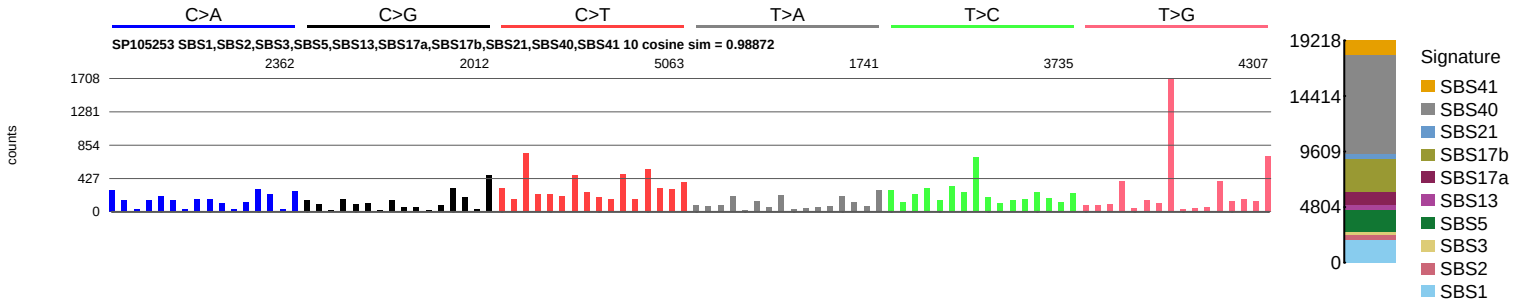
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 11 cosine sim = 0.98874
2774
1620
5028
1869
3683
4238
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
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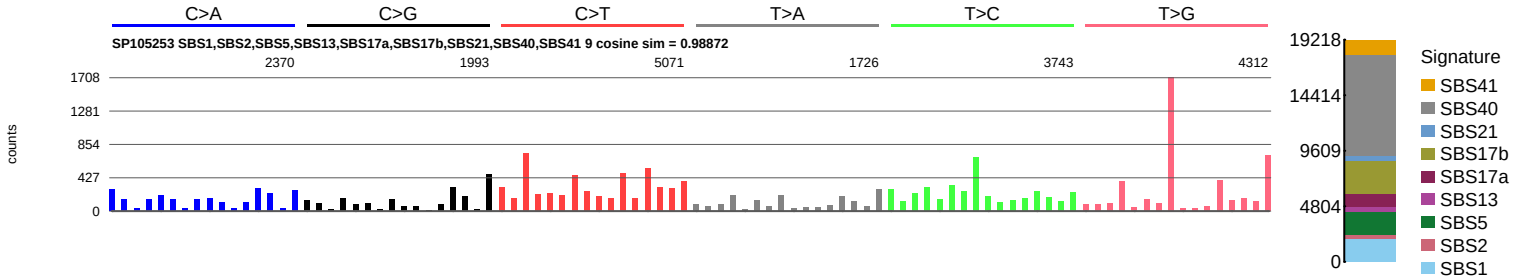
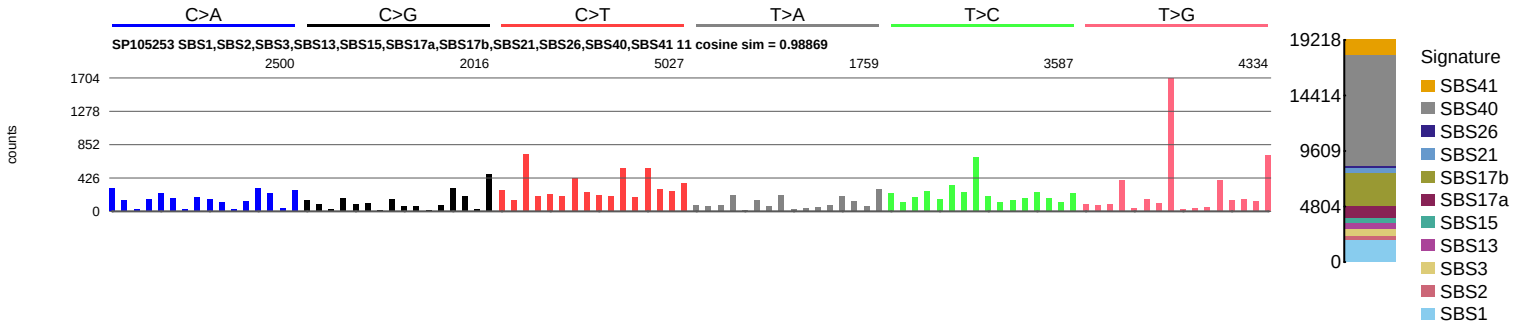
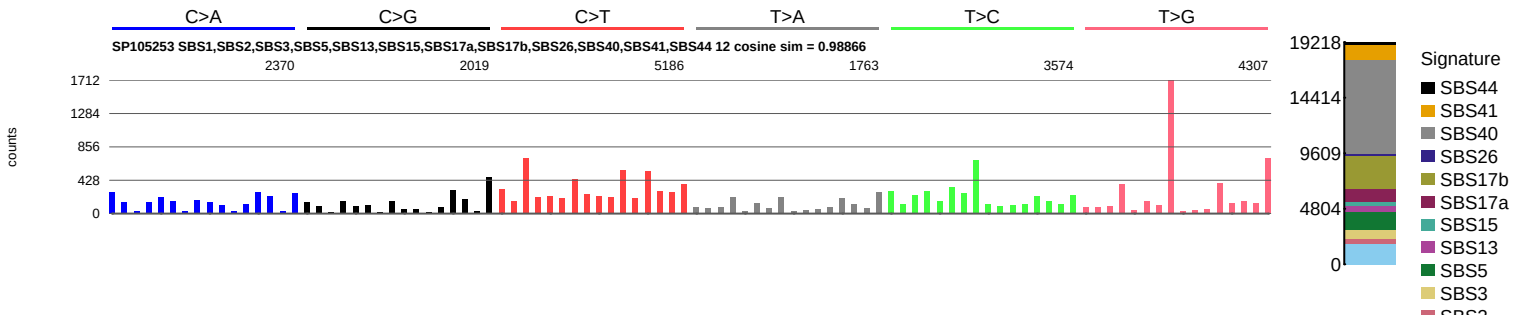
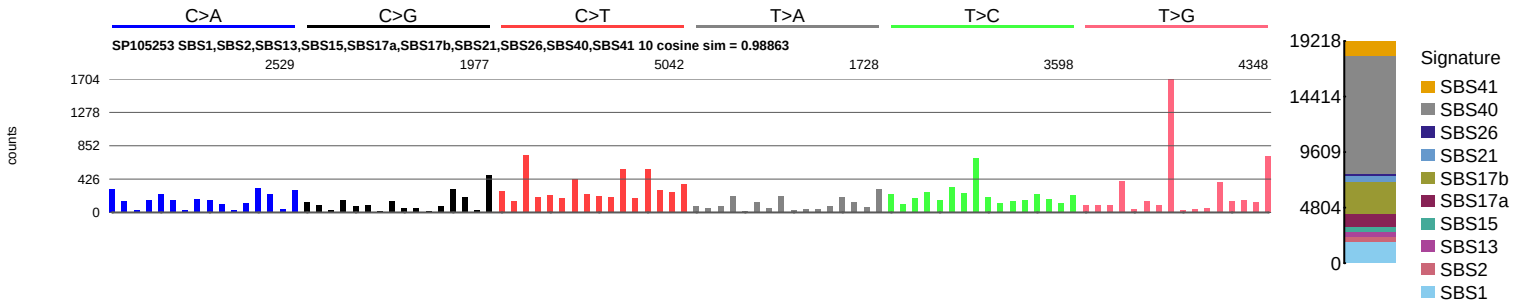
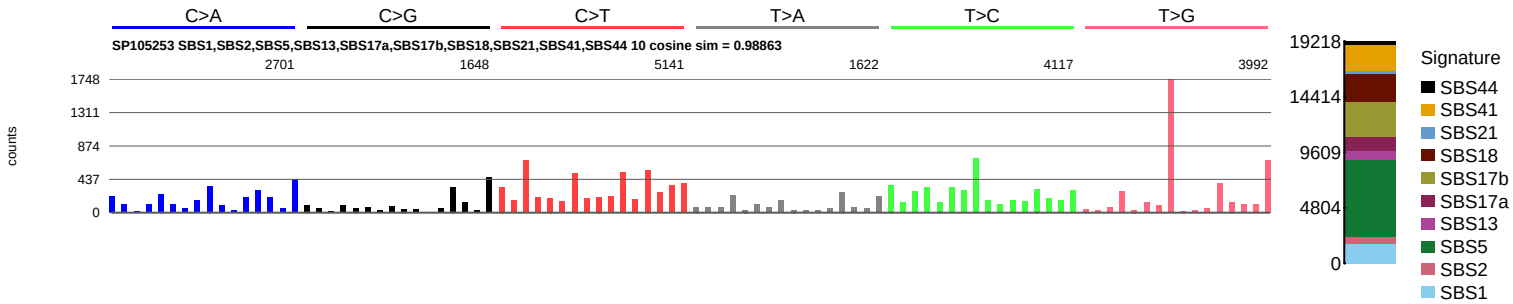
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98872
2362
2012
5063
1741
3735
4307
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

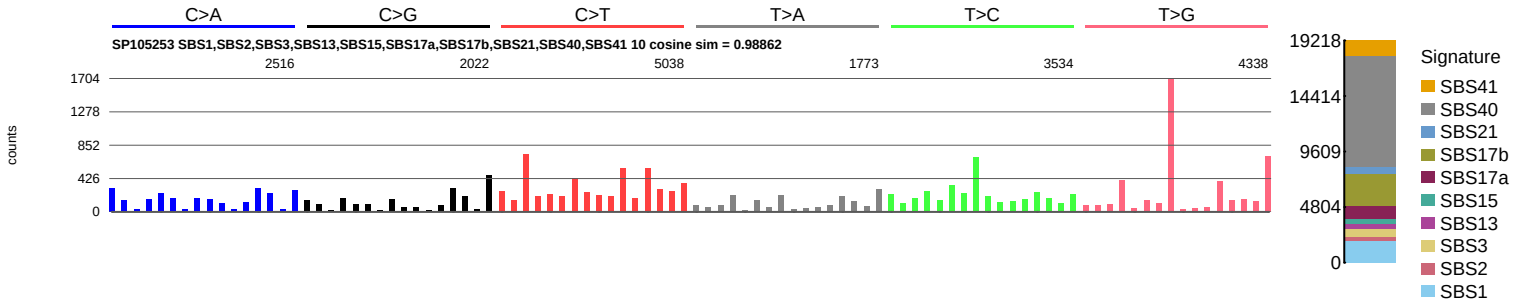

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98872
2370
1993
5071
1726
3743
4312
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

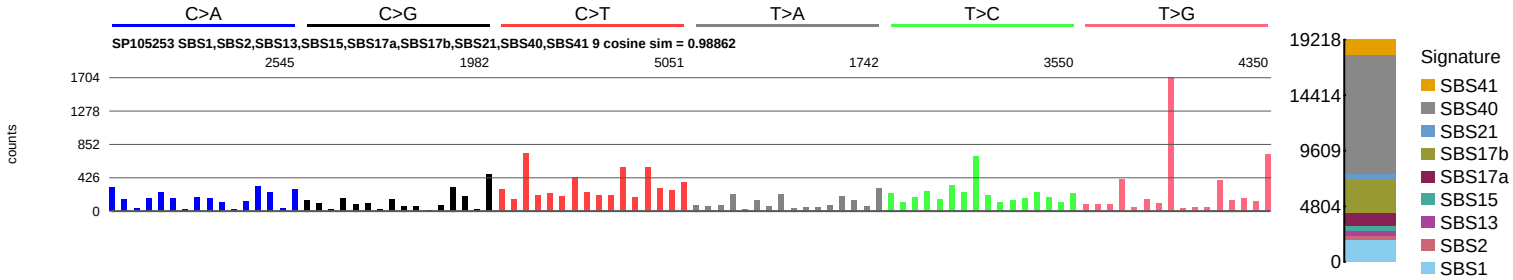

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98869
2500
2016
5027
1759
3587
4334
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

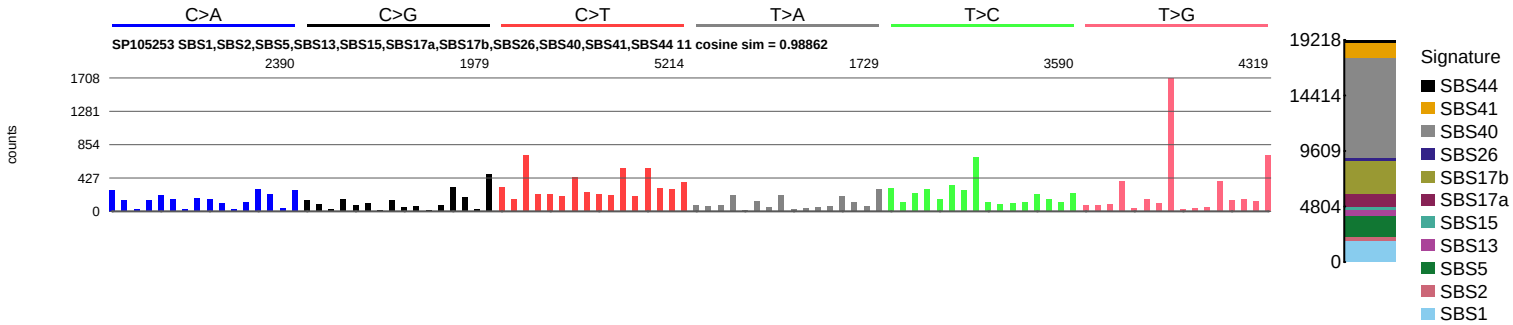

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98866
2370
2019
5186
1763
3574
4307
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3

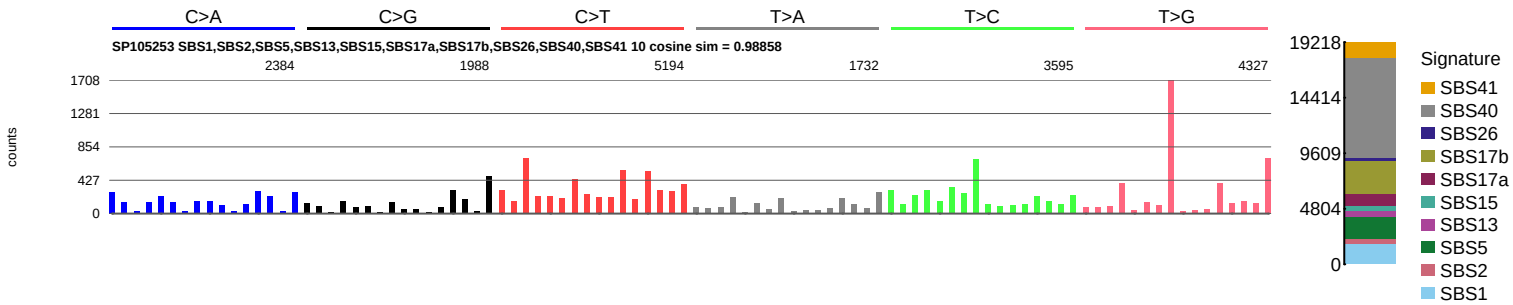

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98863
2529
1977
5042
1728
3598
4348
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98863
2701
1648
5141
1622
4117
3992
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98862
2516
2022
5038
1773
3534
4338
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98862
2545
1982
5051
1742
3550
4350
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98862
2390
1979
5214
1729
3590
4319
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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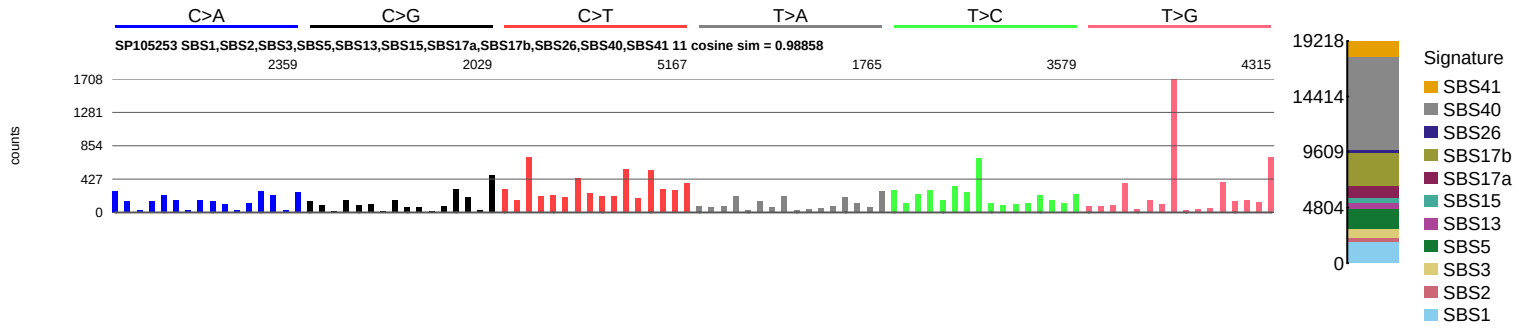
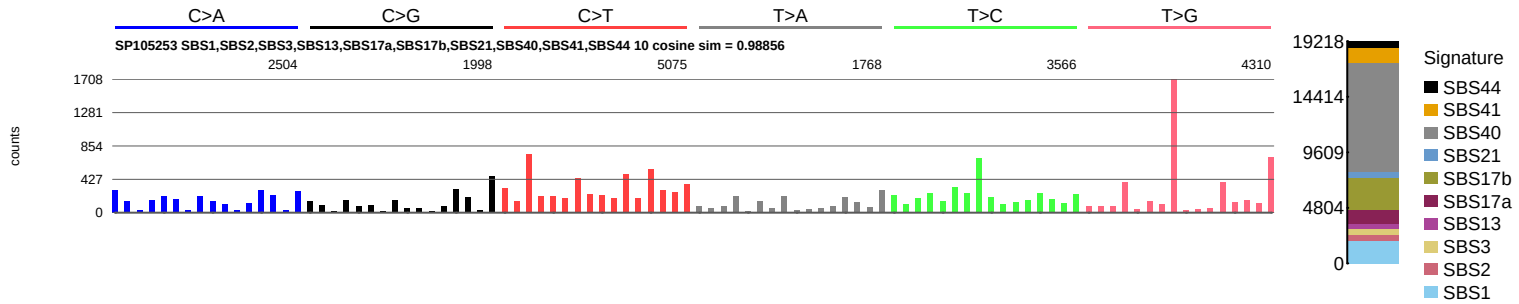
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98858
2384
1988
5194
1732
3595
4327
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 11 cosine sim = 0.98858
2359
2029
5167
1765
3579
4315
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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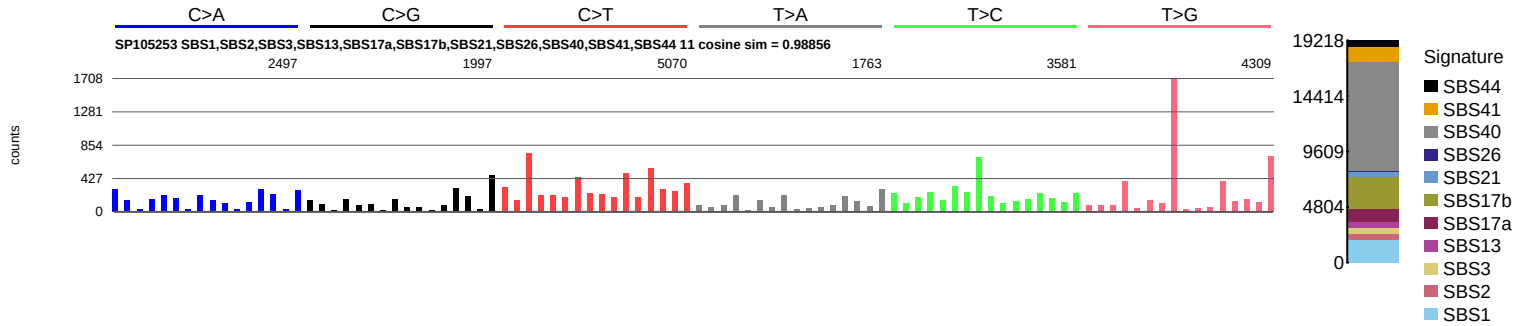
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98856
2504
1998
5075
1768
3566
4310
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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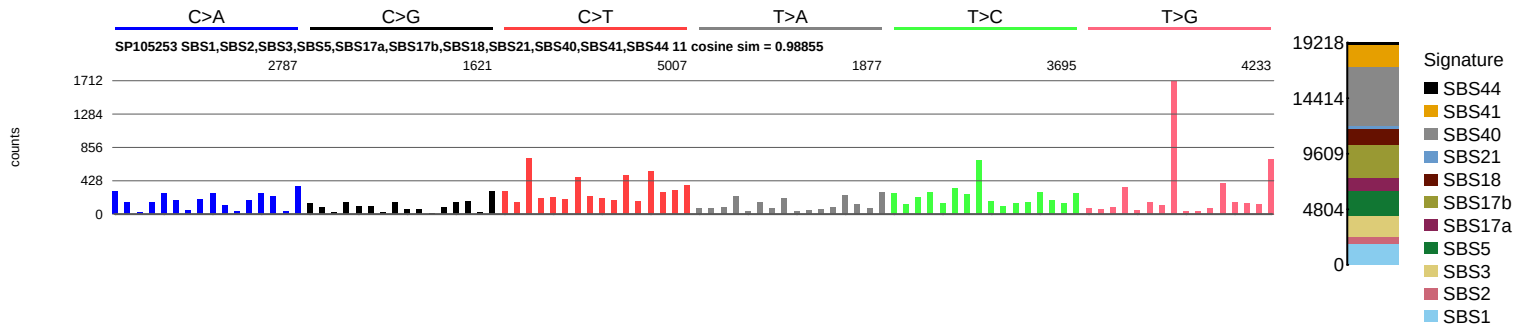
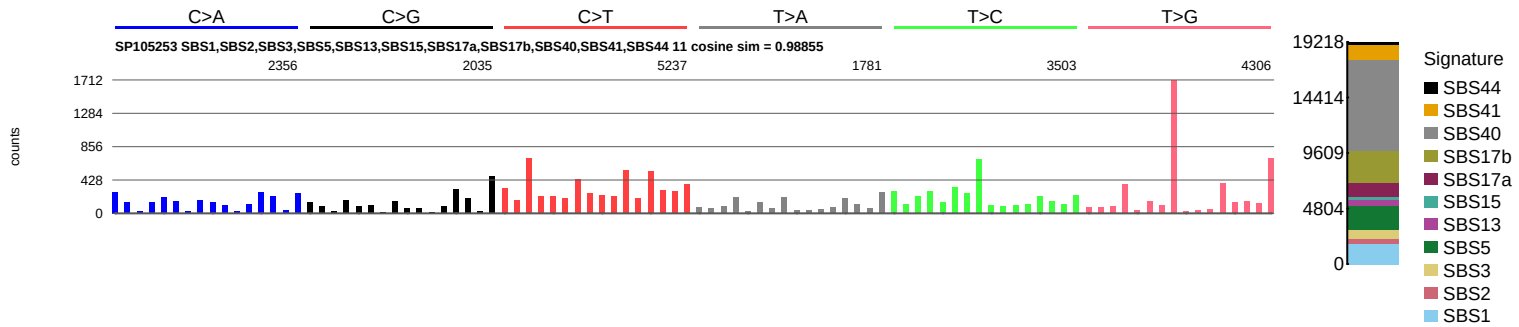
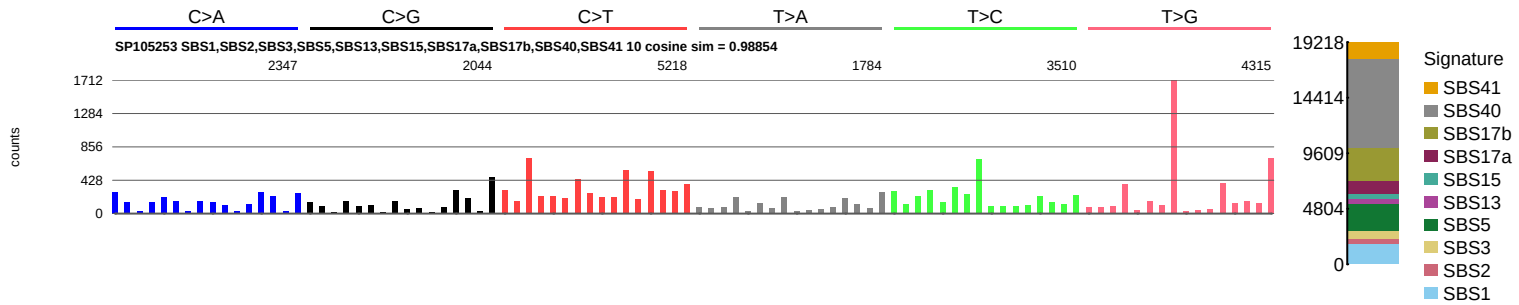
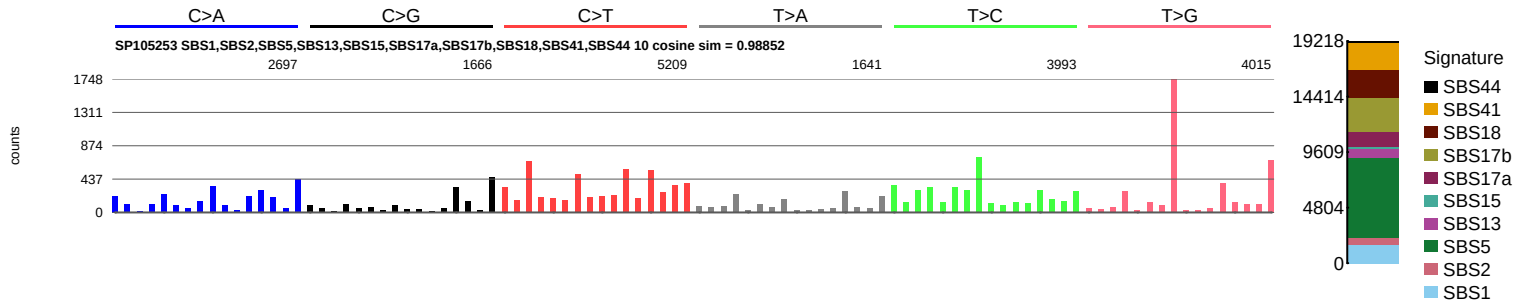
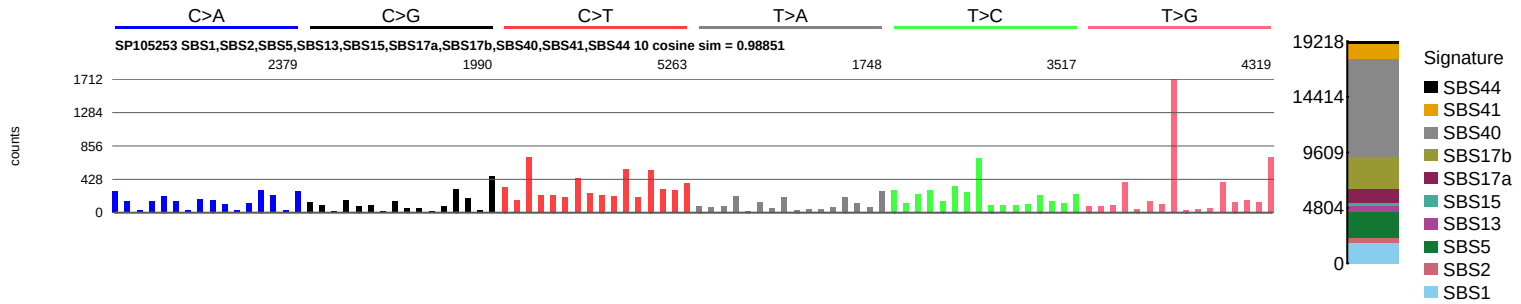
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98856
2497
1997
5070
1763
3581
4309
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

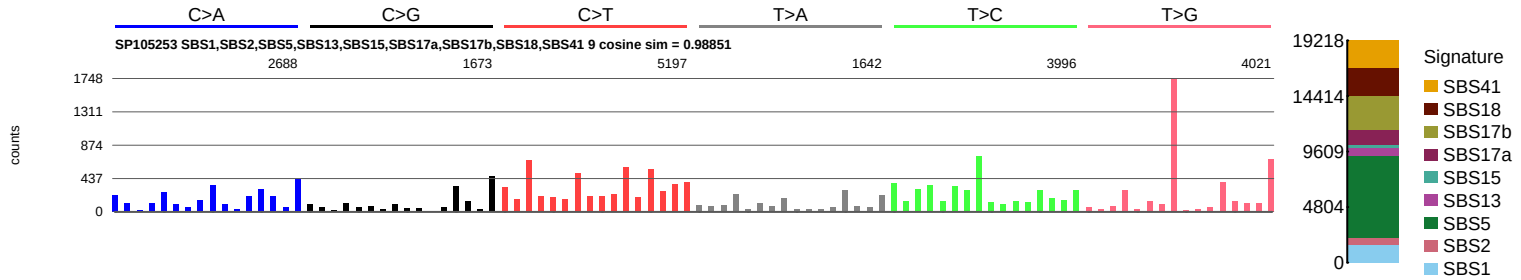

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98855
2787
1621
5007
1877
3695
4233
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 11 cosine sim = 0.98855
2356
2035
5237
1781
3503
4306
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 10 cosine sim = 0.98854
2347
2044
5218
1784
3510
4315
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98852
2697
1666
5209
1641
3993
4015
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98851
2379
1990
5263
1748
3517
4319
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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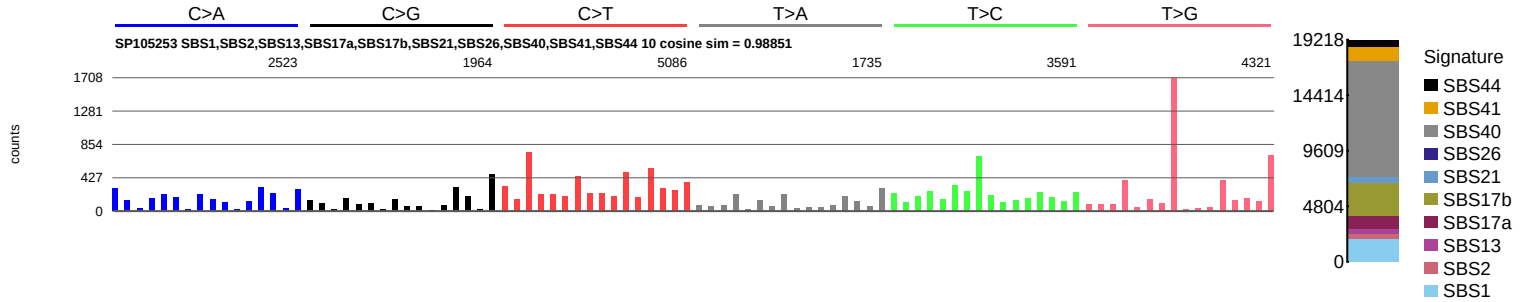
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98851
2688
1673
5197
1642
3996
4021
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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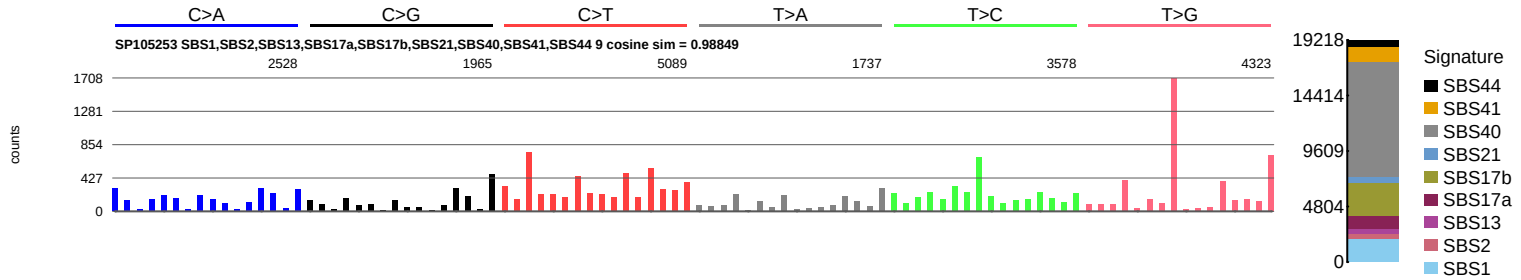
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98851
2523
1964
5086
1735
3591
4321
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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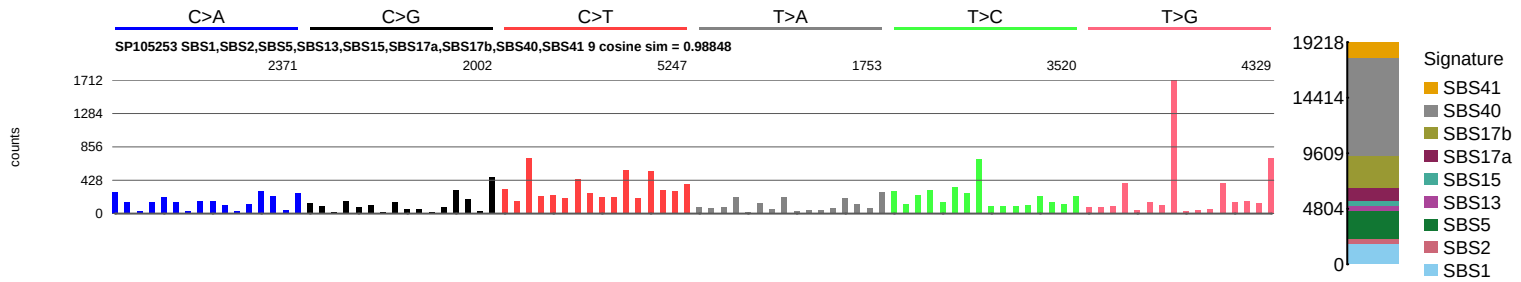
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98849
2528
1965
5089
1737
3578
4323
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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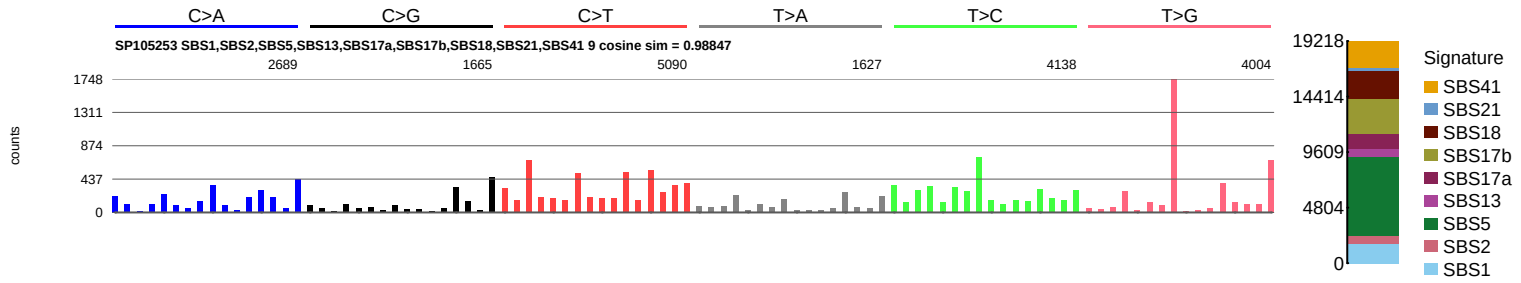
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98848
2371
2002
5247
1753
3520
4329
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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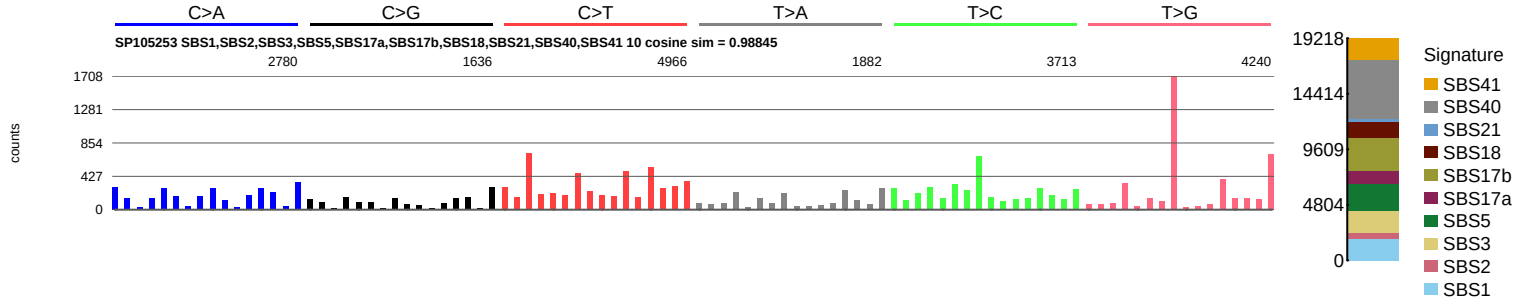
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98847
2689
1665
5090
1627
4138
4004
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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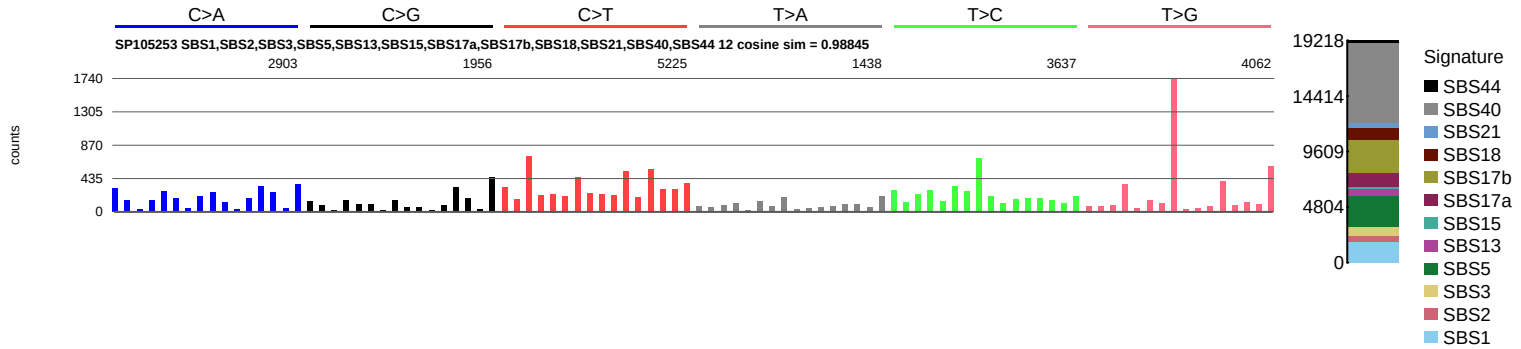
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98845
2780
1636
4966
1882
3713
4240
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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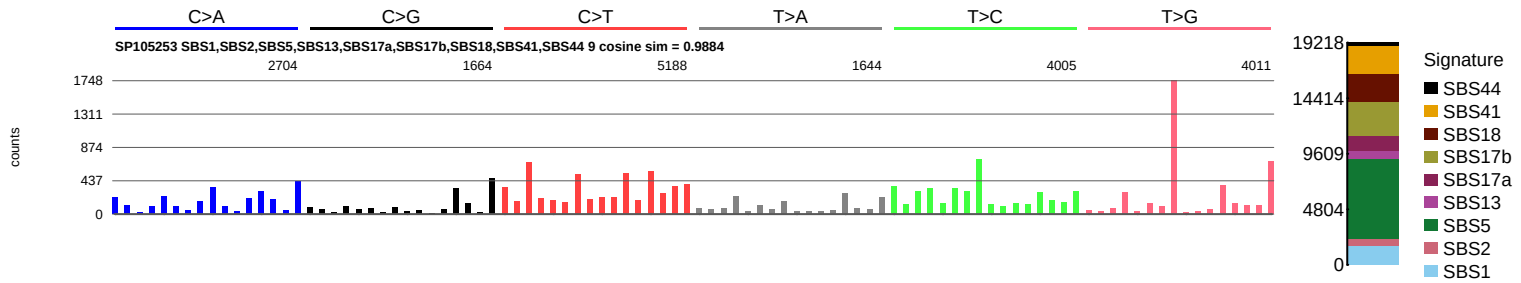
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 12 cosine sim = 0.98845
2903
1956
5225
1438
3637
4062
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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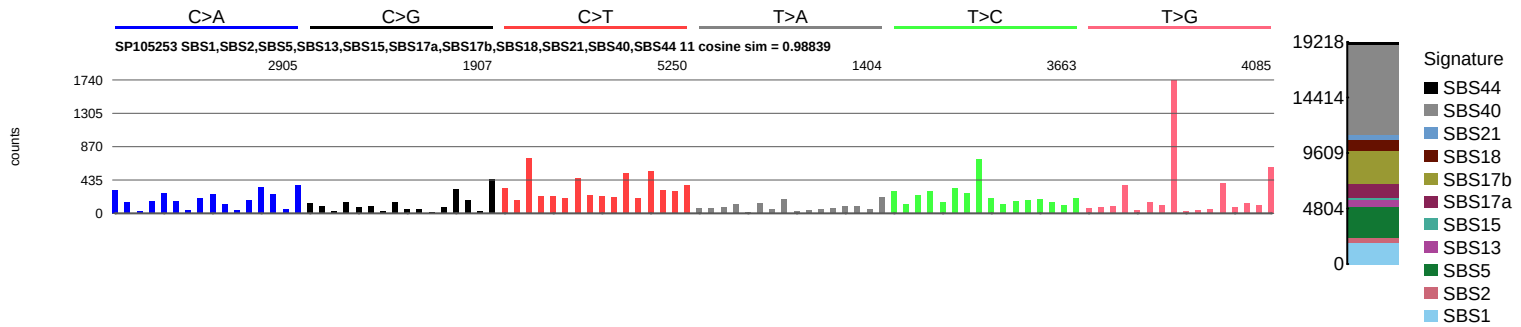
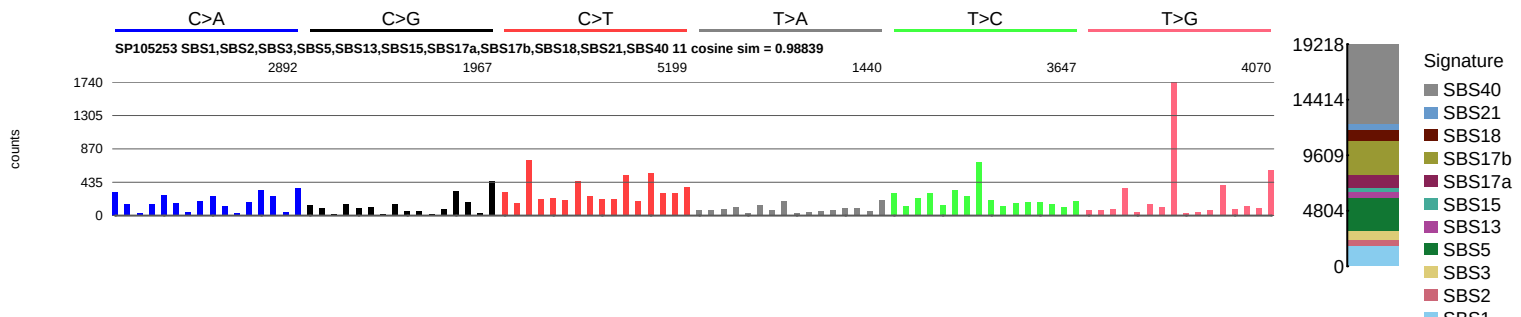
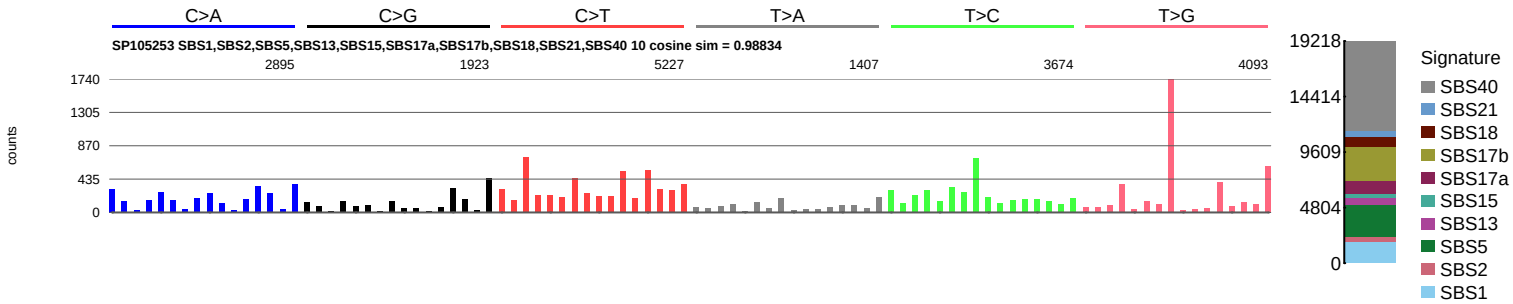
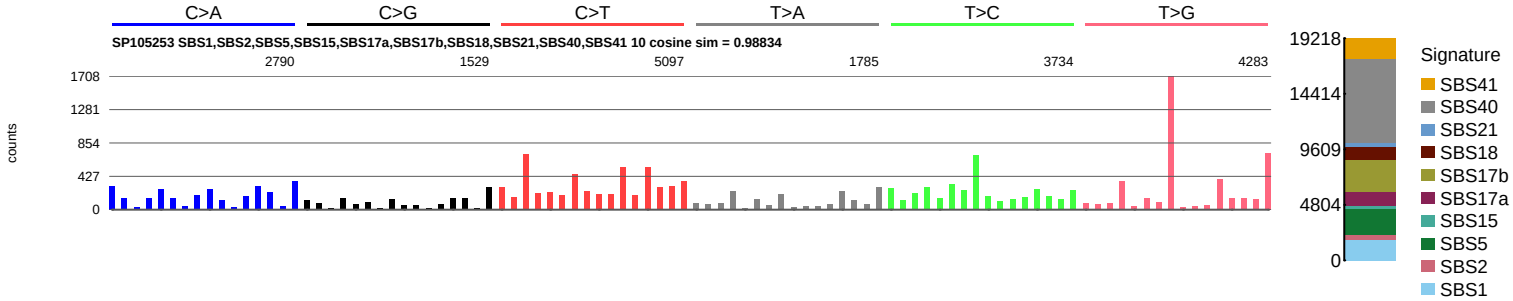
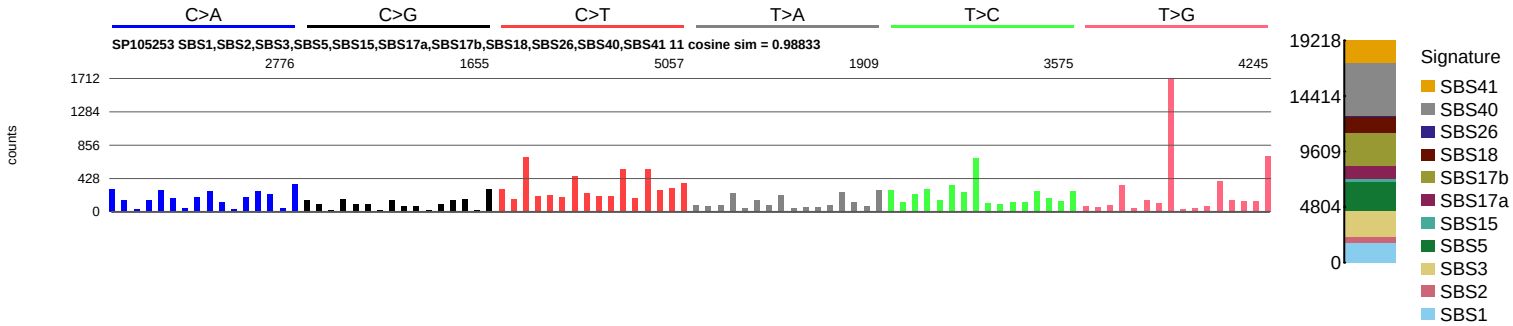
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.9884
2704
1664
5188
1644
4005
4011
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

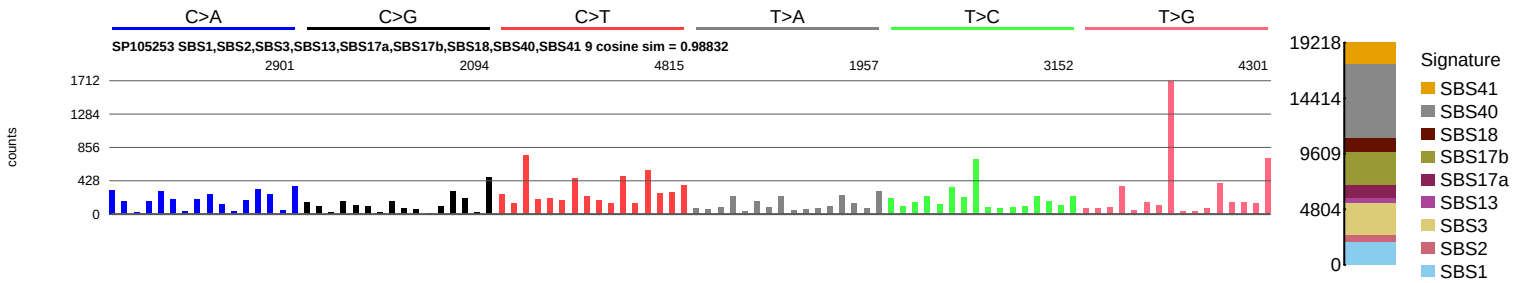

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98839
2905
1907
5250
1404
3663
4085
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

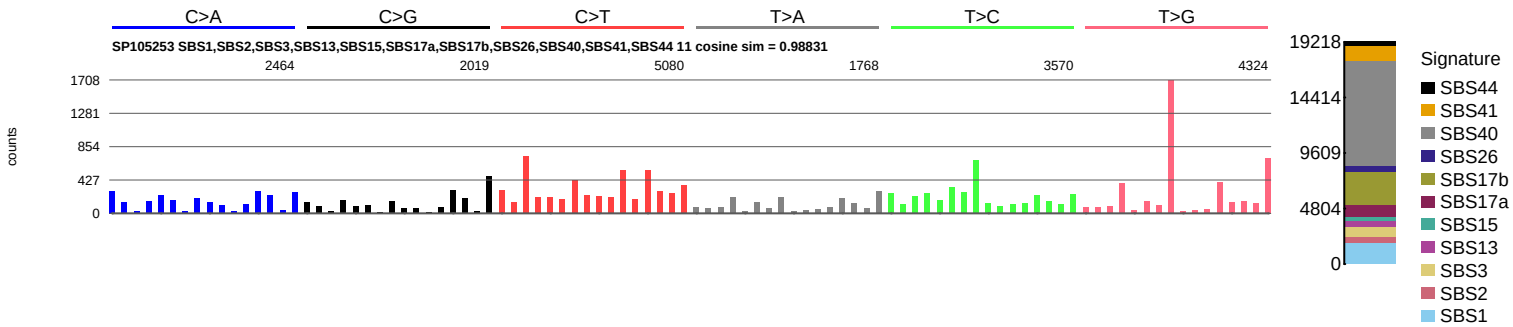

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 11 cosine sim = 0.98839
2892
1967
5199
1440
3647
4070
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

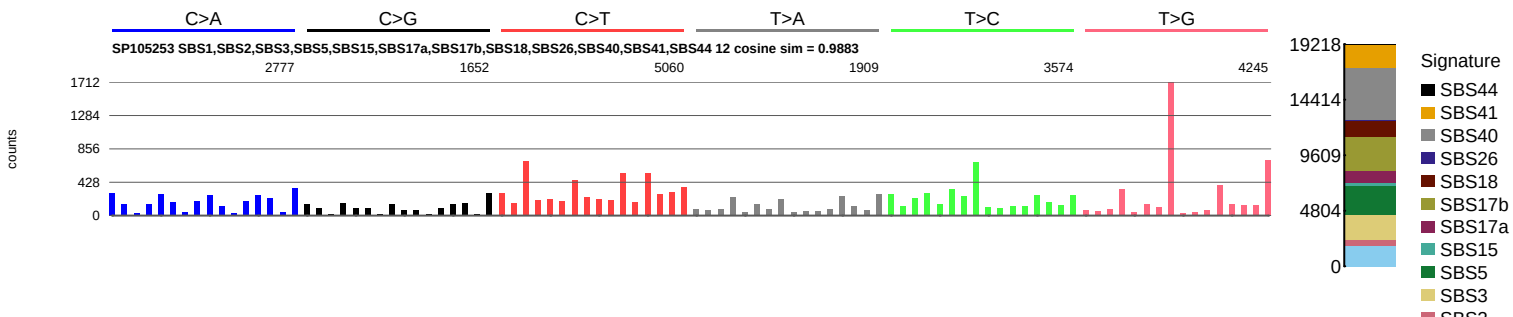

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98834
2895
1923
5227
1407
3674
4093
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98834
2790
1529
5097
1785
3734
4283
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 11 cosine sim = 0.98833
2776
1655
5057
1909
3575
4245
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98832
2901
2094
4815
1957
3152
4301
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98831
2464
2019
5080
1768
3570
4324
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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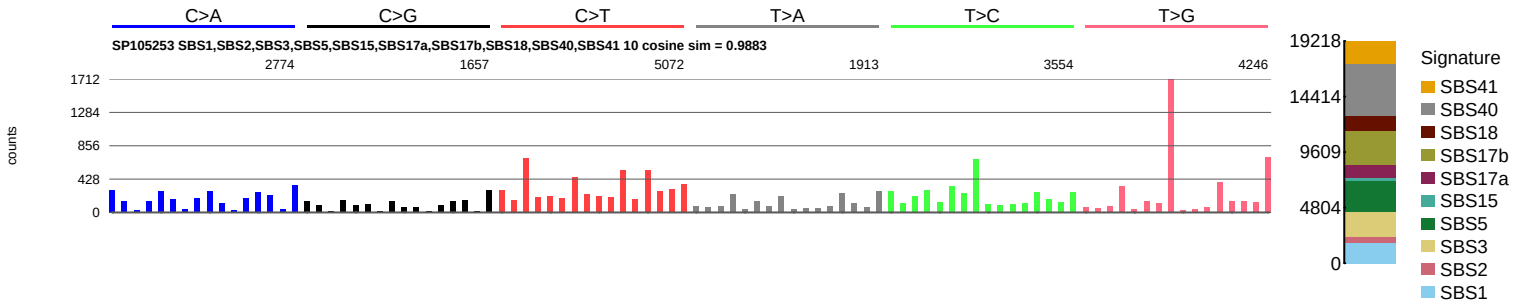
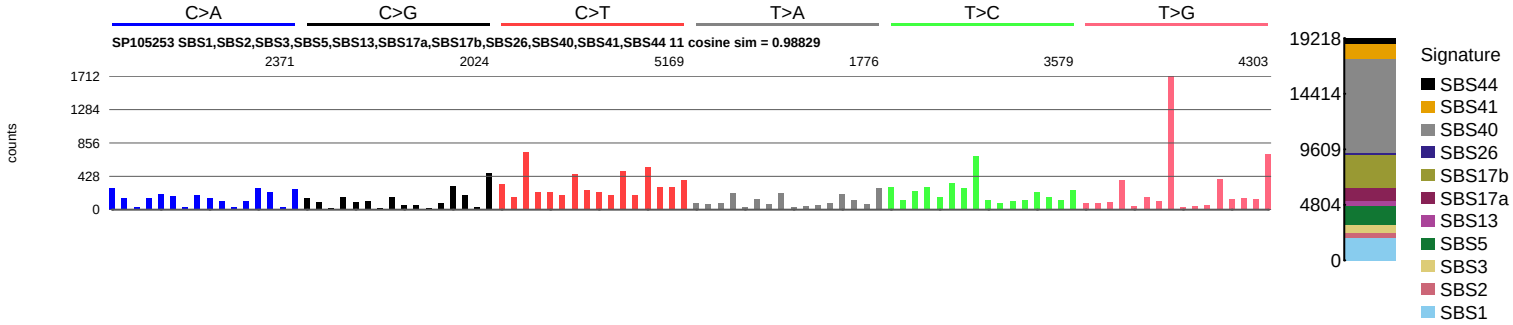
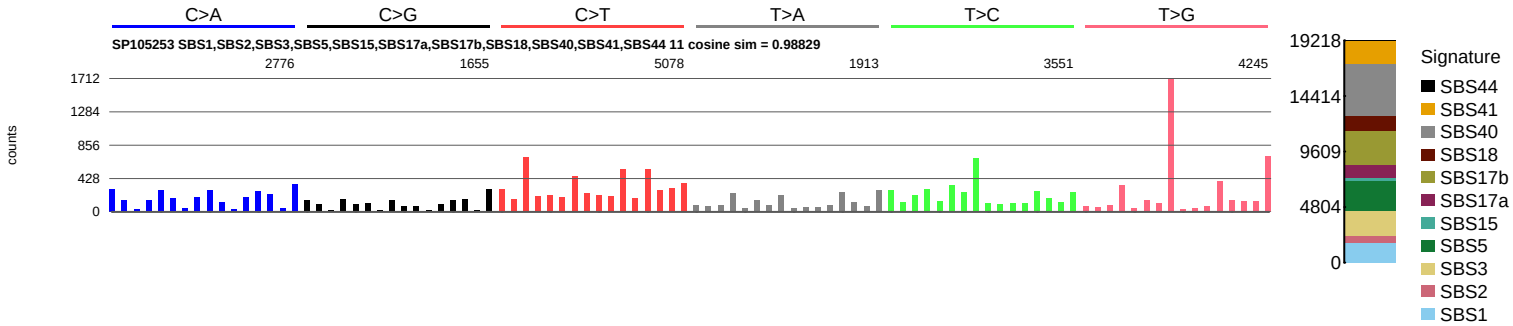
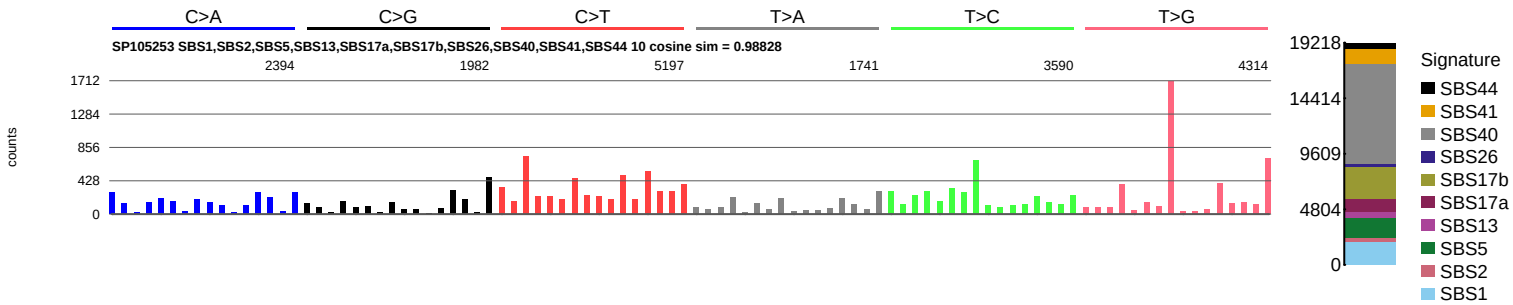
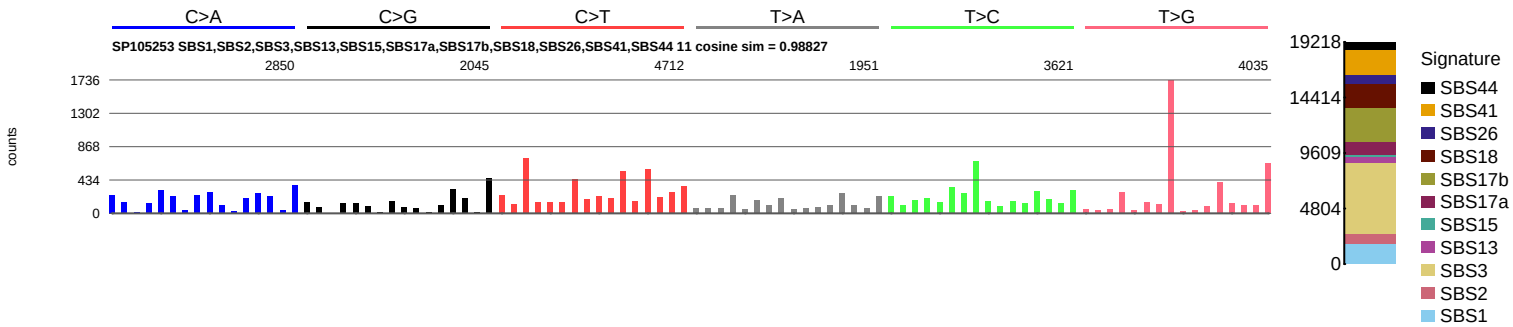
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.9883
2777
1652
5060
1909
3574
4245
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3

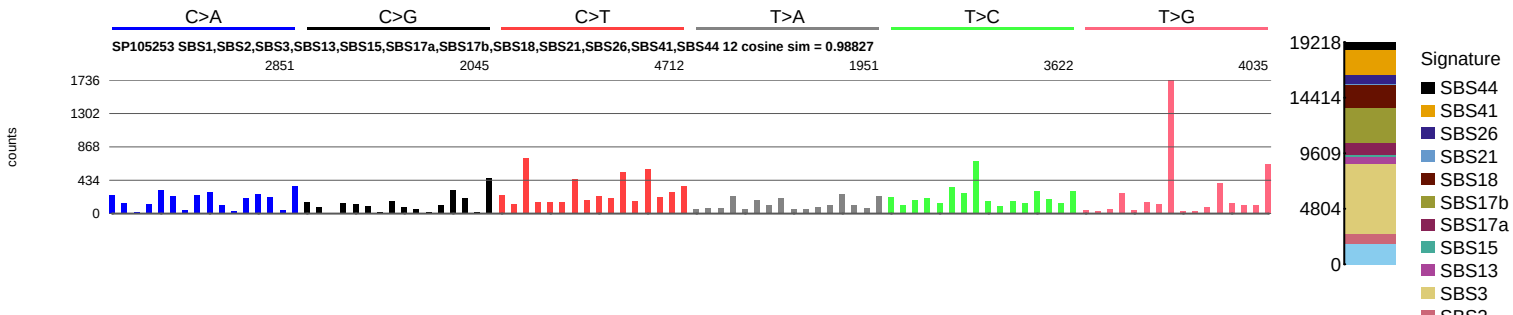

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 10 cosine sim = 0.9883
2774
1657
5072
1913
3554
4246
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

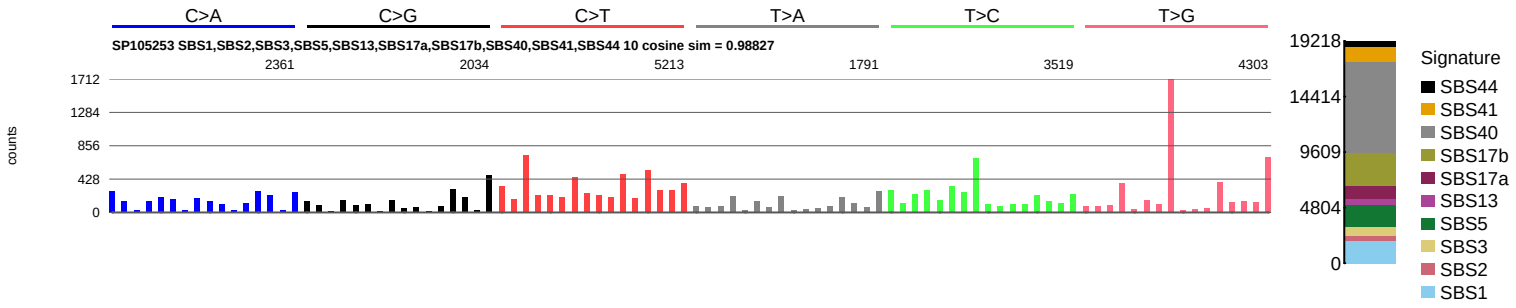

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98829
2371
2024
5169
1776
3579
4303
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

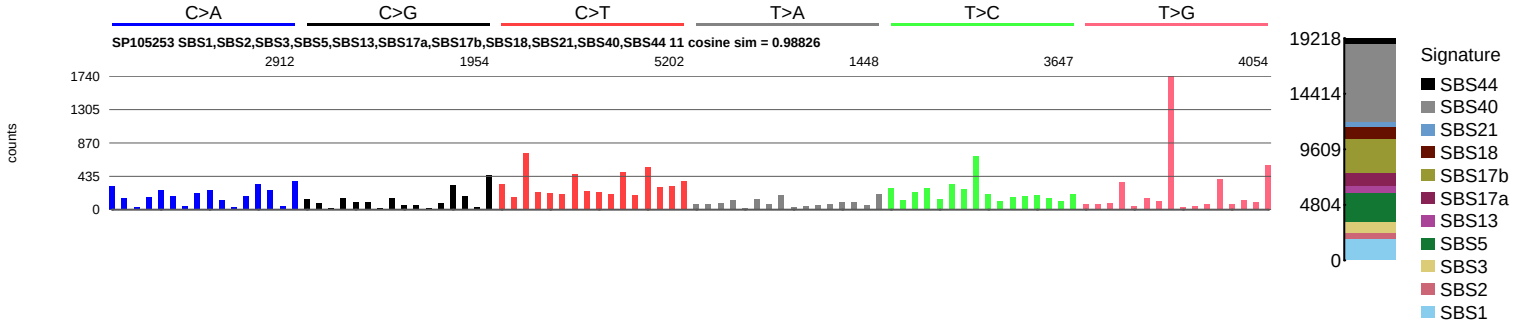

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 11 cosine sim = 0.98829
2776
1655
5078
1913
3551
4245
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

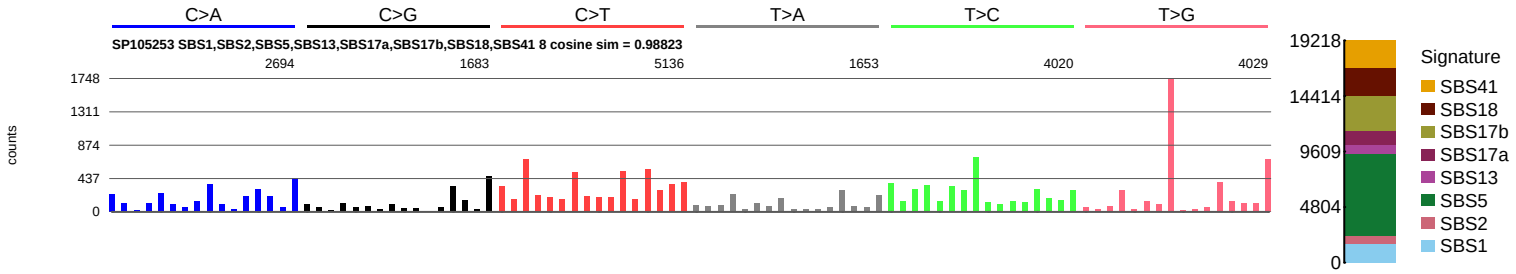

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98828
2394
1982
5197
1741
3590
4314
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

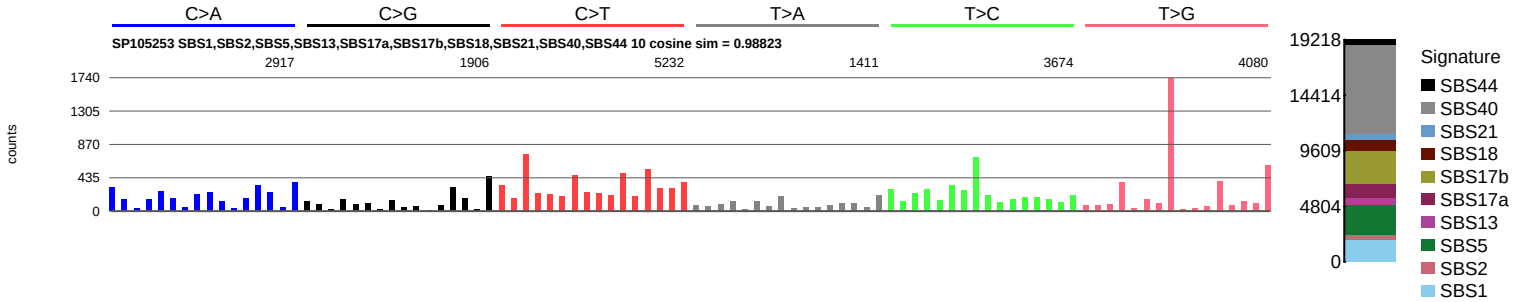

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.98827
2850
2045
4712
1951
3621
4035
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.98827
2851
2045
4712
1951
3622
4035
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98827
2361
2034
5213
1791
3519
4303
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98826
2912
1954
5202
1448
3647
4054
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98823
2694
1683
5136
1653
4020
4029
counts
1748
1311
874
437
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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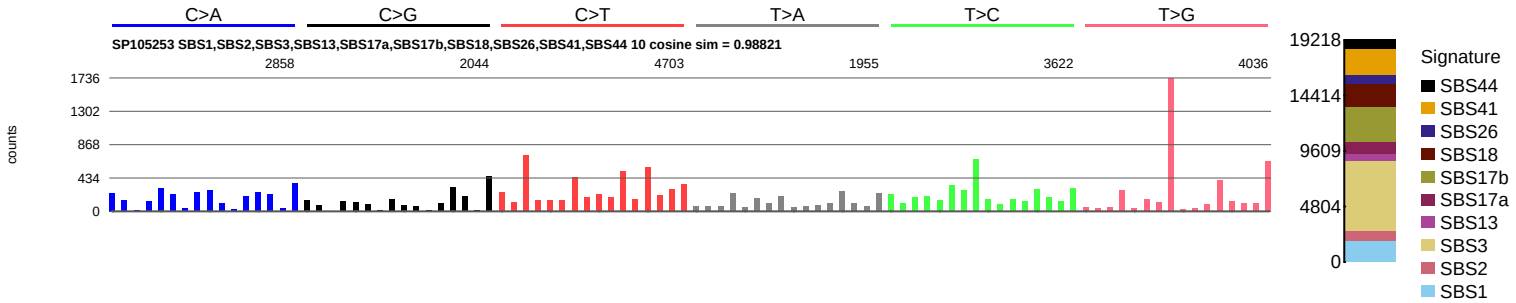
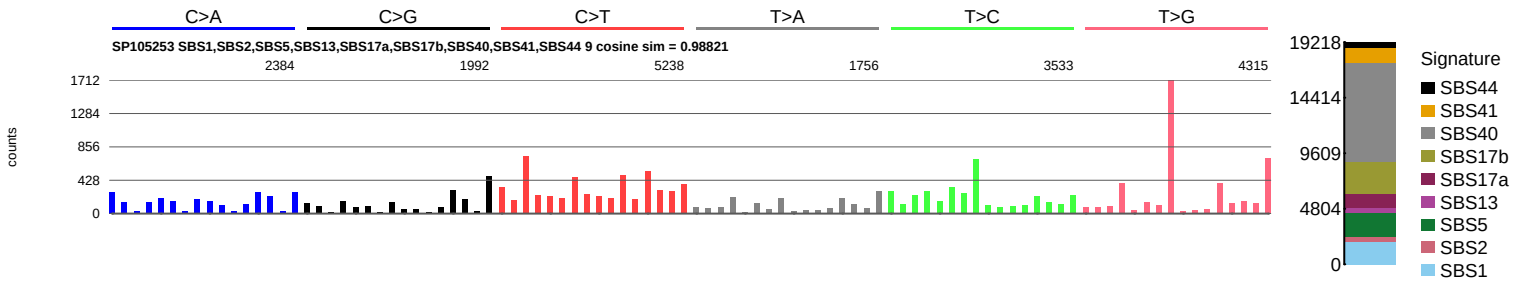
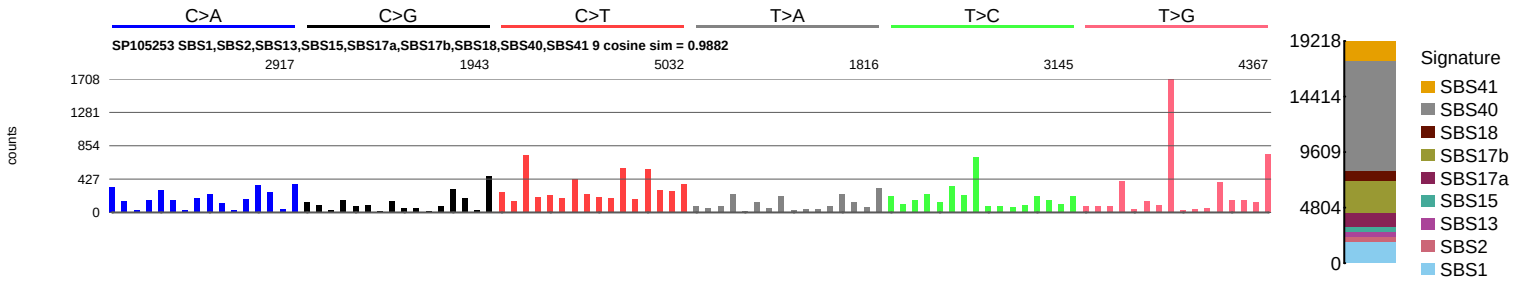
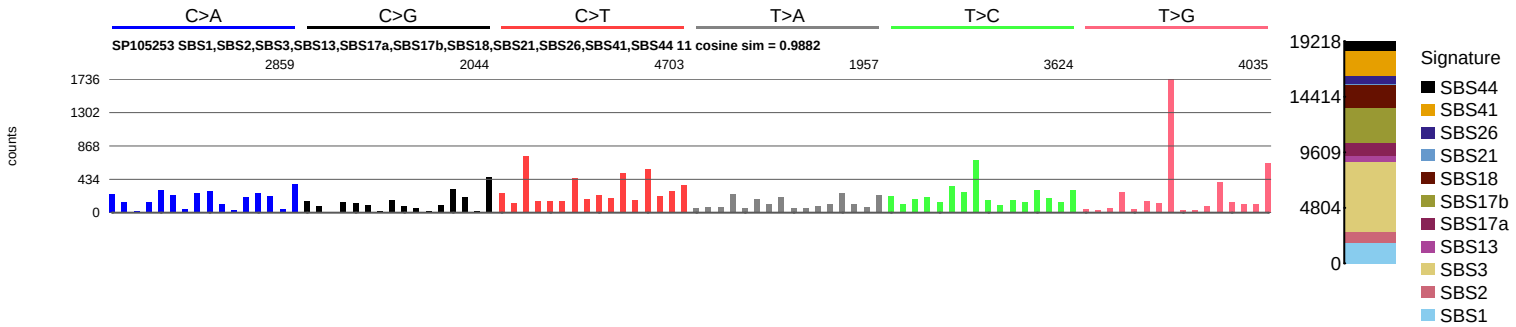
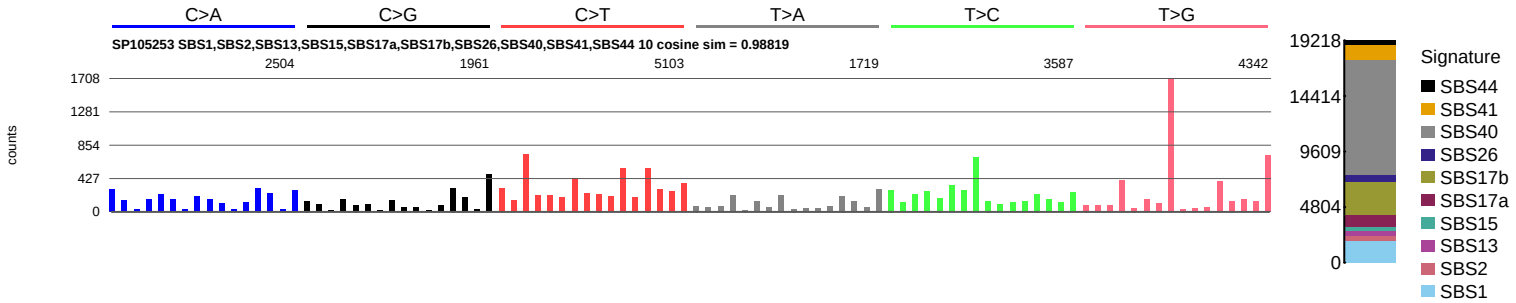
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98823
2917
1906
5232
1411
3674
4080
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

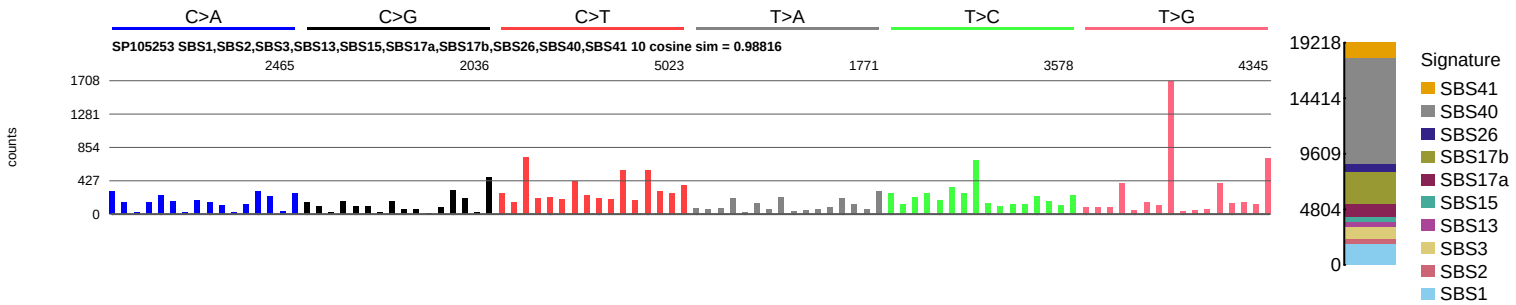

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98821
2858
2044
4703
1955
3622
4036
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

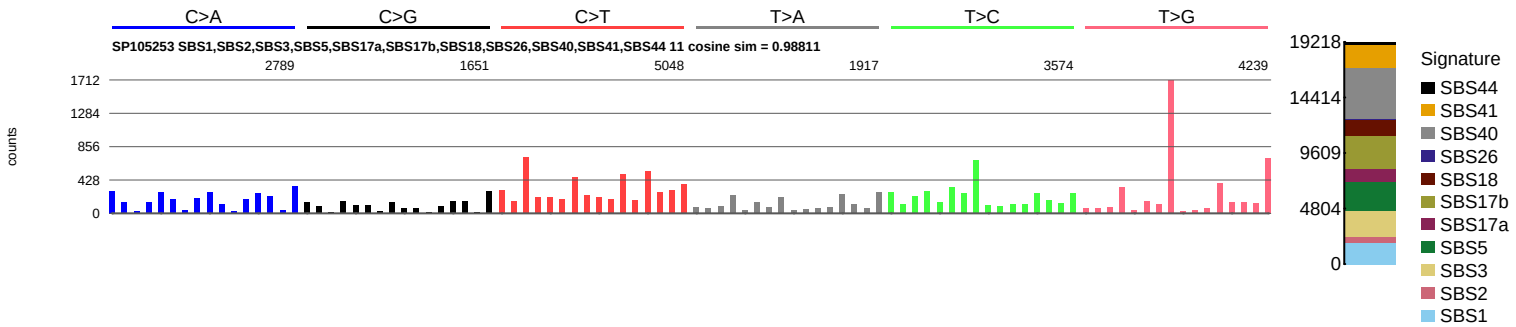

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98821
2384
1992
5238
1756
3533
4315
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

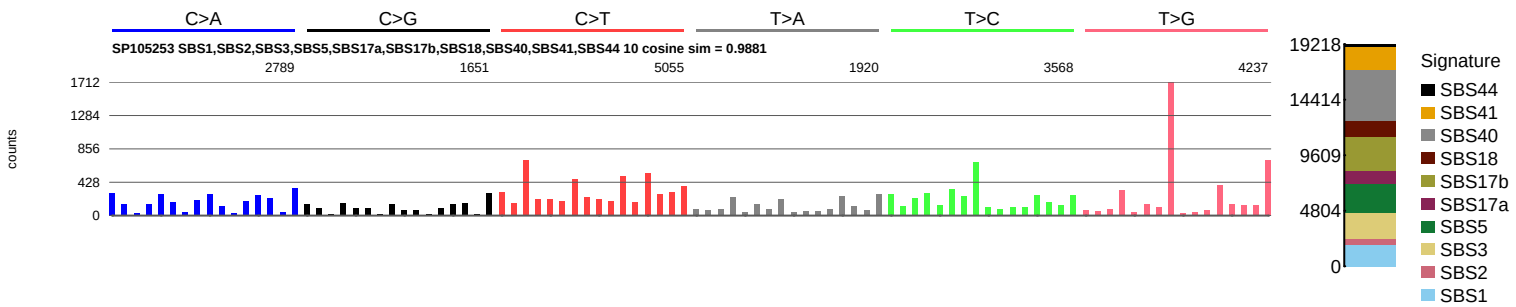

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.9882
2917
1943
5032
1816
3145
4367
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

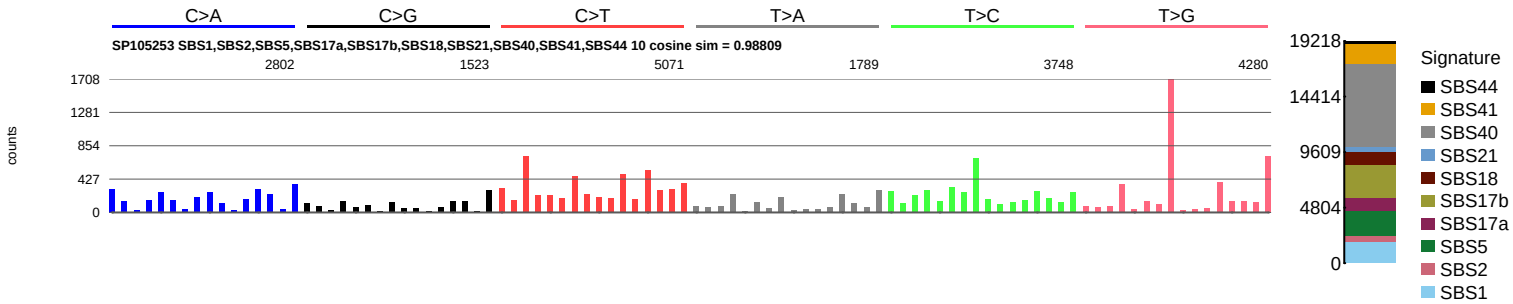

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.9882
2859
2044
4703
1957
3624
4035
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

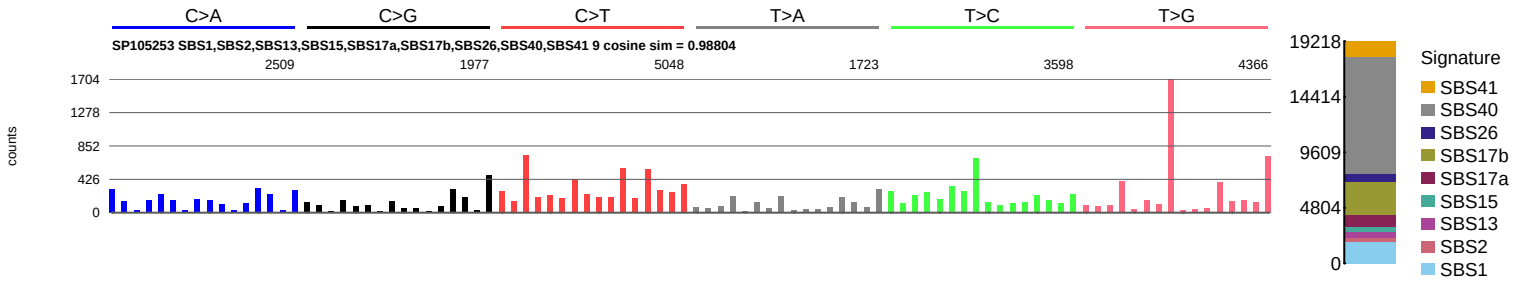

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98819
2504
1961
5103
1719
3587
4342
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

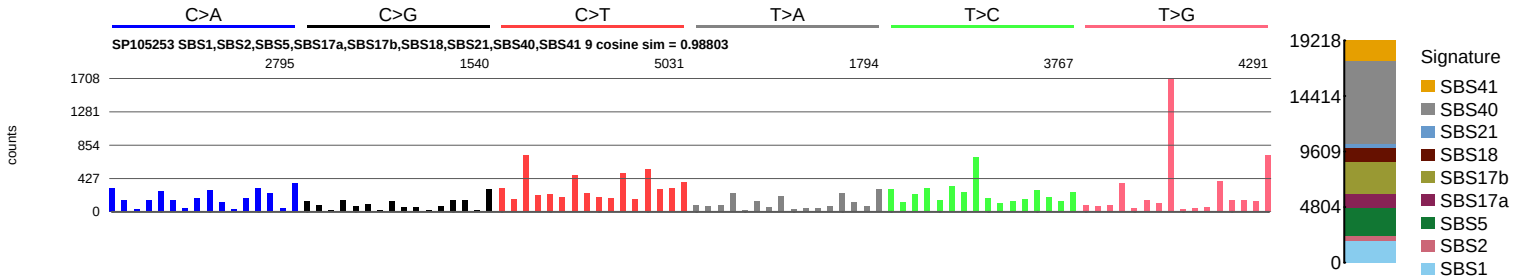

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98816
2465
2036
5023
1771
3578
4345
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

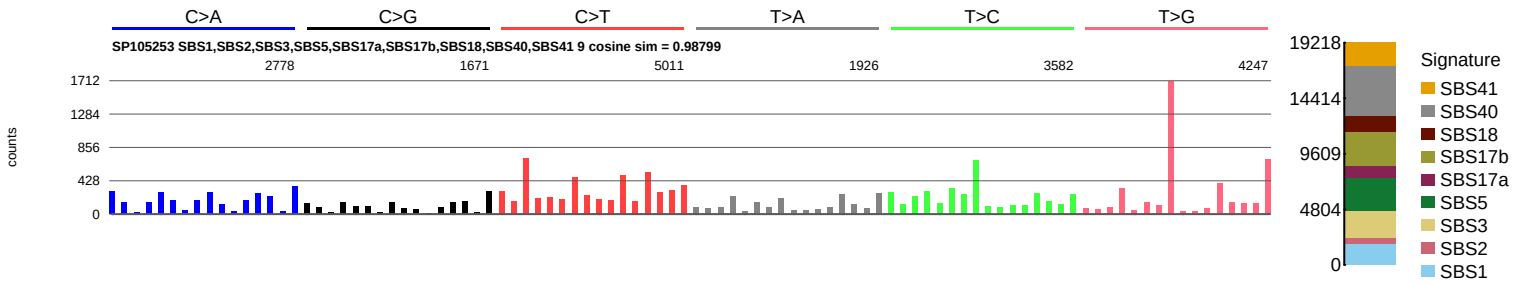

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98811
2789
1651
5048
1917
3574
4239
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

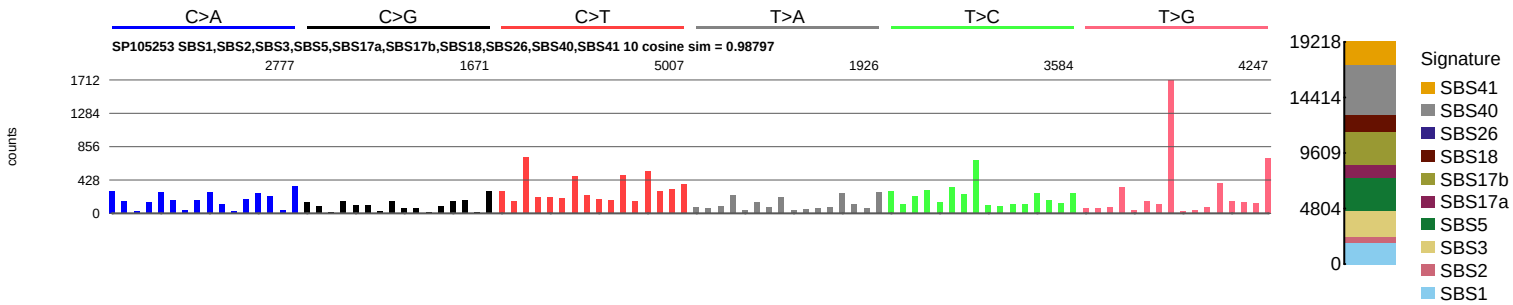

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.9881
2789
1651
5055
1920
3568
4237
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

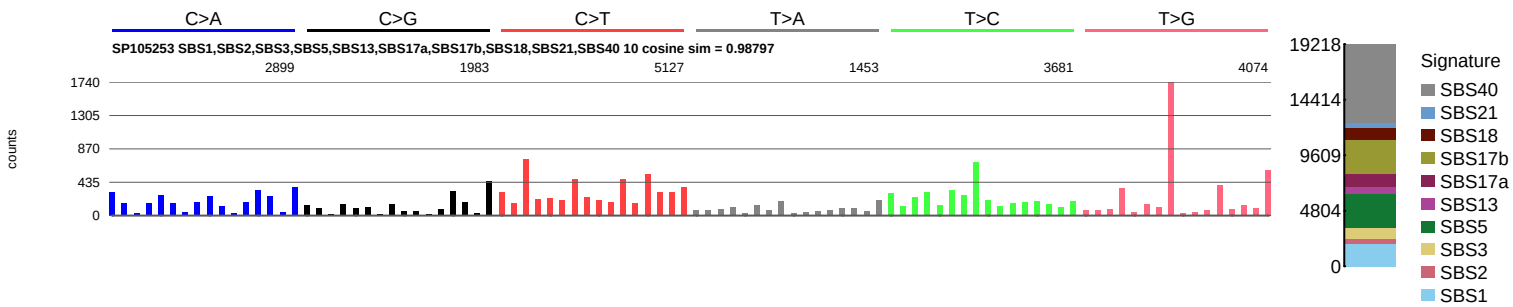

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98809
2802
1523
5071
1789
3748
4280
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

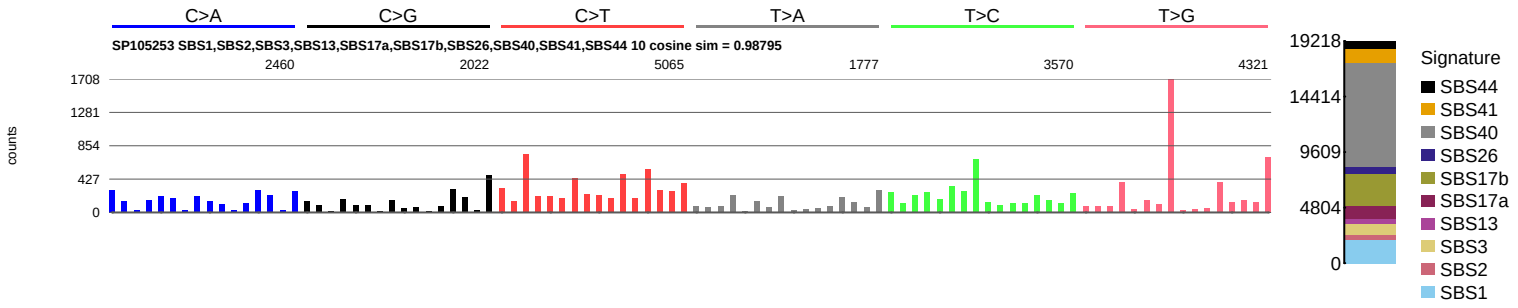

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98804
2509
1977
5048
1723
3598
4366
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

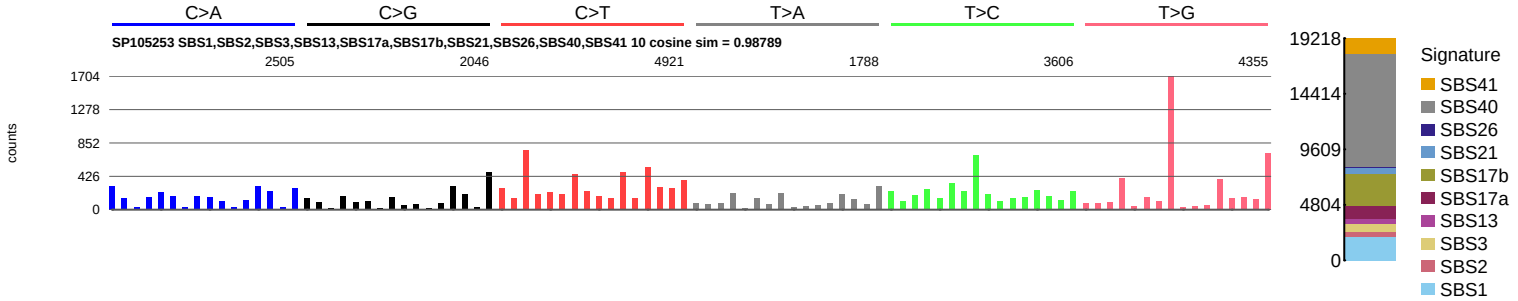

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98803
2795
1540
5031
1794
3767
4291
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

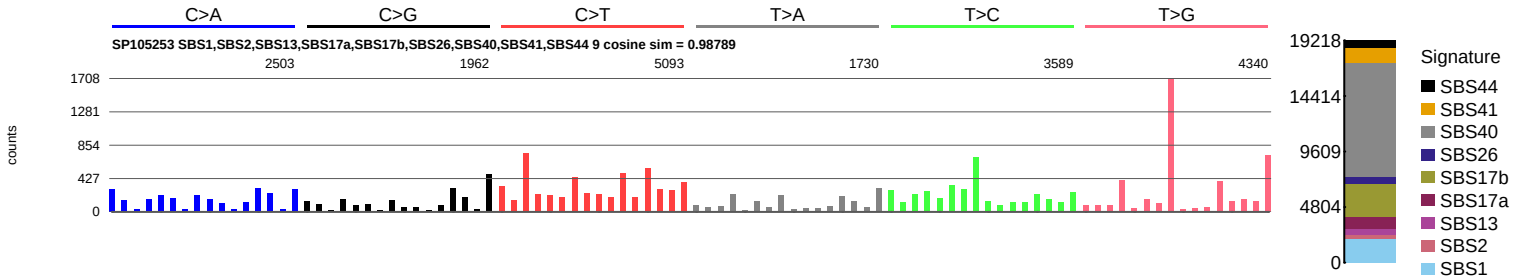

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98799
2778
1671
5011
1926
3582
4247
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

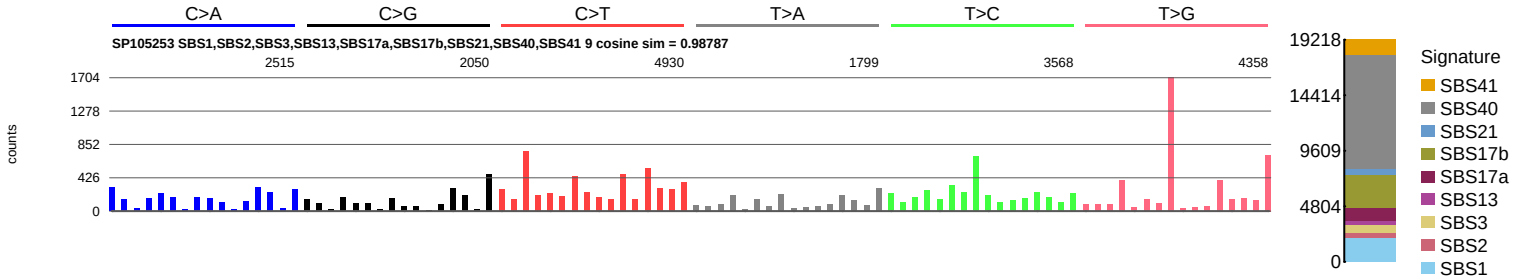

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98797
2777
1671
5007
1926
3584
4247
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

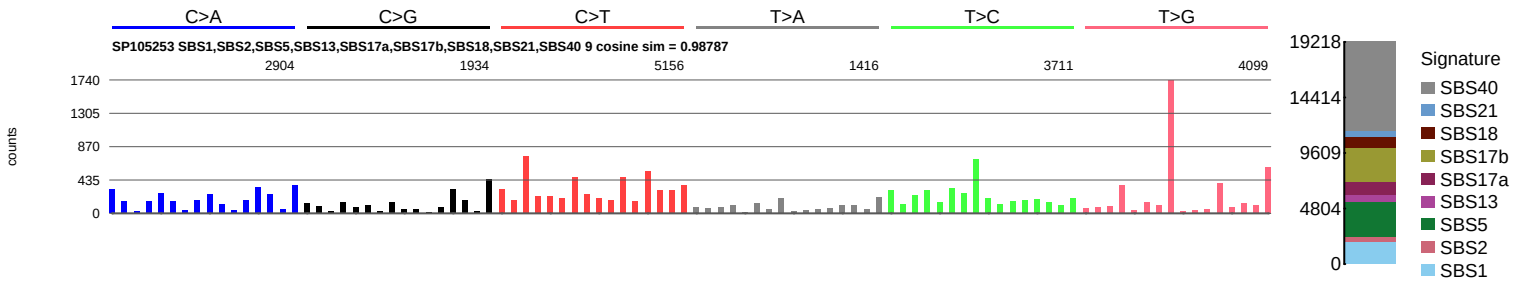

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.98797
2899
1983
5127
1453
3681
4074
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

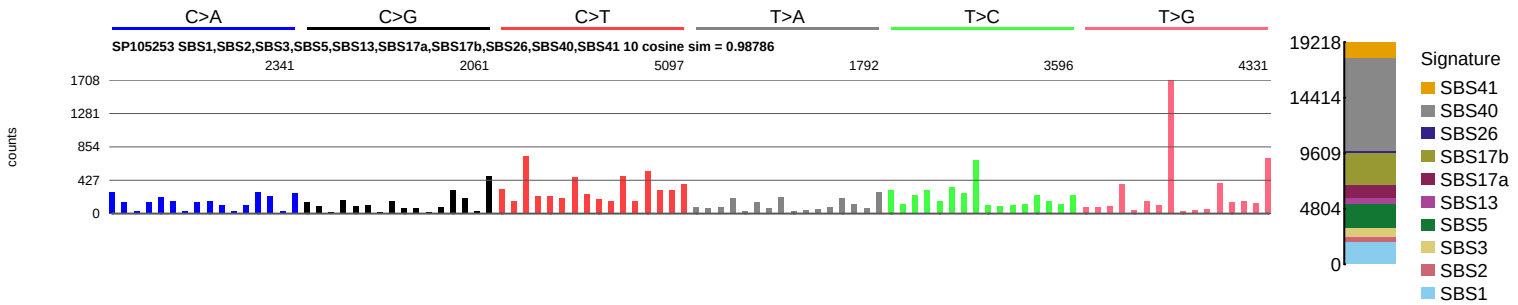

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98795
2460
2022
5065
1777
3570
4321
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

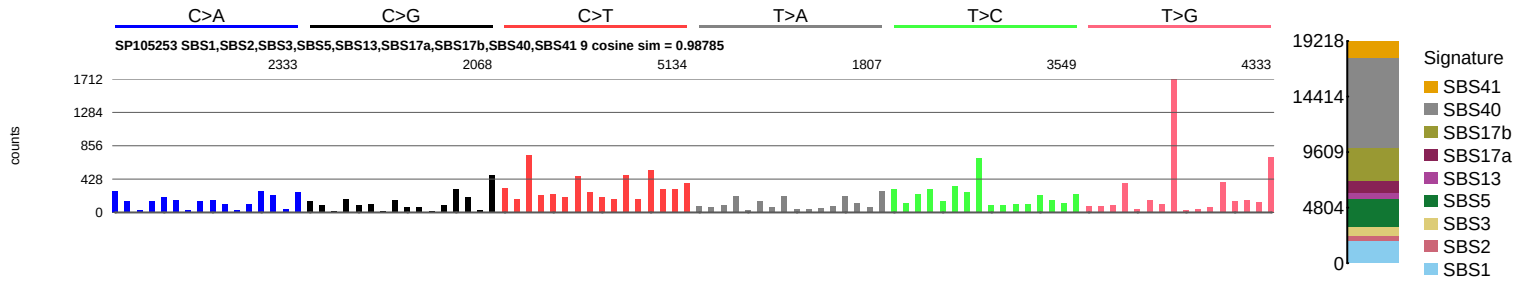

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98789
2505
2046
4921
1788
3606
4355
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

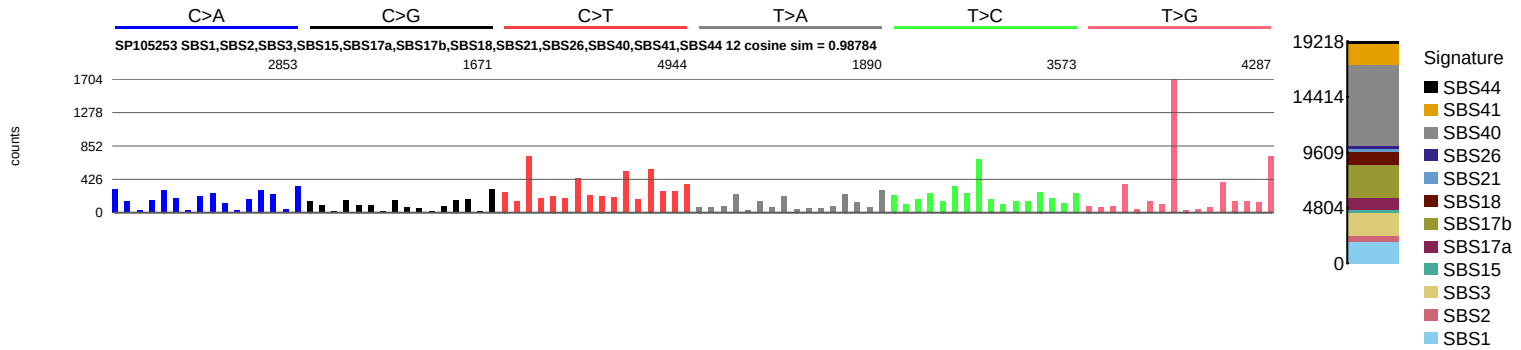

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98789
2503
1962
5093
1730
3589
4340
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98787
2515
2050
4930
1799
3568
4358
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98787
2904
1934
5156
1416
3711
4099
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98786
2341
2061
5097
1792
3596
4331
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98785
2333
2068
5134
1807
3549
4333
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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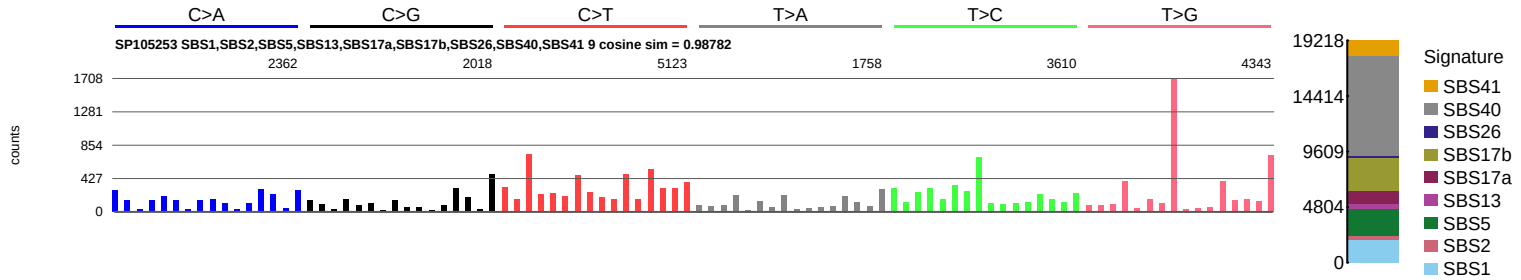
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 12 cosine sim = 0.98784
2853
1671
4944
1890
3573
4287
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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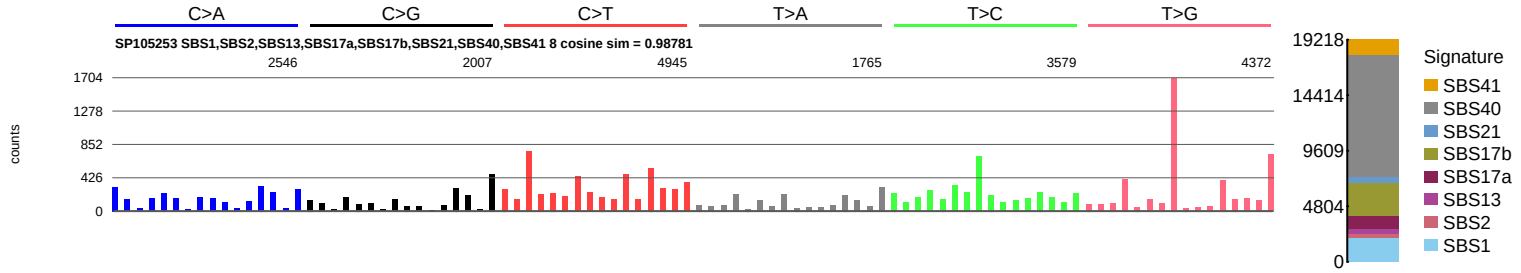
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98782
2362
2018
5123
1758
3610
4343
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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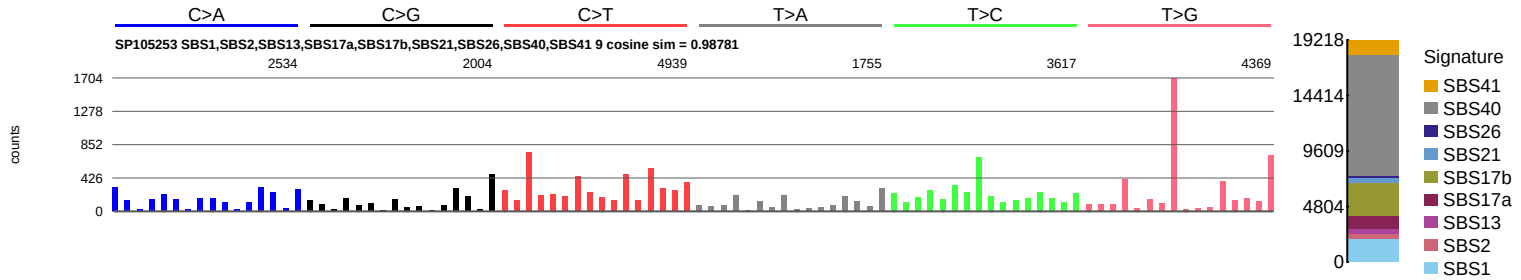
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98781
2546
2007
4945
1765
3579
4372
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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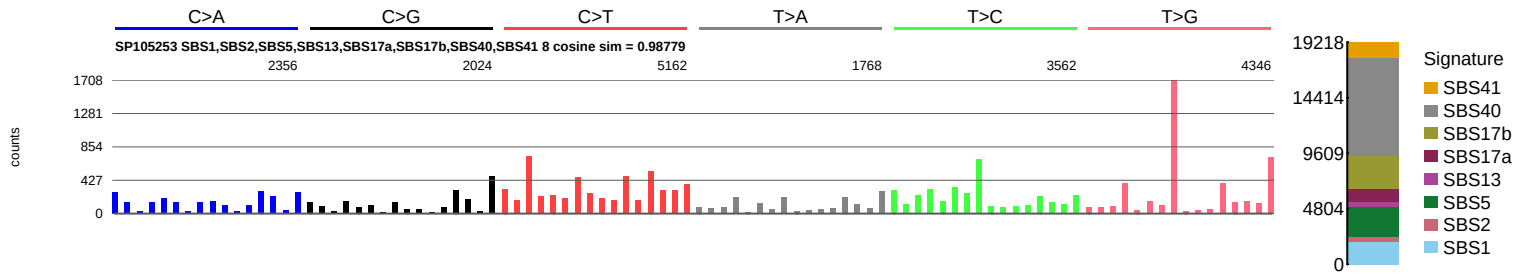
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98781
2534
2004
4939
1755
3617
4369
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1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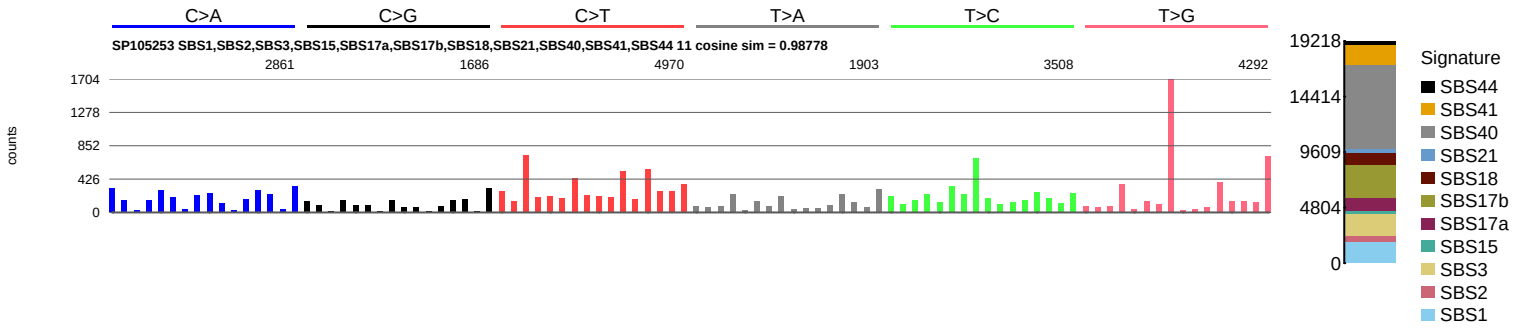
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98779
2356
2024
5162
1768
3562
4346
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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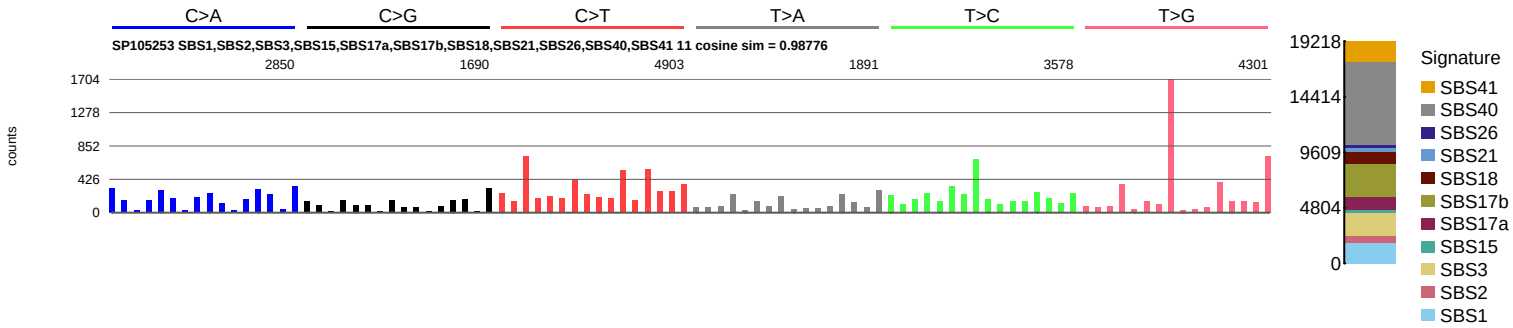
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98778
2861
1686
4970
1903
3508
4292
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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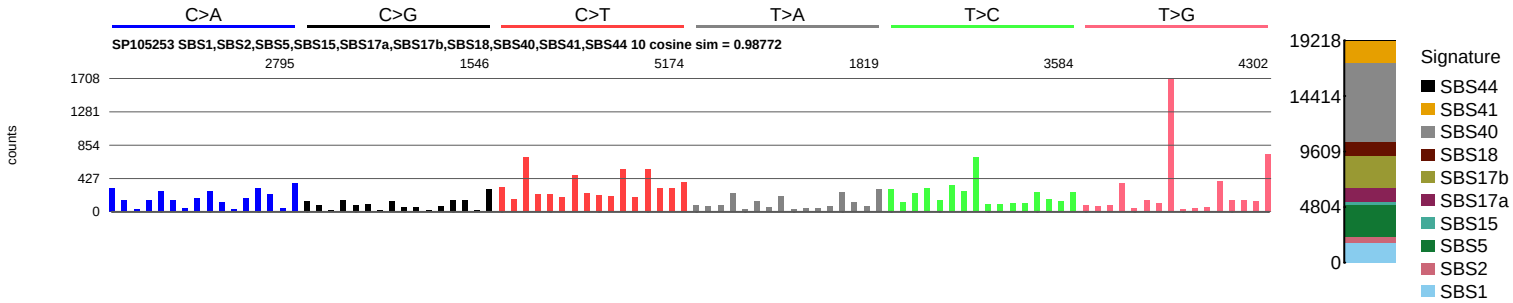
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 11 cosine sim = 0.98776
2850
1690
4903
1891
3578
4301
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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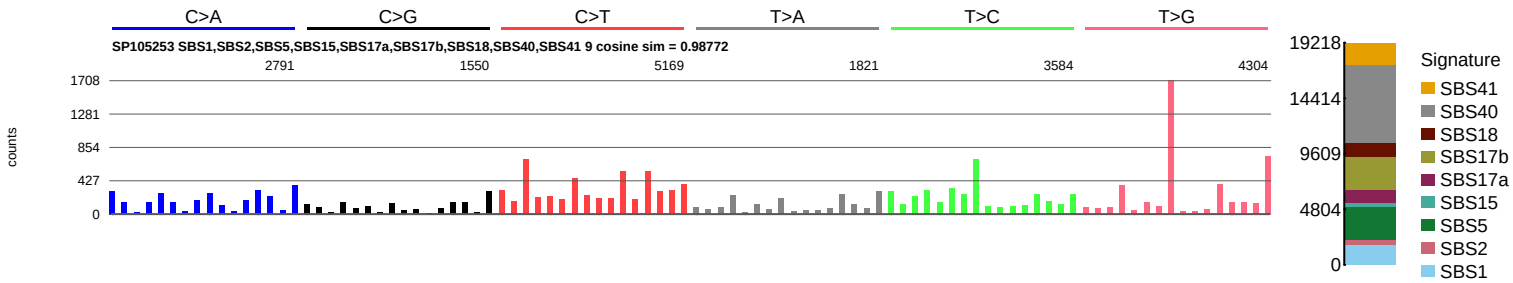
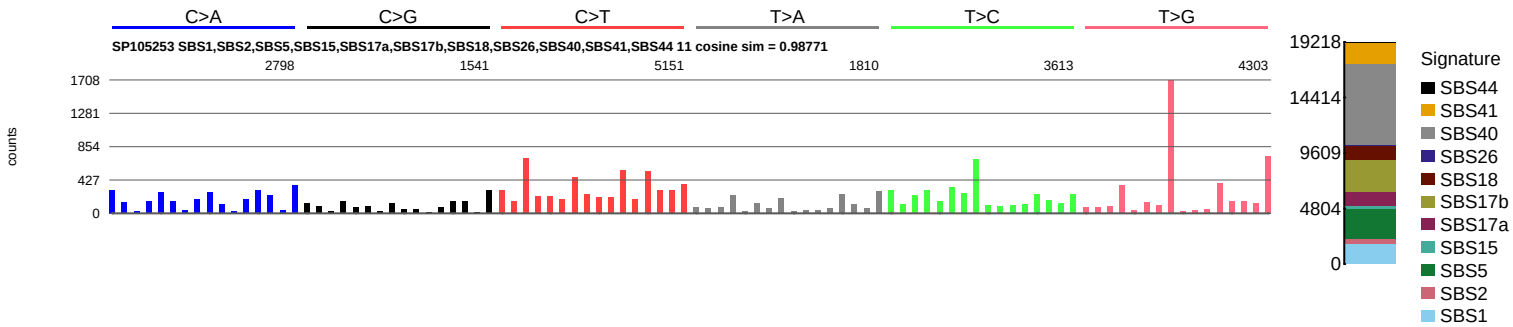
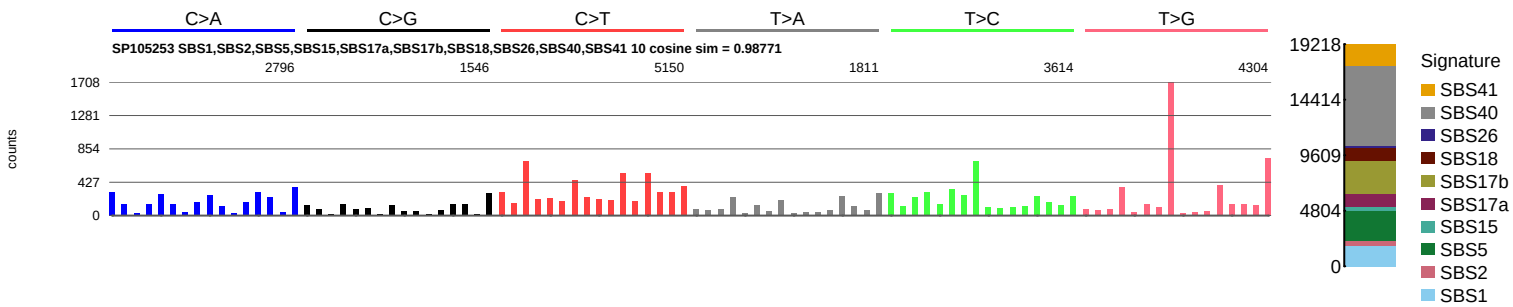
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98772
2795
1546
5174
1819
3584
4302
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98772
2791
1550
5169
1821
3584
4304
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98771
2798
1541
5151
1810
3613
4303
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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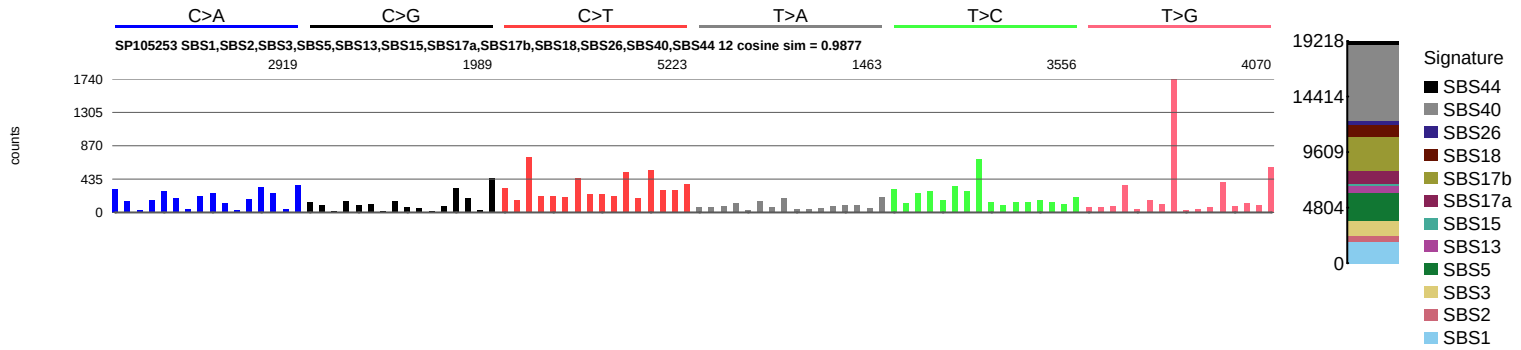
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98771
2796
1546
5150
1811
3614
4304
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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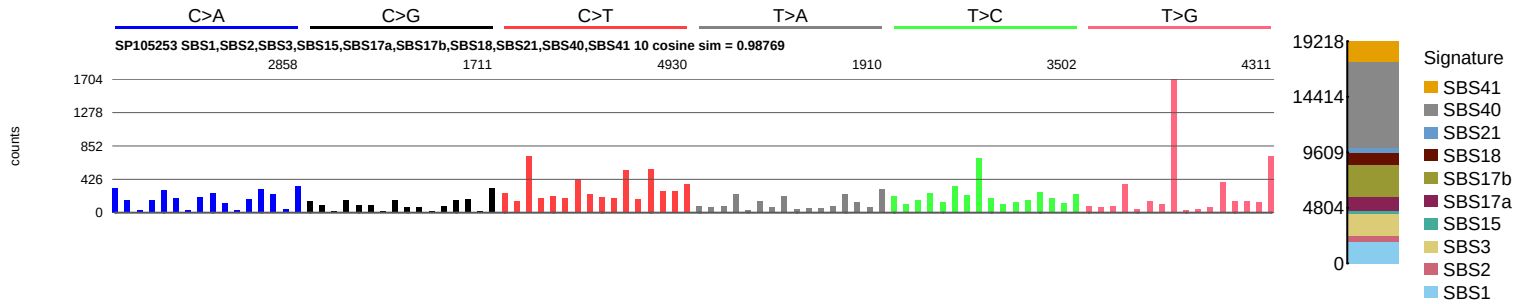
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 12 cosine sim = 0.9877
2919
1989
5223
1463
3556
4070
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
C>T
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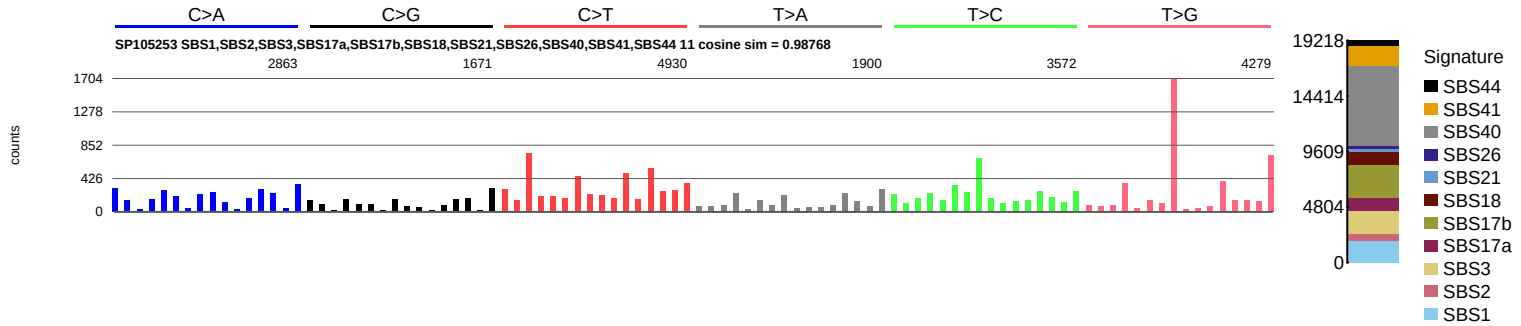
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98769
2858
1711
4930
1910
3502
4311
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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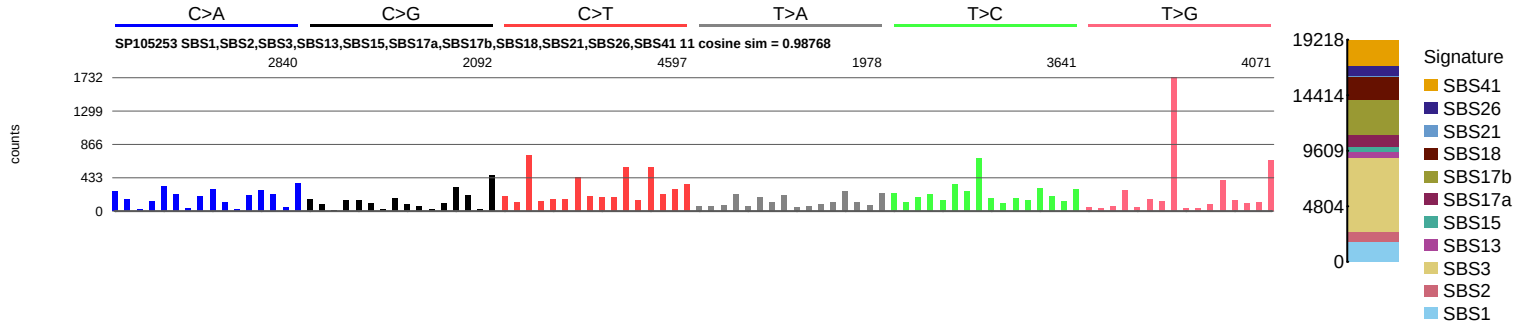
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98768
2863
1671
4930
1900
3572
4279
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.98768
2840
2092
4597
1978
3641
4071
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
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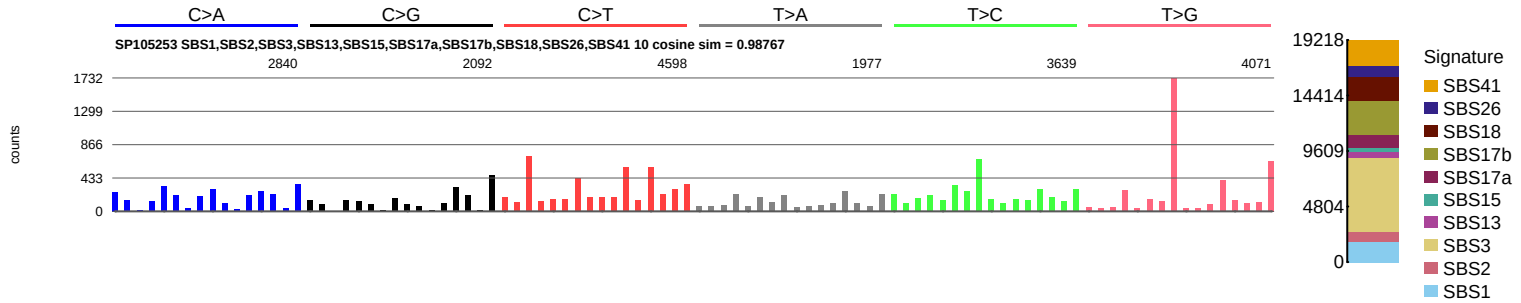
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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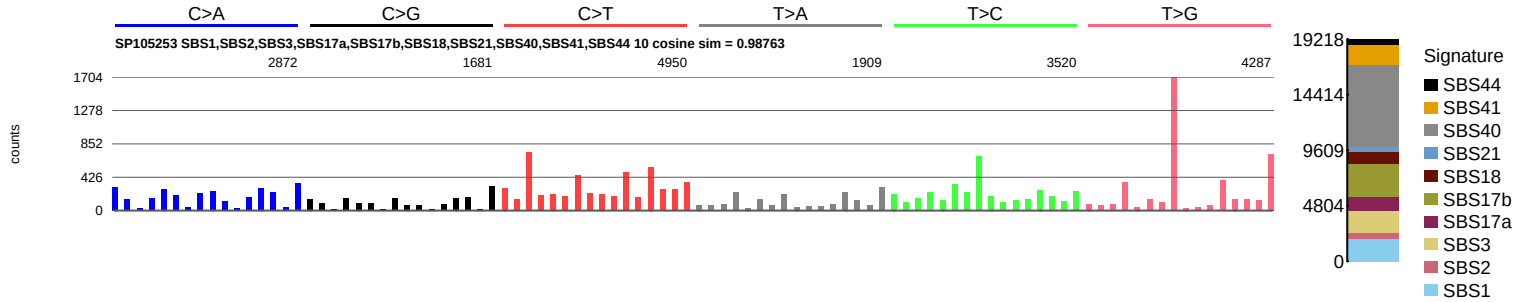
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.98767
2840
2092
4598
1977
3639
4071
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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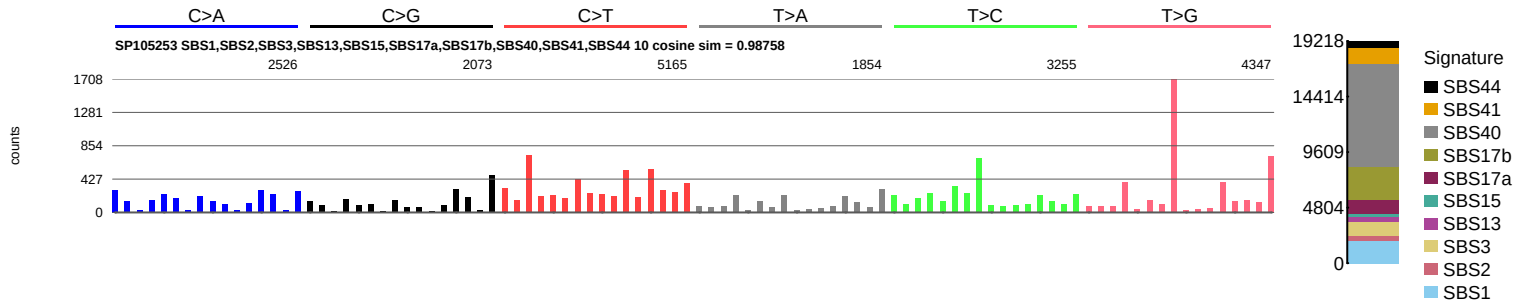
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98763
2872
1681
4950
1909
3520
4287
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98758
2526
2073
5165
1854
3255
4347
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
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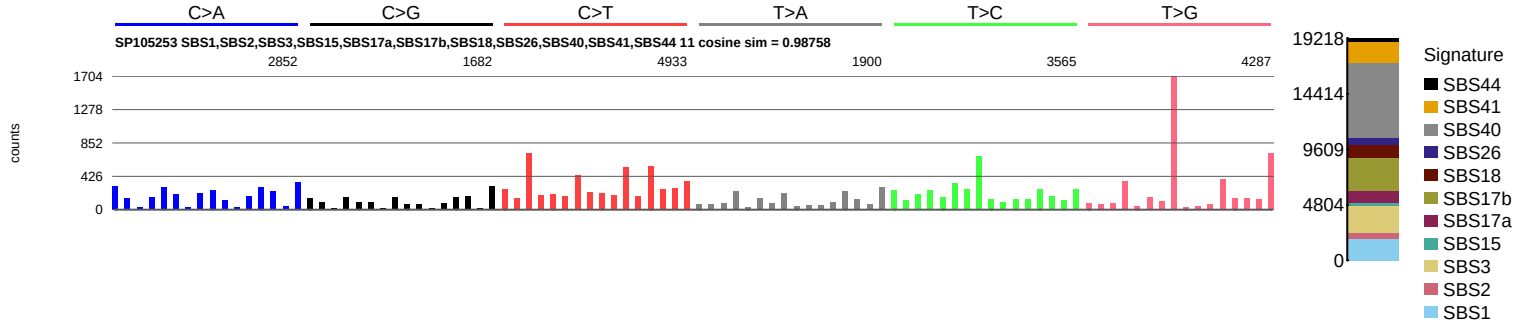
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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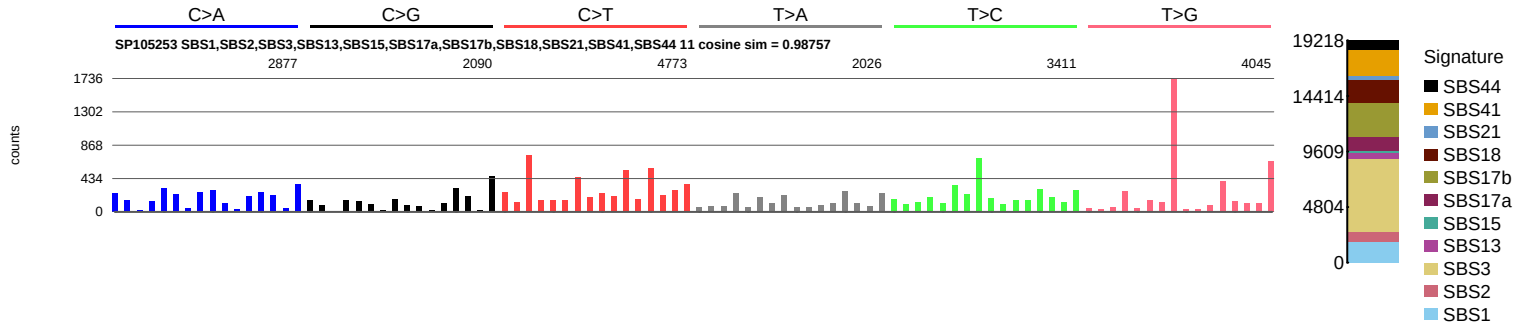
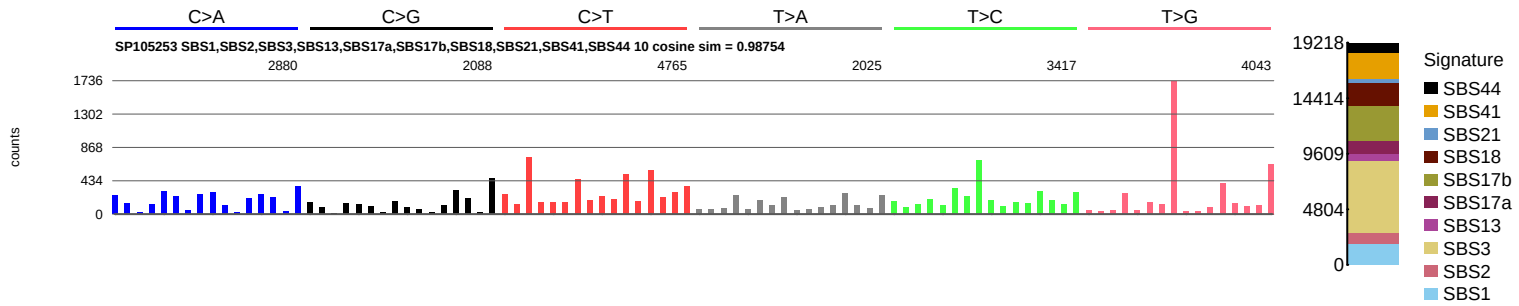
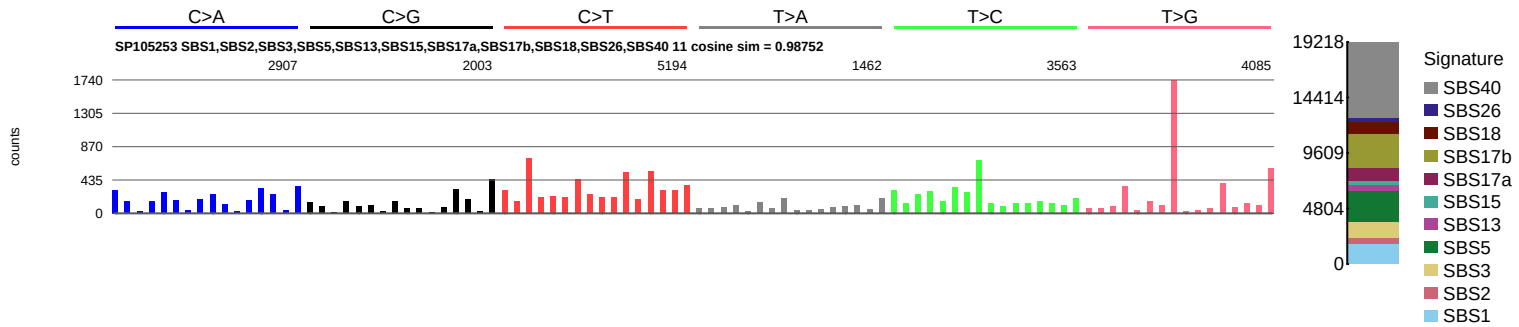
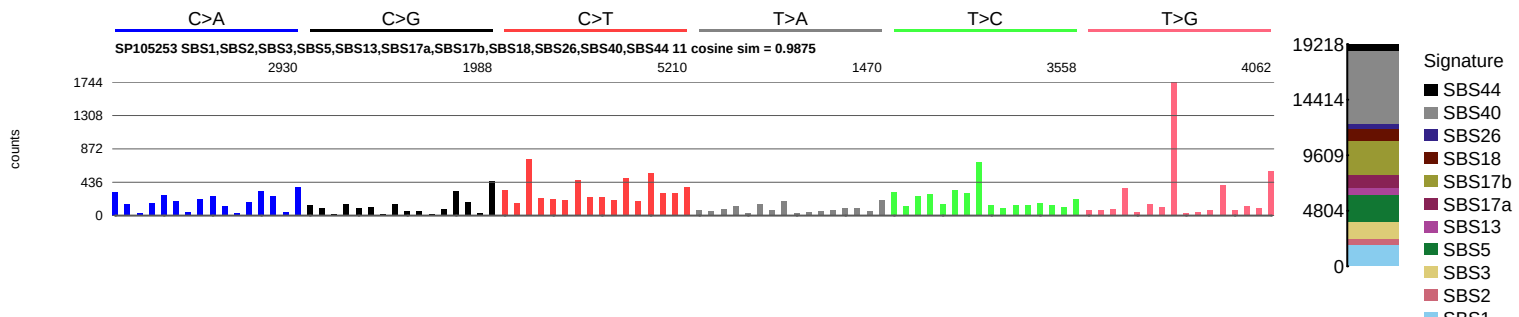
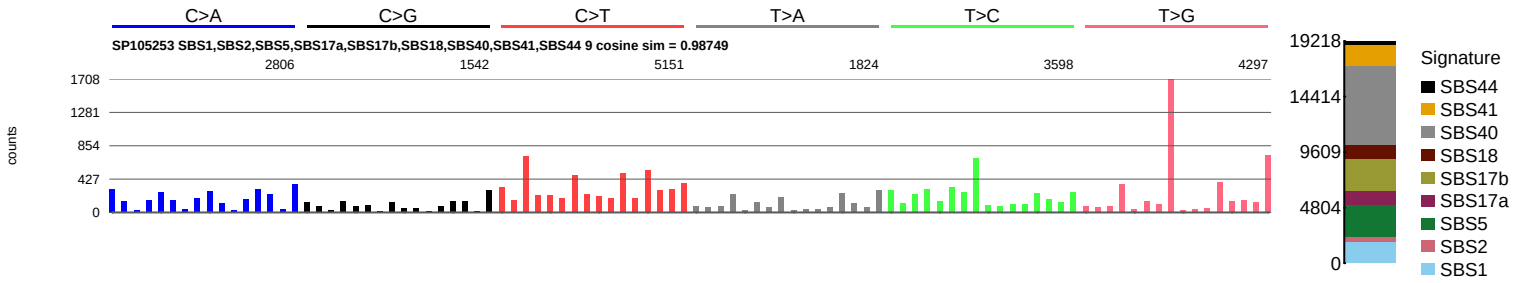
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98758
2852
1682
4933
1900
3565
4287
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 11 cosine sim = 0.98757
2877
2090
4773
2026
3411
4045
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98754
2880
2088
4765
2025
3417
4043
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 11 cosine sim = 0.98752
2907
2003
5194
1462
3563
4085
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.9875
2930
1988
5210
1470
3558
4062
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2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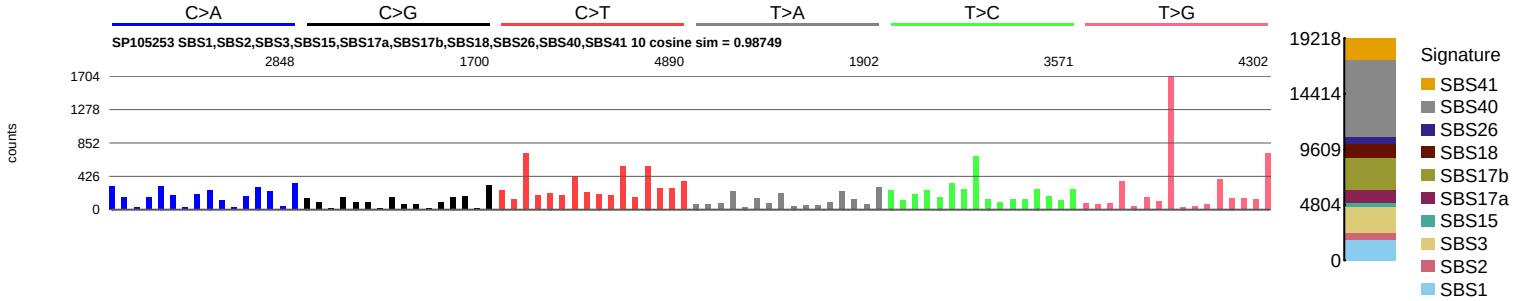
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98749
2806
1542
5151
1824
3598
4297
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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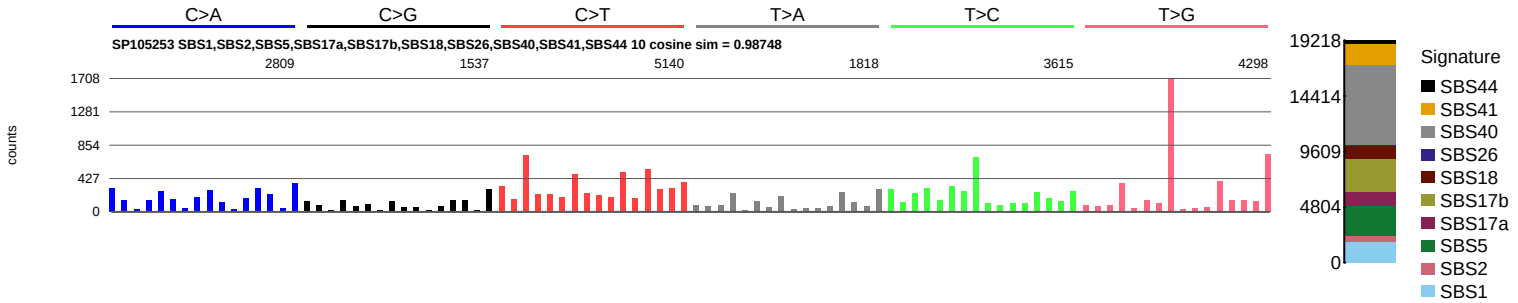
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.98749
2848
1700
4890
1902
3571
4302
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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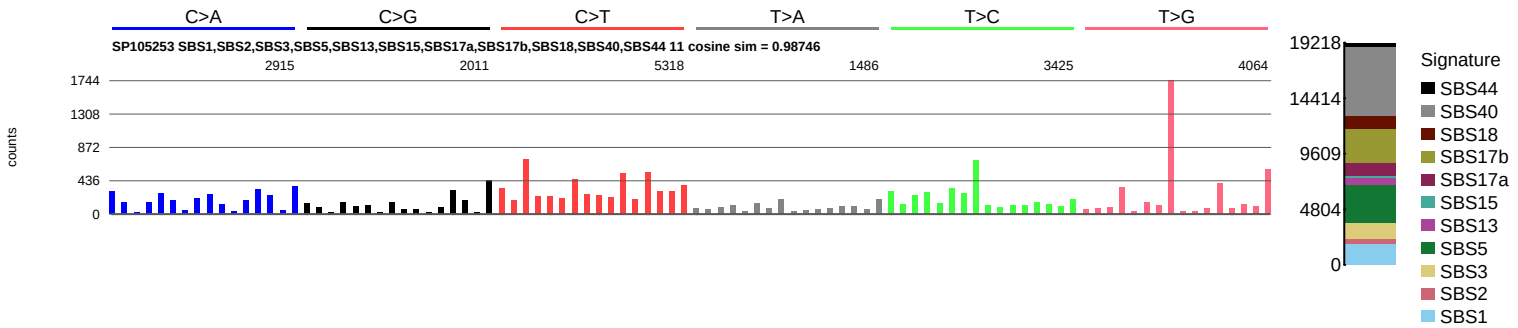
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98748
2809
1537
5140
1818
3615
4298
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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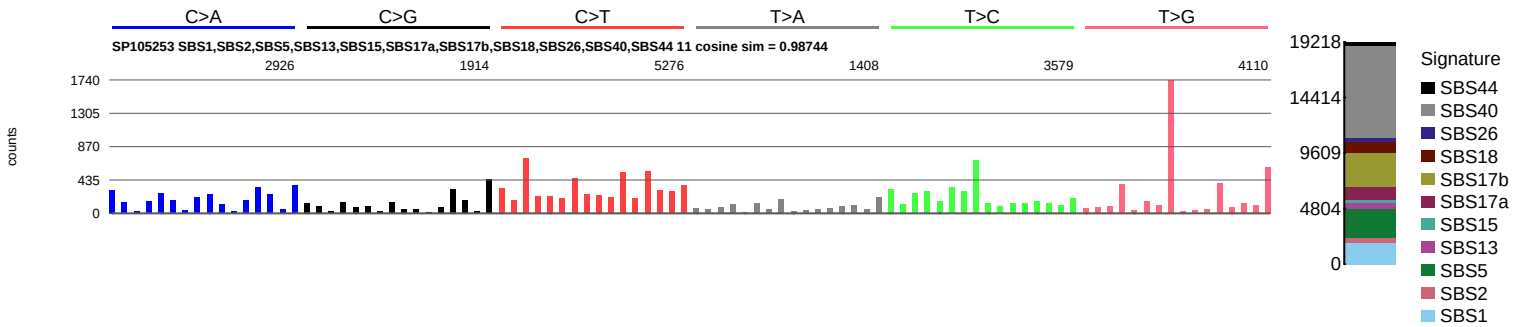
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 11 cosine sim = 0.98746
2915
2011
5318
1486
3425
4064
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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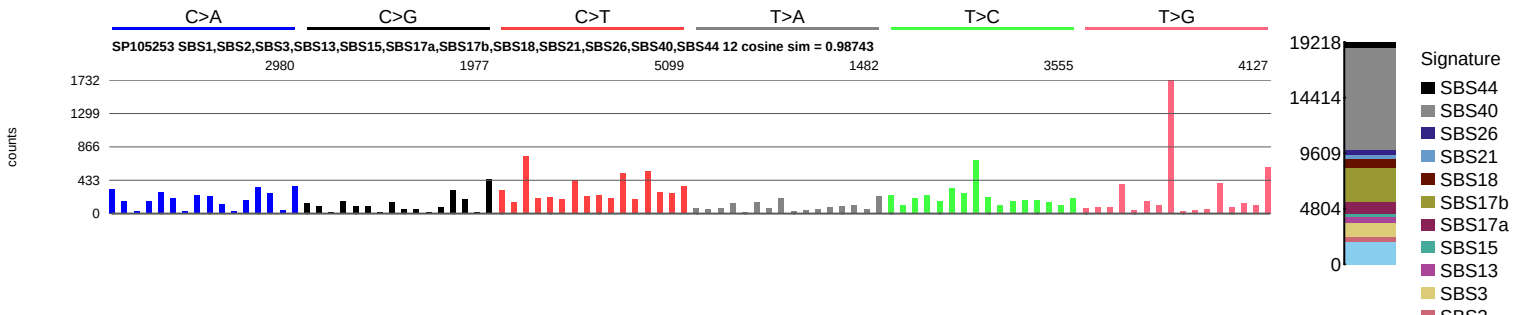
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98744
2926
1914
5276
1408
3579
4110
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
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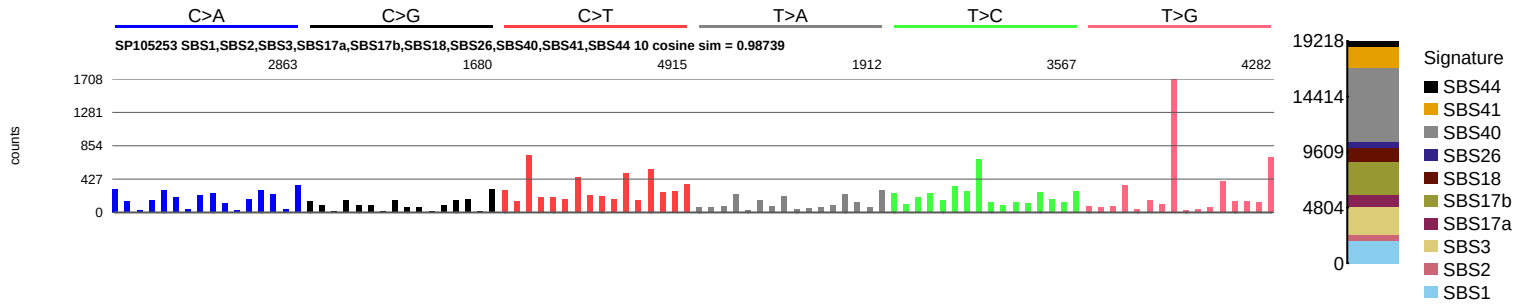
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 12 cosine sim = 0.98743
2980
1977
5099
1482
3555
4127
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3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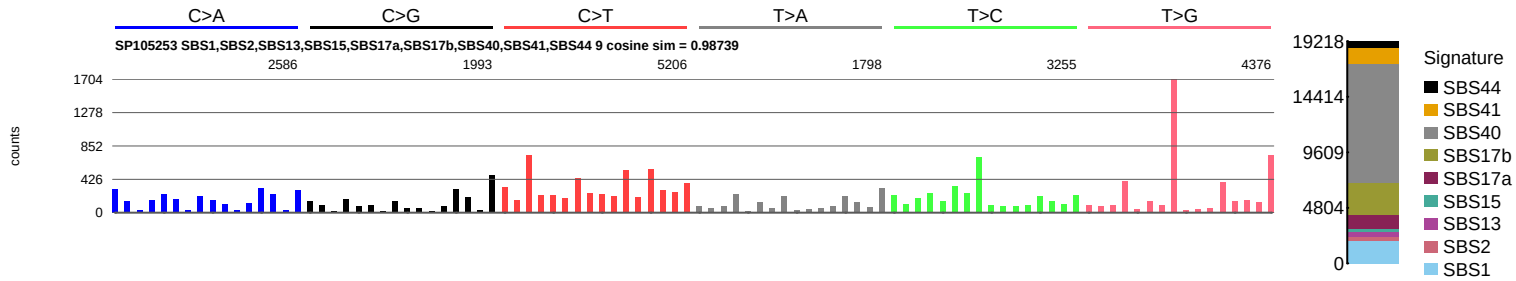
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98739
2863
1680
4915
1912
3567
4282
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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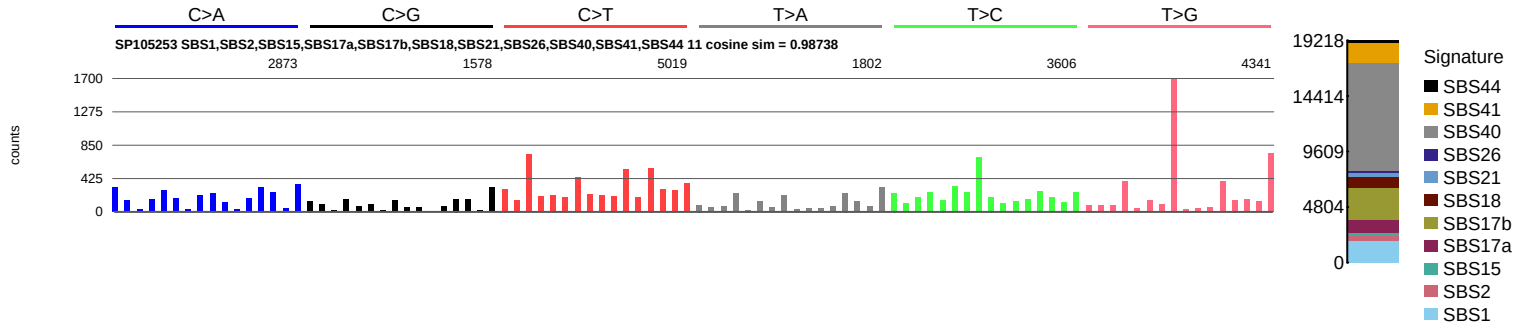
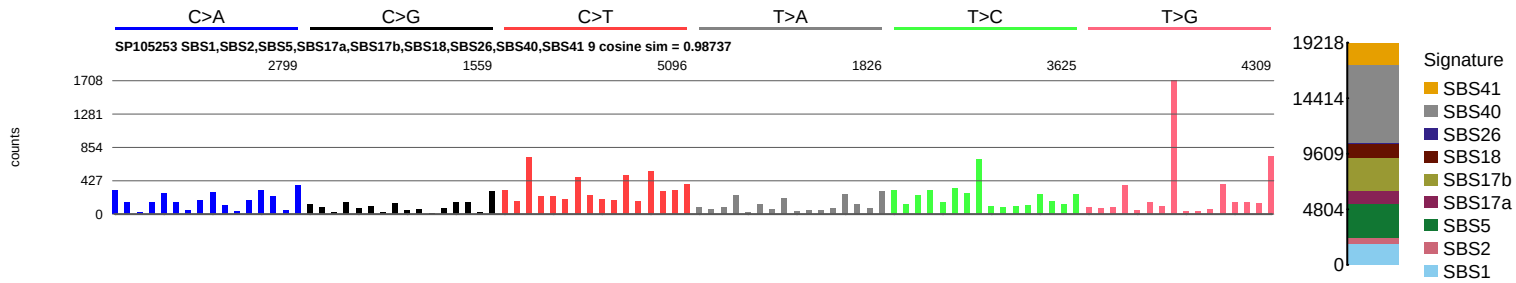
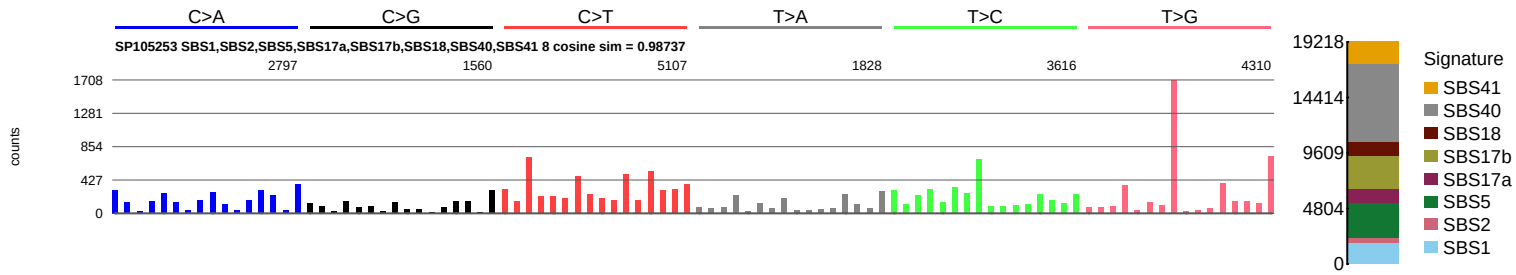
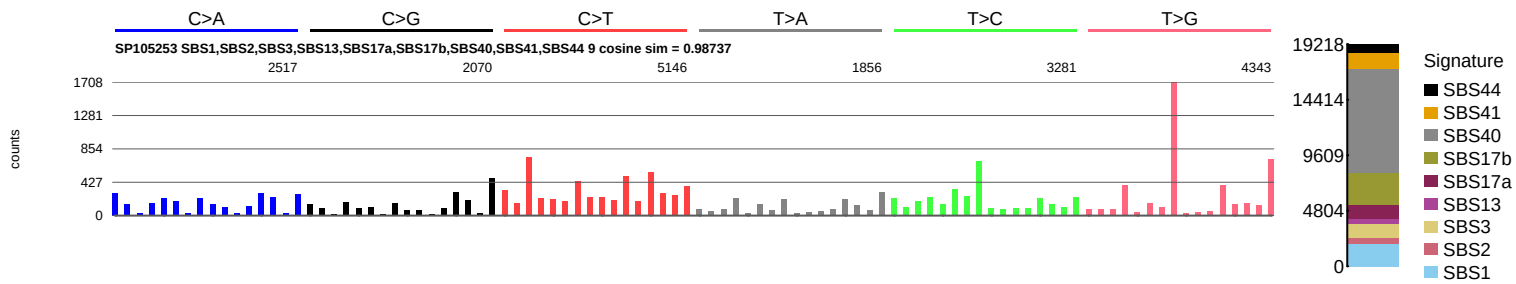
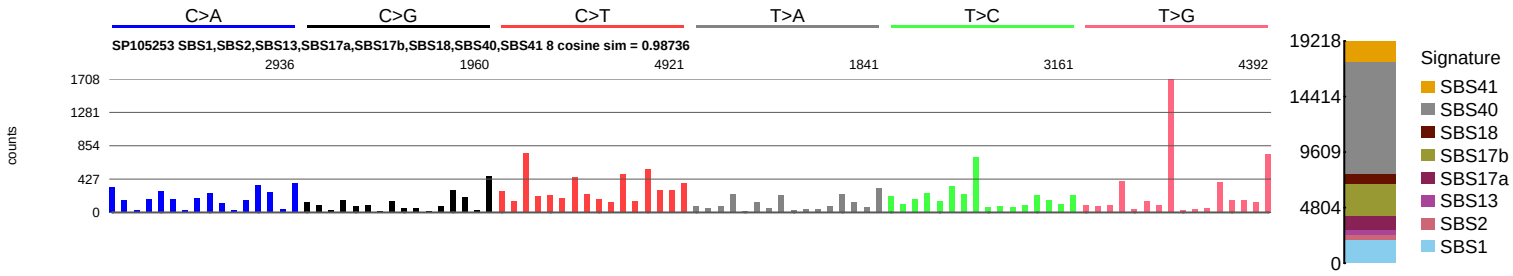
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98739
2586
1993
5206
1798
3255
4376
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

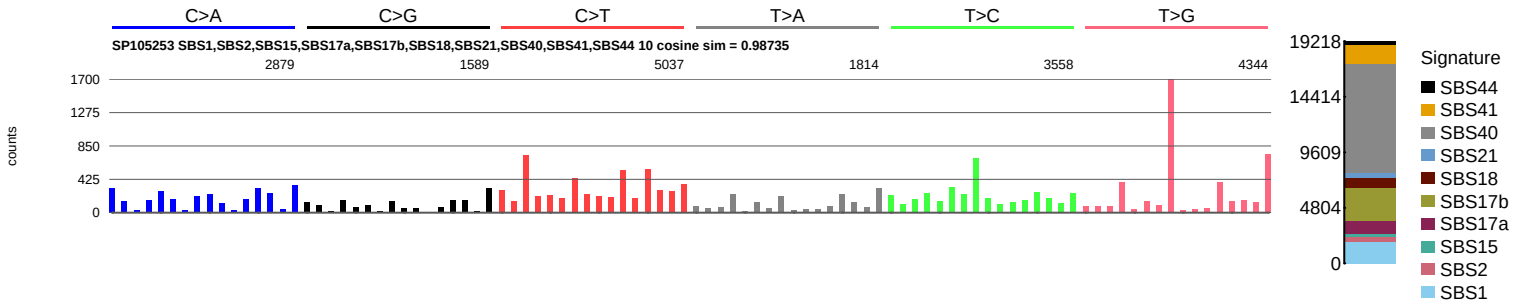

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98738
2873
1578
5019
1802
3606
4341
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

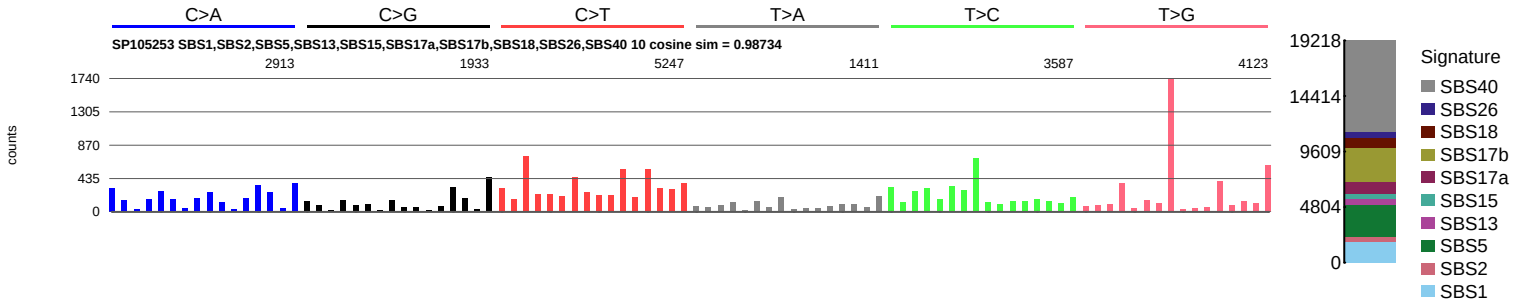

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98737
2799
1559
5096
1826
3625
4309
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98737
2797
1560
5107
1828
3616
4310
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98737
2517
2070
5146
1856
3281
4343
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98736
2936
1960
4921
1841
3161
4392
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98735
2879
1589
5037
1814
3558
4344
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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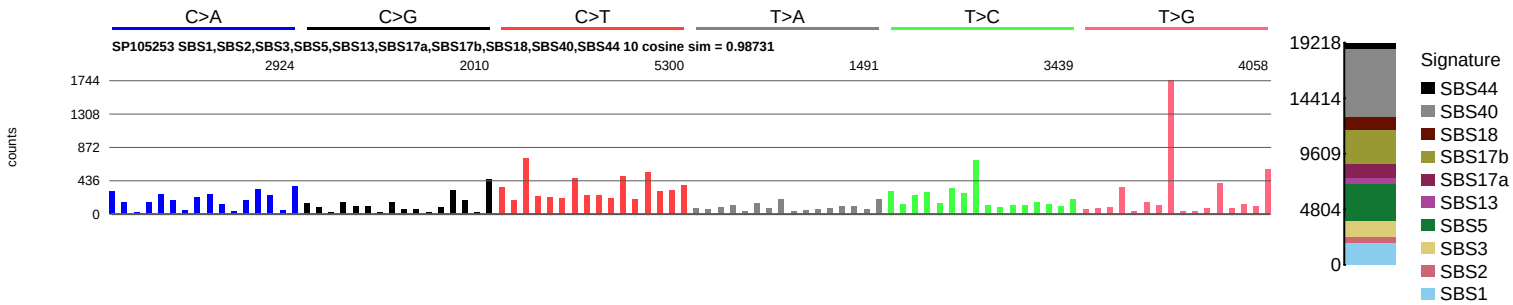
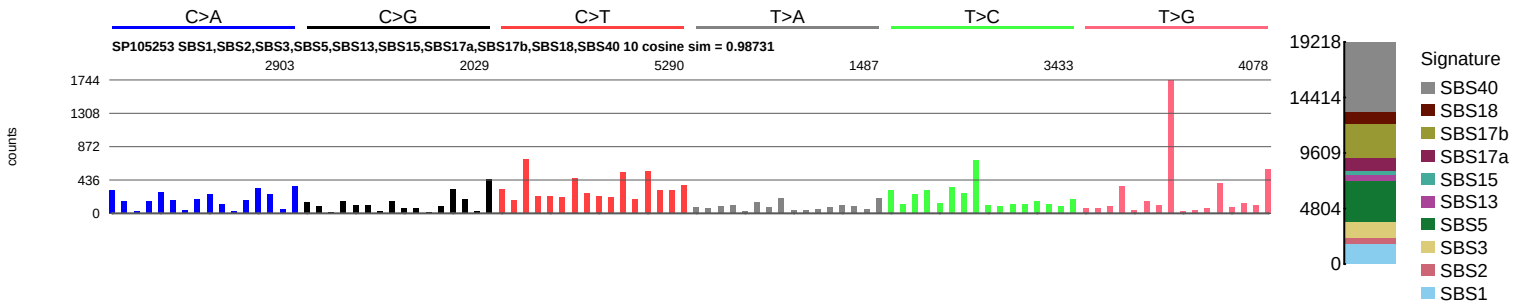
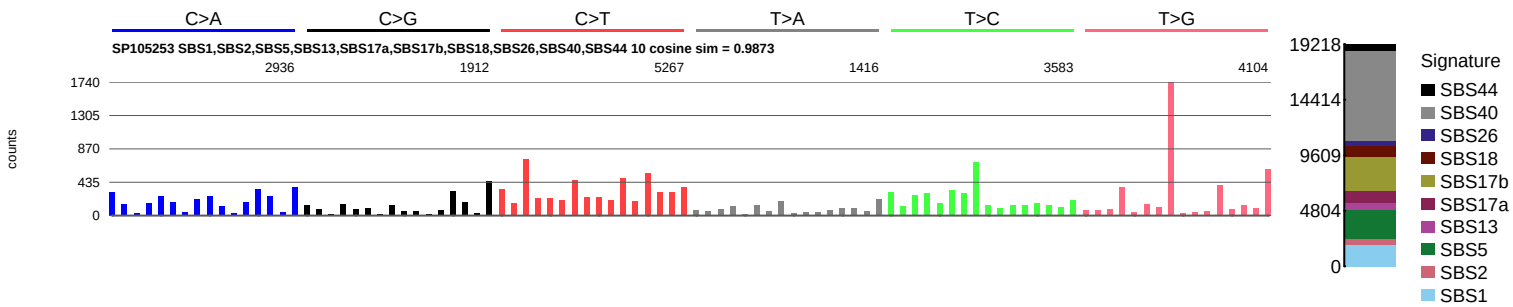
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98734
2913
1933
5247
1411
3587
4123
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98731
2924
2010
5300
1491
3439
4058
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 10 cosine sim = 0.98731
2903
2029
5290
1487
3433
4078
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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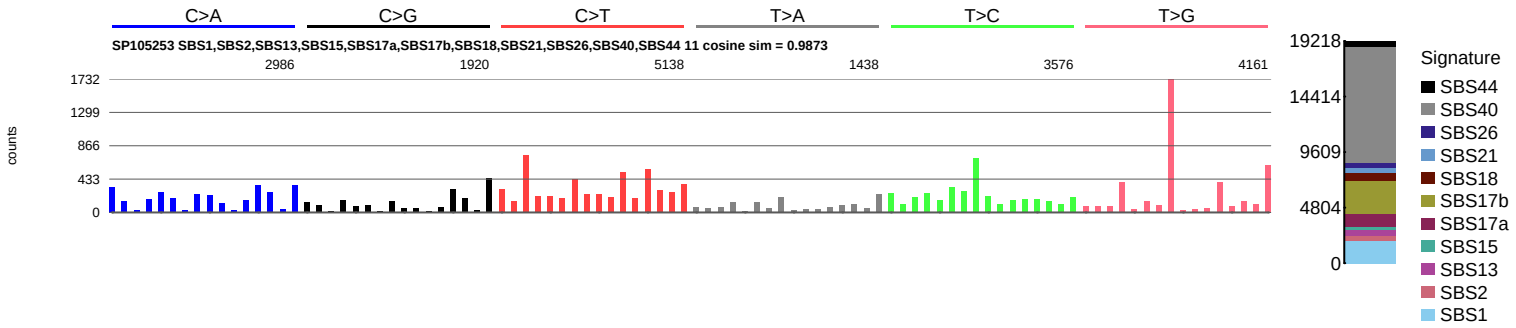
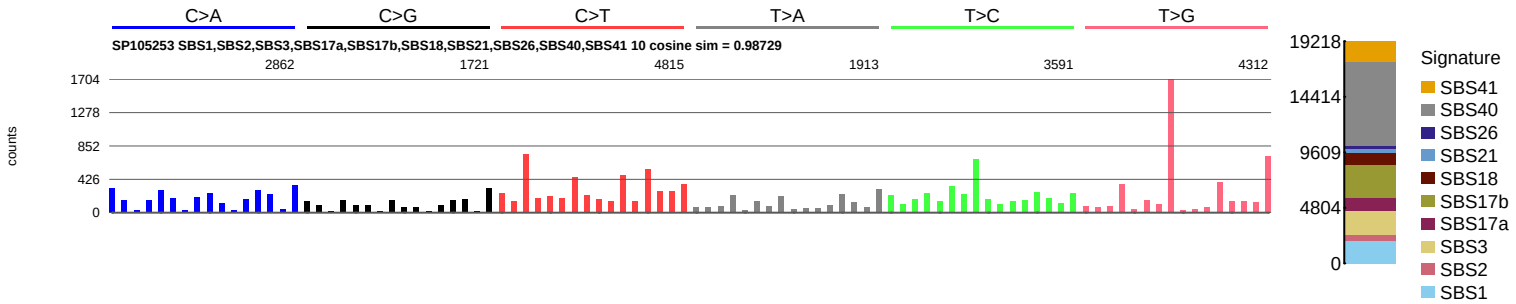
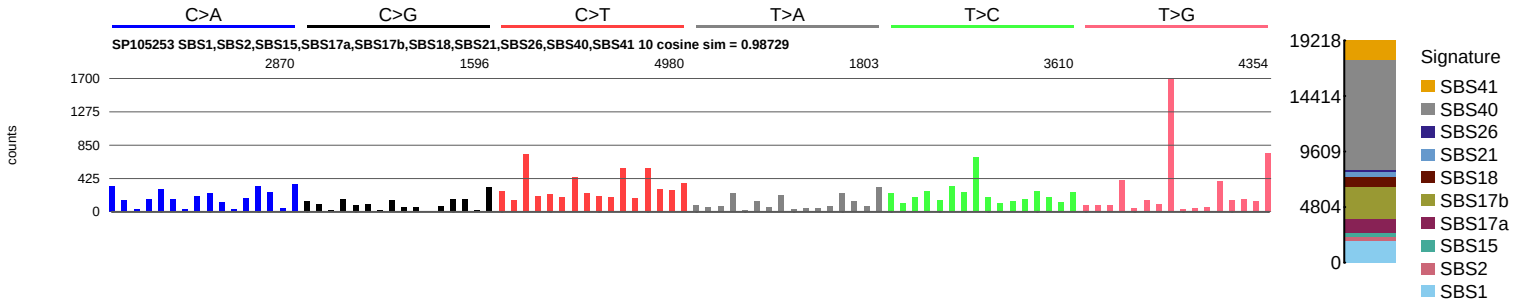
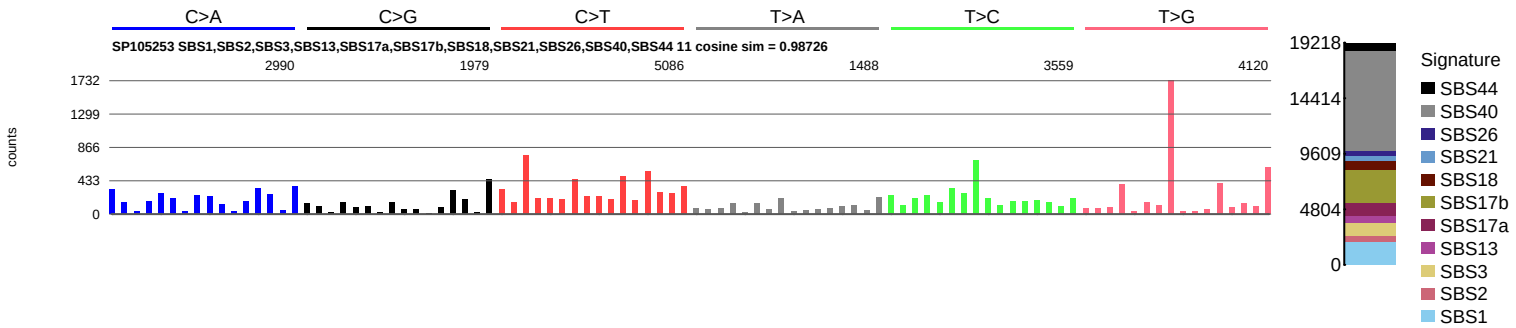
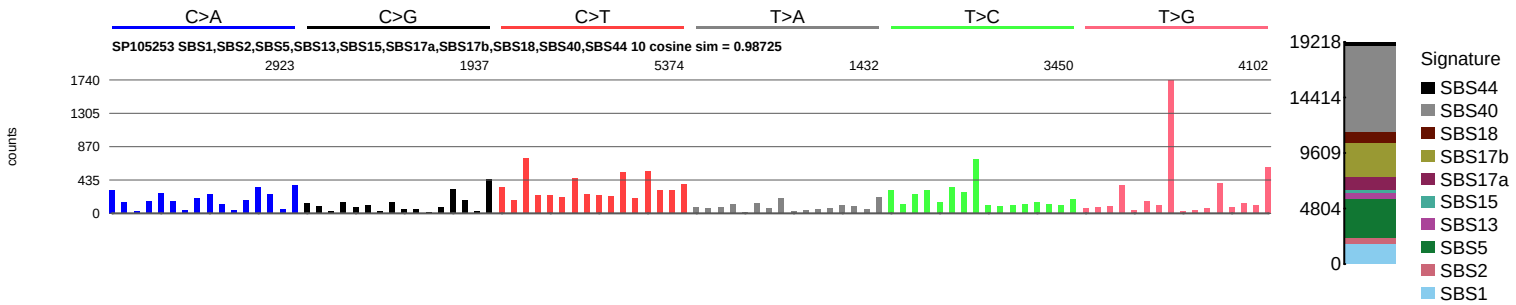
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9873
2936
1912
5267
1416
3583
4104
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

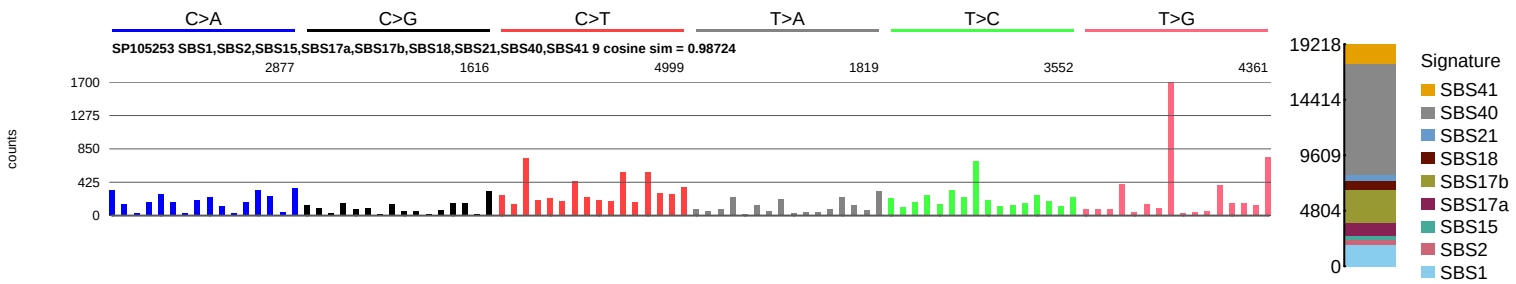

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.9873
2986
1920
5138
1438
3576
4161
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

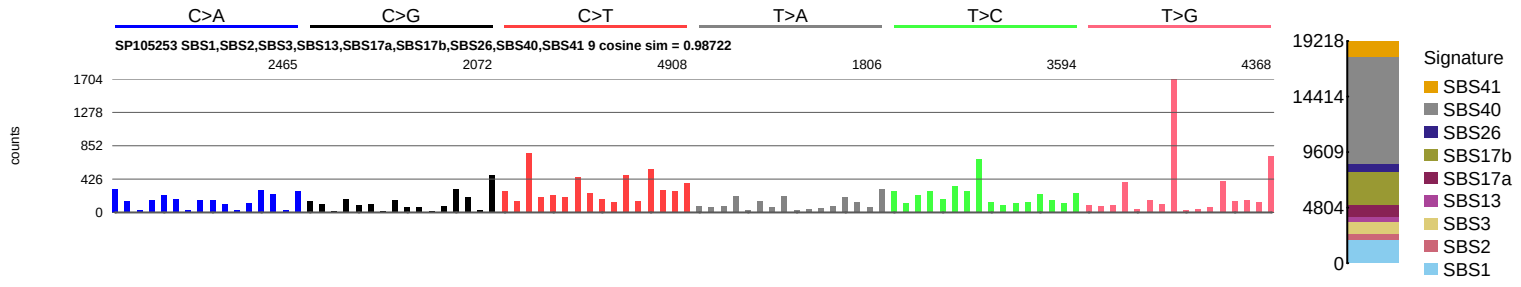

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98729
2862
1721
4815
1913
3591
4312
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

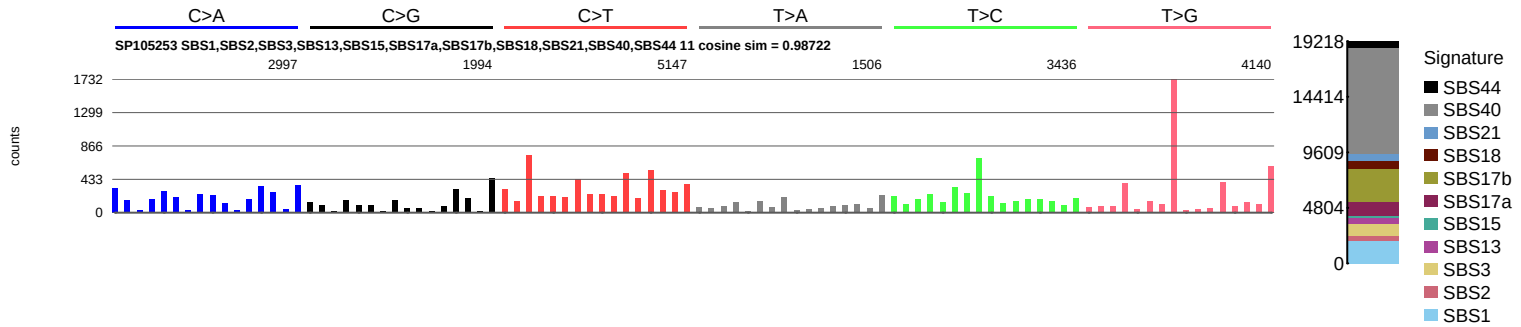

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98729
2870
1596
4980
1803
3610
4354
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

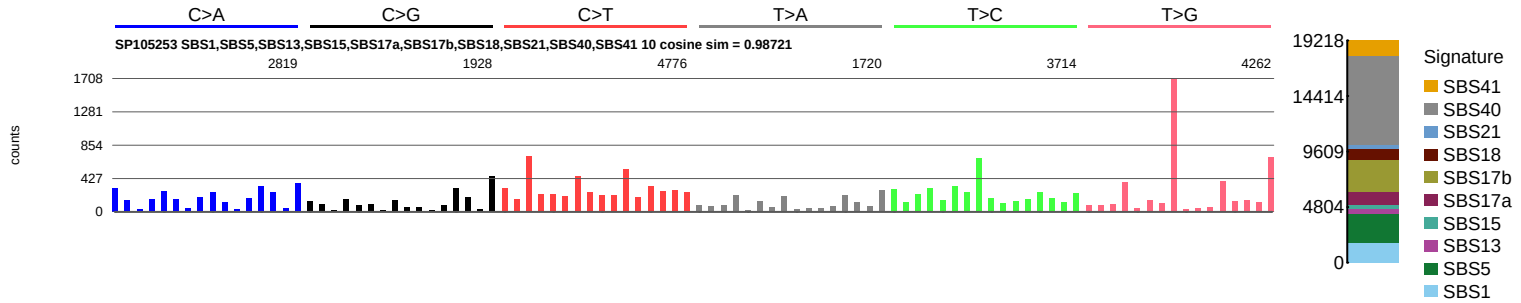

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98726
2990
1979
5086
1488
3559
4120
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

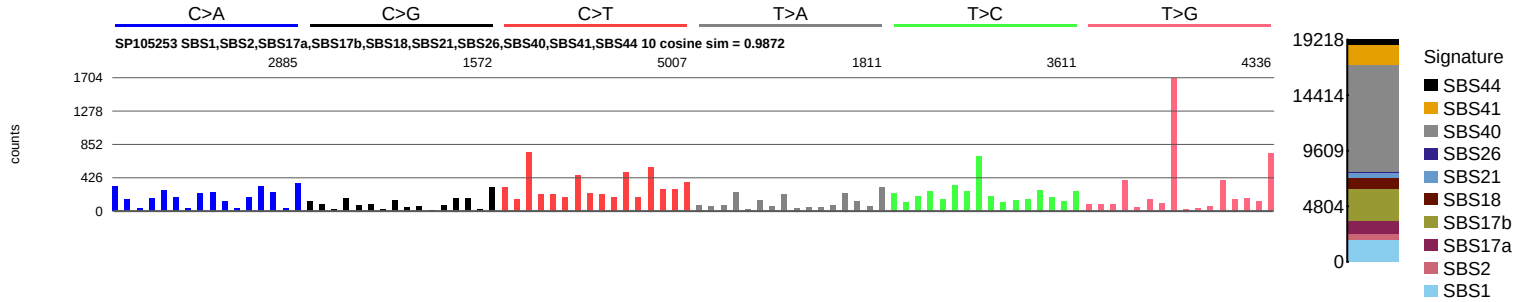

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98725
2923
1937
5374
1432
3450
4102
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

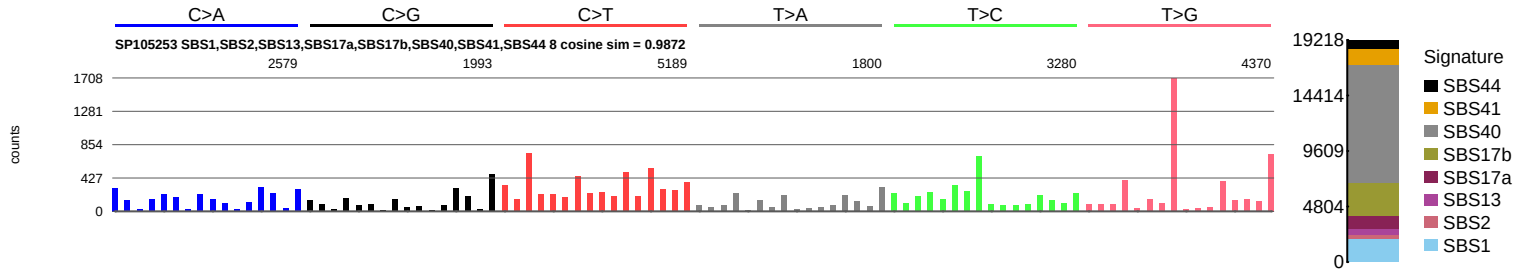

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98724
2877
1616
4999
1819
3552
4361
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

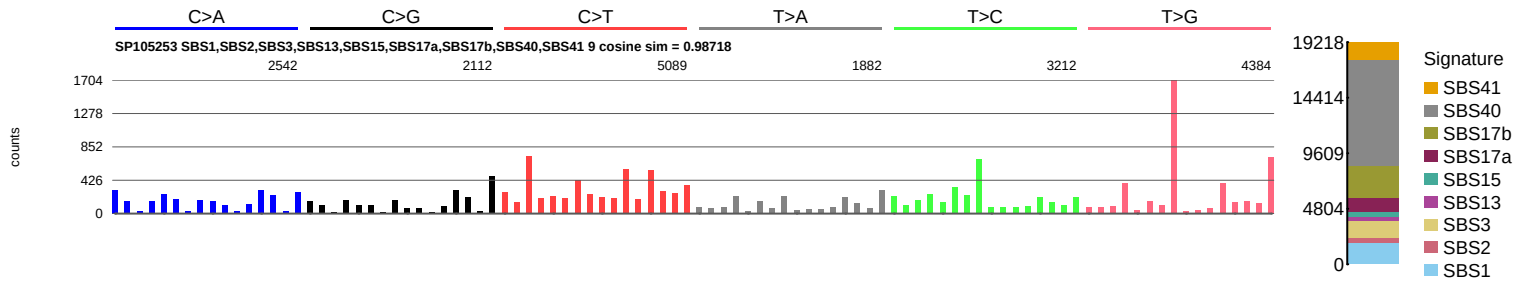

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98722
2465
2072
4908
1806
3594
4368
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

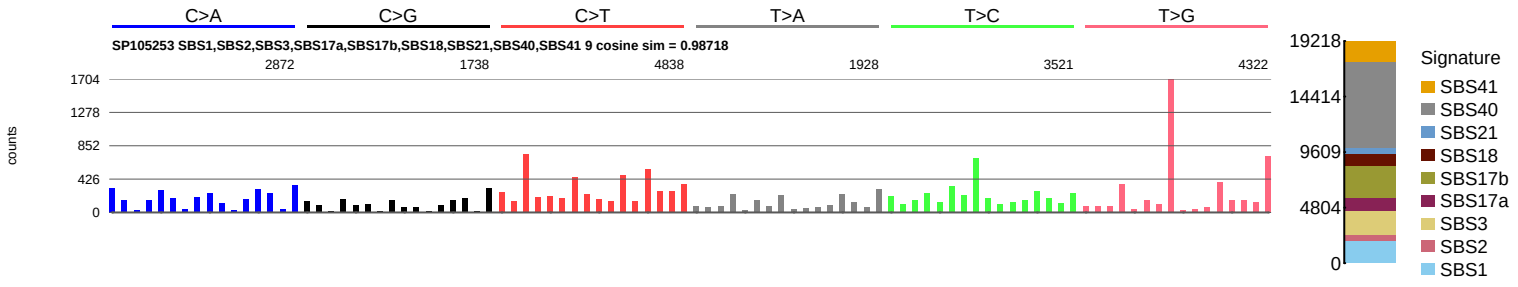

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98722
2997
1994
5147
1506
3436
4140
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

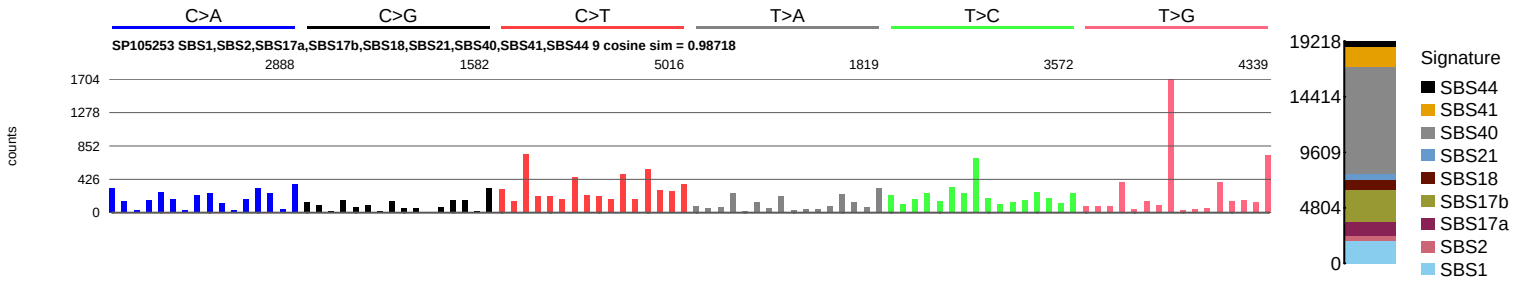

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.98721
2819
1928
4776
1720
3714
4262
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

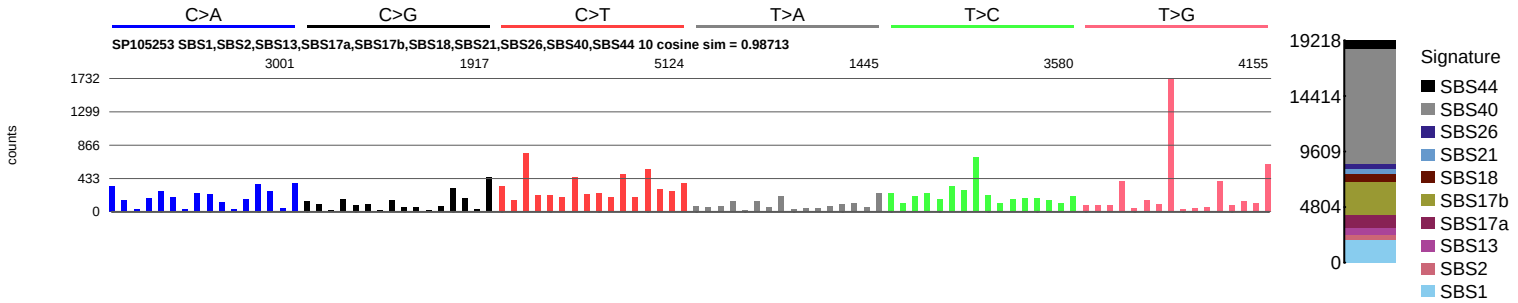

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9872
2885
1572
5007
1811
3611
4336
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

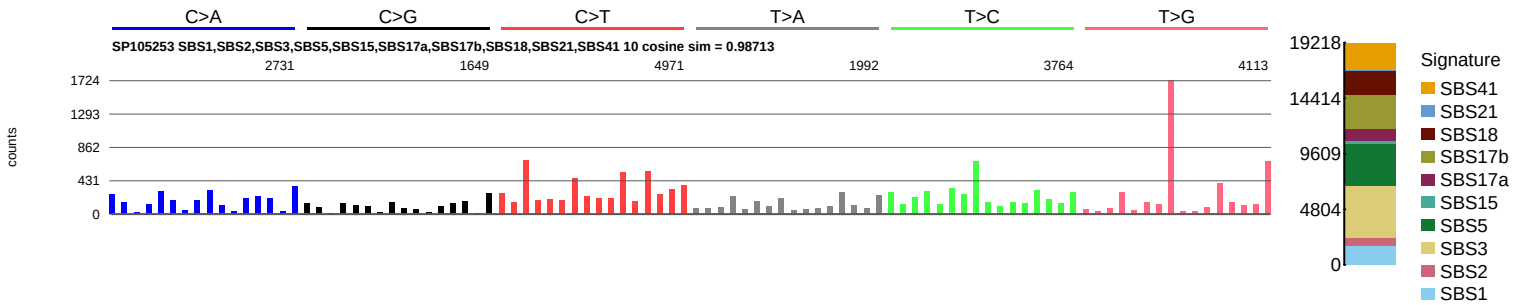

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.9872
2579
1993
5189
1800
3280
4370
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1

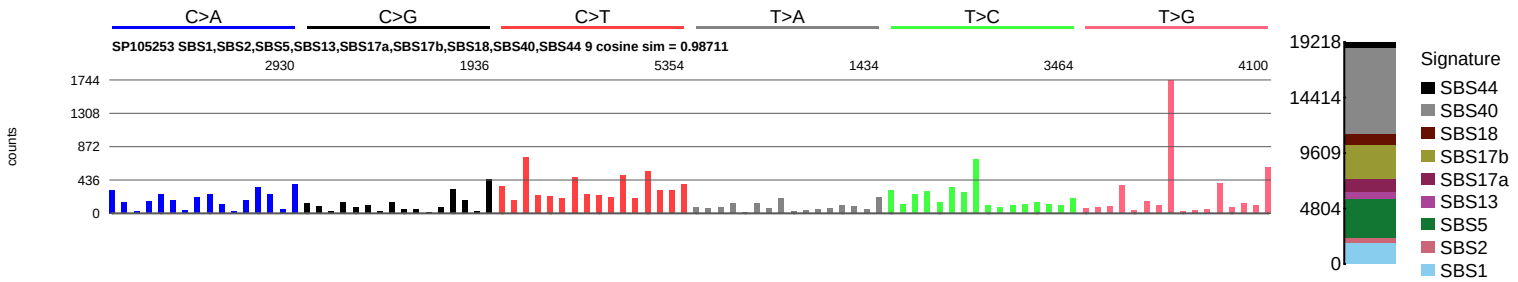

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98718
2542
2112
5089
1882
3212
4384
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

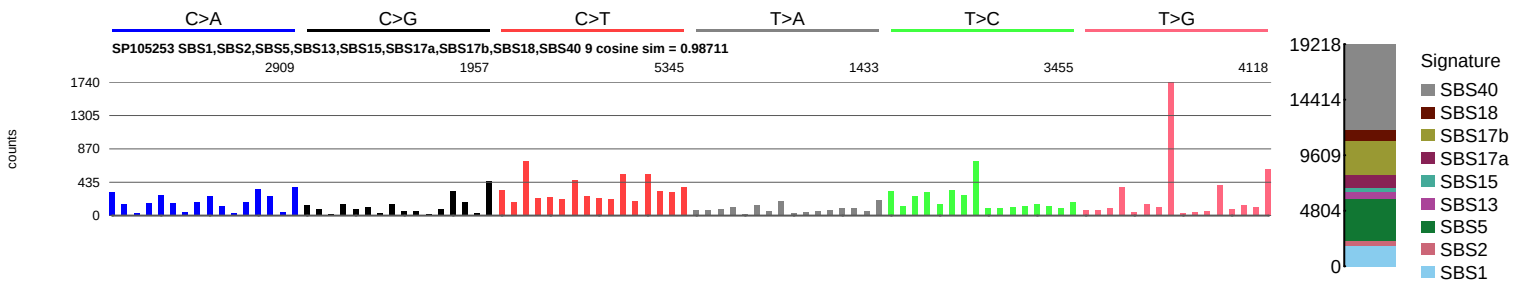

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98718
2872
1738
4838
1928
3521
4322
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

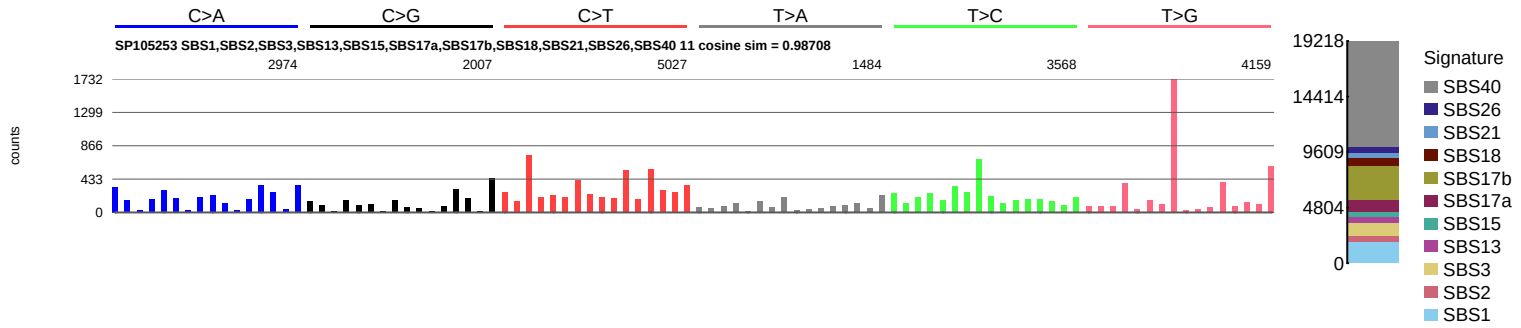

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98718
2888
1582
5016
1819
3572
4339
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

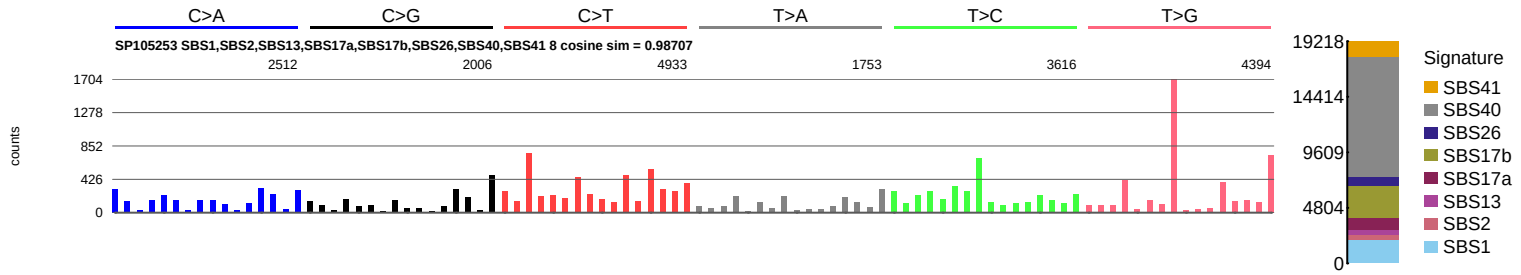

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98713
3001
1917
5124
1445
3580
4155
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

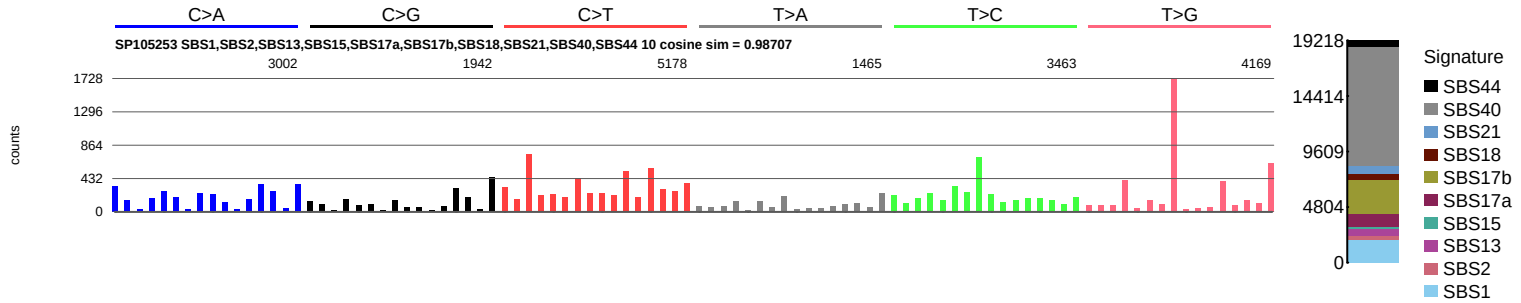

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98713
2731
1649
4971
1992
3764
4113
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

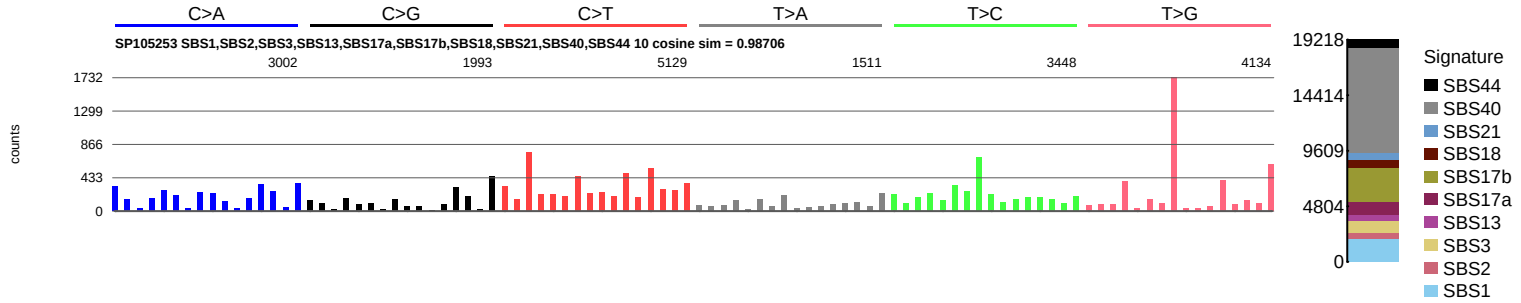

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98711
2930
1936
5354
1434
3464
4100
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

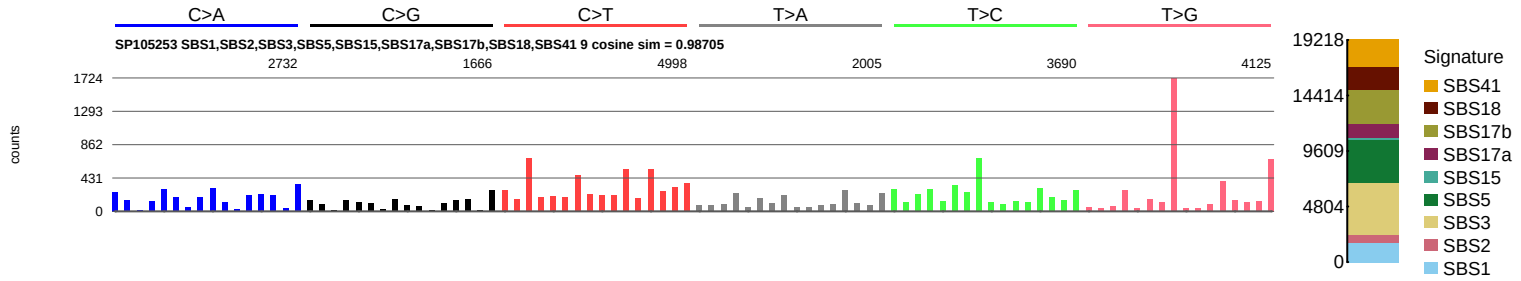

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98711
2909
1957
5345
1433
3455
4118
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

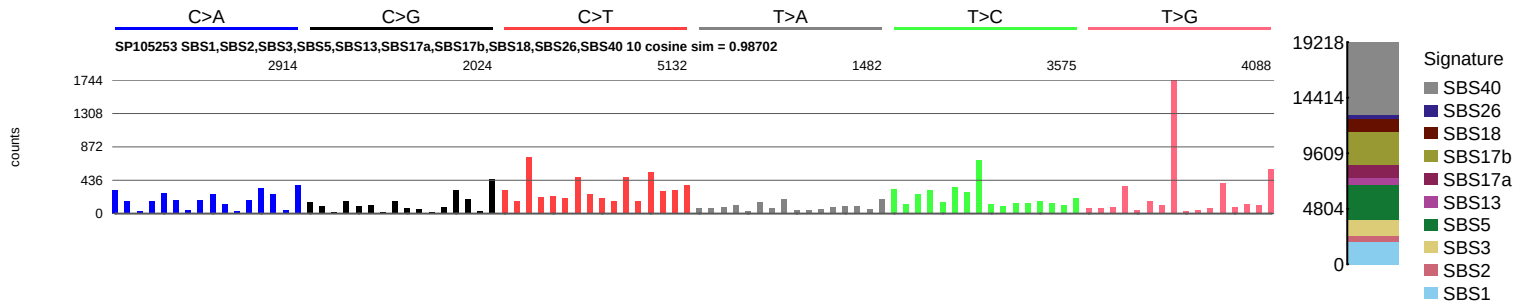

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 11 cosine sim = 0.98708
2974
2007
5027
1484
3568
4159
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

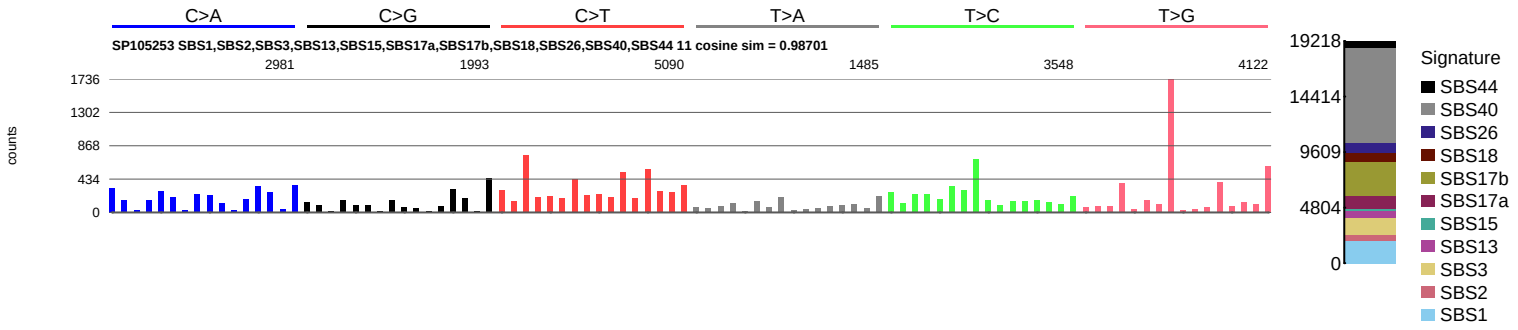

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98707
2512
2006
4933
1753
3616
4394
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

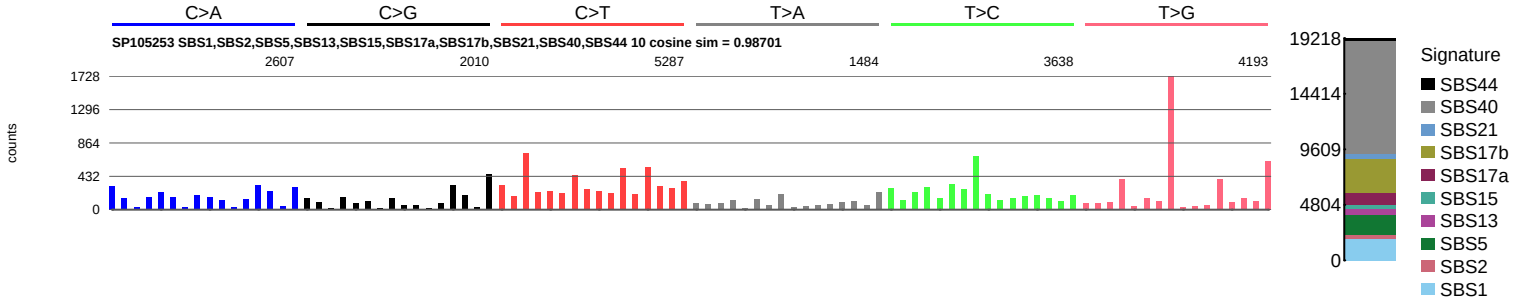

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98707
3002
1942
5178
1465
3463
4169
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

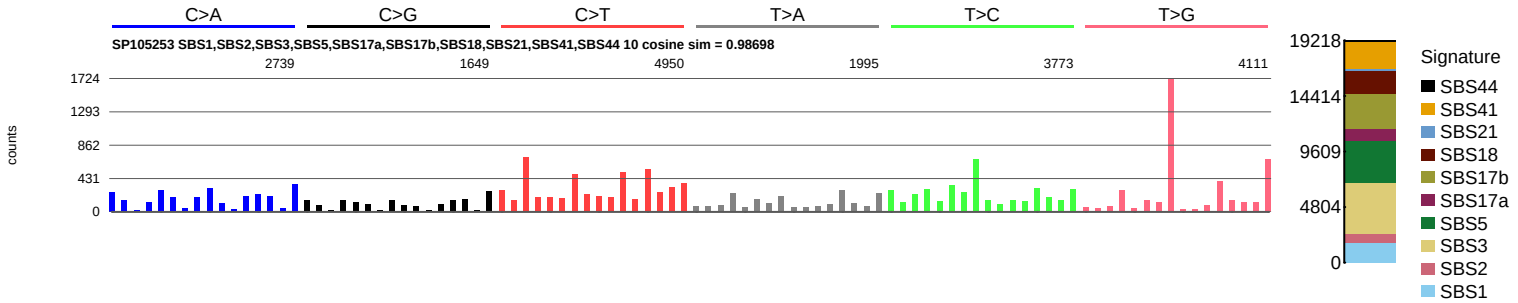

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98706
3002
1993
5129
1511
3448
4134
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

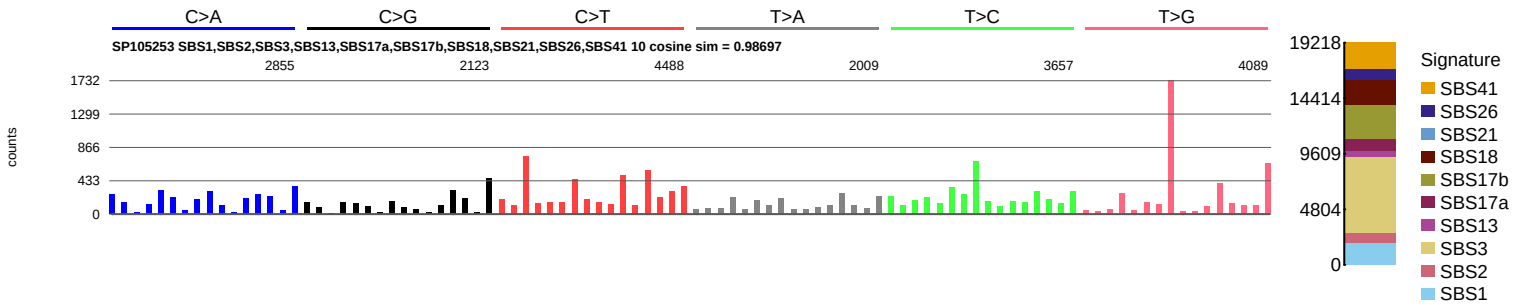

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98705
2732
1666
4998
2005
3690
4125
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

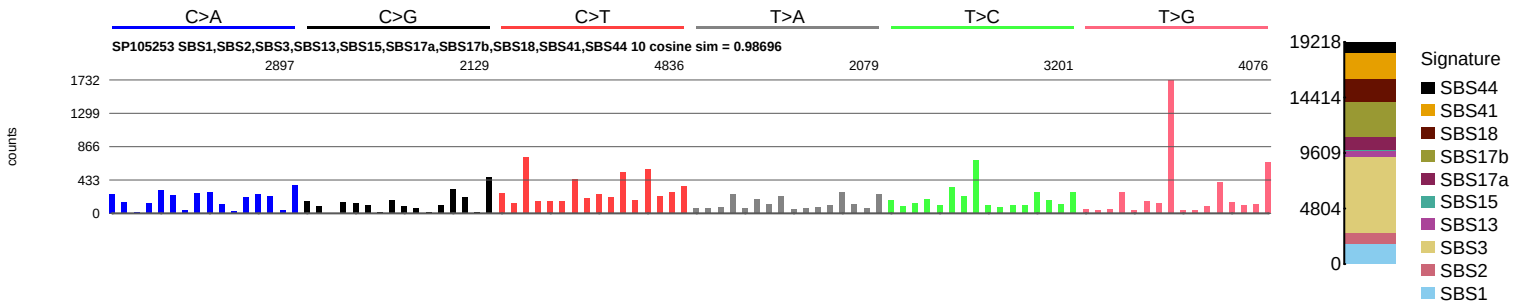

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98702
2914
2024
5132
1482
3575
4088
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

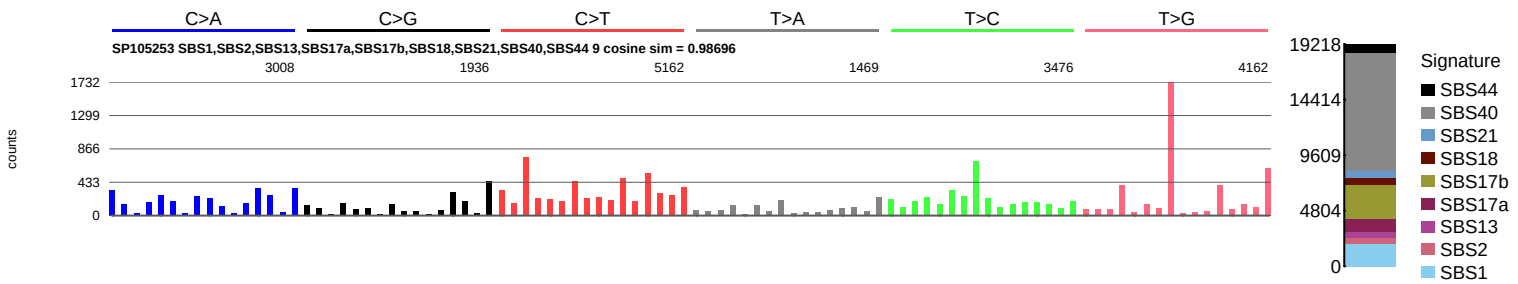

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98701
2981
1993
5090
1485
3548
4122
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98701
2607
2010
5287
1484
3638
4193
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98698
2739
1649
4950
1995
3773
4111
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98697
2855
2123
4488
2009
3657
4089
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 10 cosine sim = 0.98696
2897
2129
4836
2079
3201
4076
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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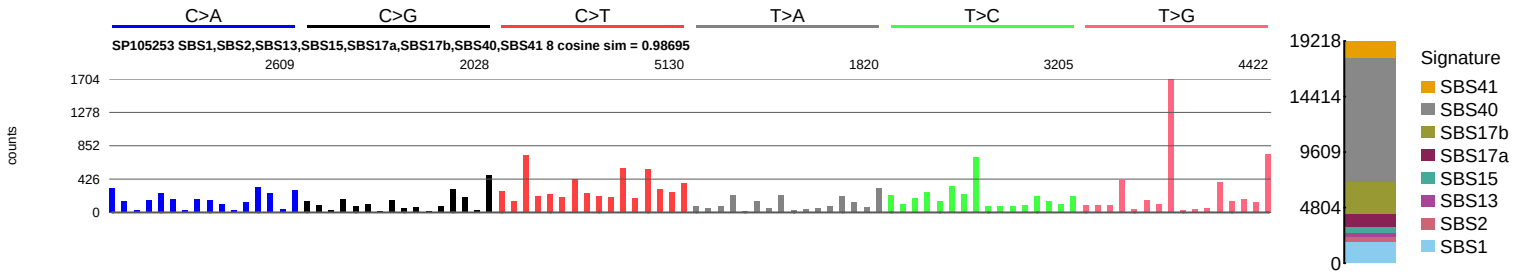
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98696
3008
1936
5162
1469
3476
4162
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>T
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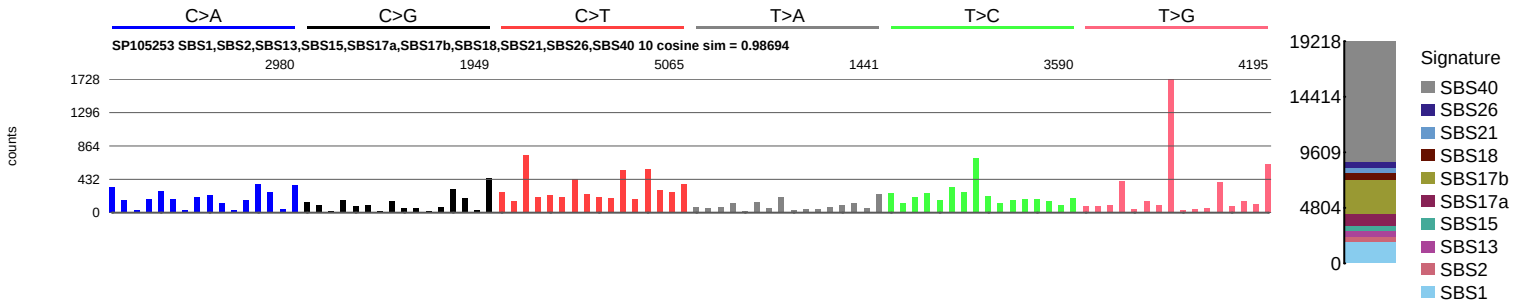
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98695
2609
2028
5130
1820
3205
4422
1704
1278
852
426
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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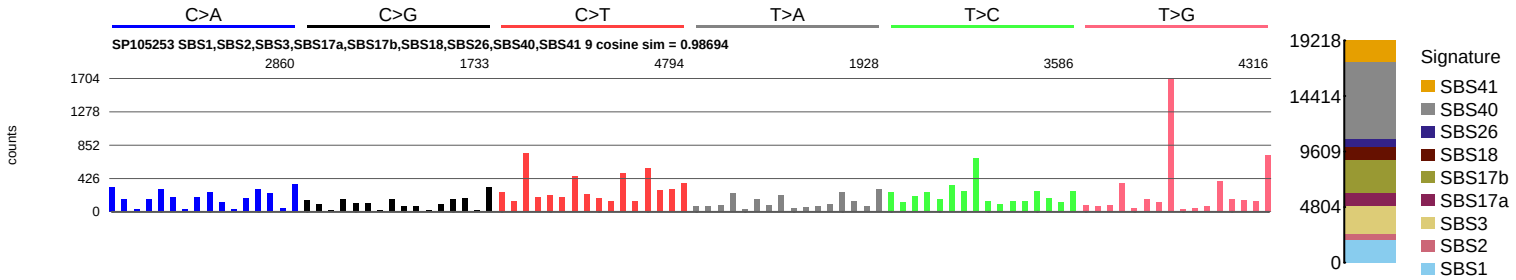
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98694
2980
1949
5065
1441
3590
4195
1728
1296
864
432
0
counts
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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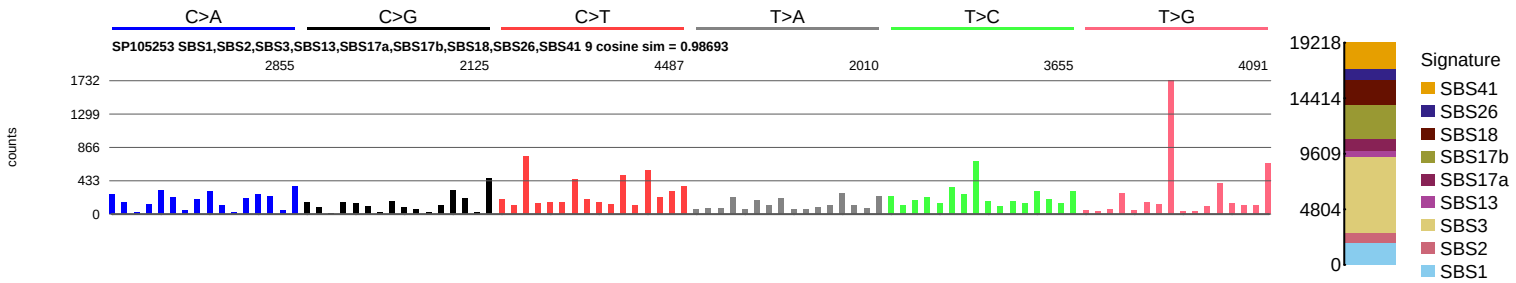
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98694
2860
1733
4794
1928
3586
4316
1704
1278
852
426
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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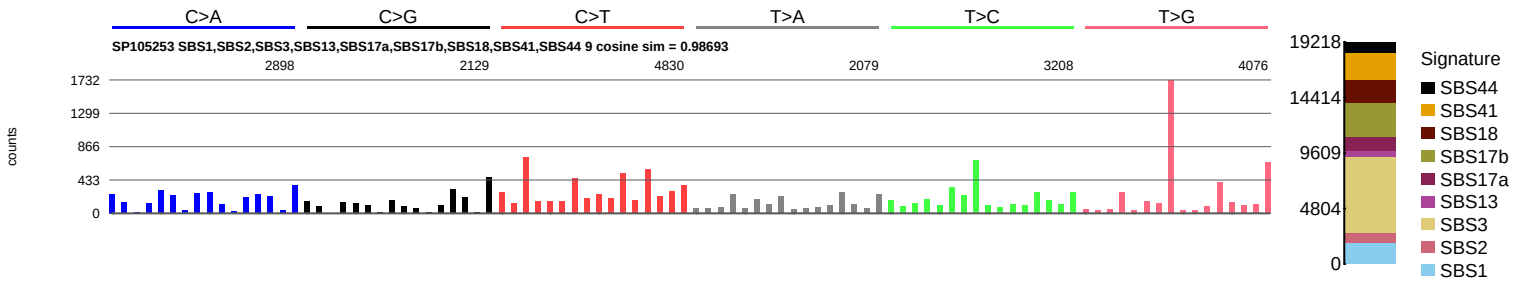
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98693
2855
2125
4487
2010
3655
4091
1732
1299
866
433
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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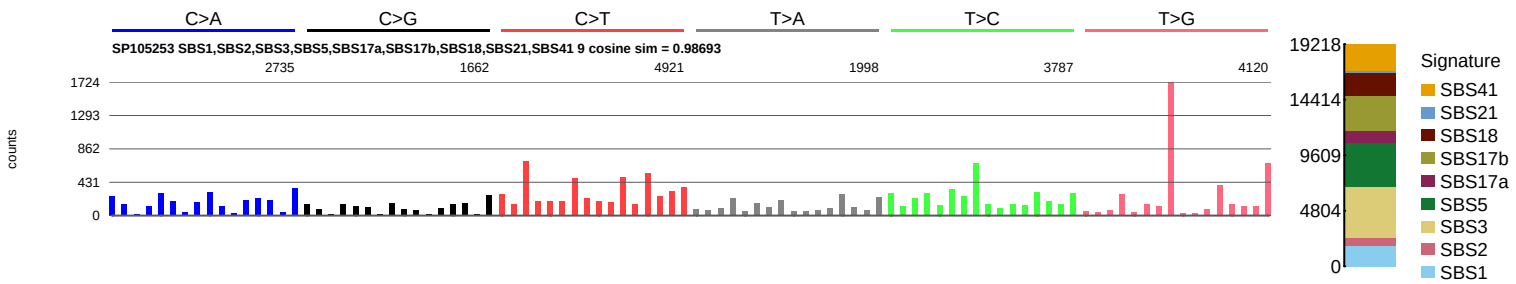
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98693
2898
2129
4830
2079
3208
4076
1732
1299
866
433
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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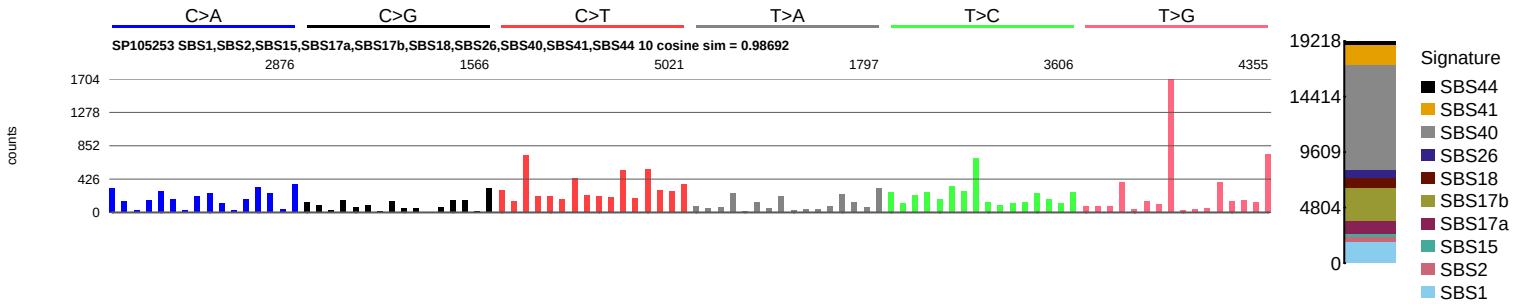
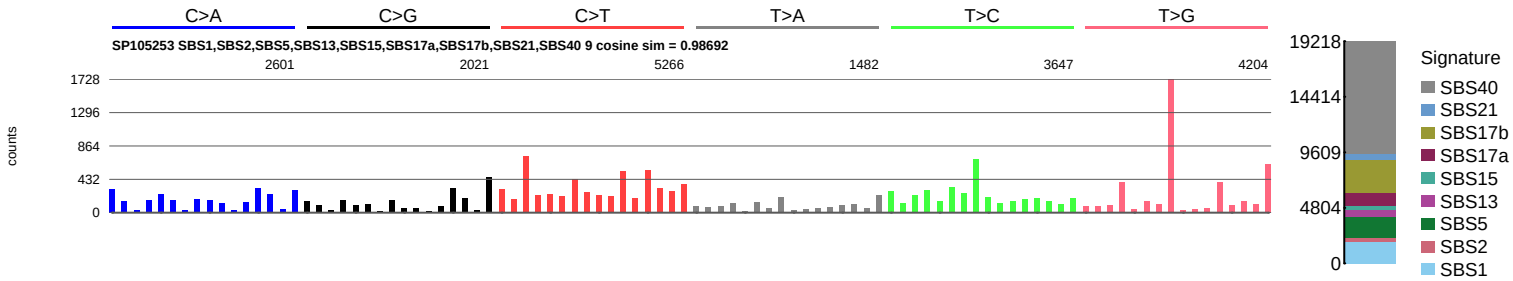
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98693
2735
1662
4921
1998
3787
4120
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98692
2876
1566
5021
1797
3606
4355
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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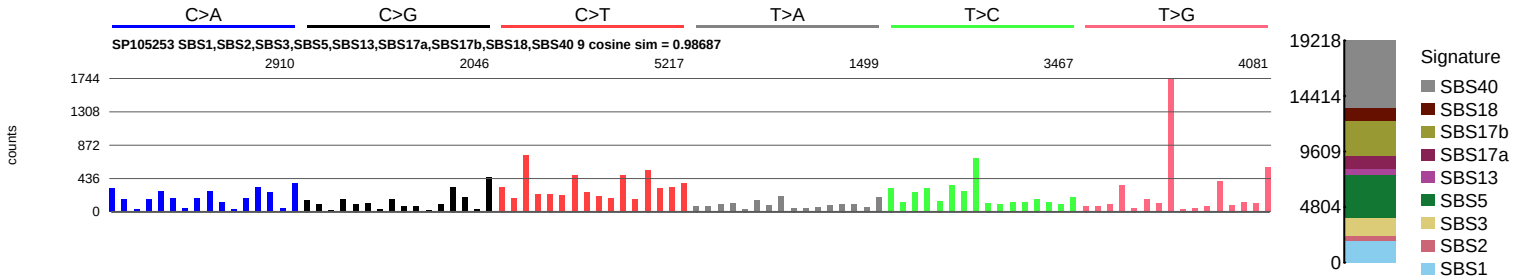
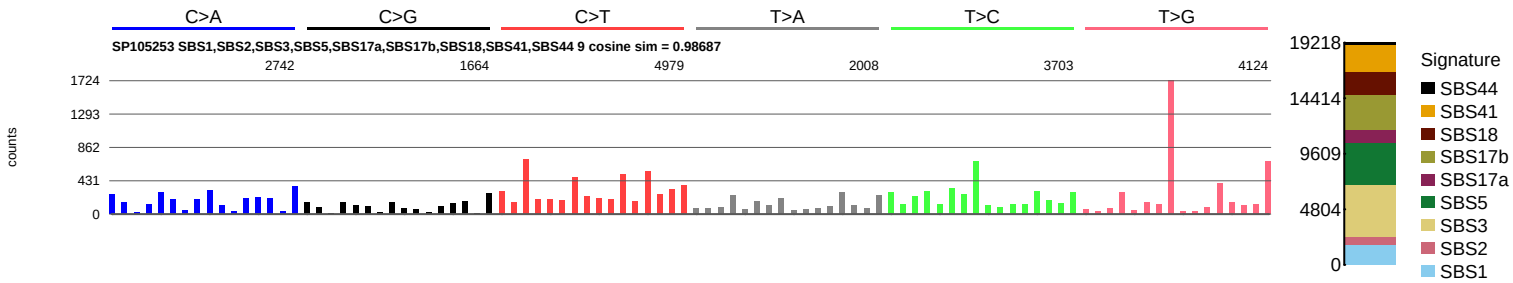
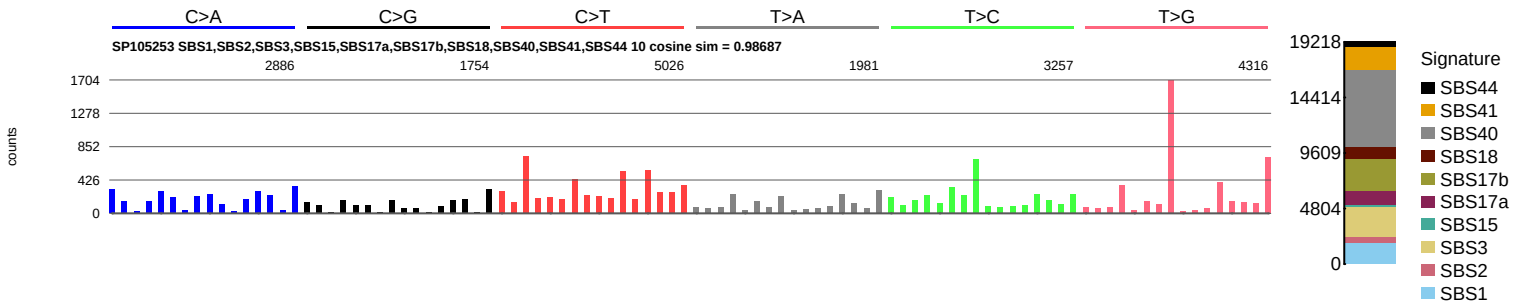
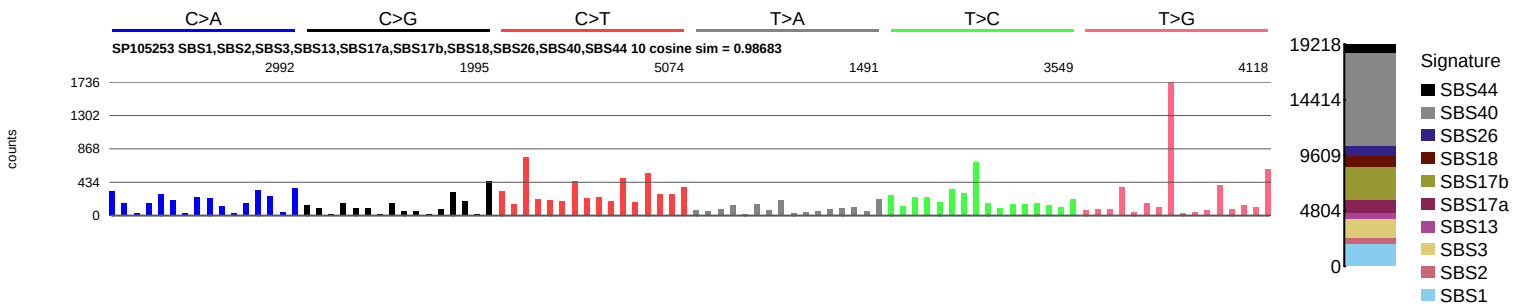
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98692
2601
2021
5266
1482
3647
4204
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98687
2910
2046
5217
1499
3467
4081
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98687
2742
1664
4979
2008
3703
4124
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 10 cosine sim = 0.98687
2886
1754
5026
1981
3257
4316
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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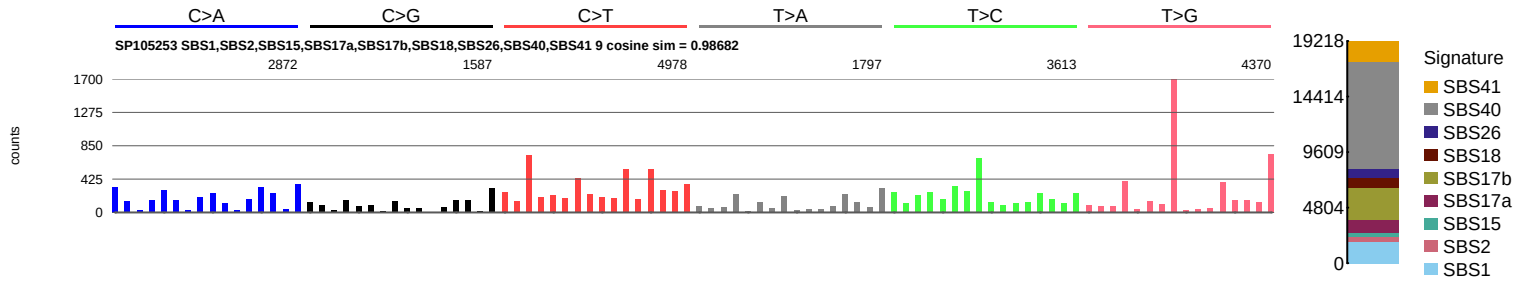
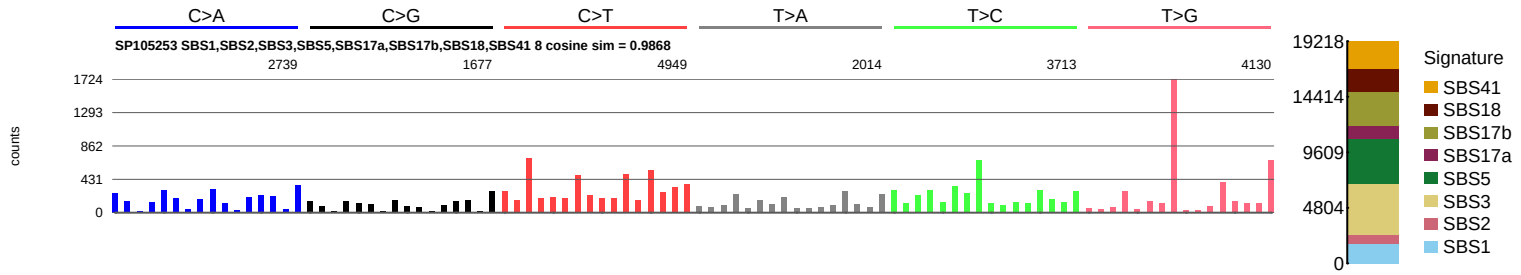
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98683
2992
1995
5074
1491
3549
4118
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98682
2872
1587
4978
1797
3613
4370
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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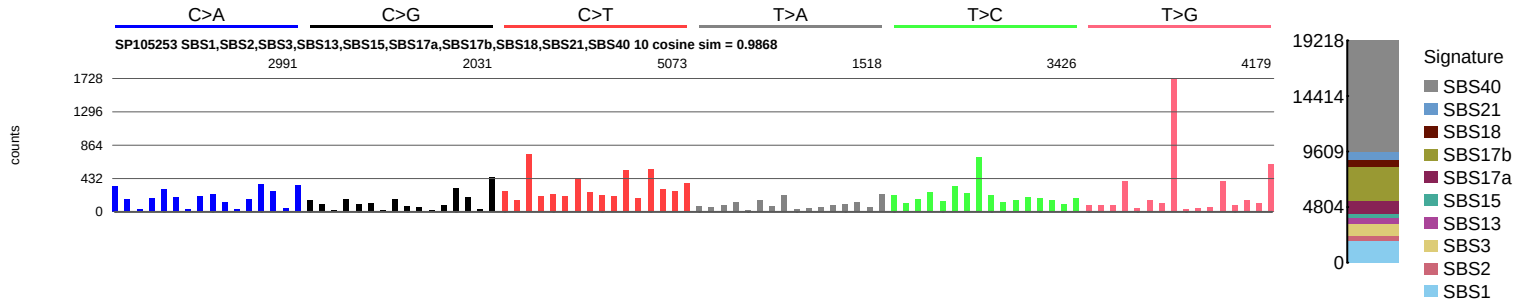
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9868
2739
1677
4949
2014
3713
4130
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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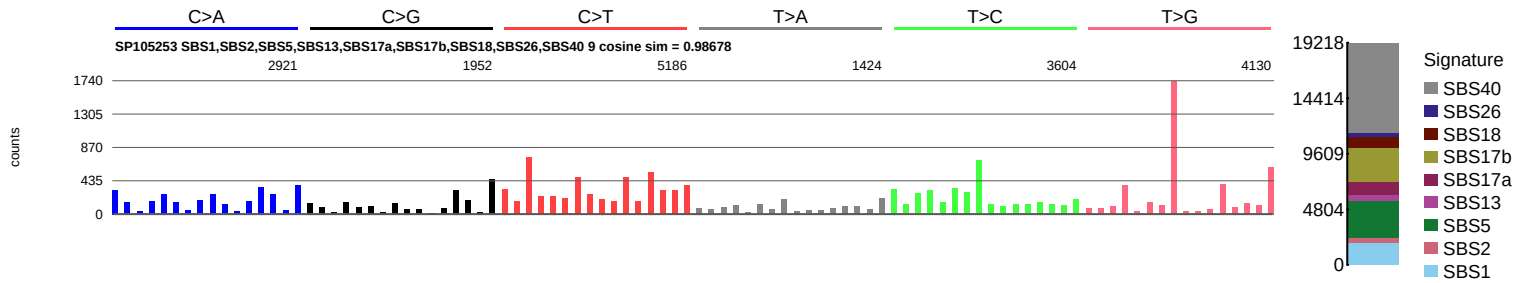
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.9868
2991
2031
5073
1518
3426
4179
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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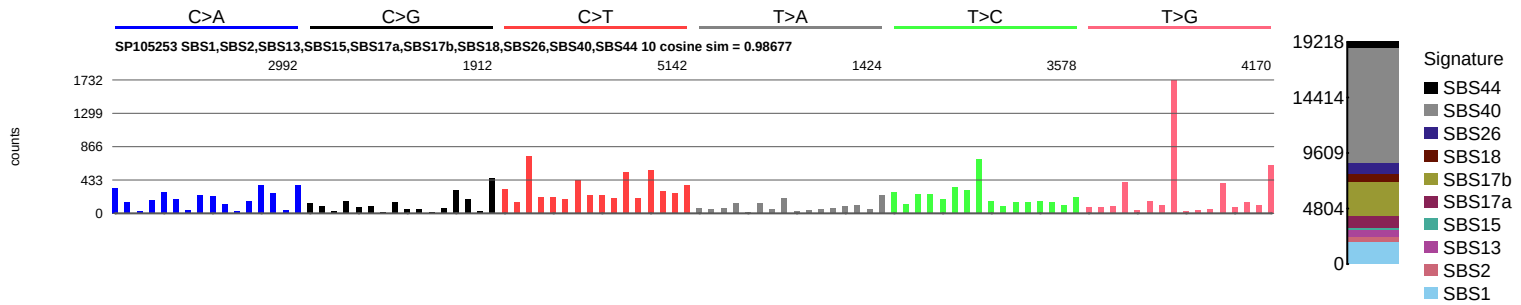
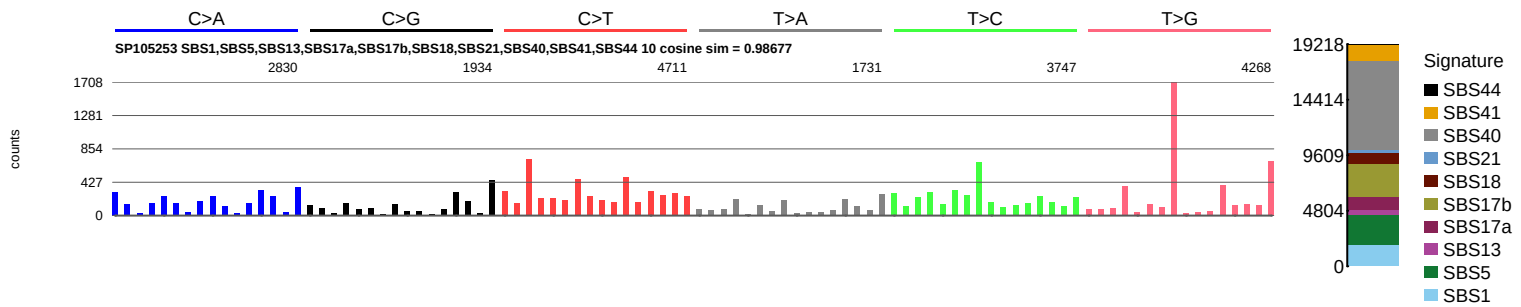
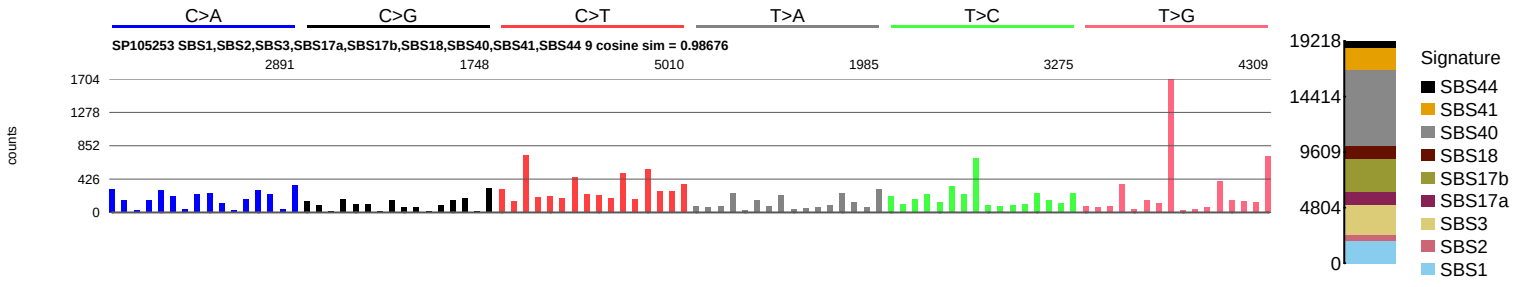
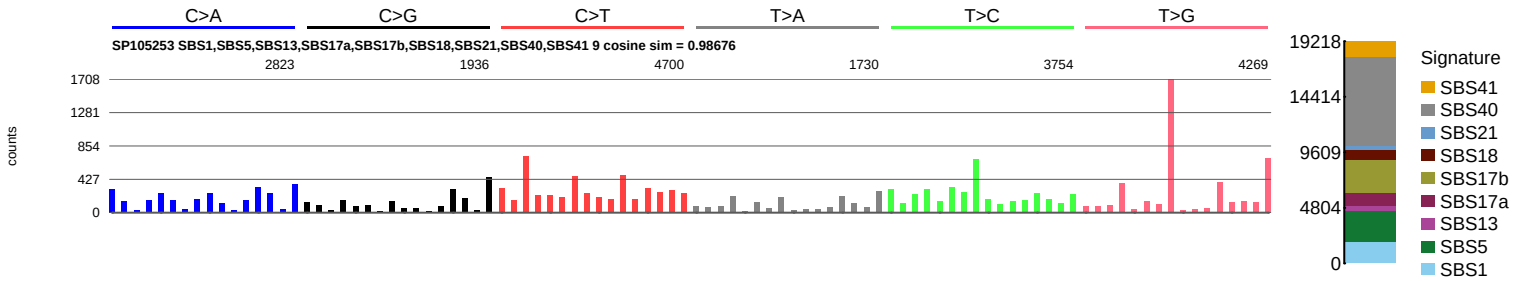
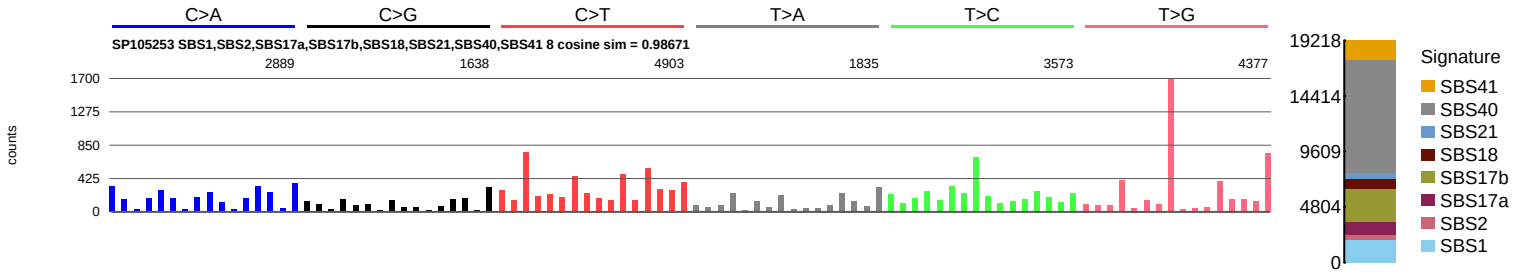
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98678
2921
1952
5186
1424
3604
4130
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

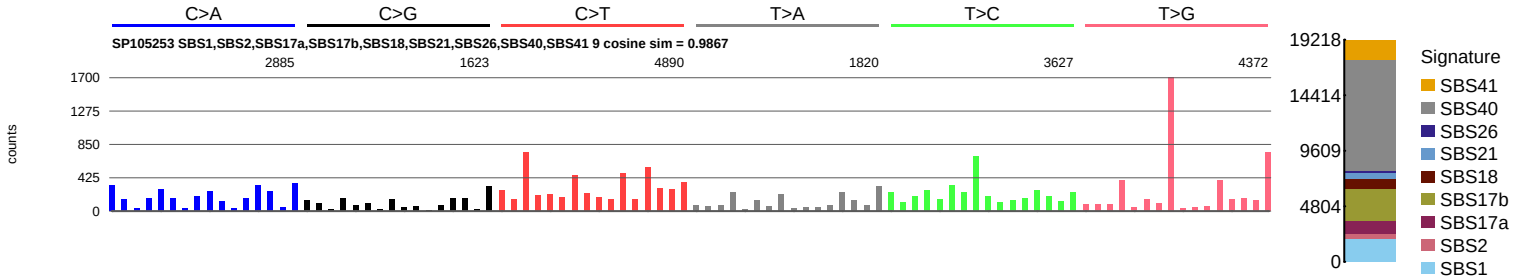

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98677
2992
1912
5142
1424
3578
4170
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

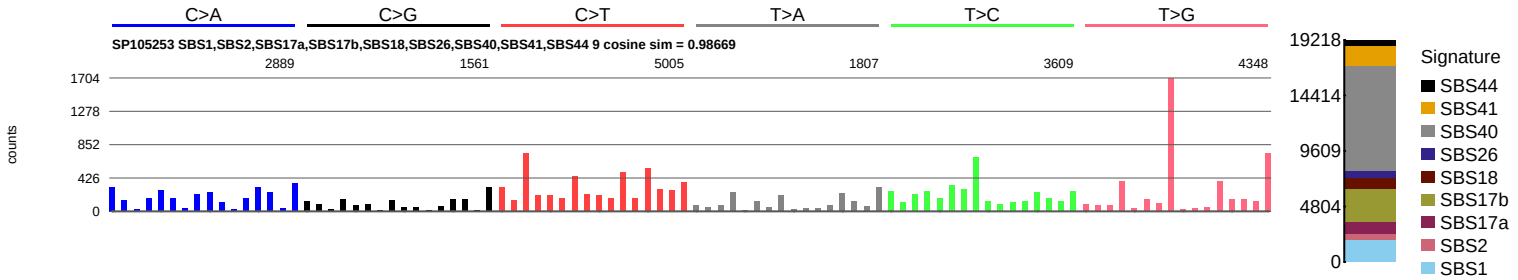

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98677
2830
1934
4711
1731
3747
4268
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

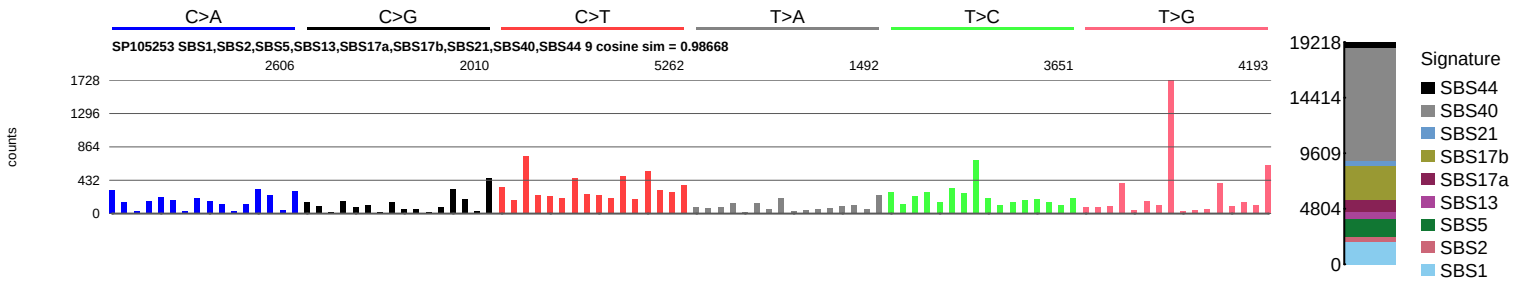

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98676
2891
1748
5010
1985
3275
4309
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98676
2823
1936
4700
1730
3754
4269
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98671
2889
1638
4903
1835
3573
4377
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.9867
2885
1623
4890
1820
3627
4372
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98669
2889
1561
5005
1807
3609
4348
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>T
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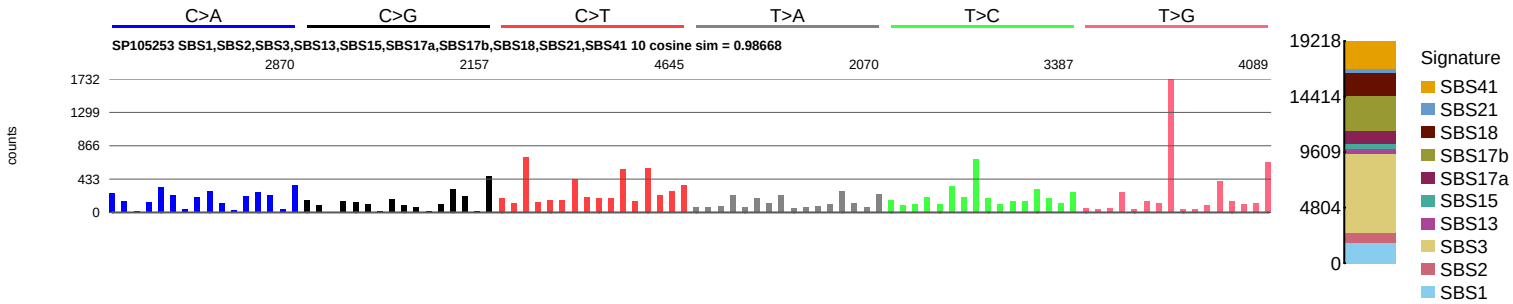
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98668
2606
2010
5262
1492
3651
4193
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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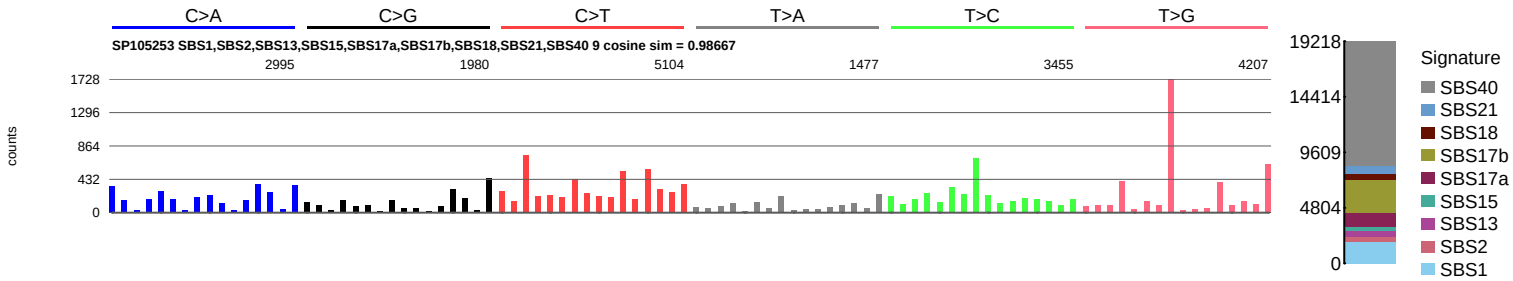
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98668
2870
2157
4645
2070
3387
4089
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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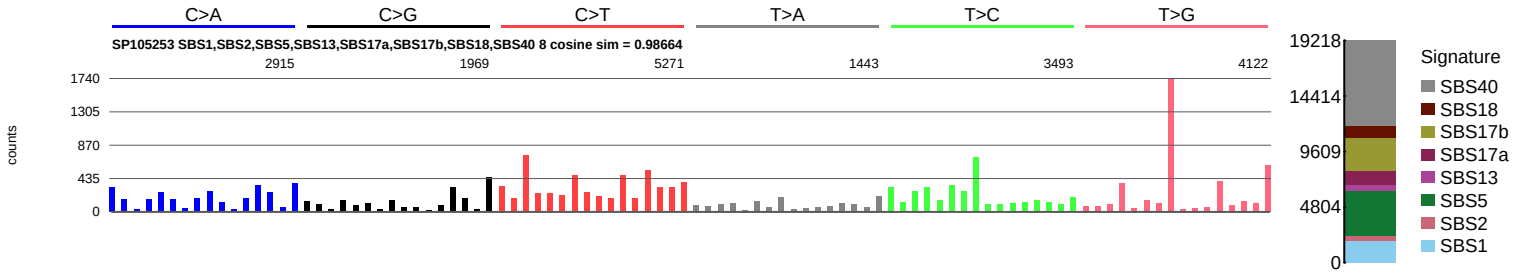
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98667
2995
1980
5104
1477
3455
4207
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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C>G
C>T
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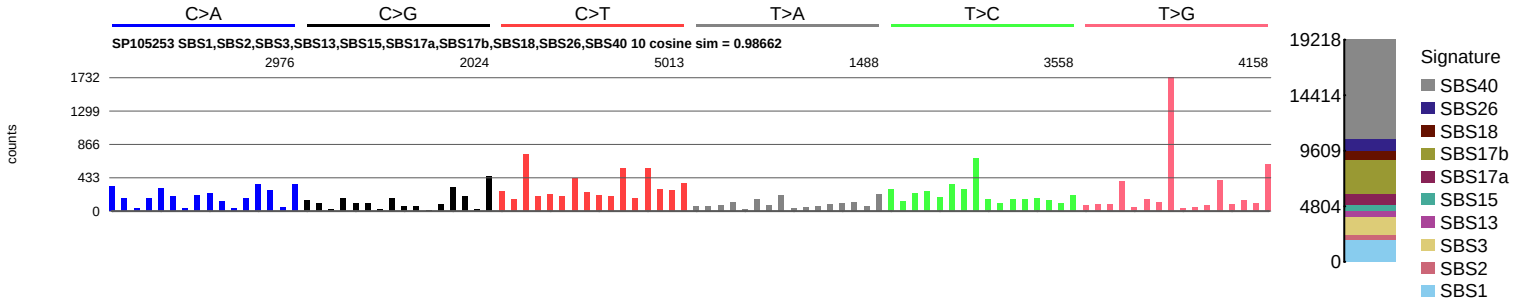
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98664
2915
1969
5271
1443
3493
4122
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>G
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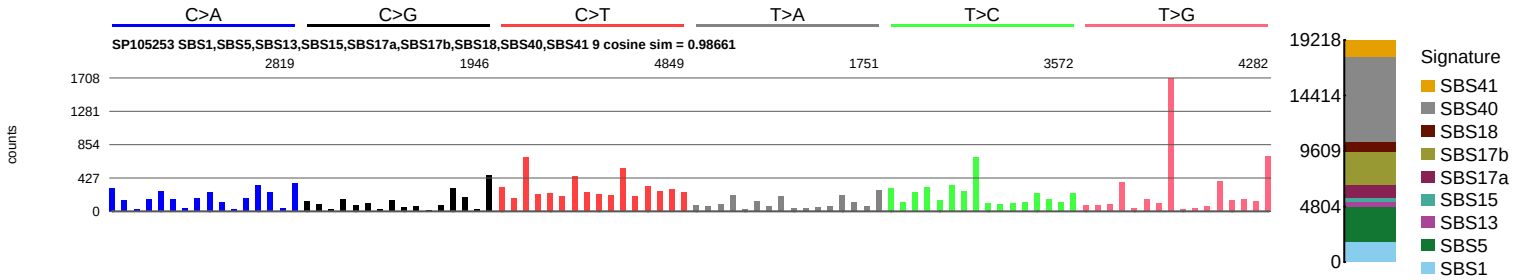
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98662
2976
2024
5013
1488
3558
4158
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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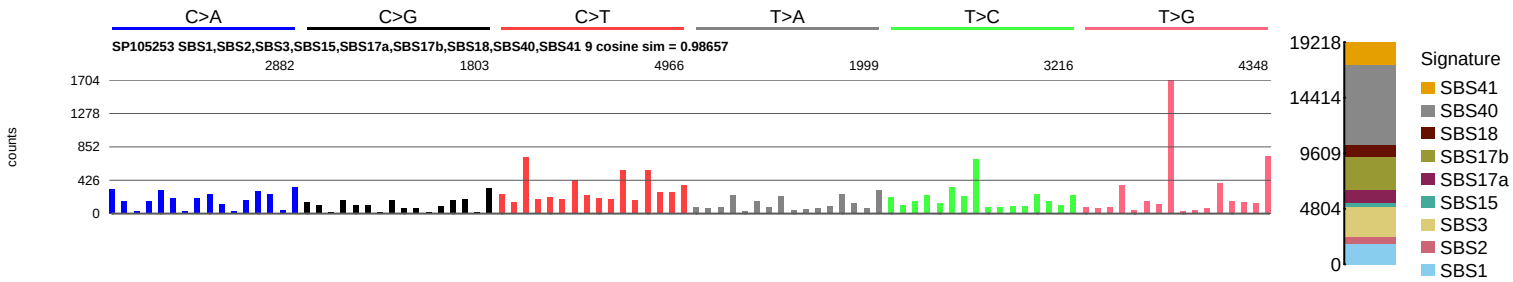
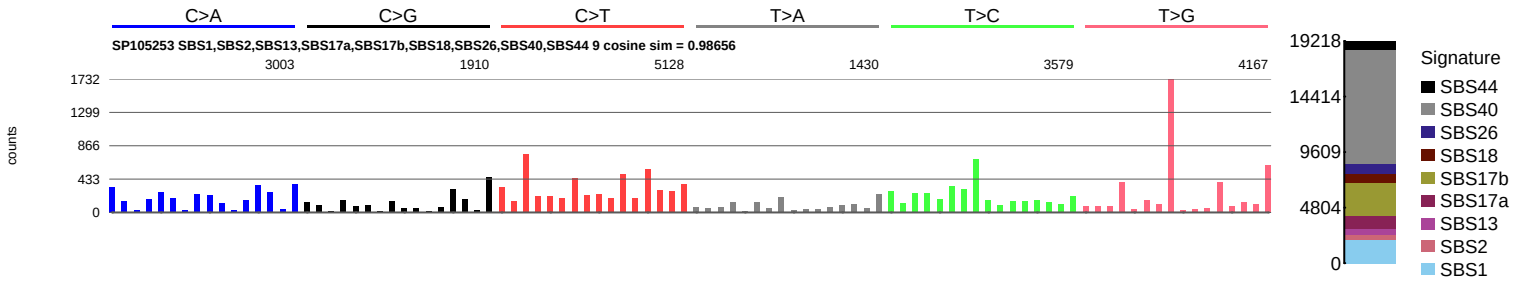
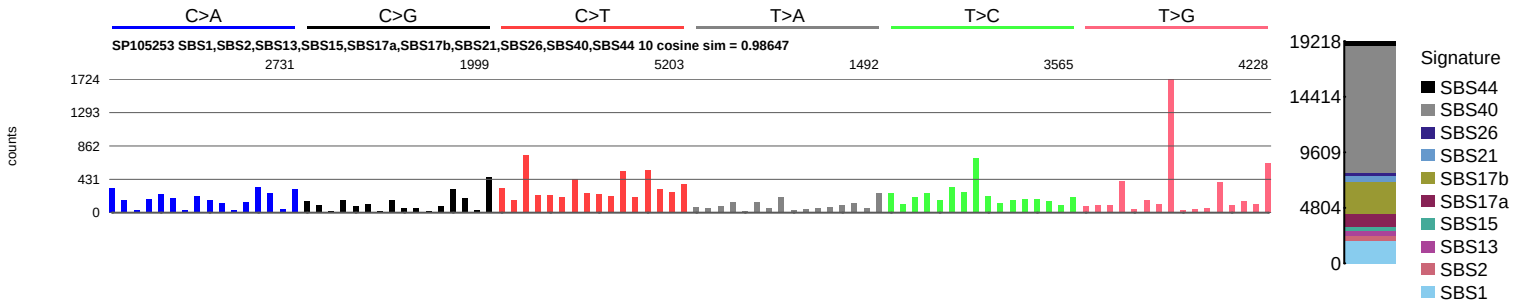
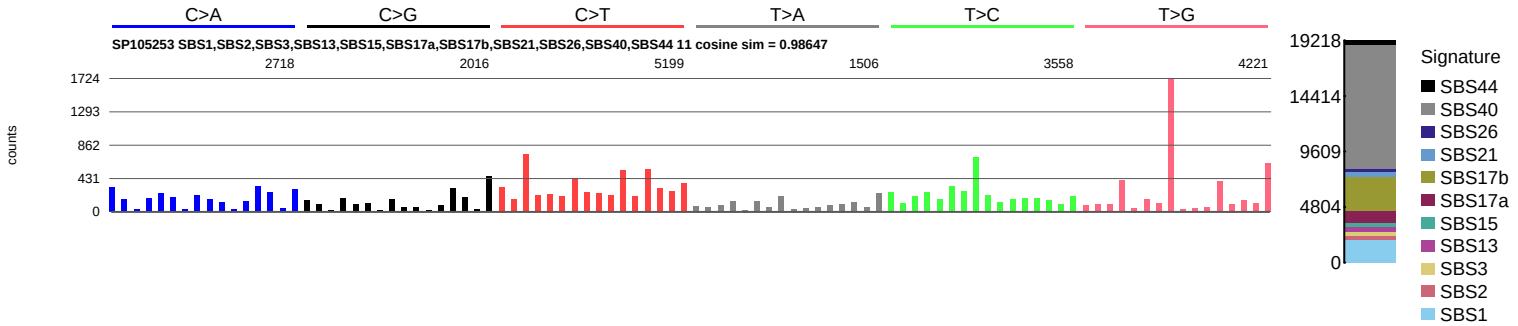
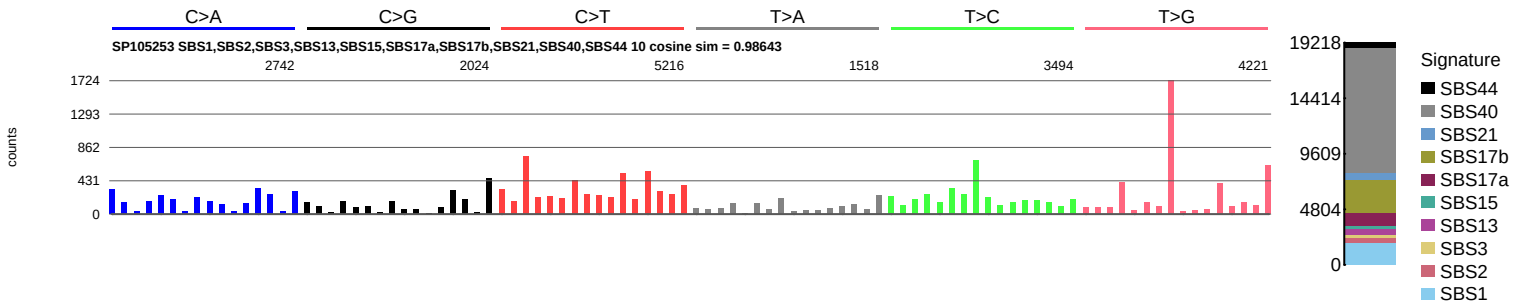
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98661
2819
1946
4849
1751
3572
4282
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

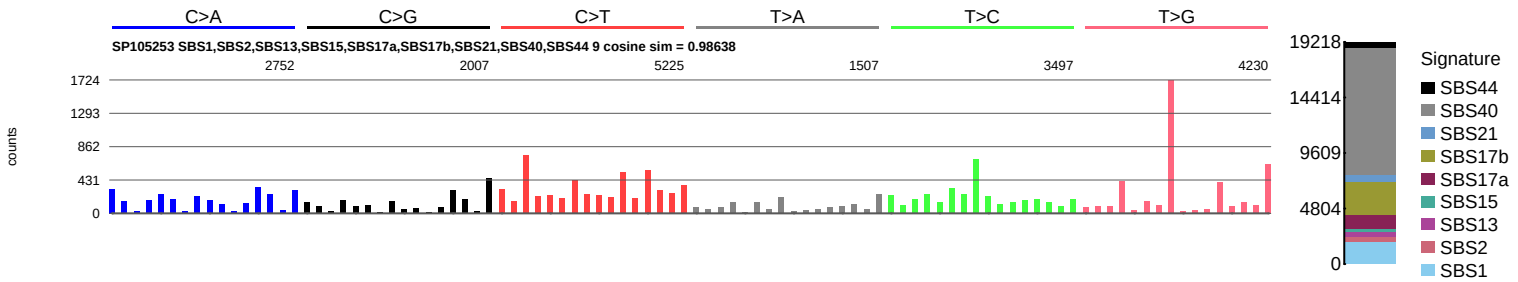

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.98657
2882
1803
4966
1999
3216
4348
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

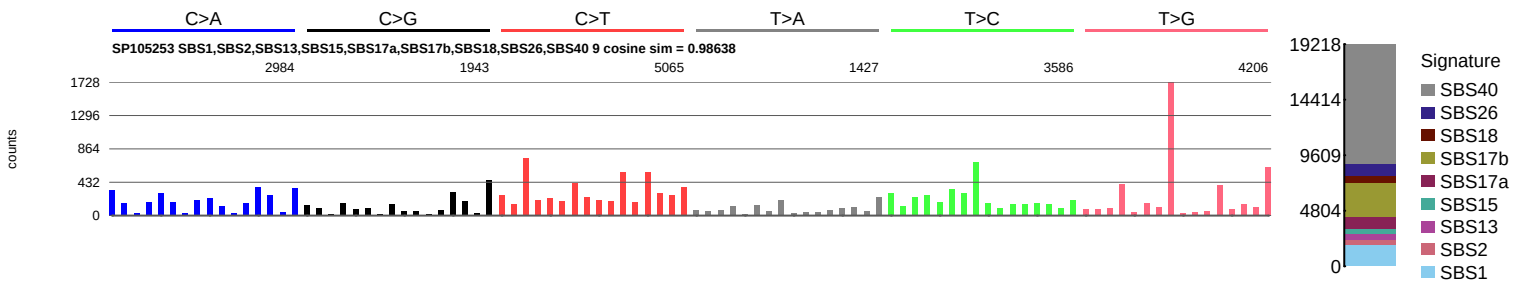

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98656
3003
1910
5128
1430
3579
4167
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

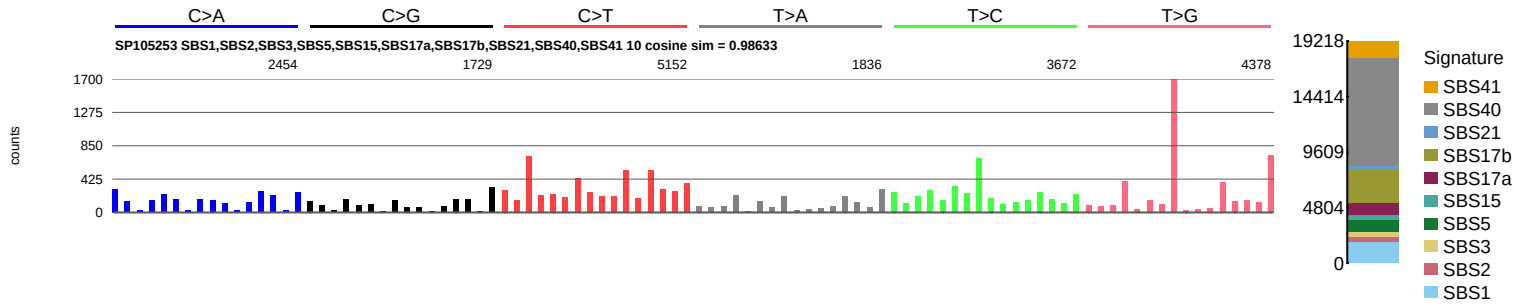

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98647
2731
1999
5203
1492
3565
4228
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

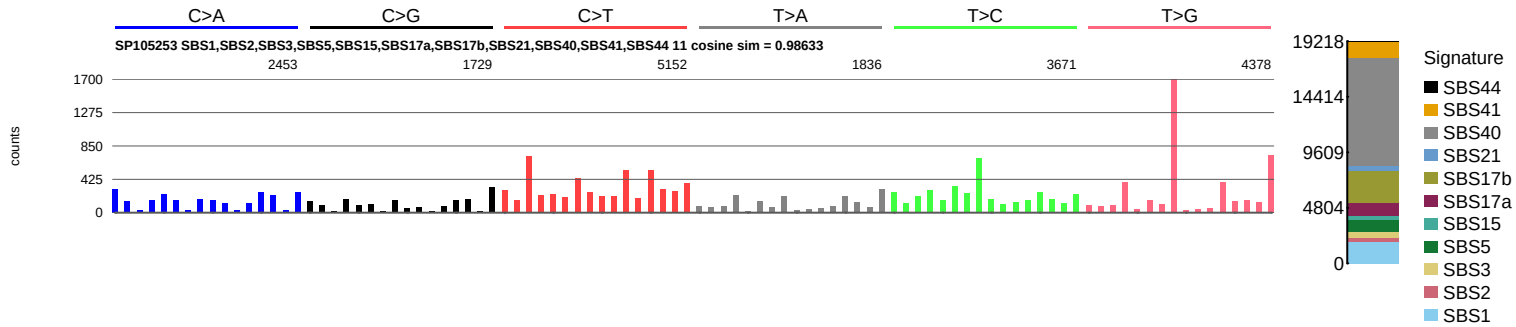

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98647
2718
2016
5199
1506
3558
4221
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

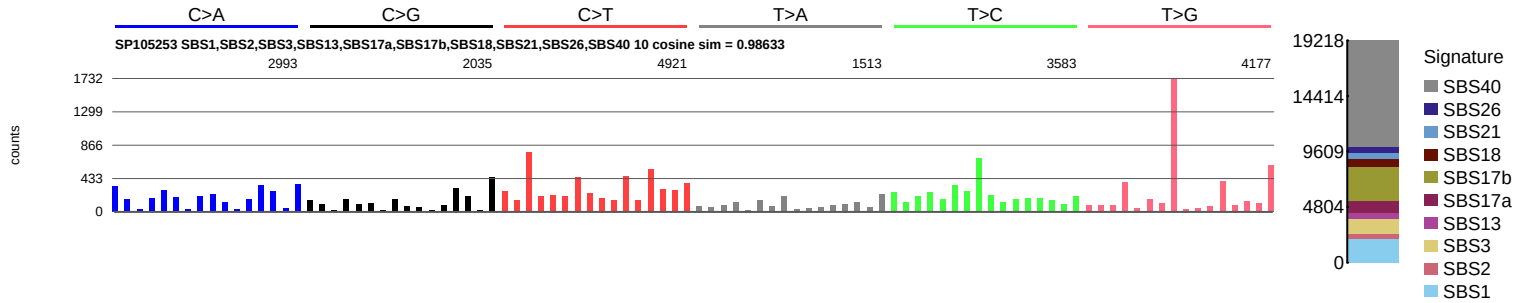

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98643
2742
2024
5216
1518
3494
4221
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98638
2752
2007
5225
1507
3497
4230
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98638
2984
1943
5065
1427
3586
4206
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 10 cosine sim = 0.98633
2454
1729
5152
1836
3672
4378
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 11 cosine sim = 0.98633
2453
1729
5152
1836
3671
4378
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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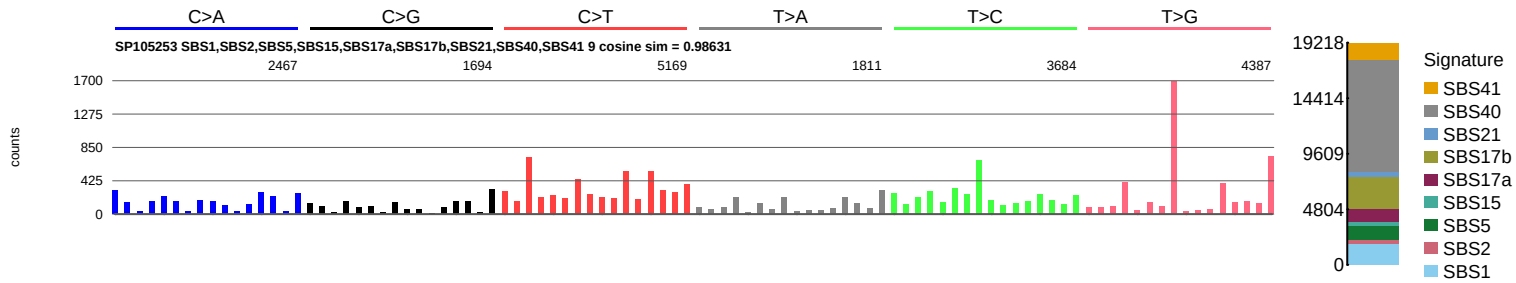
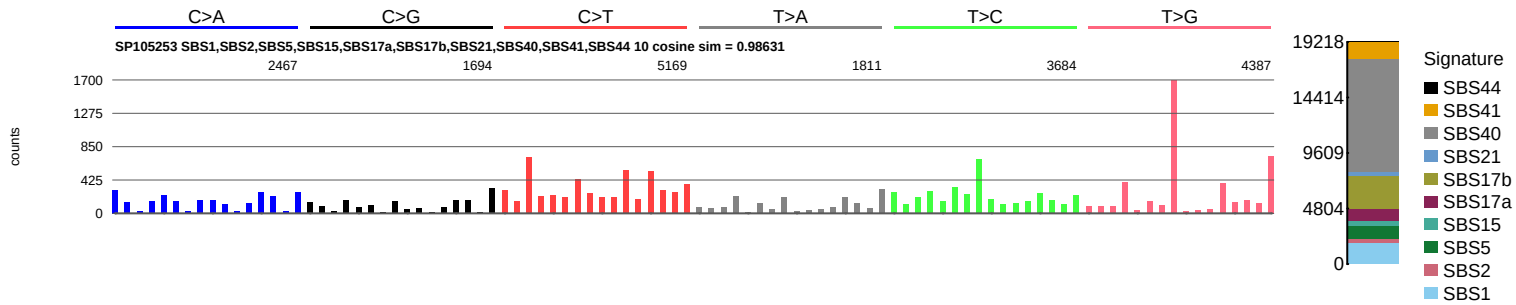
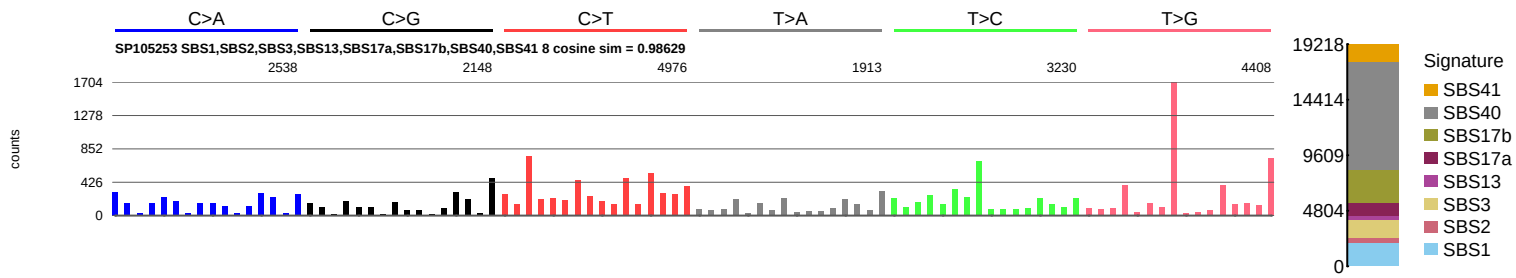
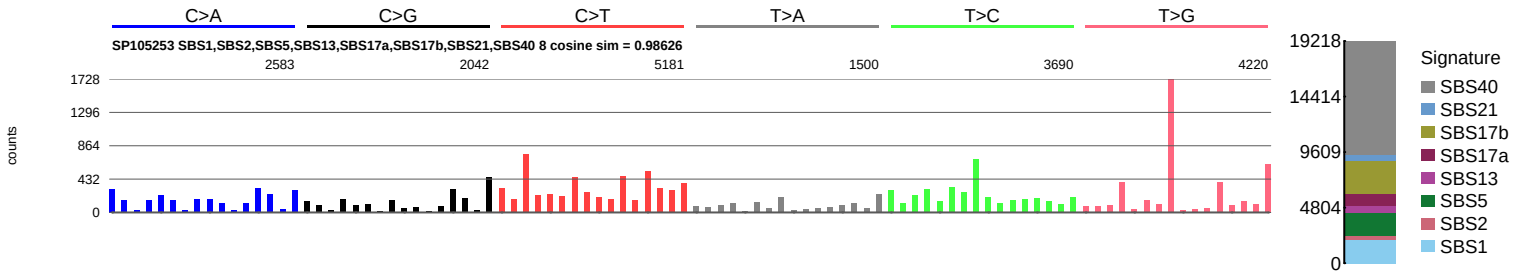
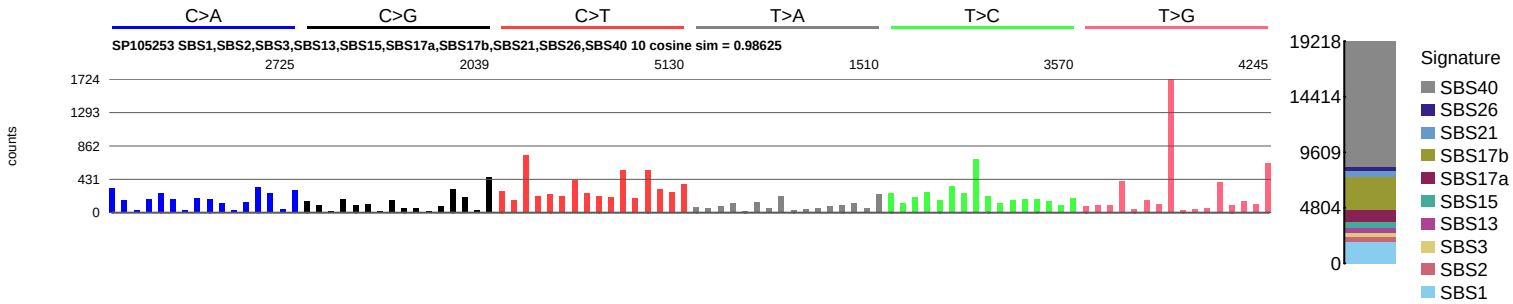
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98633
2993
2035
4921
1513
3583
4177
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

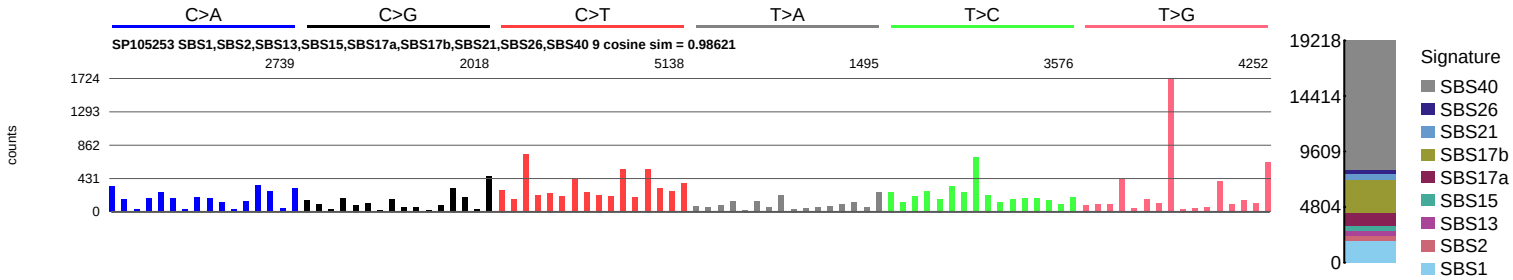

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98631
2467
1694
5169
1811
3684
4387
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

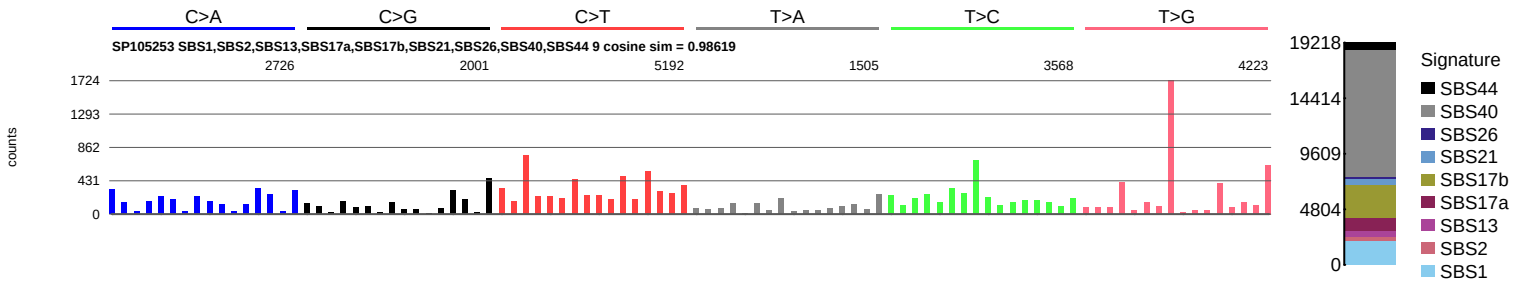

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98631
2467
1694
5169
1811
3684
4387
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

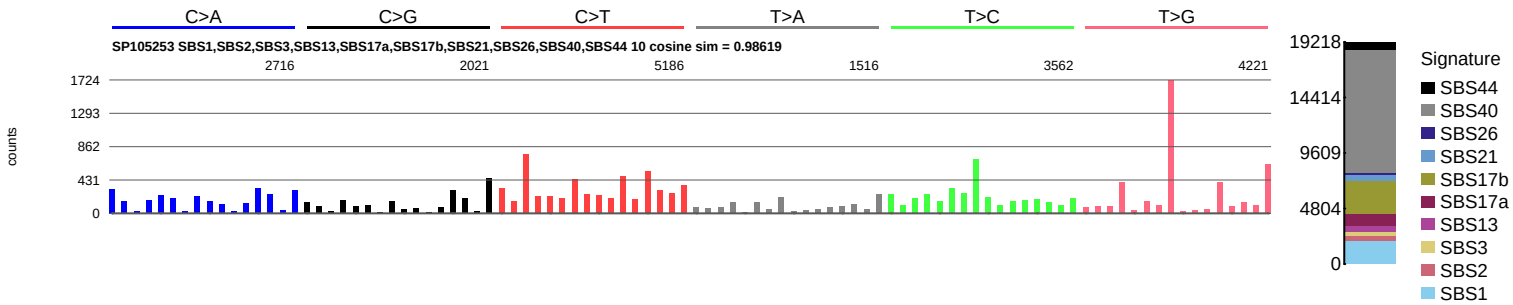

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98629
2538
2148
4976
1913
3230
4408
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98626
2583
2042
5181
1500
3690
4220
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 10 cosine sim = 0.98625
2725
2039
5130
1510
3570
4245
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98621
2739
2018
5138
1495
3576
4252
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98619
2726
2001
5192
1505
3568
4223
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98619
2716
2021
5186
1516
3562
4221
counts
1724
1293
862
431
0
19218
14414
9609
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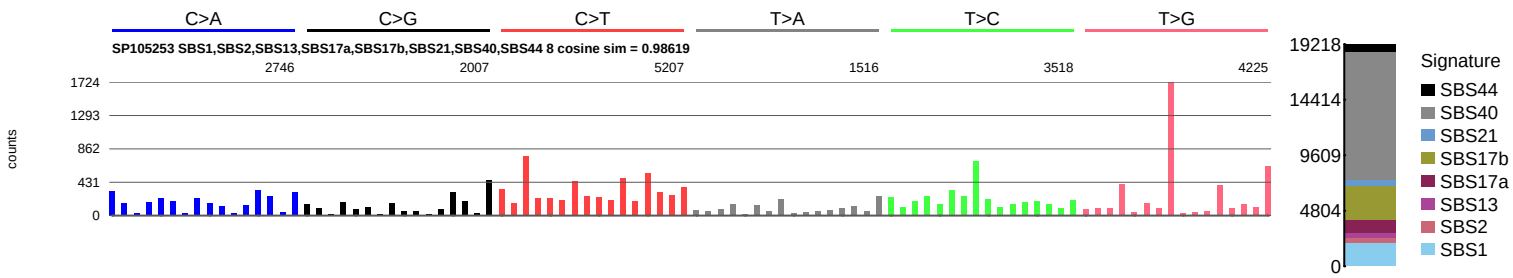
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SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98619
2746
2007
5207
1516
3518
4225
counts
1724
1293
862
431
0
19218
14414
9609
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Signature
SBS44
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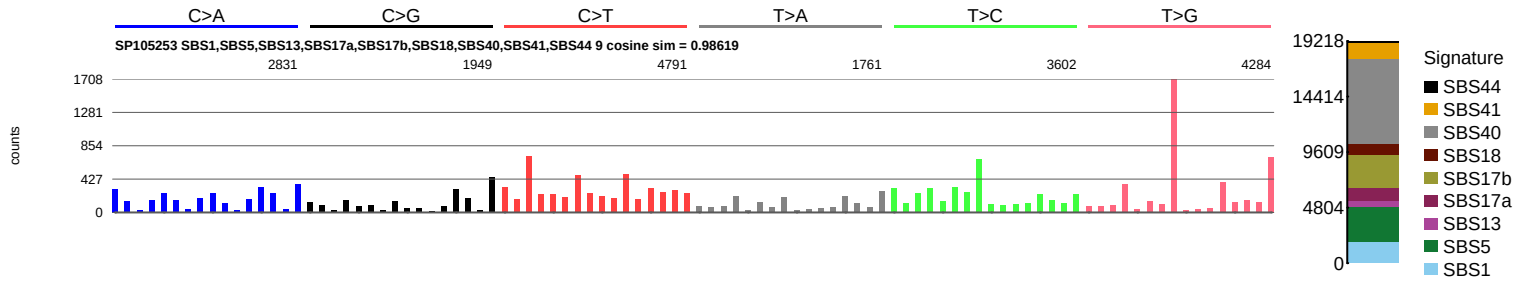
SBS1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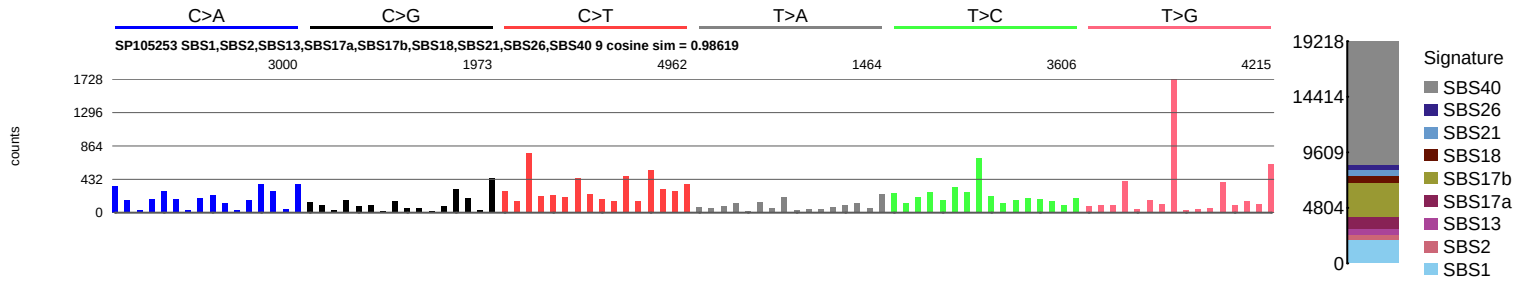
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98619
2831
1949
4791
1761
3602
4284
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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T>C
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SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98619
3000
1973
4962
1464
3606
4215
counts
1728
1296
864
432
0
19218
14414
9609
4804
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Signature
SBS40
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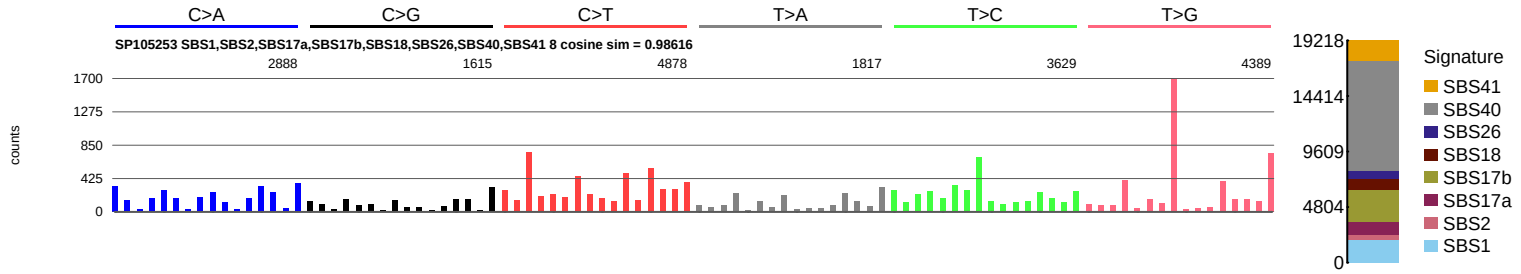
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98616
2888
1615
4878
1817
3629
4389
counts
1700
1275
850
425
0
19218
14414
9609
4804
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Signature
SBS41
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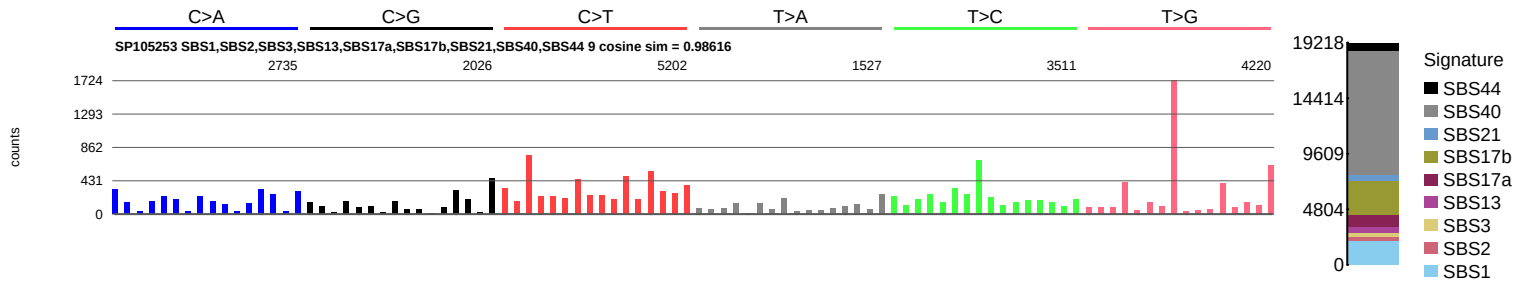
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
T>C
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SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98616
2735
2026
5202
1527
3511
4220
counts
1724
1293
862
431
0
19218
14414
9609
4804
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Signature
SBS44
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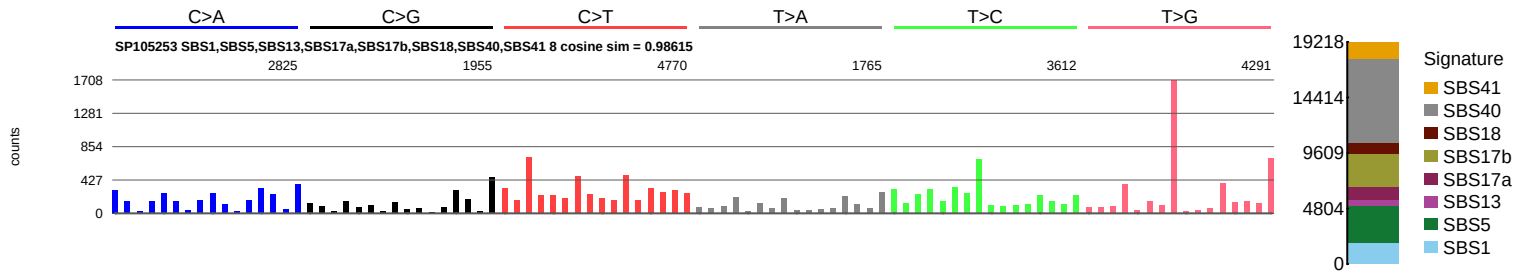
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98615
2825
1955
4770
1765
3612
4291
counts
1708
1281
854
427
0
19218
14414
9609
4804
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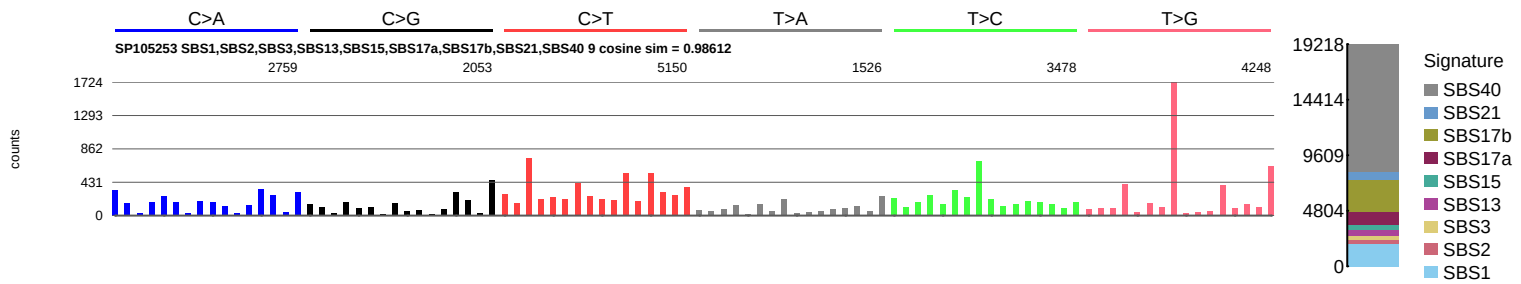
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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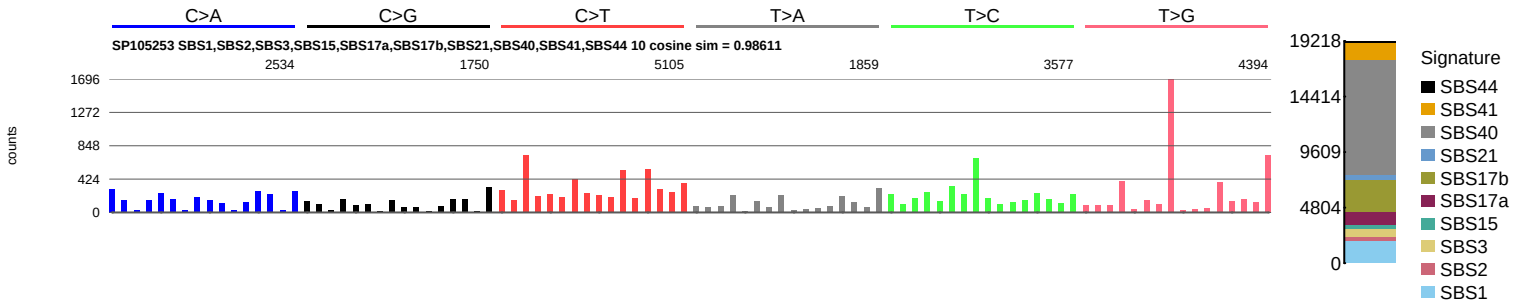
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98612
2759
2053
5150
1526
3478
4248
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98611
2534
1750
5105
1859
3577
4394
counts
1696
1272
848
424
0
19218
14414
9609
4804
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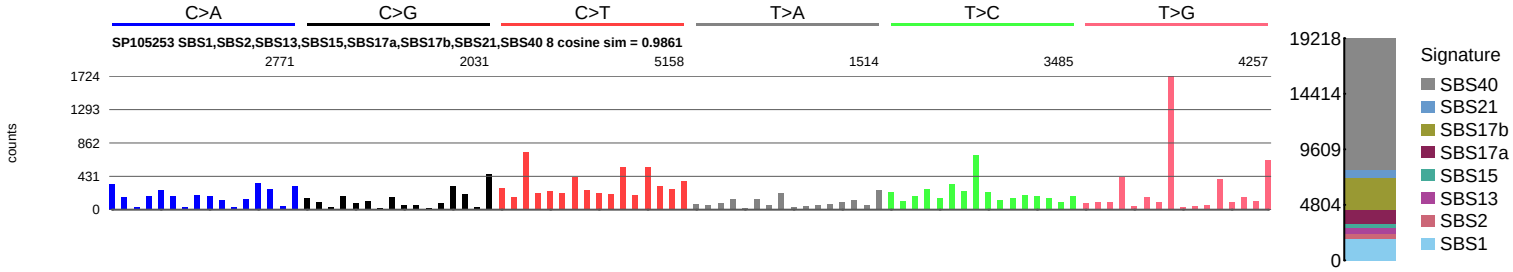
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
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T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9861
2771
2031
5158
1514
3485
4257
counts
1724
1293
862
431
0
19218
14414
9609
4804
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SBS17a
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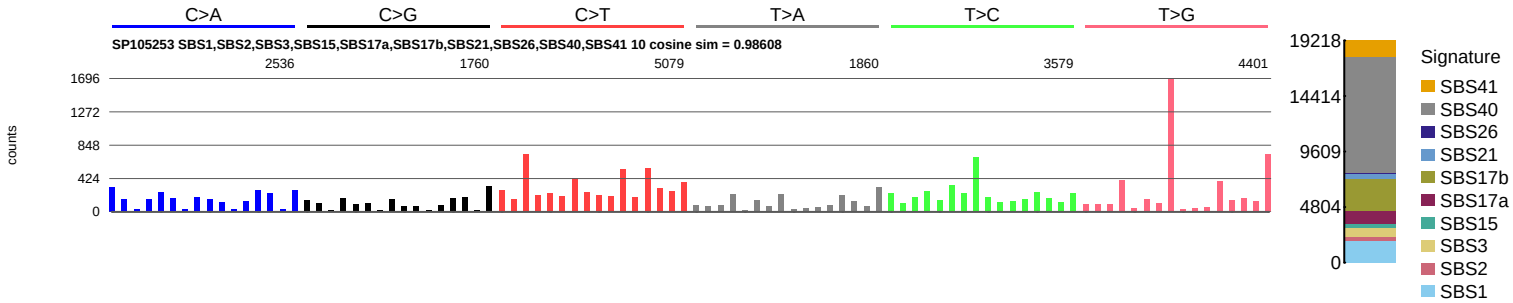
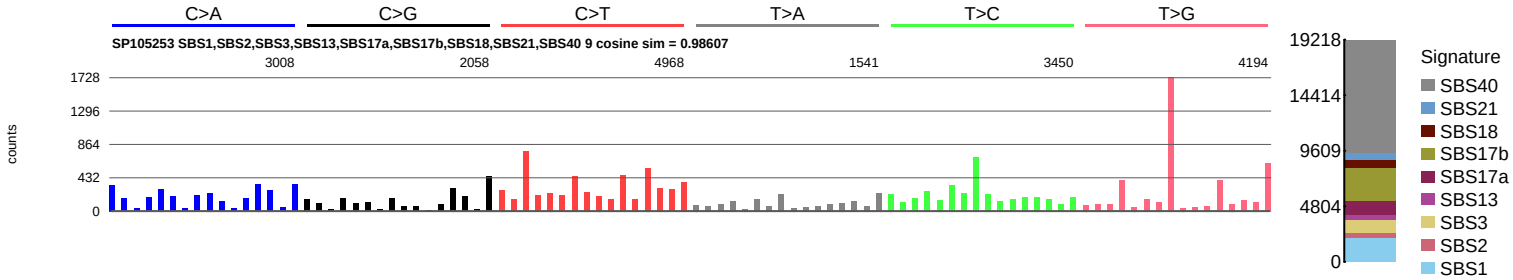
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98608
2536
1760
5079
1860
3579
4401
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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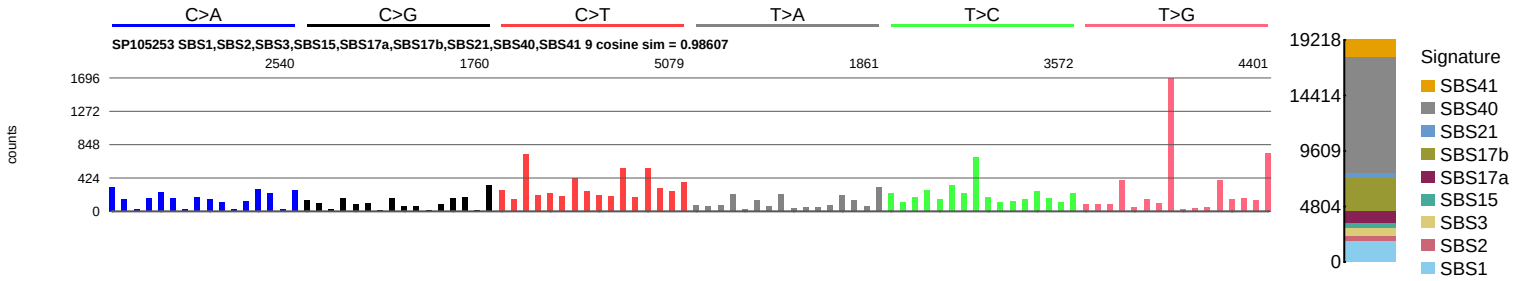
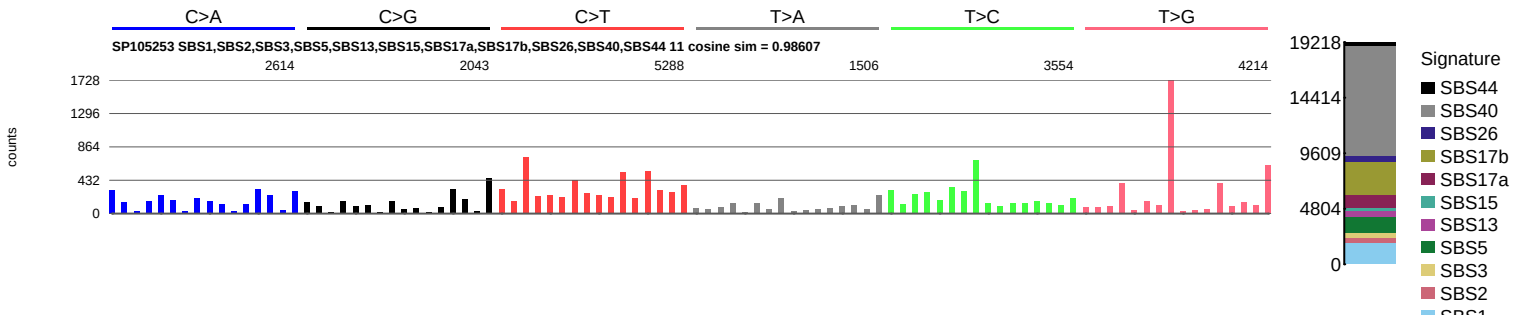
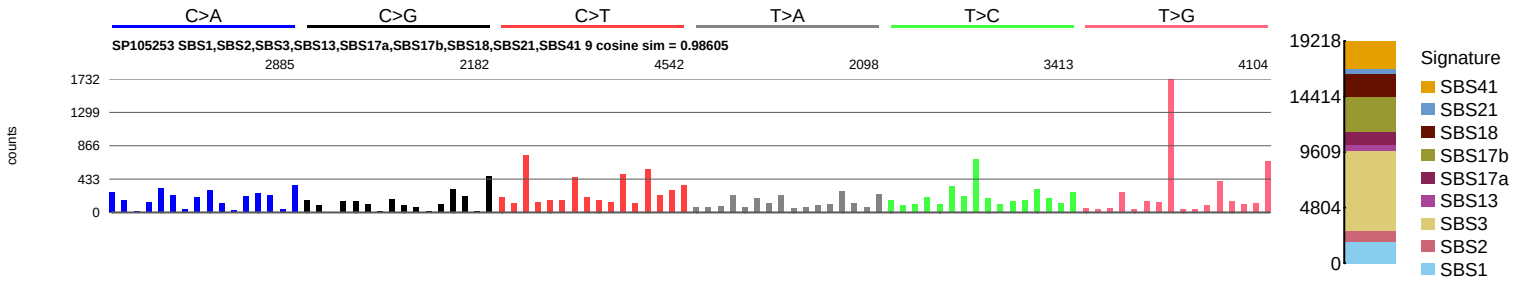
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98607
3008
2058
4968
1541
3450
4194
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98607
2540
1760
5079
1861
3572
4401
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 11 cosine sim = 0.98607
2614
2043
5288
1506
3554
4214
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98605
2885
2182
4542
2098
3413
4104
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
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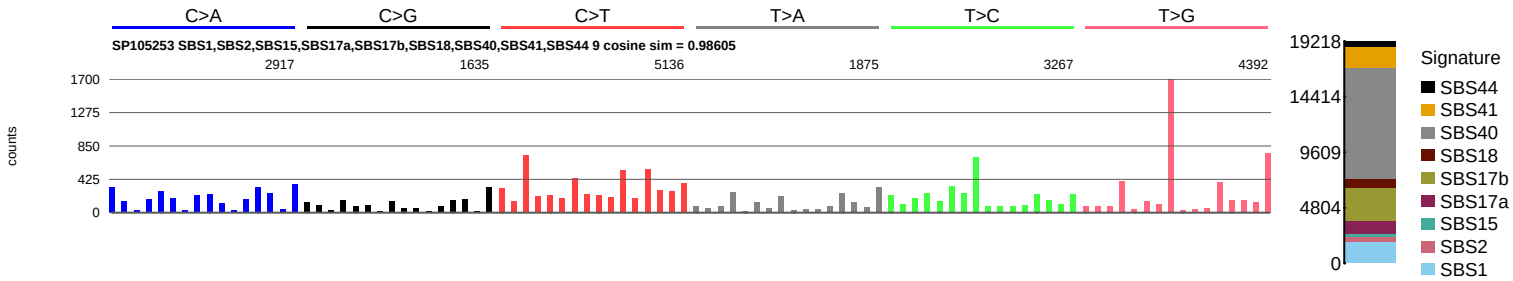
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.98605
2917
1635
5136
1875
3267
4392
counts
1700
1275
850
425
0
19218
14414
9609
4804
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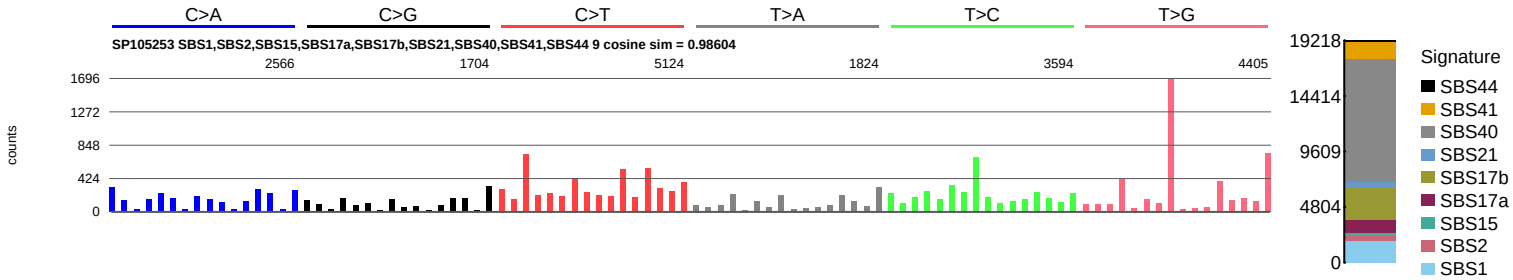
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98604
2566
1704
5124
1824
3594
4405
counts
1696
1272
848
424
0
19218
14414
9609
4804
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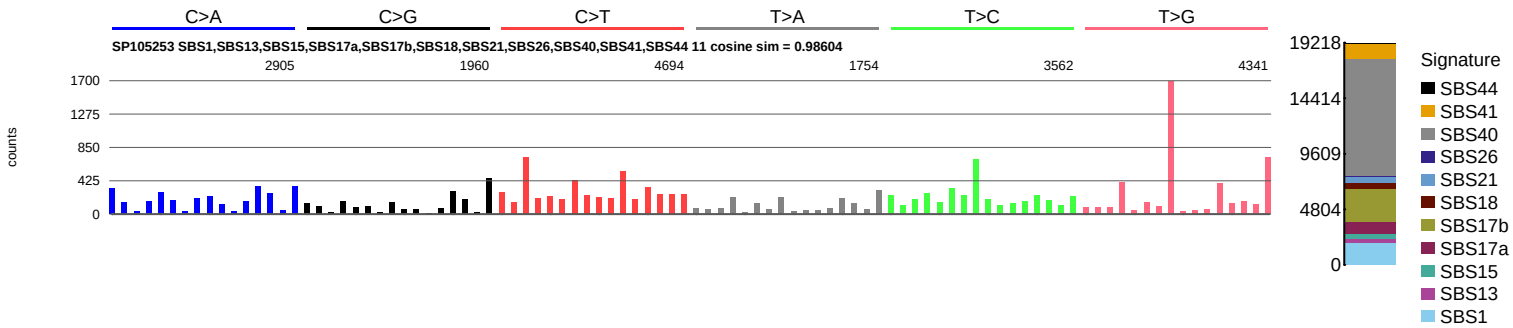
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.98604
2905
1960
4694
1754
3562
4341
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
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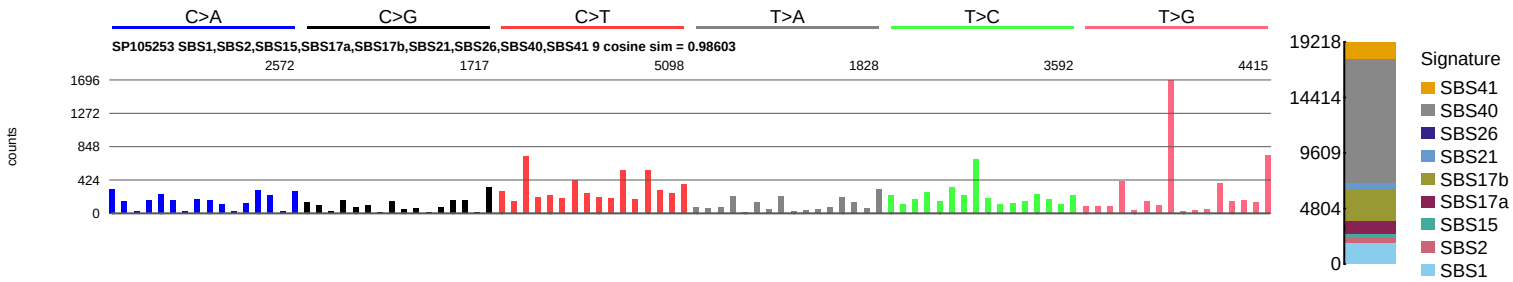
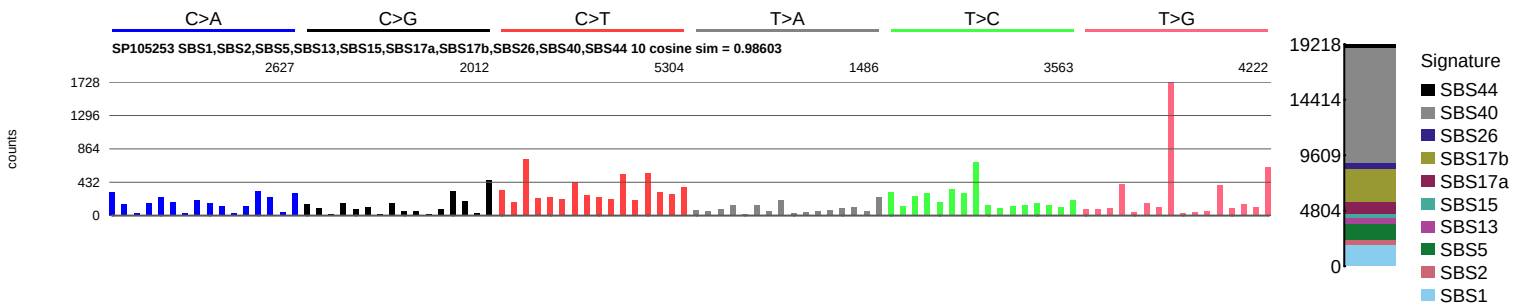
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98603
2572
1717
5098
1828
3592
4415
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98603
2627
2012
5304
1486
3563
4222
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
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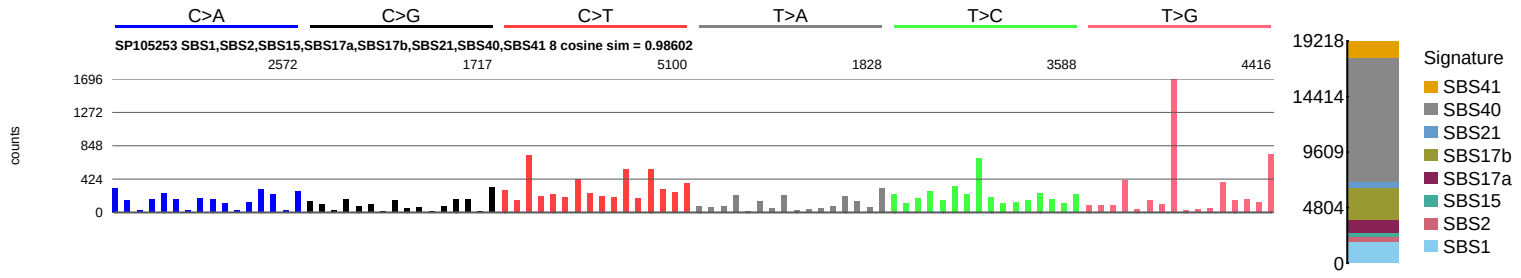
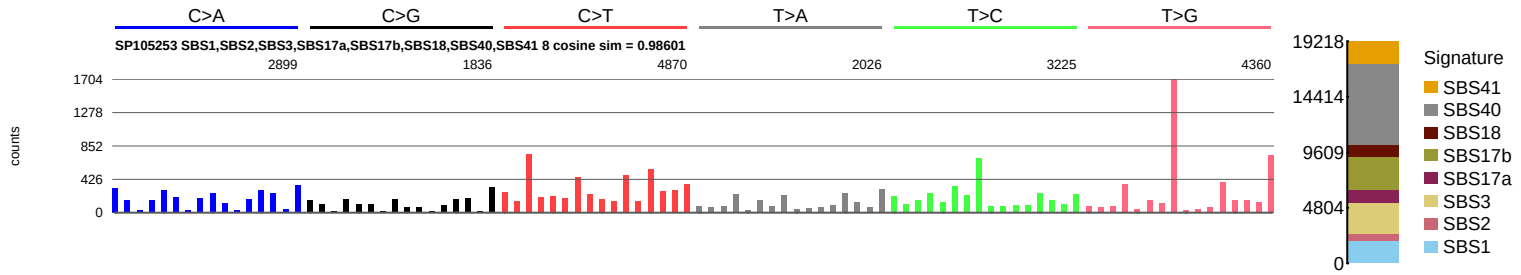
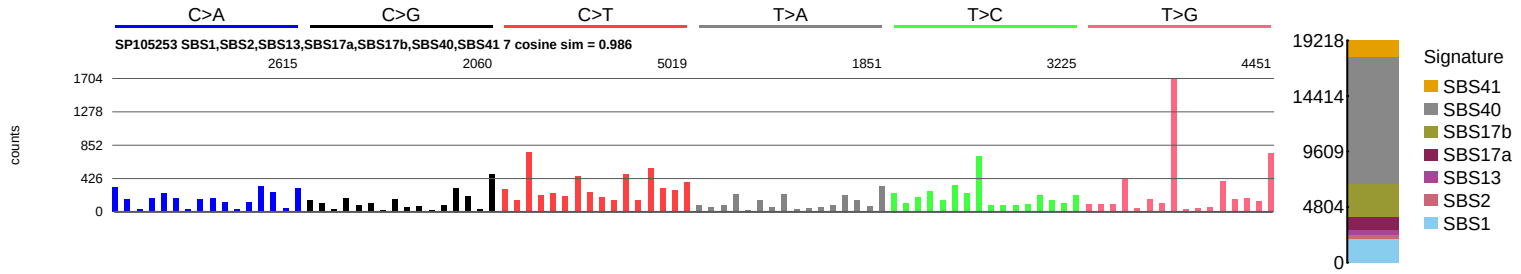
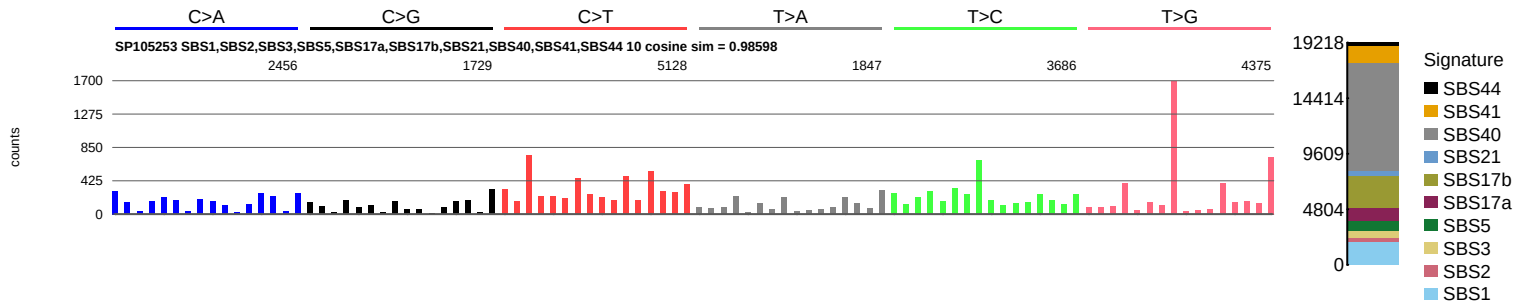
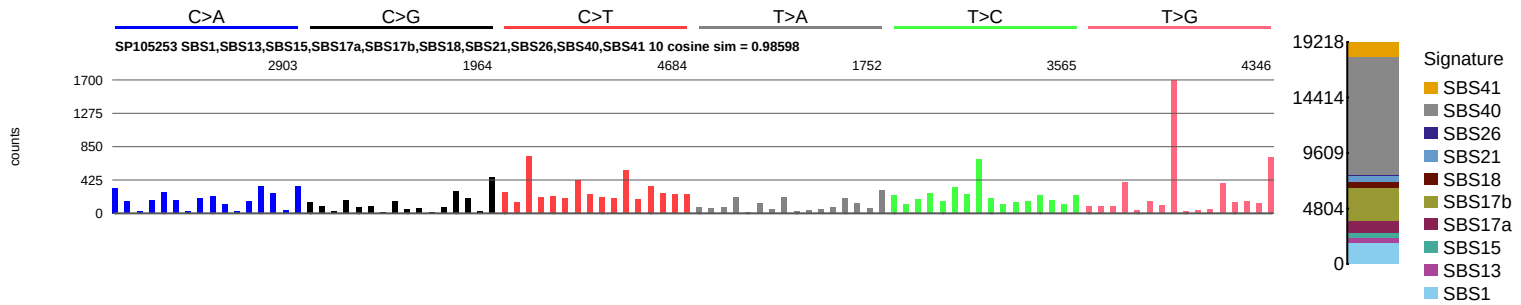
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

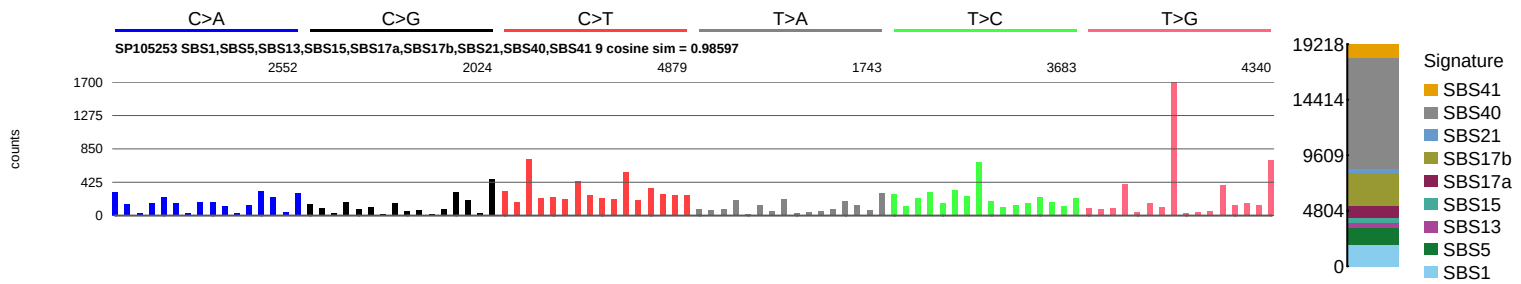

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98602
2572
1717
5100
1828
3588
4416
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98601
2899
1836
4870
2026
3225
4360
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.986
2615
2060
5019
1851
3225
4451
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98598
2456
1729
5128
1847
3686
4375
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.98598
2903
1964
4684
1752
3565
4346
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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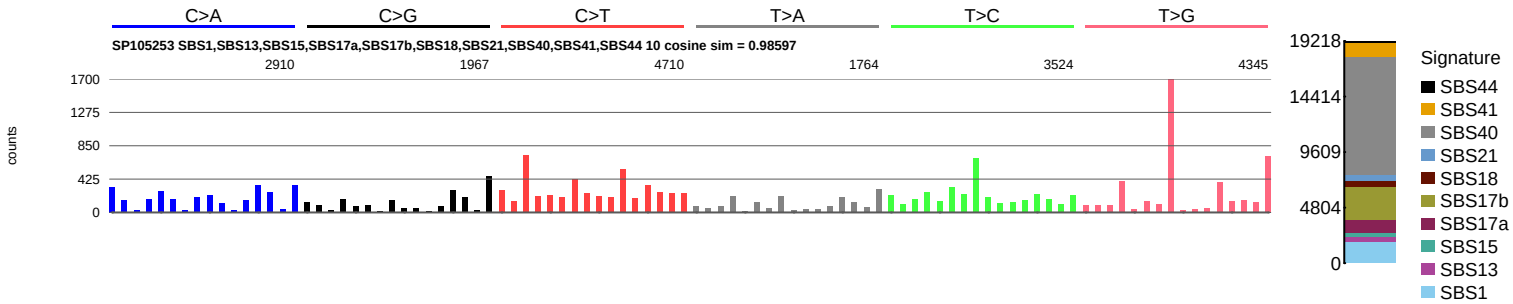
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98597
2552
2024
4879
1743
3683
4340
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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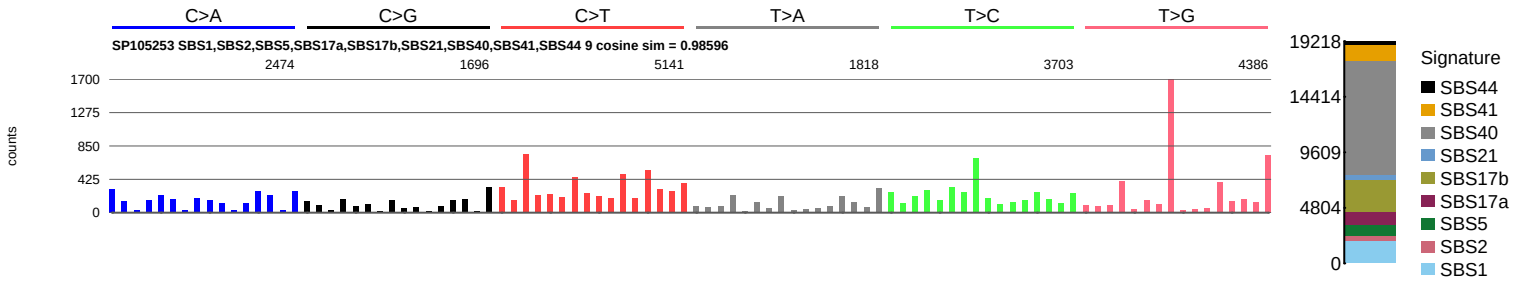
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 10 cosine sim = 0.98597
2910
1967
4710
1764
3524
4345
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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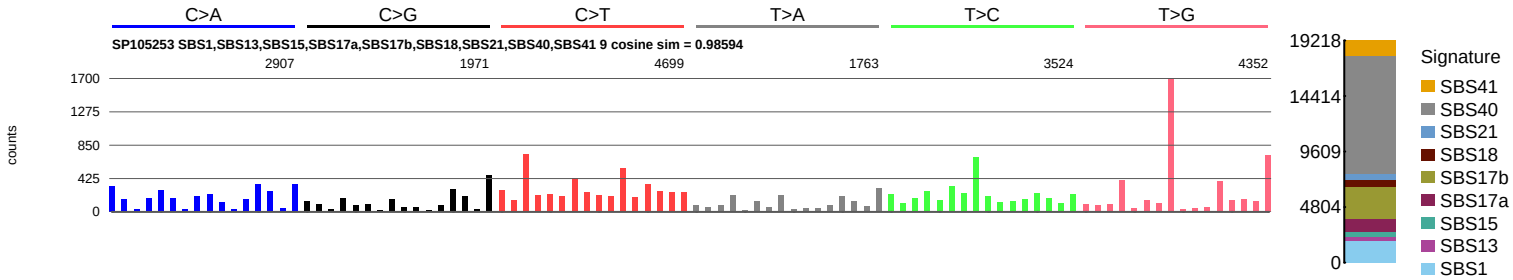
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98596
2474
1696
5141
1818
3703
4386
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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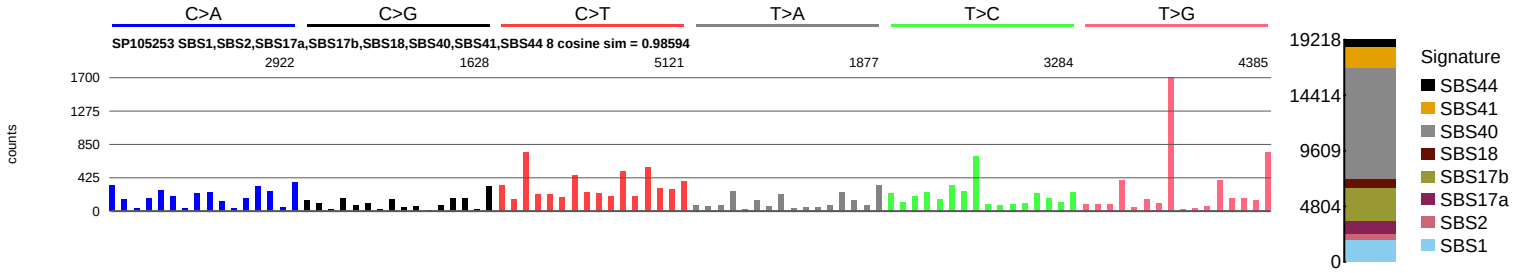
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98594
2907
1971
4699
1763
3524
4352
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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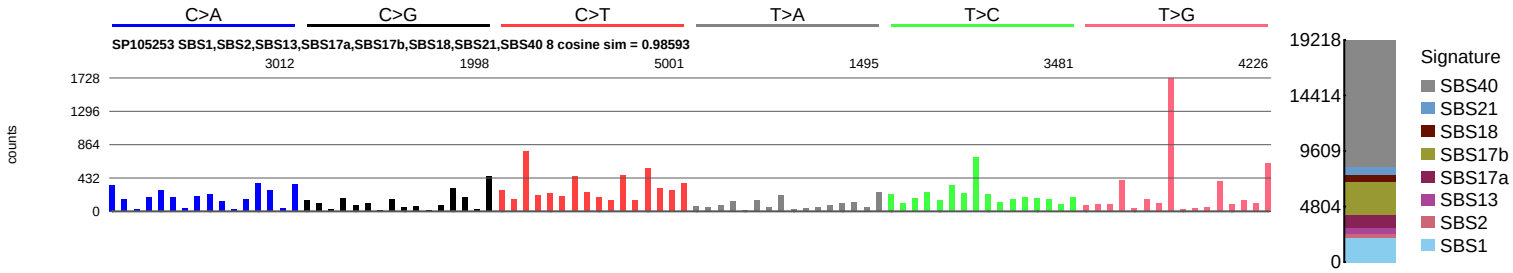
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98594
2922
1628
5121
1877
3284
4385
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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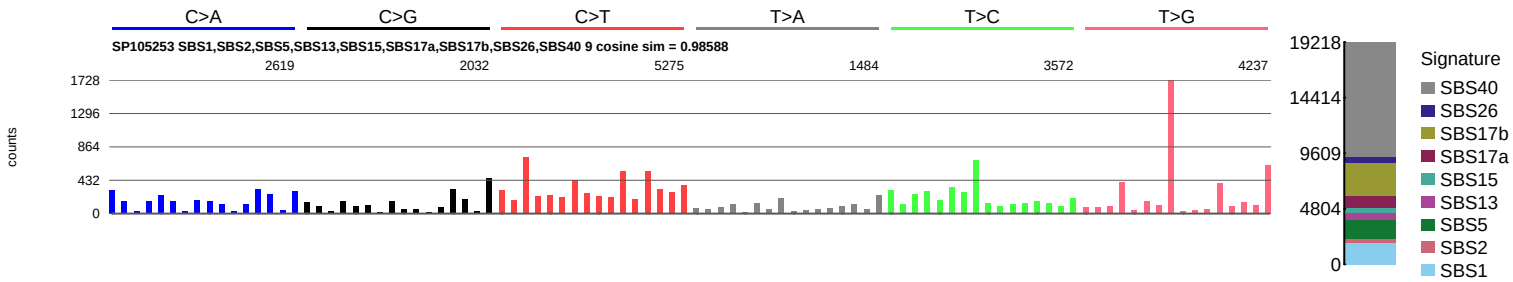
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98593
3012
1998
5001
1495
3481
4226
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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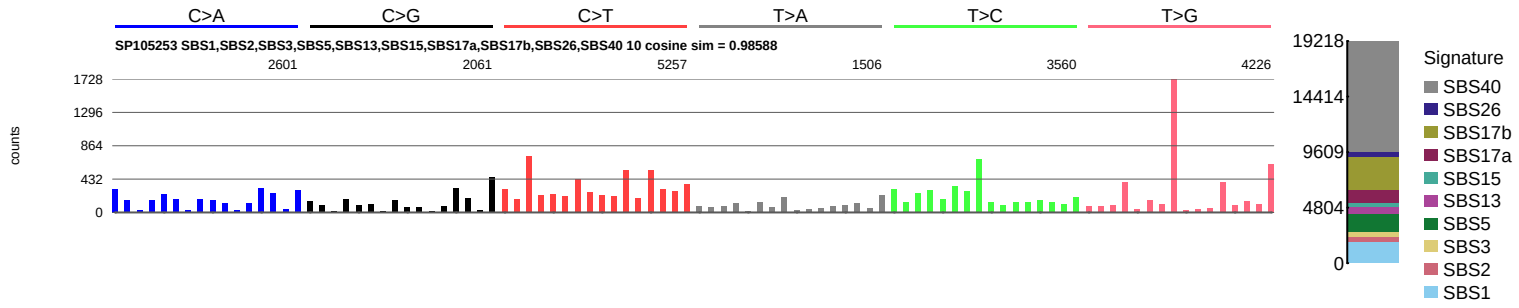
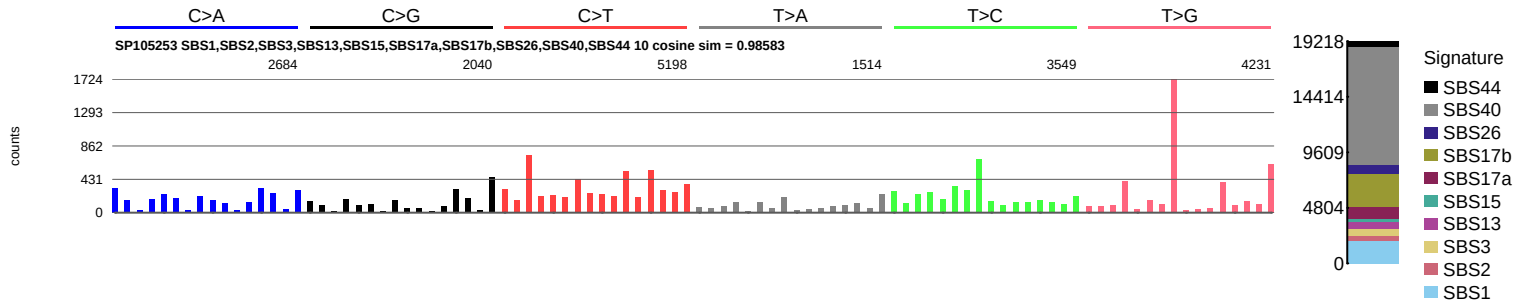
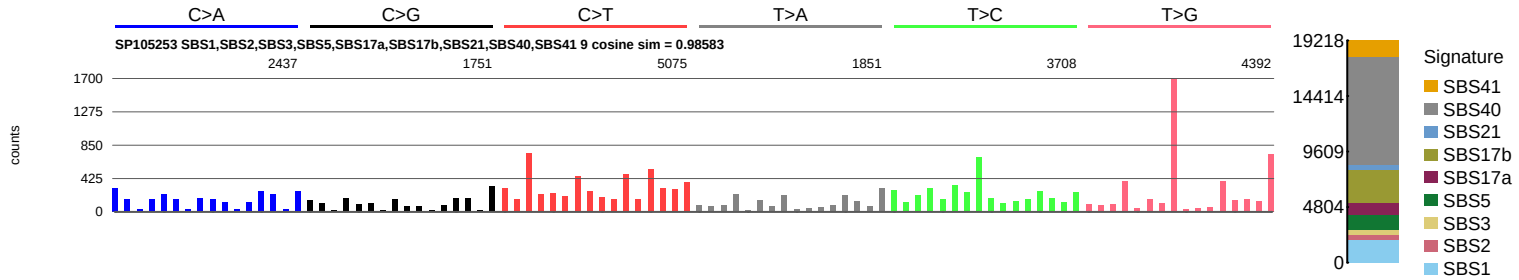
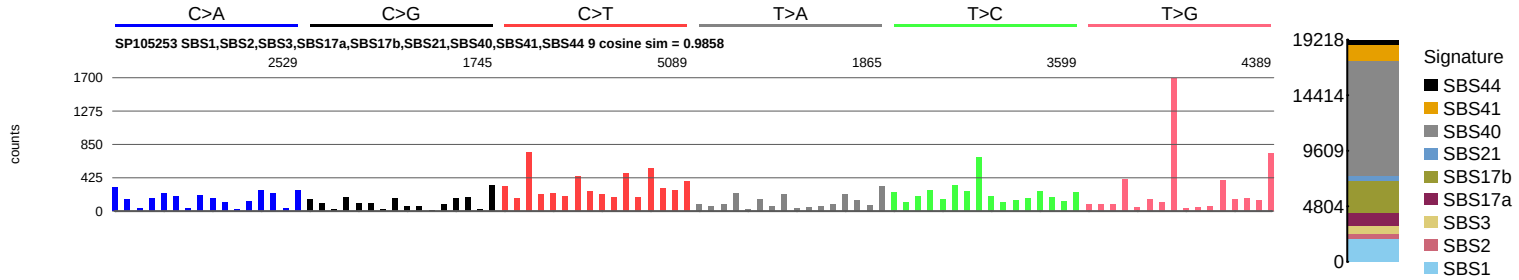
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98588
2619
2032
5275
1484
3572
4237
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 10 cosine sim = 0.98588
2601
2061
5257
1506
3560
4226
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98583
2684
2040
5198
1514
3549
4231
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 9 cosine sim = 0.98583
2437
1751
5075
1851
3708
4392
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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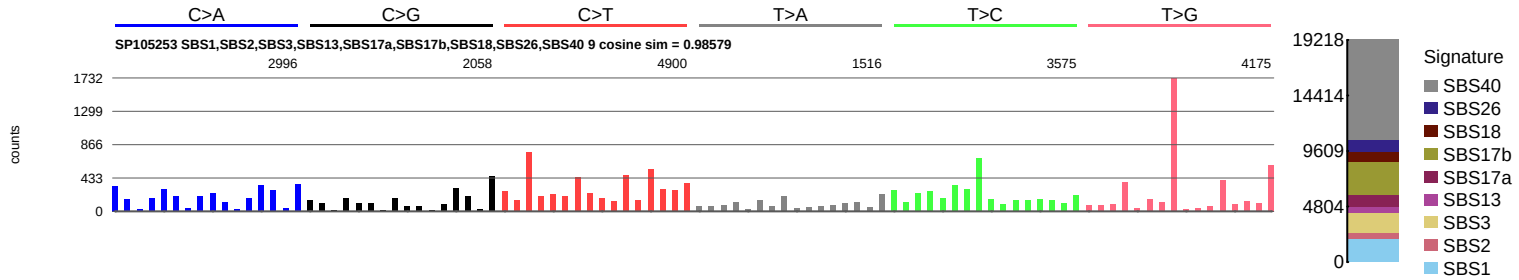
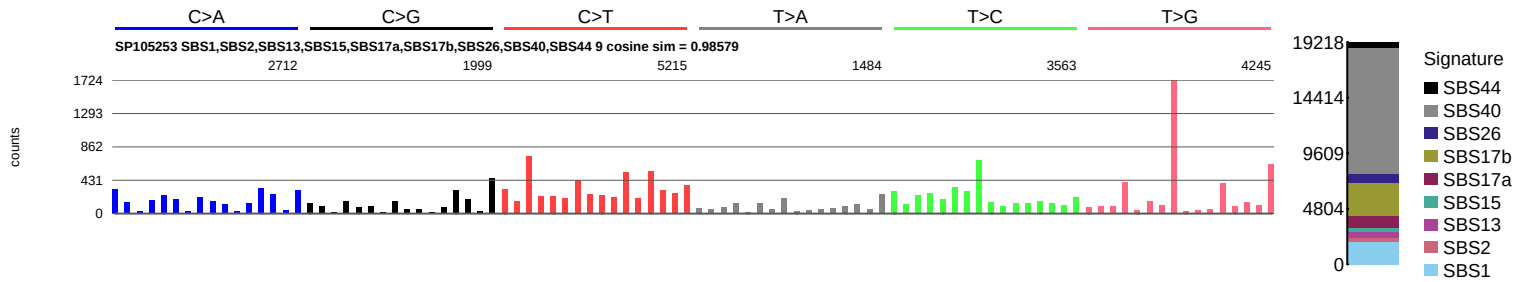
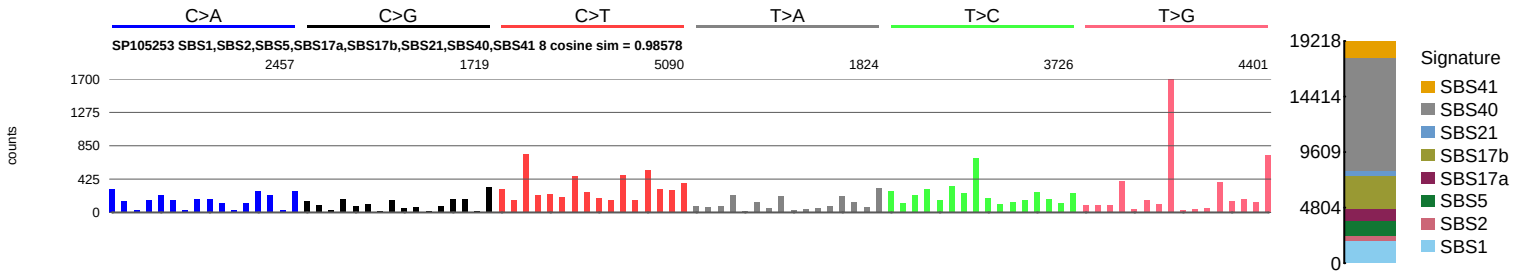
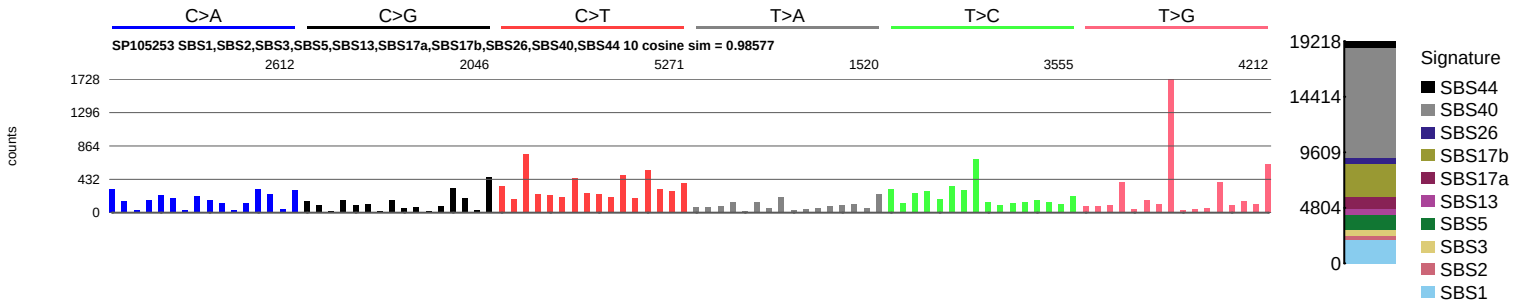
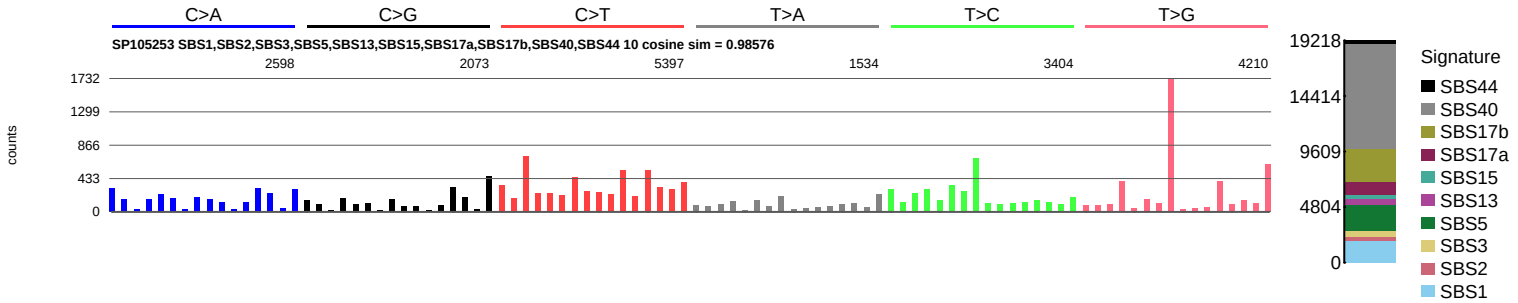
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.9858
2529
1745
5089
1865
3599
4389
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98579
2996
2058
4900
1516
3575
4175
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98579
2712
1999
5215
1484
3563
4245
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98578
2457
1719
5090
1824
3726
4401
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98577
2612
2046
5271
1520
3555
4212
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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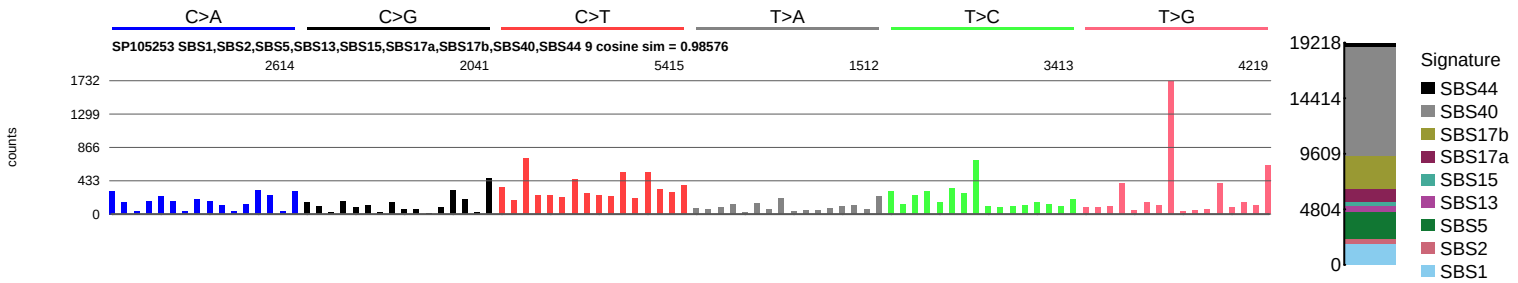
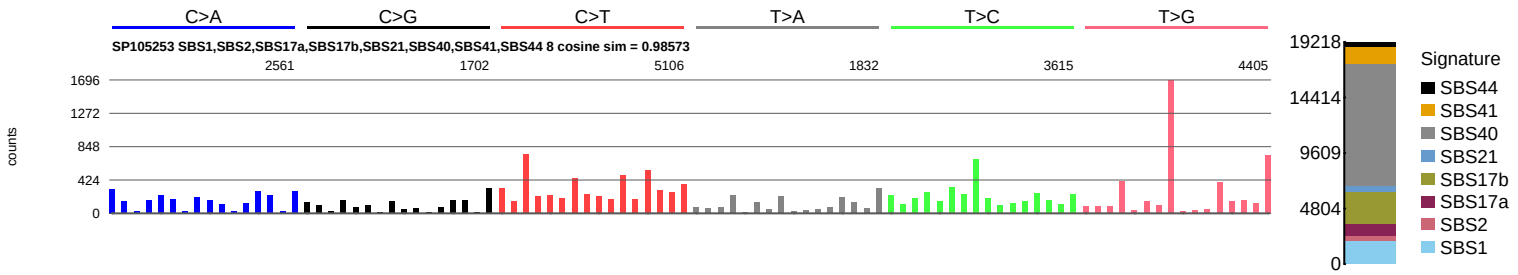
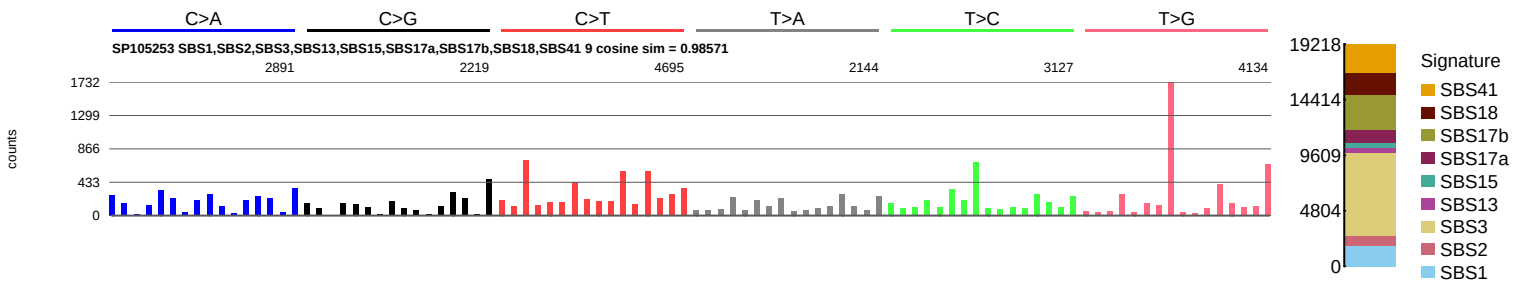
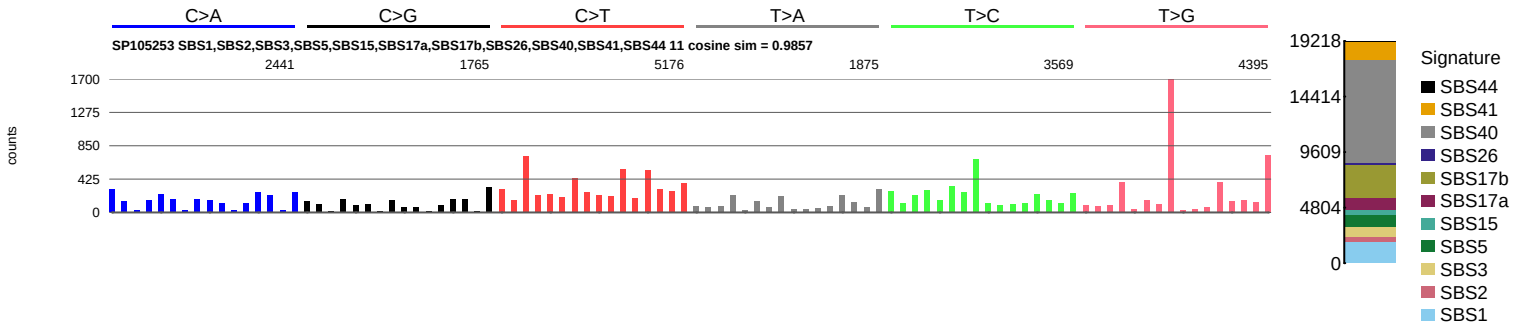
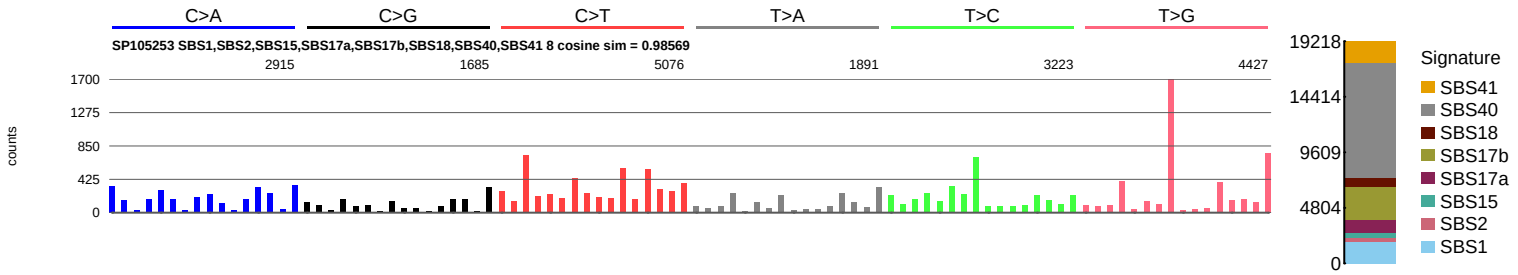
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 10 cosine sim = 0.98576
2598
2073
5397
1534
3404
4210
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

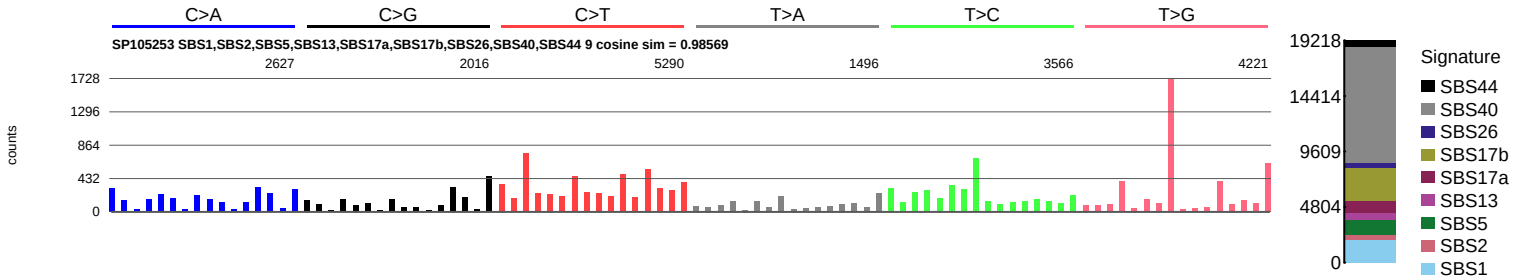

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98576
2614
2041
5415
1512
3413
4219
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98573
2561
1702
5106
1832
3615
4405
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98571
2891
2219
4695
2144
3127
4134
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 11 cosine sim = 0.9857
2441
1765
5176
1875
3569
4395
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98569
2915
1685
5076
1891
3223
4427
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98569
2627
2016
5290
1496
3566
4221
counts
1728
1296
864
432
0
19218
14414
9609
4804
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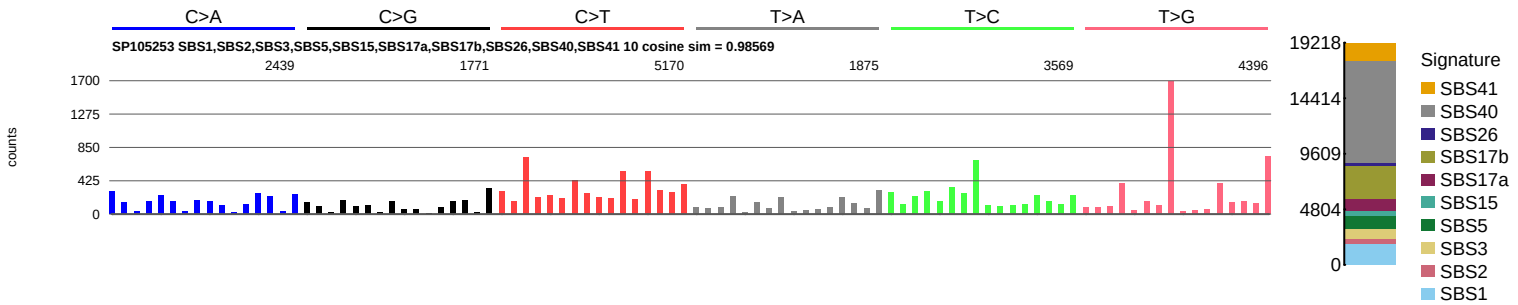
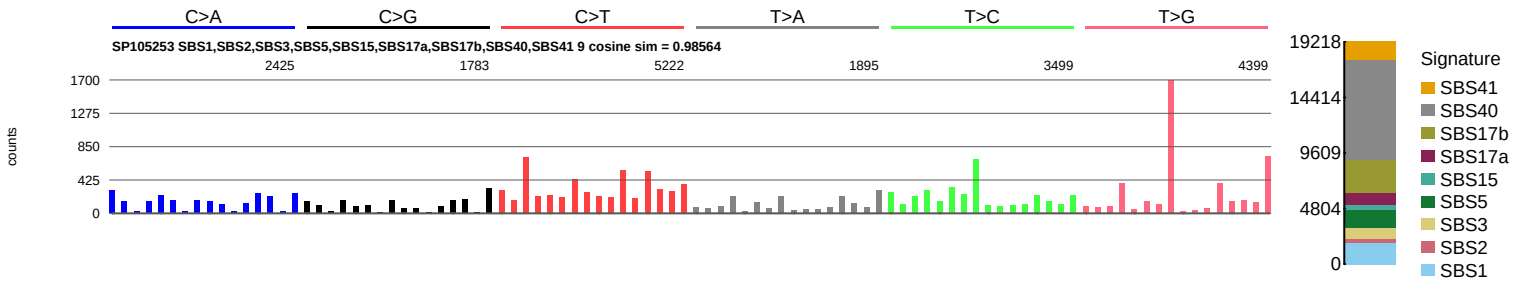
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 10 cosine sim = 0.98569
2439
1771
5170
1875
3569
4396
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 9 cosine sim = 0.98564
2425
1783
5222
1895
3499
4399
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
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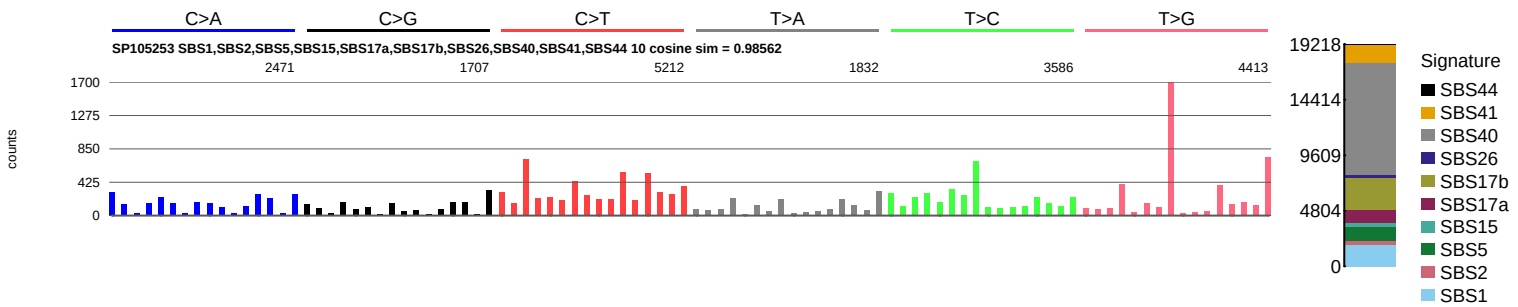
SBS1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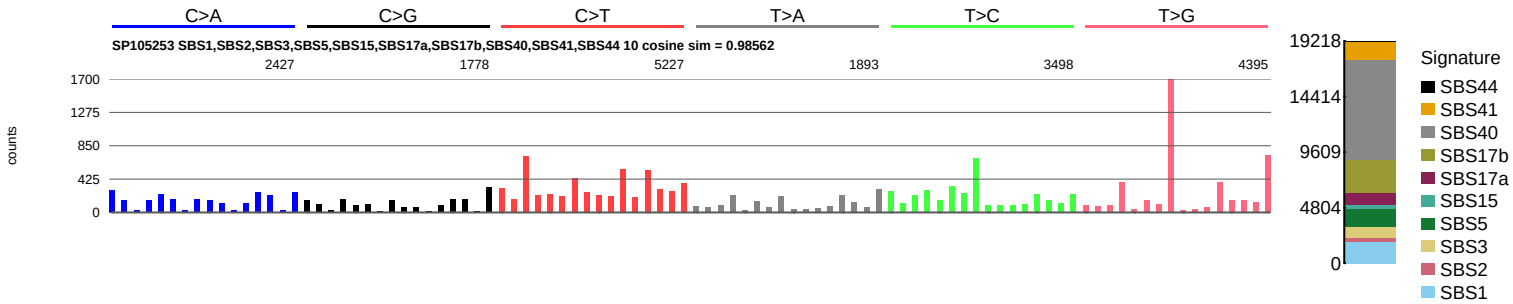
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98562
2471
1707
5212
1832
3586
4413
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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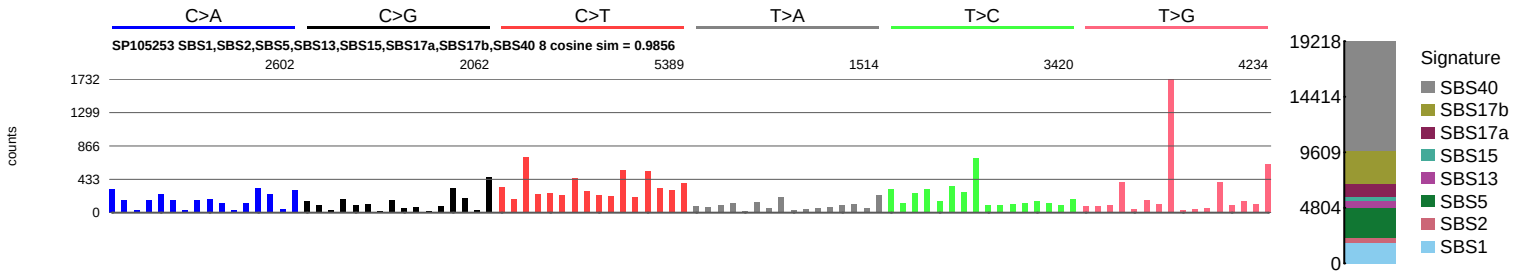
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 10 cosine sim = 0.98562
2427
1778
5227
1893
3498
4395
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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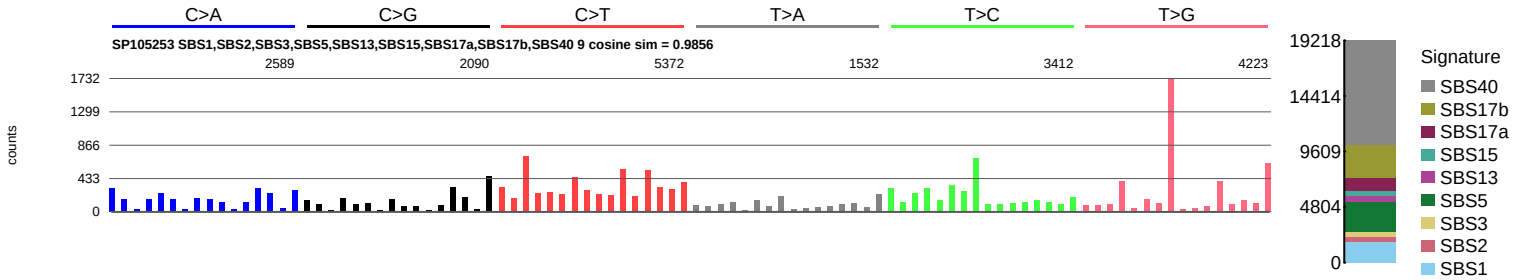
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.9856
2602
2062
5389
1514
3420
4234
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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T>A
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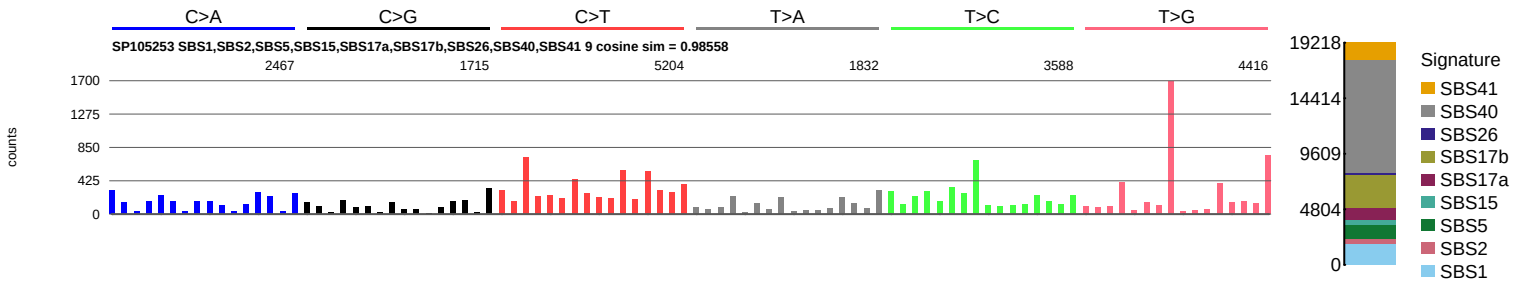
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 9 cosine sim = 0.9856
2589
2090
5372
1532
3412
4223
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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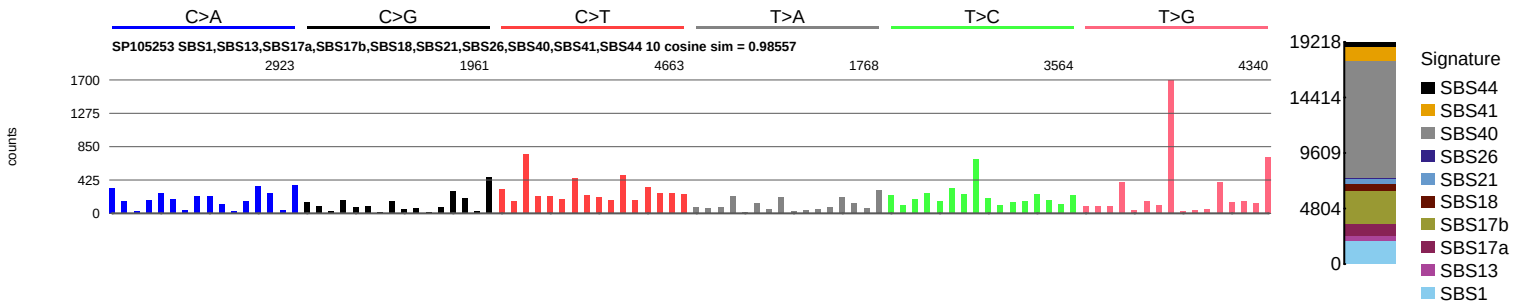
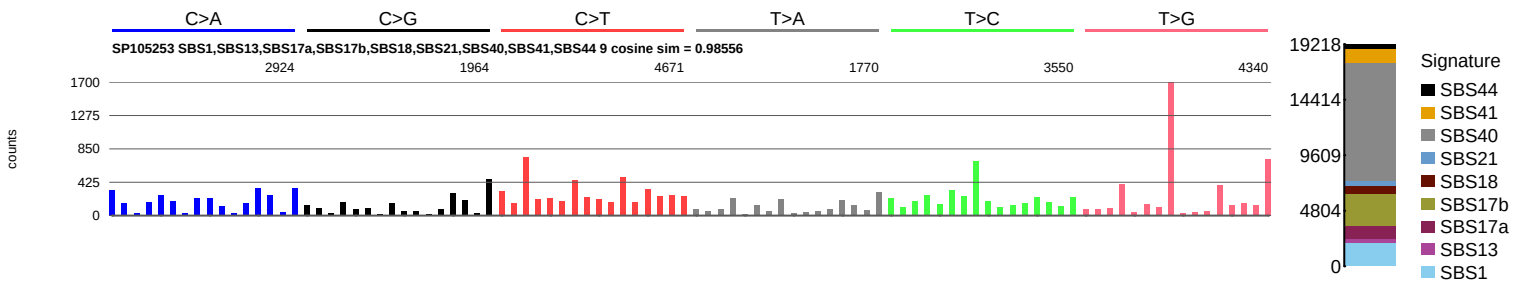
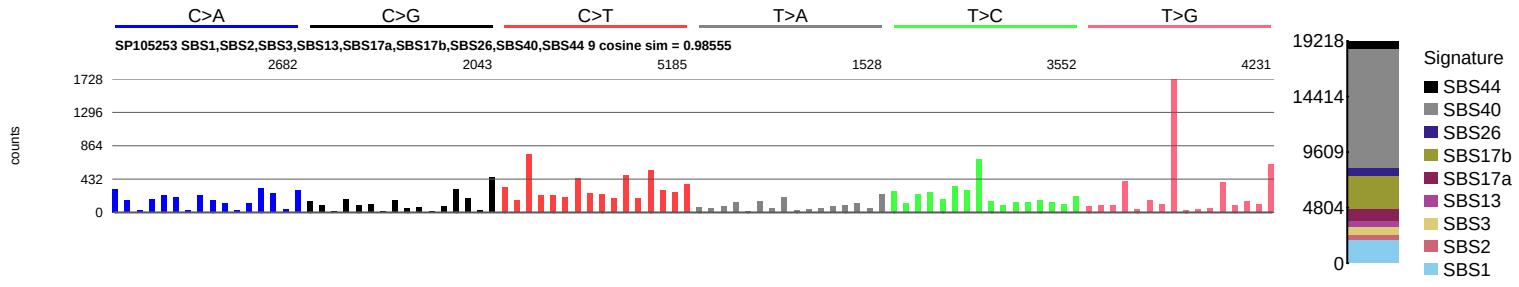
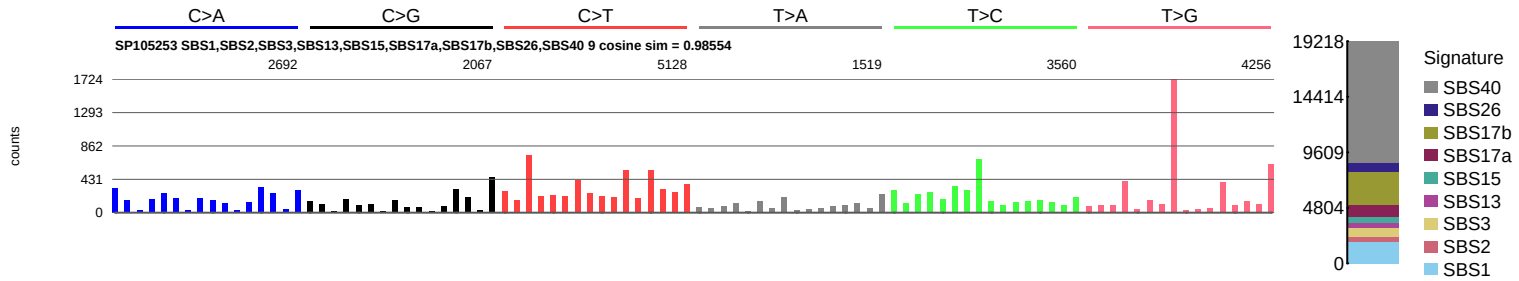
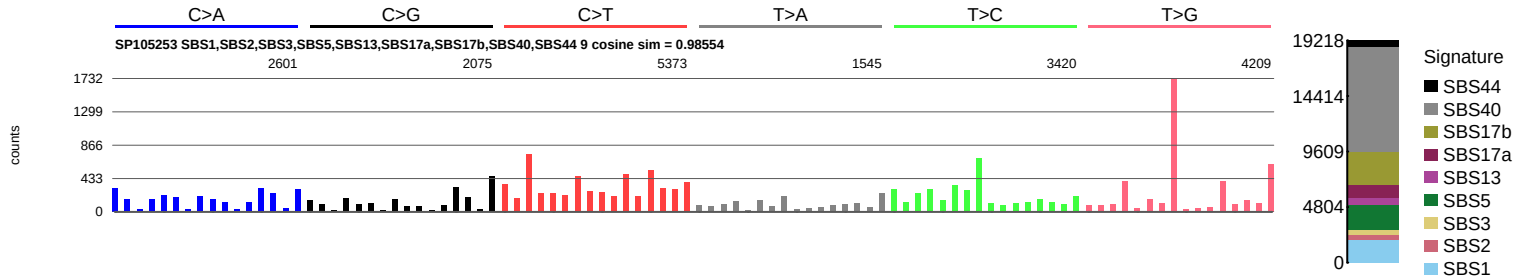
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98558
2467
1715
5204
1832
3588
4416
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

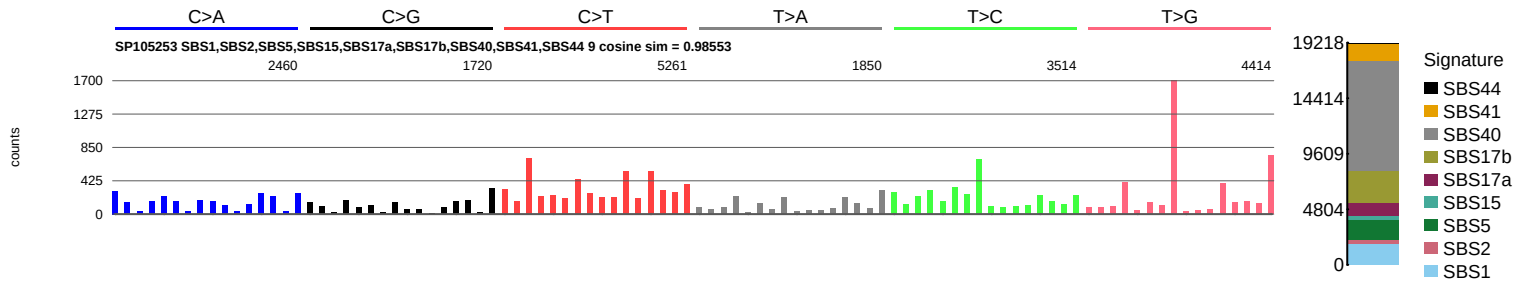

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98557
2923
1961
4663
1768
3564
4340
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

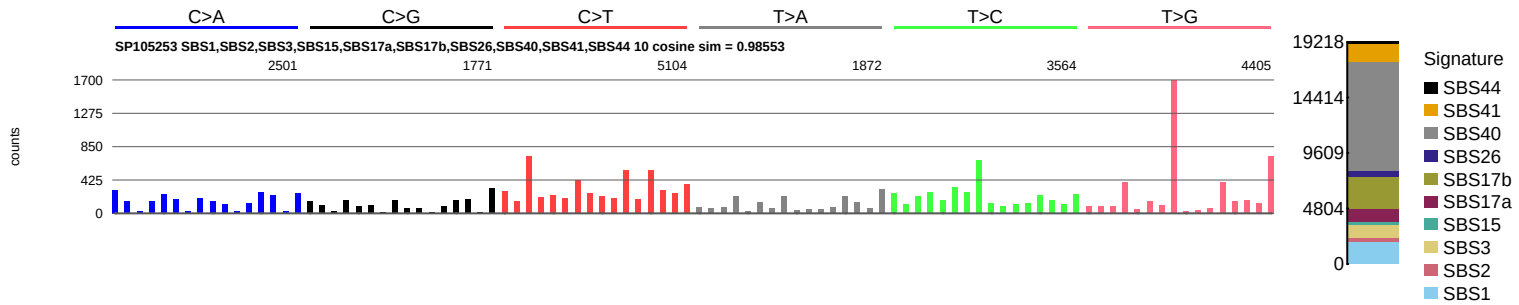

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98556
2924
1964
4671
1770
3550
4340
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

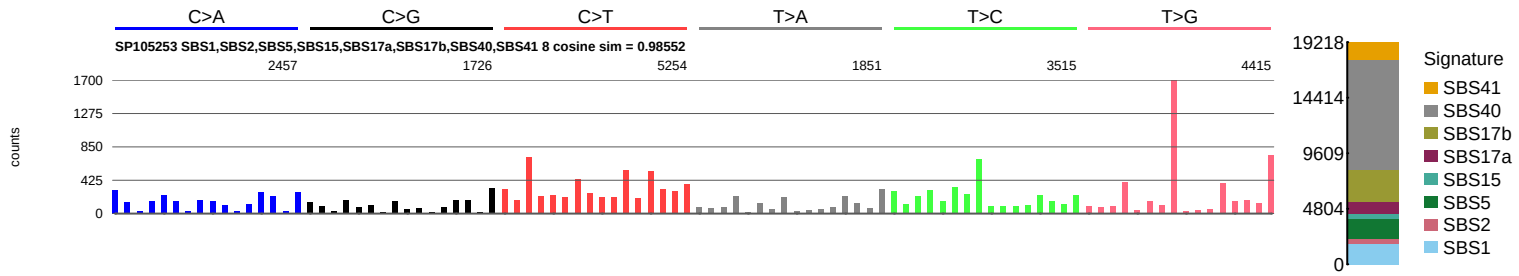

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98555
2682
2043
5185
1528
3552
4231
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

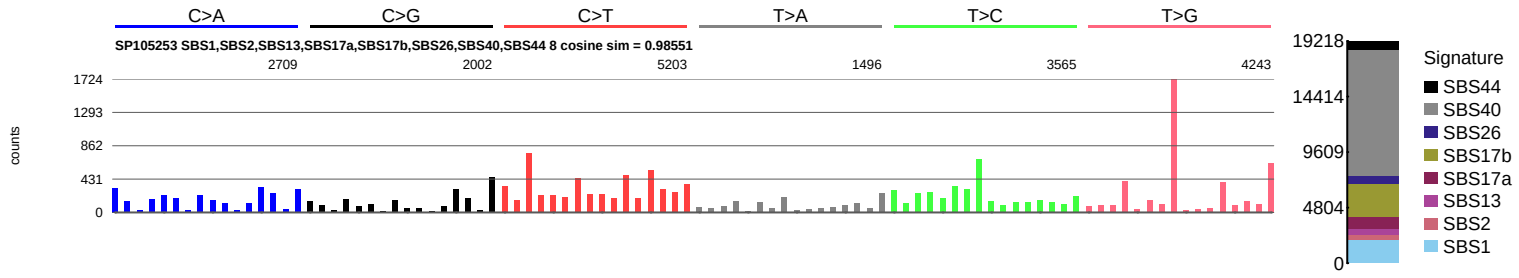

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98554
2692
2067
5128
1519
3560
4256
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

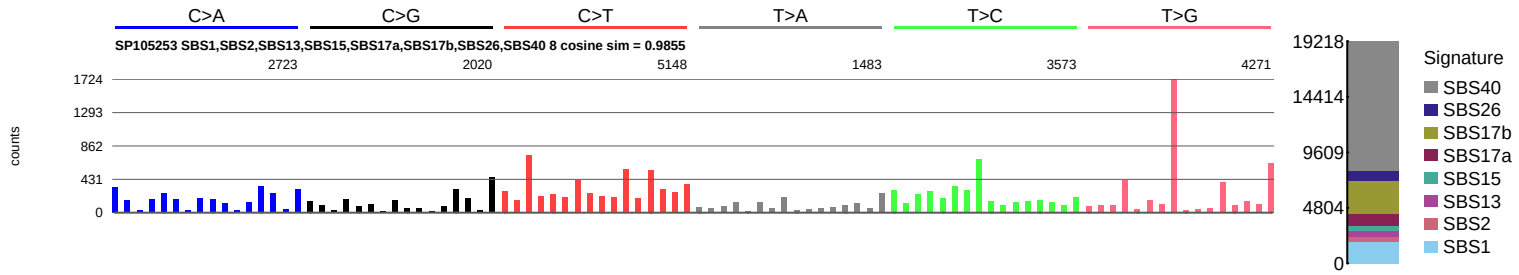

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98554
2601
2075
5373
1545
3420
4209
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98553
2460
1720
5261
1850
3514
4414
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98553
2501
1771
5104
1872
3564
4405
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98552
2457
1726
5254
1851
3515
4415
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98551
2709
2002
5203
1496
3565
4243
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1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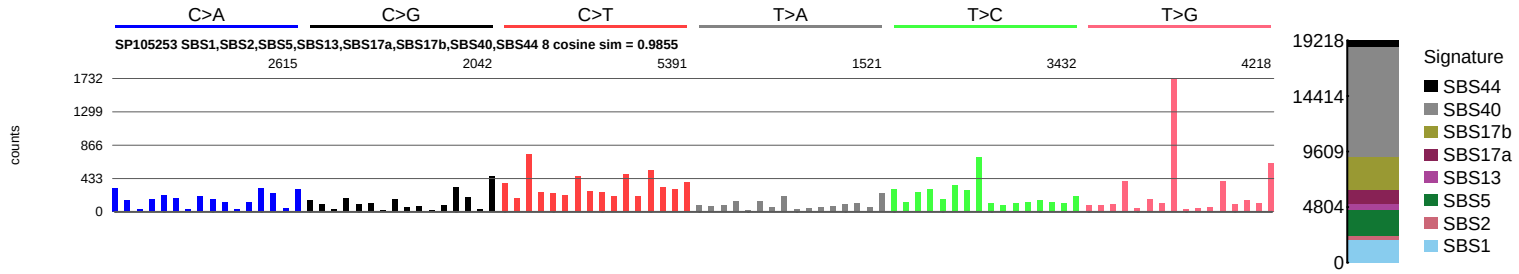
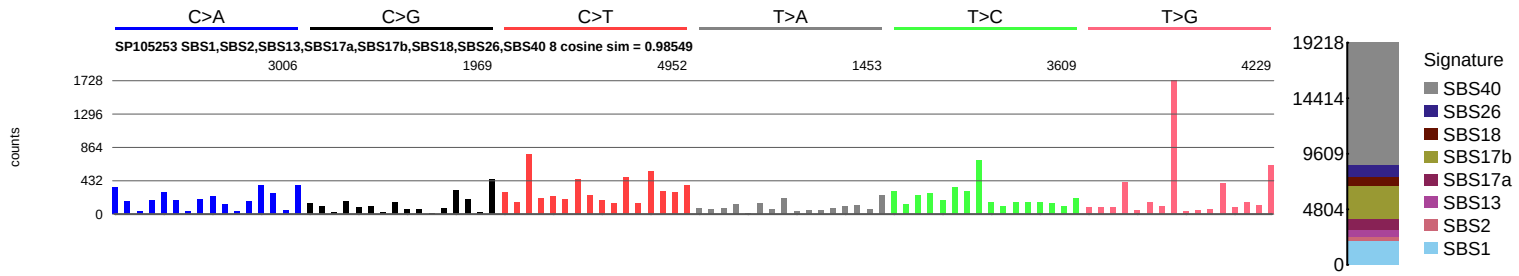
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9855
2723
2020
5148
1483
3573
4271
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.9855
2615
2042
5391
1521
3432
4218
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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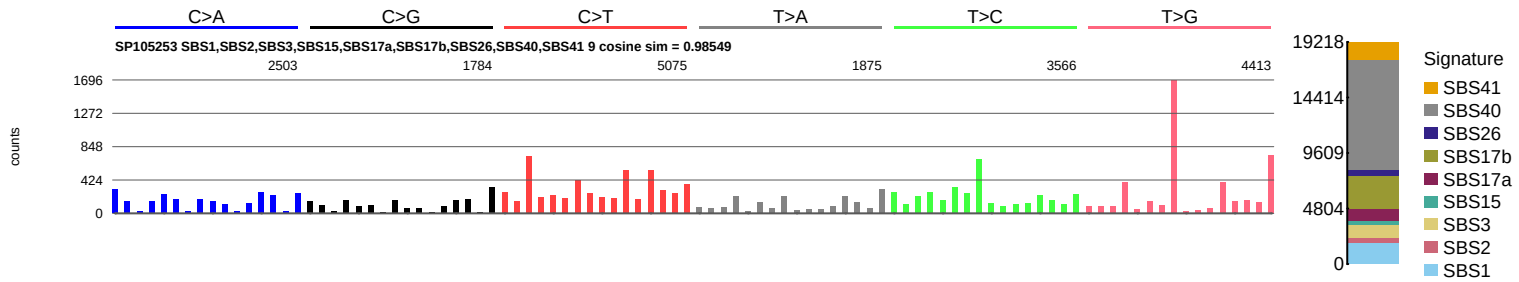
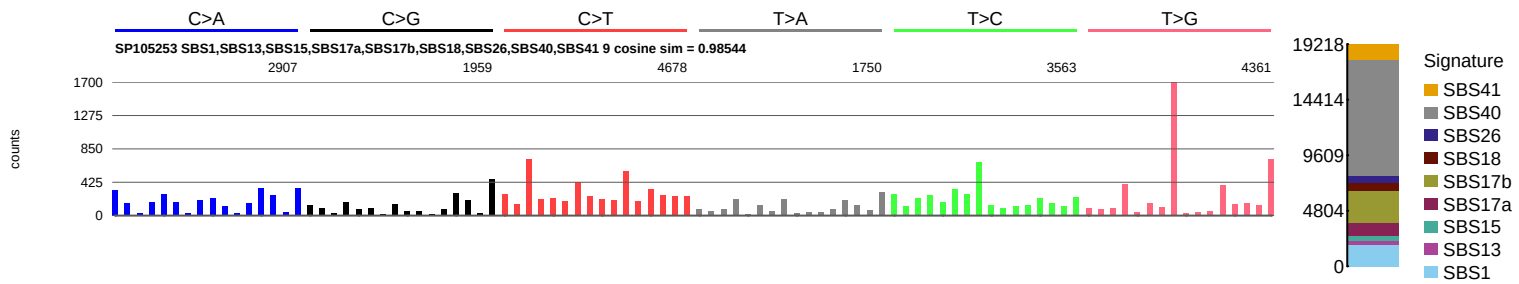
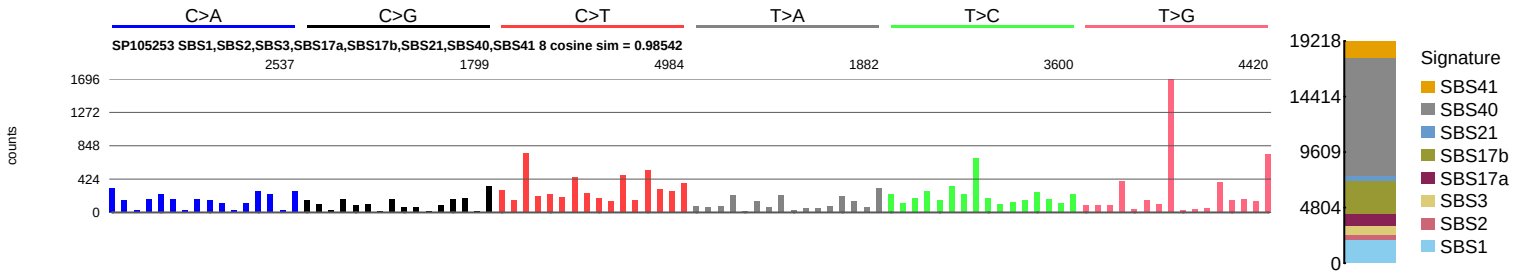
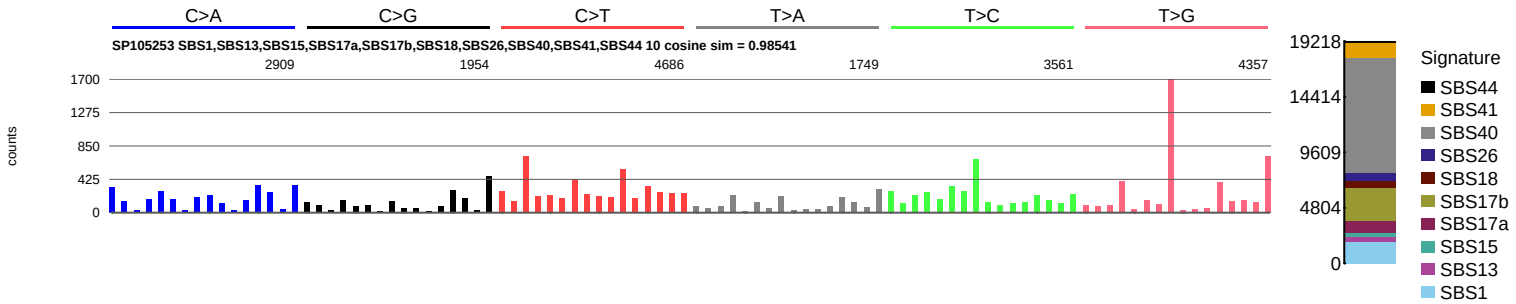
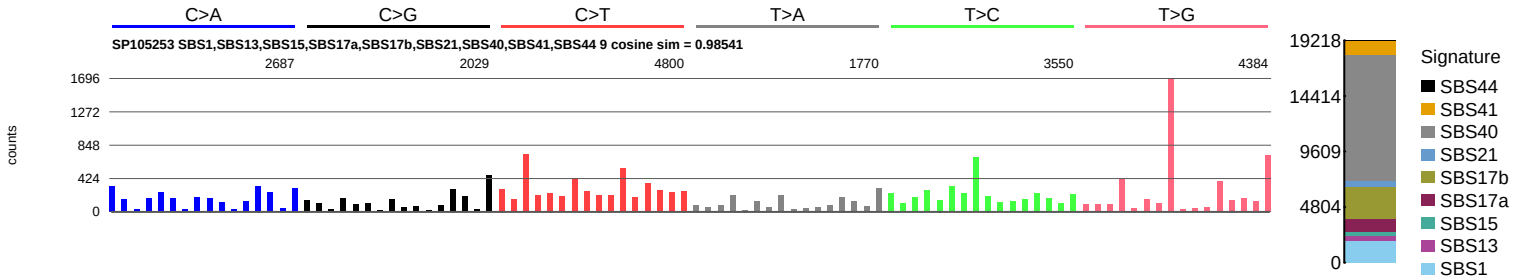
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98549
3006
1969
4952
1453
3609
4229
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1

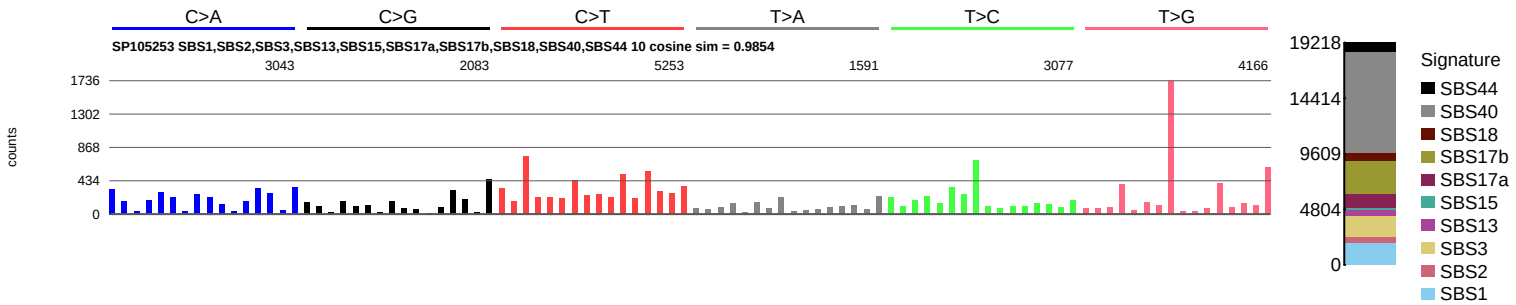

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98549
2503
1784
5075
1875
3566
4413
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

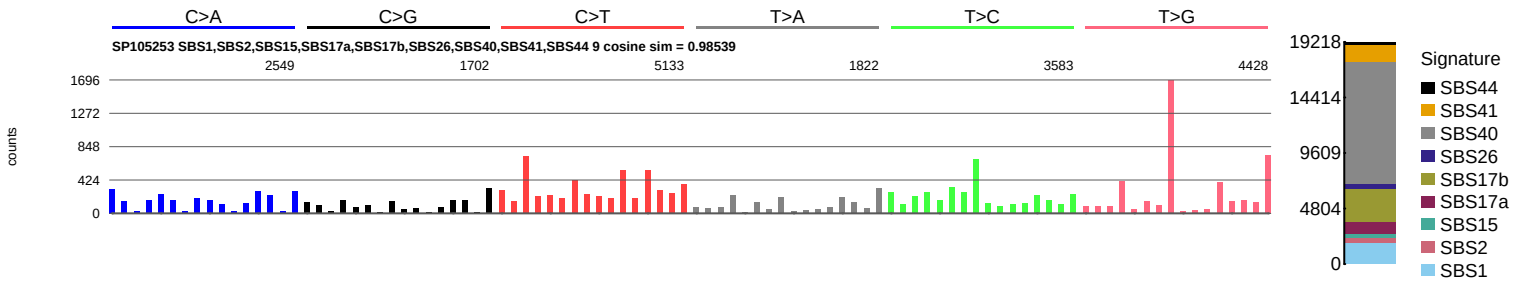

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.98544
2907
1959
4678
1750
3563
4361
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

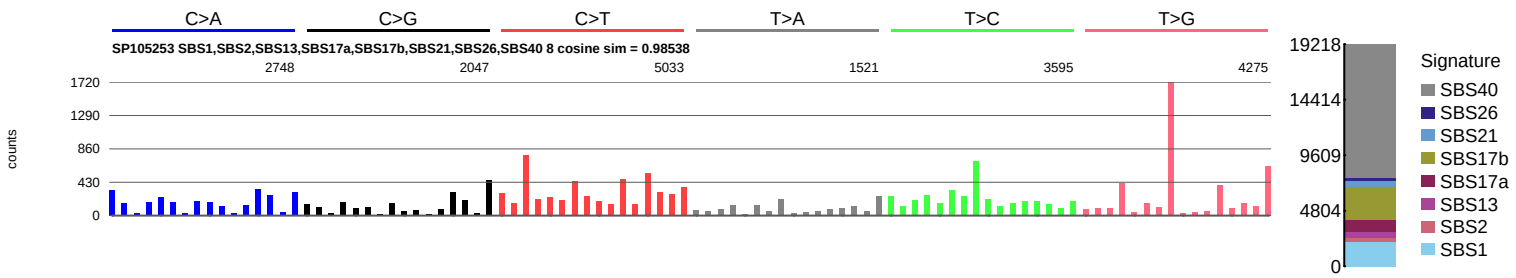

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98542
2537
1799
4984
1882
3600
4420
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98541
2909
1954
4686
1749
3561
4357
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98541
2687
2029
4800
1770
3550
4384
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.9854
3043
2083
5253
1591
3077
4166
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98539
2549
1702
5133
1822
3583
4428
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1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SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98538
2748
2047
5033
1521
3595
4275
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
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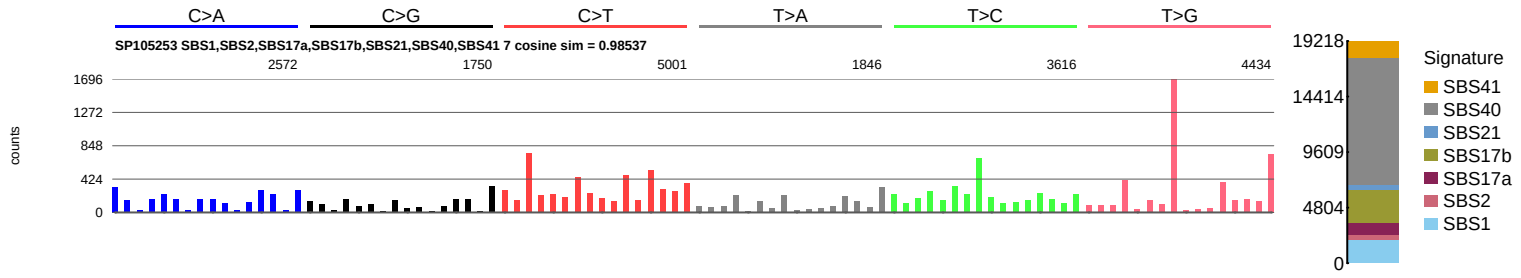
SBS13
SBS2
SBS1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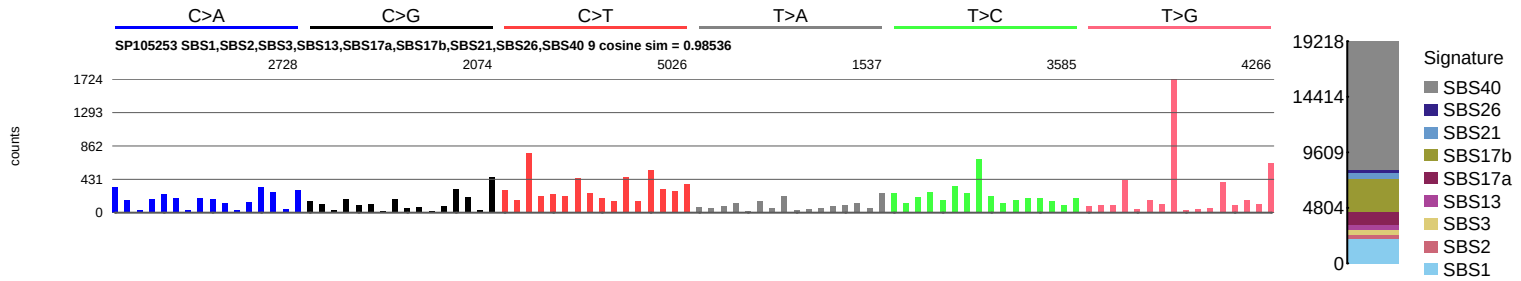
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98537
2572
1750
5001
1846
3616
4434
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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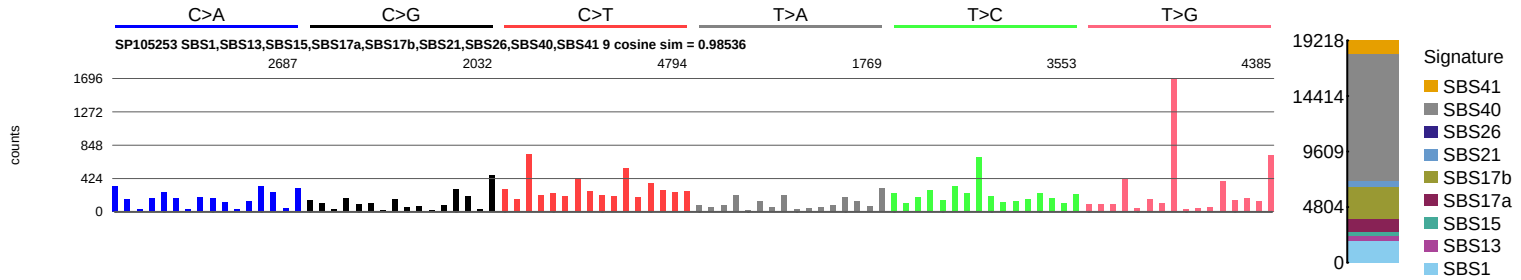
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98536
2728
2074
5026
1537
3585
4266
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>T
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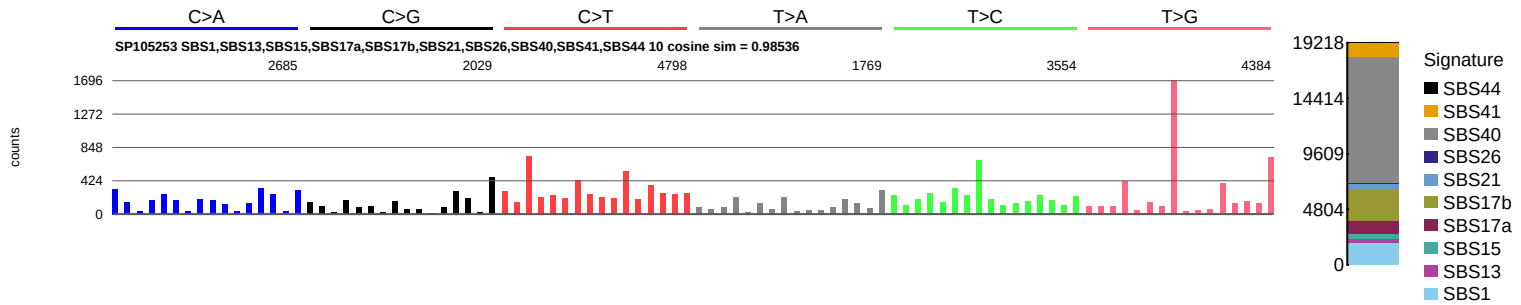
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98536
2687
2032
4794
1769
3553
4385
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.98536
2685
2029
4798
1769
3554
4384
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
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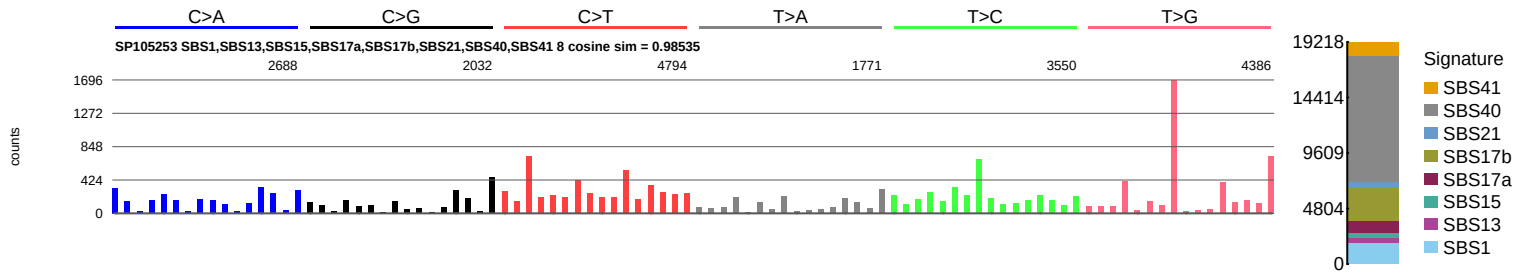
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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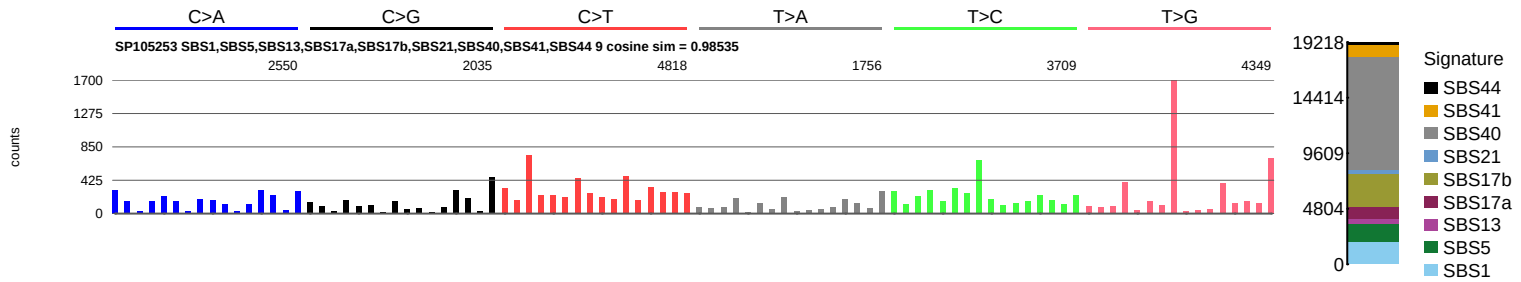
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98535
2688
2032
4794
1771
3550
4386
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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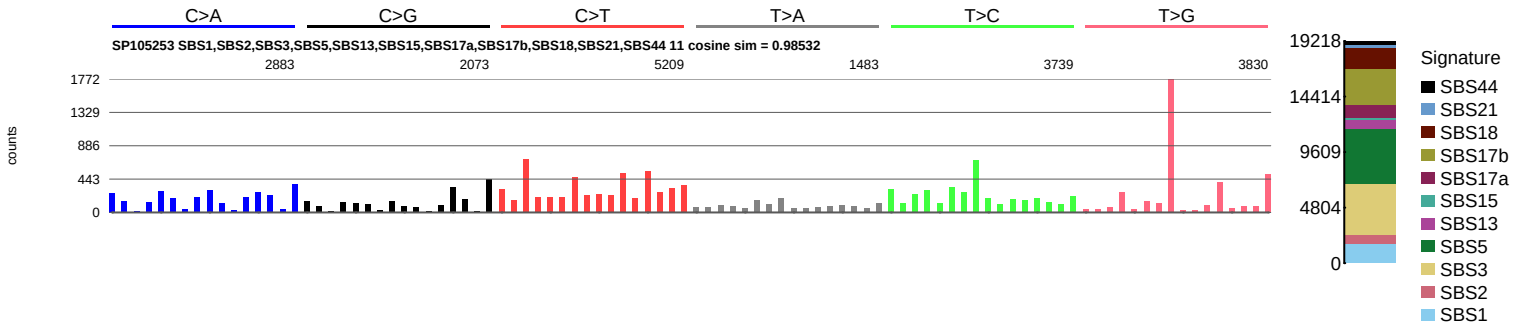
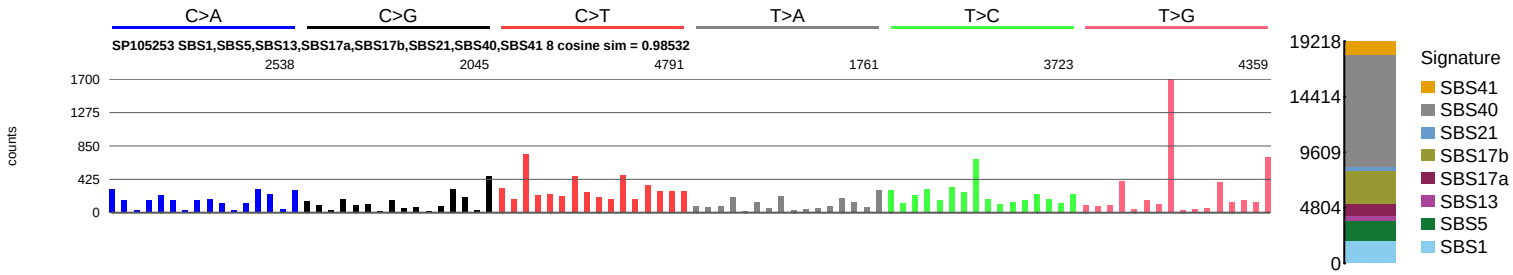
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.98535
2550
2035
4818
1756
3709
4349
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 11 cosine sim = 0.98532
2883
2073
5209
1483
3739
3830
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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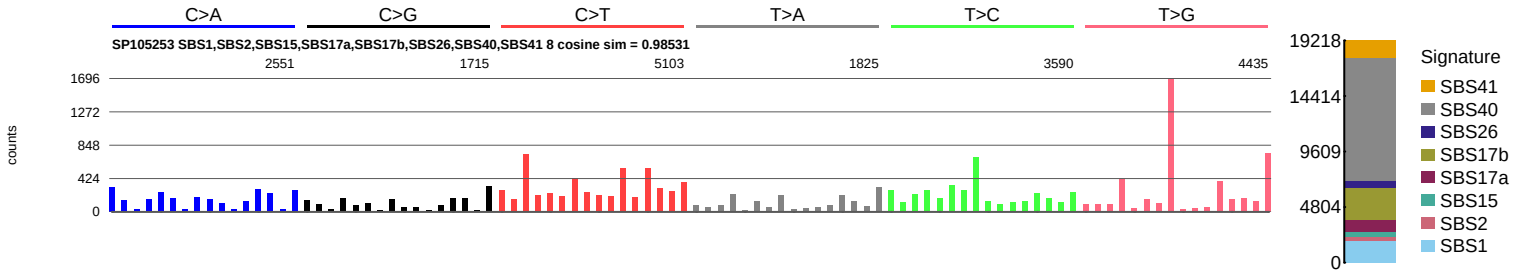
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98532
2538
2045
4791
1761
3723
4359
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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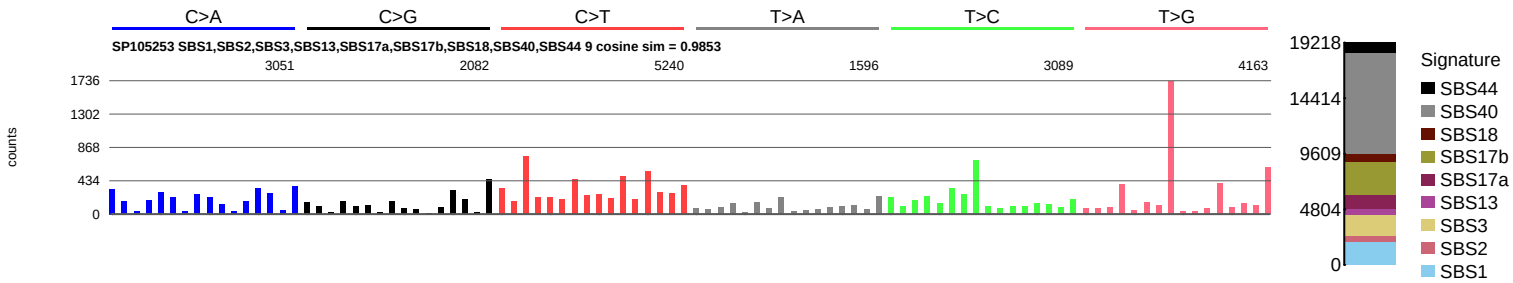
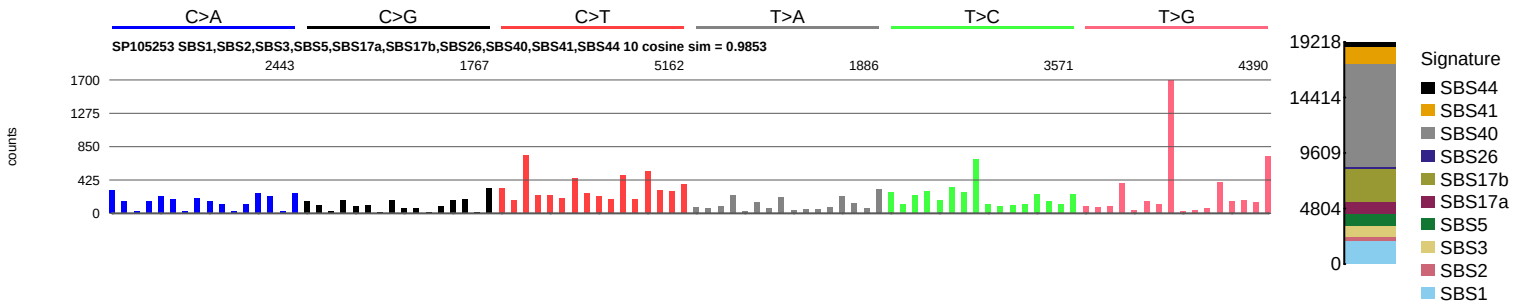
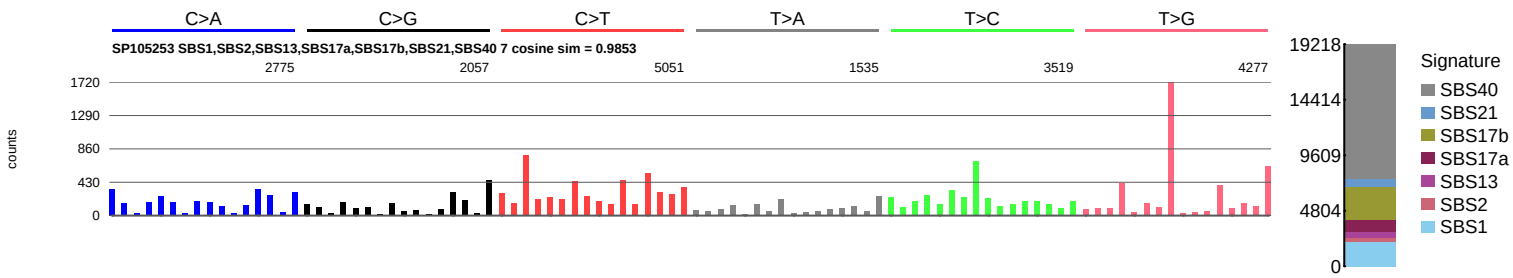
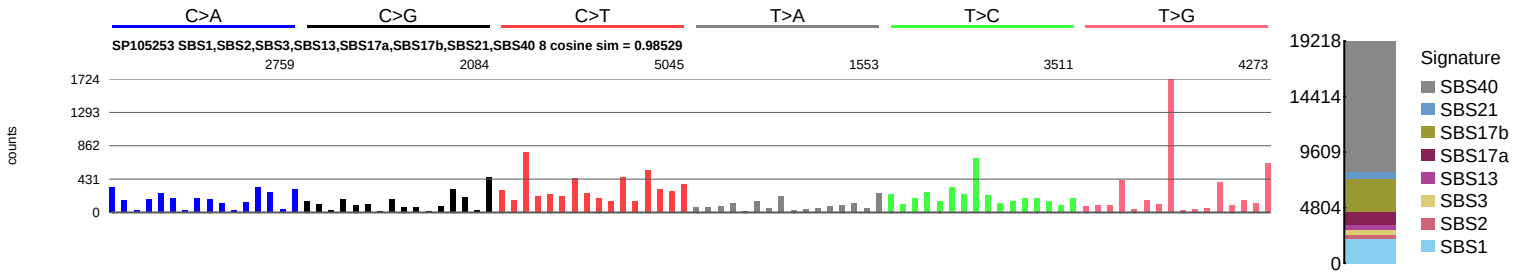
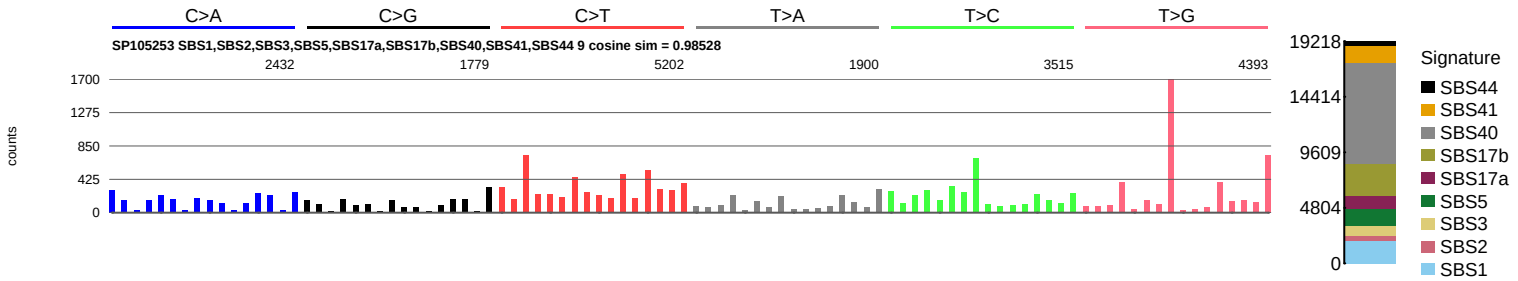
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98531
2551
1715
5103
1825
3590
4435
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9853
3051
2082
5240
1596
3089
4163
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.9853
2443
1767
5162
1886
3571
4390
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9853
2775
2057
5051
1535
3519
4277
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98529
2759
2084
5045
1553
3511
4273
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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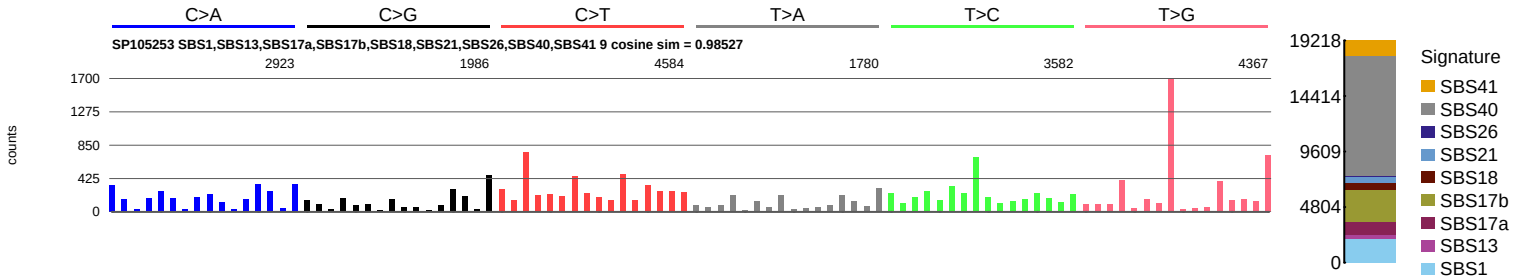
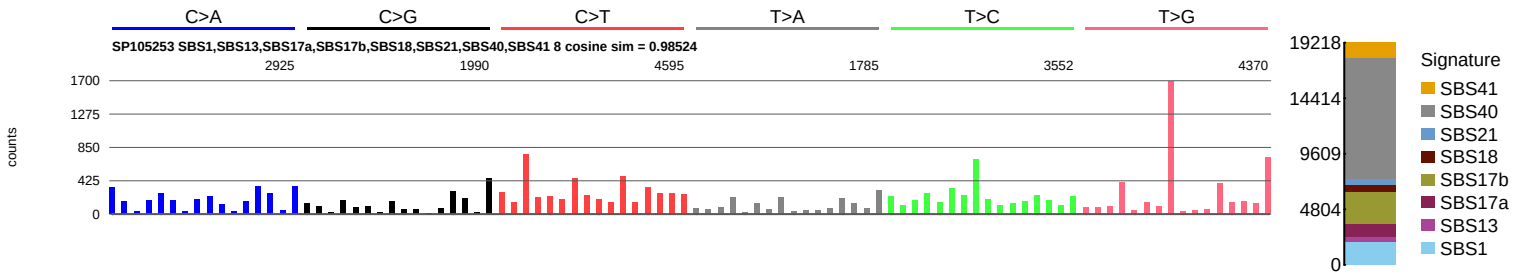
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98528
2432
1779
5202
1900
3515
4393
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.98527
2923
1986
4584
1780
3582
4367
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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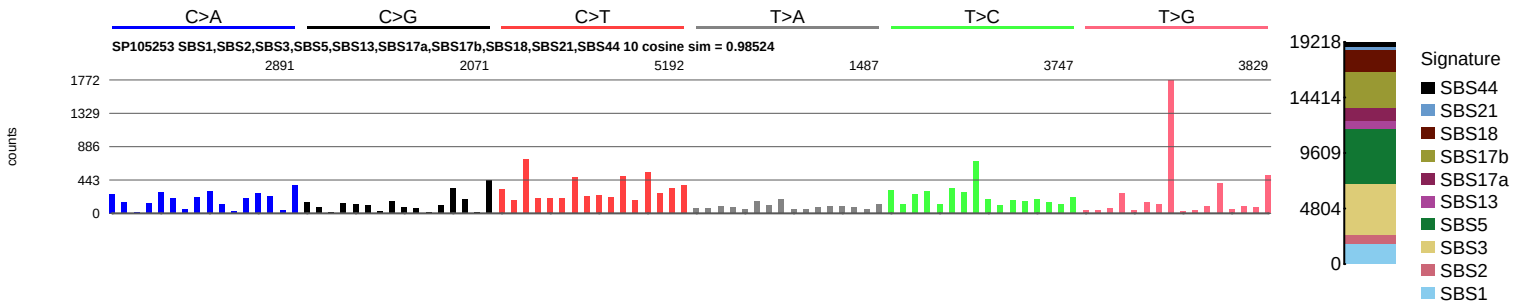
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98524
2925
1990
4595
1785
3552
4370
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98524
2891
2071
5192
1487
3747
3829
counts
1772
1329
886
443
0
19218
14414
9609
4804
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Signature
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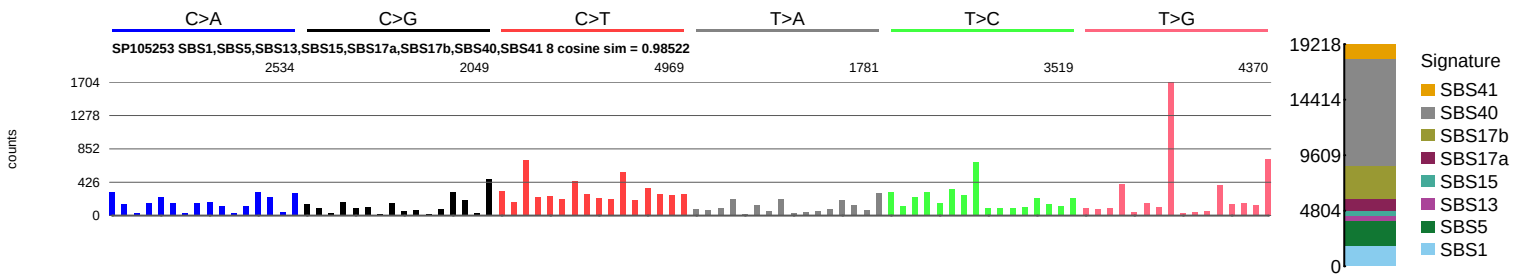
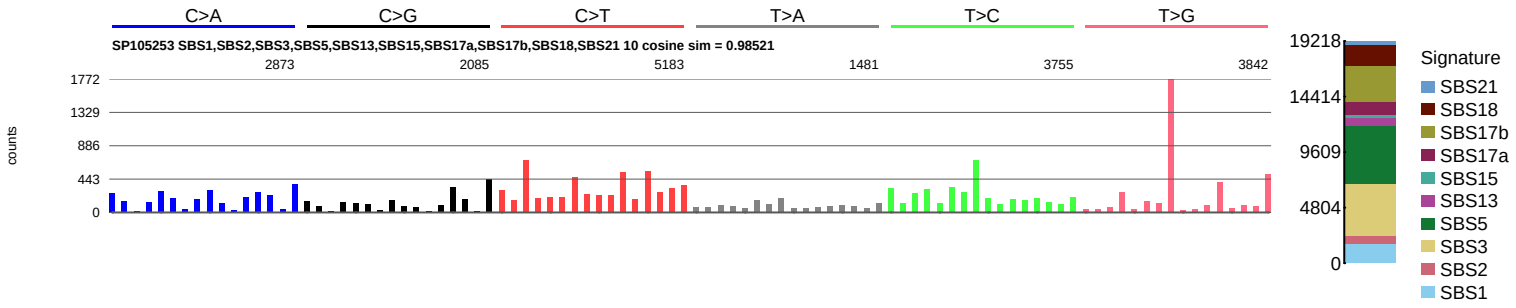
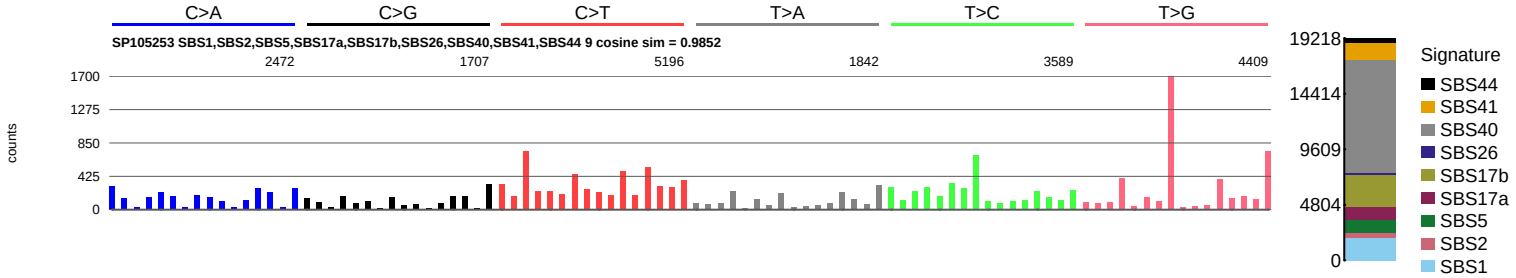
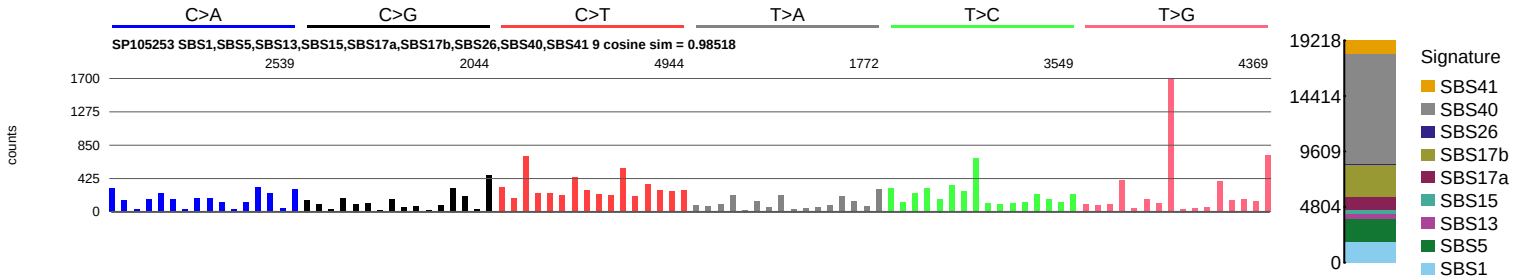
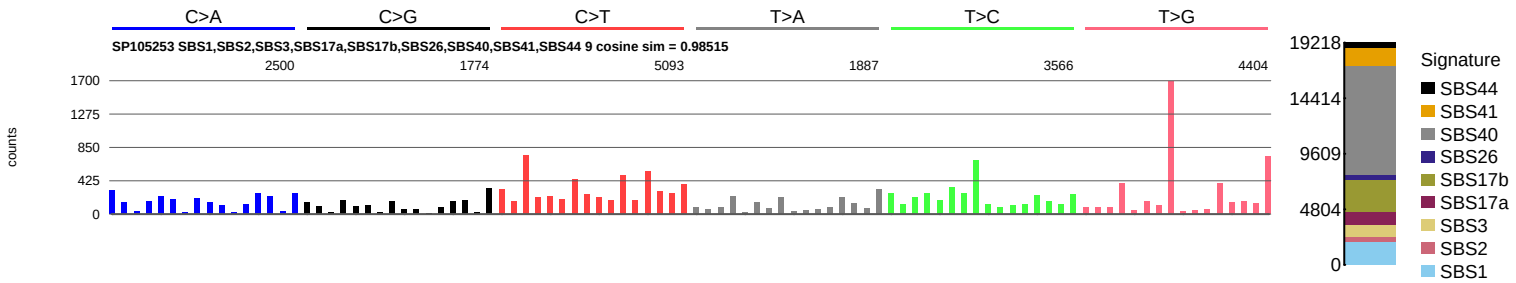
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

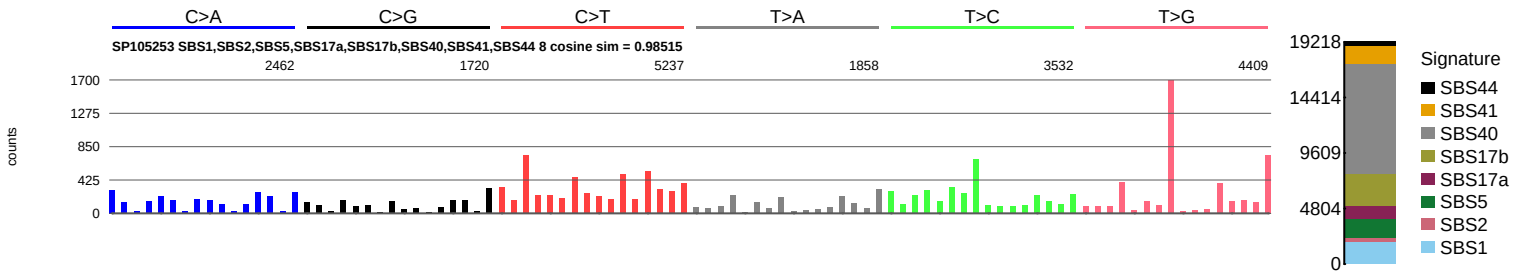

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98522
2534
2049
4969
1781
3519
4370
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

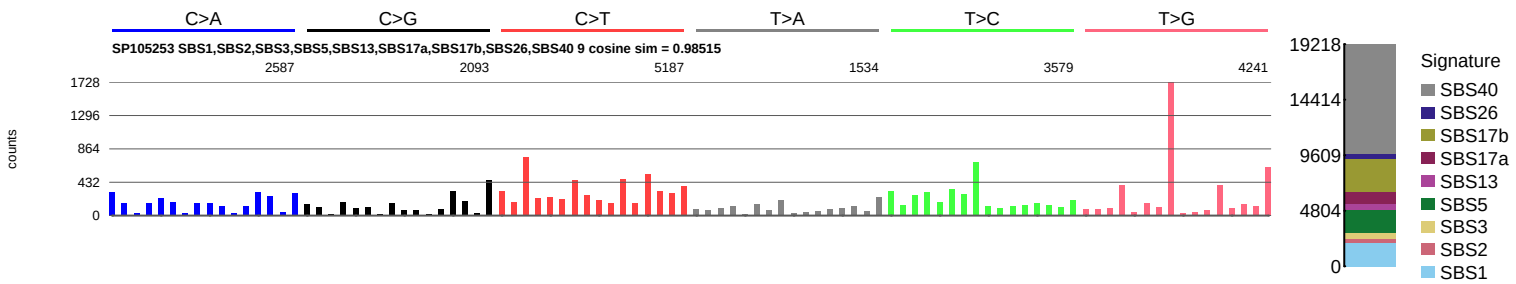

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98521
2873
2085
5183
1481
3755
3842
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.9852
2472
1707
5196
1842
3589
4409
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98518
2539
2044
4944
1772
3549
4369
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98515
2500
1774
5093
1887
3566
4404
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98515
2462
1720
5237
1858
3532
4409
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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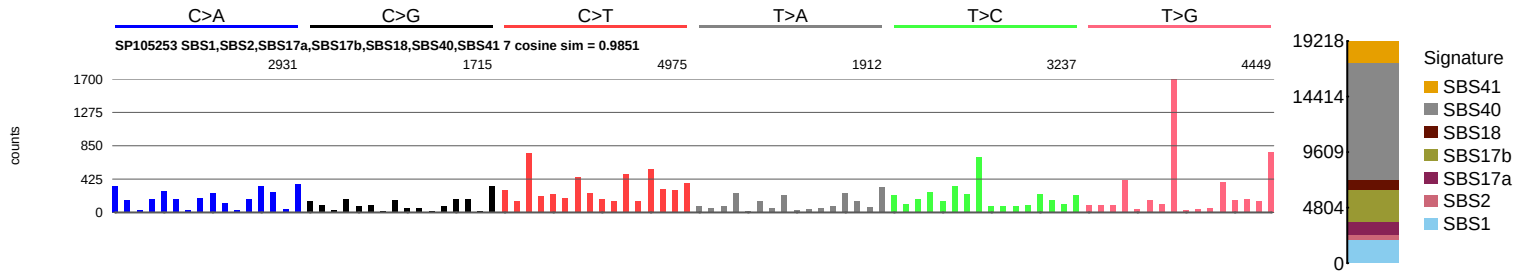
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.98515
2587
2093
5187
1534
3579
4241
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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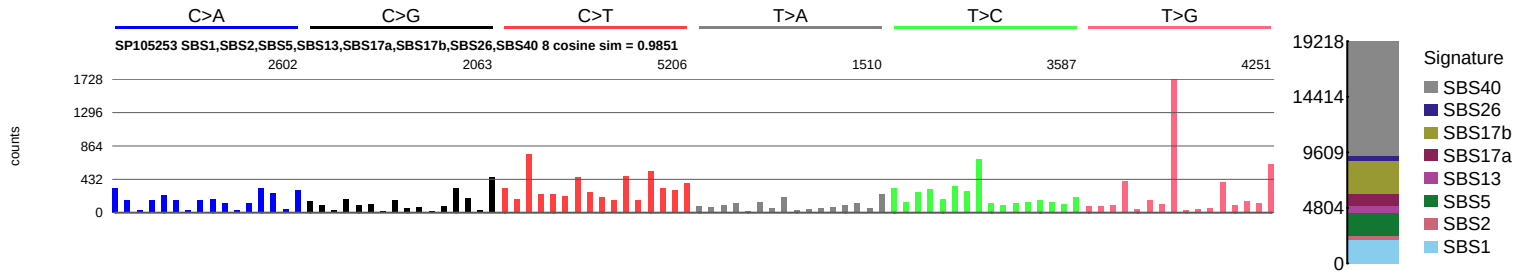
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.9851
2931
1715
4975
1912
3237
4449
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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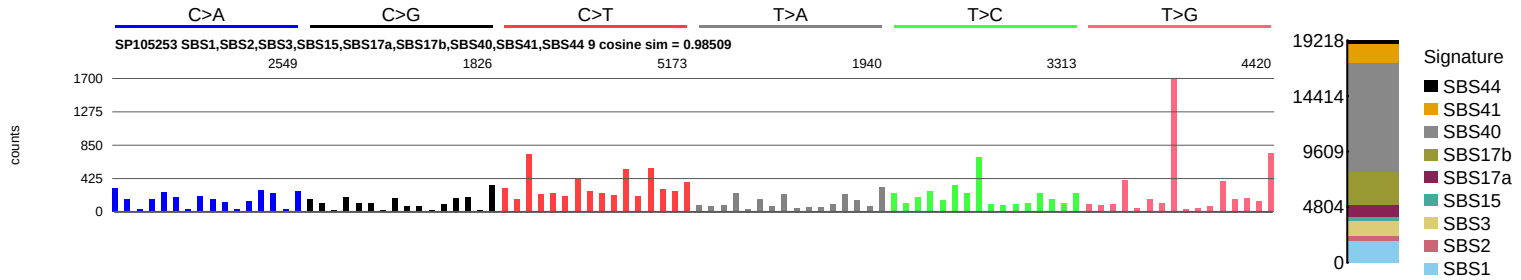
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.9851
2602
2063
5206
1510
3587
4251
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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C>T
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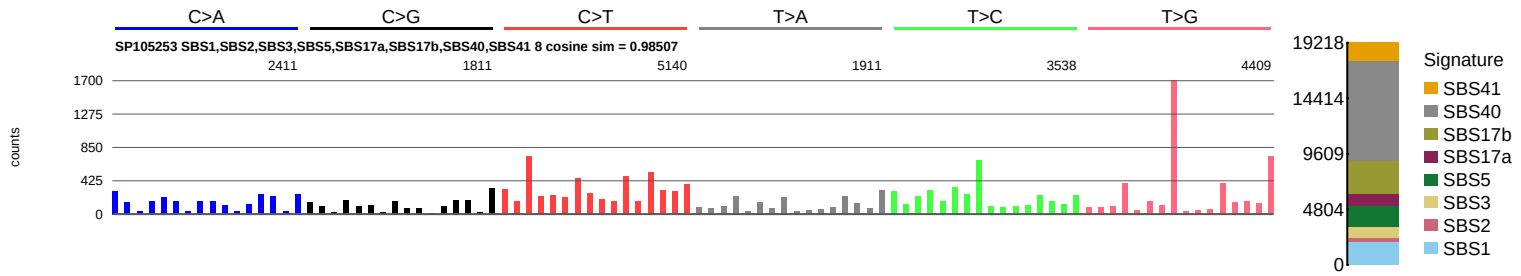
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 9 cosine sim = 0.98509
2549
1826
5173
1940
3313
4420
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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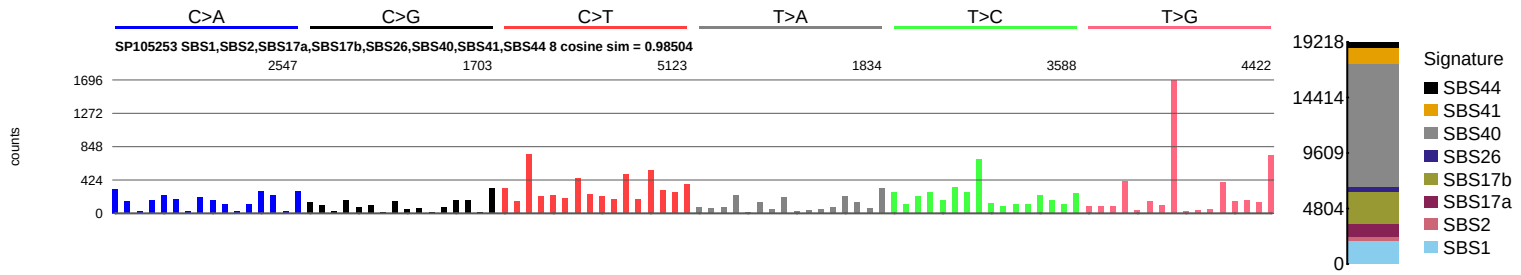
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98507
2411
1811
5140
1911
3538
4409
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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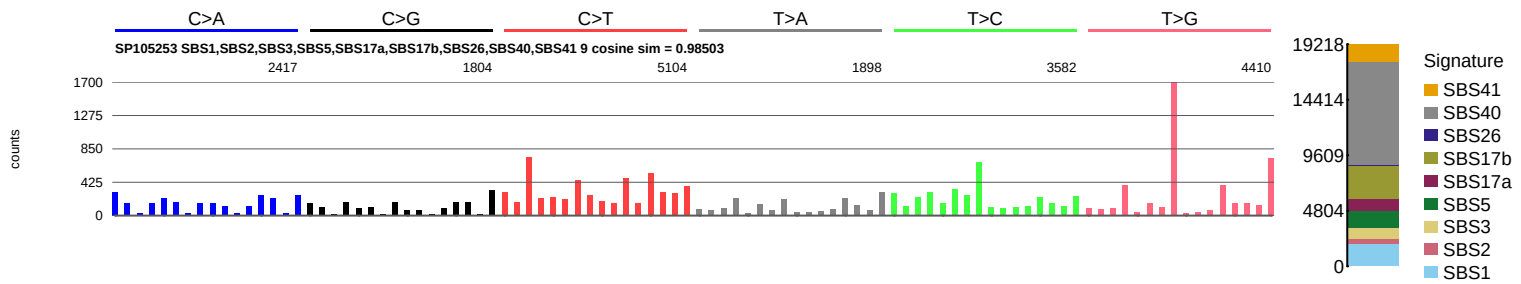
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98504
2547
1703
5123
1834
3588
4422
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1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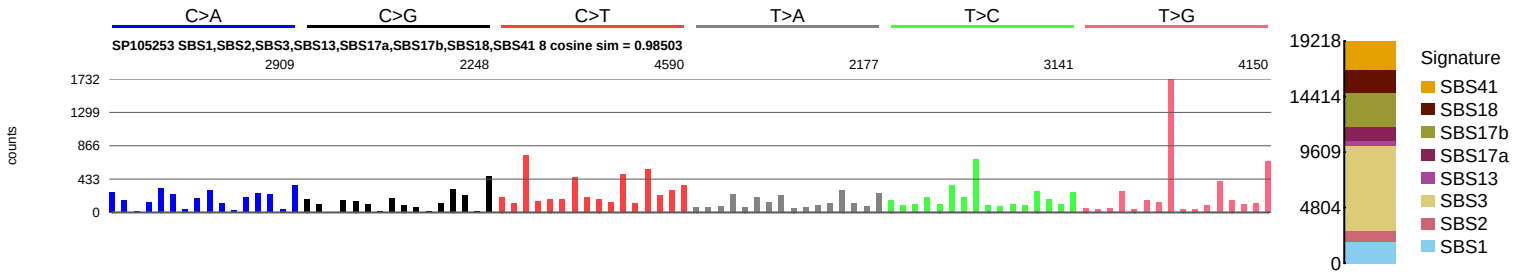
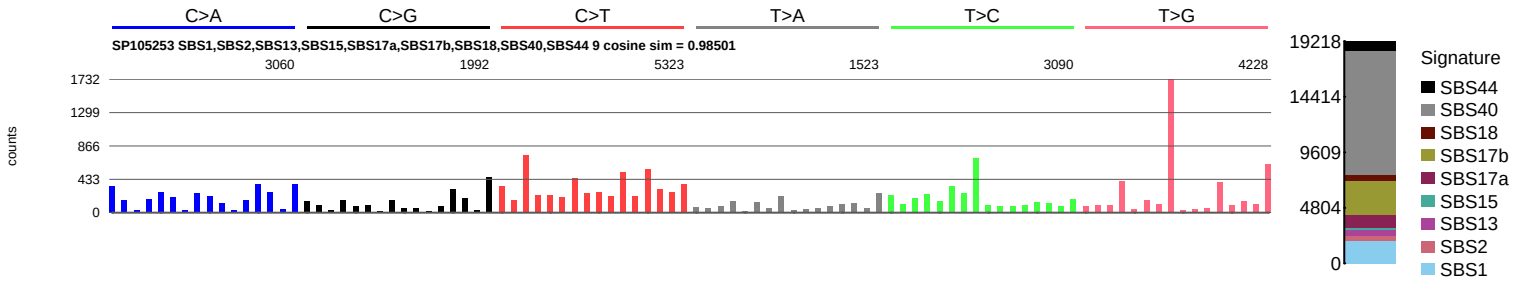
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 9 cosine sim = 0.98503
2417
1804
5104
1898
3582
4410
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98503
2909
2248
4590
2177
3141
4150
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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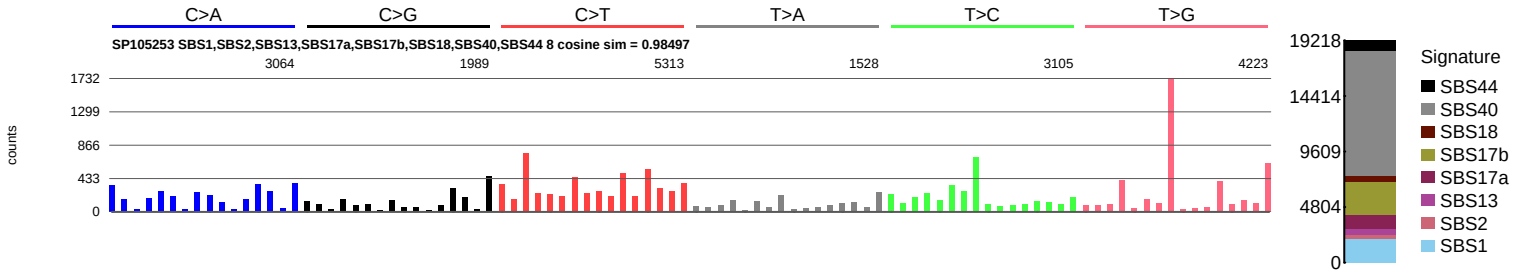
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98501
3060
1992
5323
1523
3090
4228
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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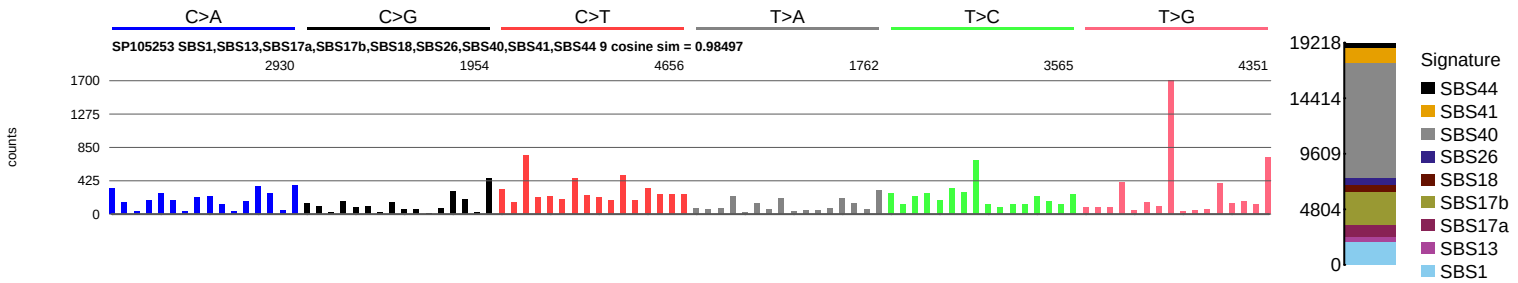
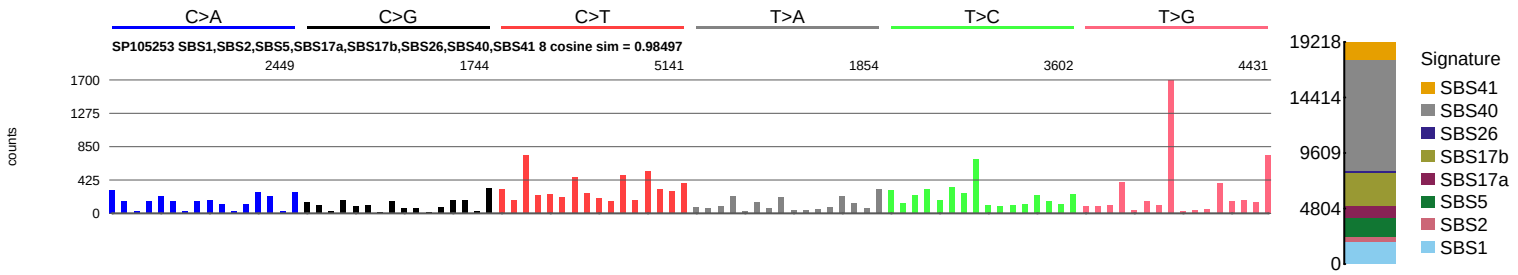
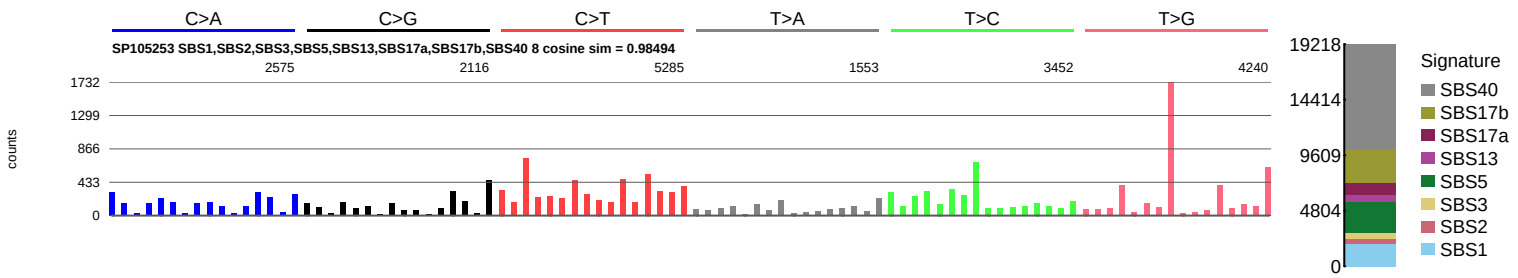
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98497
3064
1989
5313
1528
3105
4223
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98497
2930
1954
4656
1762
3565
4351
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98497
2449
1744
5141
1854
3602
4431
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1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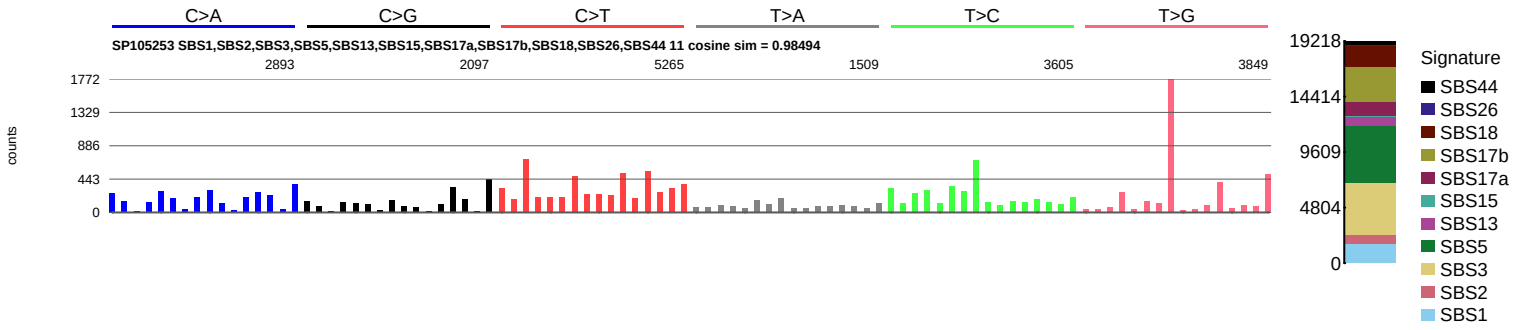
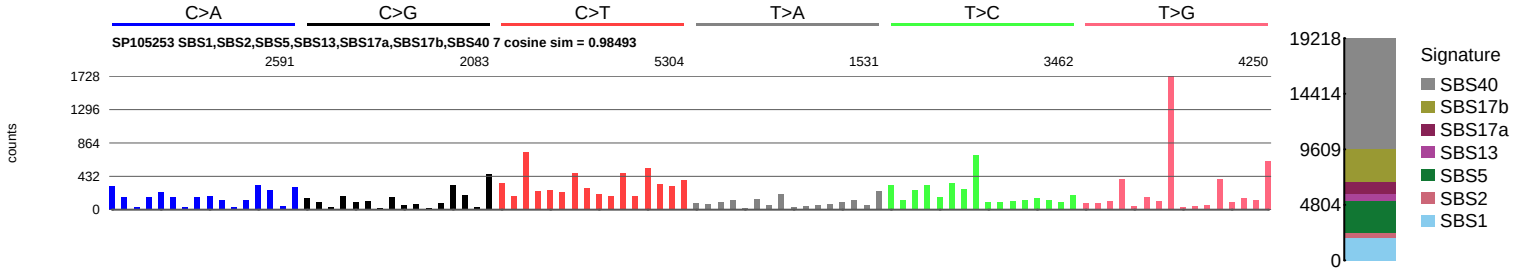
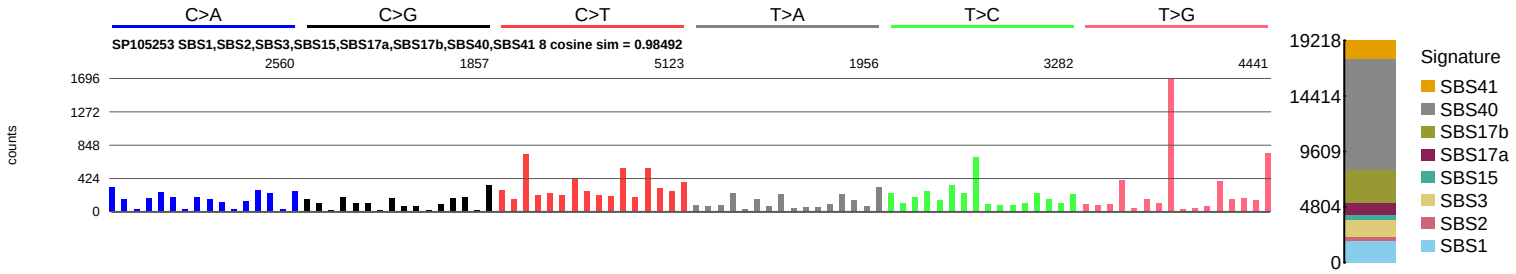
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98494
2575
2116
5285
1553
3452
4240
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 11 cosine sim = 0.98494
2893
2097
5265
1509
3605
3849
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98493
2591
2083
5304
1531
3462
4250
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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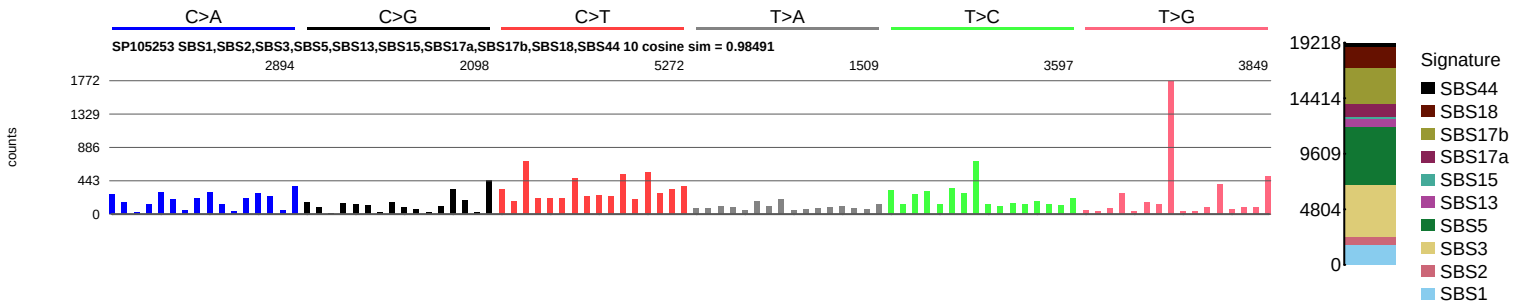
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 8 cosine sim = 0.98492
2560
1857
5123
1956
3282
4441
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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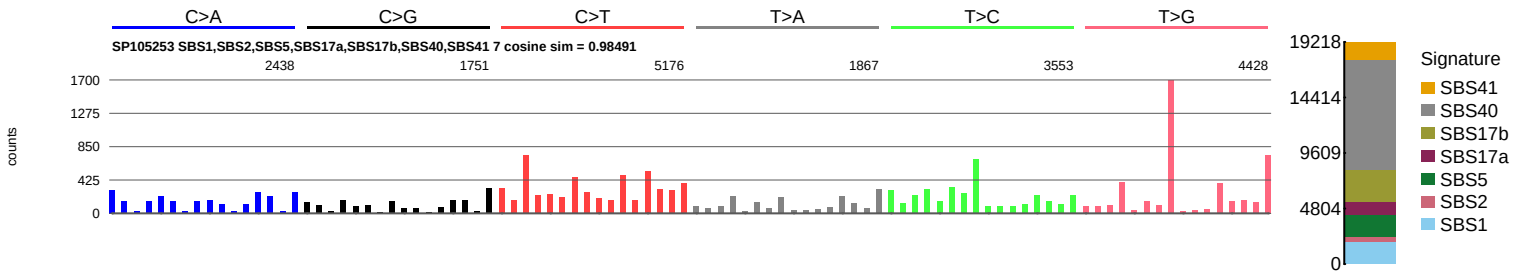
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 10 cosine sim = 0.98491
2894
2098
5272
1509
3597
3849
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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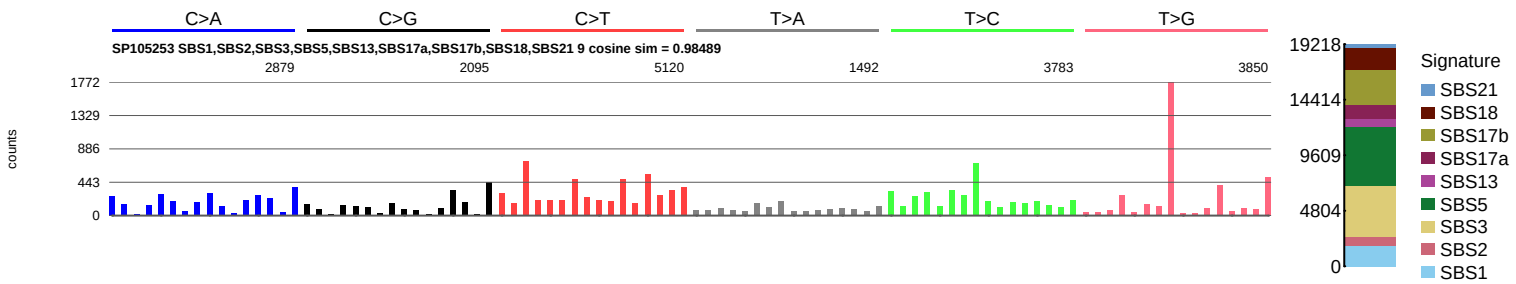
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98491
2438
1751
5176
1867
3553
4428
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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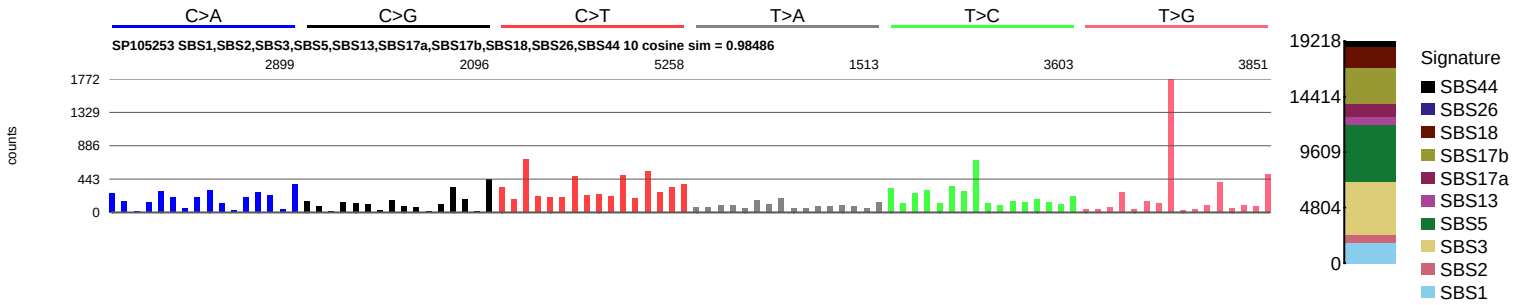
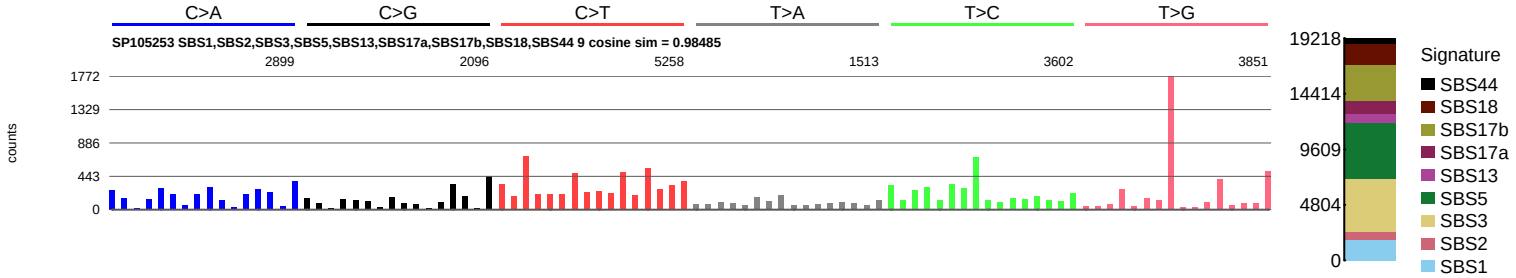
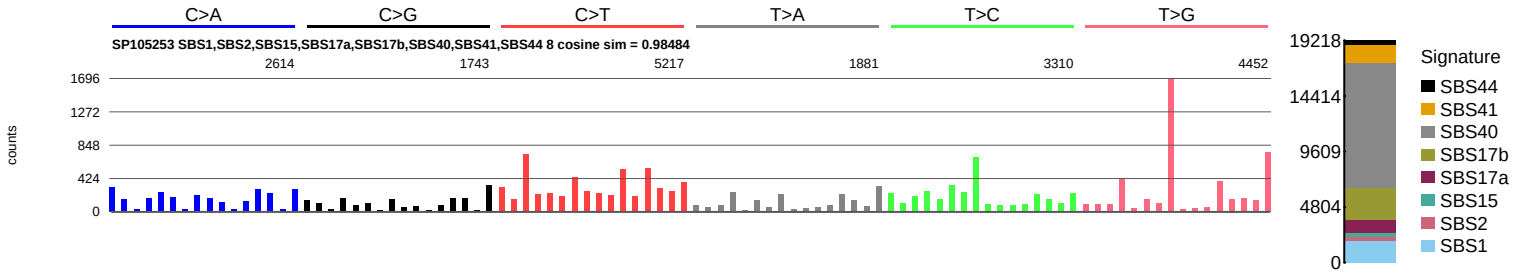
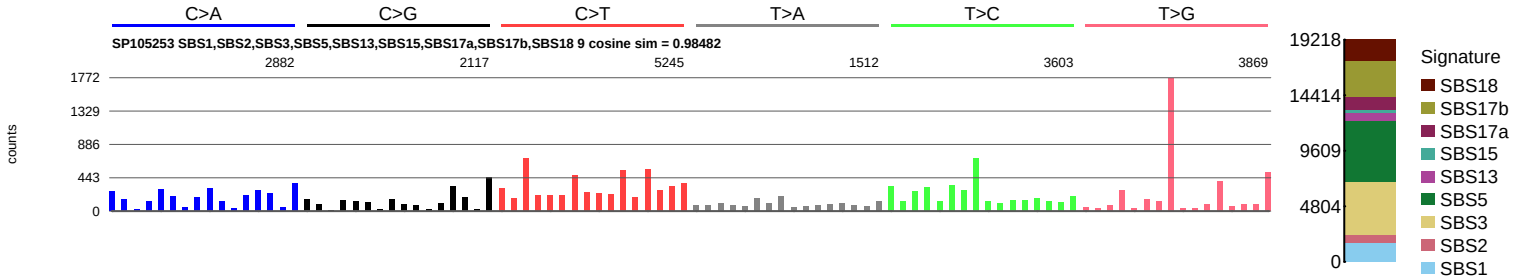
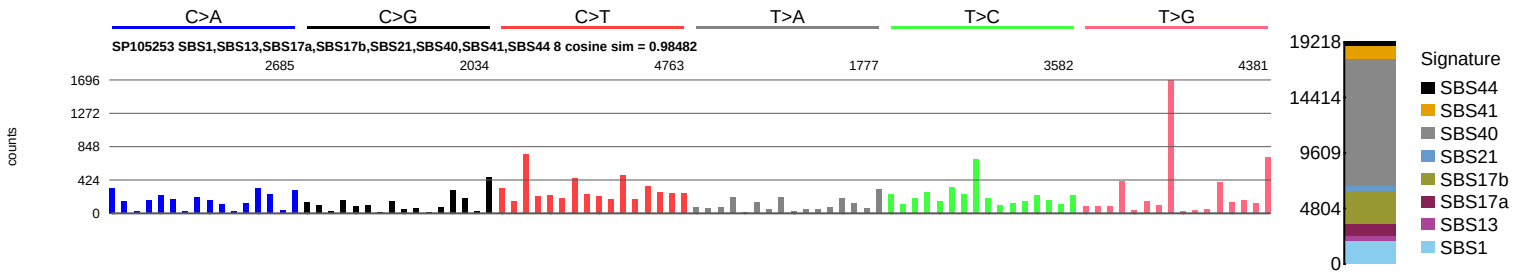
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98489
2879
2095
5120
1492
3783
3850
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

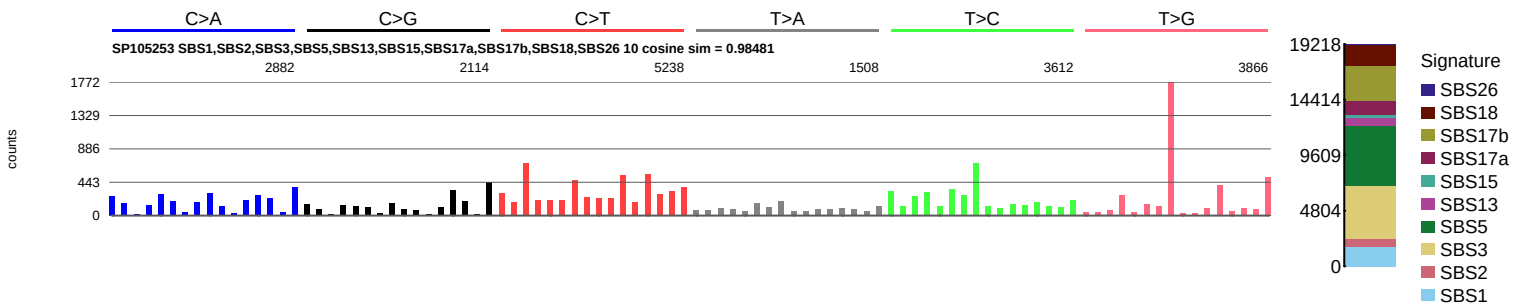

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98486
2899
2096
5258
1513
3603
3851
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

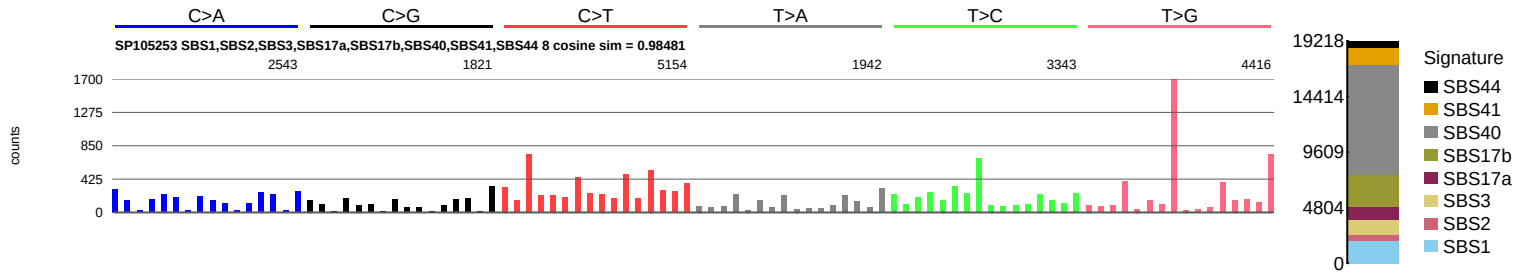

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98485
2899
2096
5258
1513
3602
3851
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98484
2614
1743
5217
1881
3310
4452
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98482
2882
2117
5245
1512
3603
3869
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98482
2685
2034
4763
1777
3582
4381
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98481
2882
2114
5238
1508
3612
3866
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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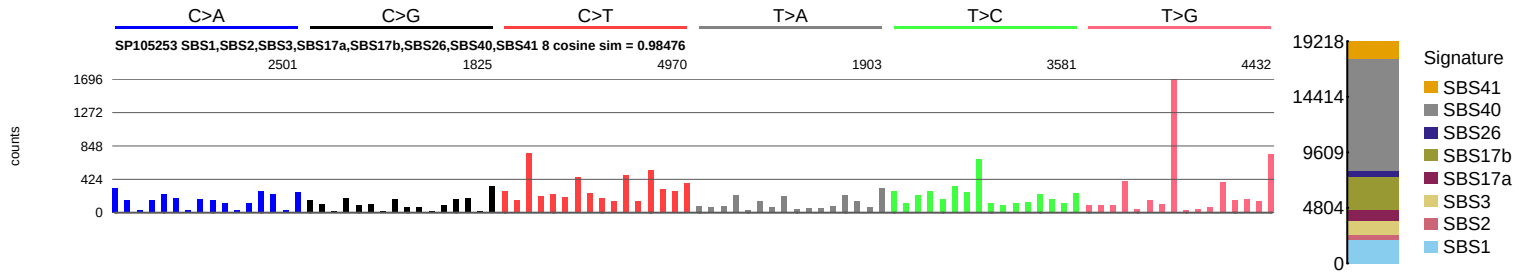
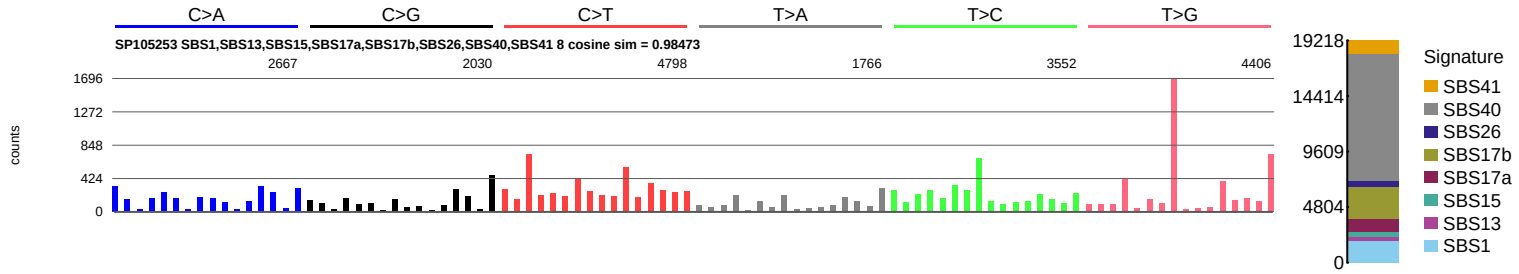
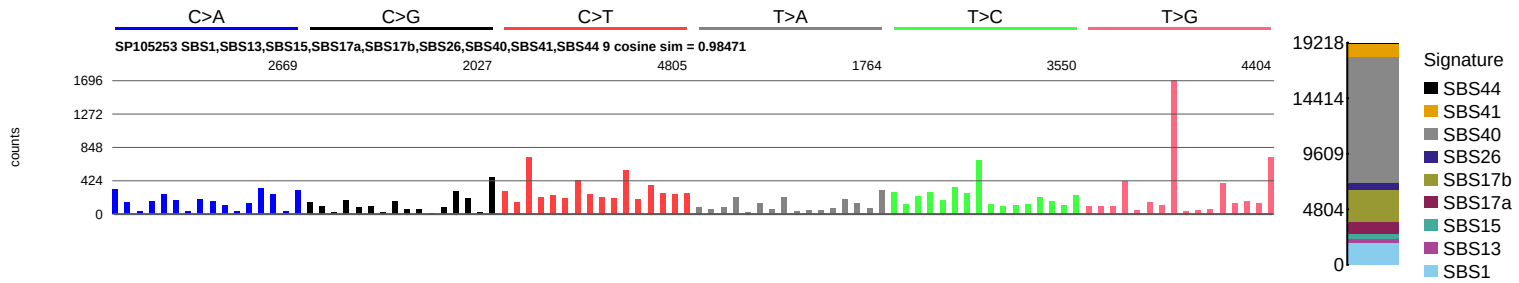
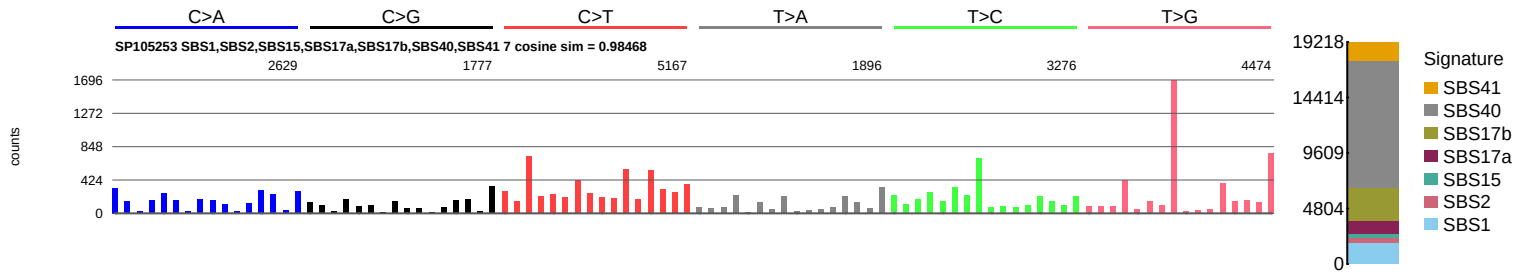
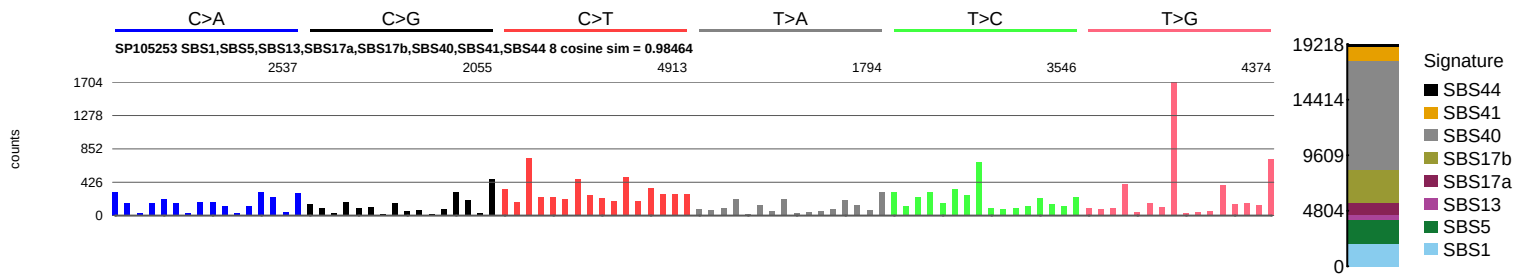
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98481
2543
1821
5154
1942
3343
4416
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

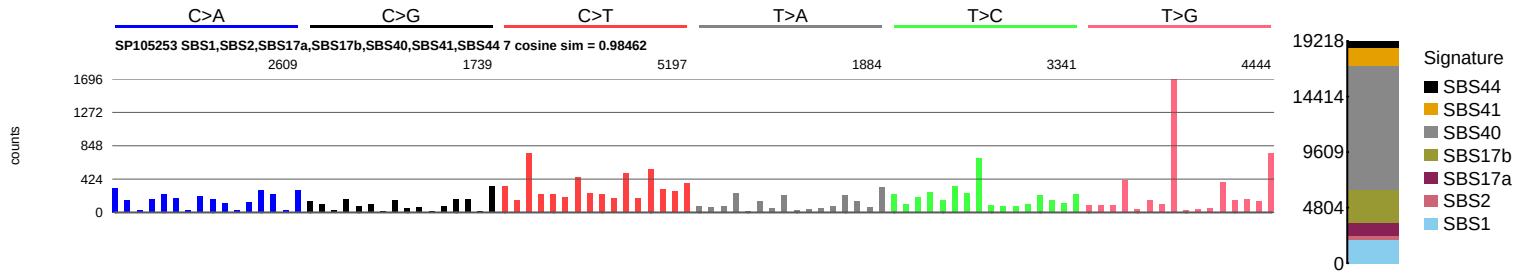

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98476
2501
1825
4970
1903
3581
4432
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

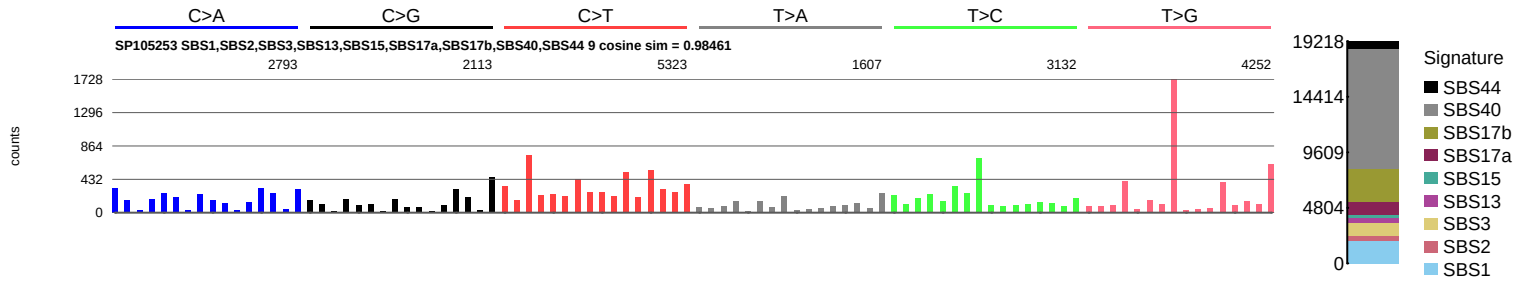

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98473
2667
2030
4798
1766
3552
4406
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

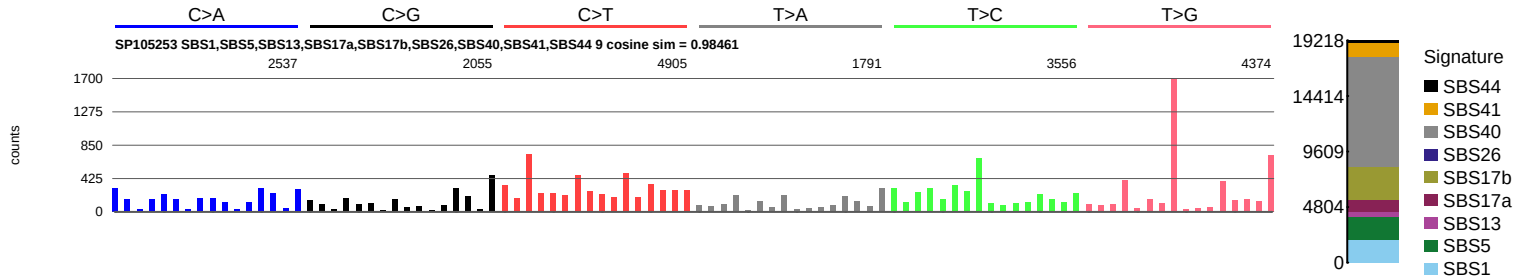

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98471
2669
2027
4805
1764
3550
4404
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

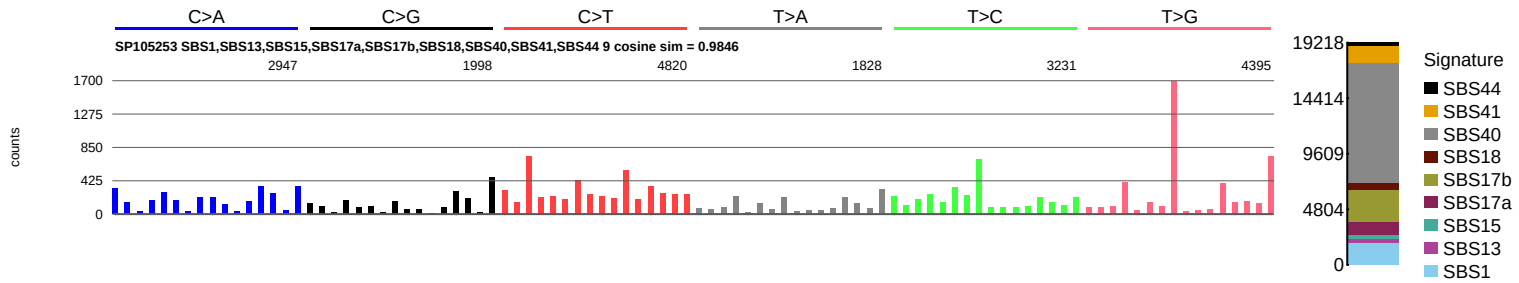

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98468
2629
1777
5167
1896
3276
4474
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1

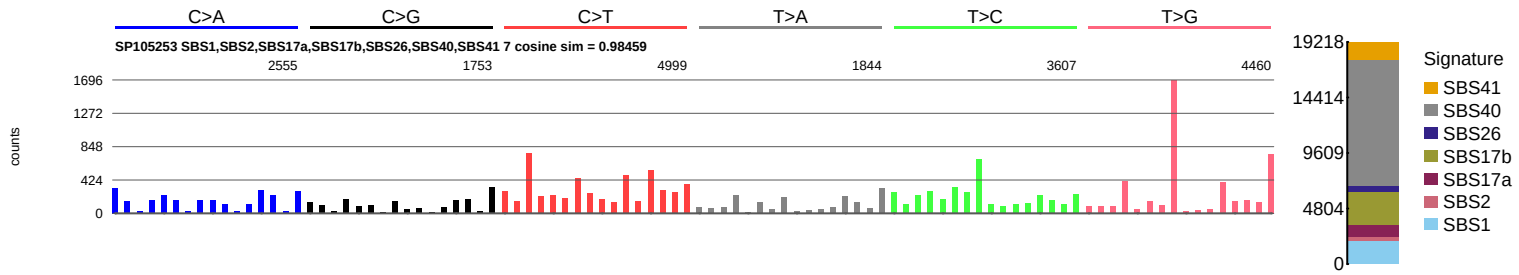

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98464
2537
2055
4913
1794
3546
4374
counts
1704
1278
852
426
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

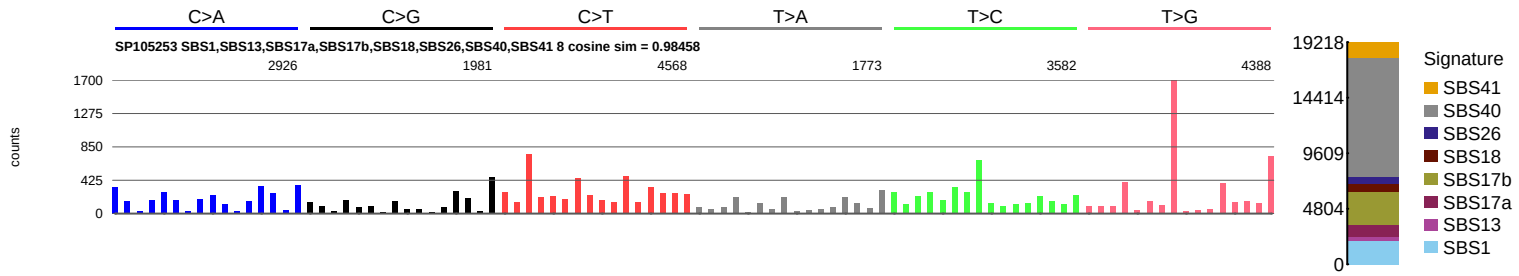

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98462
2609
1739
5197
1884
3341
4444
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1

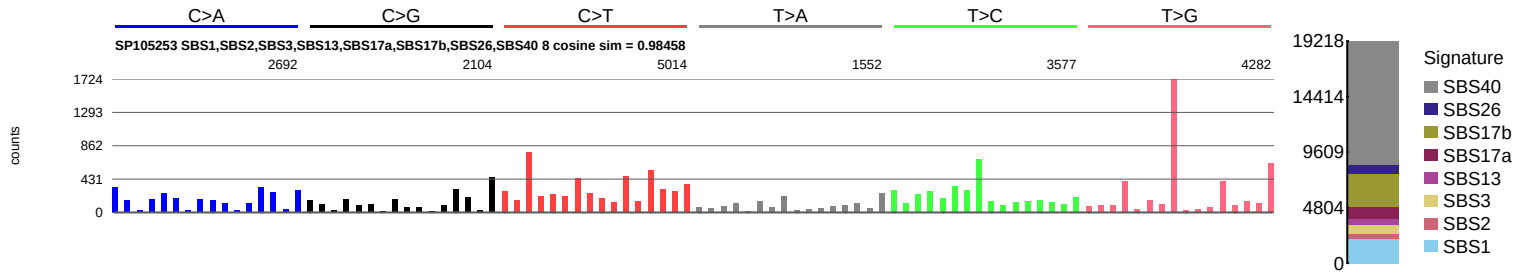

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98461
2793
2113
5323
1607
3132
4252
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

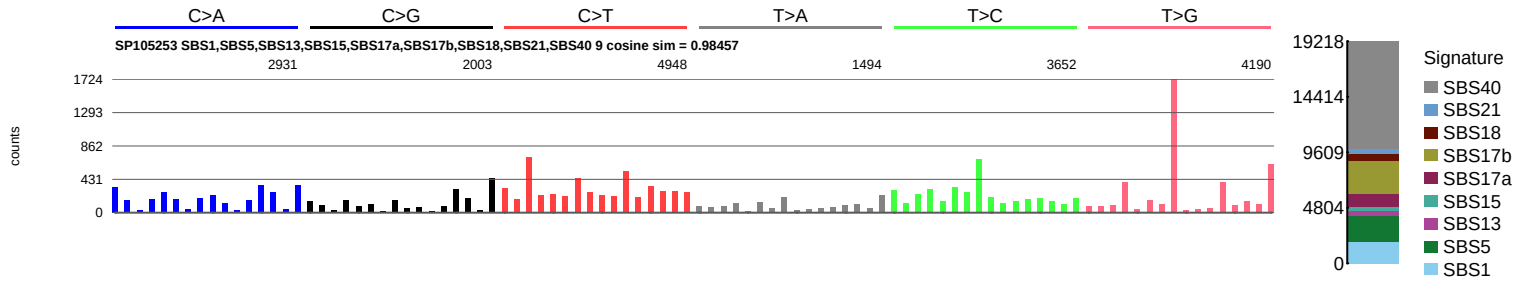

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.98461
2537
2055
4905
1791
3556
4374
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

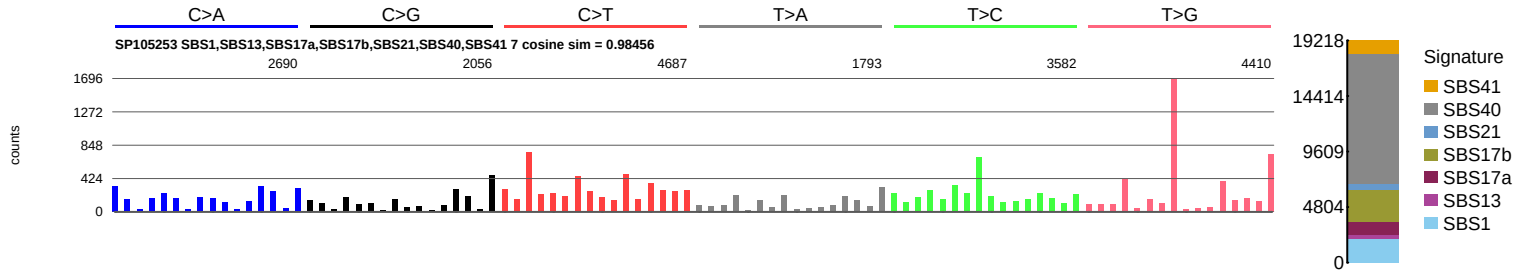

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.9846
2947
1998
4820
1828
3231
4395
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

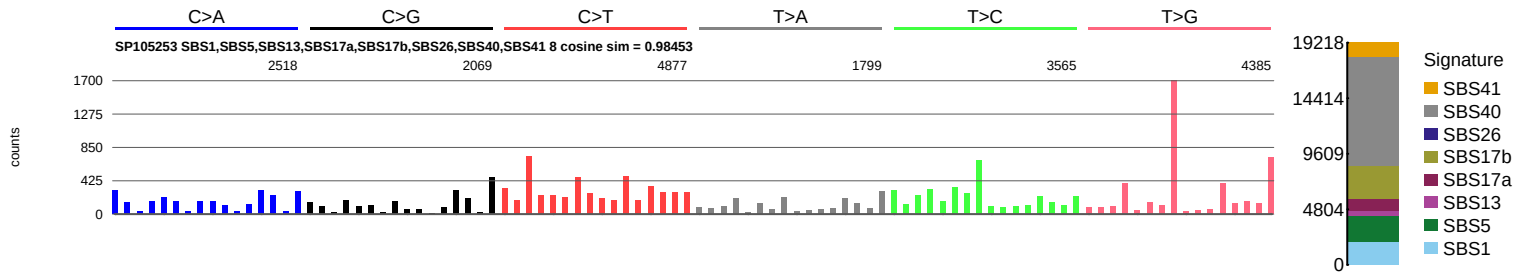

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98459
2555
1753
4999
1844
3607
4460
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1

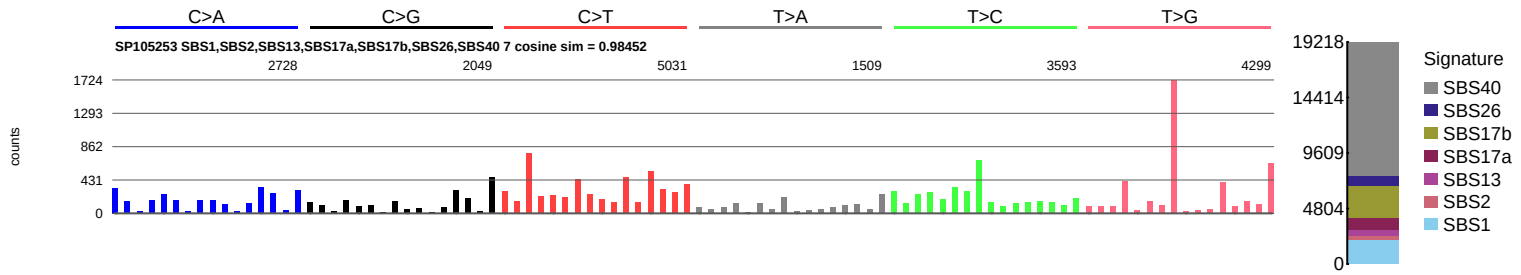

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98458
2926
1981
4568
1773
3582
4388
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

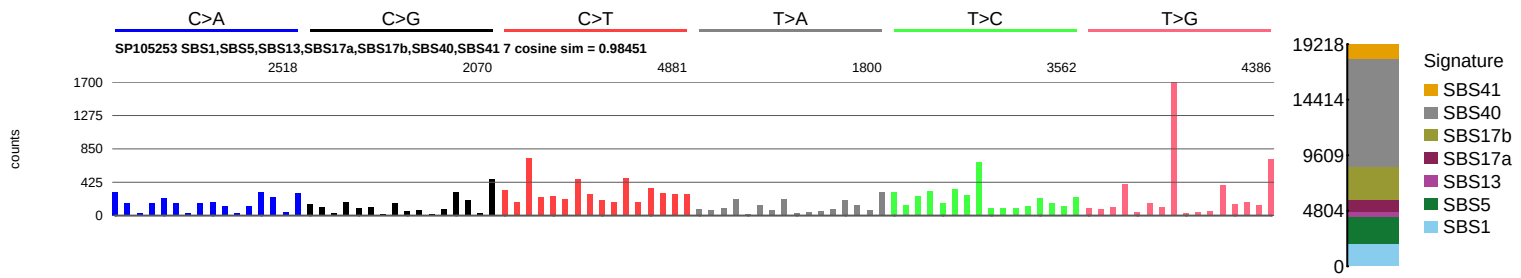

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98458
2692
2104
5014
1552
3577
4282
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

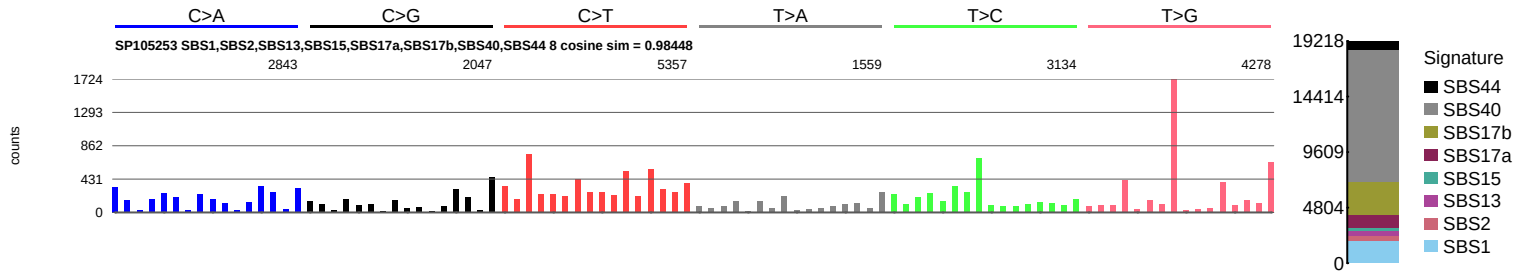

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98457
2931
2003
4948
1494
3652
4190
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

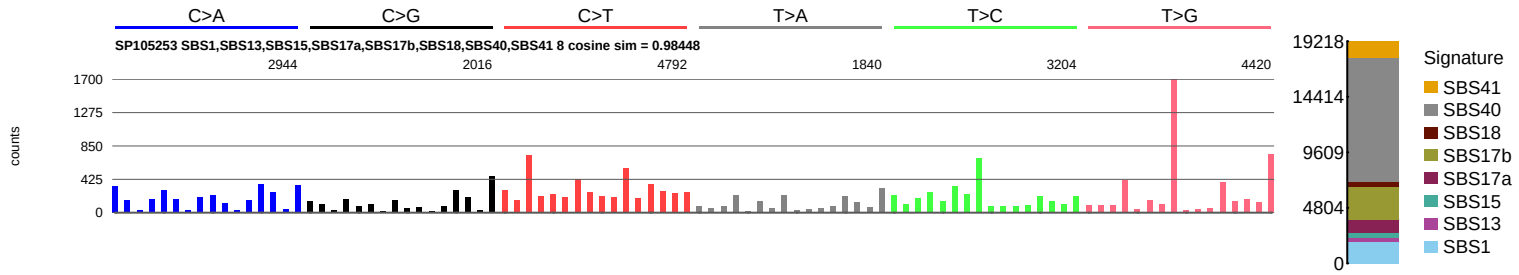

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98456
2690
2056
4687
1793
3582
4410
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

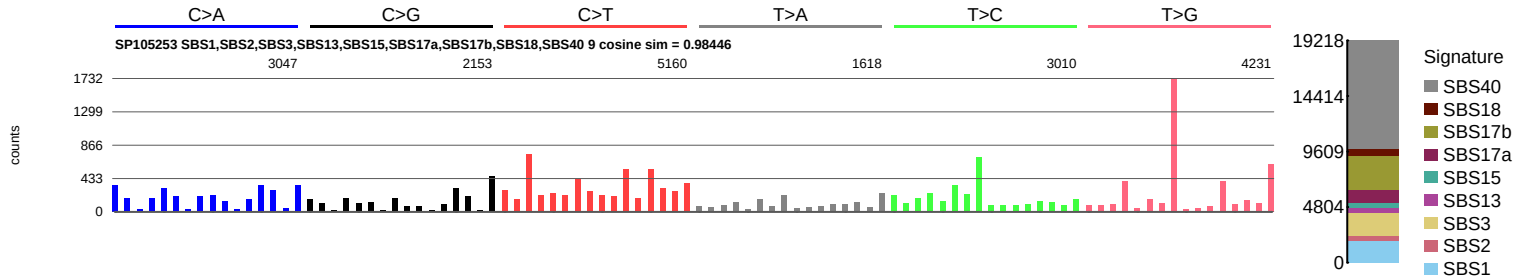

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98453
2518
2069
4877
1799
3565
4385
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

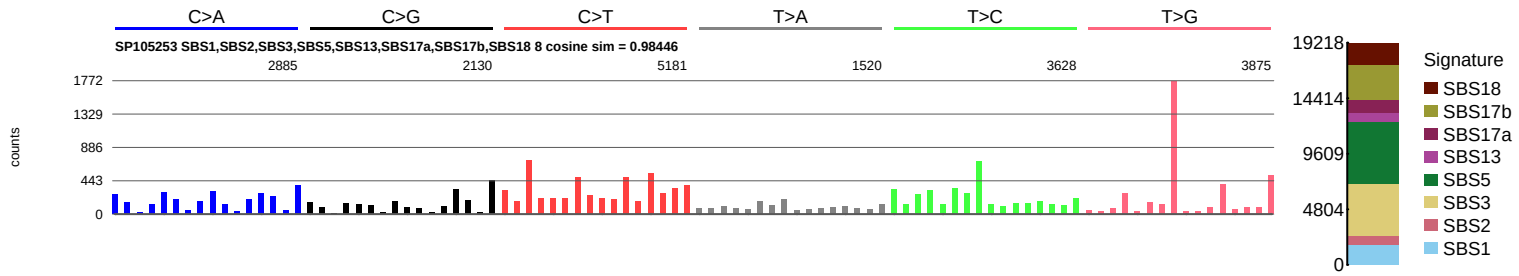

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98452
2728
2049
5031
1509
3593
4299
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS2
SBS1

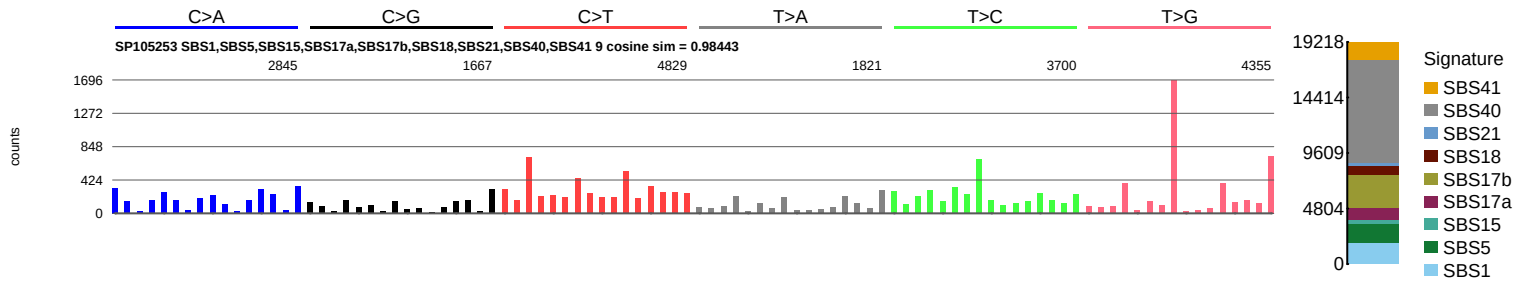

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98451
2518
2070
4881
1800
3562
4386
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

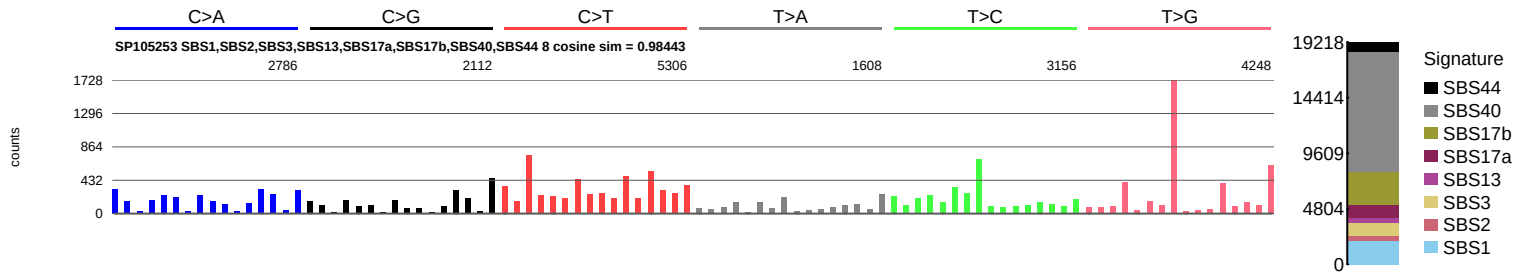

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98448
2843
2047
5357
1559
3134
4278
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

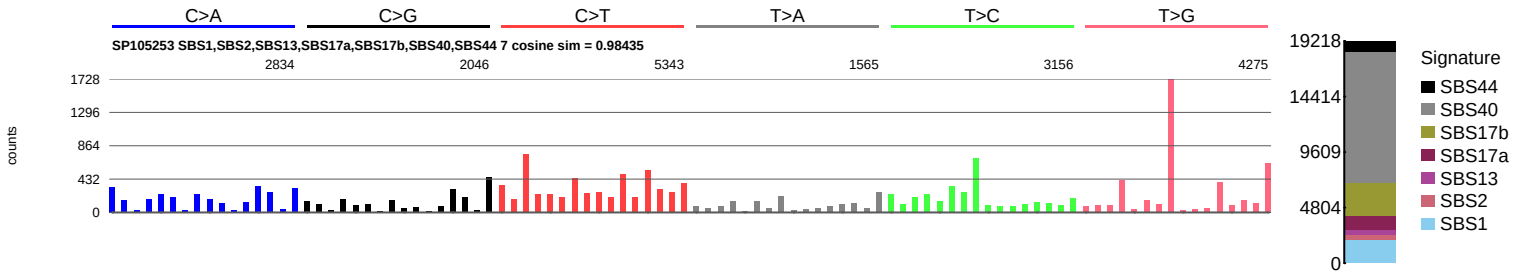

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.98448
2944
2016
4792
1840
3204
4420
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98446
3047
2153
5160
1618
3010
4231
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98446
2885
2130
5181
1520
3628
3875
counts
1772
1329
886
443
0
19218
14414
9609
4804
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.98443
2845
1667
4829
1821
3700
4355
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98443
2786
2112
5306
1608
3156
4248
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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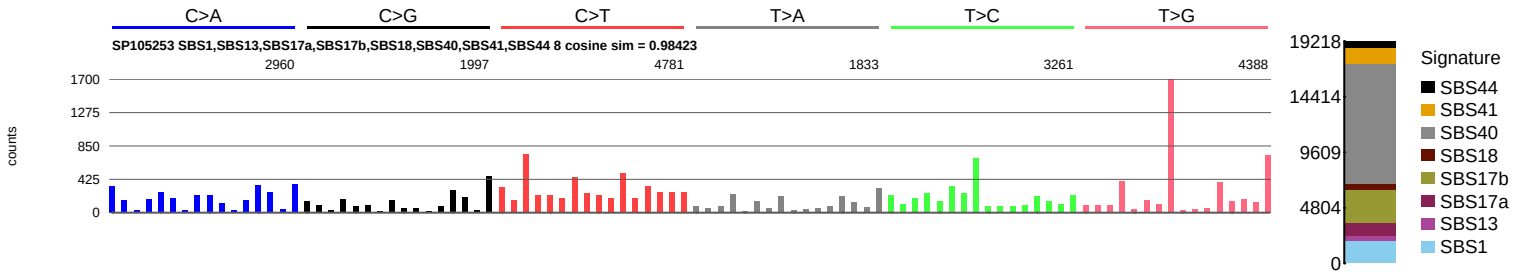
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98435
2834
2046
5343
1565
3156
4275
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
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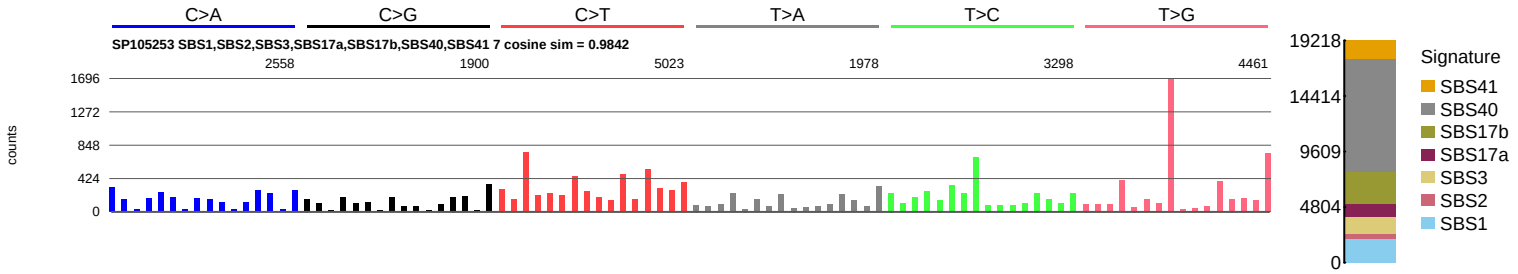
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98423
2960
1997
4781
1833
3261
4388
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
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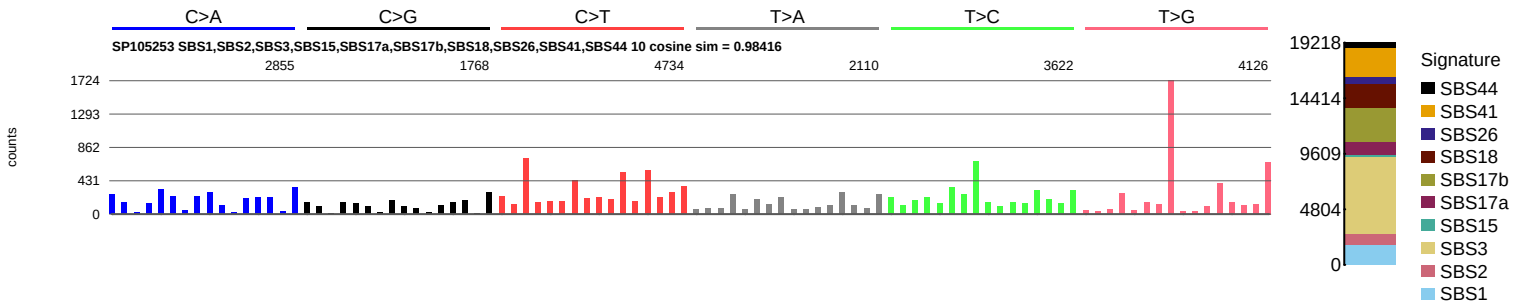
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.9842
2558
1900
5023
1978
3298
4461
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>T
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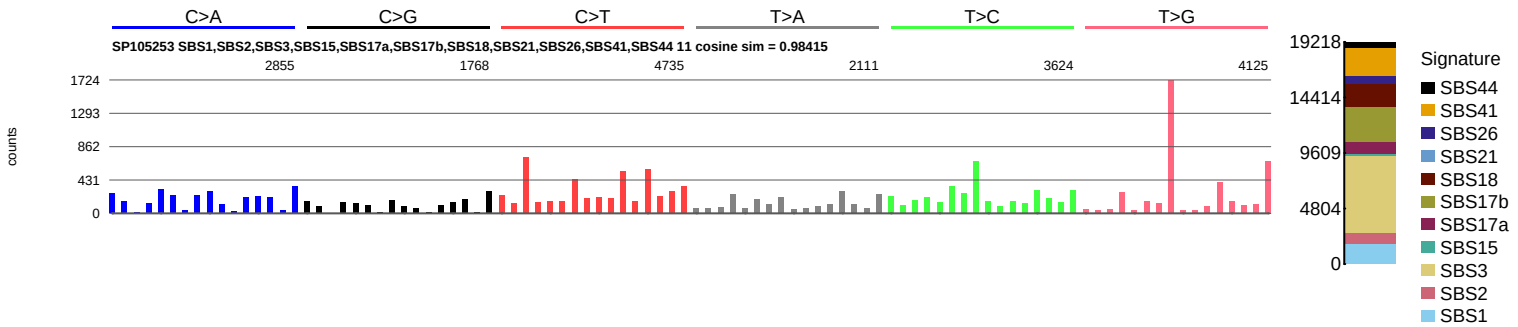
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98416
2855
1768
4734
2110
3622
4126
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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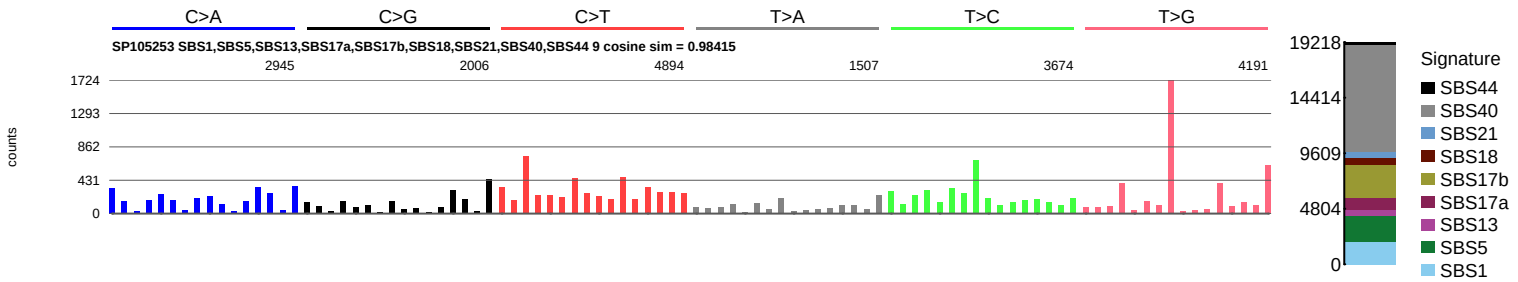
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98415
2855
1768
4735
2111
3624
4125
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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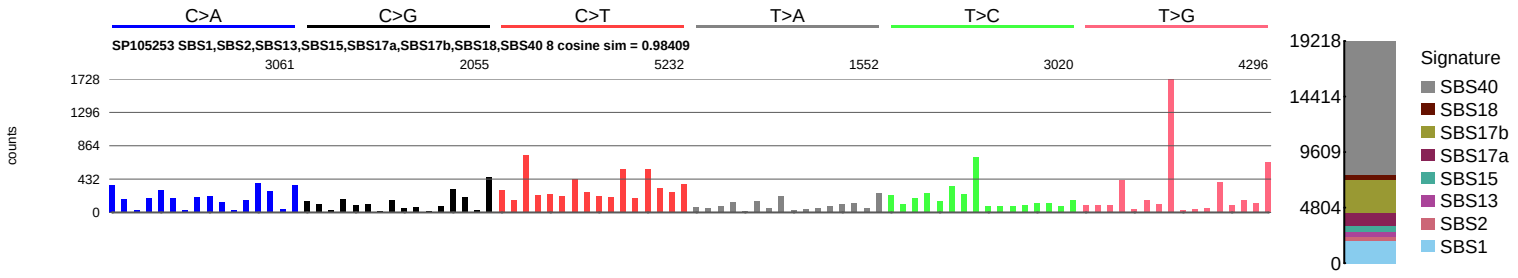
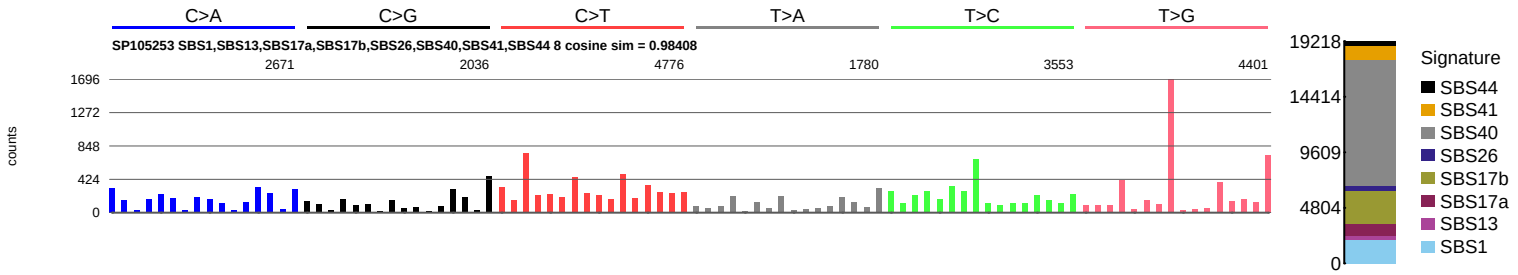
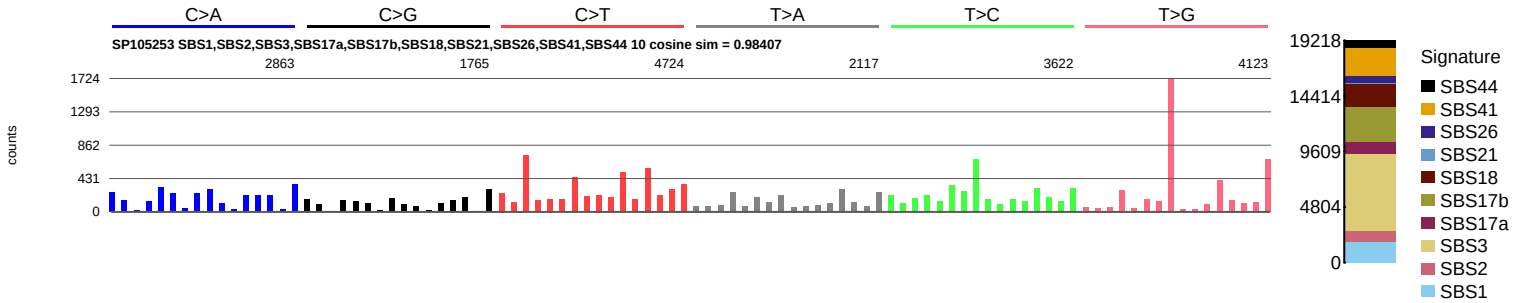
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98415
2945
2006
4894
1507
3674
4191
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98409
3061
2055
5232
1552
3020
4296
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98408
2671
2036
4776
1780
3553
4401
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1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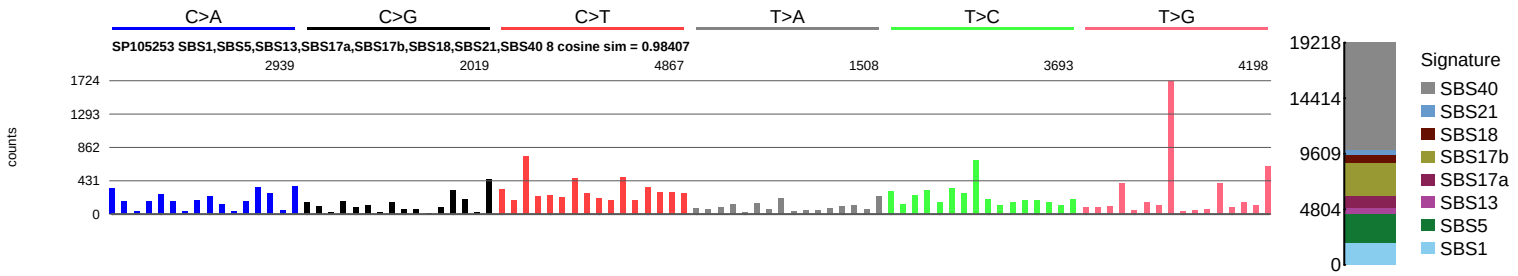
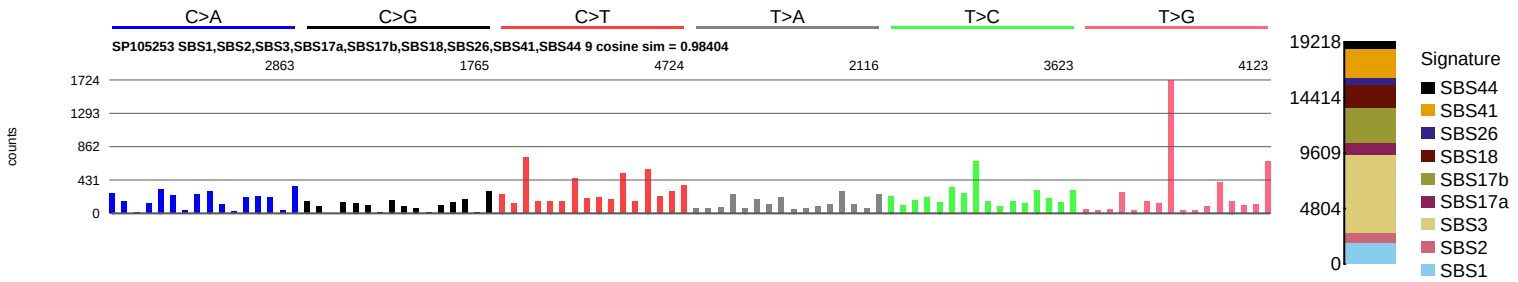
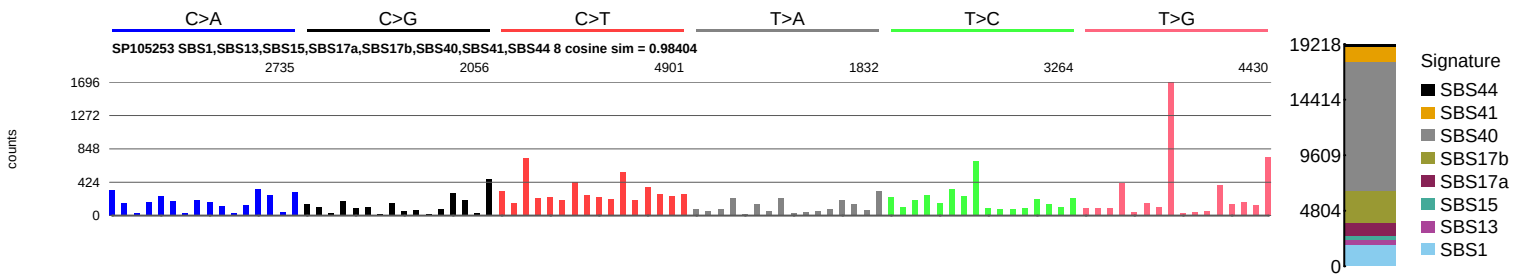
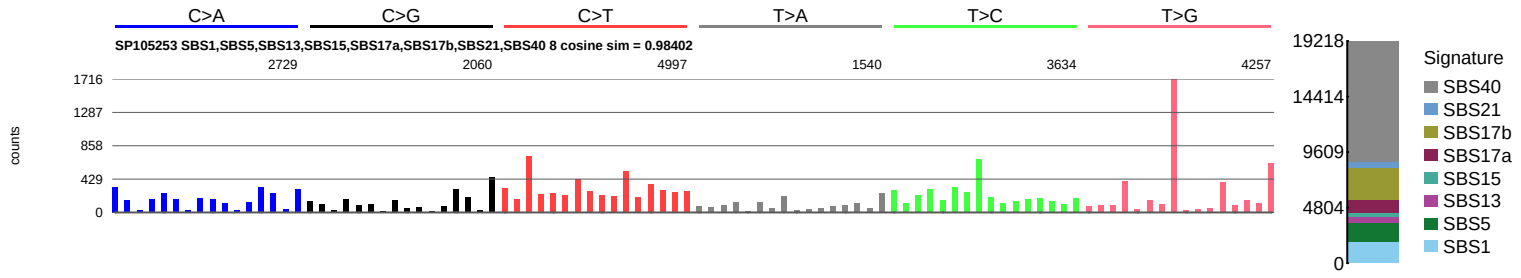
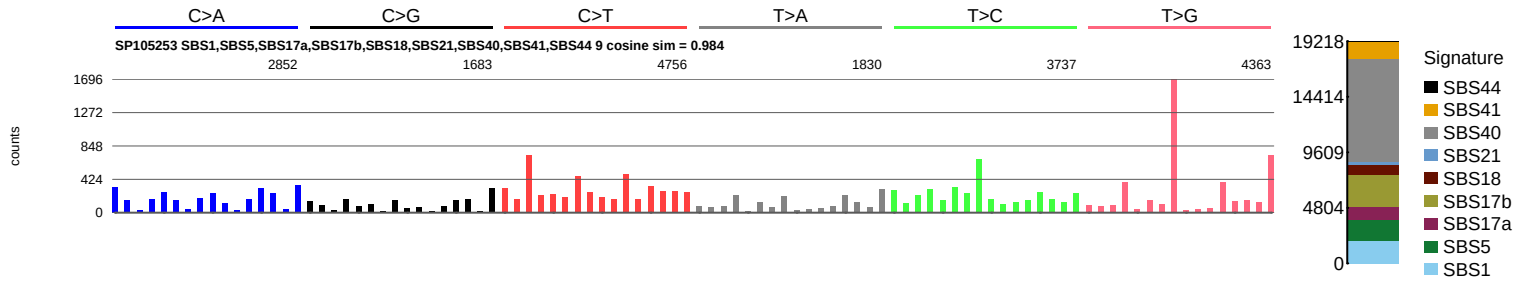
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98407
2863
1765
4724
2117
3622
4123
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

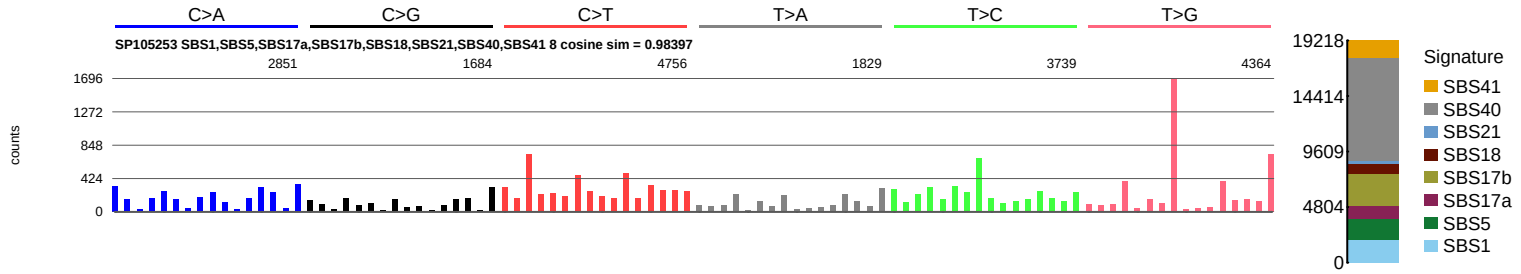

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98407
2939
2019
4867
1508
3693
4198
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

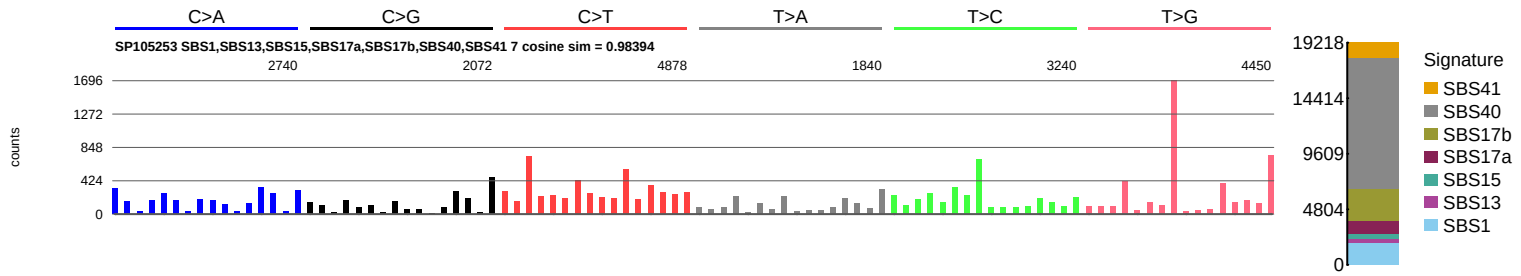

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98404
2863
1765
4724
2116
3623
4123
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

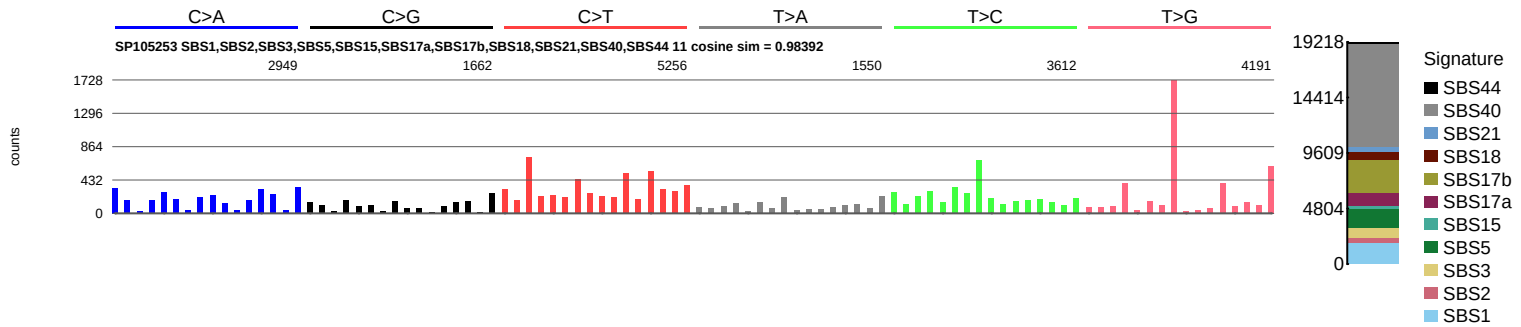

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.98404
2735
2056
4901
1832
3264
4430
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

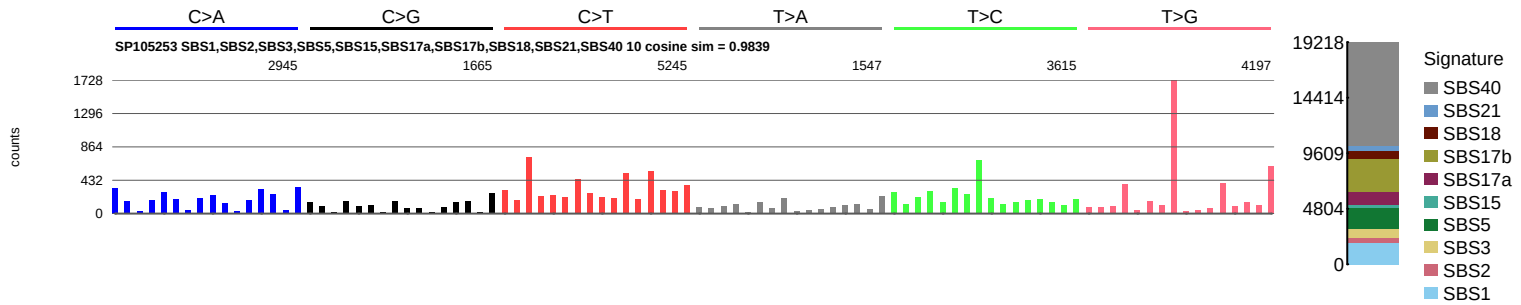

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98402
2729
2060
4997
1540
3634
4257
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

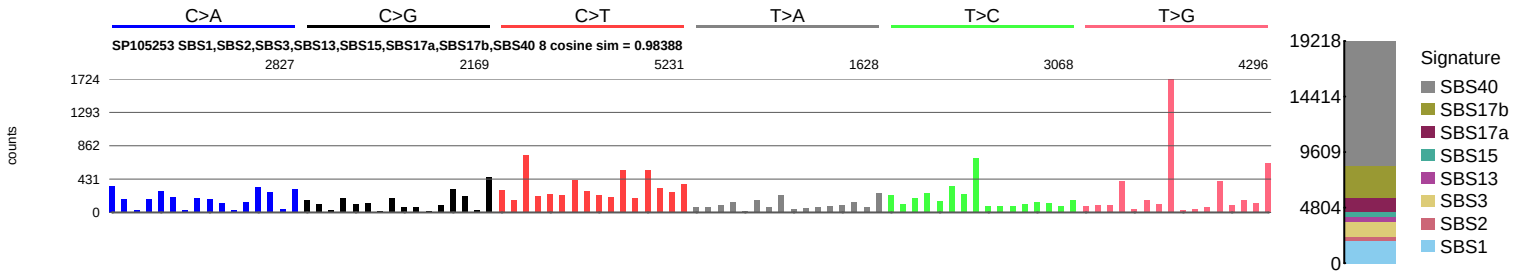

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.984
2852
1683
4756
1830
3737
4363
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

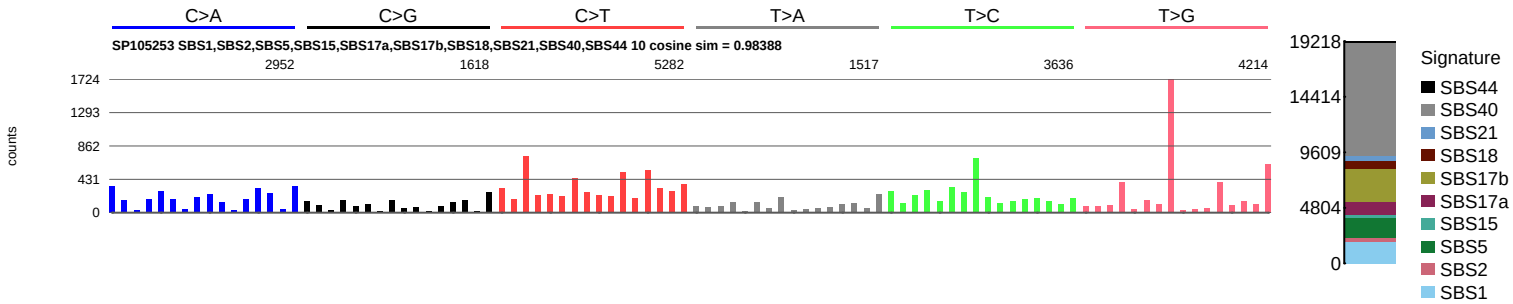

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98397
2851
1684
4756
1829
3739
4364
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

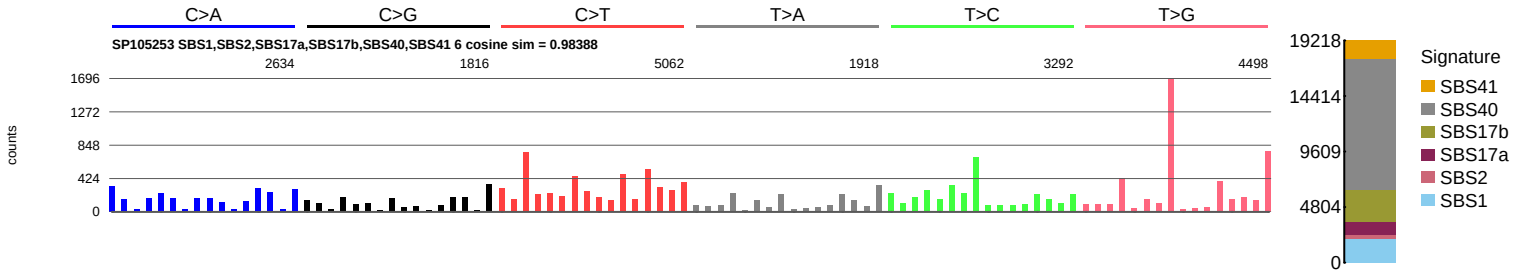

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.98394
2740
2072
4878
1840
3240
4450
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

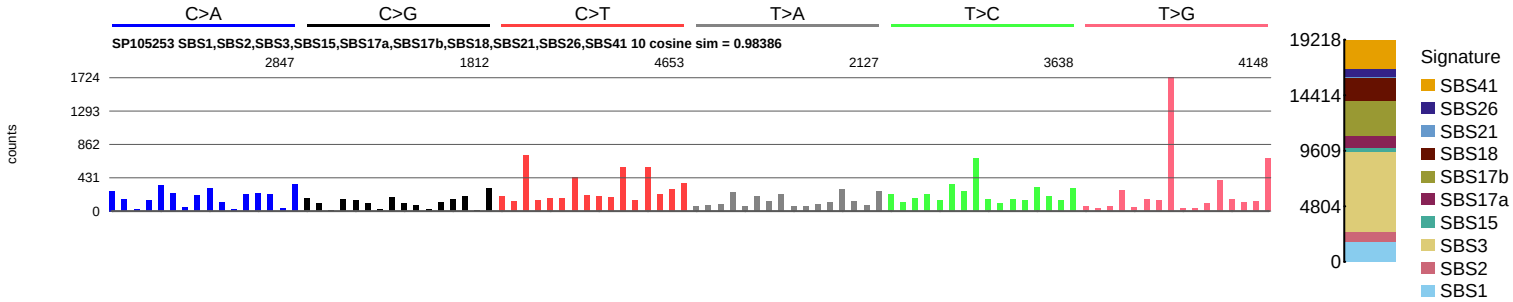

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 11 cosine sim = 0.98392
2949
1662
5256
1550
3612
4191
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

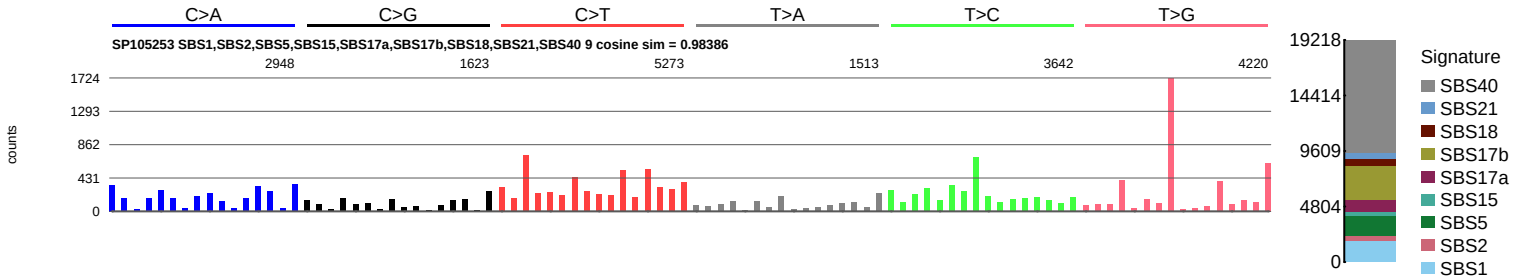

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 10 cosine sim = 0.9839
2945
1665
5245
1547
3615
4197
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

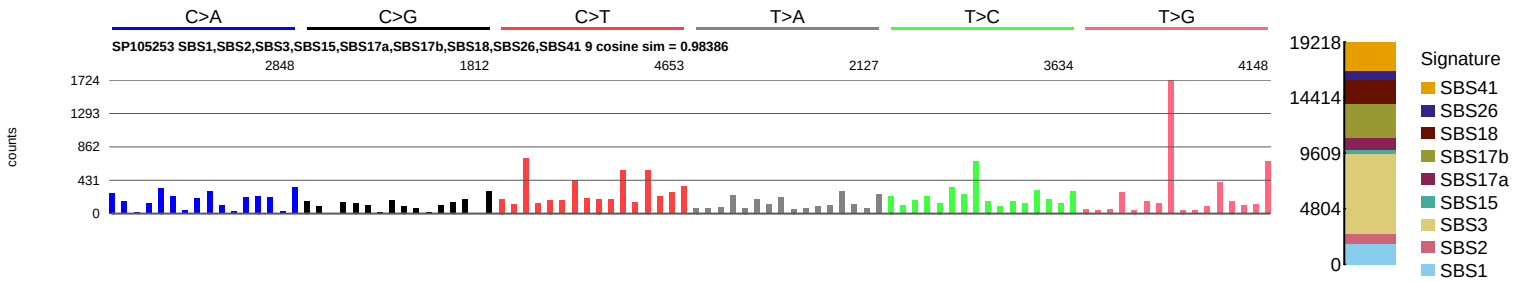

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98388
2827
2169
5231
1628
3068
4296
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1

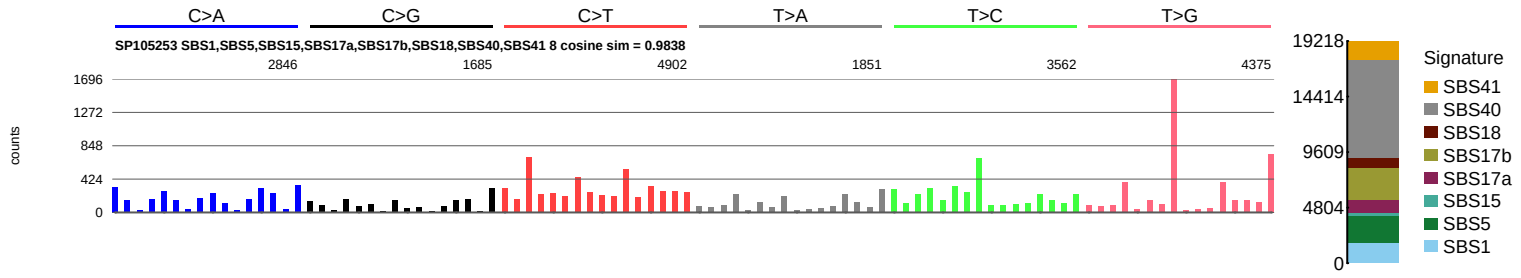

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98388
2952
1618
5282
1517
3636
4214
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98388
2634
1816
5062
1918
3292
4498
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98386
2847
1812
4653
2127
3638
4148
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98386
2948
1623
5273
1513
3642
4220
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98386
2848
1812
4653
2127
3634
4148
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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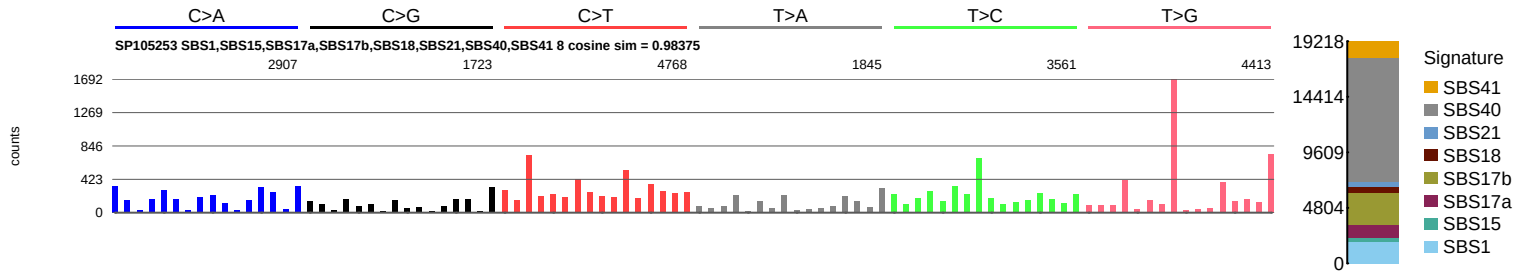
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9838
2846
1685
4902
1851
3562
4375
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>C
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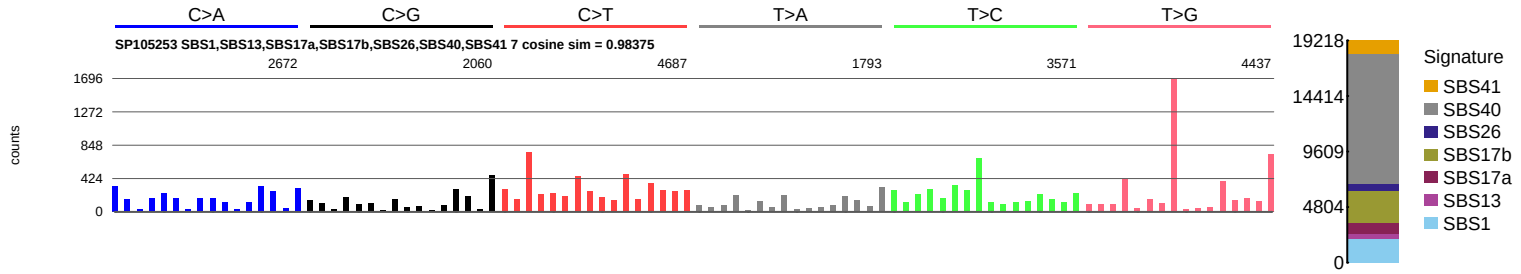
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.98375
2907
1723
4768
1845
3561
4413
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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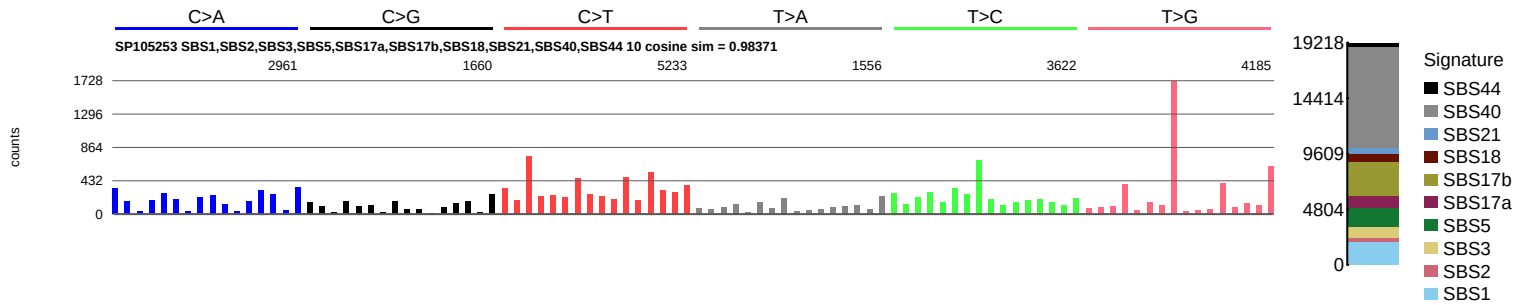
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98375
2672
2060
4687
1793
3571
4437
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1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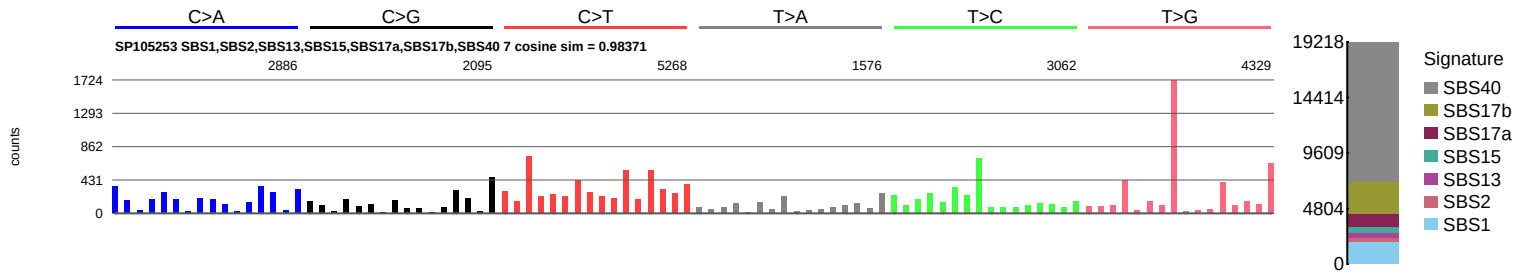
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98371
2961
1660
5233
1556
3622
4185
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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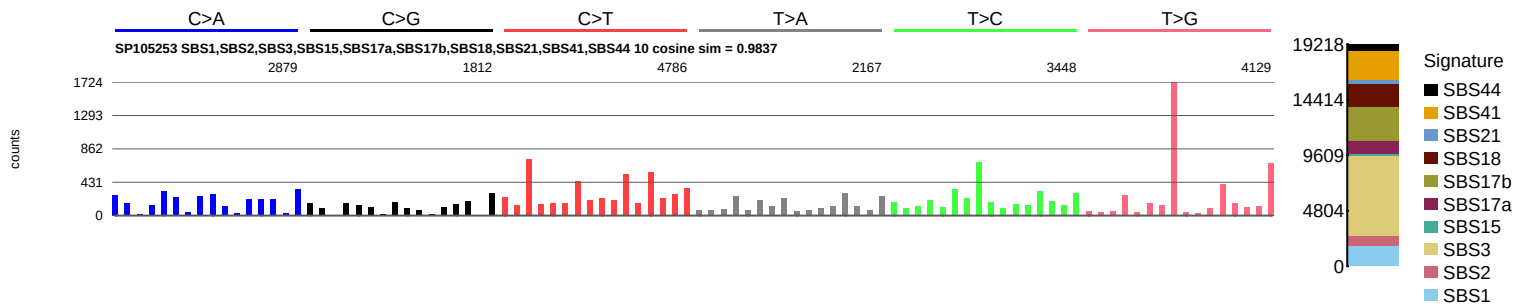
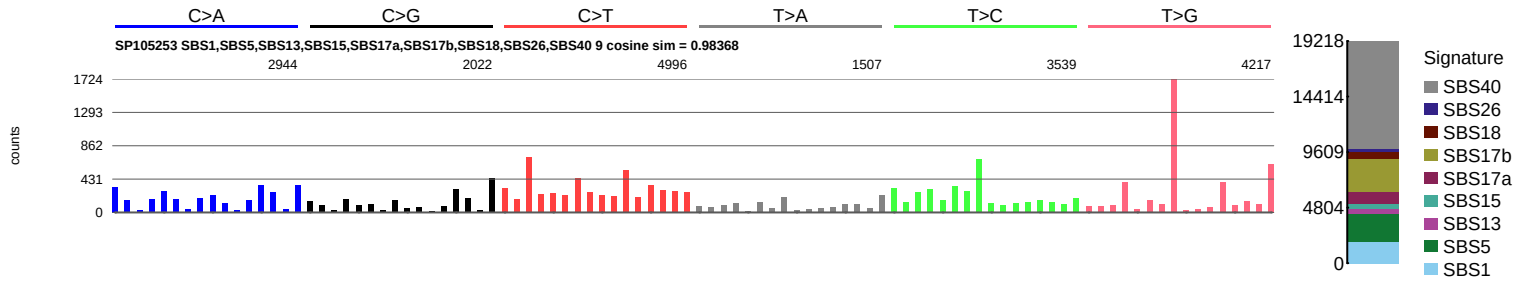
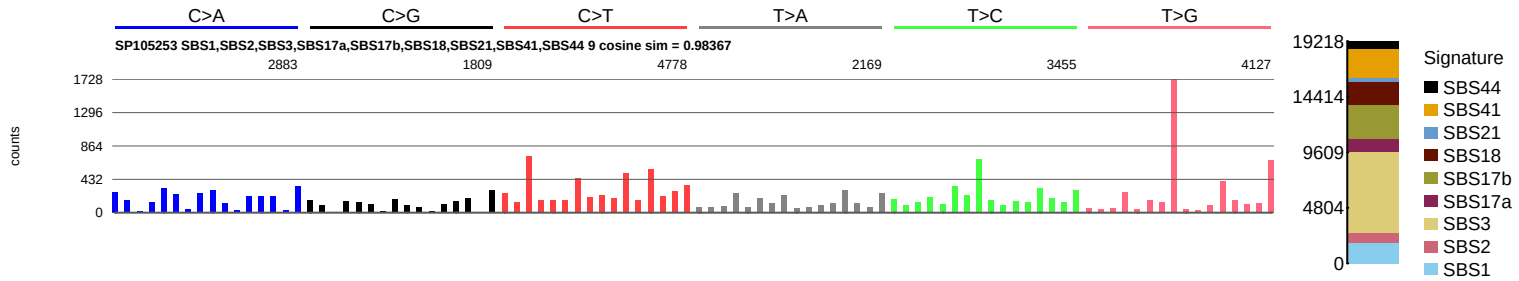
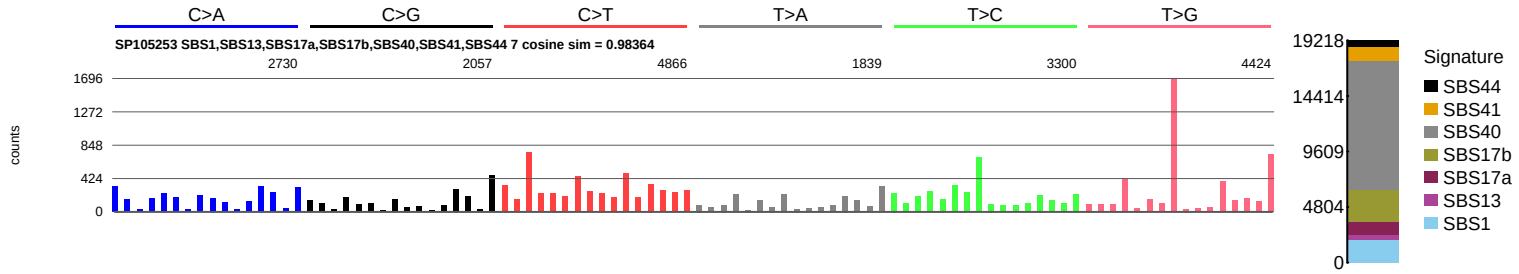
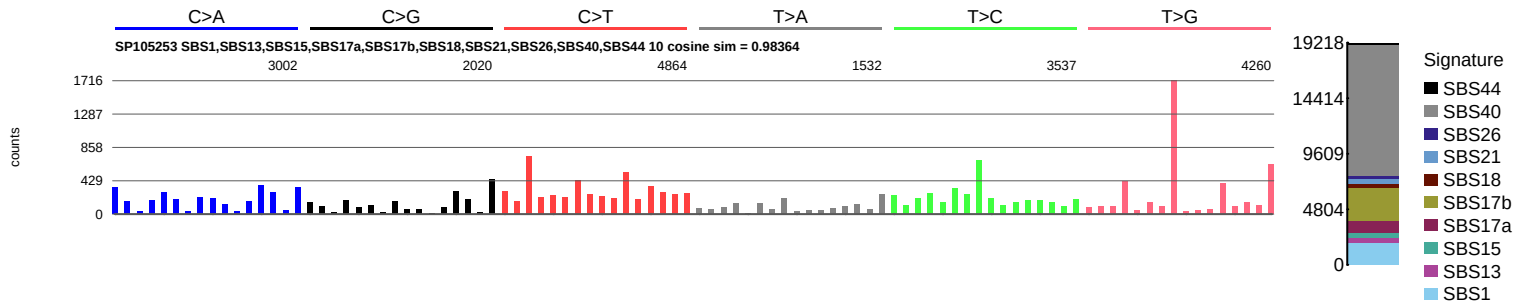
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98371
2886
2095
5268
1576
3062
4329
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

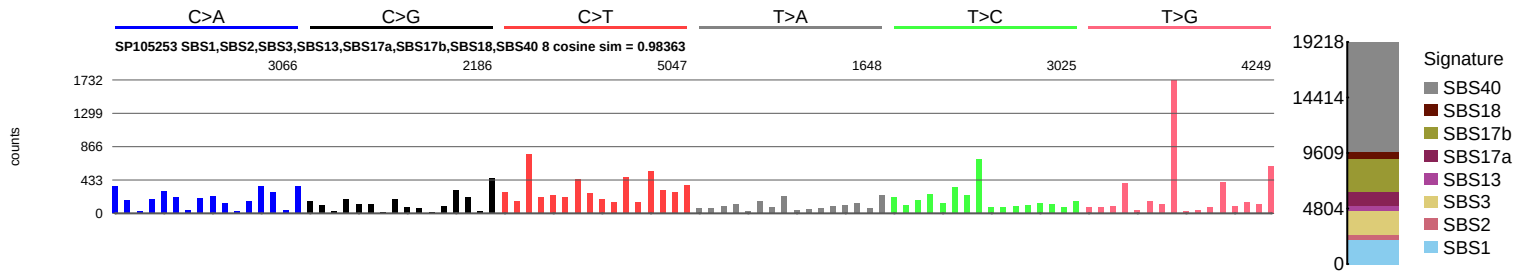

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9837
2879
1812
4786
2167
3448
4129
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

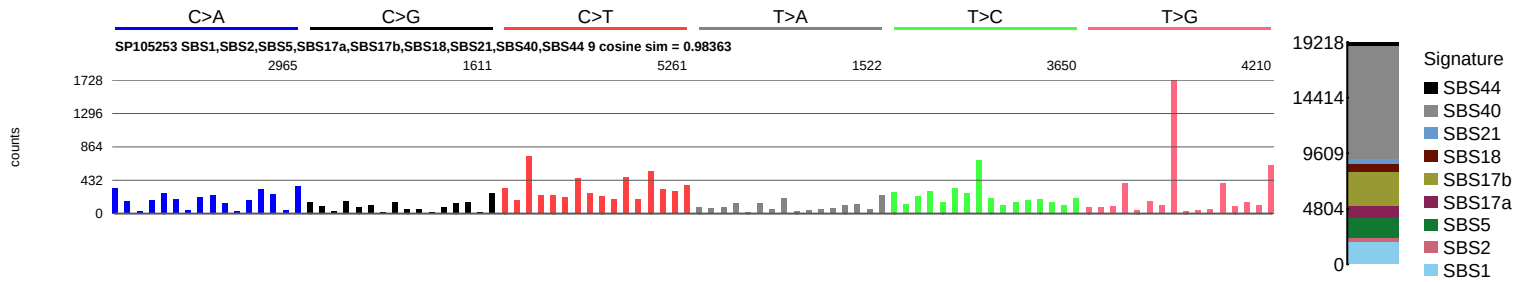

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98368
2944
2022
4996
1507
3539
4217
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

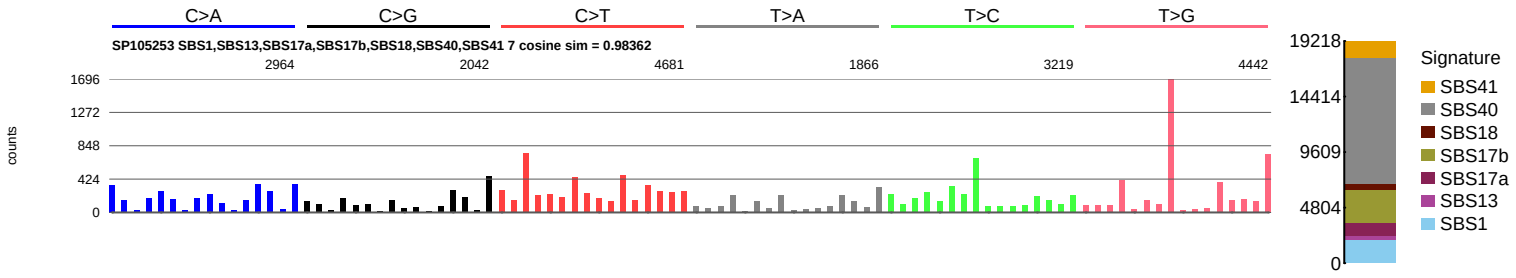

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98367
2883
1809
4778
2169
3455
4127
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

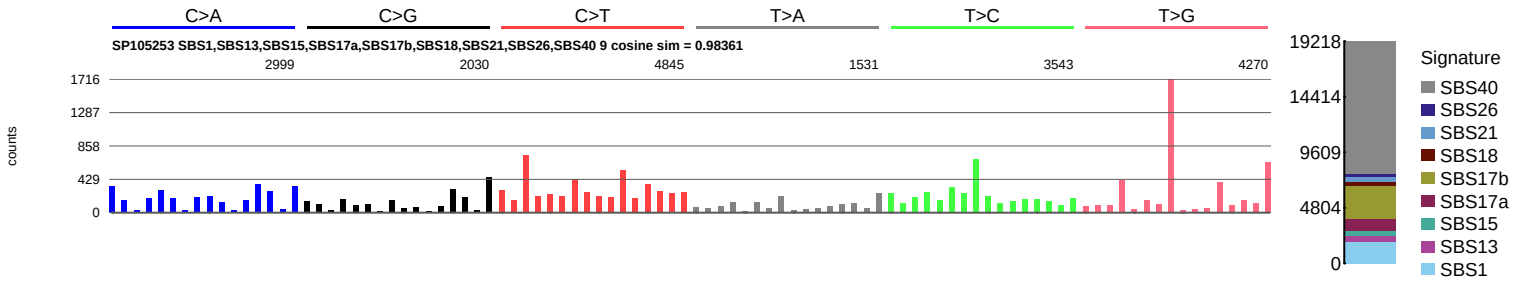

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98364
2730
2057
4866
1839
3300
4424
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS1

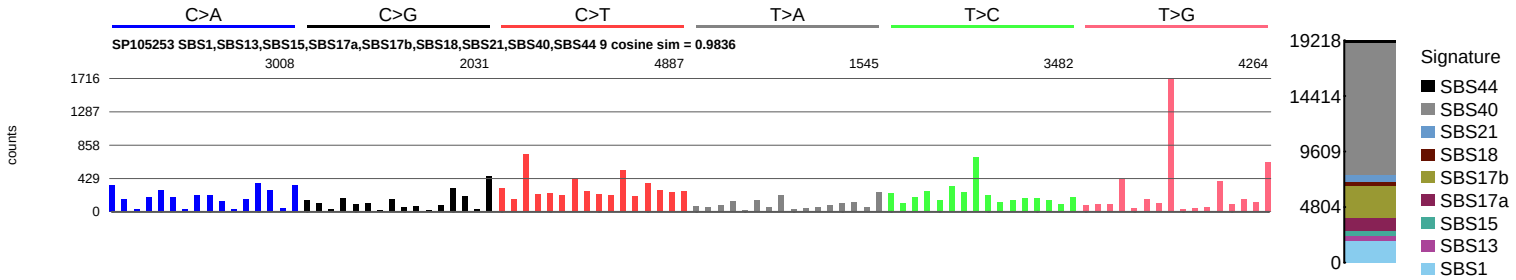

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98364
3002
2020
4864
1532
3537
4260
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

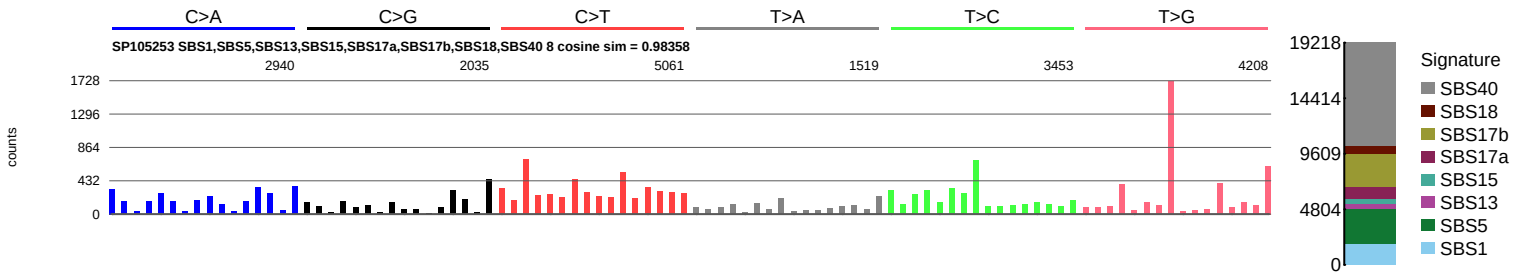

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98363
3066
2186
5047
1648
3025
4249
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

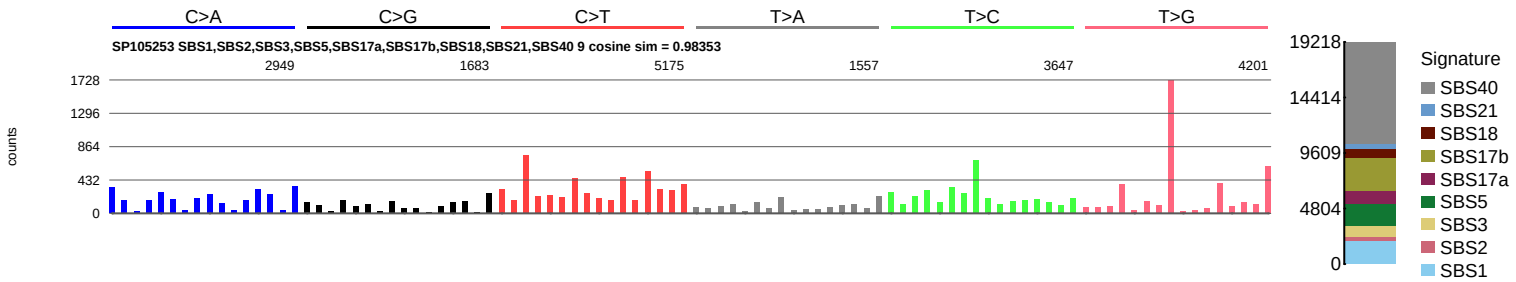

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98363
2965
1611
5261
1522
3650
4210
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98362
2964
2042
4681
1866
3219
4442
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98361
2999
2030
4845
1531
3543
4270
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.9836
3008
2031
4887
1545
3482
4264
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98358
2940
2035
5061
1519
3453
4208
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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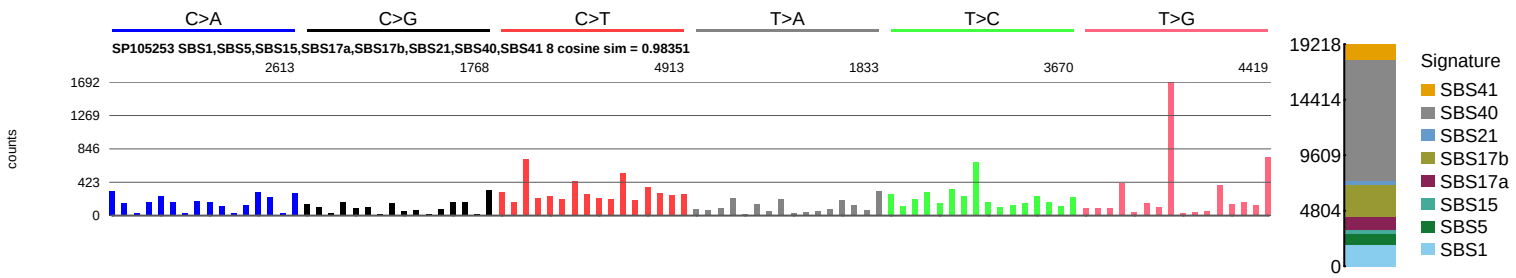
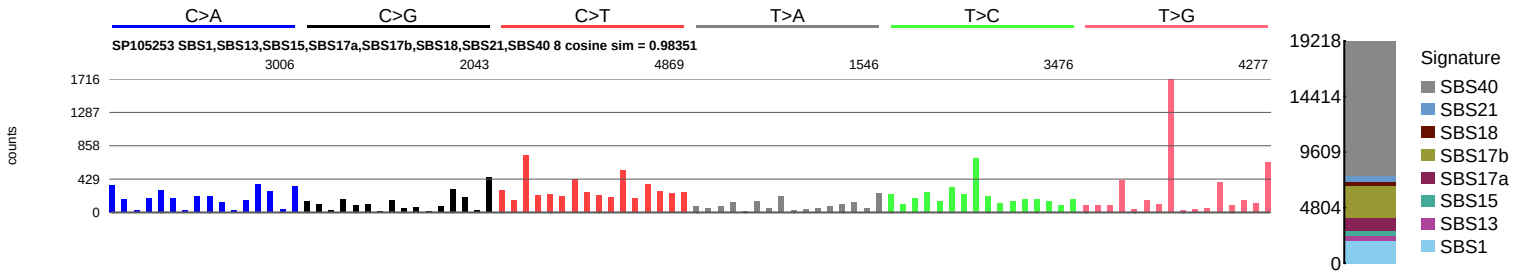
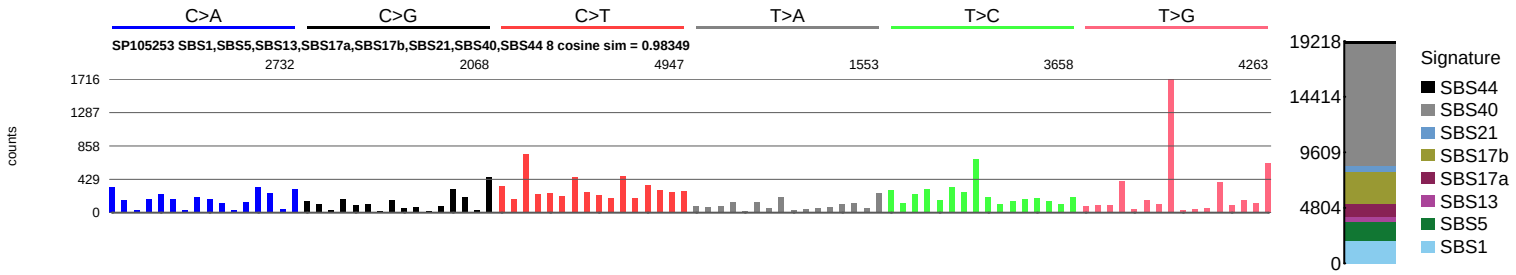
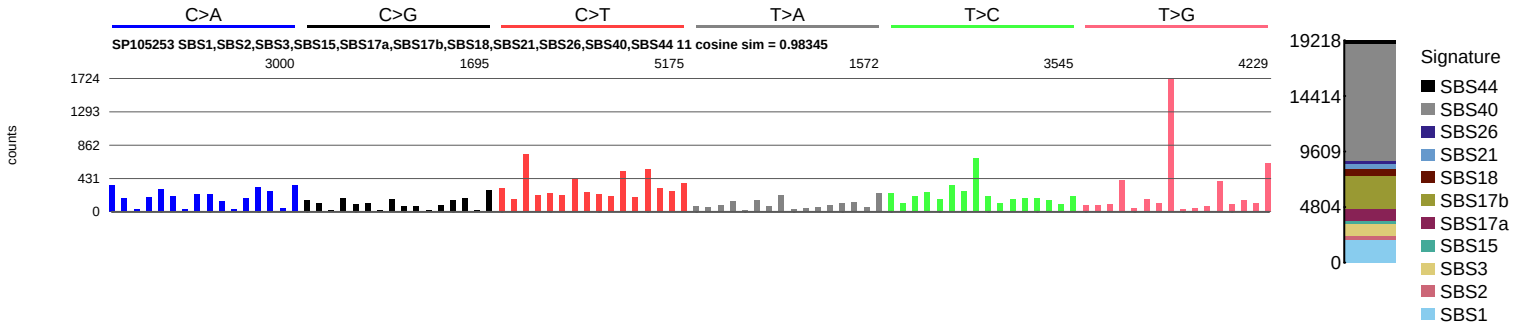
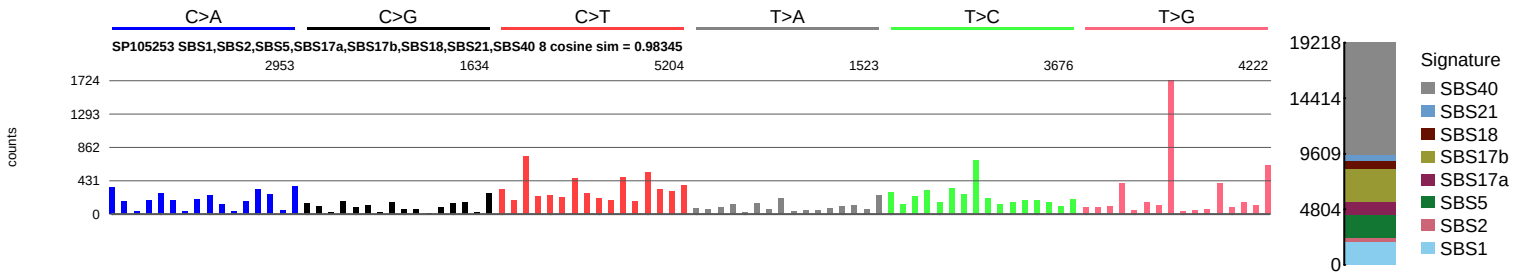
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98353
2949
1683
5175
1557
3647
4201
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

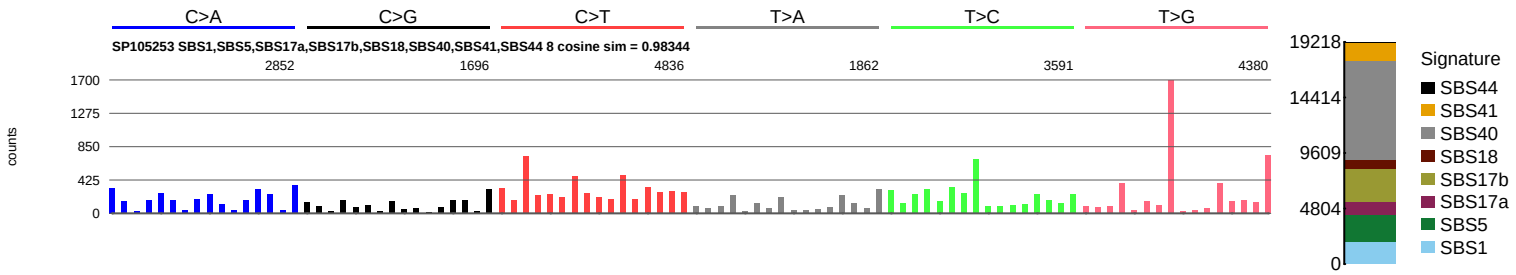

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.98351
2613
1768
4913
1833
3670
4419
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1

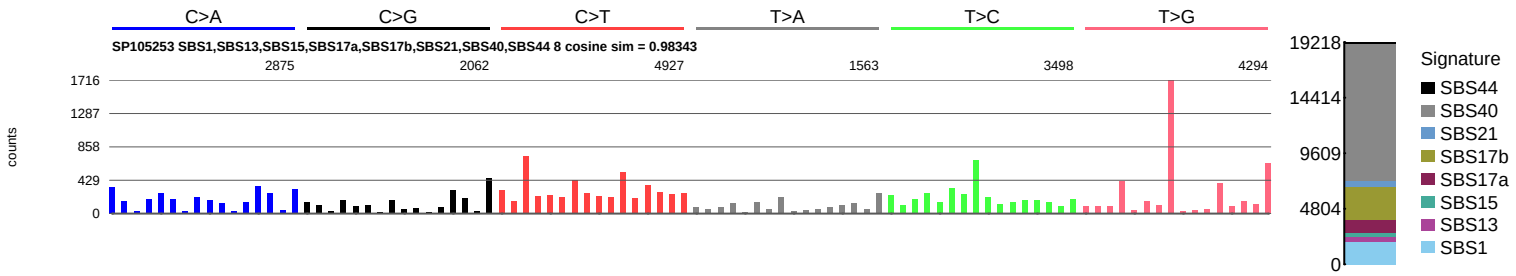

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98351
3006
2043
4869
1546
3476
4277
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

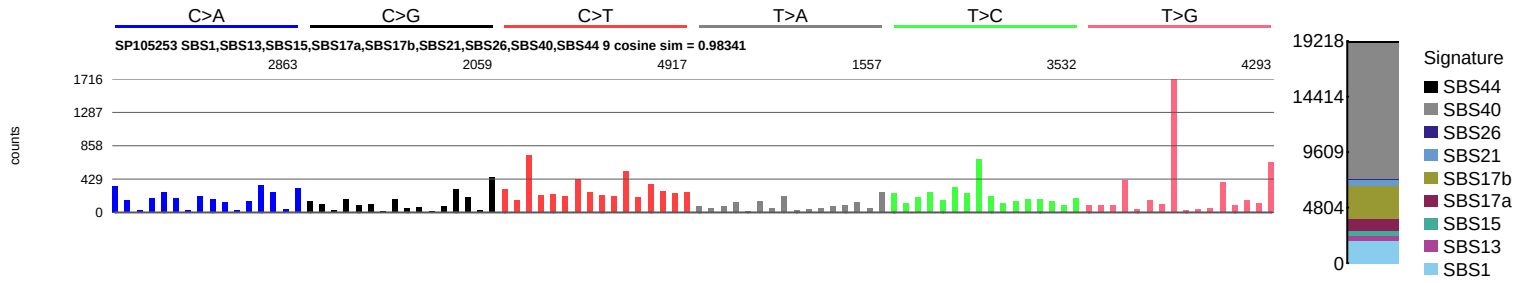

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98349
2732
2068
4947
1553
3658
4263
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1

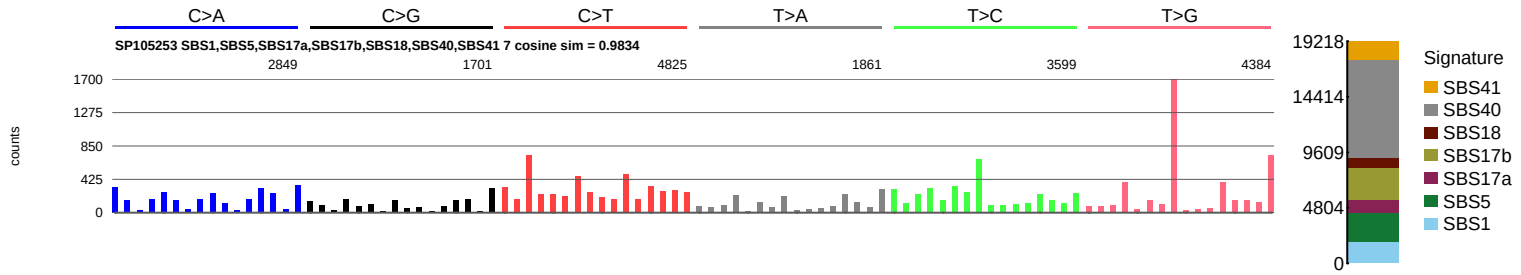

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 11 cosine sim = 0.98345
3000
1695
5175
1572
3545
4229
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98345
2953
1634
5204
1523
3676
4222
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98344
2852
1696
4836
1862
3591
4380
counts
1700
1275
850
425
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98343
2875
2062
4927
1563
3498
4294
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98341
2863
2059
4917
1557
3532
4293
1716
1287
858
429
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
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T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.9834
2849
1701
4825
1861
3599
4384
1700
1275
850
425
0
counts
19218
14414
9609
4804
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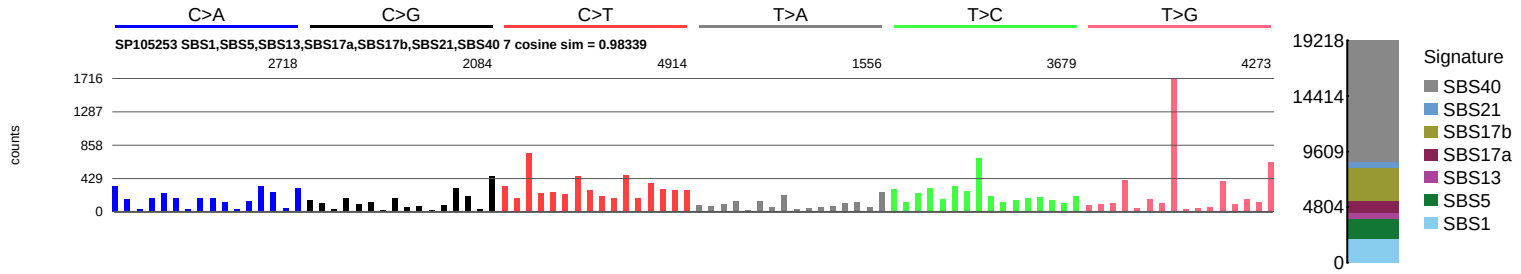
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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C>G
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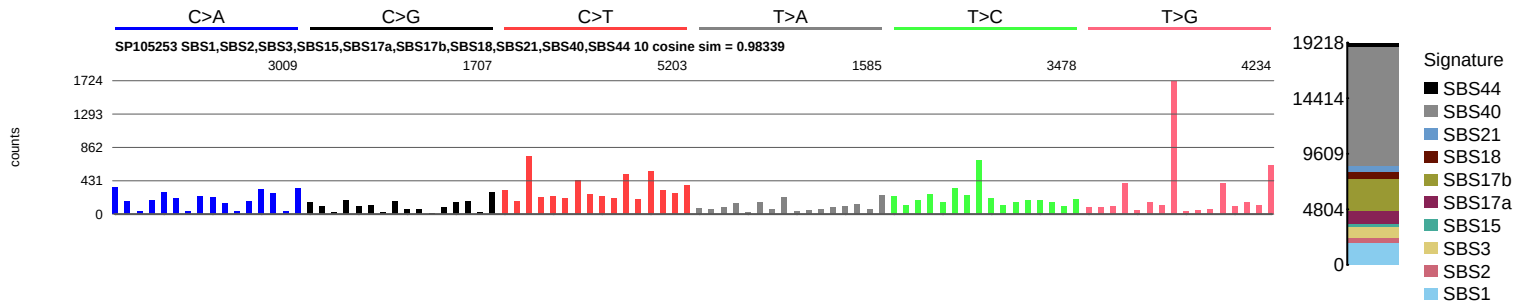
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98339
2718
2084
4914
1556
3679
4273
1716
1287
858
429
0
counts
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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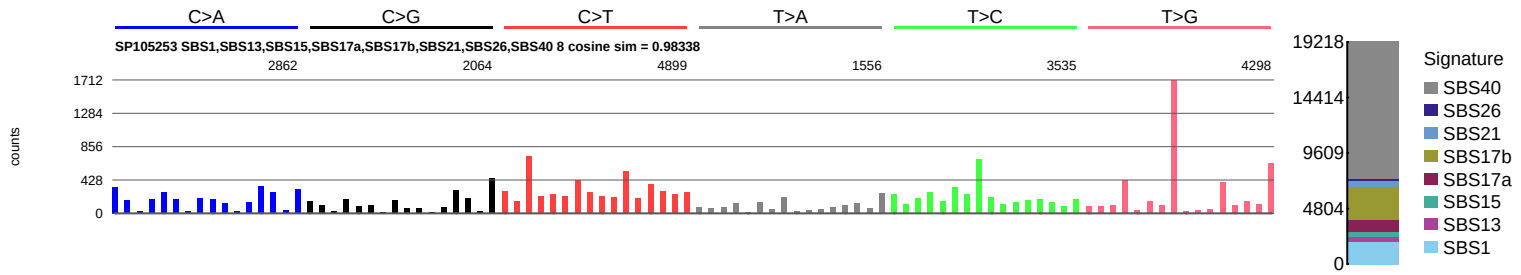
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 10 cosine sim = 0.98339
3009
1707
5203
1585
3478
4234
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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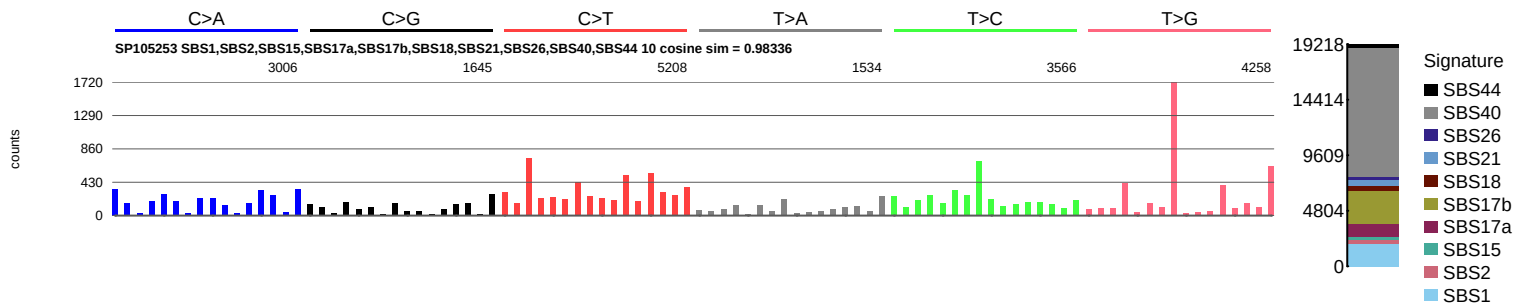
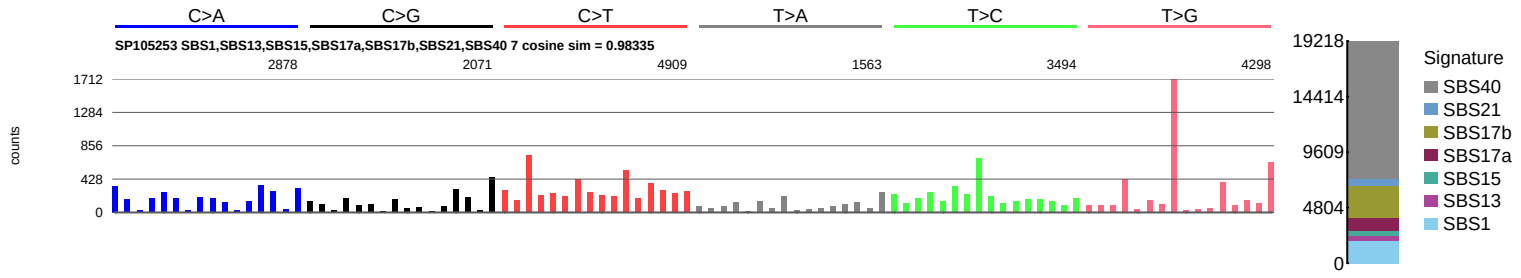
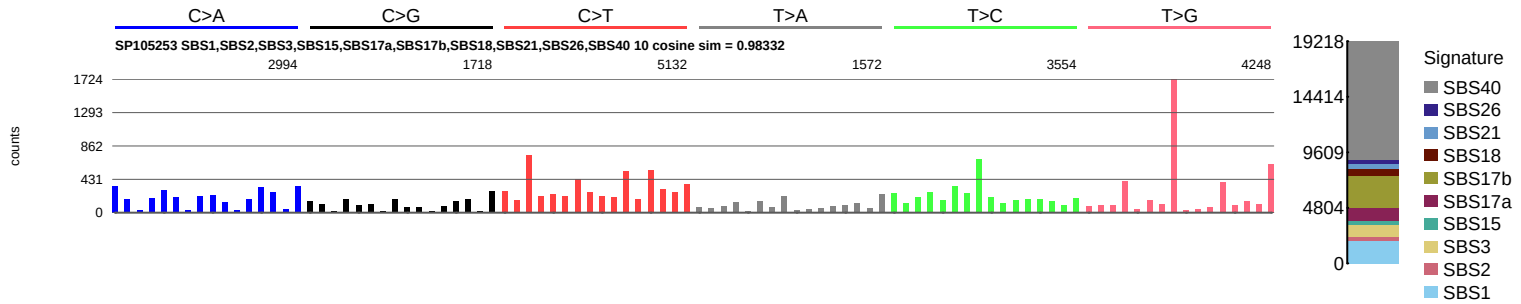
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98338
2862
2064
4899
1556
3535
4298
1712
1284
856
428
0
counts
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98336
3006
1645
5208
1534
3566
4258
1720
1290
860
430
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98335
2878
2071
4909
1563
3494
4298
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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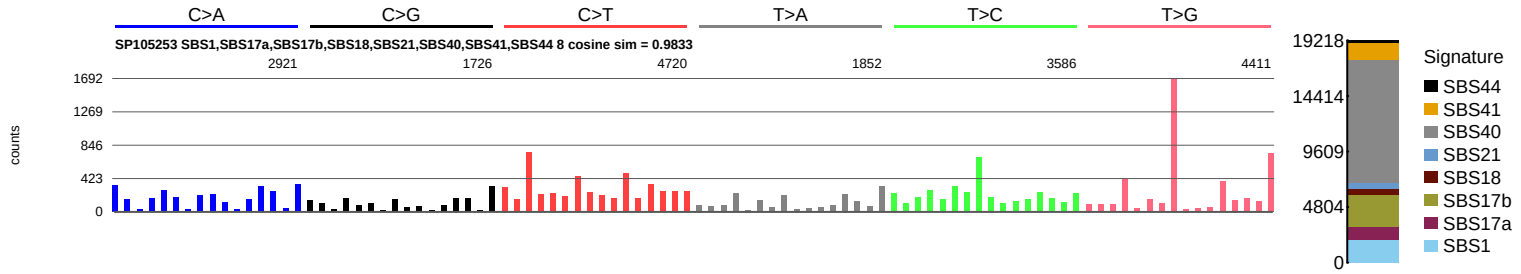
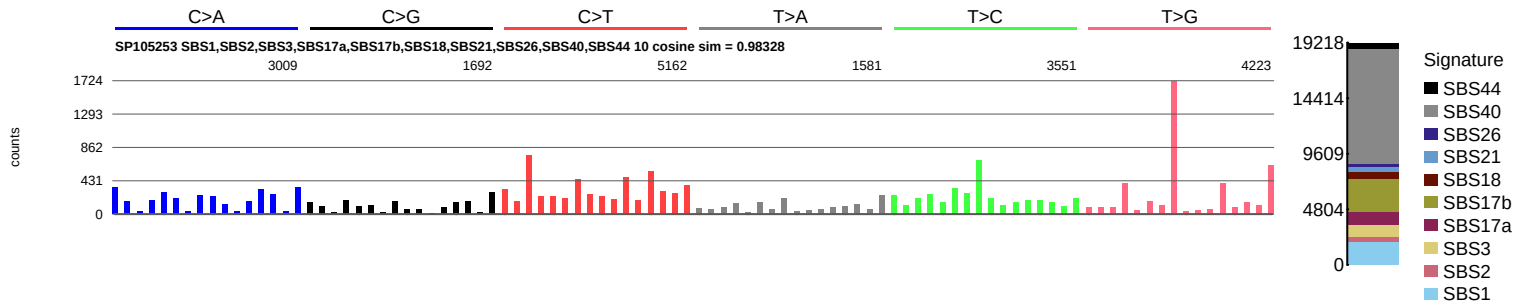
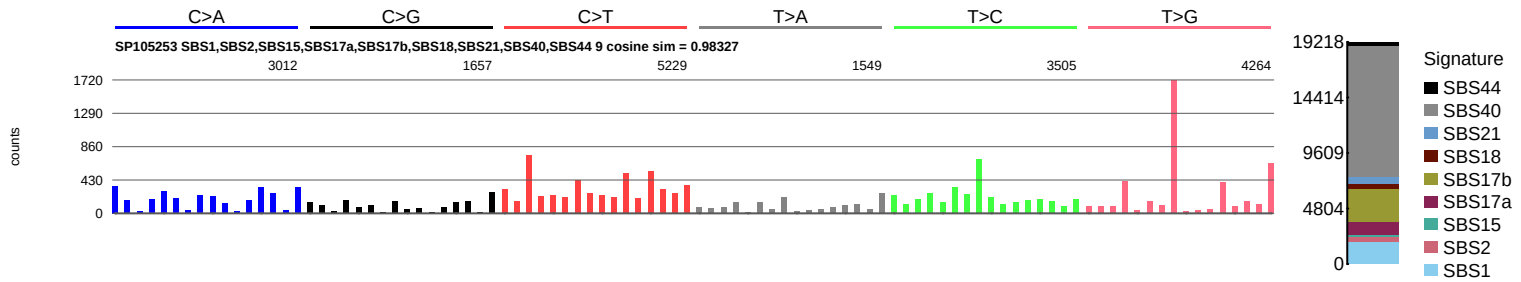
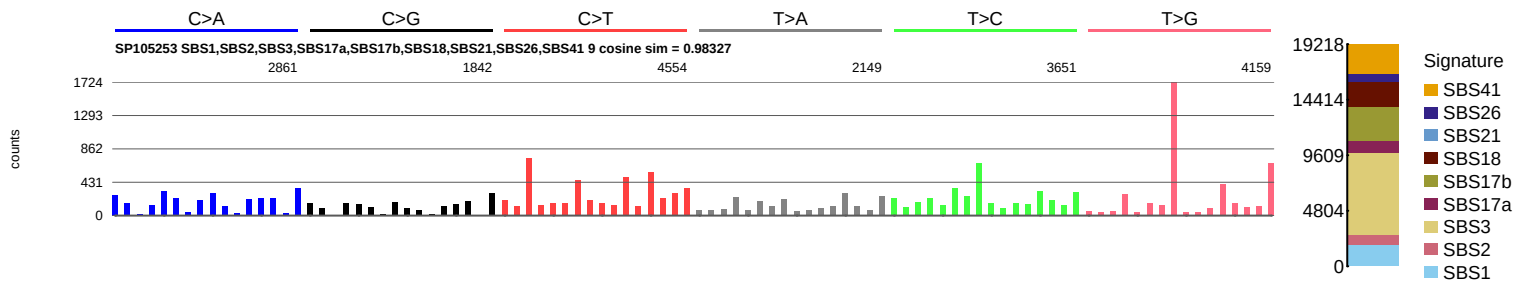
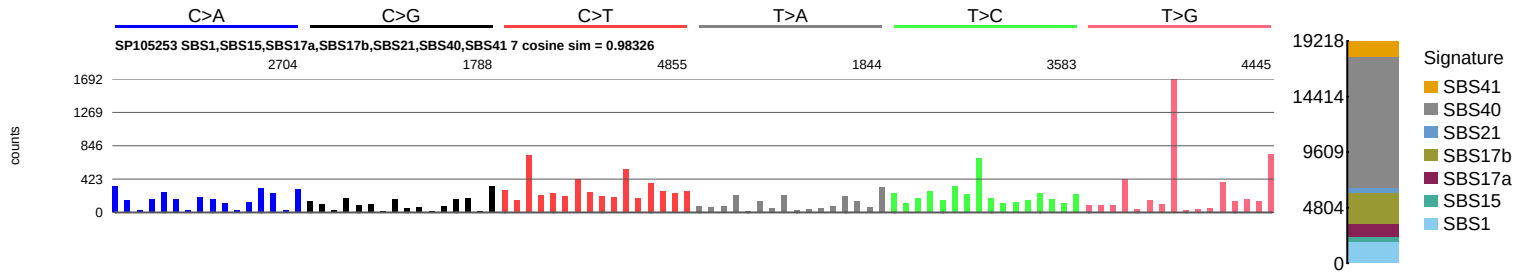
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98332
2994
1718
5132
1572
3554
4248
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

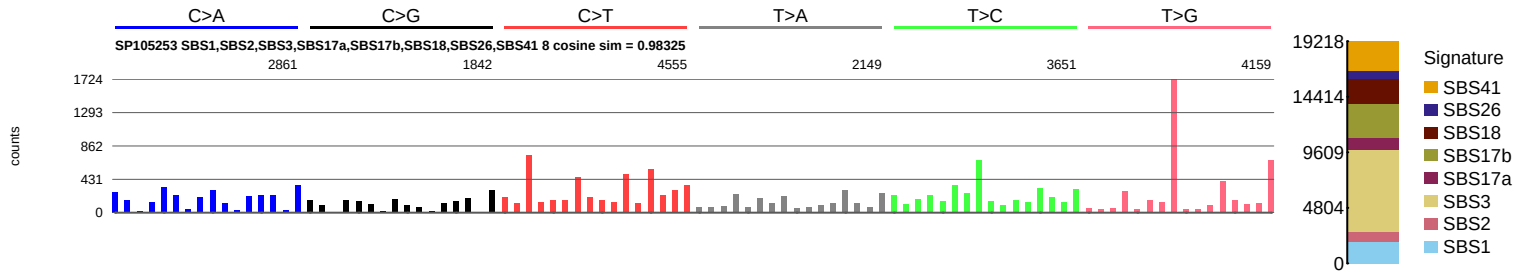

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.9833
2921
1726
4720
1852
3586
4411
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

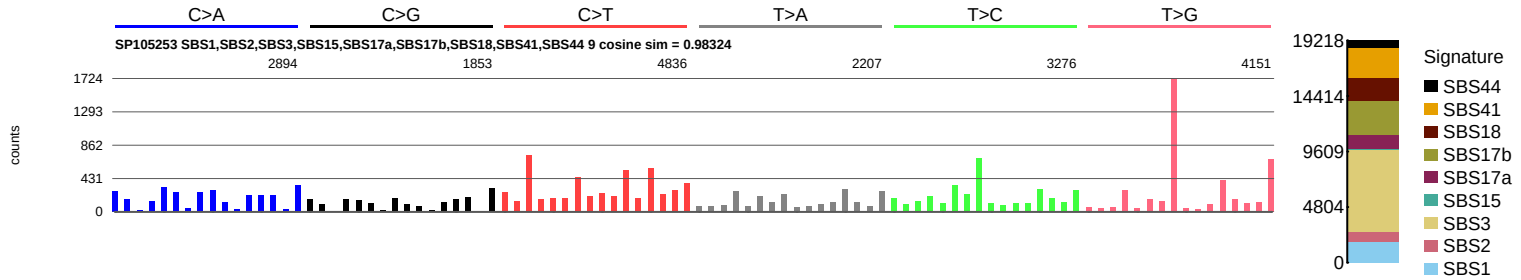

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98328
3009
1692
5162
1581
3551
4223
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

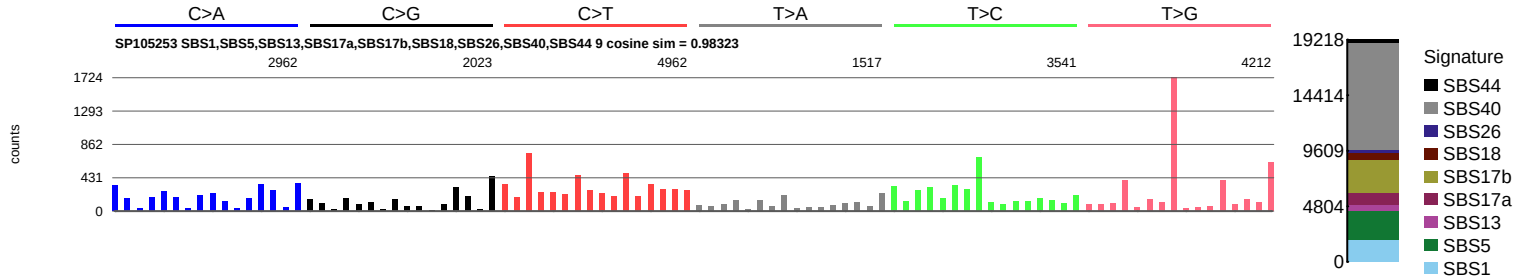

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98327
3012
1657
5229
1549
3505
4264
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98327
2861
1842
4554
2149
3651
4159
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98326
2704
1788
4855
1844
3583
4445
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98325
2861
1842
4555
2149
3651
4159
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.98324
2894
1853
4836
2207
3276
4151
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98323
2962
2023
4962
1517
3541
4212
counts
1724
1293
862
431
0
19218
14414
9609
4804
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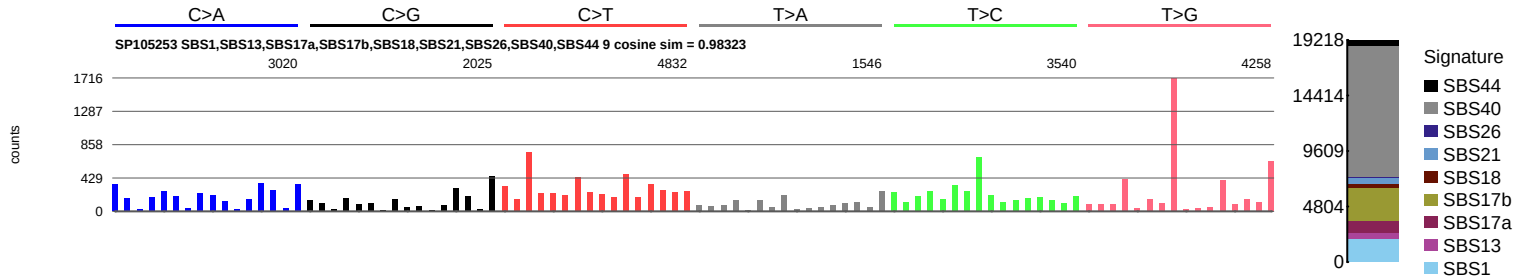
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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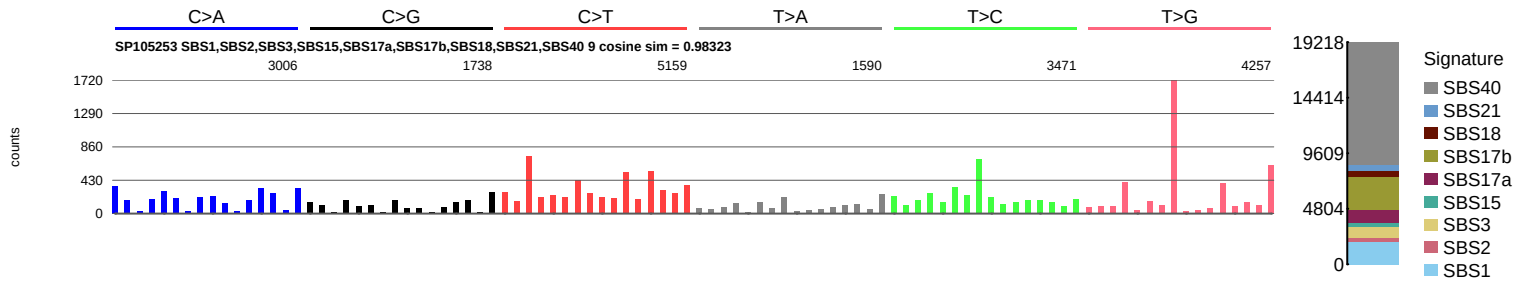
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98323
3020
2025
4832
1546
3540
4258
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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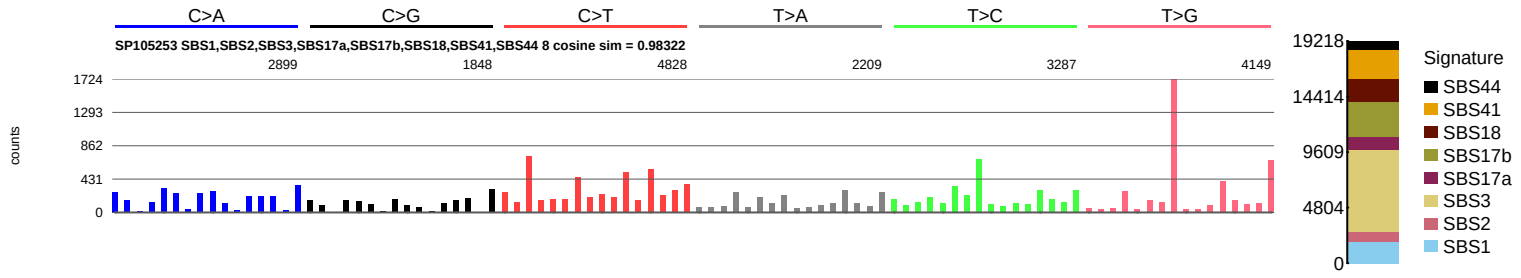
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98323
3006
1738
5159
1590
3471
4257
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98322
2899
1848
4828
2209
3287
4149
counts
1724
1293
862
431
0
19218
14414
9609
4804
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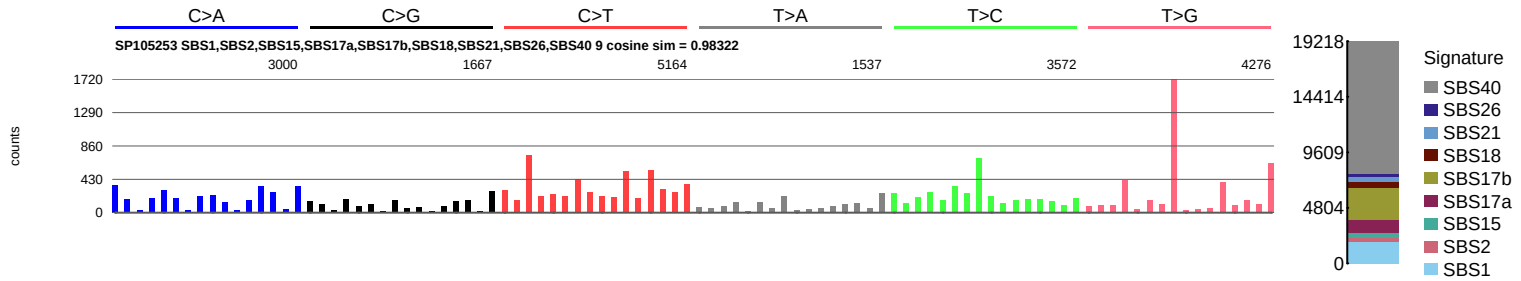
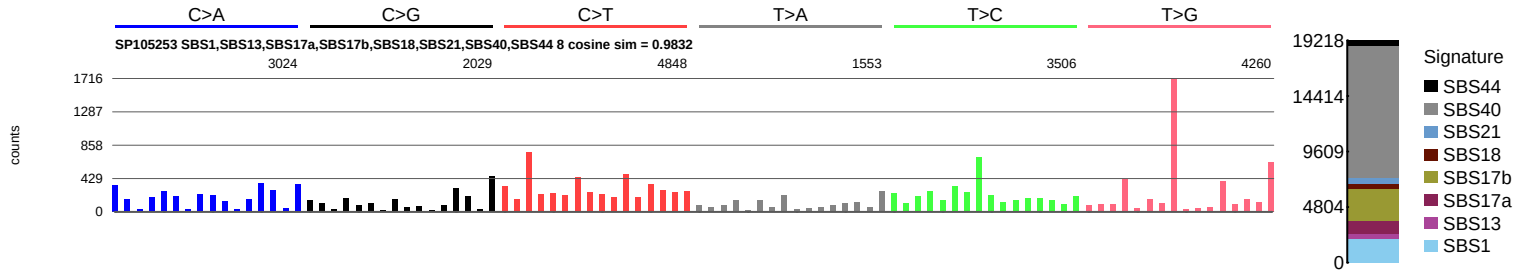
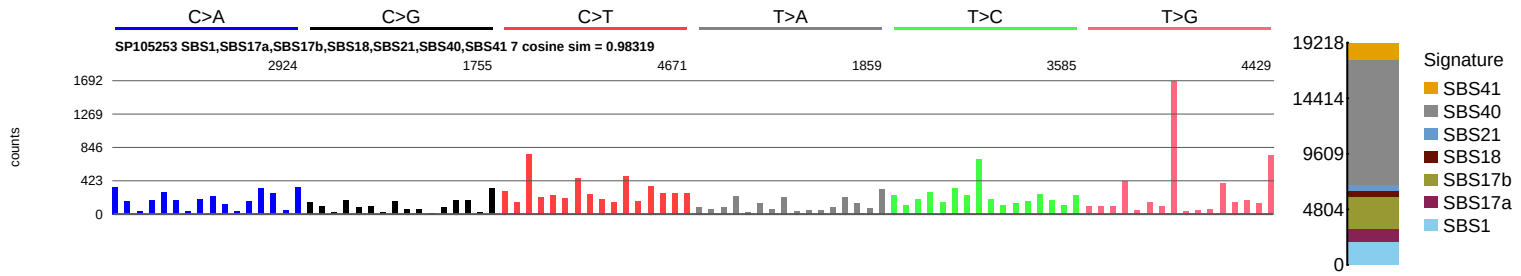
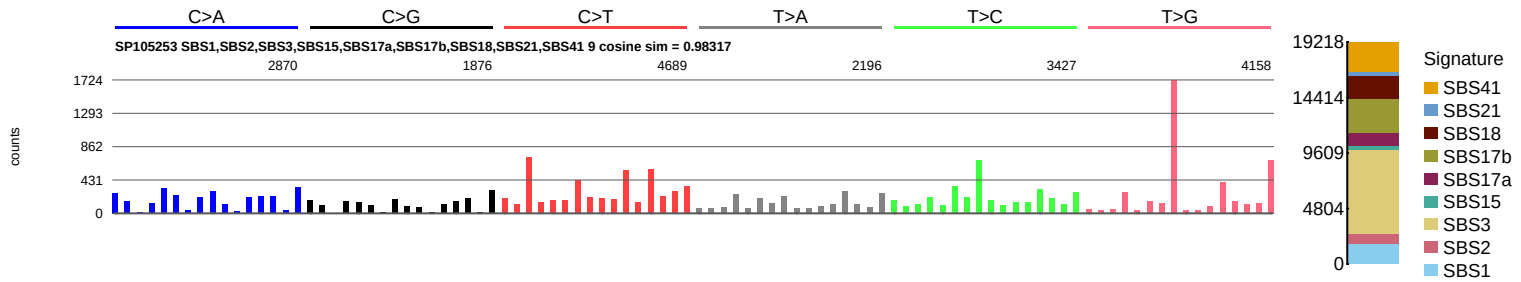
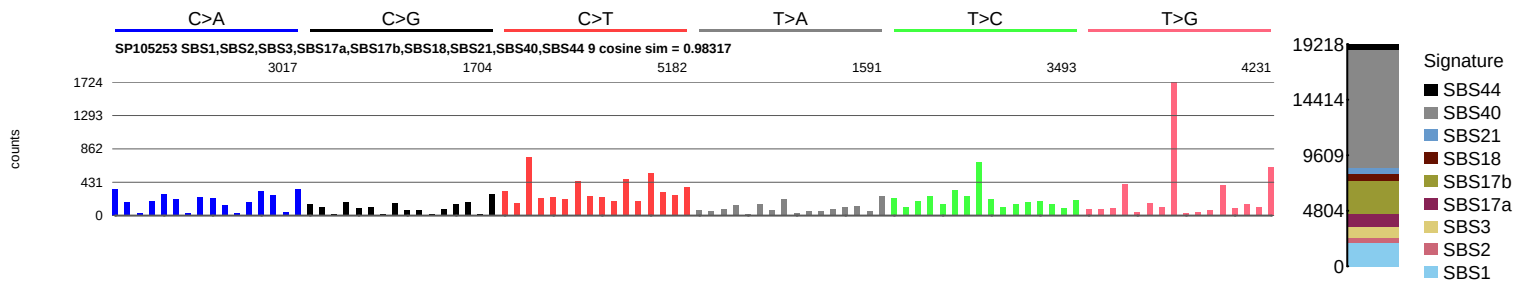
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

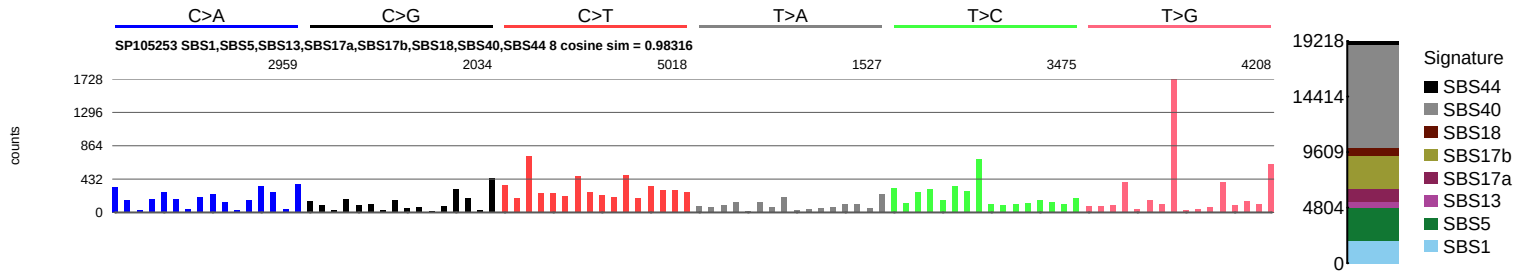

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98322
3000
1667
5164
1537
3572
4276
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9832
3024
2029
4848
1553
3506
4260
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.98319
2924
1755
4671
1859
3585
4429
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98317
2870
1876
4689
2196
3427
4158
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98317
3017
1704
5182
1591
3493
4231
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98316
2959
2034
5018
1527
3475
4208
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
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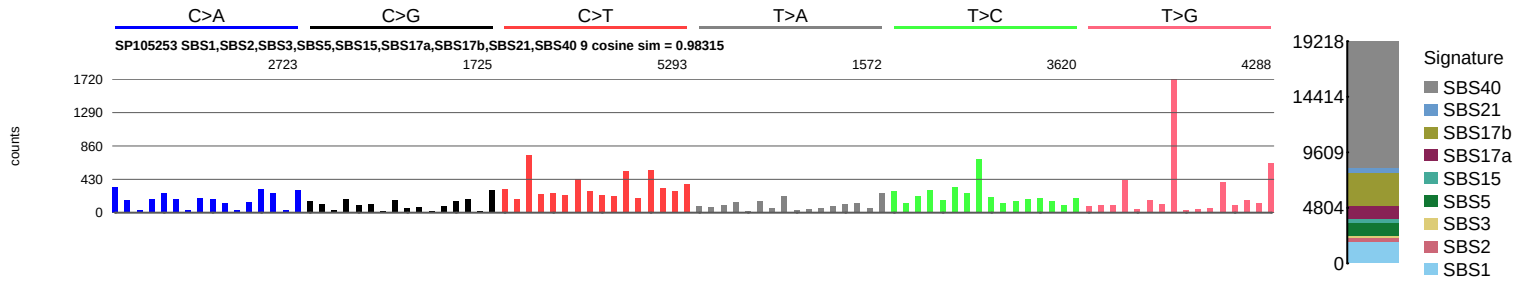
SBS13
SBS5
SBS1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SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 9 cosine sim = 0.98315
2723
1725
5293
1572
3620
4288
counts
1720
1290
860
430
0
19218
14414
9609
4804
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Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
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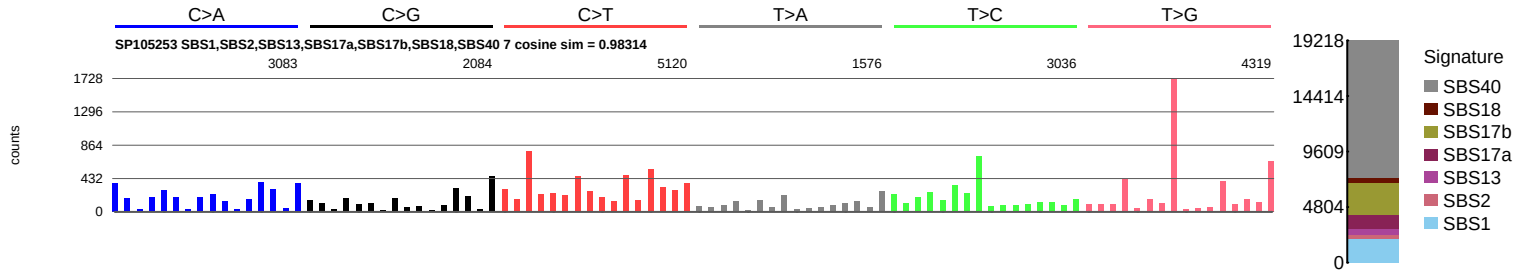
SBS2
SBS1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C>G
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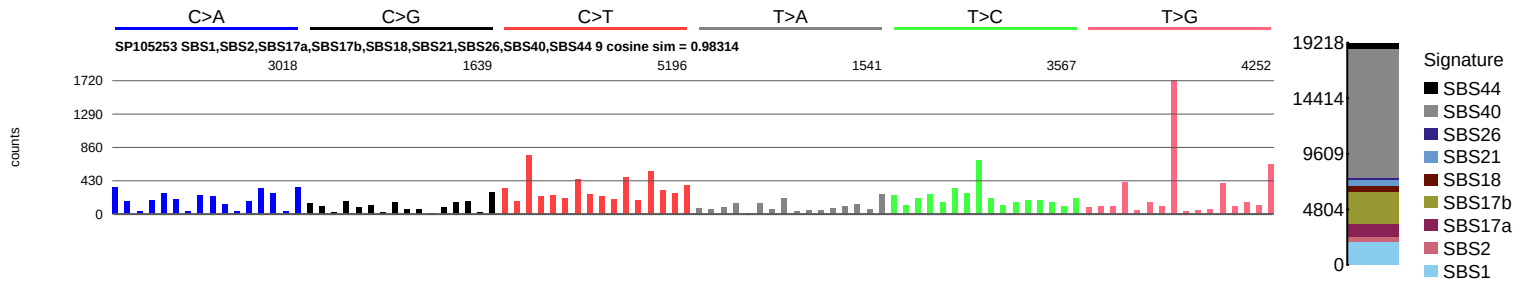
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98314
3083
2084
5120
1576
3036
4319
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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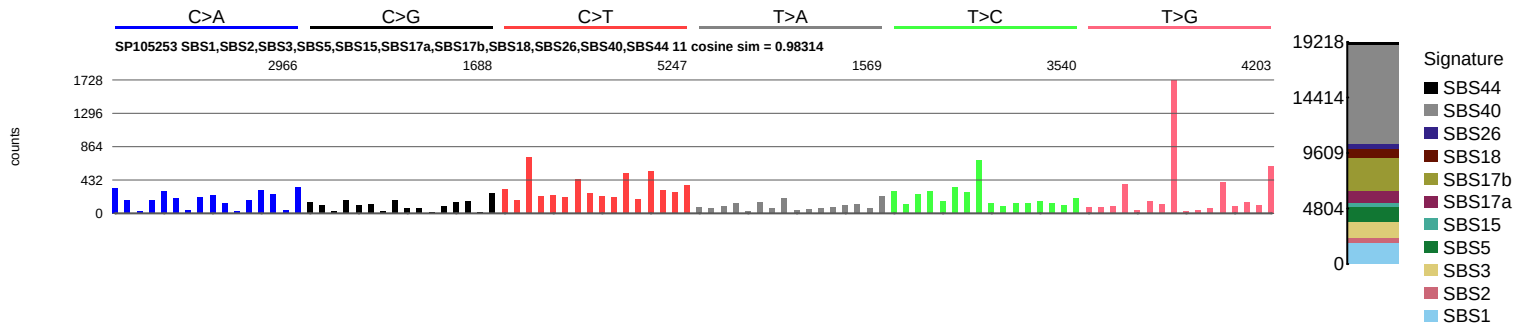
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98314
3018
1639
5196
1541
3567
4252
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
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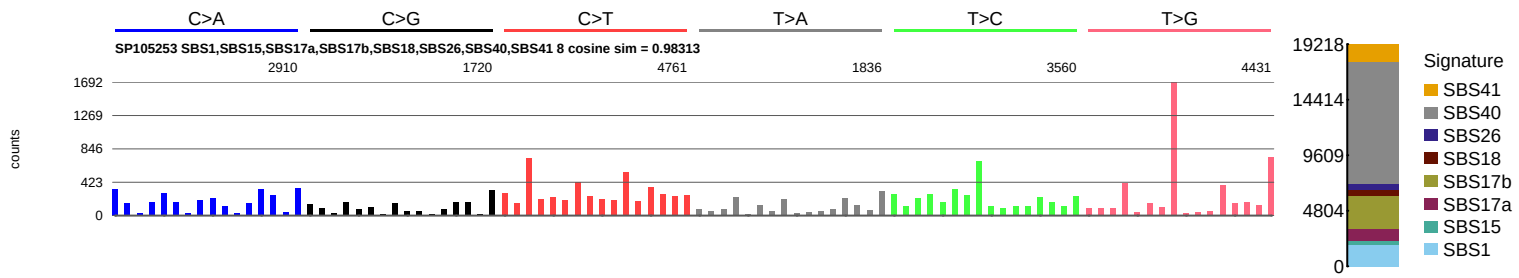
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98314
2966
1688
5247
1569
3540
4203
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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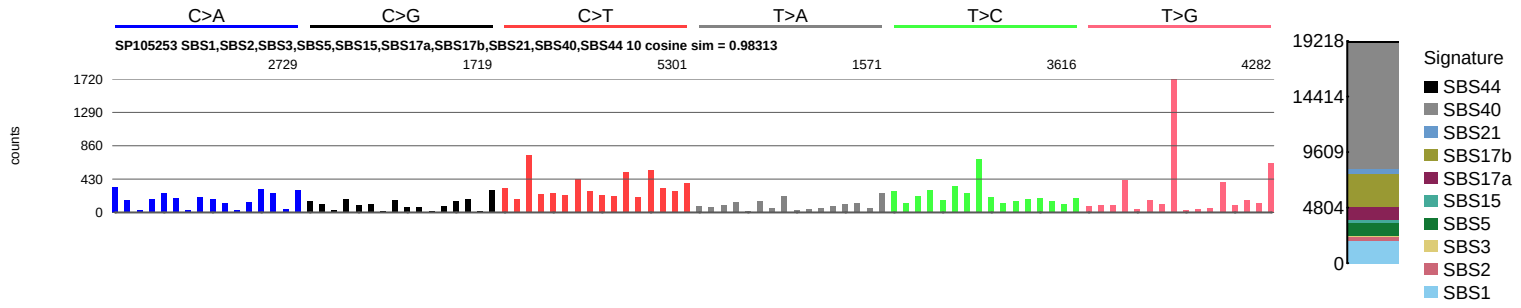
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.98313
2910
1720
4761
1836
3560
4431
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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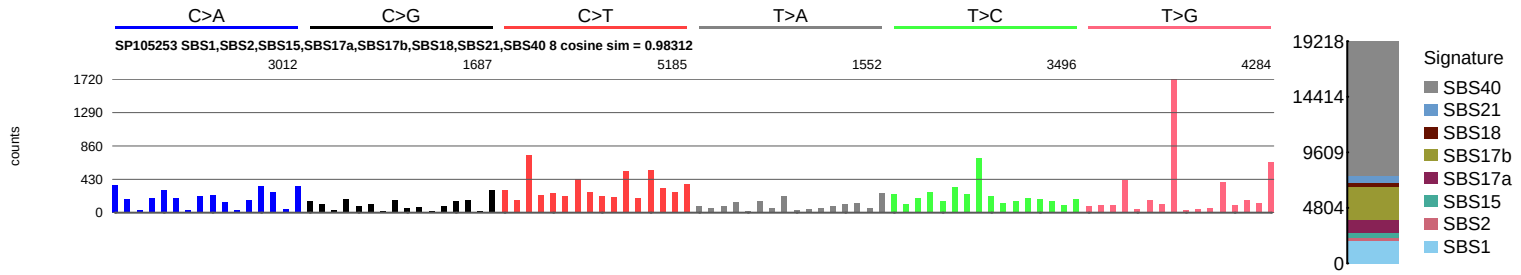
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 10 cosine sim = 0.98313
2729
1719
5301
1571
3616
4282
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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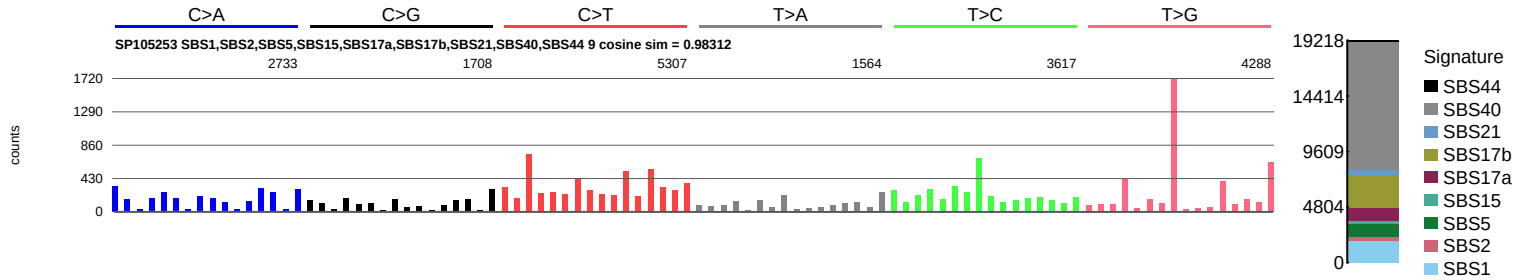
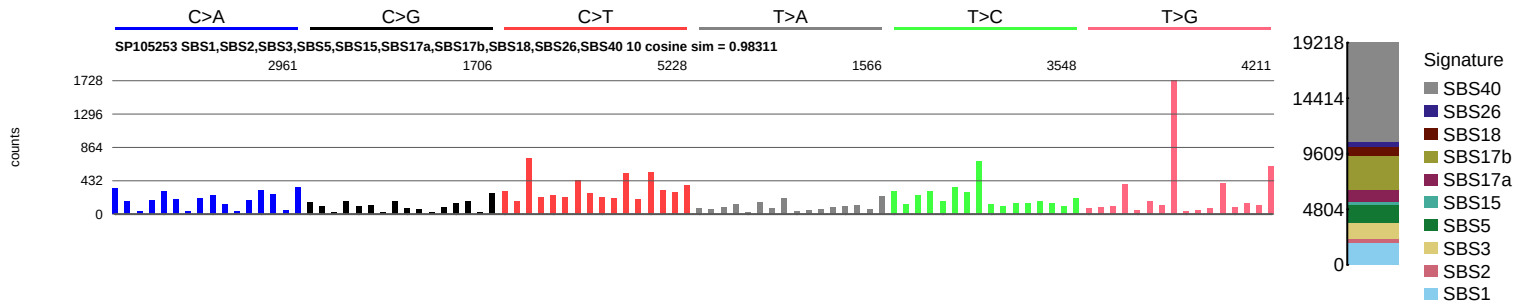
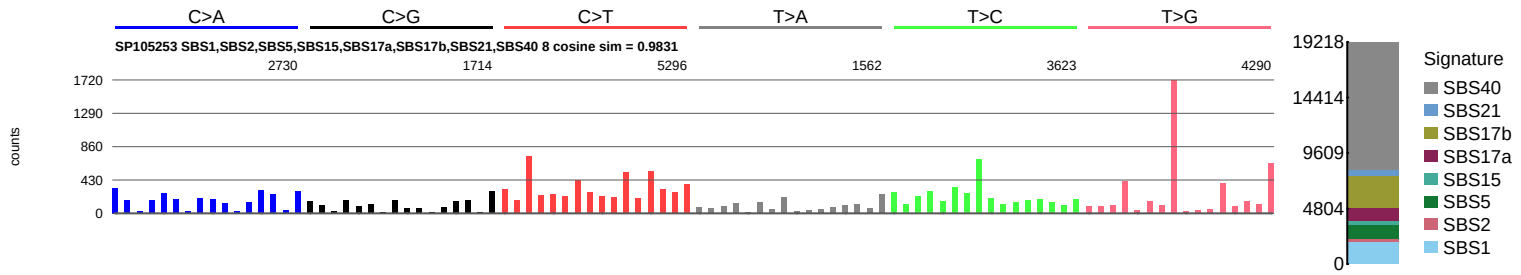
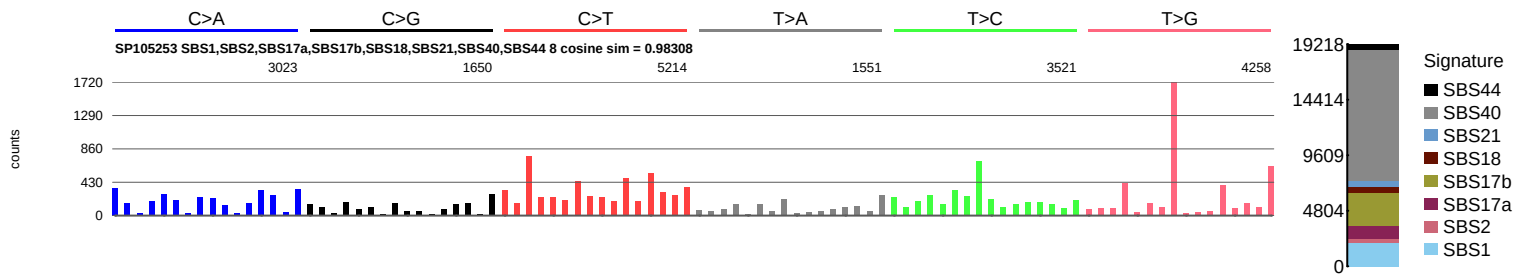
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98312
3012
1687
5185
1552
3496
4284
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98312
2733
1708
5307
1564
3617
4288
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98311
2961
1706
5228
1566
3548
4211
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.9831
2730
1714
5296
1562
3623
4290
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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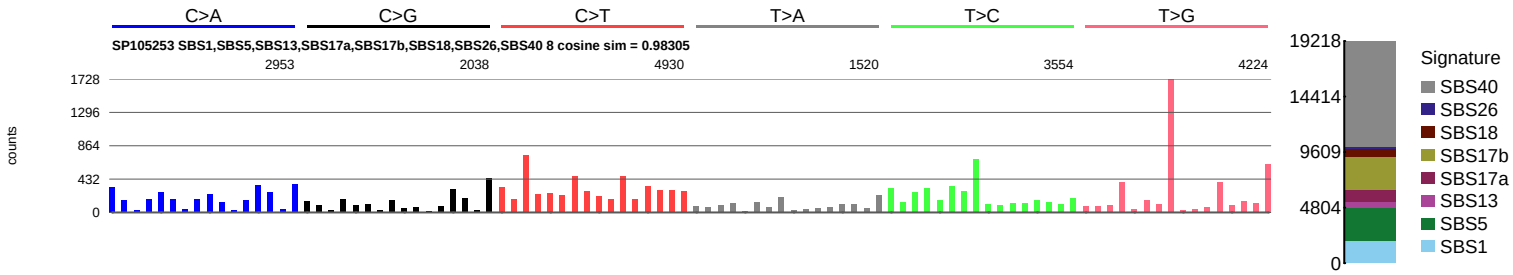
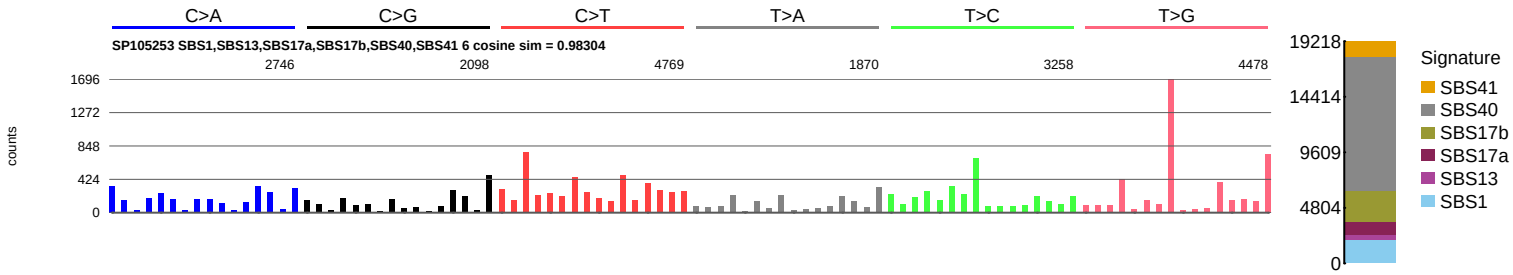
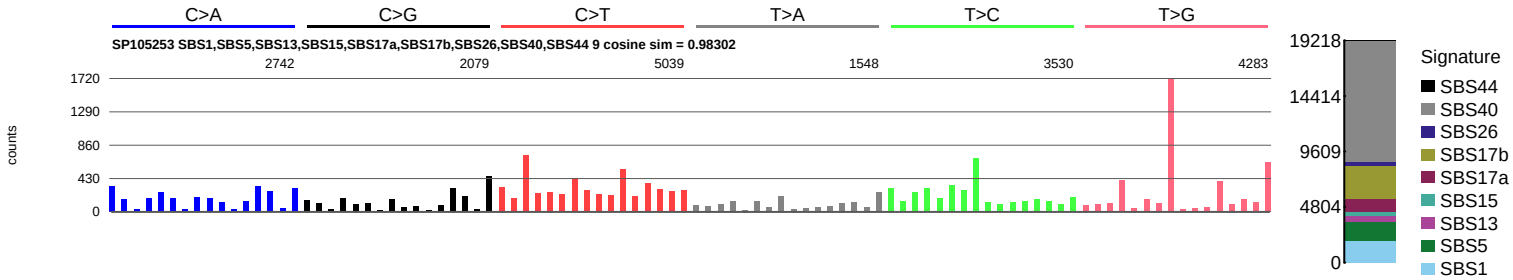
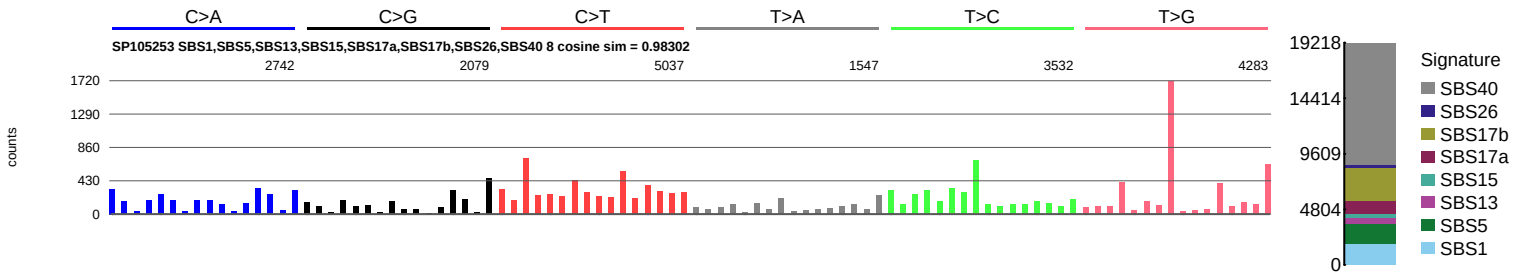
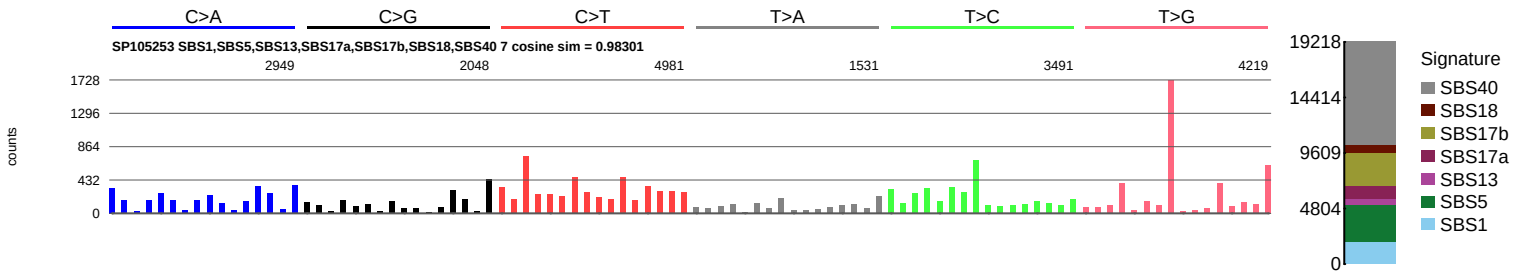
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98308
3023
1650
5214
1551
3521
4258
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98305
2953
2038
4930
1520
3554
4224
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98304
2746
2098
4769
1870
3258
4478
counts
1696
1272
848
424
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98302
2742
2079
5039
1548
3530
4283
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98302
2742
2079
5037
1547
3532
4283
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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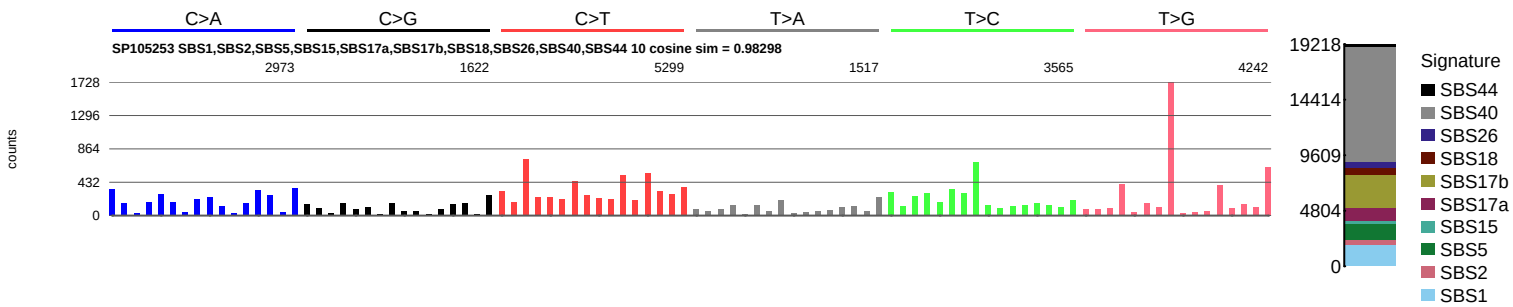
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98301
2949
2048
4981
1531
3491
4219
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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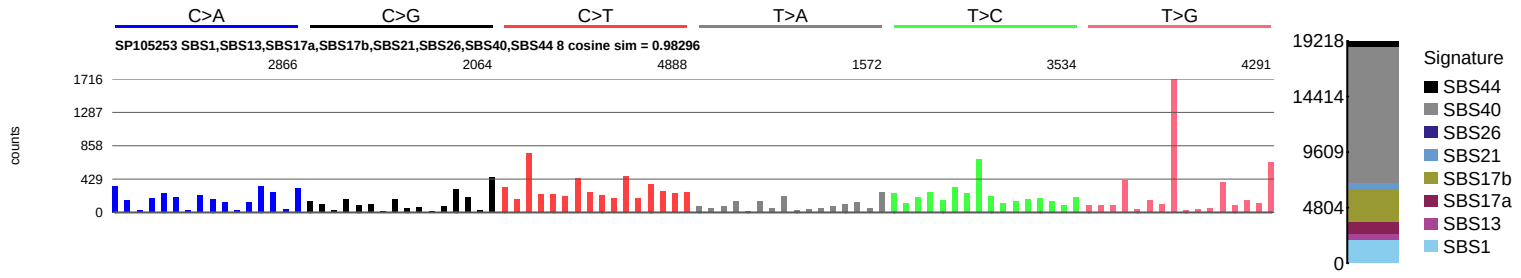
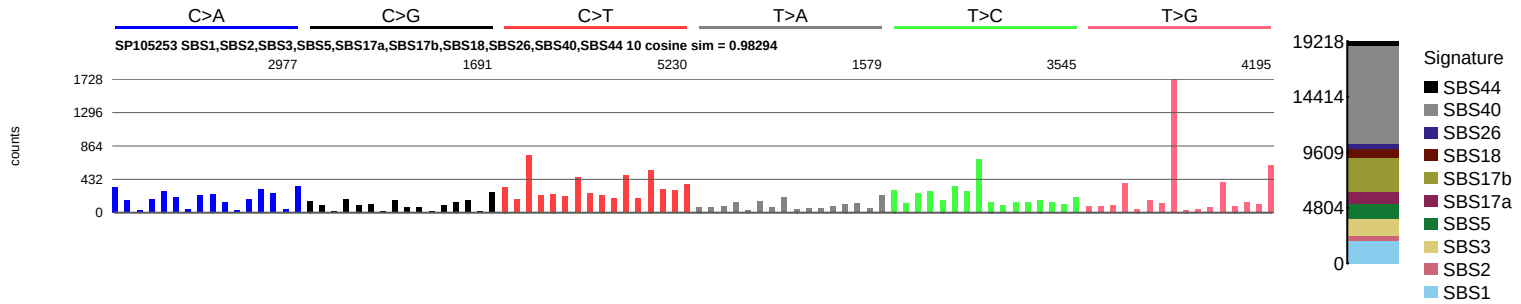
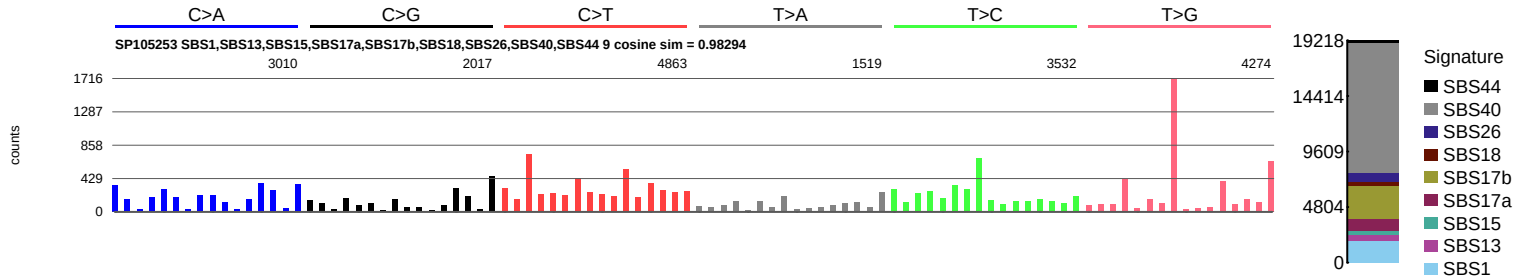
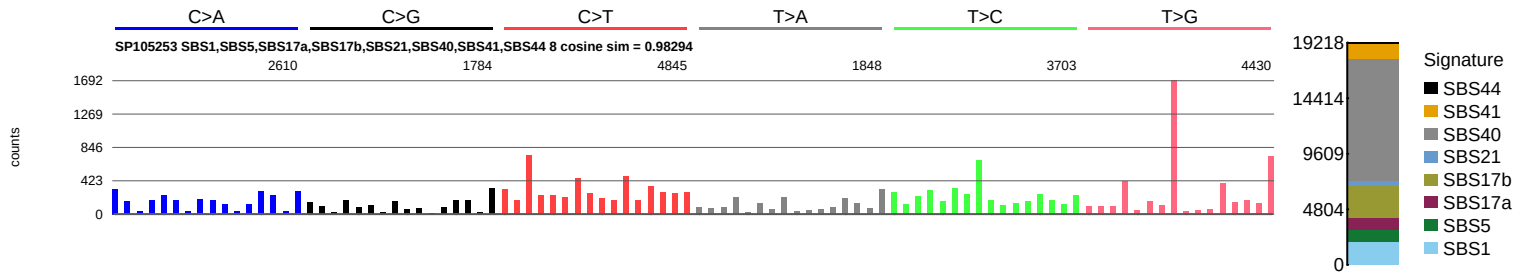
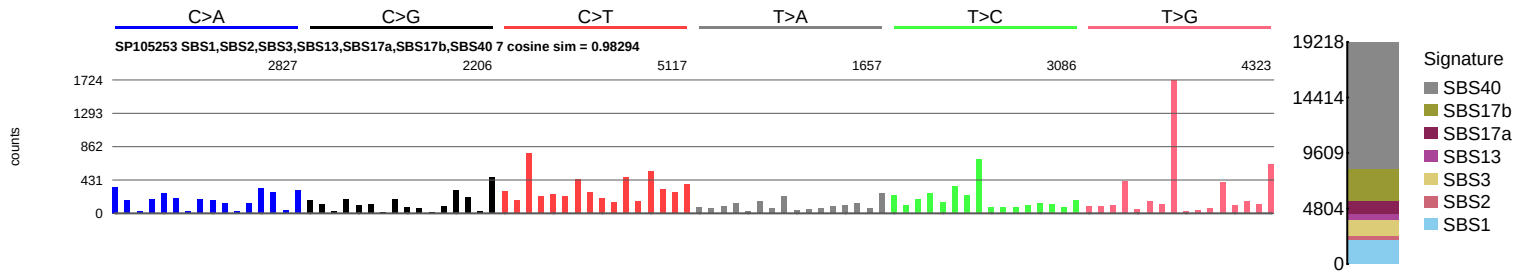
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98298
2973
1622
5299
1517
3565
4242
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

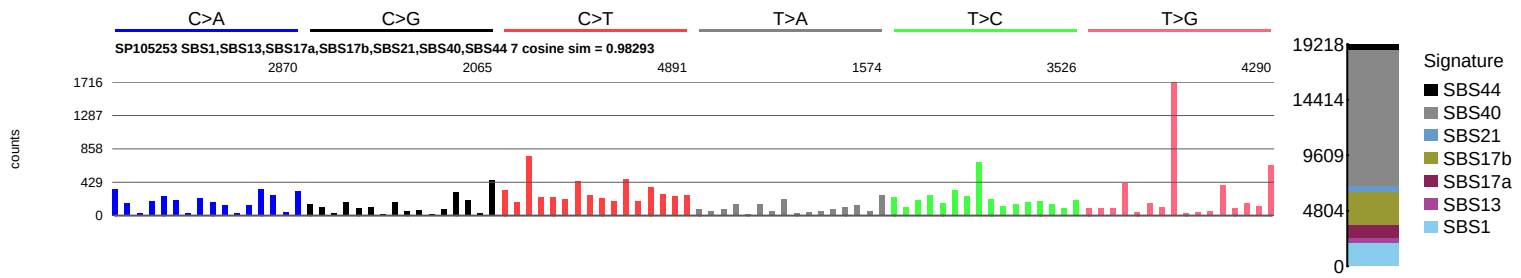

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98296
2866
2064
4888
1572
3534
4291
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1

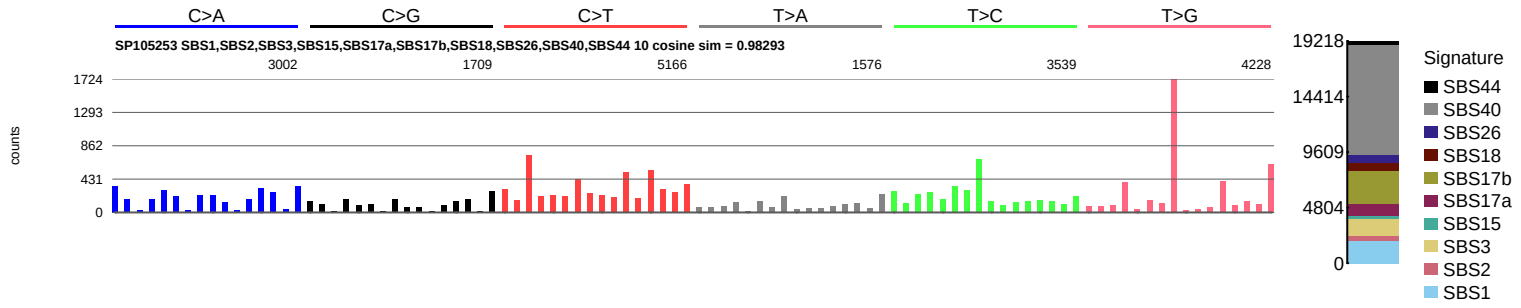

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98294
2977
1691
5230
1579
3545
4195
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

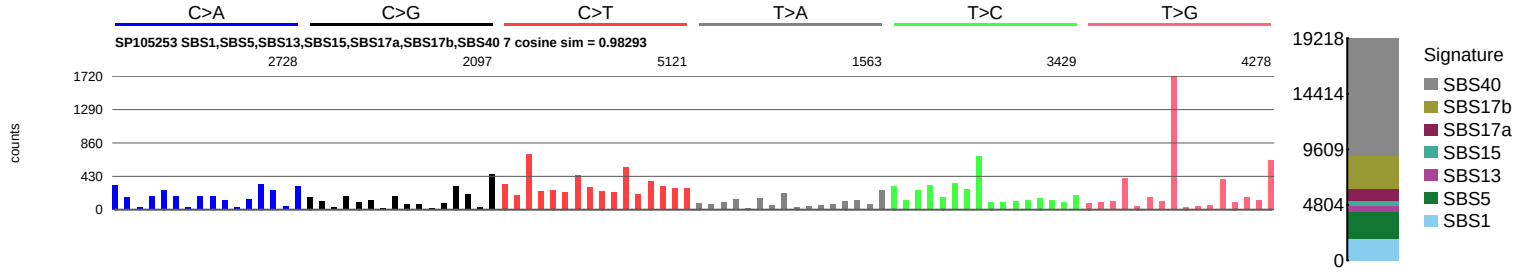

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98294
3010
2017
4863
1519
3532
4274
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.98294
2610
1784
4845
1848
3703
4430
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98294
2827
2206
5117
1657
3086
4323
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98293
2870
2065
4891
1574
3526
4290
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98293
3002
1709
5166
1576
3539
4228
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
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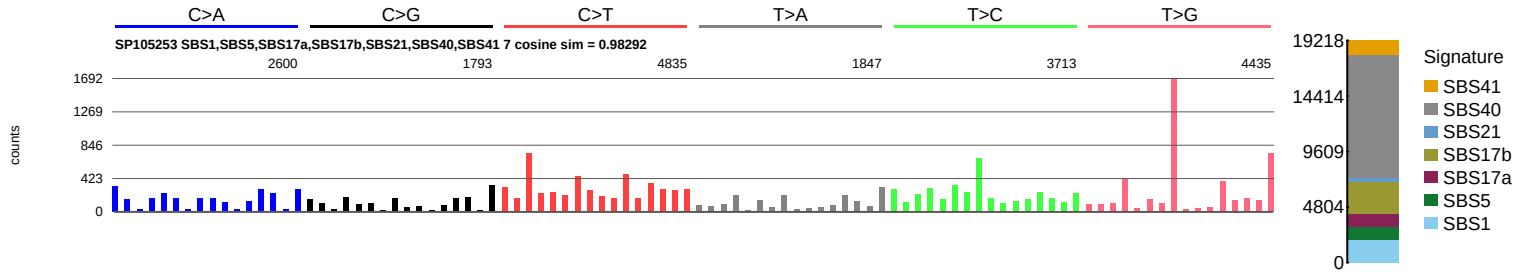
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98293
2728
2097
5121
1563
3429
4278
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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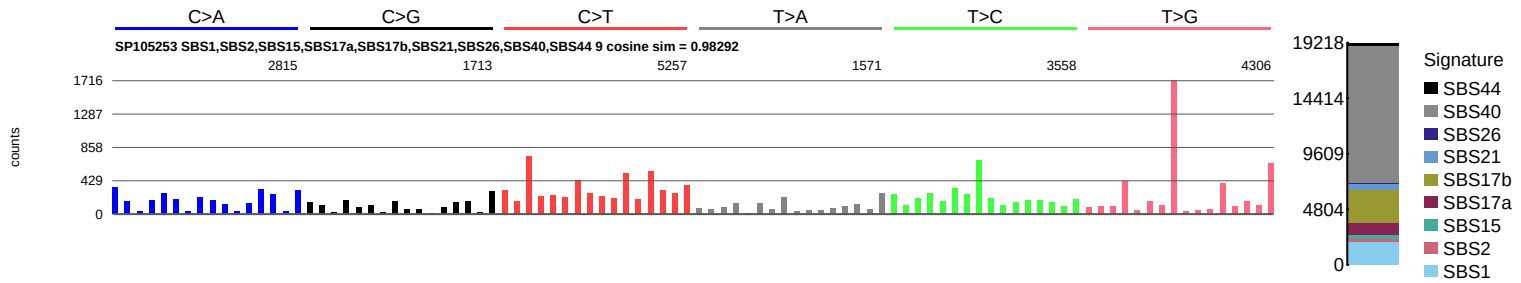
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.98292
2600
1793
4835
1847
3713
4435
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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C>T
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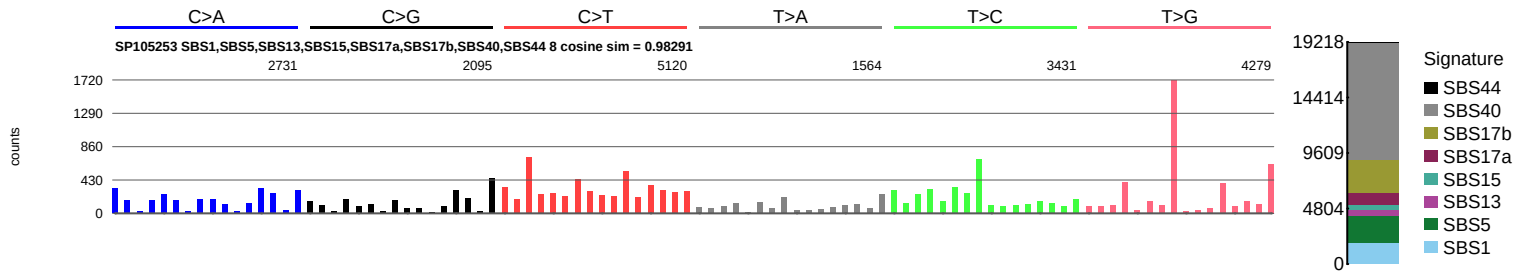
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98292
2815
1713
5257
1571
3558
4306
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1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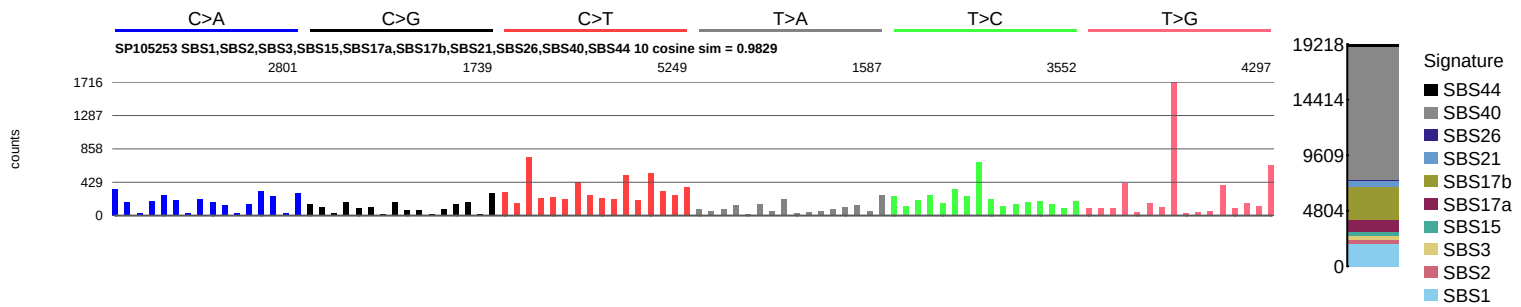
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98291
2731
2095
5120
1564
3431
4279
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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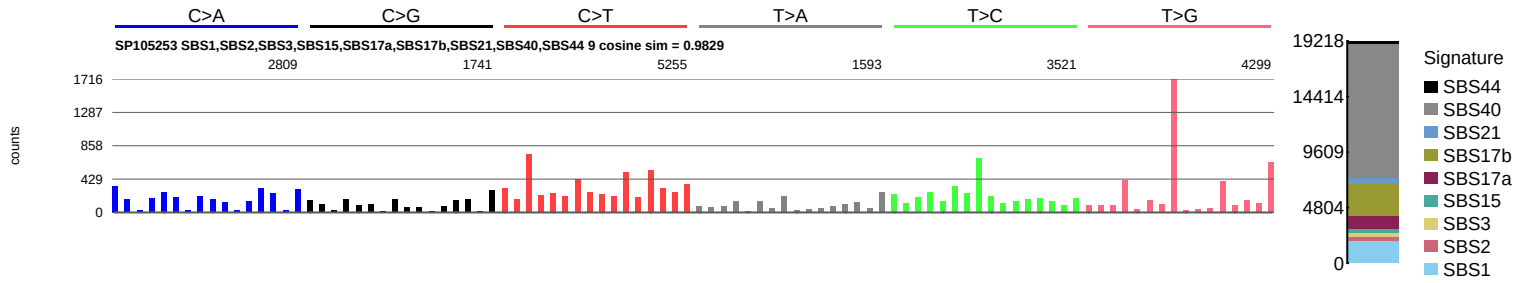
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.9829
2801
1739
5249
1587
3552
4297
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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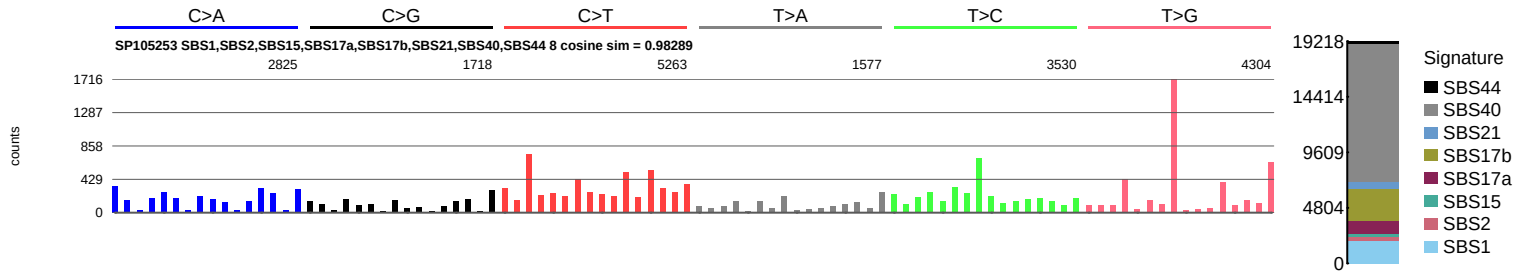
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.9829
2809
1741
5255
1593
3521
4299
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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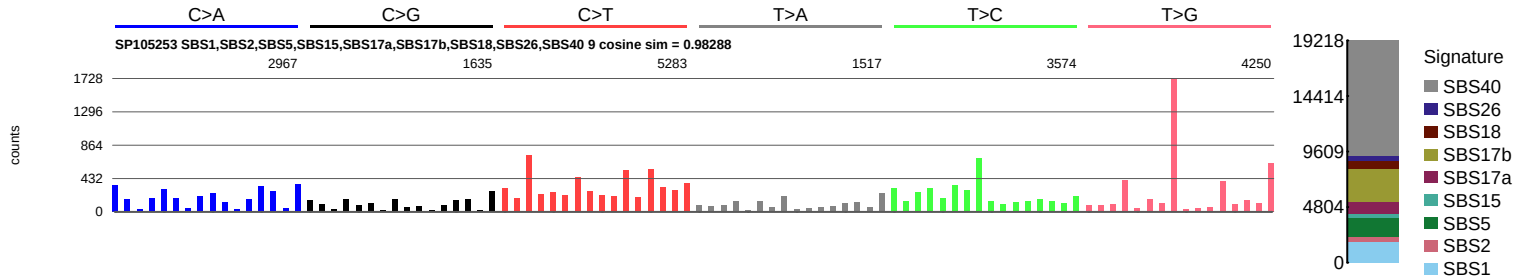
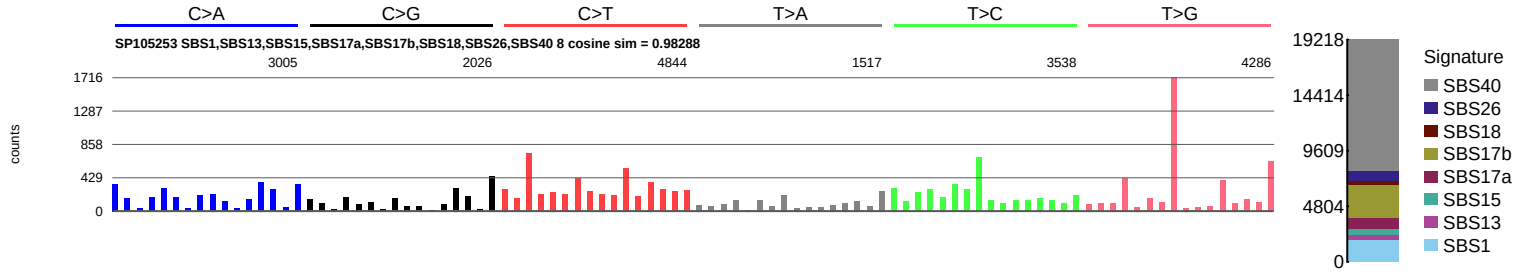
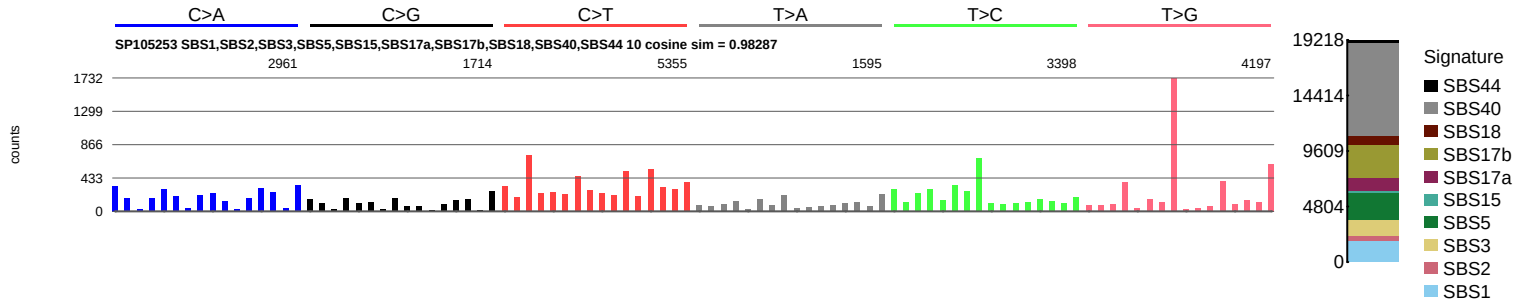
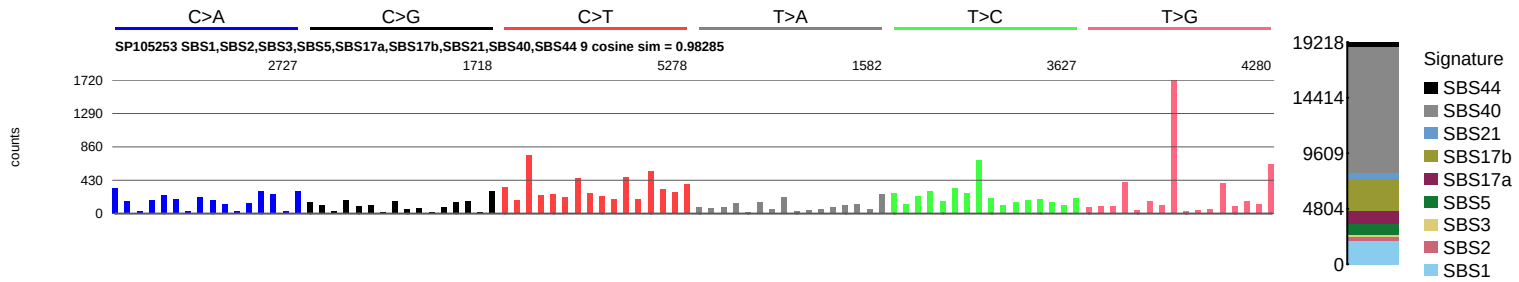
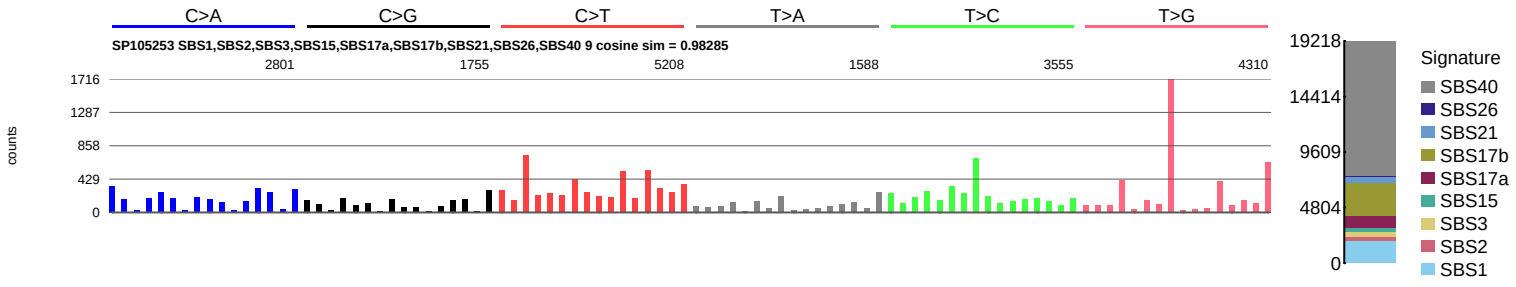
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98289
2825
1718
5263
1577
3530
4304
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1

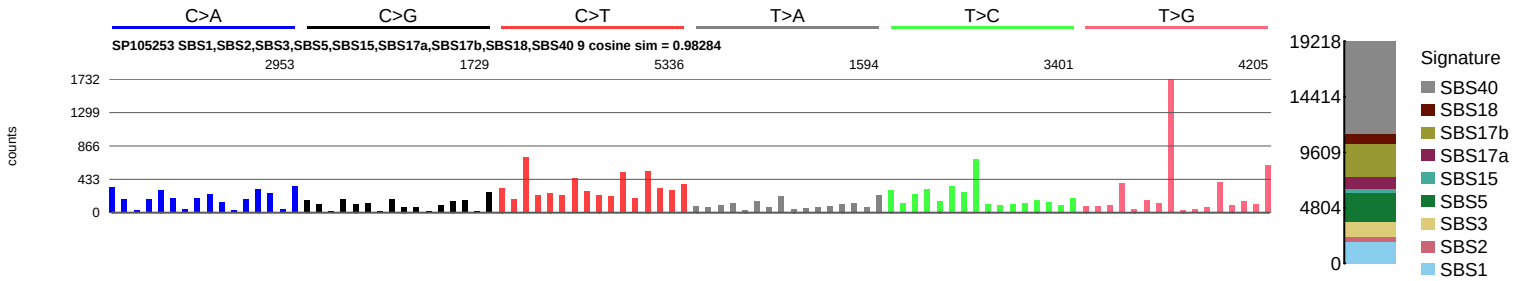

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98288
2967
1635
5283
1517
3574
4250
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

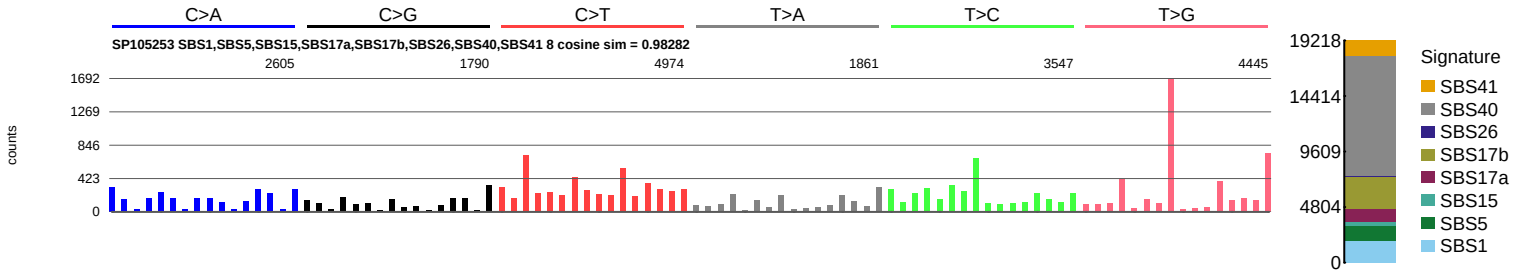

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98288
3005
2026
4844
1517
3538
4286
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

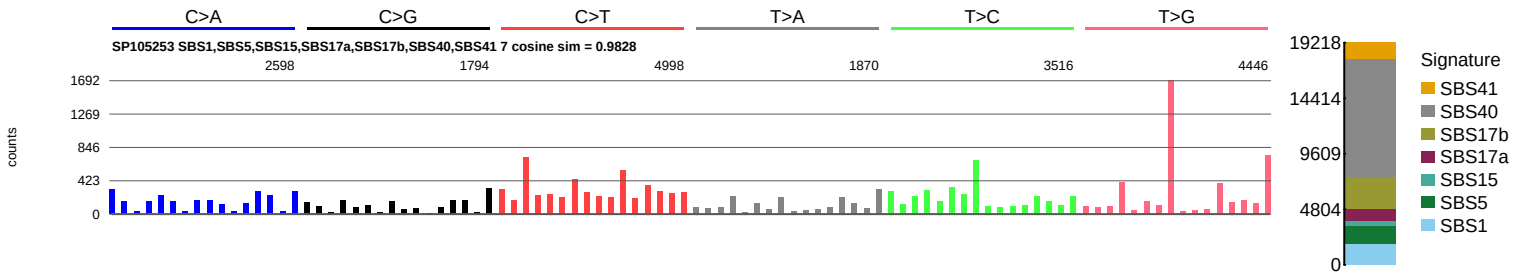

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98287
2961
1714
5355
1595
3398
4197
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

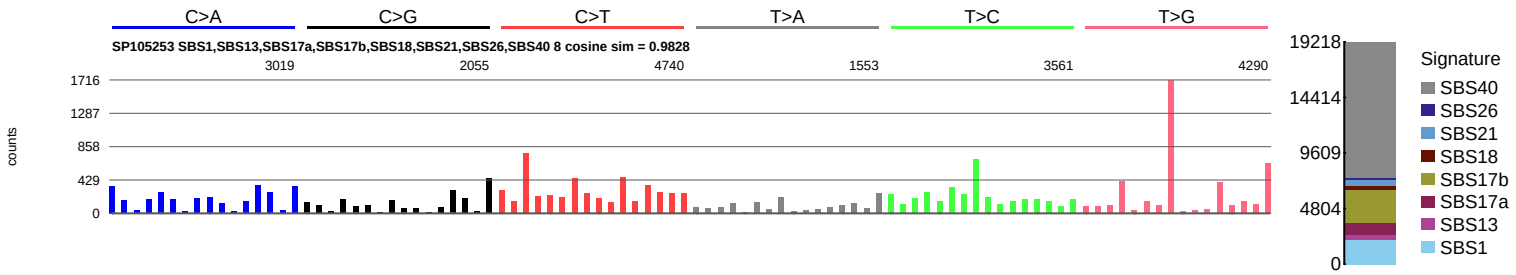

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 9 cosine sim = 0.98285
2727
1718
5278
1582
3627
4280
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

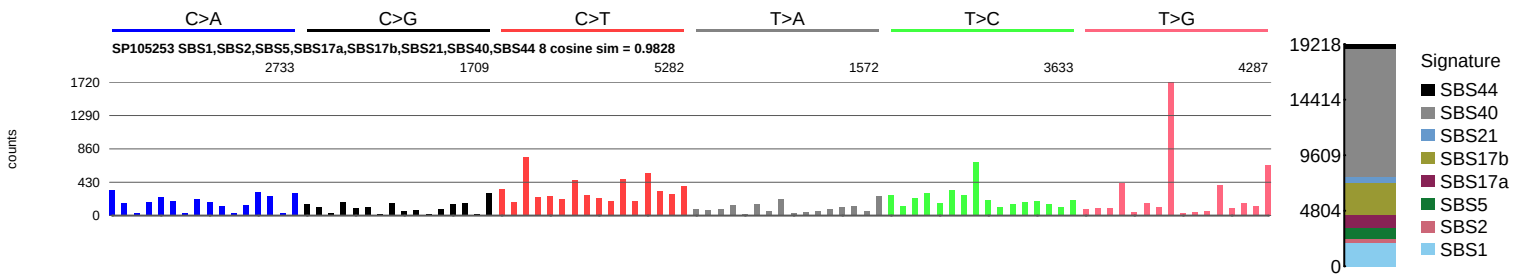

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 9 cosine sim = 0.98285
2801
1755
5208
1588
3555
4310
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

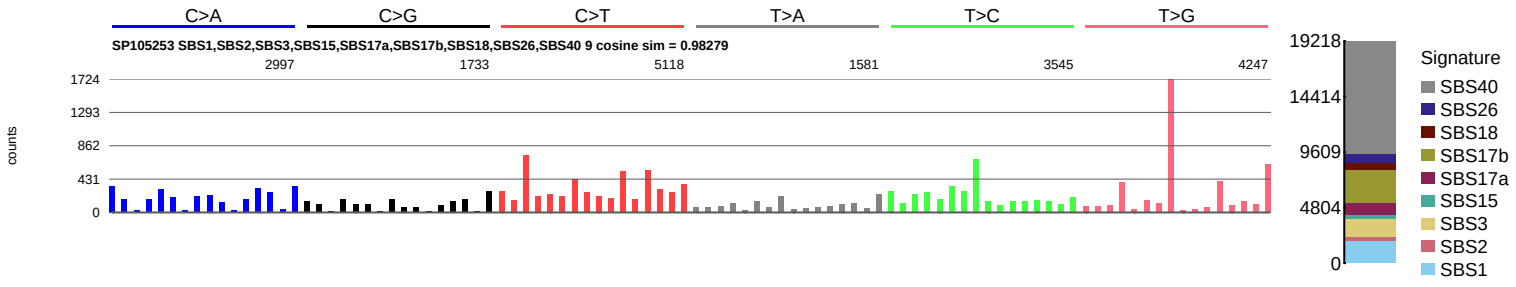

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98284
2953
1729
5336
1594
3401
4205
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

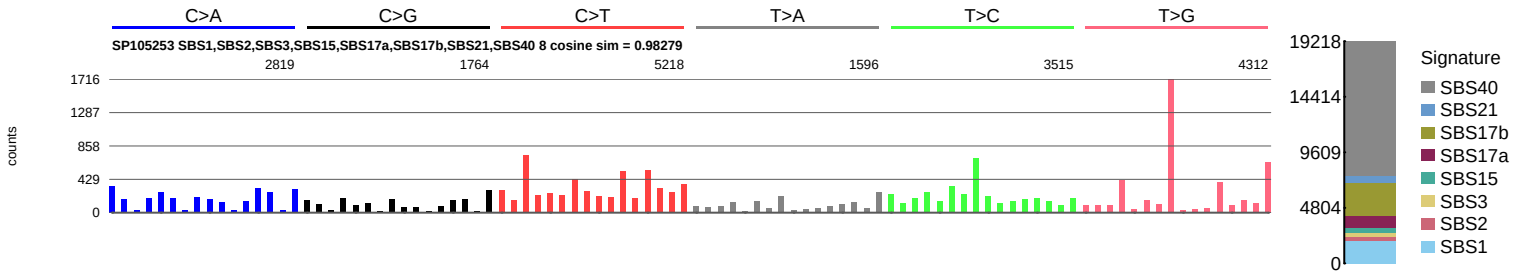

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.98282
2605
1790
4974
1861
3547
4445
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1

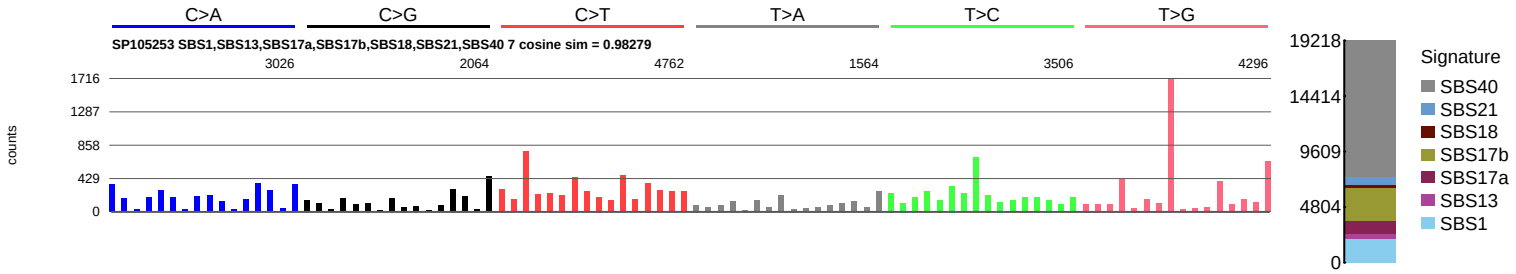

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.9828
2598
1794
4998
1870
3516
4446
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1

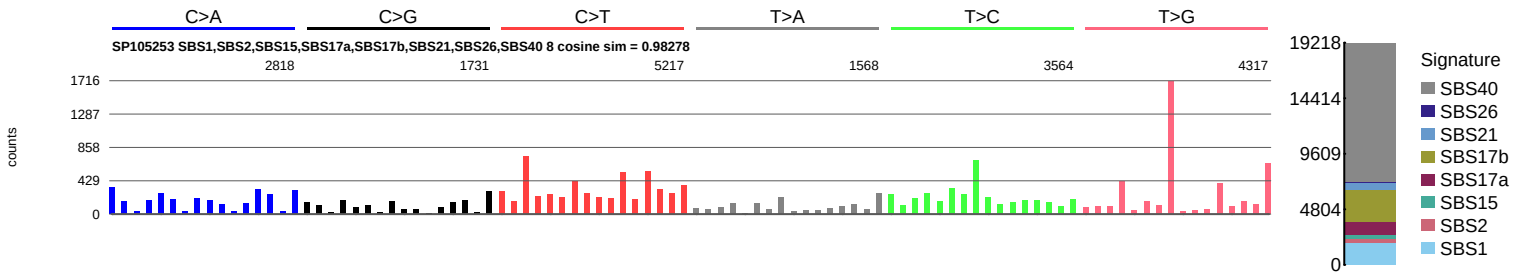

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.9828
3019
2055
4740
1553
3561
4290
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

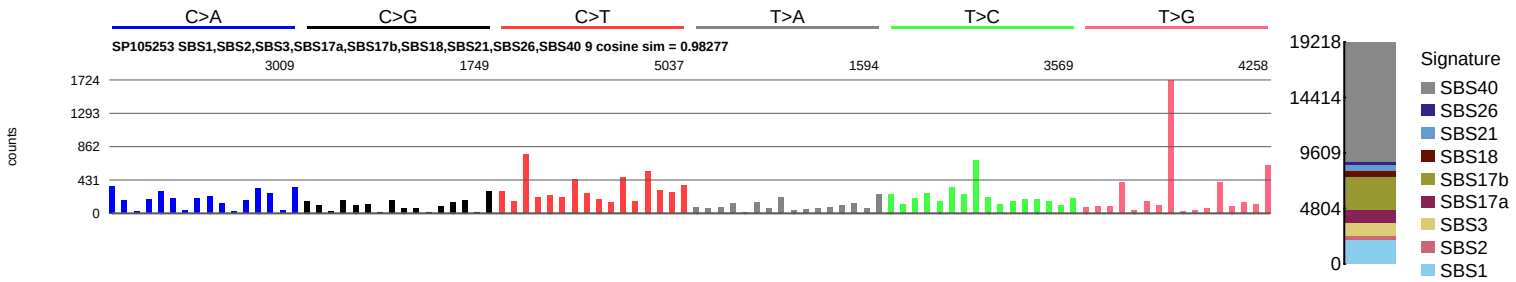

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.9828
2733
1709
5282
1572
3633
4287
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1

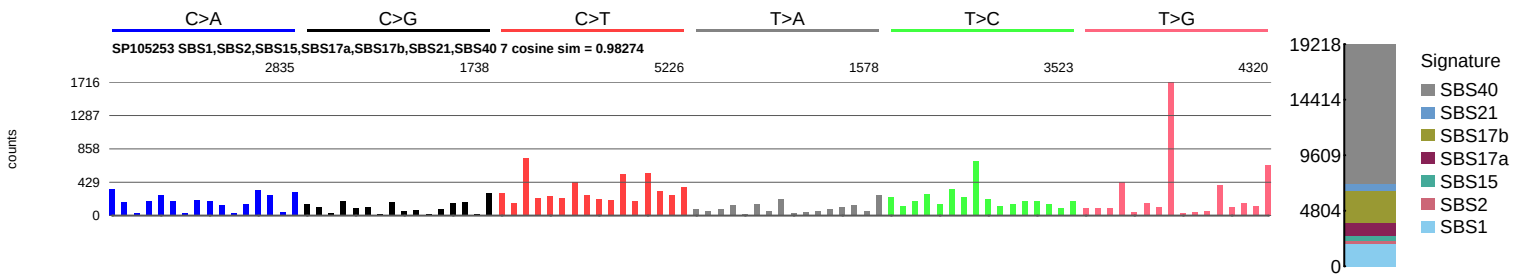

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98279
2997
1733
5118
1581
3545
4247
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

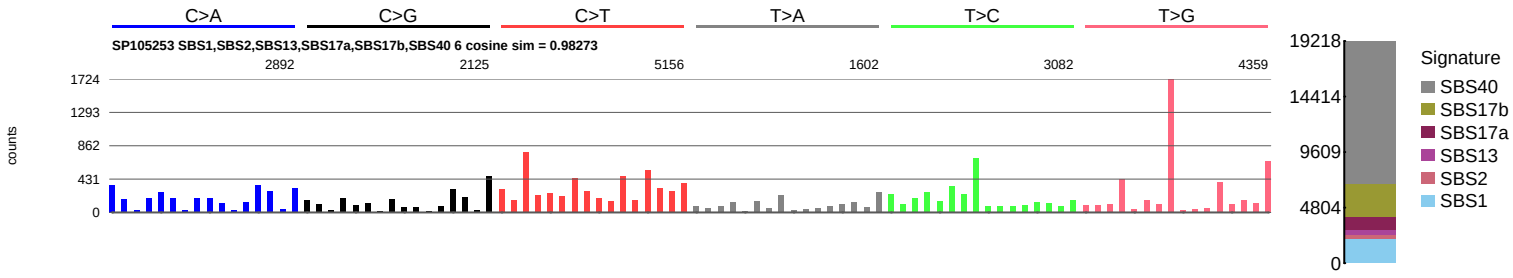

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98279
2819
1764
5218
1596
3515
4312
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

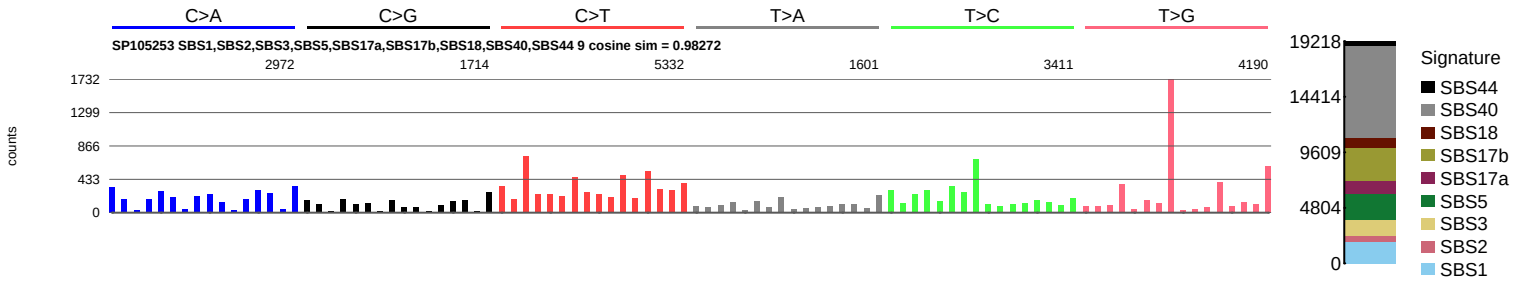

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98279
3026
2064
4762
1564
3506
4296
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1

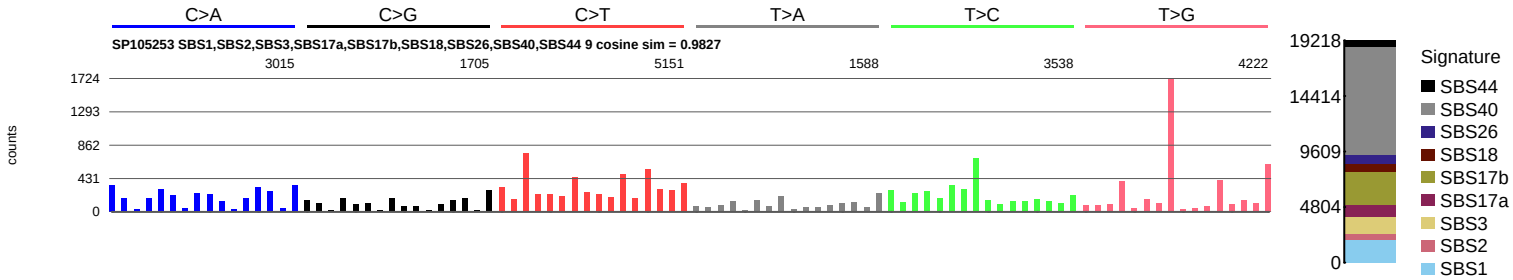

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98278
2818
1731
5217
1568
3564
4317
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98277
3009
1749
5037
1594
3569
4258
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98274
2835
1738
5226
1578
3523
4320
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98273
2892
2125
5156
1602
3082
4359
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98272
2972
1714
5332
1601
3411
4190
1732
1299
866
433
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9827
3015
1705
5151
1588
3538
4222
1724
1293
862
431
0
counts
19218
14414
9609
4804
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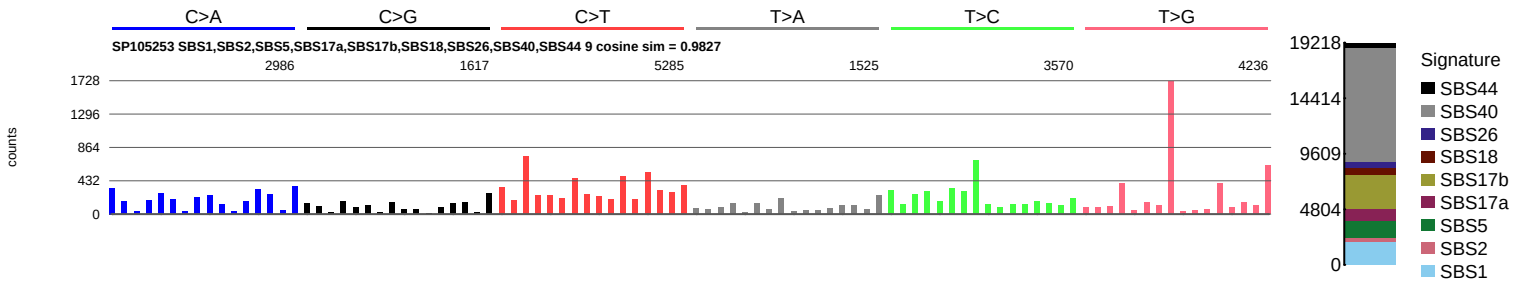
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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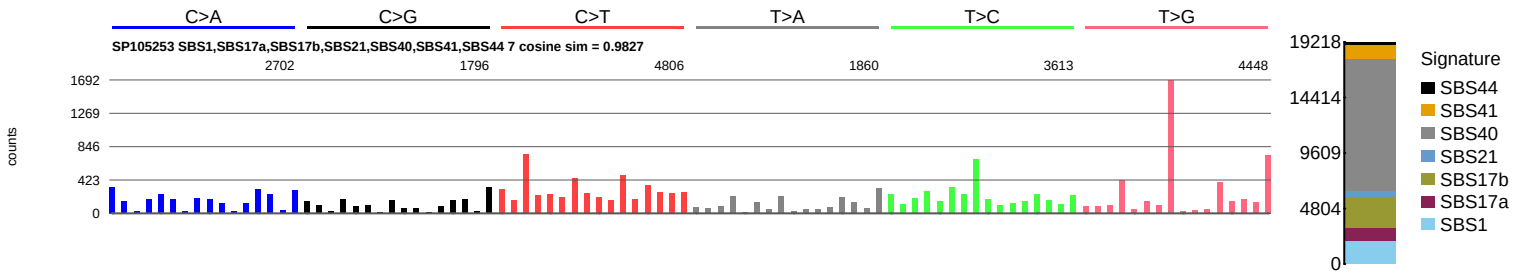
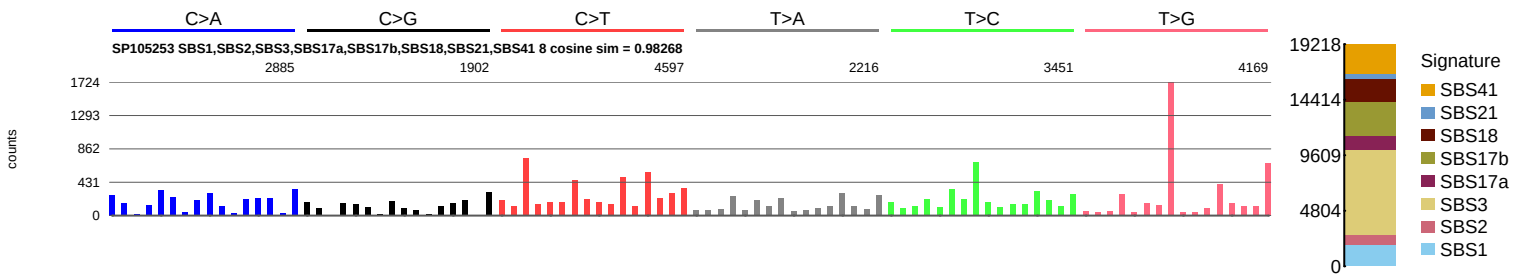
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9827
2986
1617
5285
1525
3570
4236
1728
1296
864
432
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.9827
2702
1796
4806
1860
3613
4448
1692
1269
846
423
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1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C>G
C>T
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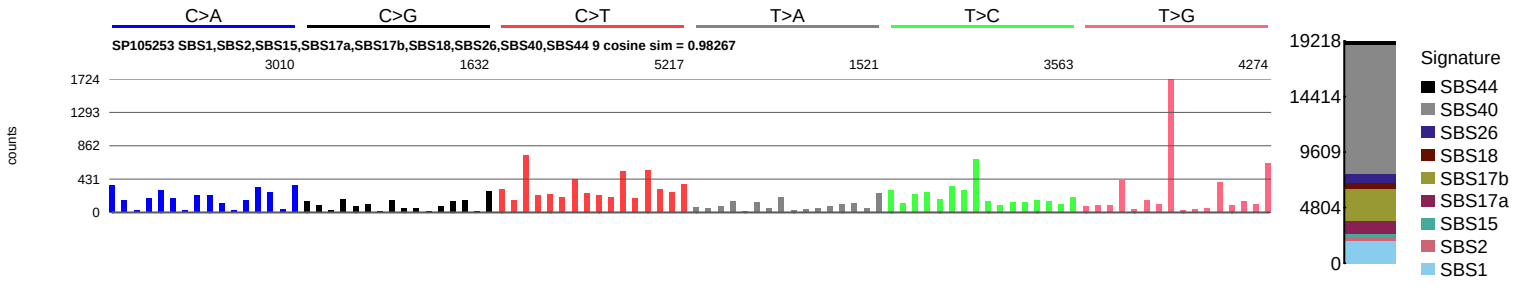
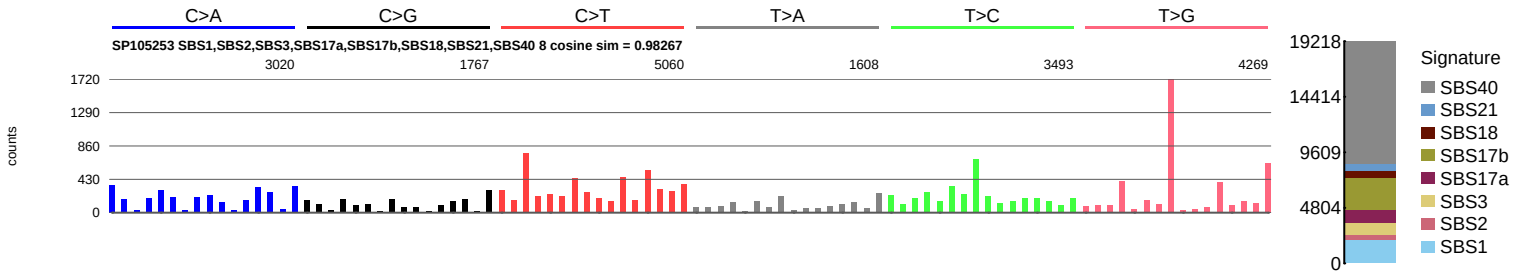
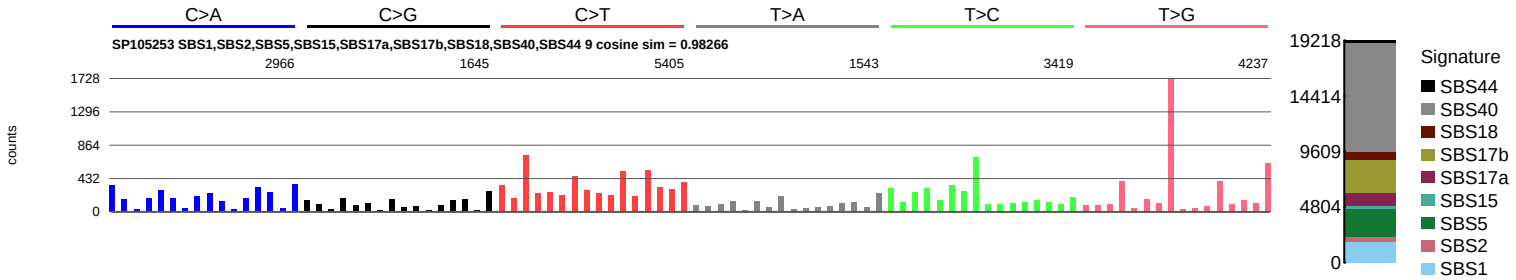
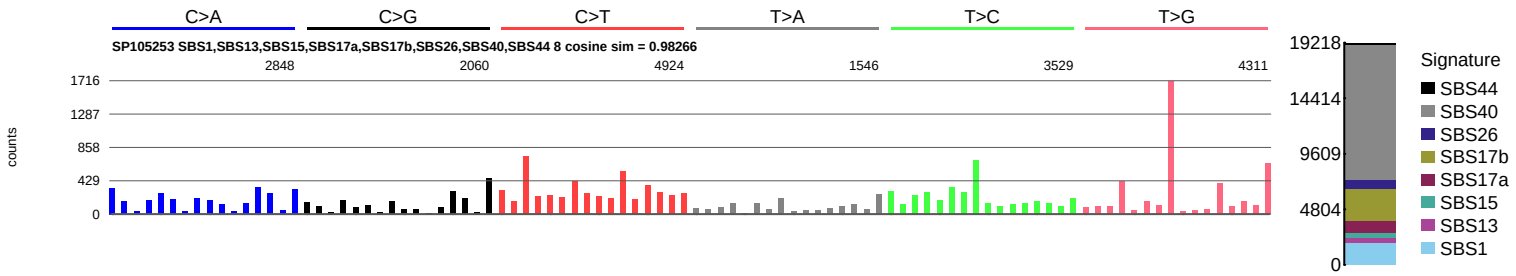
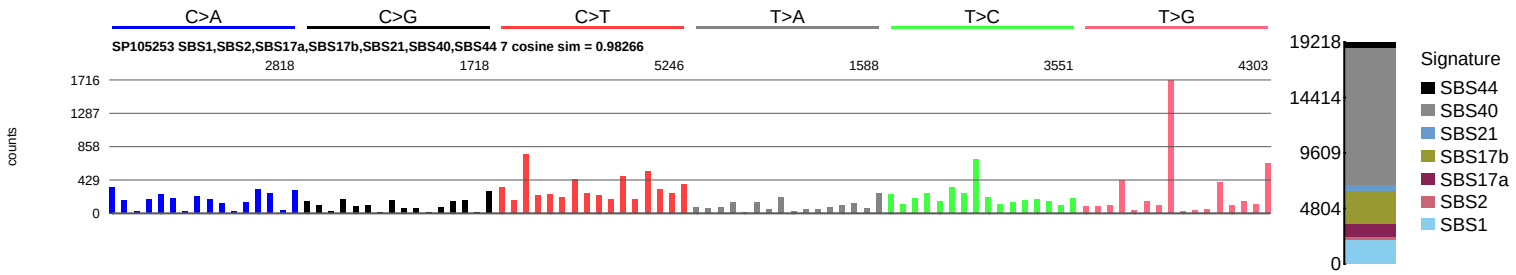
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98268
2885
1902
4597
2216
3451
4169
1724
1293
862
431
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

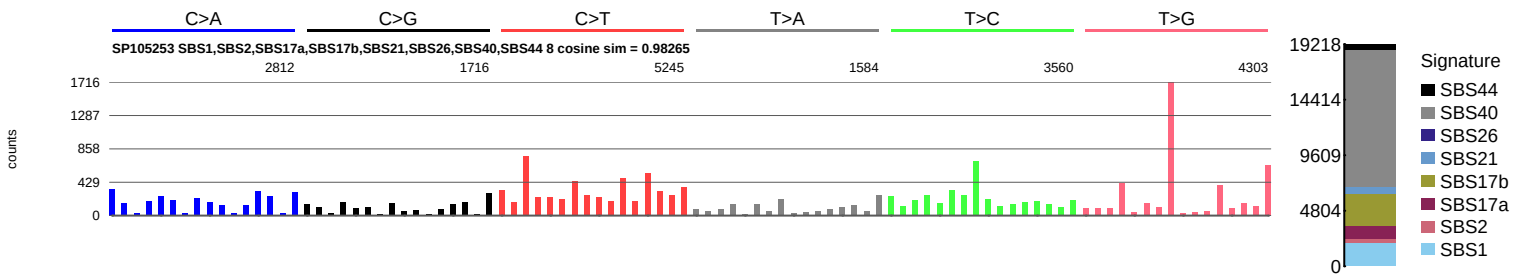

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98267
3010
1632
5217
1521
3563
4274
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

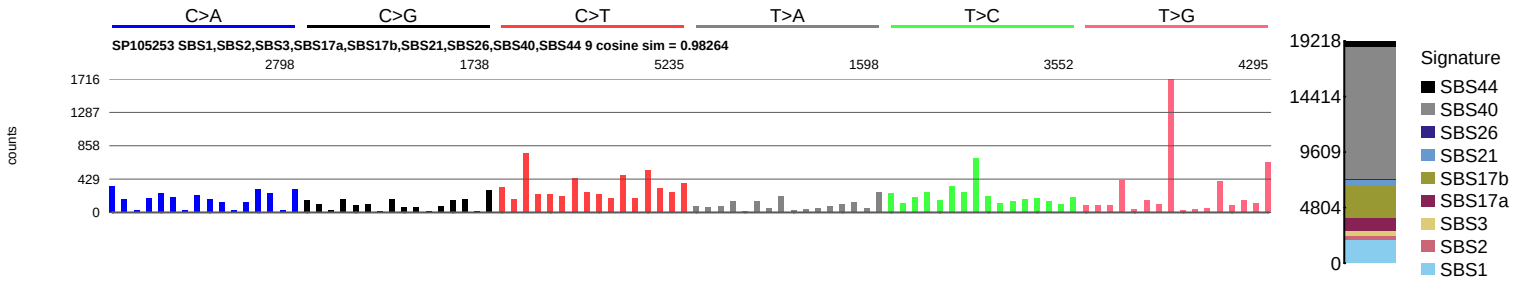

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98267
3020
1767
5060
1608
3493
4269
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

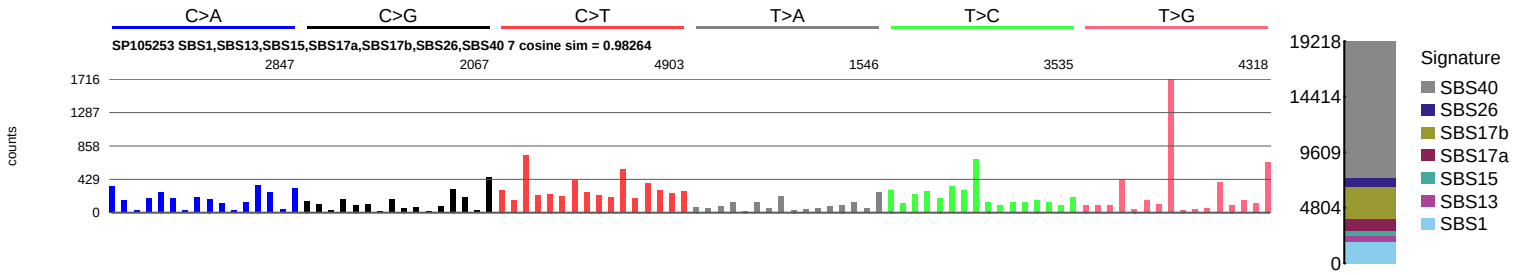

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98266
2966
1645
5405
1543
3419
4237
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

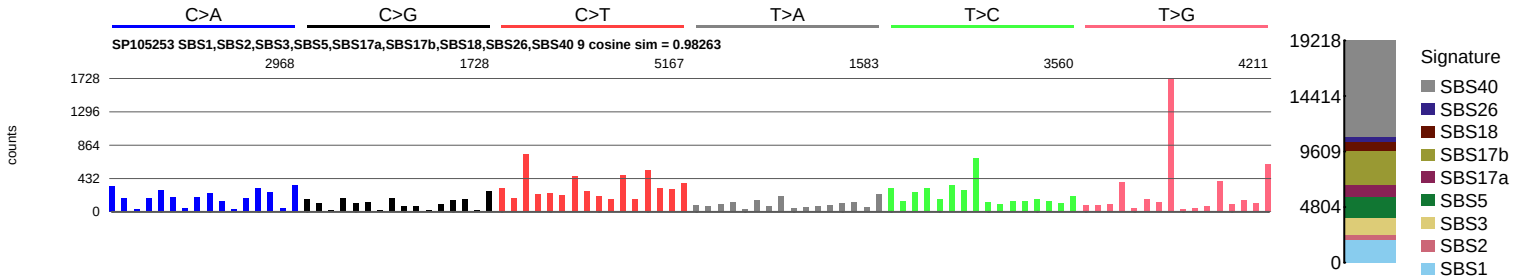

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98266
2848
2060
4924
1546
3529
4311
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1

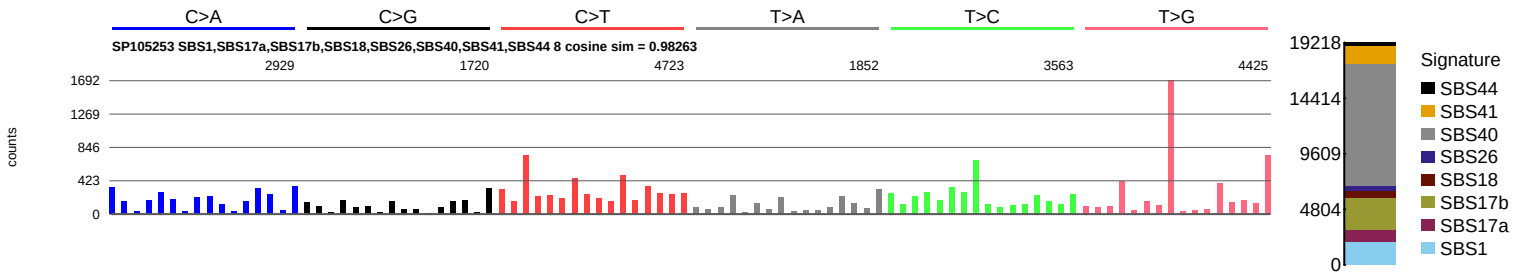

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98266
2818
1718
5246
1588
3551
4303
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1

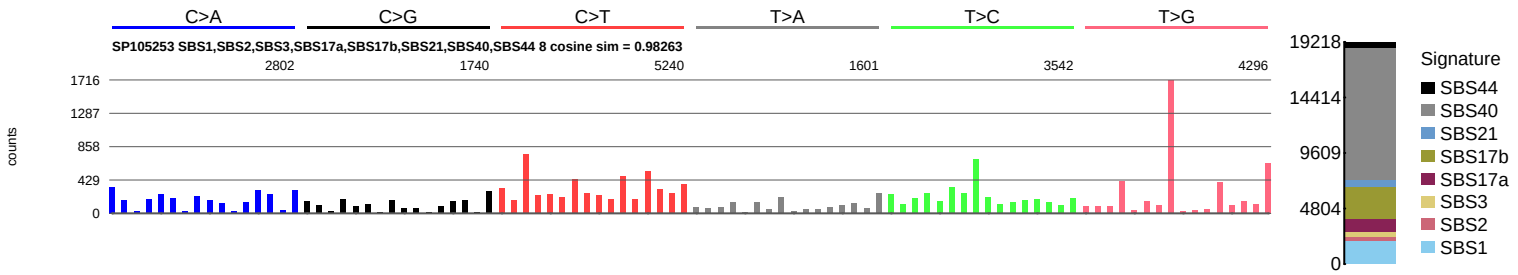

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98265
2812
1716
5245
1584
3560
4303
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1

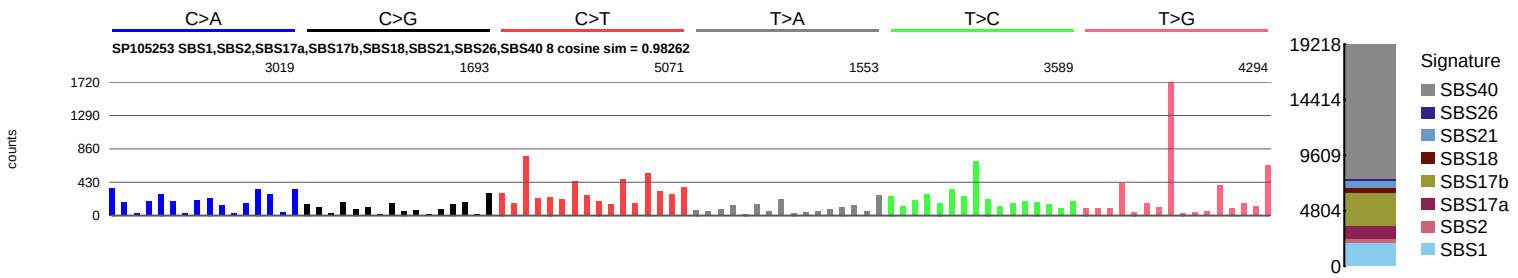

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98264
2798
1738
5235
1598
3552
4295
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98264
2847
2067
4903
1546
3535
4318
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98263
2968
1728
5167
1583
3560
4211
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.98263
2929
1720
4723
1852
3563
4425
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98263
2802
1740
5240
1601
3542
4296
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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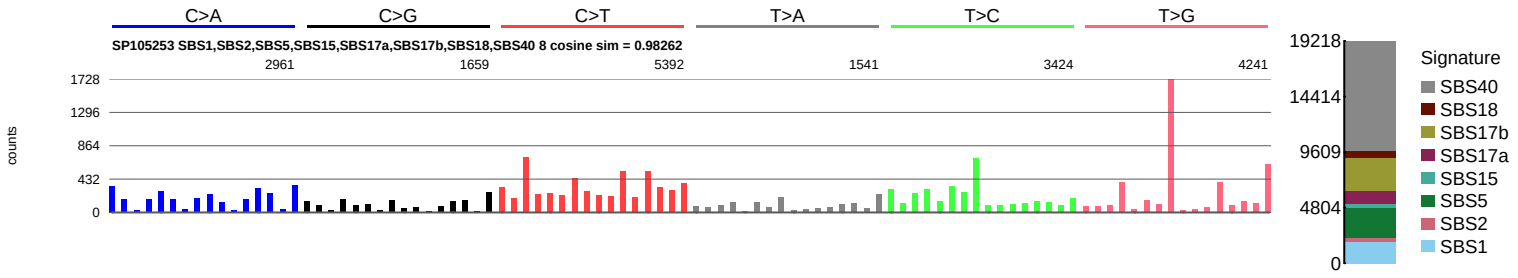
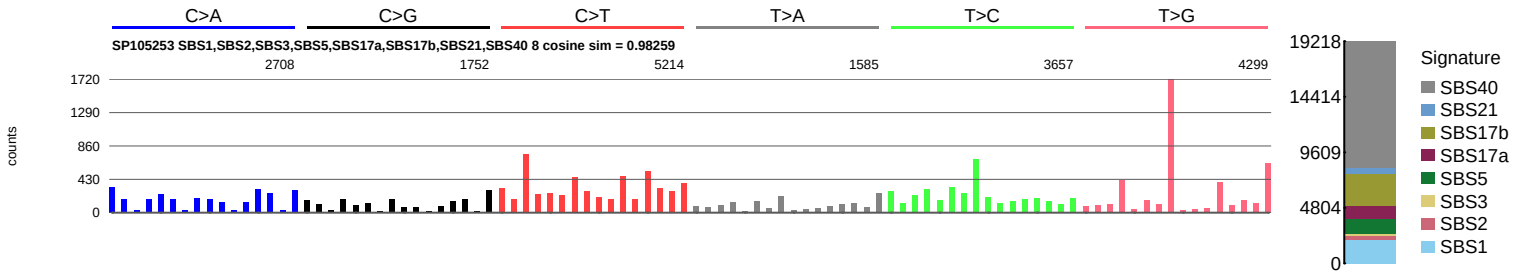
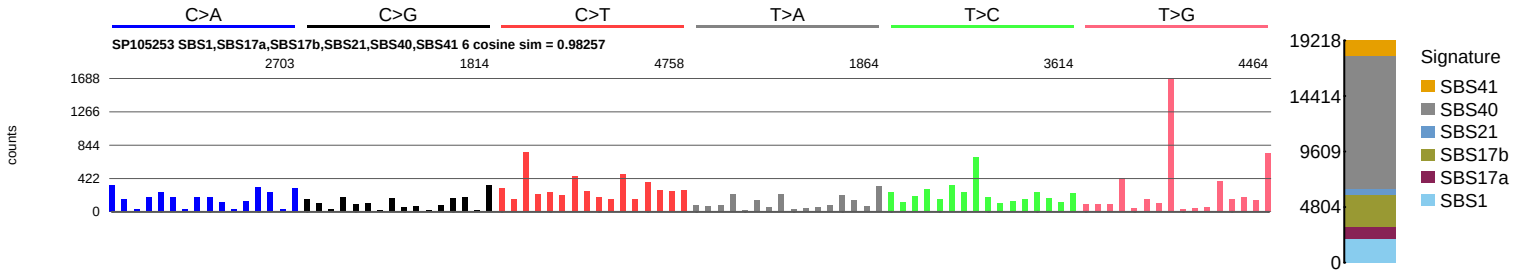
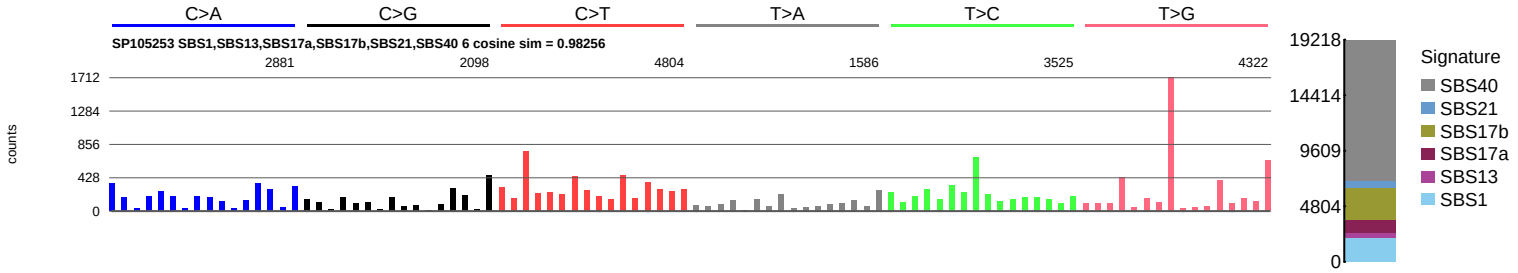
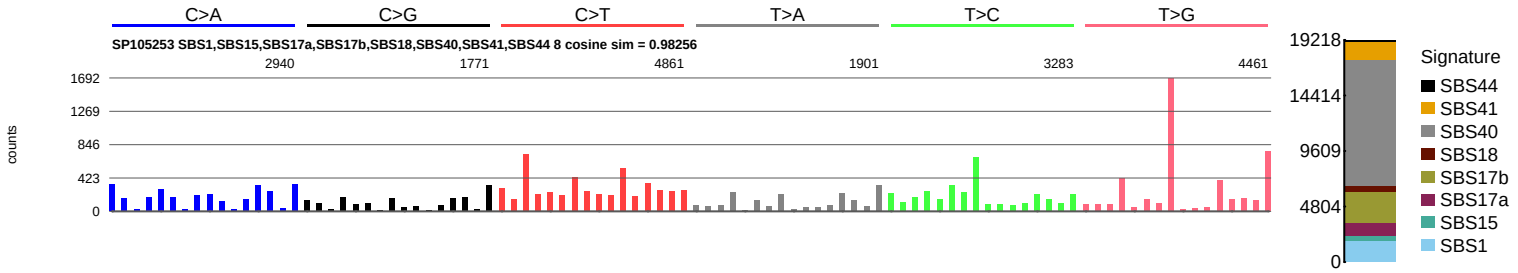
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98262
3019
1693
5071
1553
3589
4294
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

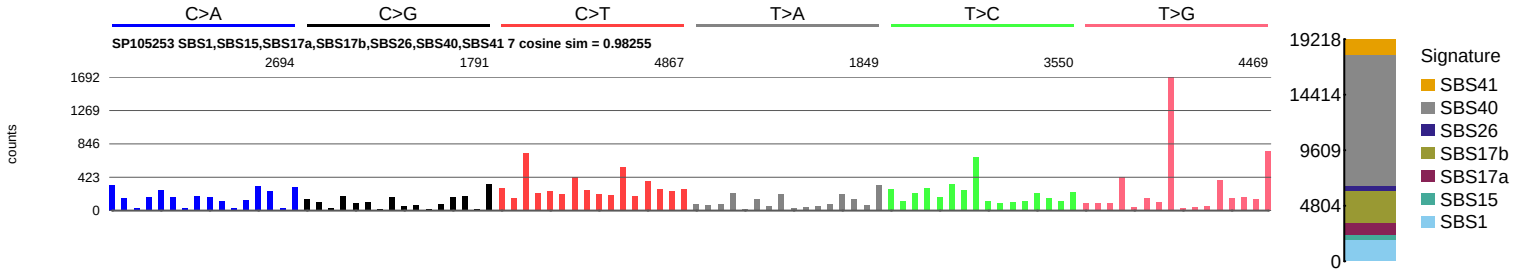

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98262
2961
1659
5392
1541
3424
4241
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

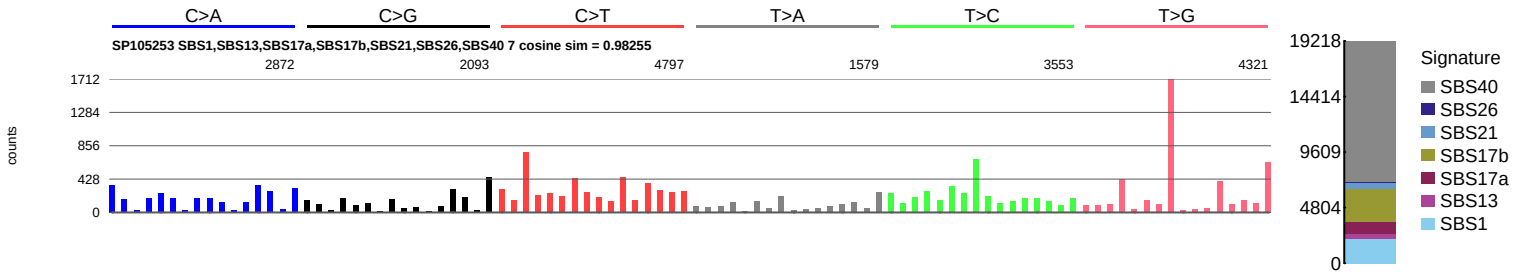

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98259
2708
1752
5214
1585
3657
4299
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

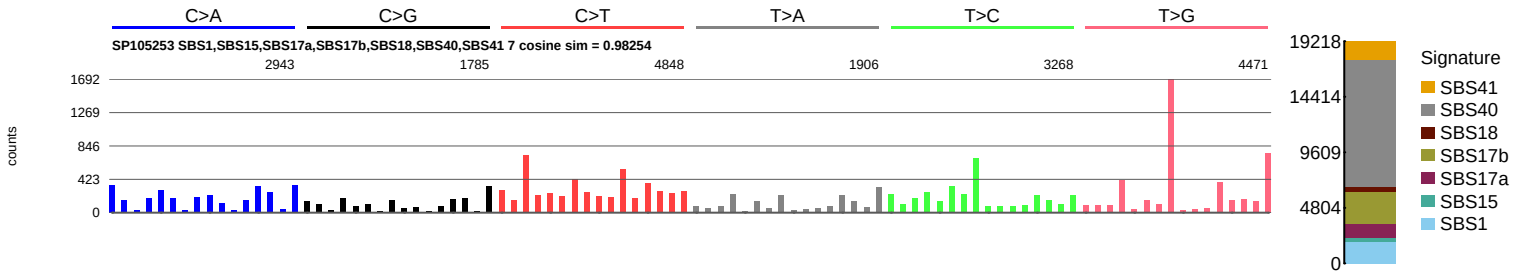

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.98257
2703
1814
4758
1864
3614
4464
counts
1688
1266
844
422
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1

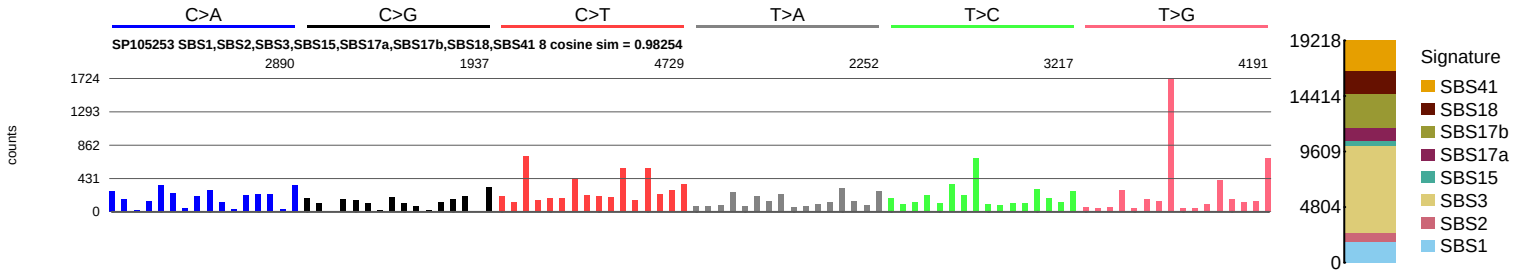

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98256
2881
2098
4804
1586
3525
4322
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.98256
2940
1771
4861
1901
3283
4461
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.98255
2694
1791
4867
1849
3550
4469
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98255
2872
2093
4797
1579
3553
4321
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.98254
2943
1785
4848
1906
3268
4471
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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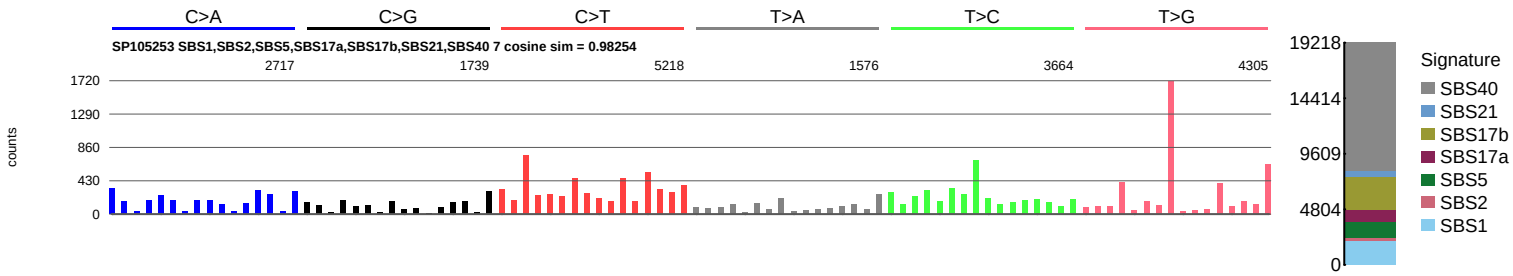
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98254
2890
1937
4729
2252
3217
4191
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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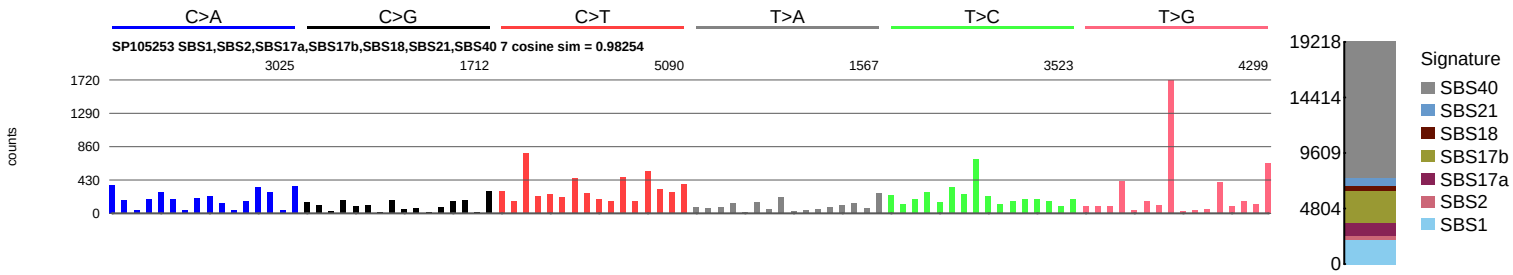
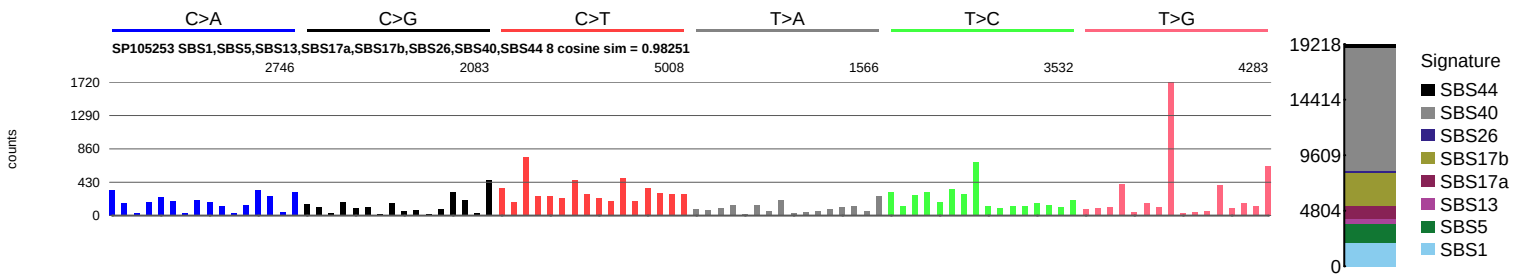
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98254
2717
1739
5218
1576
3664
4305
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98254
3025
1712
5090
1567
3523
4299
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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C>G
C>T
T>A
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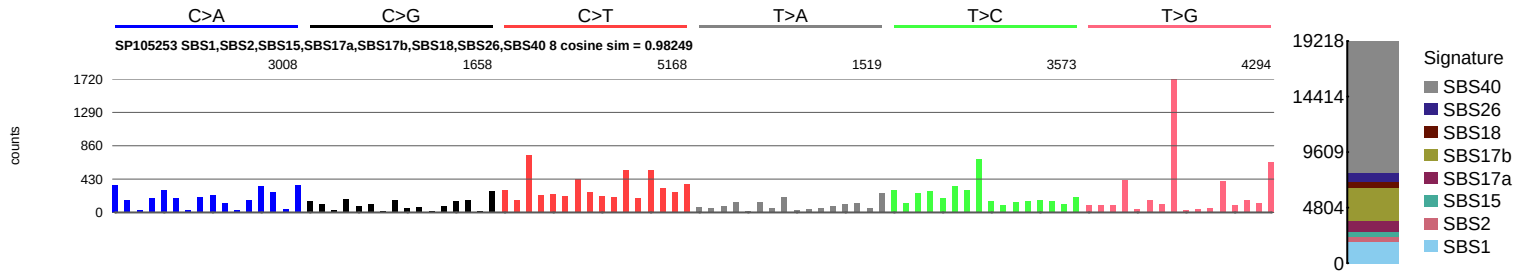
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98251
2746
2083
5008
1566
3532
4283
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1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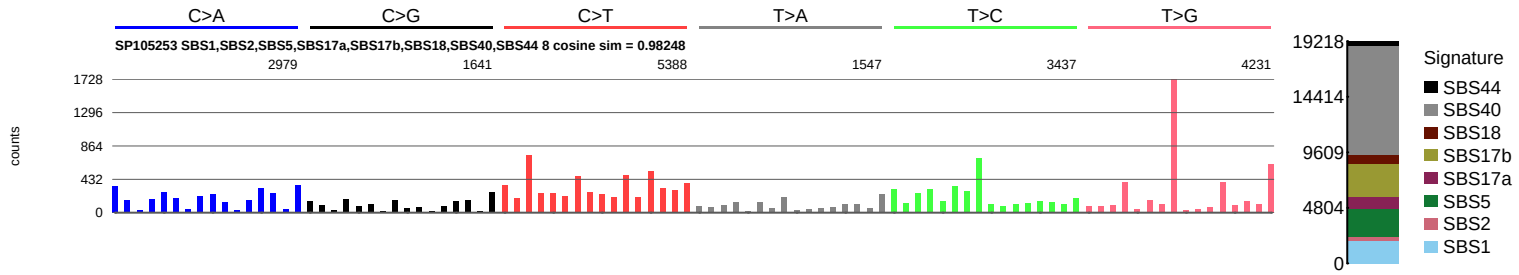
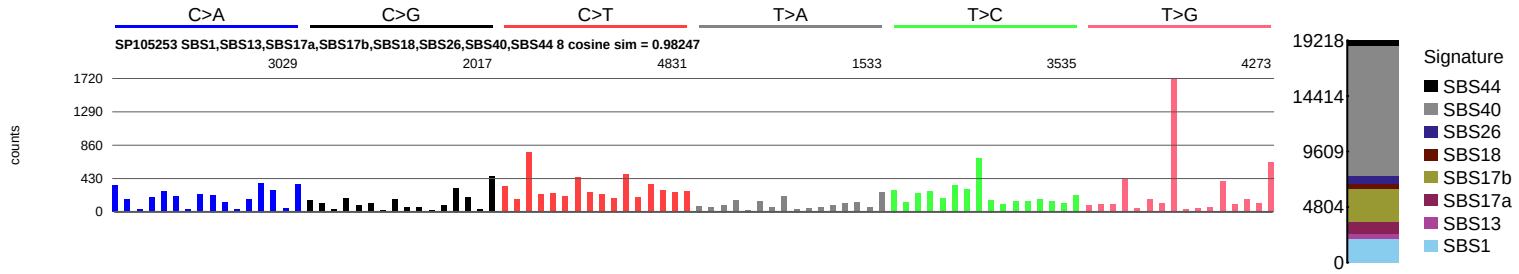
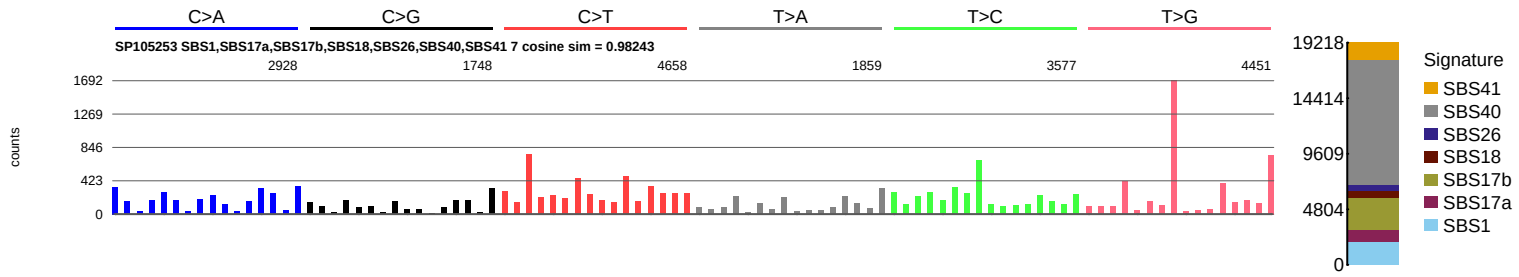
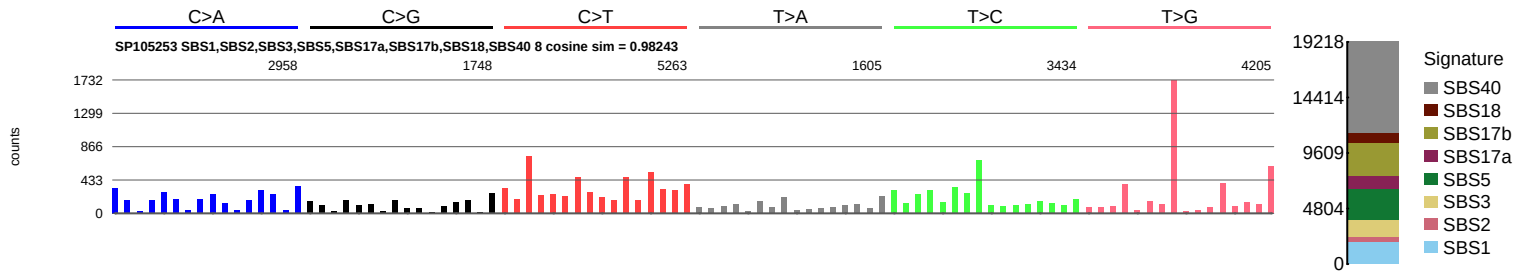
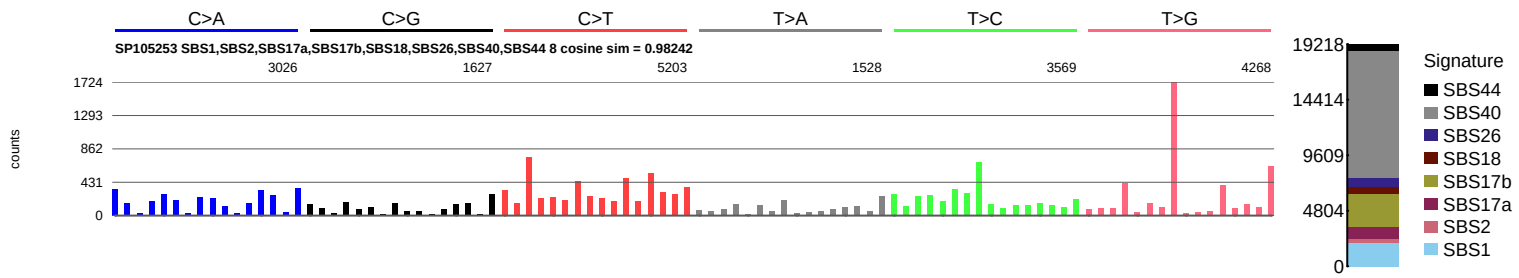
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98249
3008
1658
5168
1519
3573
4294
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1

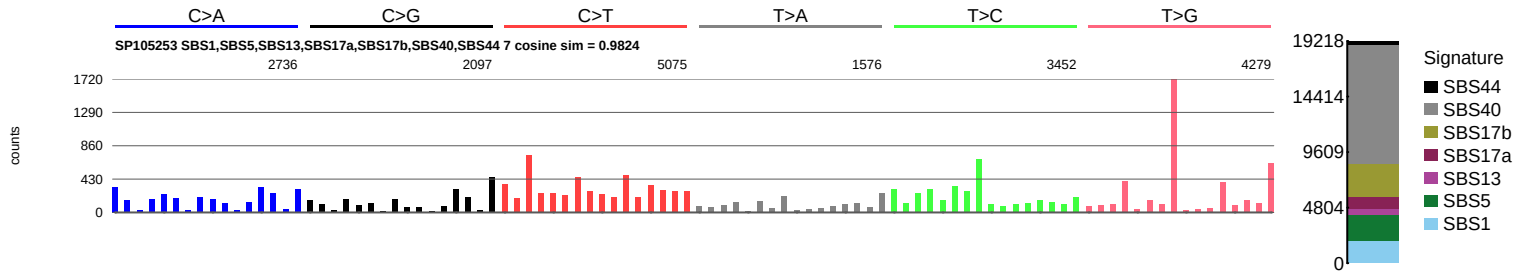

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98248
2979
1641
5388
1547
3437
4231
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

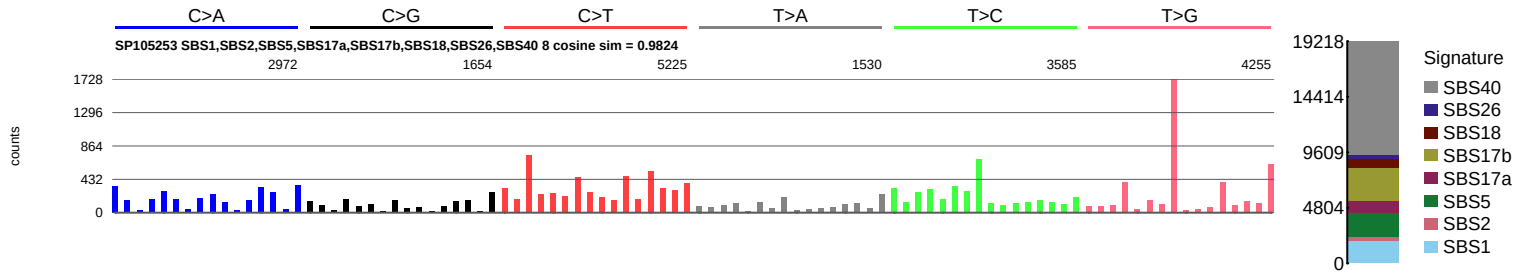

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98247
3029
2017
4831
1533
3535
4273
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

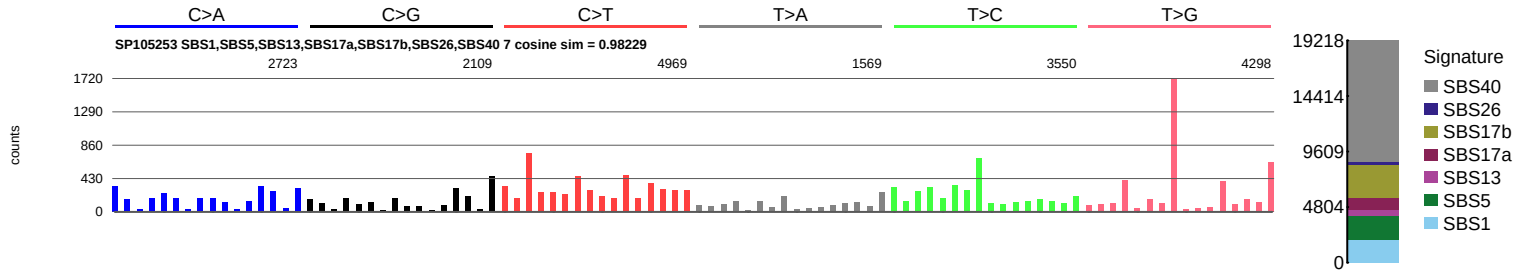

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.98243
2928
1748
4658
1859
3577
4451
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1

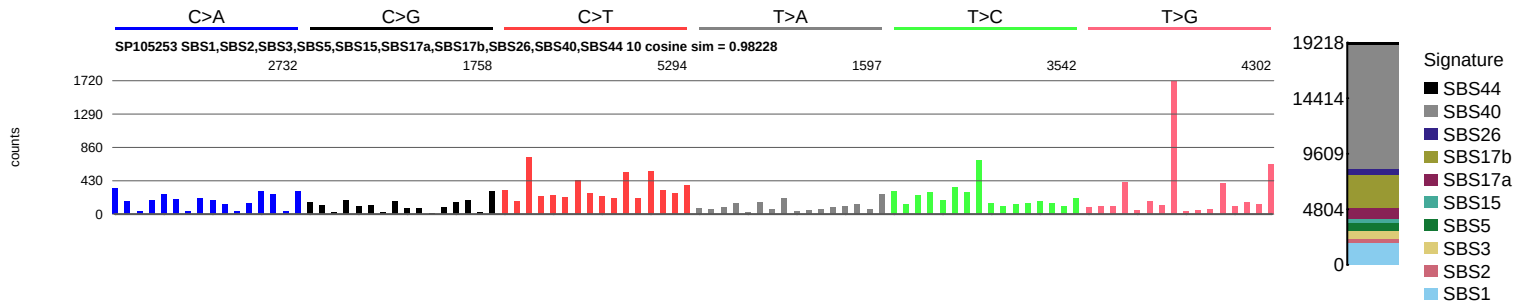

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98243
2958
1748
5263
1605
3434
4205
counts
1732
1299
866
433
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

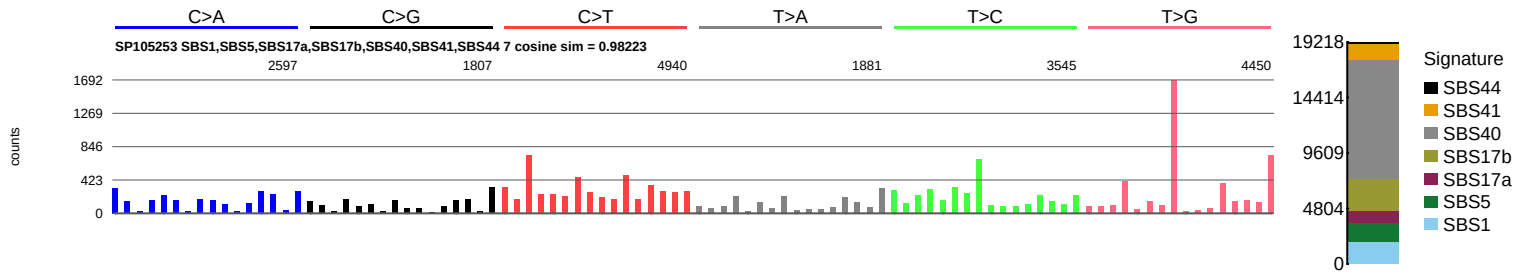

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98242
3026
1627
5203
1528
3569
4268
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1

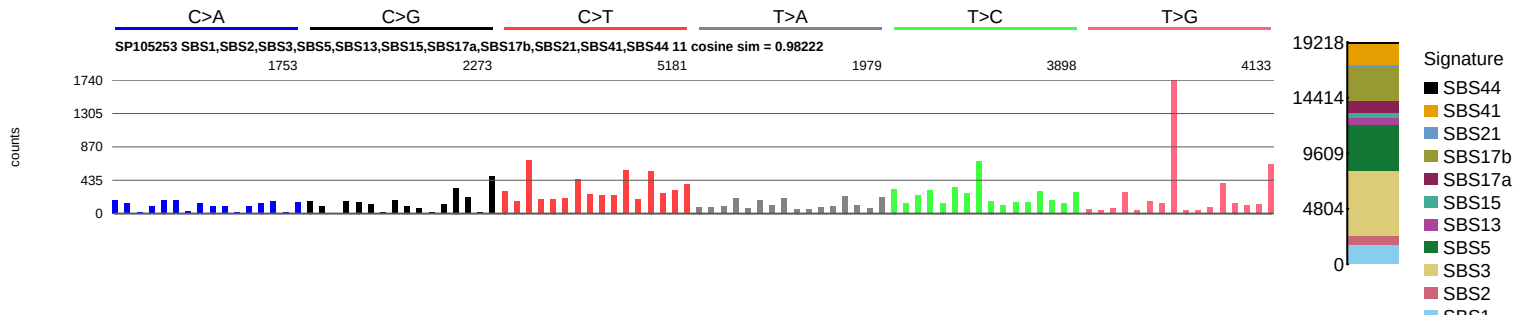

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.9824
2736
2097
5075
1576
3452
4279
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1

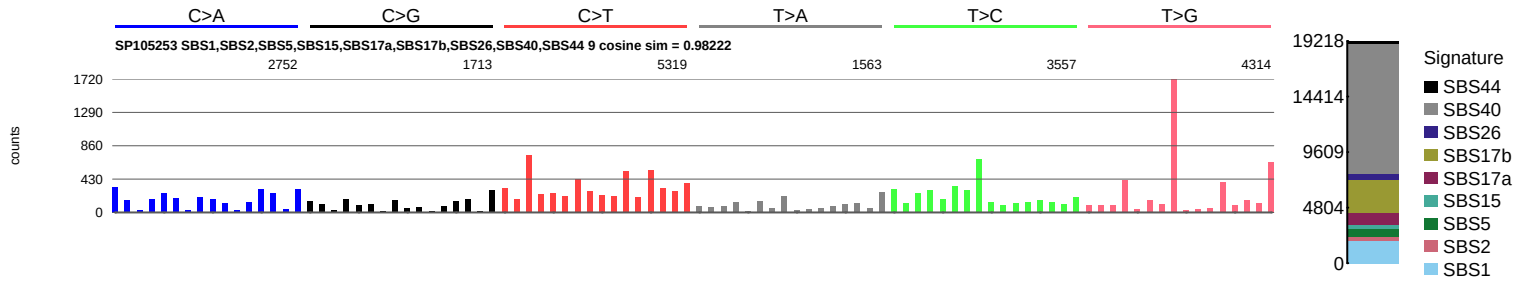

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.9824
2972
1654
5225
1530
3585
4255
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

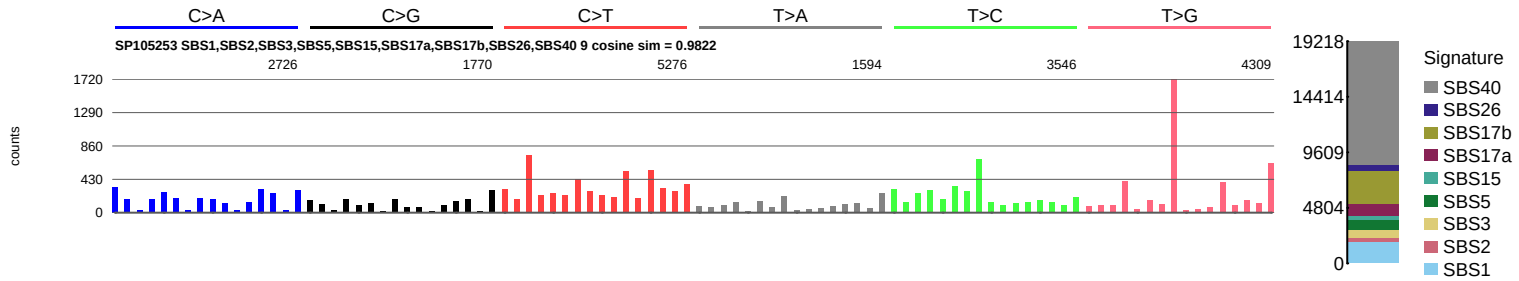

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98229
2723
2109
4969
1569
3550
4298
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1

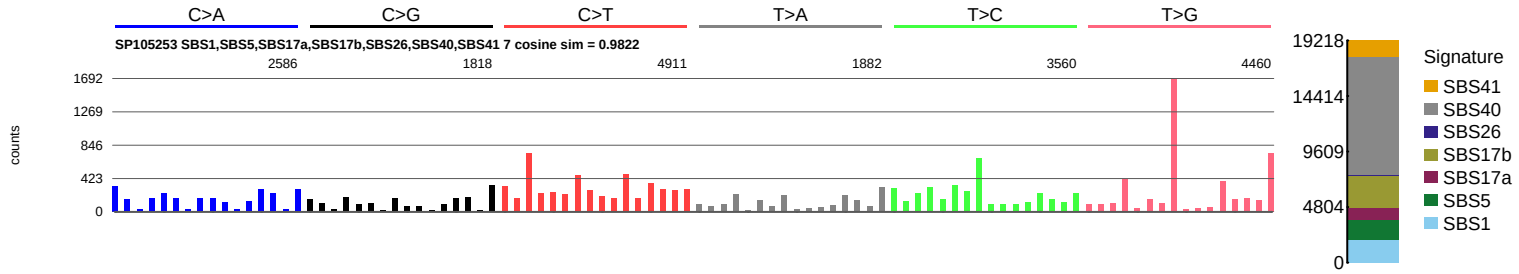

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 10 cosine sim = 0.98228
2732
1758
5294
1597
3542
4302
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

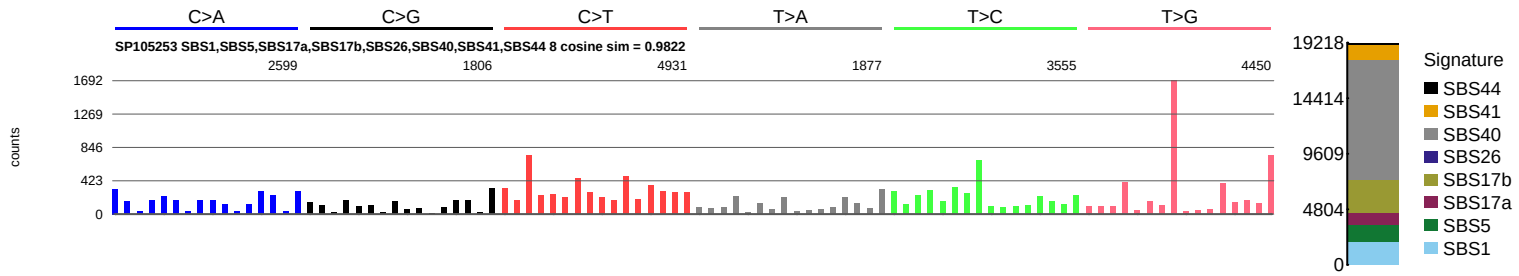

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98223
2597
1807
4940
1881
3545
4450
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1

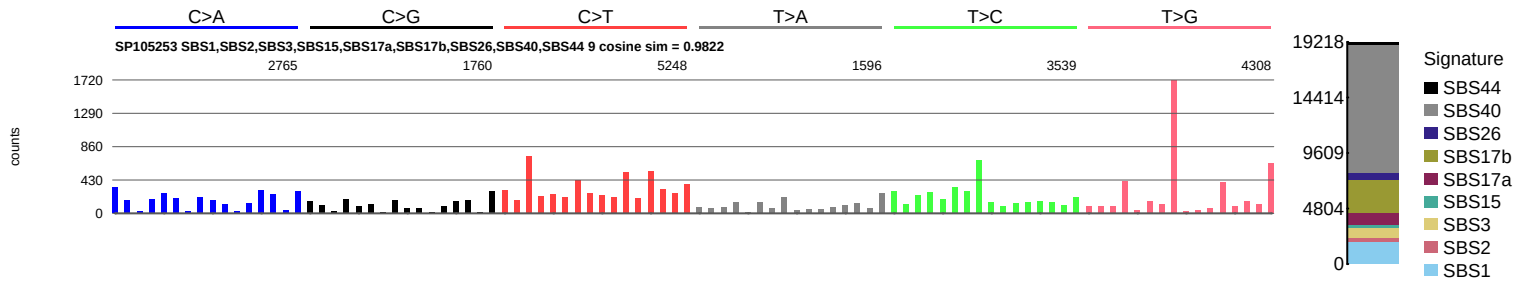

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 11 cosine sim = 0.98222
1753
2273
5181
1979
3898
4133
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

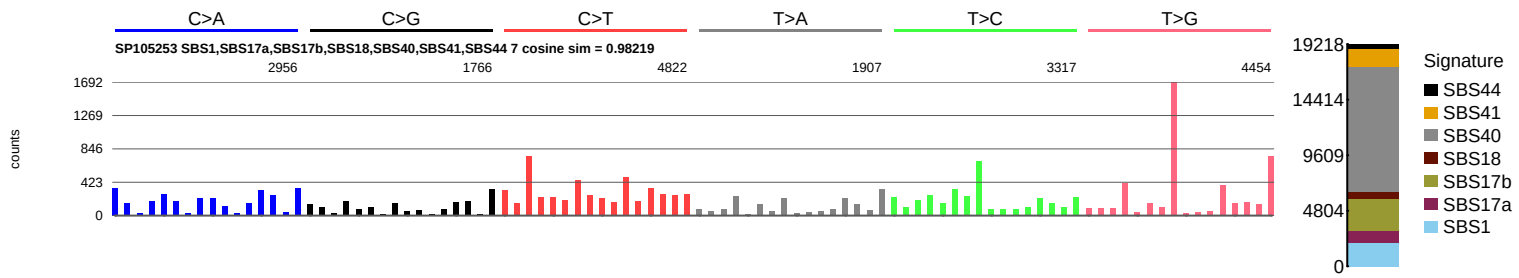

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98222
2752
1713
5319
1563
3557
4314
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

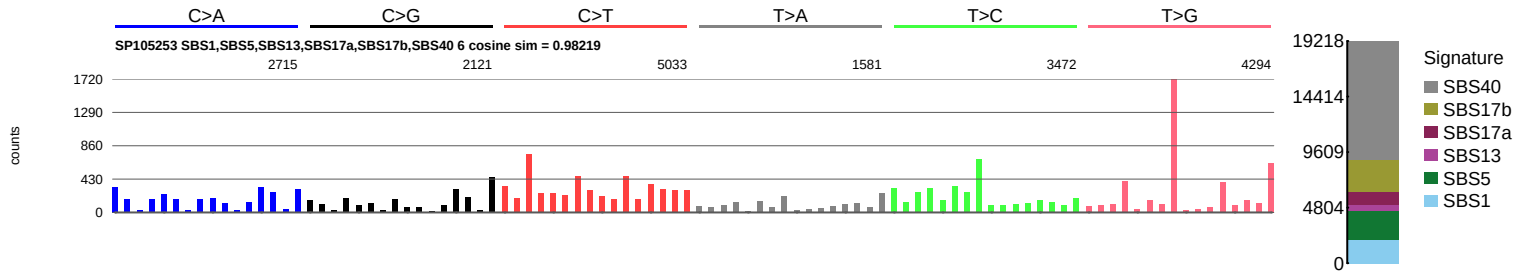

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 9 cosine sim = 0.9822
2726
1770
5276
1594
3546
4309
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

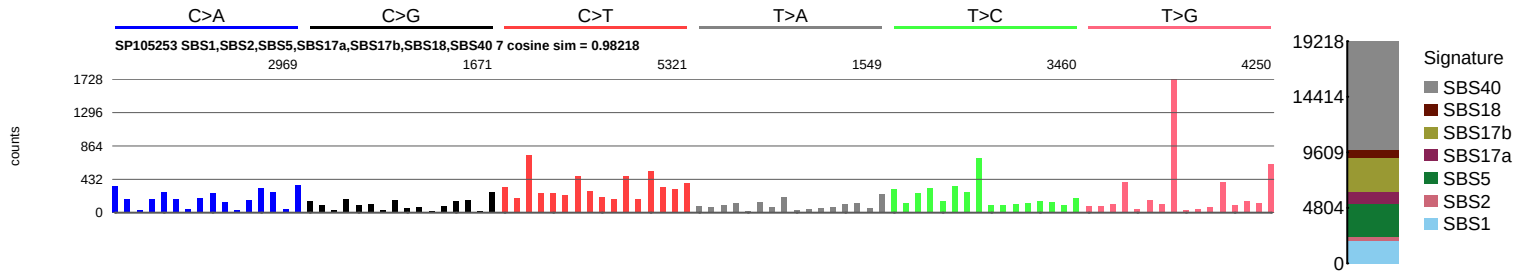

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.9822
2586
1818
4911
1882
3560
4460
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

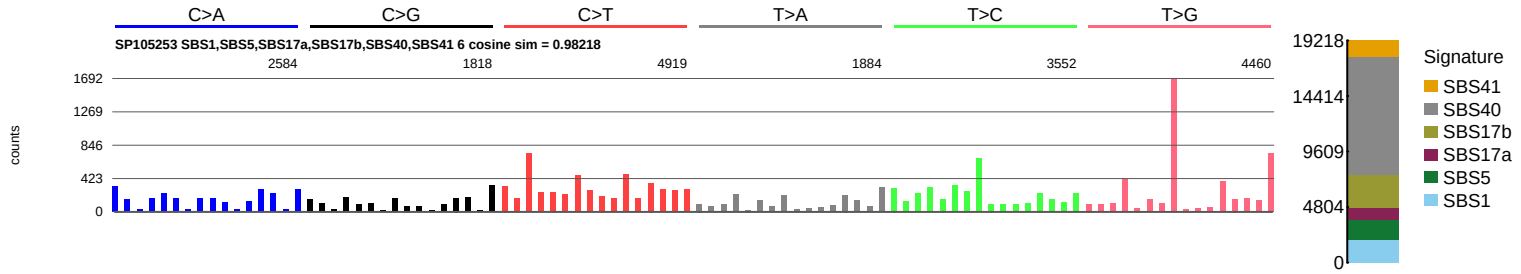

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.9822
2599
1806
4931
1877
3555
4450
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1

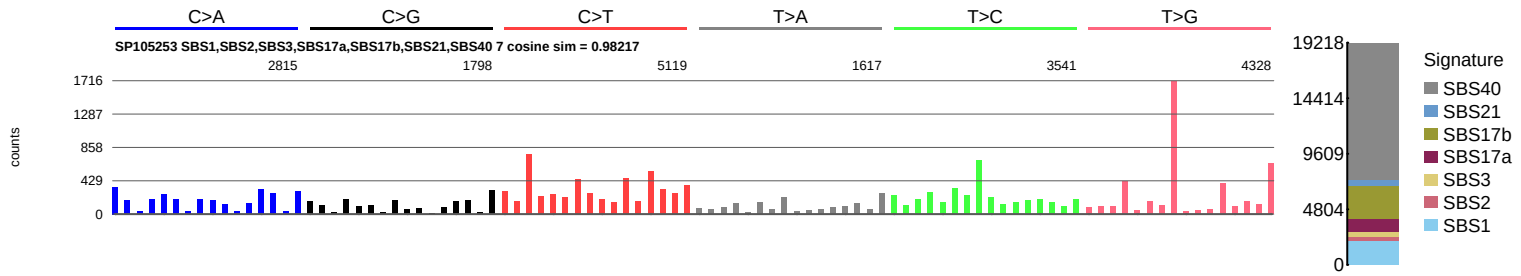

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.9822
2765
1760
5248
1596
3539
4308
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.98219
2956
1766
4822
1907
3317
4454
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98219
2715
2121
5033
1581
3472
4294
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98218
2969
1671
5321
1549
3460
4250
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98218
2584
1818
4919
1884
3552
4460
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1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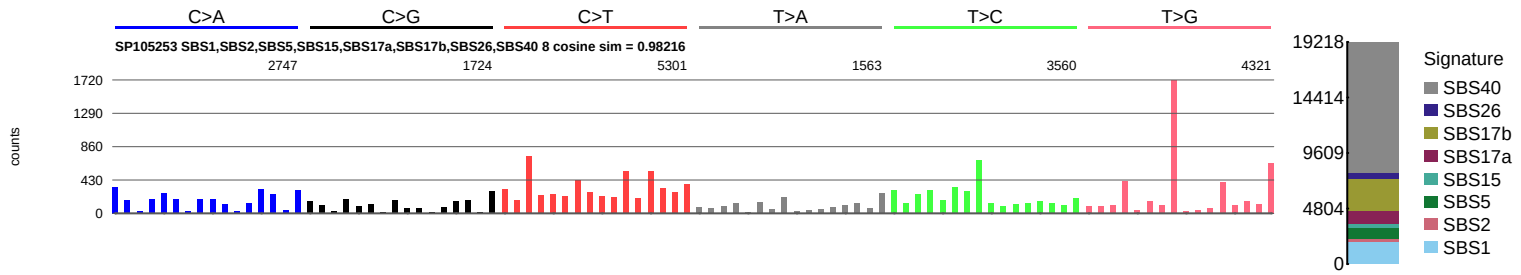
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98217
2815
1798
5119
1617
3541
4328
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>T
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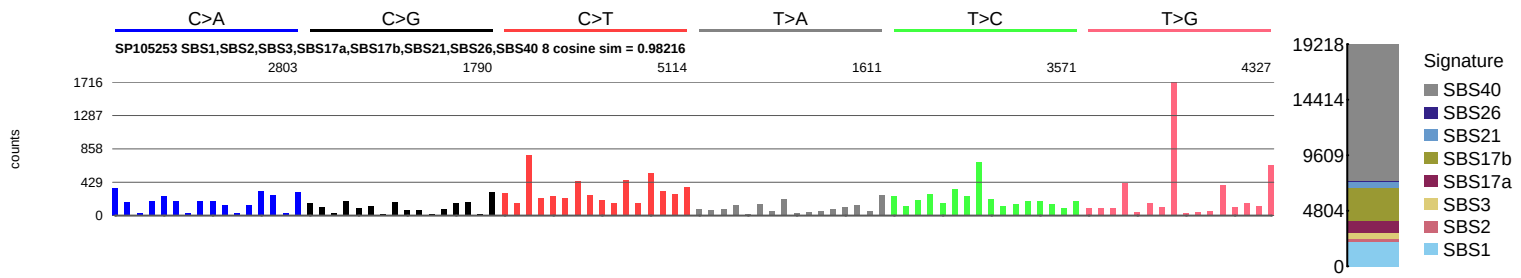
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98216
2747
1724
5301
1563
3560
4321
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1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C>G
C>T
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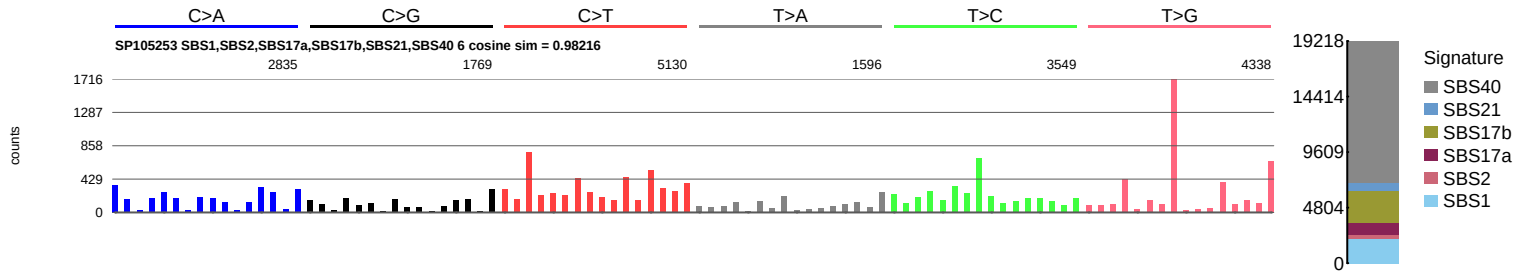
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98216
2803
1790
5114
1611
3571
4327
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1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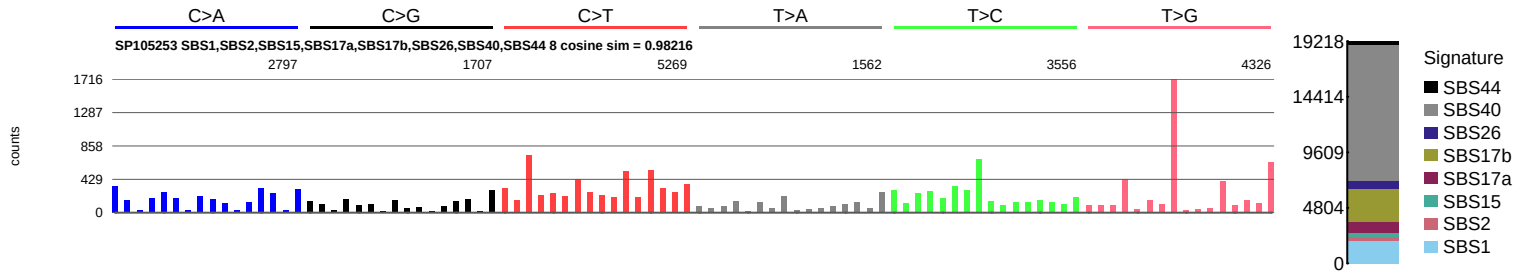
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98216
2835
1769
5130
1596
3549
4338
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS2
SBS1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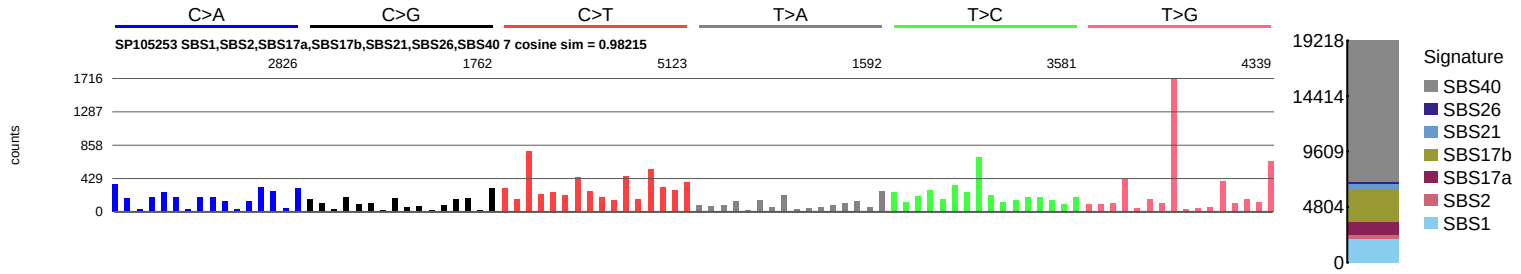
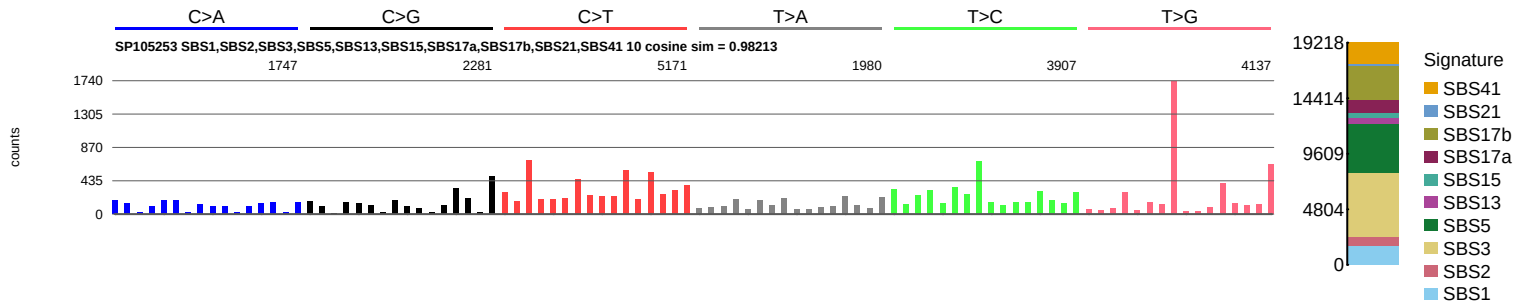
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98216
2797
1707
5269
1562
3556
4326
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98215
2826
1762
5123
1592
3581
4339
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS2
SBS1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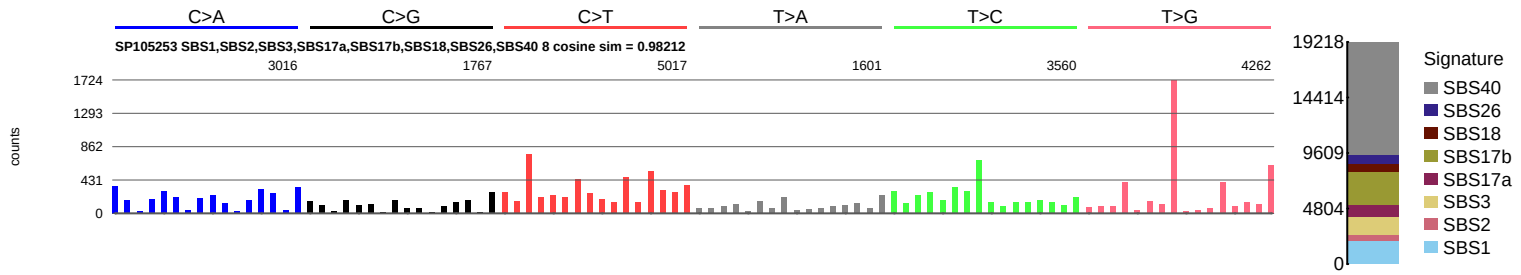
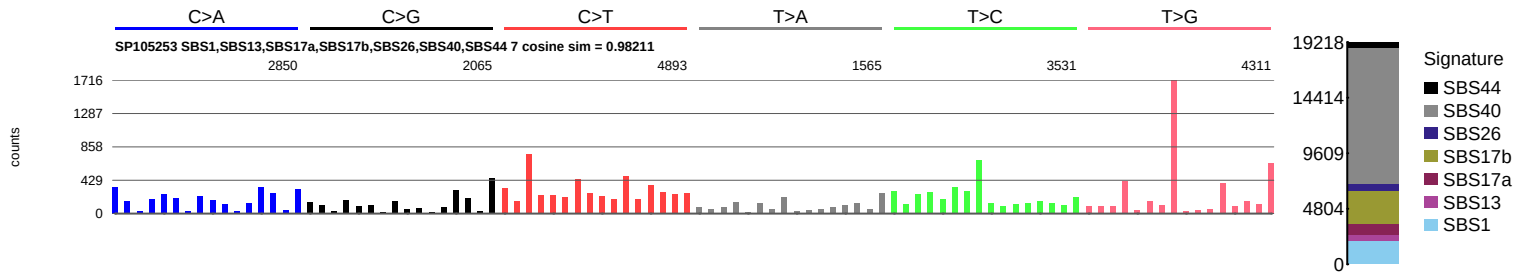
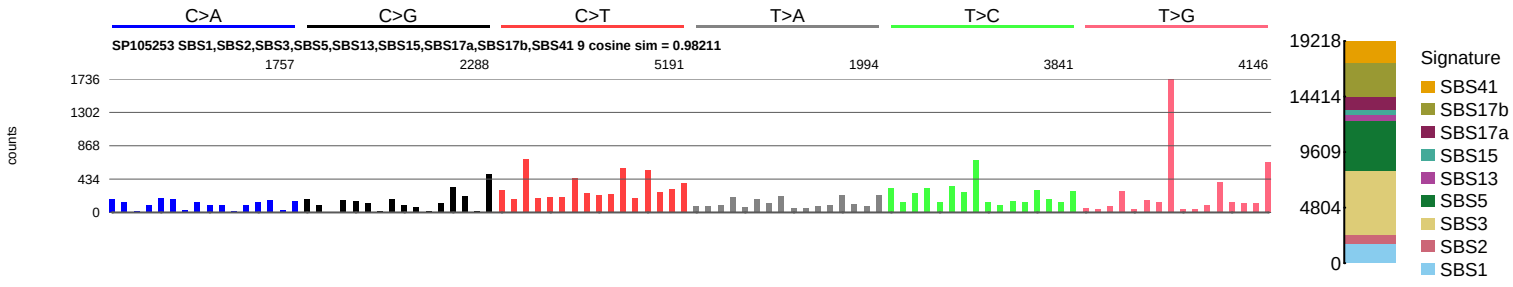
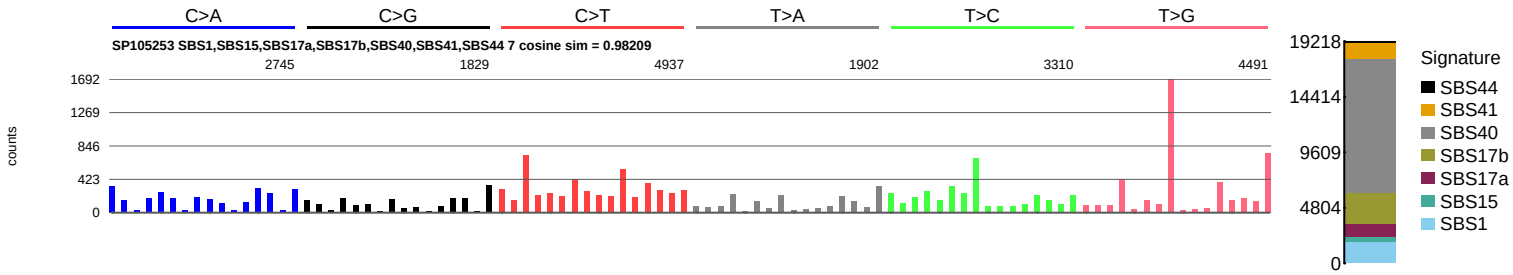
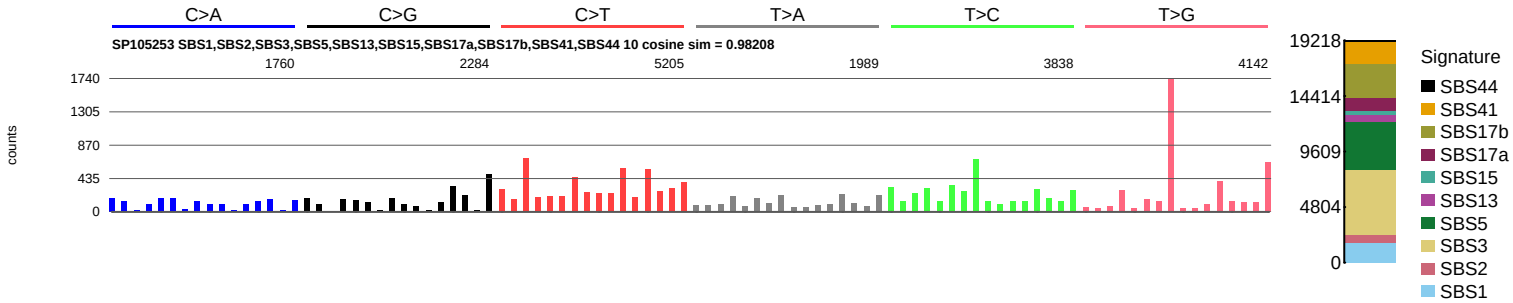
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.98213
1747
2281
5171
1980
3907
4137
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

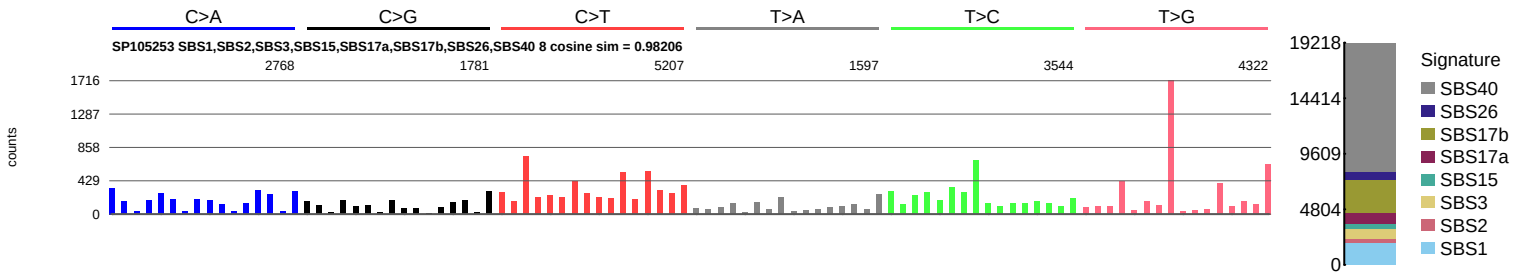

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98212
3016
1767
5017
1601
3560
4262
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98211
2850
2065
4893
1565
3531
4311
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98211
1757
2288
5191
1994
3841
4146
1736
1302
868
434
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.98209
2745
1829
4937
1902
3310
4491
1692
1269
846
423
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41,SBS44 10 cosine sim = 0.98208
1760
2284
5205
1989
3838
4142
1740
1305
870
435
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98206
2768
1781
5207
1597
3544
4322
1716
1287
858
429
0
counts
19218
14414
9609
4804
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Signature
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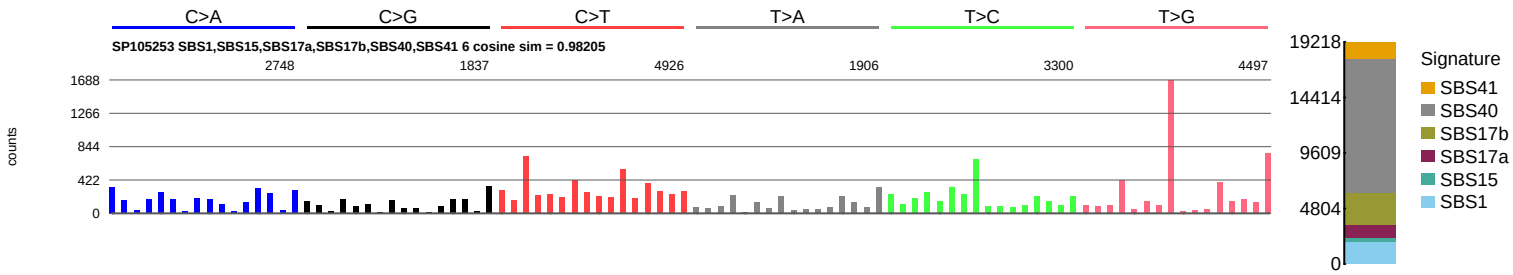
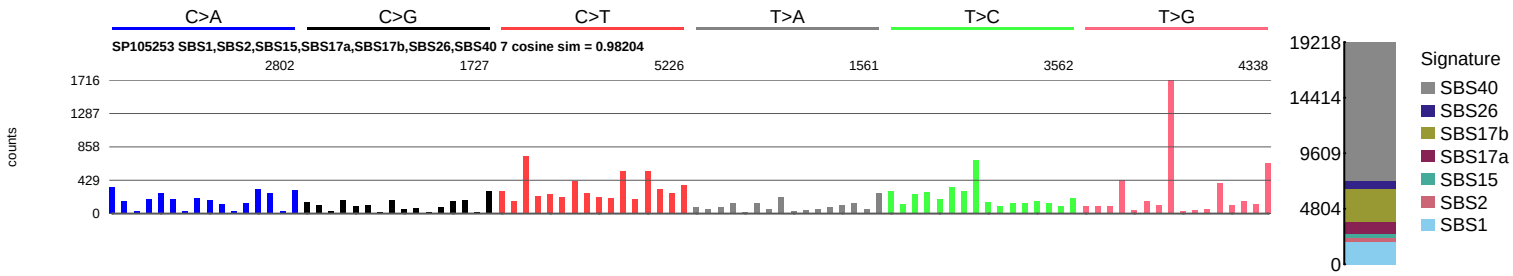
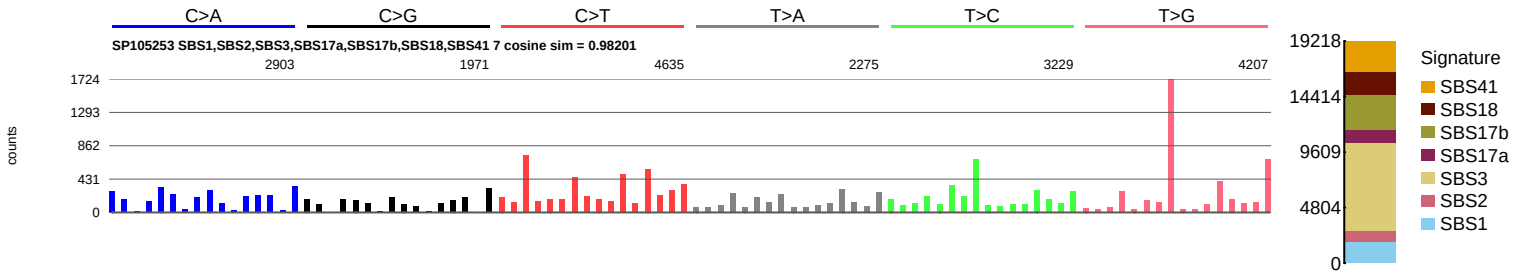
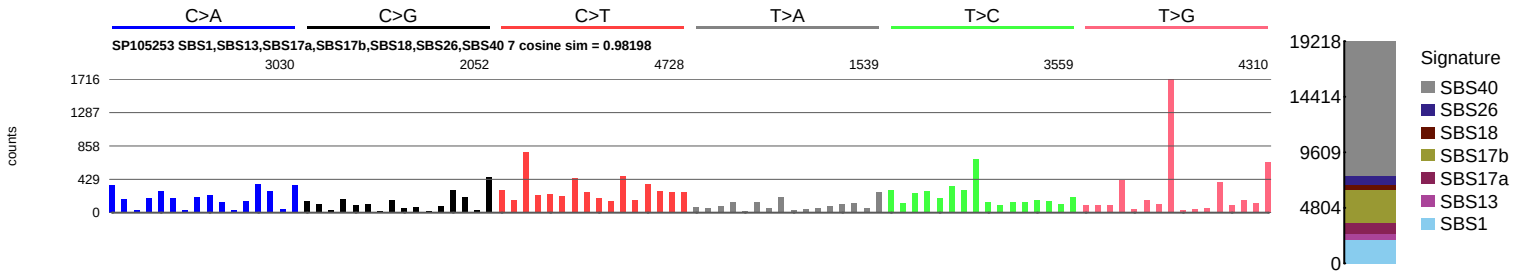
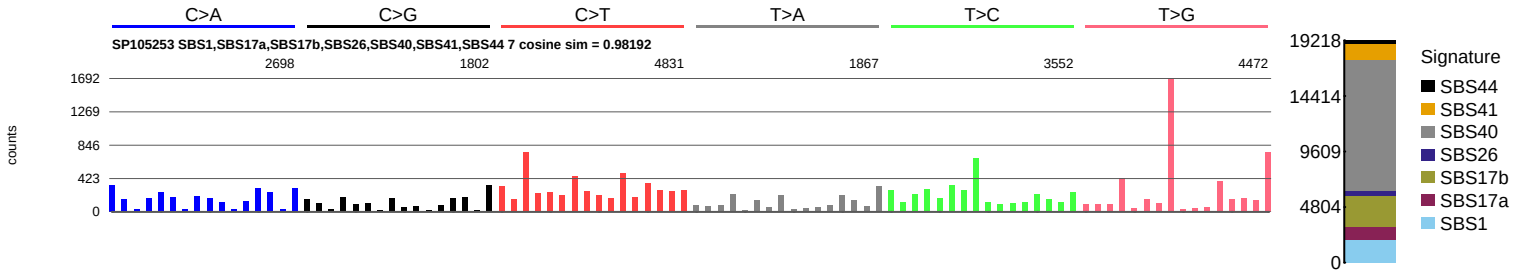
SBS3
SBS2
SBS1

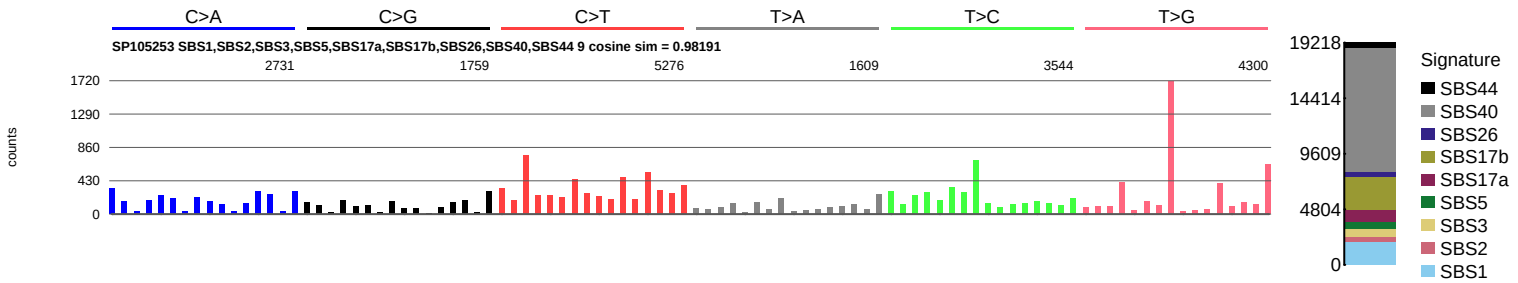

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.98205
2748
1837
4926
1906
3300
4497
1688
1266
844
422
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1

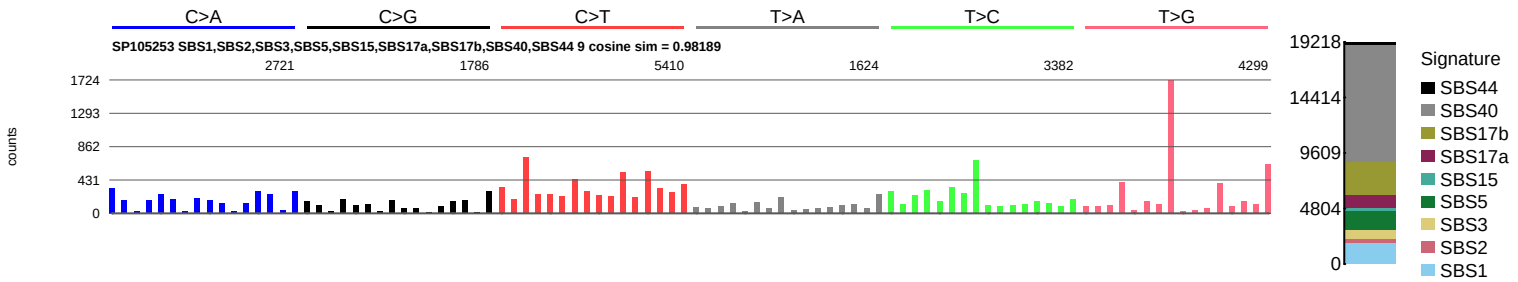

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98204
2802
1727
5226
1561
3562
4338
1716
1287
858
429
0
counts
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS2
SBS1

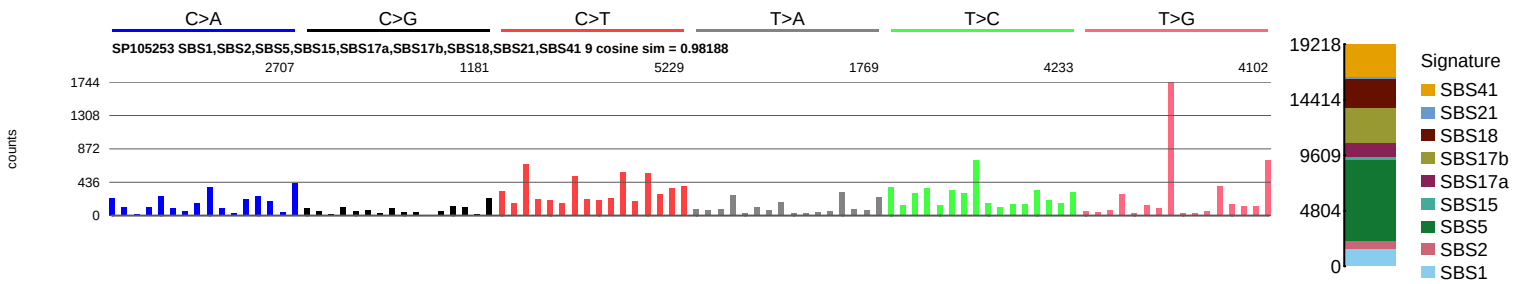

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98201
2903
1971
4635
2275
3229
4207
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

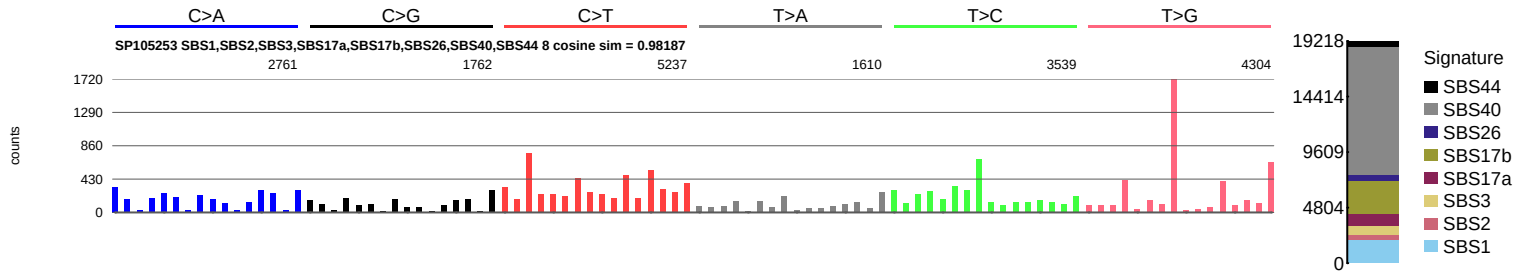

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98198
3030
2052
4728
1539
3559
4310
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

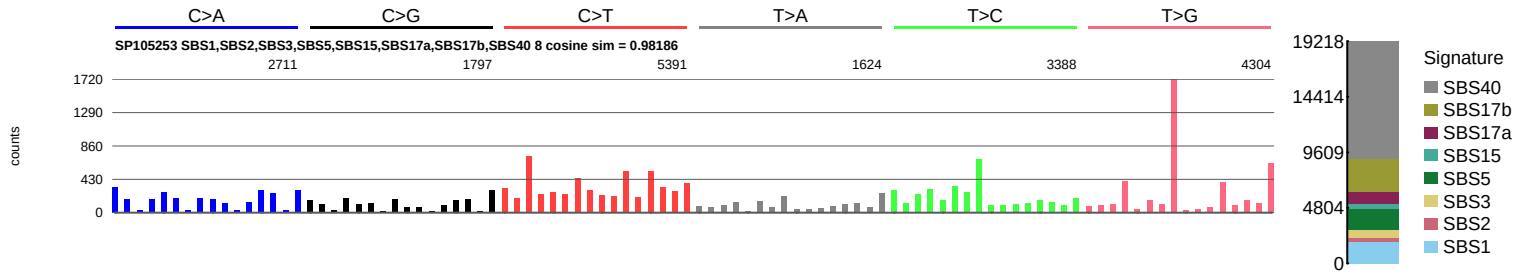

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.98192
2698
1802
4831
1867
3552
4472
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1

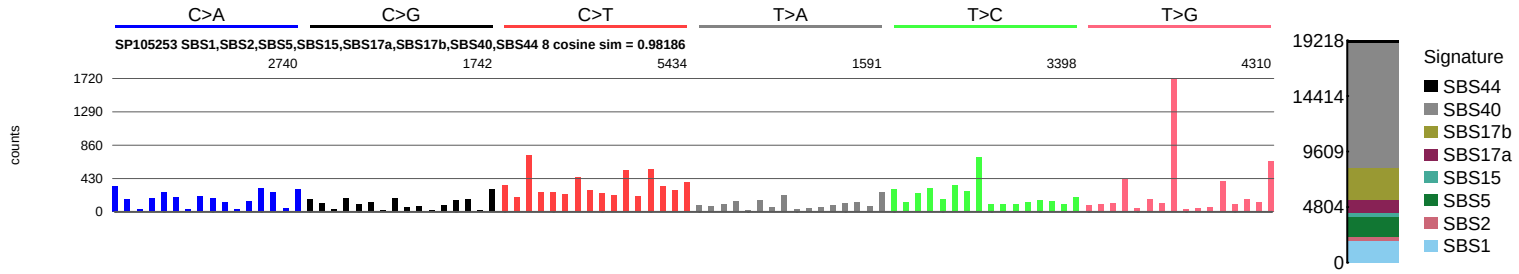

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98191
2731
1759
5276
1609
3544
4300
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98189
2721
1786
5410
1624
3382
4299
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98188
2707
1181
5229
1769
4233
4102
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98187
2761
1762
5237
1610
3539
4304
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98186
2711
1797
5391
1624
3388
4304
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98186
2740
1742
5434
1591
3398
4310
counts
1720
1290
860
430
0
19218
14414
9609
4804
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Signature
SBS44
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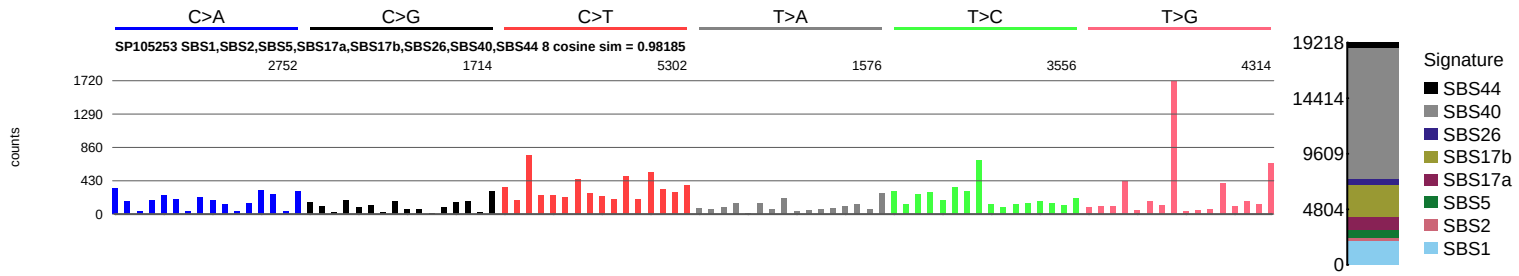
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98185
2752
1714
5302
1576
3556
4314
counts
1720
1290
860
430
0
19218
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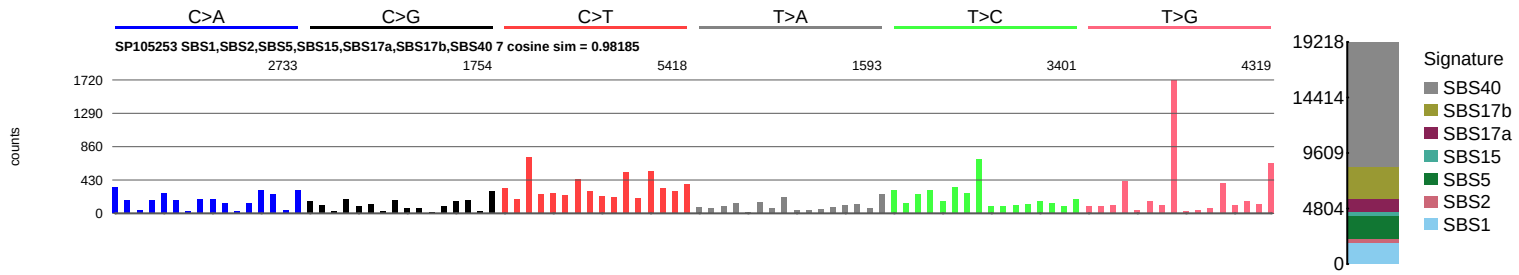
SBS1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SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98185
2733
1754
5418
1593
3401
4319
counts
1720
1290
860
430
0
19218
14414
9609
4804
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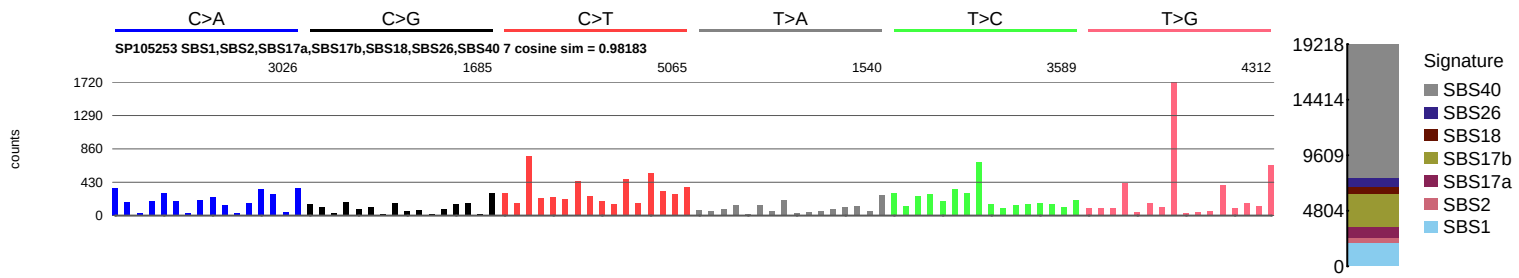
SBS1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C>G
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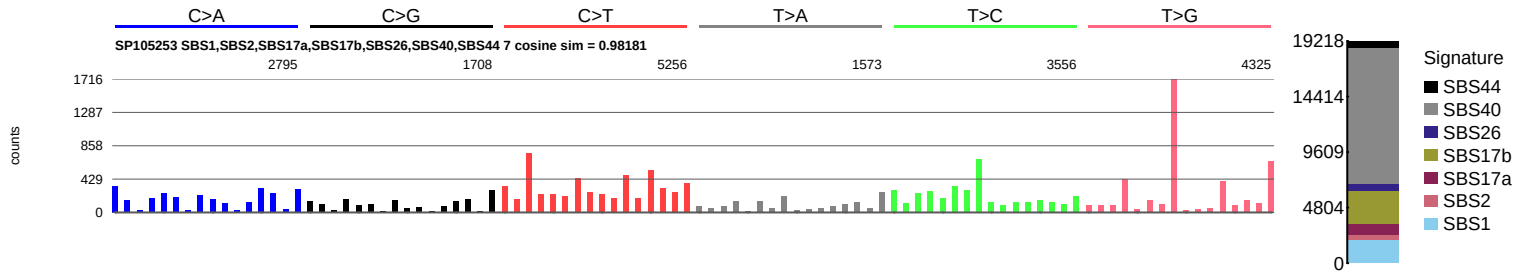
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98183
3026
1685
5065
1540
3589
4312
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
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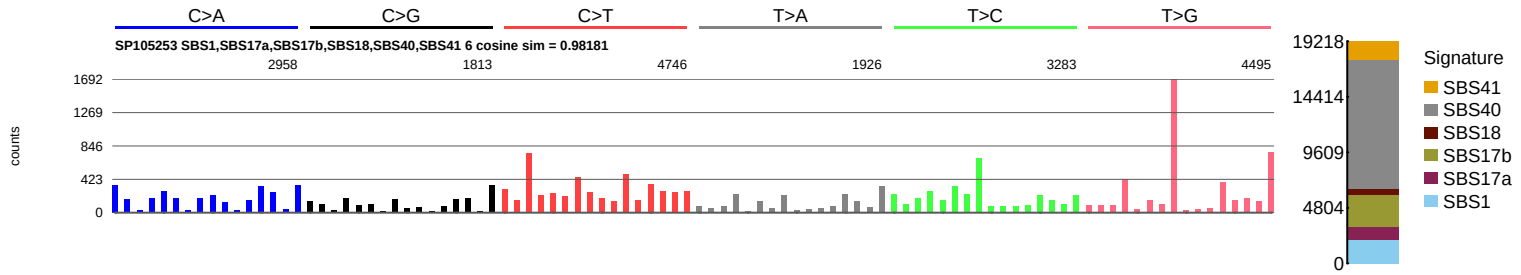
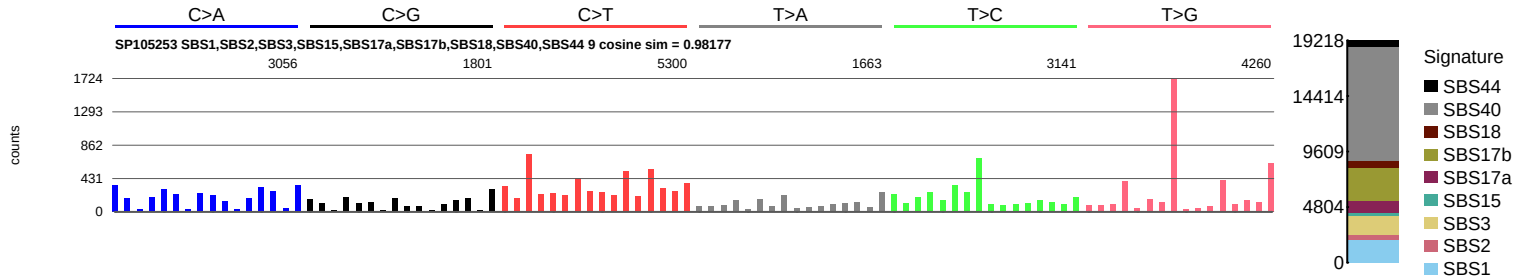
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98181
2795
1708
5256
1573
3556
4325
1716
1287
858
429
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.98181
2958
1813
4746
1926
3283
4495
1692
1269
846
423
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98177
3056
1801
5300
1663
3141
4260
1724
1293
862
431
0
counts
19218
14414
9609
4804
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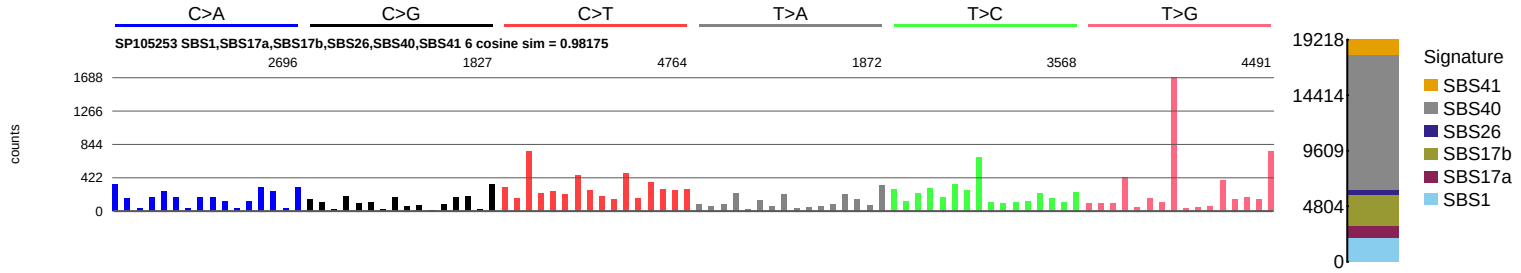
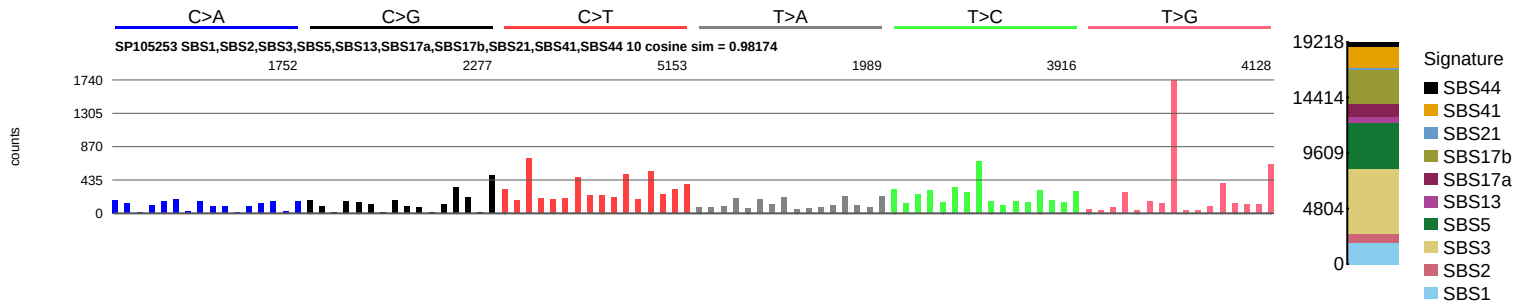
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.98175
2696
1827
4764
1872
3568
4491
1688
1266
844
422
0
counts
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
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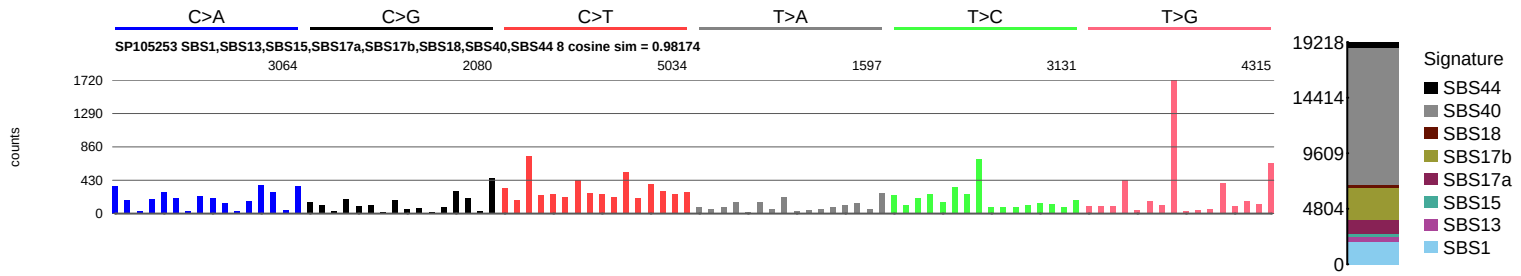
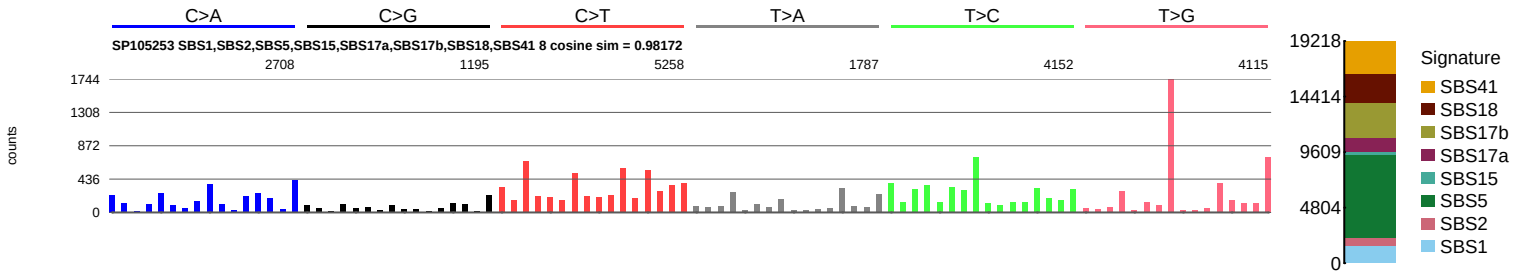
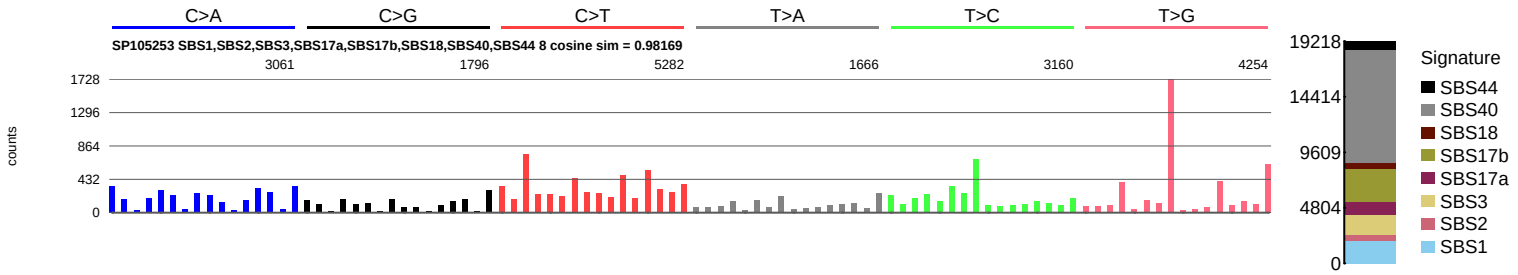
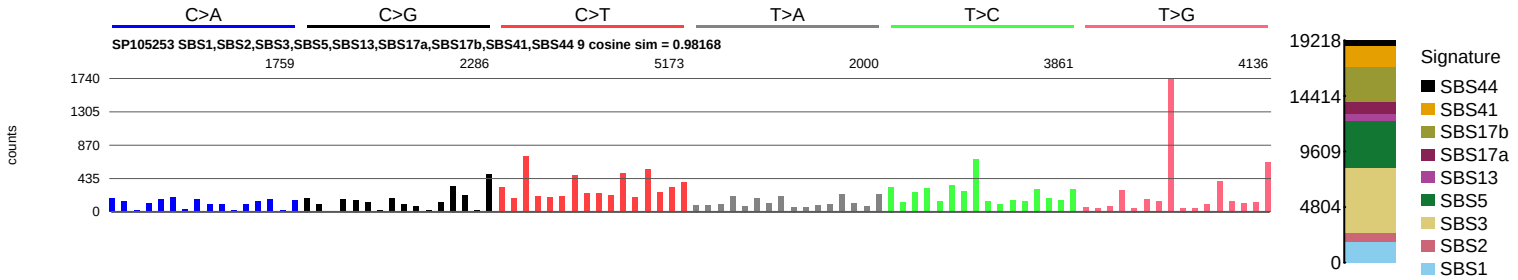
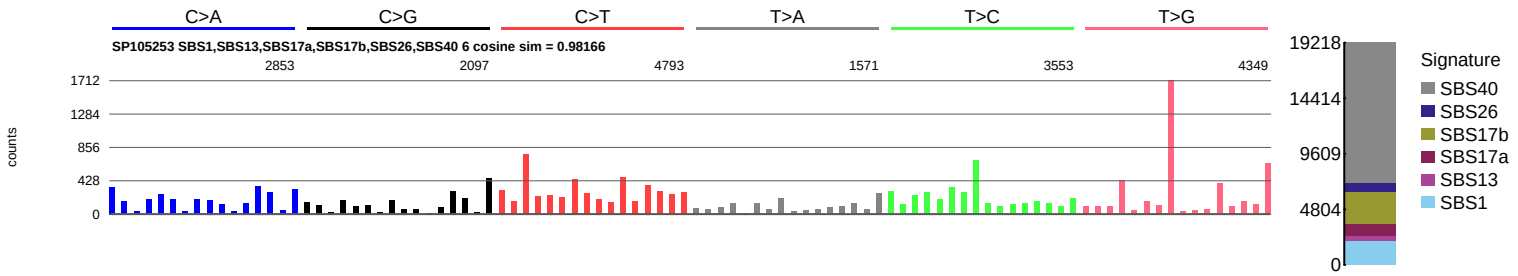
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41,SBS44 10 cosine sim = 0.98174
1752
2277
5153
1989
3916
4128
1740
1305
870
435
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

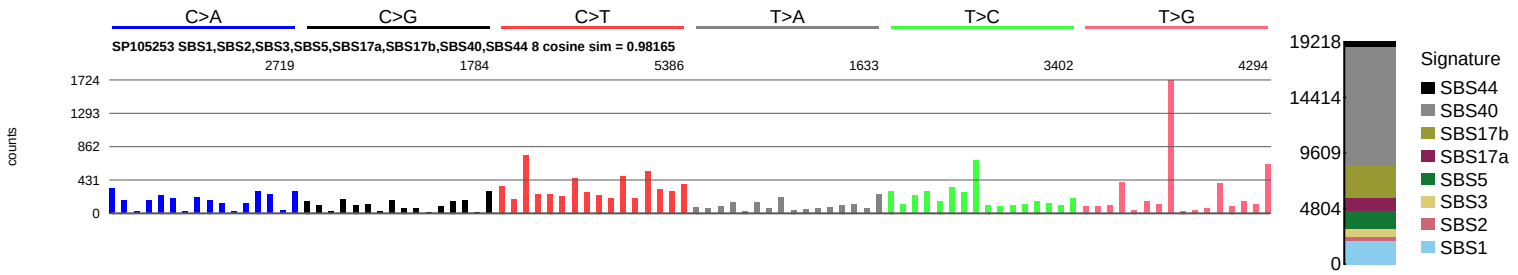

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98174
3064
2080
5034
1597
3131
4315
1720
1290
860
430
0
counts
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

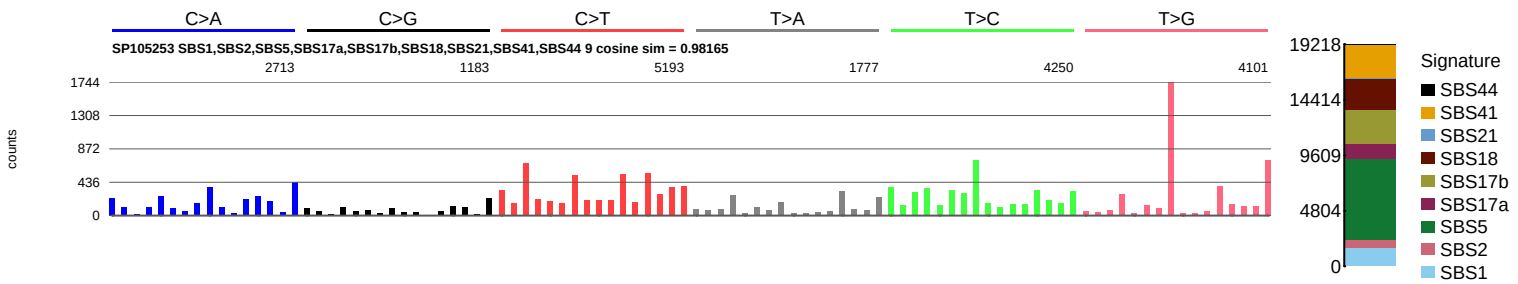

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98172
2708
1195
5258
1787
4152
4115
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

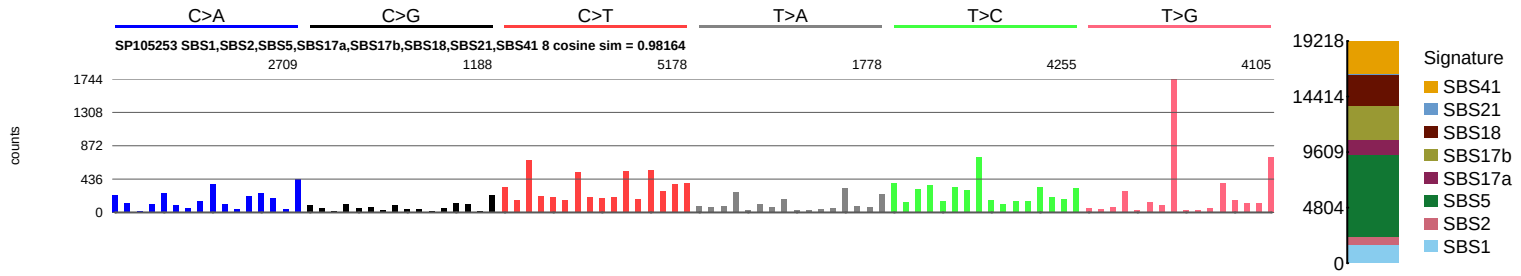

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98169
3061
1796
5282
1666
3160
4254
counts
1728
1296
864
432
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

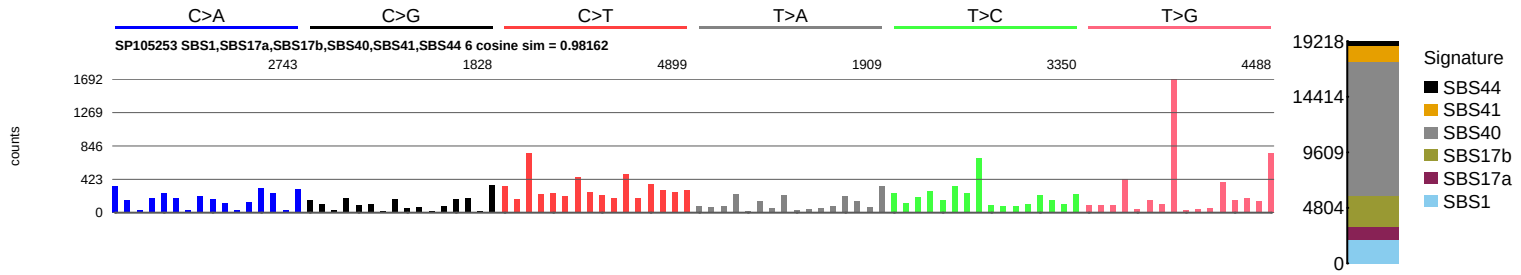

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98168
1759
2286
5173
2000
3861
4136
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98166
2853
2097
4793
1571
3553
4349
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98165
2719
1784
5386
1633
3402
4294
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98165
2713
1183
5193
1777
4250
4101
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98164
2709
1188
5178
1778
4255
4105
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1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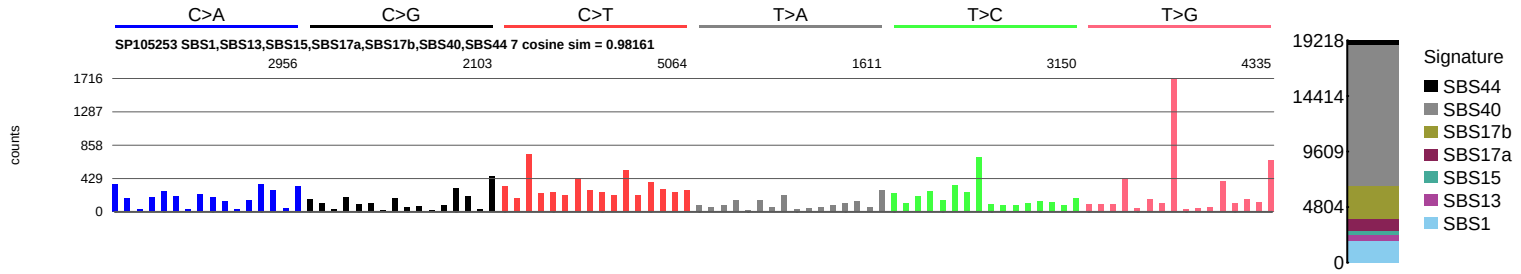
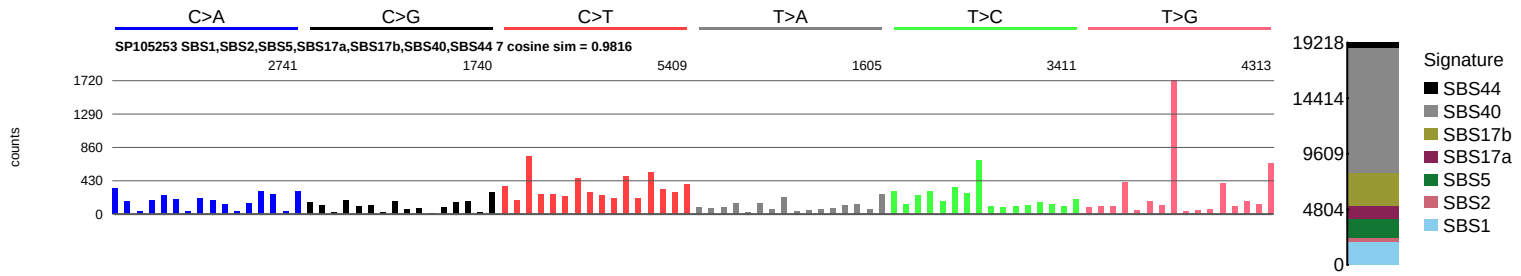
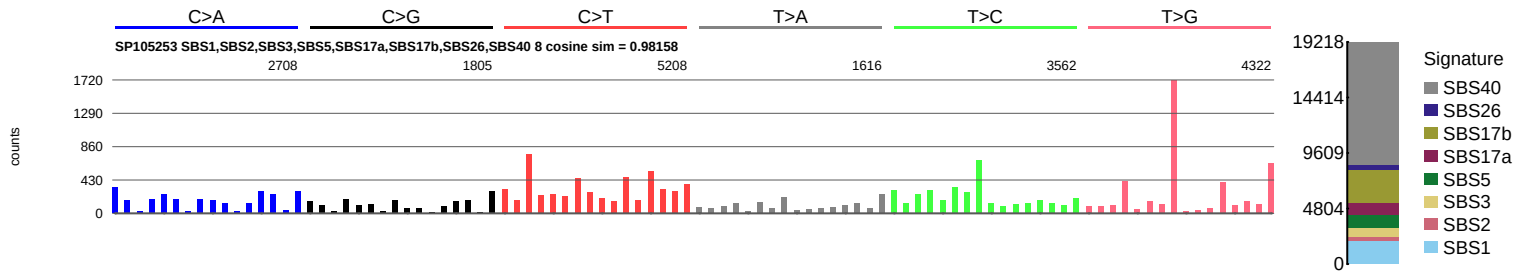
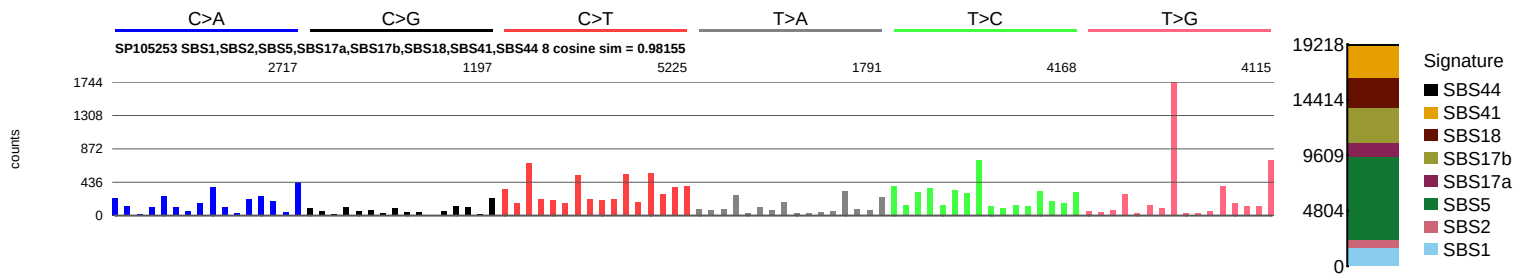
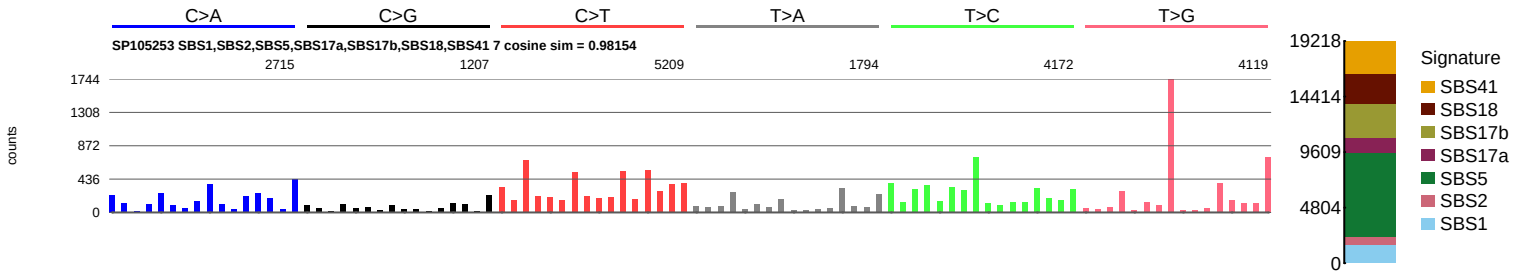
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.98162
2743
1828
4899
1909
3350
4488
counts
1692
1269
846
423
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1

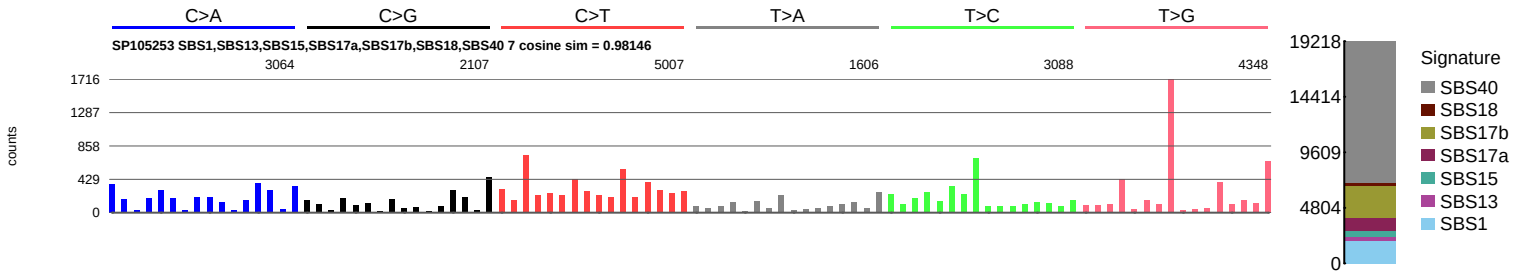

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98161
2956
2103
5064
1611
3150
4335
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1

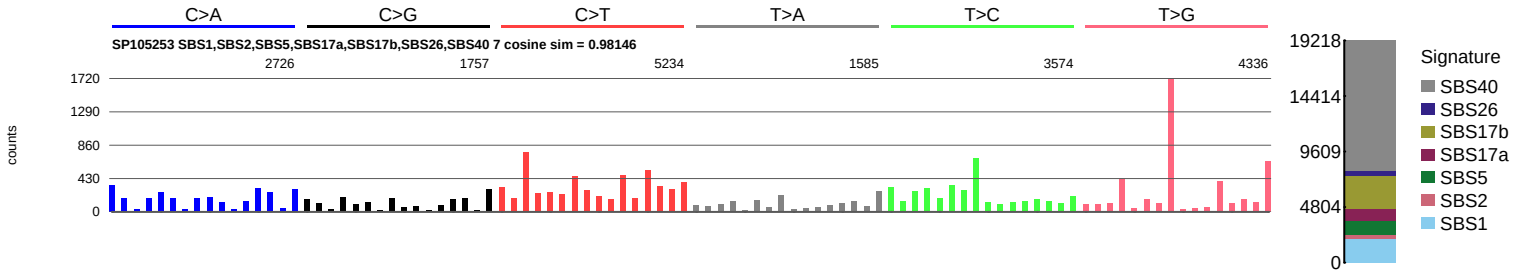

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.9816
2741
1740
5409
1605
3411
4313
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1

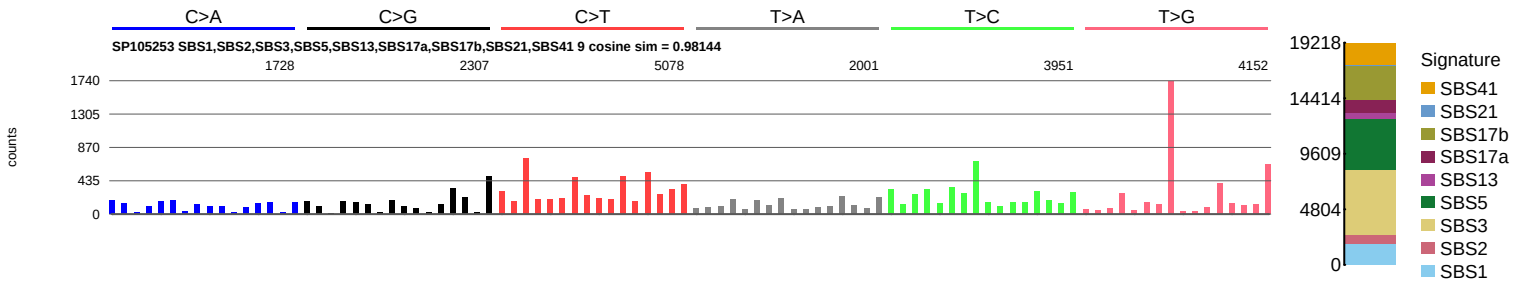

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98158
2708
1805
5208
1616
3562
4322
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

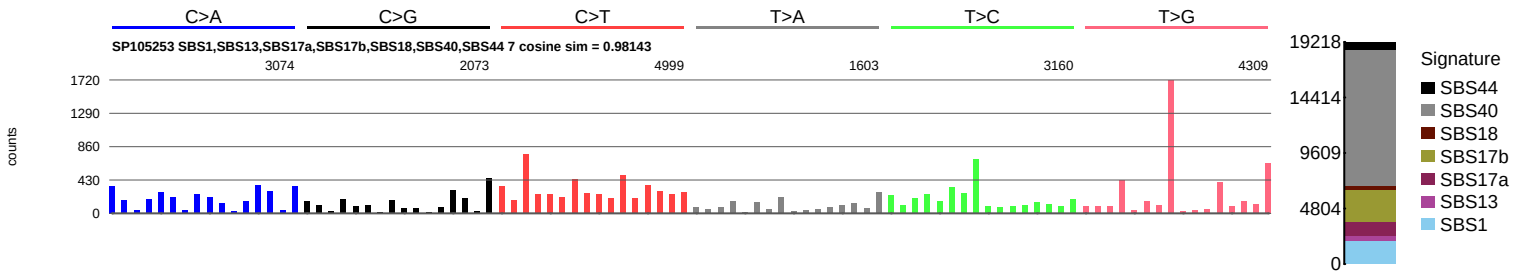

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98155
2717
1197
5225
1791
4168
4115
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98154
2715
1207
5209
1794
4172
4119
counts
1744
1308
872
436
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98146
3064
2107
5007
1606
3088
4348
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98146
2726
1757
5234
1585
3574
4336
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98144
1728
2307
5078
2001
3951
4152
counts
1740
1305
870
435
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98143
3074
2073
4999
1603
3160
4309
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
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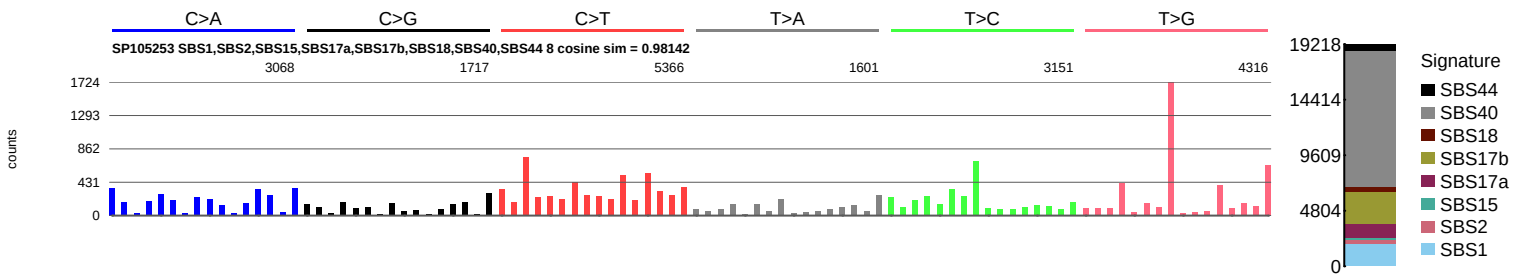
SBS1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SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98142
3068
1717
5366
1601
3151
4316
counts
1724
1293
862
431
0
19218
14414
9609
4804
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SBS40
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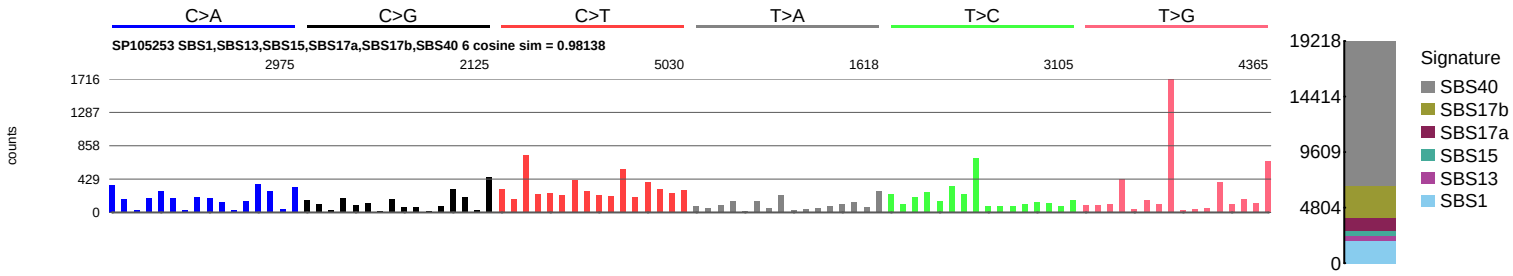
SBS15
SBS2
SBS1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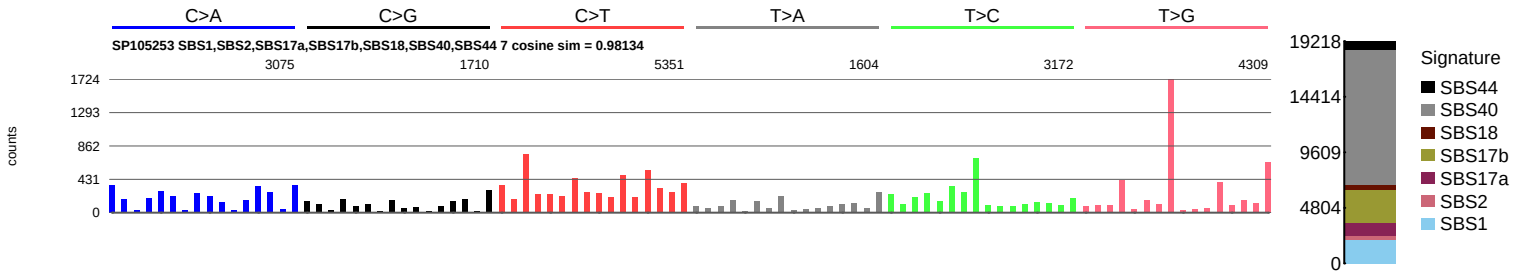
SP105253 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98138
2975
2125
5030
1618
3105
4365
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1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SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98134
3075
1710
5351
1604
3172
4309
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
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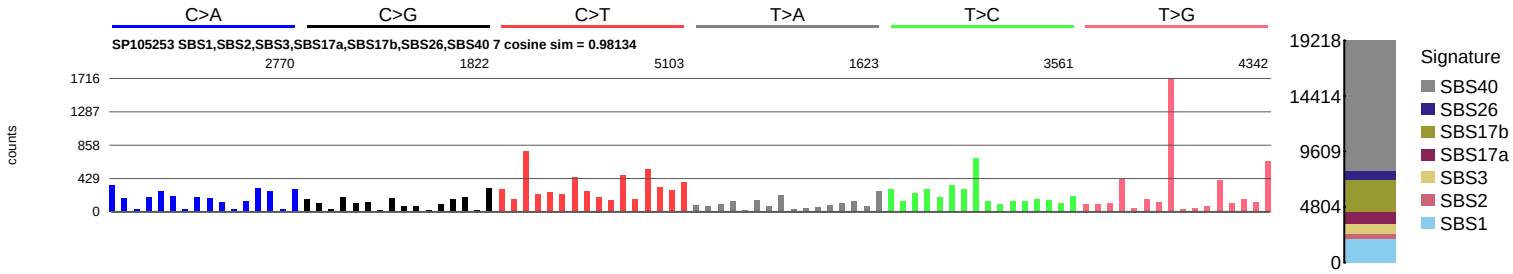
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1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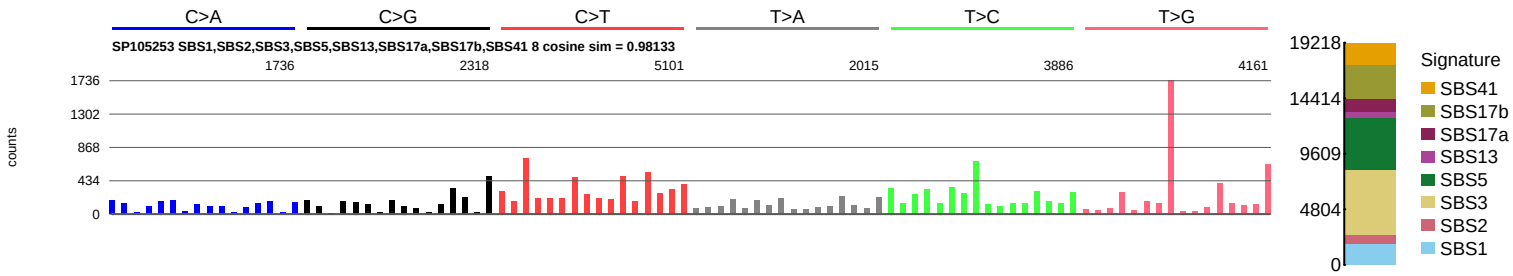
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98134
2770
1822
5103
1623
3561
4342
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1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T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98133
1736
2318
5101
2015
3886
4161
counts
1736
1302
868
434
0
19218
14414
9609
4804
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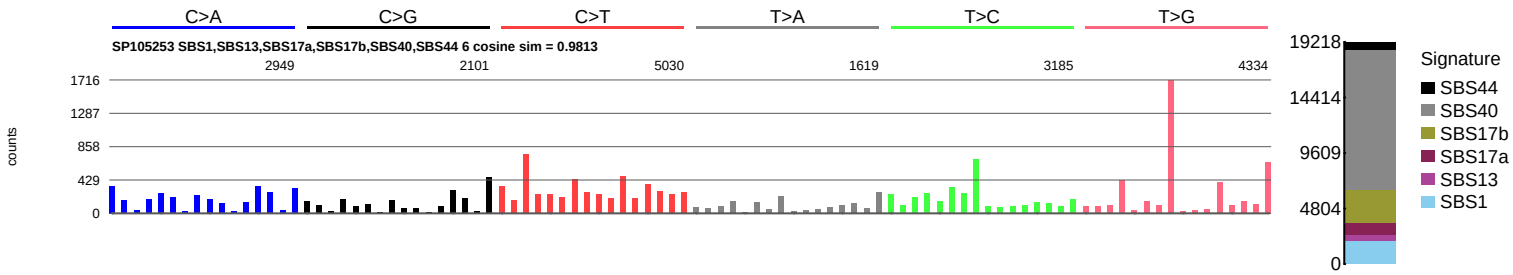
SBS2
SBS1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C>G
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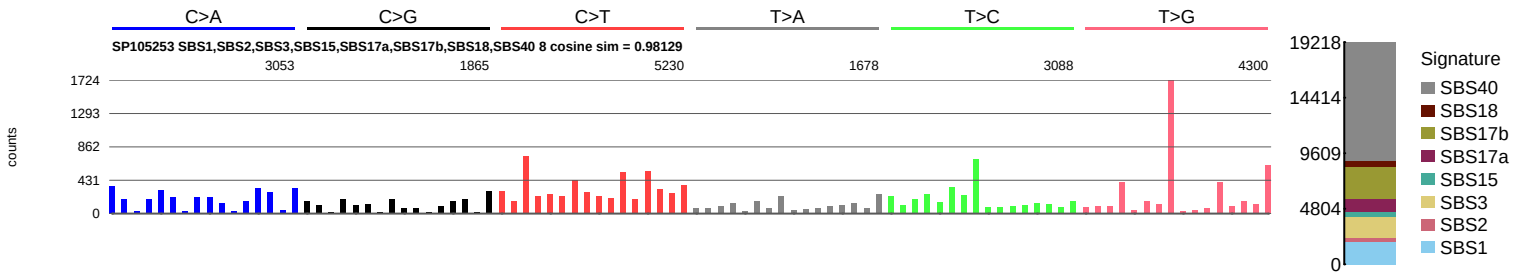
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.9813
2949
2101
5030
1619
3185
4334
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS1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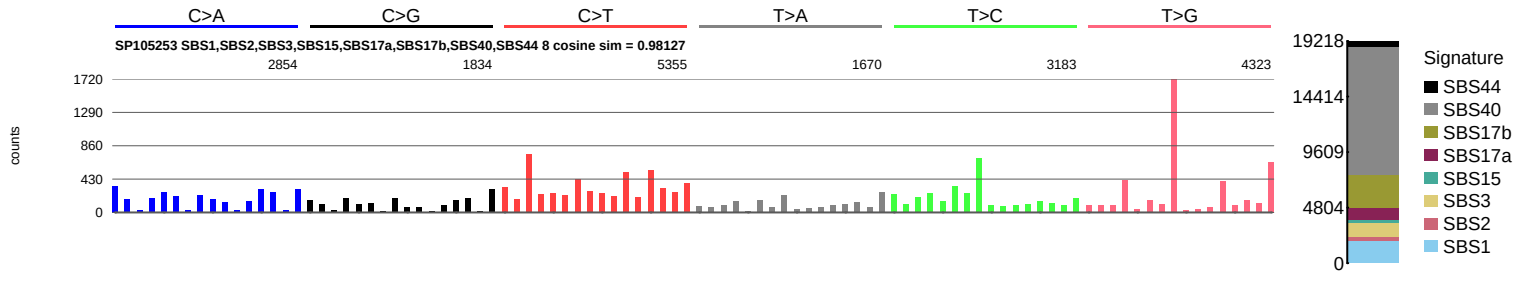
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98129
3053
1865
5230
1678
3088
4300
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>G
C>T
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T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98127
2854
1834
5355
1670
3183
4323
counts
1720
1290
860
430
0
19218
14414
9609
4804
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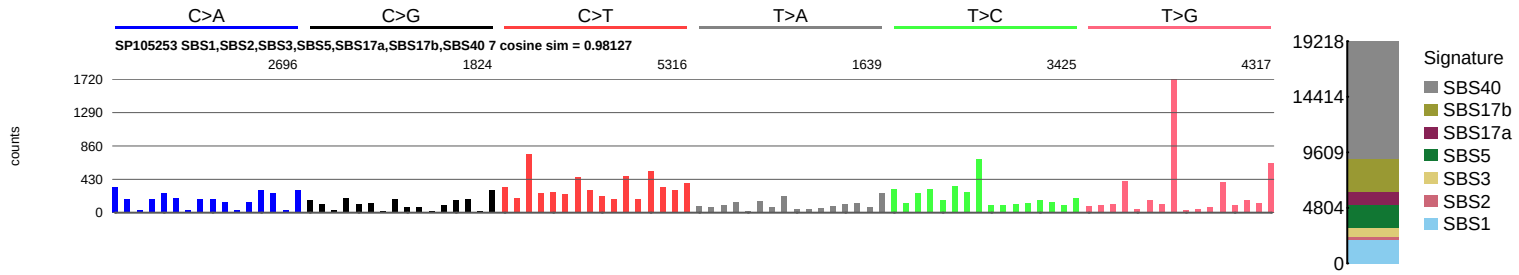
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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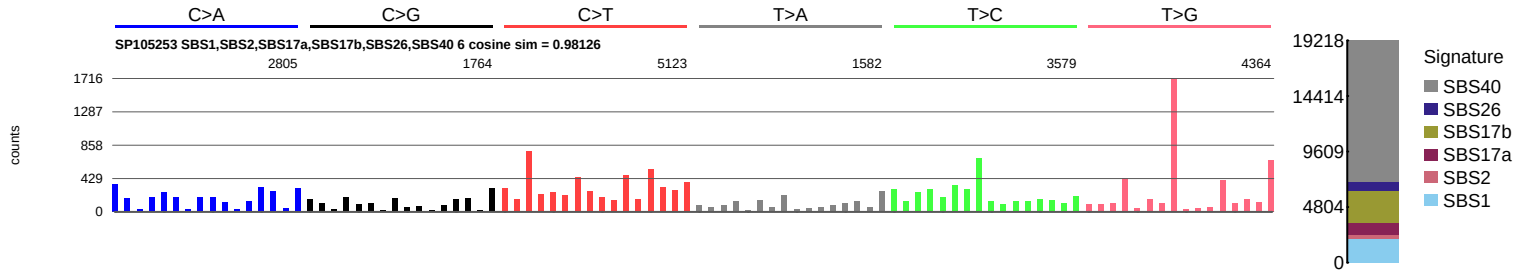
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98127
2696
1824
5316
1639
3425
4317
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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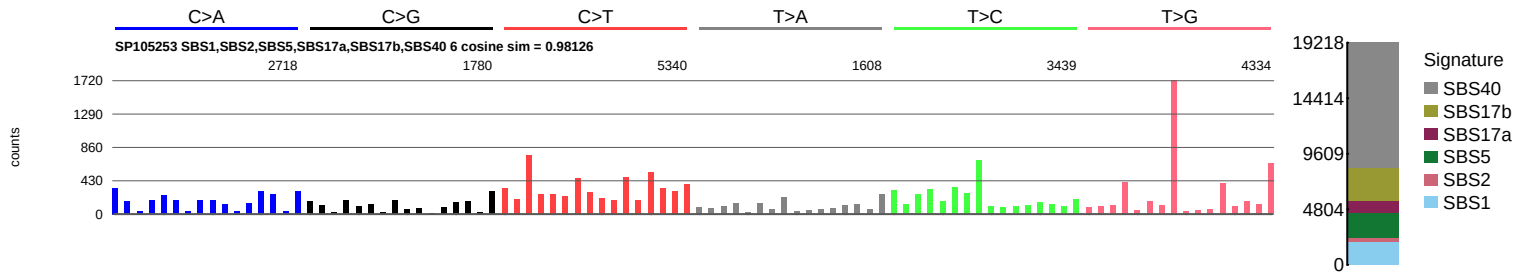
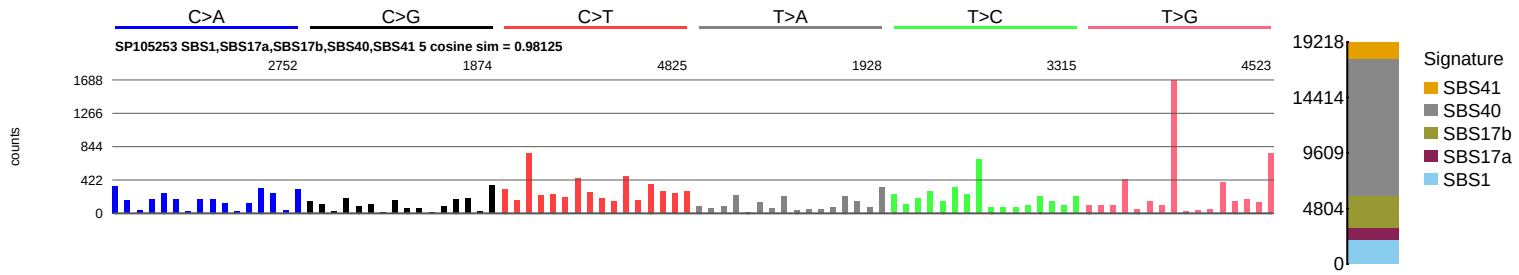
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98126
2805
1764
5123
1582
3579
4364
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98126
2718
1780
5340
1608
3439
4334
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS2
SBS1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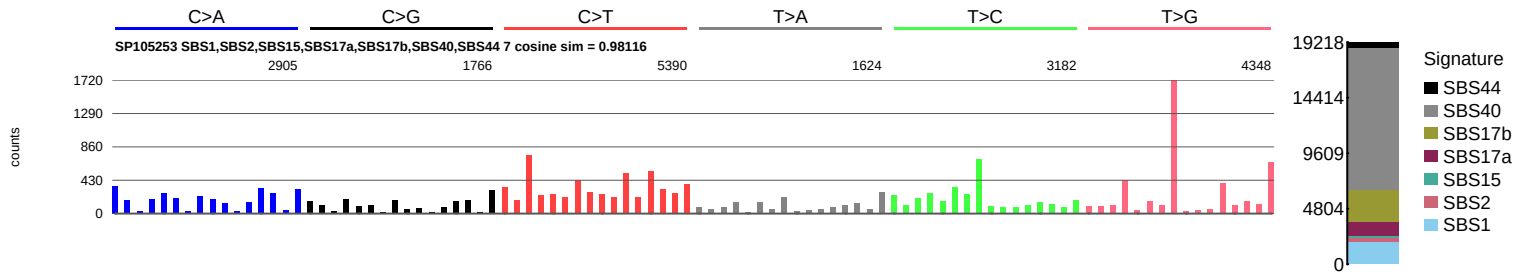
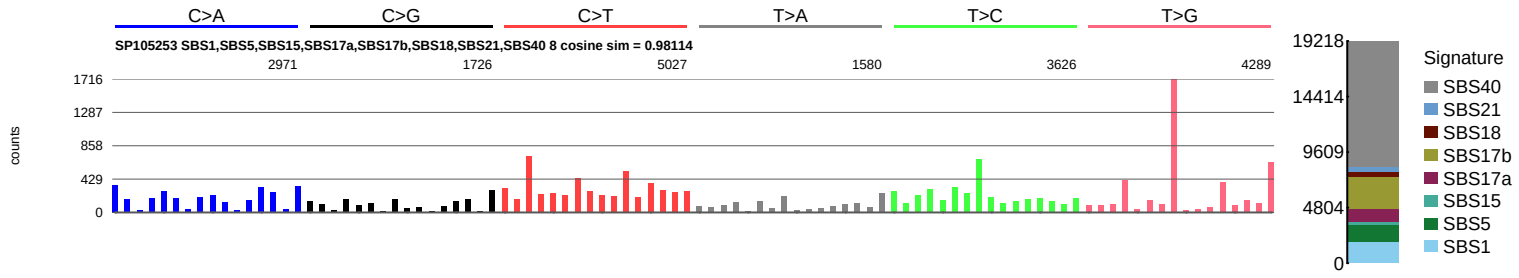
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.98125
2752
1874
4825
1928
3315
4523
counts
1688
1266
844
422
0
19218
14414
9609
4804
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98116
2905
1766
5390
1624
3182
4348
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1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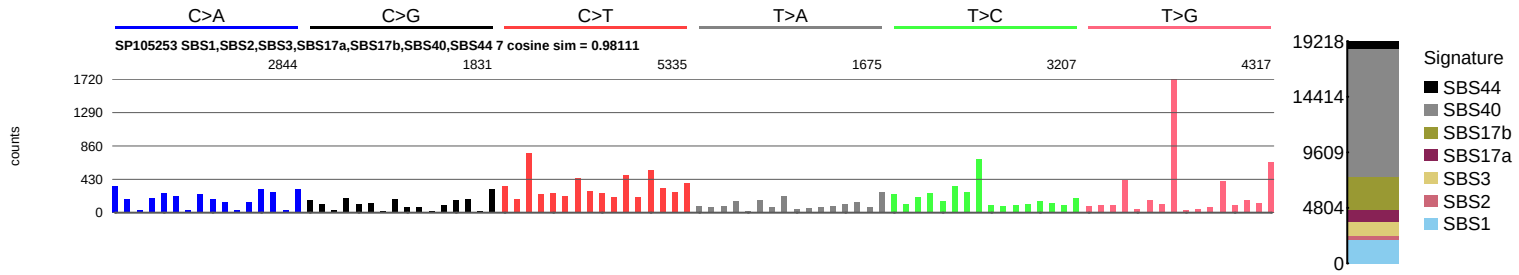
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98114
2971
1726
5027
1580
3626
4289
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98111
2844
1831
5335
1675
3207
4317
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
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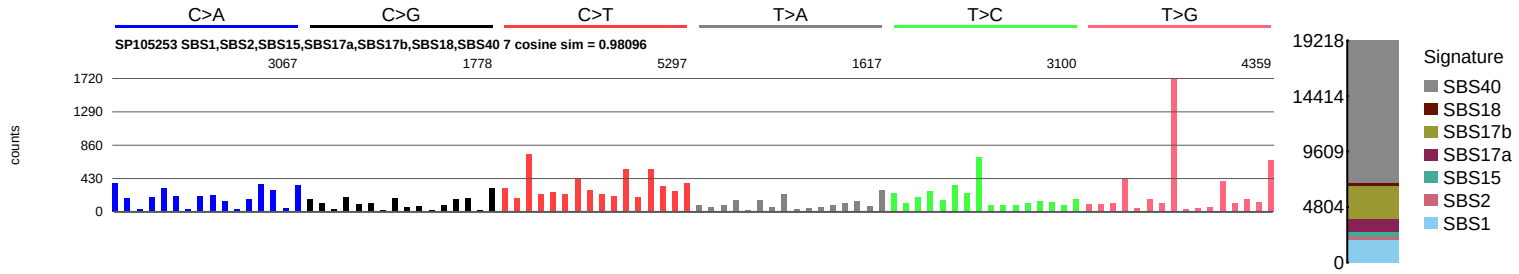
SBS44
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1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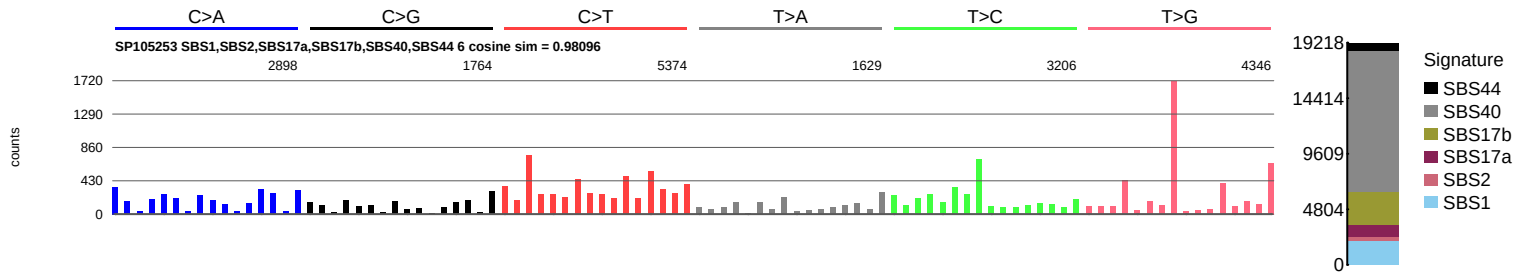
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98096
3067
1778
5297
1617
3100
4359
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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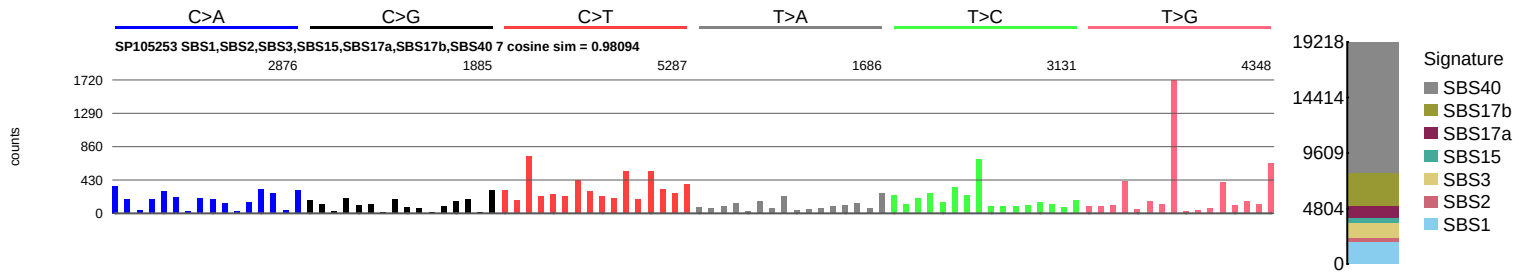
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98096
2898
1764
5374
1629
3206
4346
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS2
SBS1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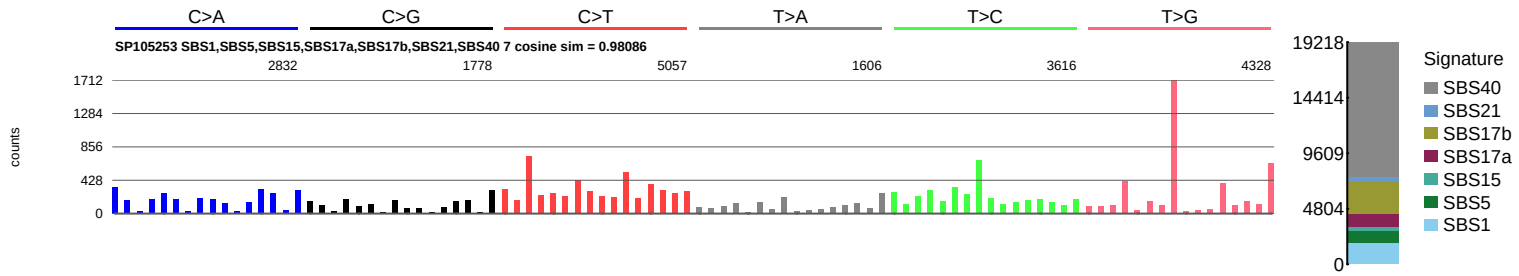
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98094
2876
1885
5287
1686
3131
4348
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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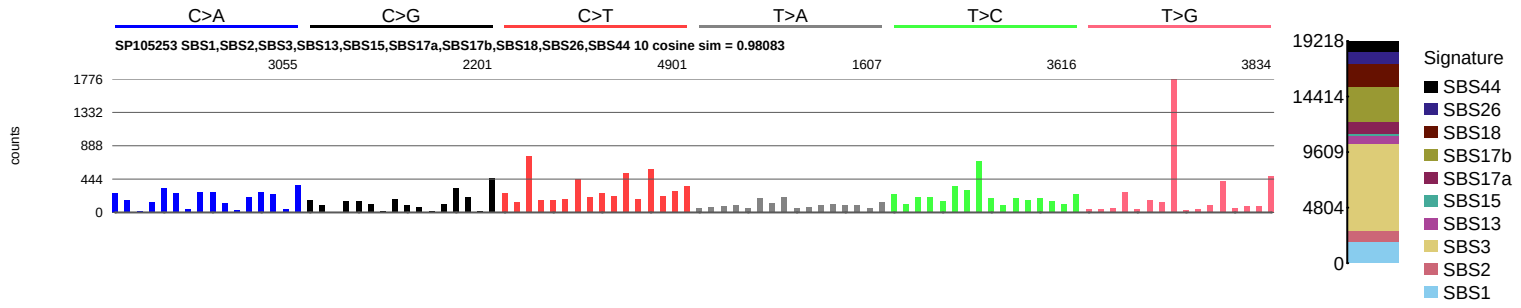
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98086
2832
1778
5057
1606
3616
4328
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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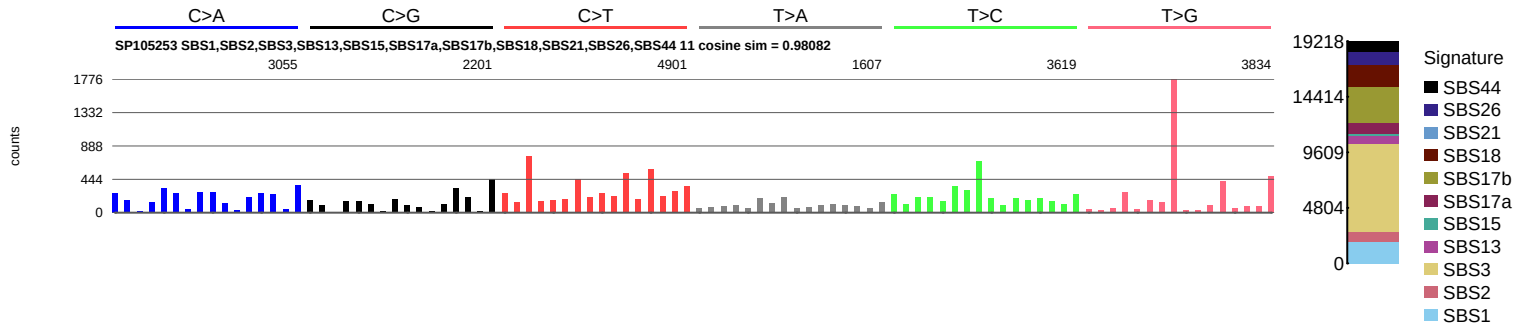
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98083
3055
2201
4901
1607
3616
3834
counts
1776
1332
888
444
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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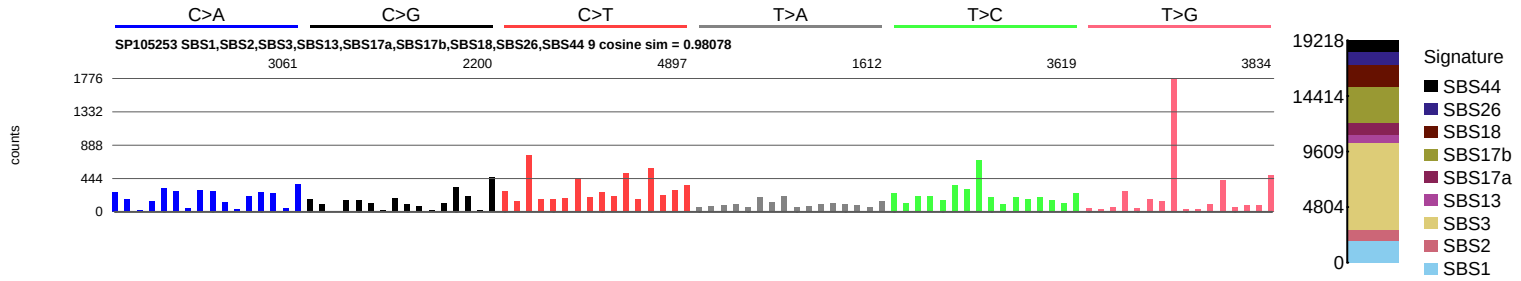
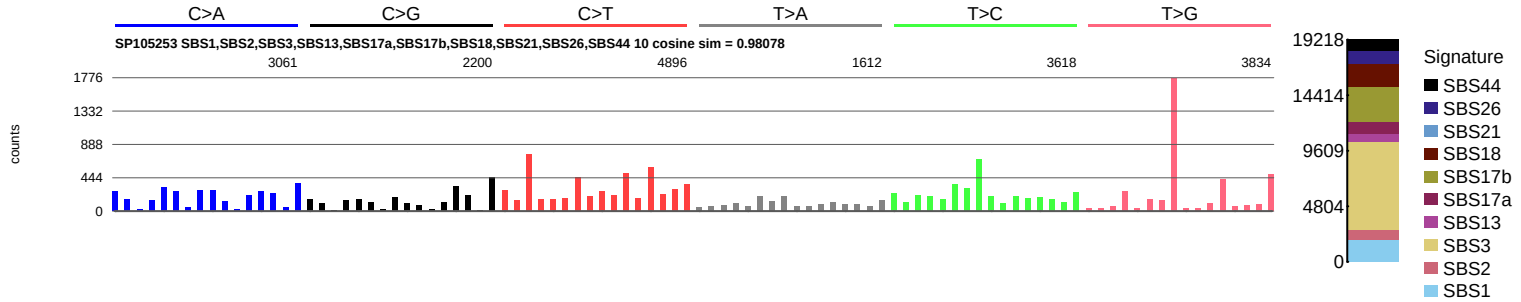
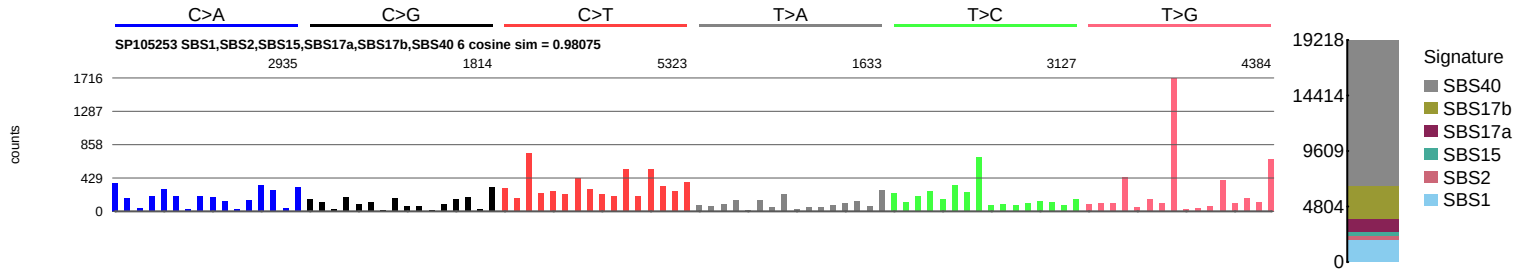
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98082
3055
2201
4901
1607
3619
3834
counts
1776
1332
888
444
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98078
3061
2200
4897
1612
3619
3834
counts
1776
1332
888
444
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98078
3061
2200
4896
1612
3618
3834
counts
1776
1332
888
444
0
19218
14414
9609
4804
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98075
2935
1814
5323
1633
3127
4384
counts
1716
1287
858
429
0
19218
14414
9609
4804
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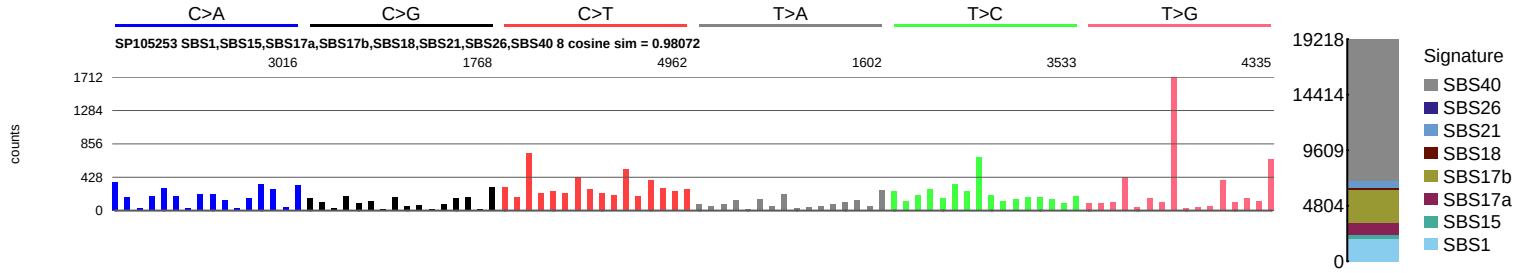
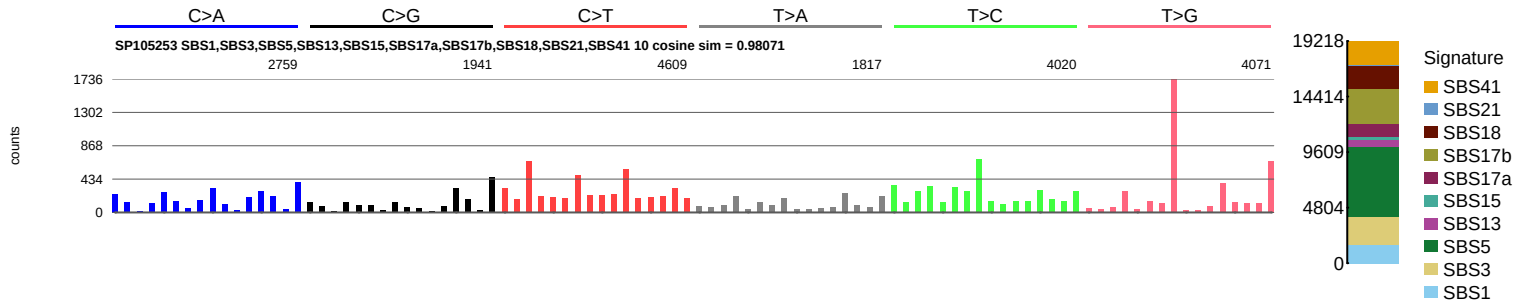
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98072
3016
1768
4962
1602
3533
4335
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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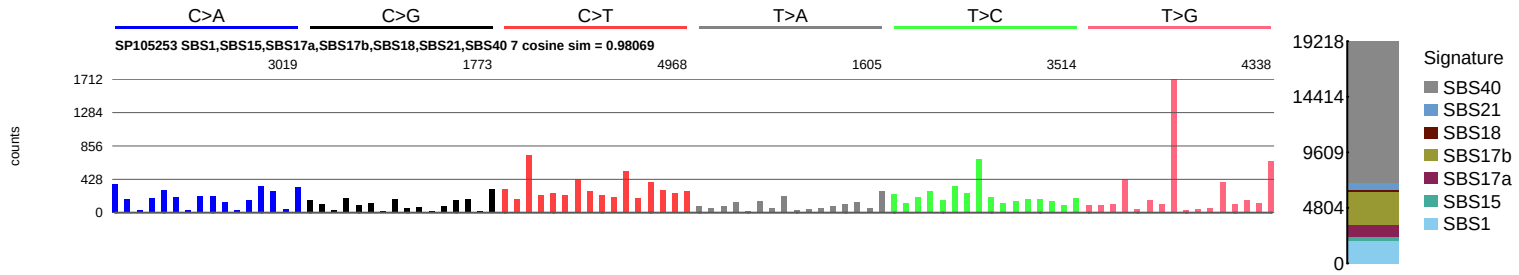
T>G
SP105253 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.98071
2759
1941
4609
1817
4020
4071
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
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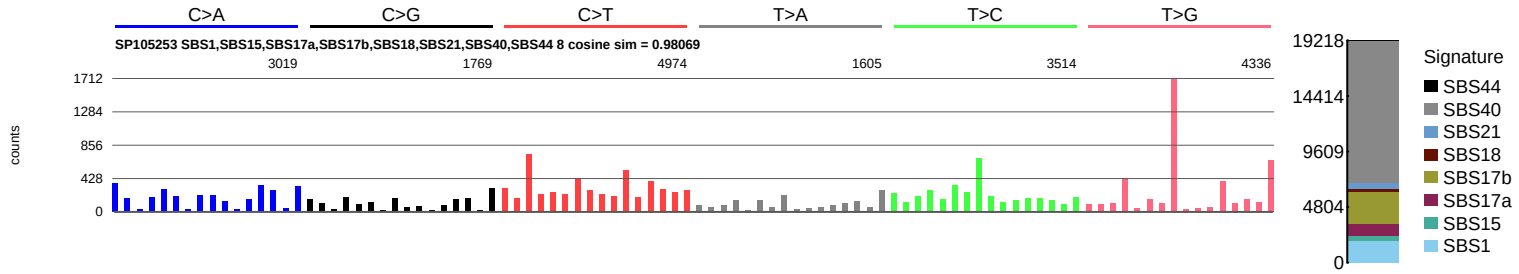
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98069
3019
1773
4968
1605
3514
4338
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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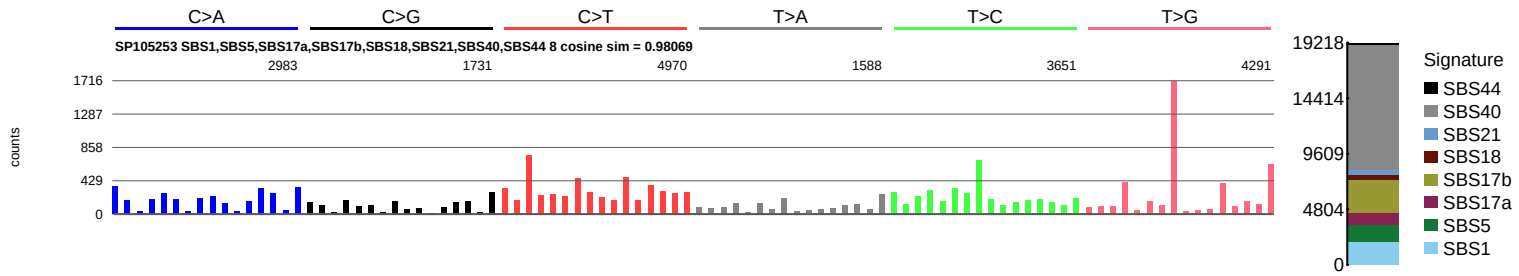
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98069
3019
1769
4974
1605
3514
4336
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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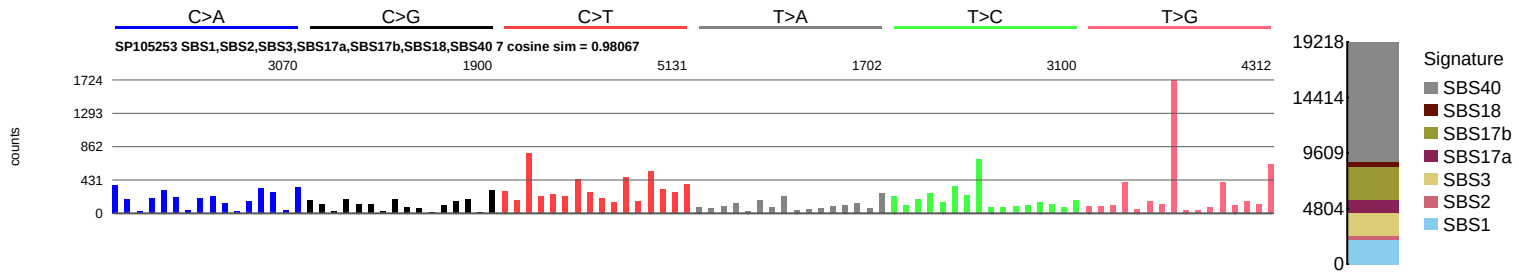
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.98069
2983
1731
4970
1588
3651
4291
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
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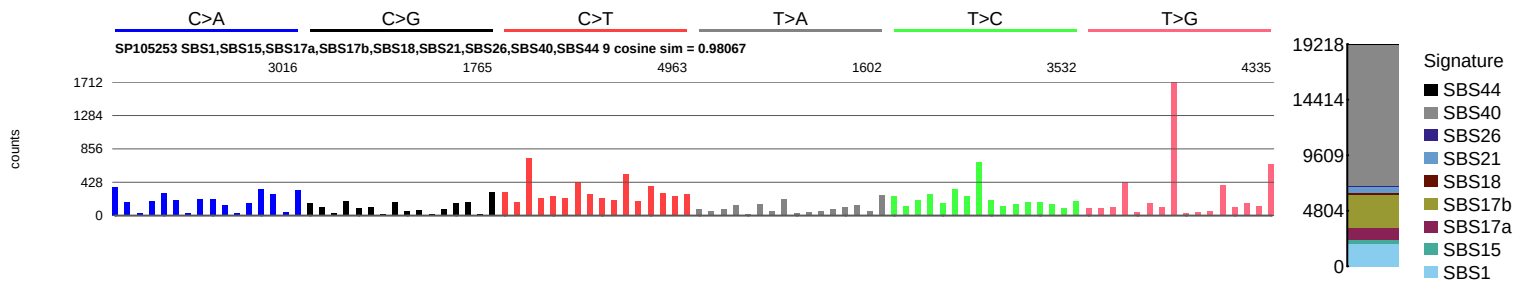
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98067
3070
1900
5131
1702
3100
4312
counts
1724
1293
862
431
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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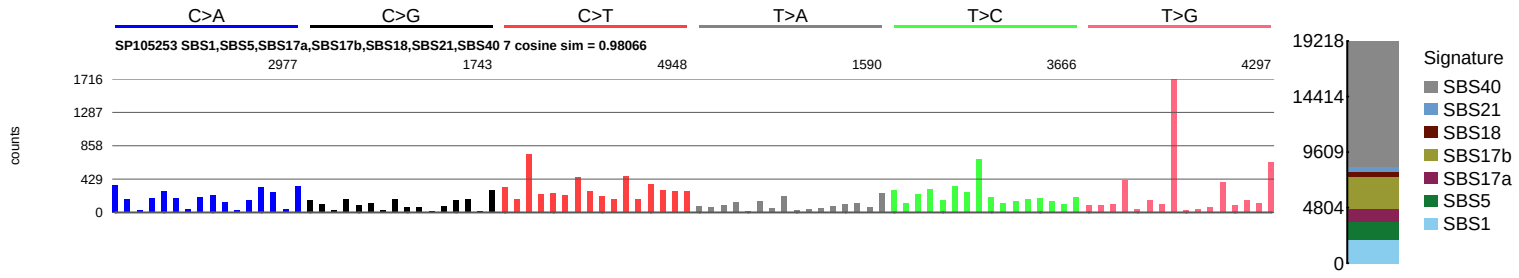
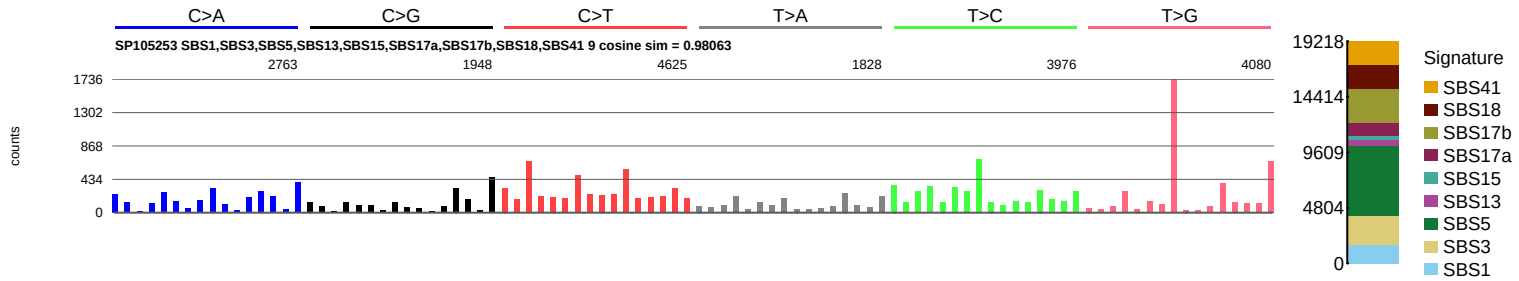
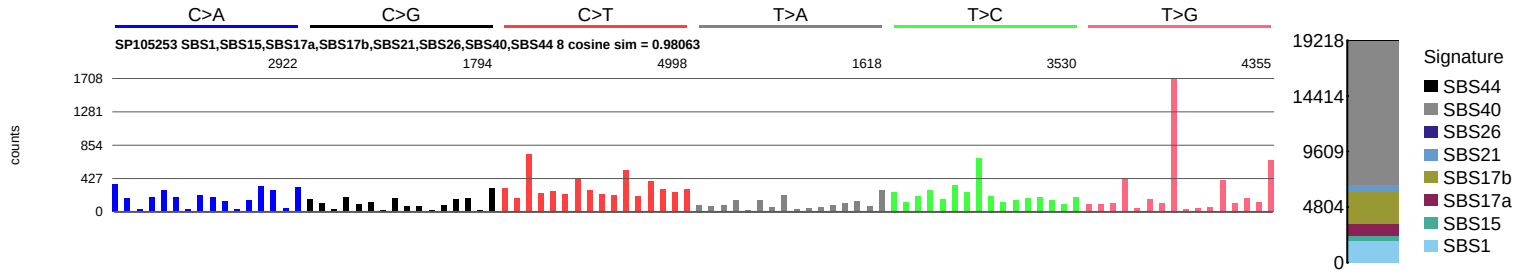
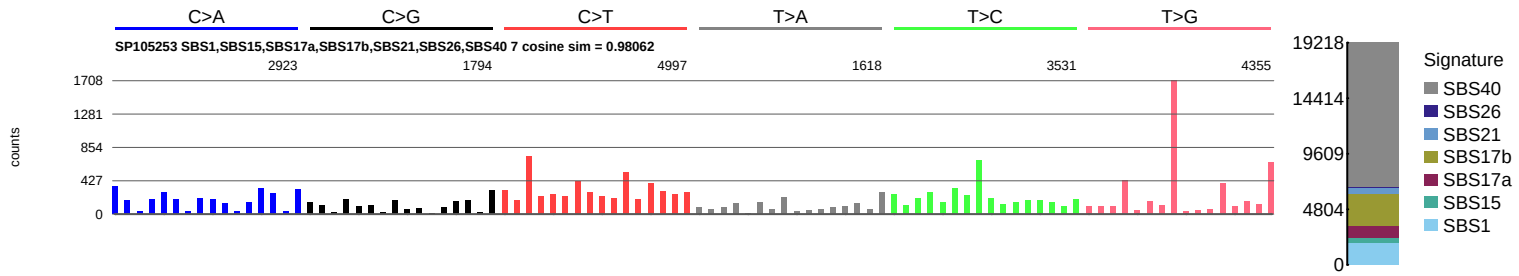
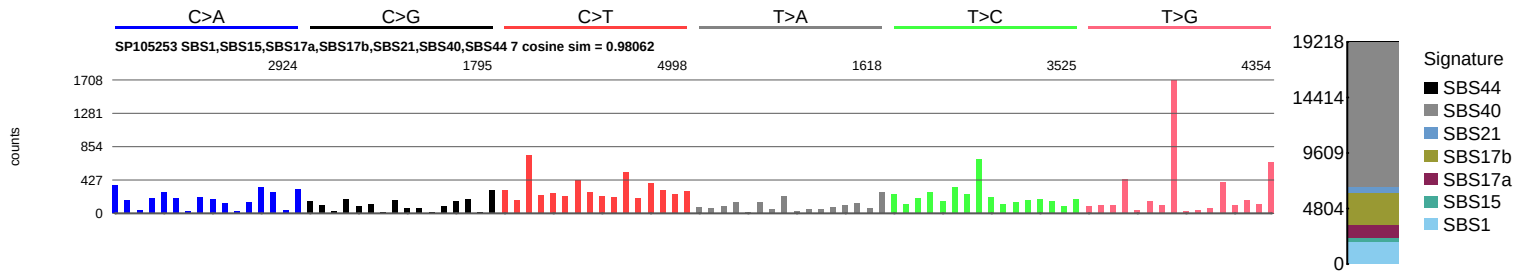
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98067
3016
1765
4963
1602
3532
4335
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

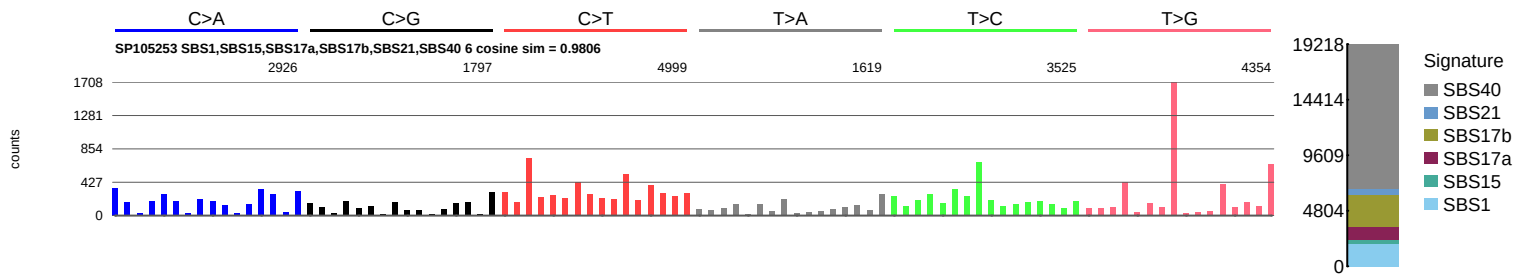

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98066
2977
1743
4948
1590
3666
4297
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

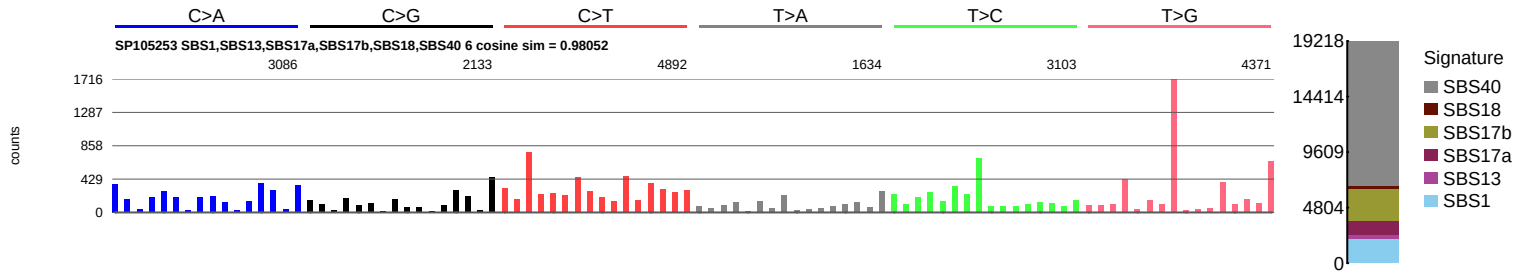

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.98063
2763
1948
4625
1828
3976
4080
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

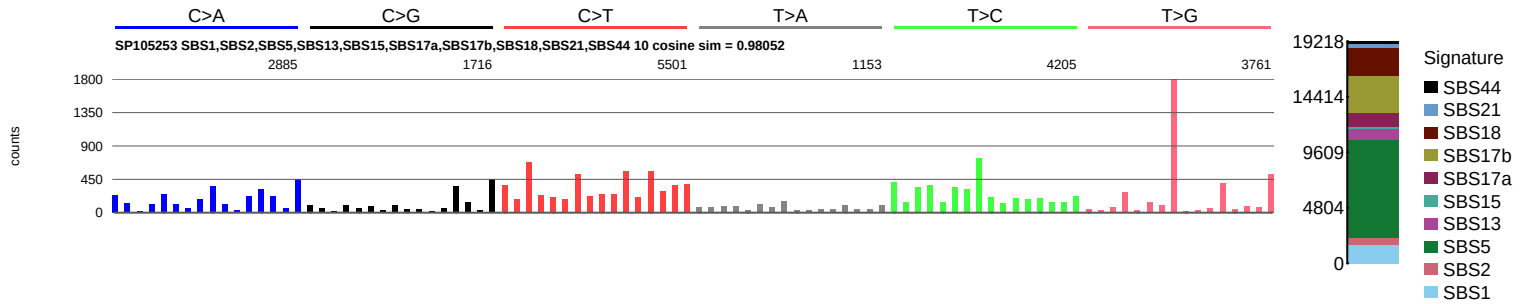

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98063
2922
1794
4998
1618
3530
4355
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

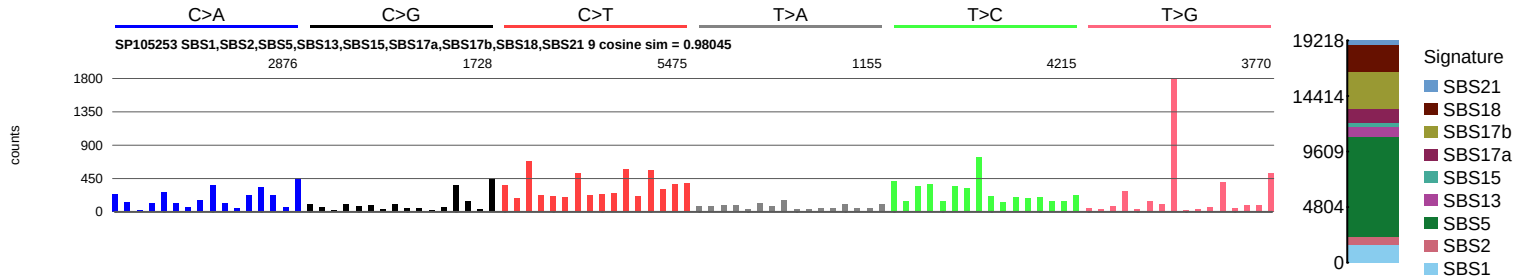

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98062
2923
1794
4997
1618
3531
4355
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1

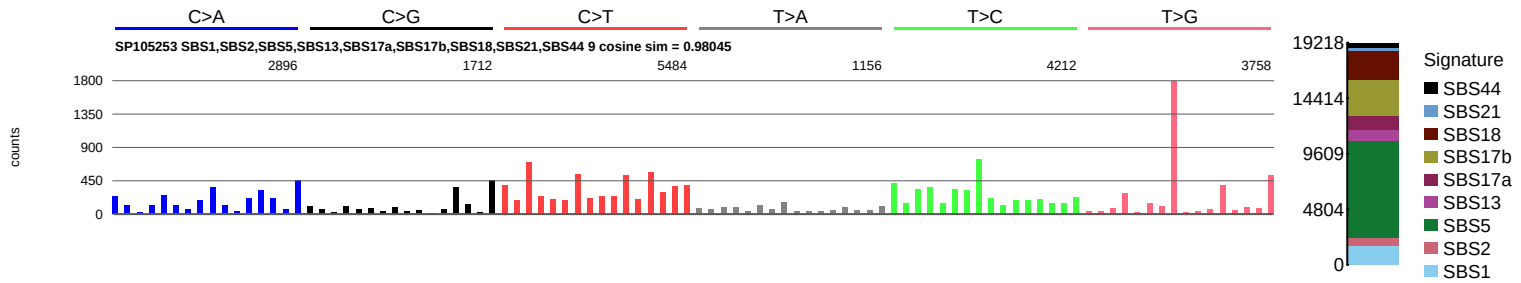

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98062
2924
1795
4998
1618
3525
4354
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

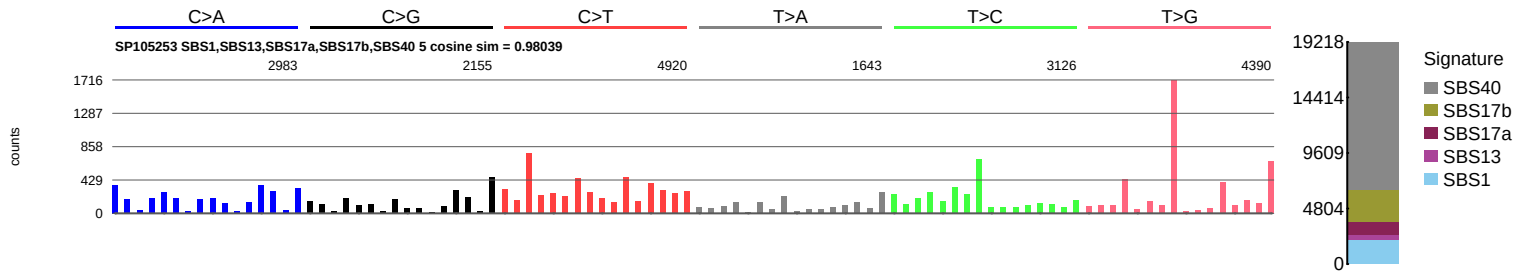

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.9806
2926
1797
4999
1619
3525
4354
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1

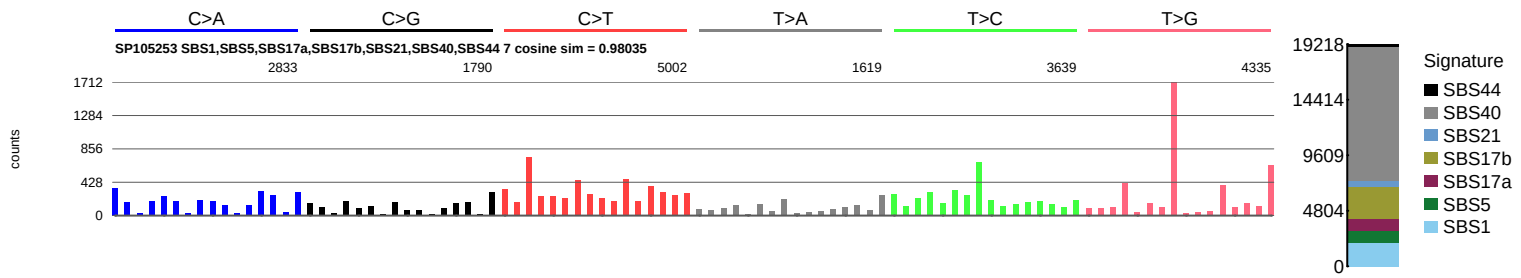

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98052
3086
2133
4892
1634
3103
4371
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98052
2885
1716
5501
1153
4205
3761
counts
1800
1350
900
450
0
19218
14414
9609
4804
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98045
2876
1728
5475
1155
4215
3770
counts
1800
1350
900
450
0
19218
14414
9609
4804
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98045
2896
1712
5484
1156
4212
3758
counts
1800
1350
900
450
0
19218
14414
9609
4804
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS13,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98039
2983
2155
4920
1643
3126
4390
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS1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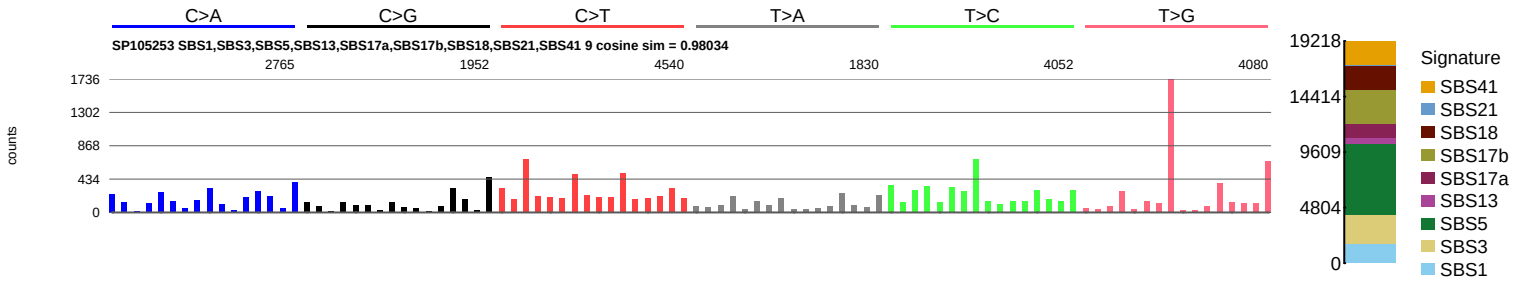
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98035
2833
1790
5002
1619
3639
4335
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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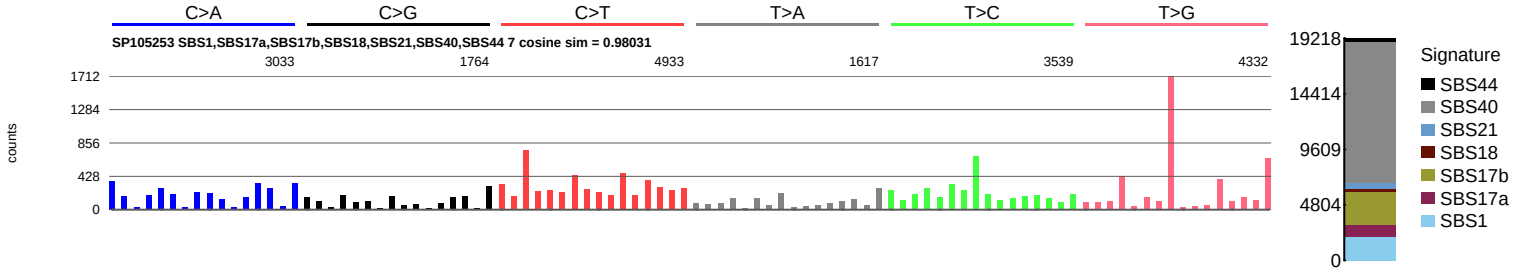
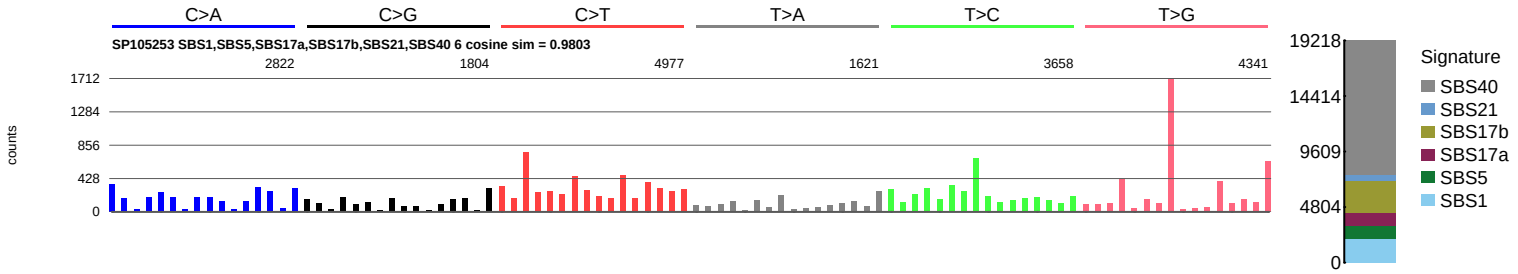
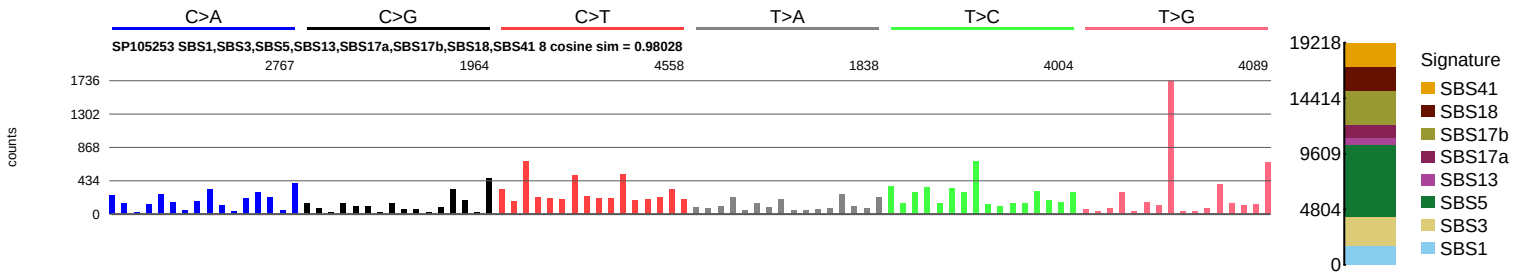
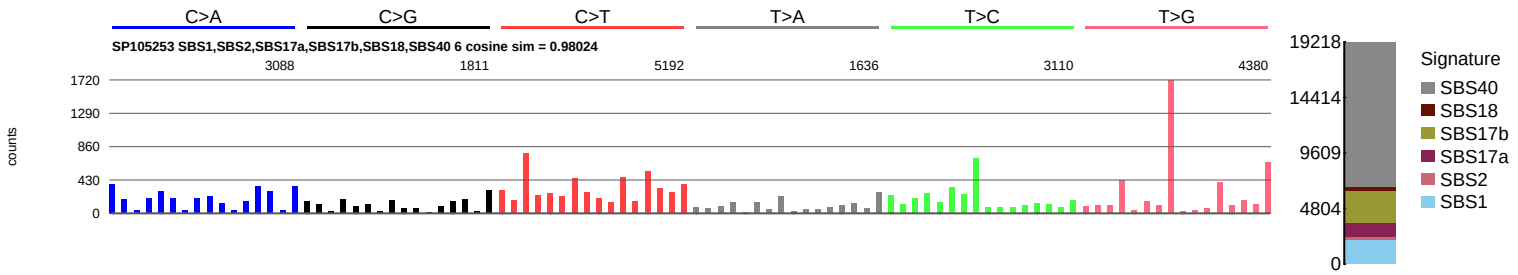
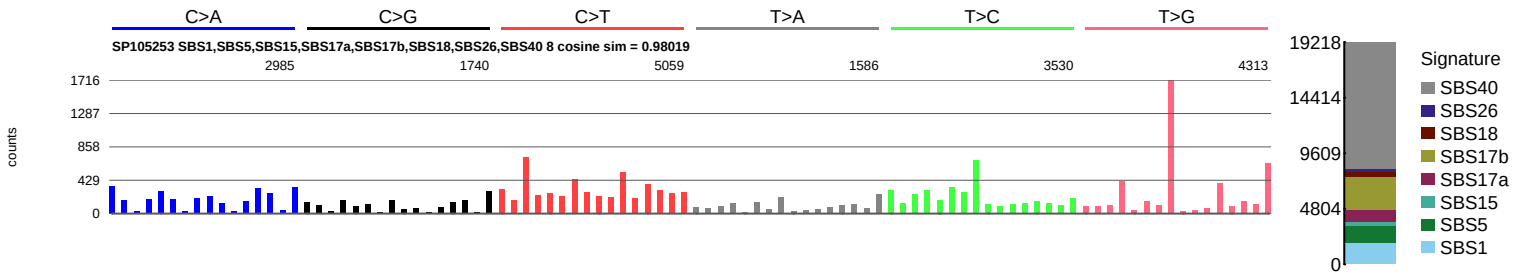
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98034
2765
1952
4540
1830
4052
4080
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

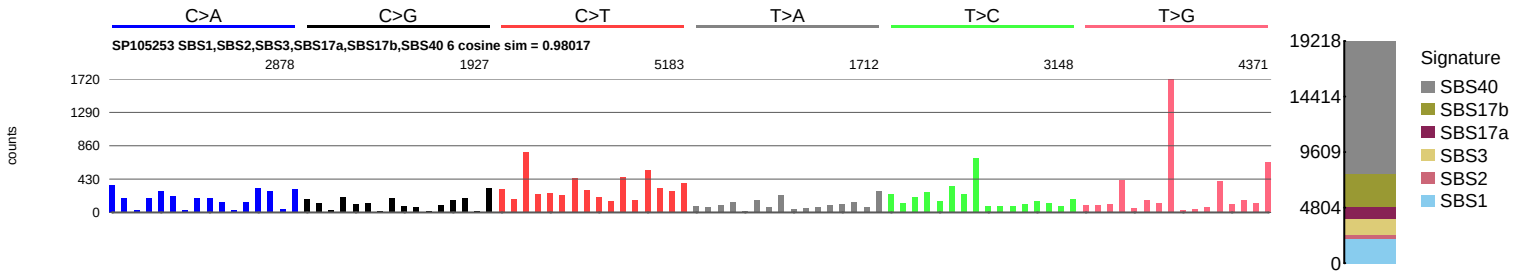

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98031
3033
1764
4933
1617
3539
4332
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

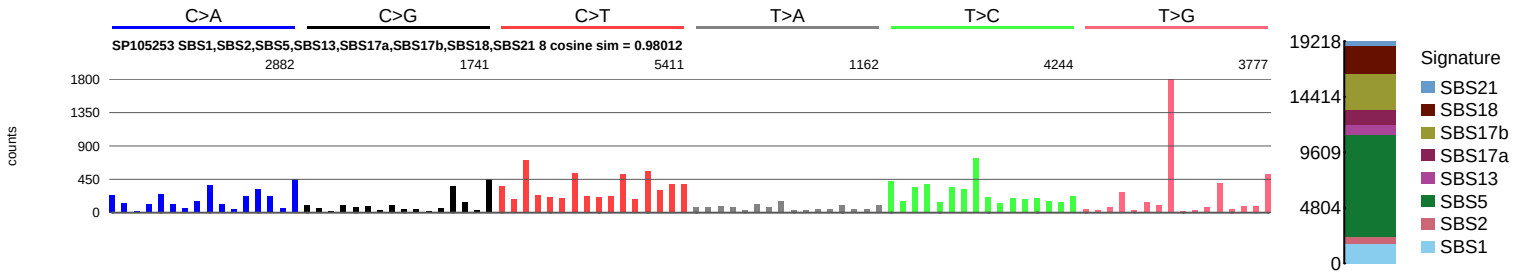

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.9803
2822
1804
4977
1621
3658
4341
counts
1712
1284
856
428
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

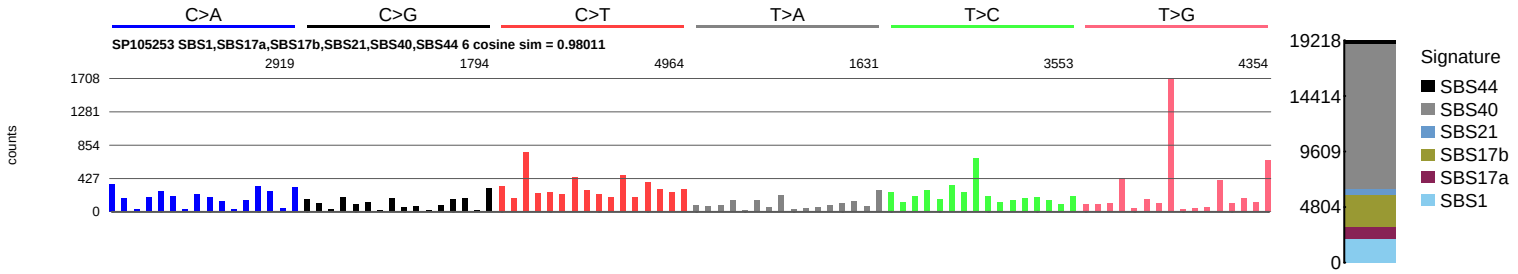

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98028
2767
1964
4558
1838
4004
4089
counts
1736
1302
868
434
0
19218
14414
9609
4804
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

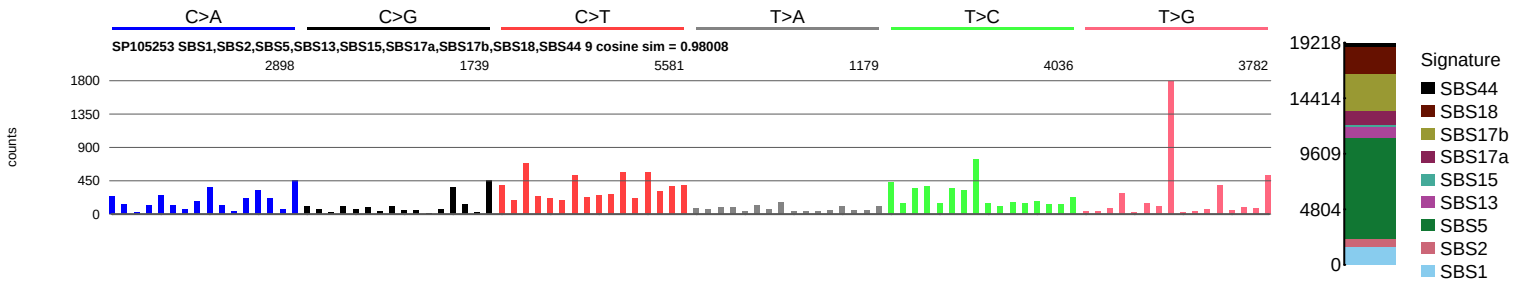

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98024
3088
1811
5192
1636
3110
4380
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS2
SBS1

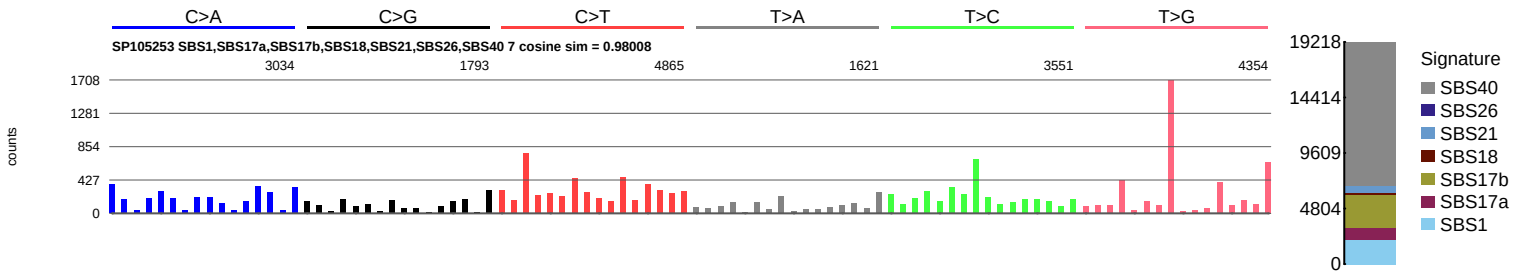

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98019
2985
1740
5059
1586
3530
4313
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

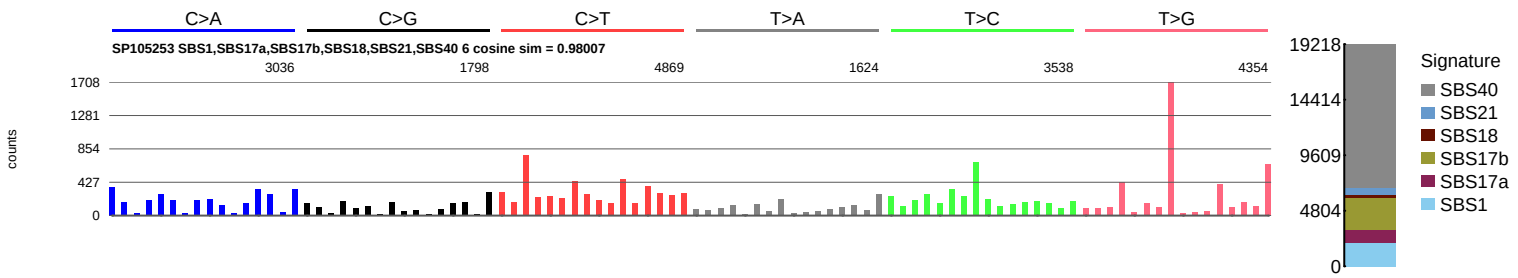

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98017
2878
1927
5183
1712
3148
4371
counts
1720
1290
860
430
0
19218
14414
9609
4804
0
Signature
SBS40
SBS17b
SBS17a
SBS3
SBS2
SBS1

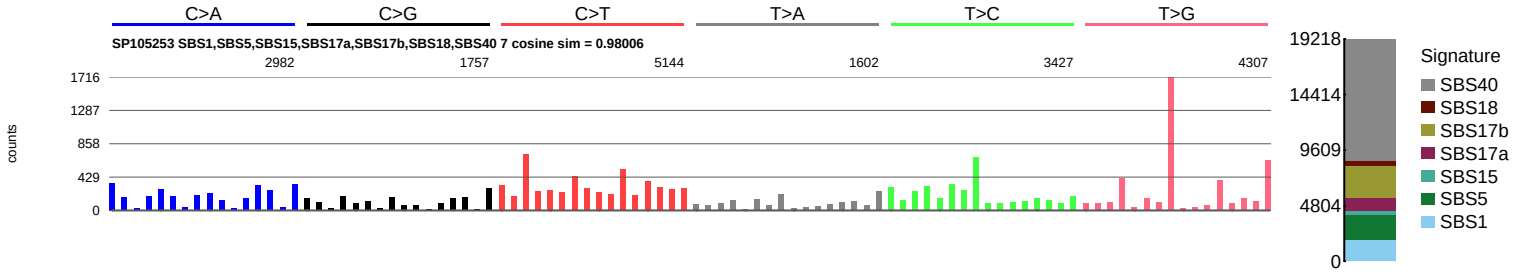

C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98012
2882
1741
5411
1162
4244
3777
counts
1800
1350
900
450
0
19218
14414
9609
4804
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98011
2919
1794
4964
1631
3553
4354
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98008
2898
1739
5581
1179
4036
3782
counts
1800
1350
900
450
0
19218
14414
9609
4804
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98008
3034
1793
4865
1621
3551
4354
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98007
3036
1798
4869
1624
3538
4354
counts
1708
1281
854
427
0
19218
14414
9609
4804
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP105253 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98006
2982
1757
5144
1602
3427
4307
counts
1716
1287
858
429
0
19218
14414
9609
4804
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1